

Goldbach Programme

Master File

Research knowledge base for future models

Factorization-graph programme for the binary Goldbach conjecture: the canonical selector $p_0(N) = \min\{p \in \mathbb{P} : p \geq N/2\}$, the stopped traversal, Canonical Root Dynamics (CRD), the complete theorem registry, the current proof frontier, and the full legacy failure log together with the post-v4.81 continuation through 2026-09-07H. The source checkpoints are preserved losslessly in appendices, without duplicating an already identical prefix.

This is not a publication draft. It is a project bible. It contains no marketing language and no suggestion that Goldbach is proved anywhere. Binary Goldbach is **OPEN**. Universal success of the canonical root is **OPEN**. The *old finite-selected-transcript* p_0 architecture is **SUPERSEDED**; complete-support and multigeneration p_0 reachability remain live.

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Chapter 1

AI Navigation Map

HOW TO USE THIS DOCUMENT

If you are a model receiving only this file, read in this order.

1. **Chapter 2** (three pages) — what is true right now. Do not skip it; several beliefs that were live in earlier versions of the corpus are dead.
2. **Chapter 3** — the notation is fixed. One symbol has one meaning throughout. Do not silently re-define $h, m, Q, R, T, U, J, \eta, \nu, \gamma$.
3. **Chapter 4** plus §11.8 — the theorem registries. Before using any result, look up its status and dependencies there.
4. **Chapter 9** — the only place where new work should start.
5. **Chapters 13 and 14** — read *before* proposing anything. The legacy log has 239 entries and the post-v4.81 continuation is integrated through run 2026-09-07H, including the rank/mirror audit and the same- N finite-view results.

Three rules that are violated most often.

1. Failure of the canonical traversal is *not* global Goldbach failure. Never invoke a theorem whose hypothesis is $\text{Good}_N = \emptyset$ under p_0 -failure alone. FAIL-UPPERHALF-SCOPE-001
2. A successful-world witness does not refute a statement conditioned on a closed failed world; a noncanonical root does not refute a canonical statement. FAIL-F1-DOMAIN-MIX-001
3. Never quote a count without its domain. The unit-complement row $N - r = 1$ is a separate class and is excluded everywhere. FAIL-UNIT-COMPLEMENT-001, FAIL-CENSUS-DOMAIN-001

Never raise a status. **OPEN**, **CONJECTURE**, **EMPIRICAL PATTERN**, **VERIFIED COMPUTATION** and **PROVED** are different things and stay different. A finite scan with zero counterexamples is never promoted to a theorem. If you prove something, add a new registry row; do not edit an old one.

IF YOU NEED X $\rightarrow$ GO TO SECTION Y

If you need ...	Go to ...
definitions (all of them)	Chapter 3: notation registry §3.1, objects §3.2, valuation layer §3.7
p_0 — definition, geometry, why it matters	§3.6 (definition), Chapter 6 (whole programme), §6.5 (what is and is not special)
BFS / stopped traversal — exact algorithm	§3.3 (definition), §3.4 (runnable reference implementation), §3.5 (historical variants)

If you need ...	Go to ...
EAC — exceptional autonomous component	§3.8; orientation warning in §3.8; prior-art boundary §16.2
canonical failed world	§3.9 (definition $R_N(r)$, closure, unique source); fixed point B_N^{low} in §3.10
CRD — Canonical Root Dynamics	Chapter 8; epistemic convention §8.1; architecture wall §8.2
C0 / C0a — small-gap canonical obligation	§8.3
C0b — canonical reflection-defect ledger	§8.4; branch targets TI/TD/TS/TF in §8.5
P1 / P1BR — branchwise production upgrade	§9.4
L1 — actual layer realisation	§9.5
E1 — escape or coupling	§9.6
W1 — simultaneous clean witnesses	§9.7
K1 — source-sensitive capacity	§9.8
X1 — canonical failure exclusion	§9.9
RPD — repeated prime-power redundancy	§9.10
computational evidence — all of it, with ranges	Chapter 12; reproducible generator in Appendix A
failed approaches — curated narrative	Chapter 13
full failure log / later source record — 239 legacy entries plus post-v4.81 deltas	Chapter 14; index §14.1
literature	Chapter 16; what each external theorem does and does not give
current frontier	§11.6 first (binding 2026-09-07H override); then Chapter 11, Chapter 10, and the historical frontiers in Chapter 9.
exact product locks (T_S , T_R , overlap, orthogonality, full- Q)	§5.3
residue no-gos ($q \geq 5$ saturation, mod 3, topology)	§5.4
FRESH/SMOOTH , smoothness tax, first-front conservation and split	§7.3
lossless prime-power root ledger , collision budget, weighted cut	§7.4
Exact-D pair compensation , primitive normal form, gap filters, branch $k = 1$	§7.5
sparse geometric core lock (FFS layer)	§7.6

If you need ...	Go to ...
surviving lemmas of killed theories (CMFT, CGST, CSD, CAPT, RSD, rough- P_2 , H1-FP)	§13.3
rejected literature — what was tried and why it failed	§16.3 ; the 52-source audit §16.3.2 ; the frontier salvage audit §16.4.1 ; named missing lemmas §16.5
scope classes A / B / C / D — is this really about p_0 ?	§5.1 ; full classification table there
what to work on next — triage rules and tiers	§15.1 ; the five best moves in §18.4
Euler defect — and why its normalisation matters	§5.7
shadow / retention arithmetic	§5.5
path invariants (Bézout, winding)	§5.6
$T = UJ$, Euler defect , η , ν	§3.7 , §5.7
D_q , M_q , FRESH/SMOOTH	§7.1
B_{abs} , B_{re} , incidence ledger	§7.2
genealogical defect rank κ_{gen}	§6.1
boundary filtration	§6.2
phase diagram $\theta = h/G$, Type I/II/III	§6.3
bidirectional rank ρ_0	§6.4
six special N (128, 1718, 1862, 1928, 2200, 6142)	§6.6
escape margin G_{esc}	§6.7
post-CRD successor theories (CMFT, CGST, CSD, CAPT, RSD)	§13.2
parity barrier / why it is the real wall	§13.4
historical / superseded results	Chapter 17
what not to try again	Chapter 18

Status vocabulary

Status	Meaning
PROVED	A proof is reproduced here or was independently reconstructed from definitions in the latest hostile audit, and no later correction is known.
PROVED WITH REPAIR	Proved after an explicit repair to hypotheses or to a step. The repair is always stated.

Status	Meaning
AUDITED IDENTITY	An exact identity that survived independent reconstruction on a declared domain, but has not been promoted to canonical PROVED .
VERIFIED COMPUTATION	An exact finite computation with a stated range and provenance. <i>Never</i> evidence for a universal statement.
EMPIRICAL PATTERN	A pattern observed in data, with no causal isolation. Not a theorem, not evidence for one.
CONJECTURE	A precisely stated conjecture, usually with computational support.
OPEN	A precisely stated target for which no proof is known.
KILLED	Killed by a proof, a domain correction, or an exact counterexample. Not usable as a lemma.
SUPERSEDED	Retained historically; not a live target. Its correct lemmas may still be used, its programme may not be restarted.
SCOPE-CORRECTED	The statement survives on a narrower domain; what died is a use outside that domain. Used inside the imported failure log, where it is the entry-level analogue of PROVED WITH REPAIR , and for imported results whose domain was narrowed.
<i>Two statuses carried over verbatim from the sources, so that no import is silently promoted:</i>	
PROVED – PENDING RED TEAM	The source states a complete proof but records that the result has not been adversarially audited ([BIB] PROVED-PENDING-REDTEAM). Stronger than CONJECTURE , weaker than PROVED . Do not upgrade it by citing it.
NEEDS VERIFICATION	The source states the result but flags an unverified step or an unverified uniform dependence ([BIB] NEEDS VERIFICATION). Treat as CONJECTURE until the flagged step is discharged.

Status mapping used when importing from the sources. [BIB]/[CORE]
 PROVED → **PROVED**; PROVED-PENDING-REDTEAM → **PROVED – PENDING RED TEAM**;
 PROVED-PENDING-INDEPENDENT-AUDIT → **PROVED – PENDING RED TEAM**; NEEDS
 VERIFICATION → **NEEDS VERIFICATION**; TRANSFERRED-PROVED → **PROVED**; SOURCE
 CLAIM: NEW-PROVED and SOURCE: PROVED-PENDING → **PROVED – PENDING RED TEAM**;
 CONDITIONAL → **CONJECTURE** with the condition stated; COMPUTATIONAL/EXPERIMENT
 → **VERIFIED COMPUTATION**; HEURISTIC → **EMPIRICAL PATTERN**; WORKING → **OPEN**;
 REJECTED/DISPROVED/ABANDONED → **KILLED**; CORRECTED → **SCOPE-CORRECTED**; RETIRED →
SUPERSEDED. The mapping never raises a status.

Chapter 2

Current State of the Project

Binding state after the 2026-09-07H continuation. The project studies binary Goldbach through the factorization graph $p \rightarrow q$ when $q \mid N-p$, the canonical selector $p_0(N) = \min\{p \in \mathbb{P} : p \geq N/2\}$, and now the full rank/mirror spectrum of upper roots. Runs E–H sharpen the scope wall: for every fixed rank essentially every odd first-row integer can occur; even the pair (rank, exact mirror) does not determine the eventual verdict; and arbitrarily prescribed finite prime-label valuations can be synchronised across a common same- N prefix $p_0, \dots, p_K$. Conversely, complete factorisations cannot keep reusing one fixed finite prime alphabet throughout a sufficiently long early-rank prefix. These facts do *not* control the moving closed BAD basin B_N . The binding open problem is therefore to link the actual root position/selecting rule to the *complete, N -dependent* BAD landscape. No proof of binary Goldbach, universal p_0 reachability, or universal composite-success for any noncanonical survivor rank is known.

2.1 Top-level status

Statement	Status	Where
Binary Goldbach: every even $N \geq 4$ is a sum of two primes	OPEN	external; §16.2
Every even $N \geq 6$ with $N/2$ composite has $\text{Good}_N \neq \emptyset$ where the canonical traversal from p_0 terminates in SUCCESS	OPEN	Chapter 9
The old p_0 finite-selected-transcript architecture	SUPERSEDED	§8.2; complete-support reachability remains open
Canonical failure has an exact fixed-point characterisation	PROVED	§3.10
Exactly six $N \leq 6 \cdot 10^4$ have $B_N^{\text{low}} \neq \emptyset$	VERIFIED COMPUTATION	§6.6
p_0 is a genuinely distinguished vertex (boundary, genealogy, uniqueness)	PROVED	Chapter 6
A universal dynamic explanation of the empirical reliability of p_0	OPEN	§11.7
Every upper root with composite complement fails exactly when its complete first support lies in the maximal active closed BAD basin B_N	PROVED	§11.3

Statement	Status	Where
For each fixed rank k and odd mirror $m \geq 3$, some even N realises $N - p_k(N) = m$; fixed (k, m) fibres are nevertheless finite and effective	PROVED	§11.4
The ranks 10, 14, 26, 28, 30, 62, 63, 71, 74 are universally safe on the composite-complement domain	OPEN	exact bounded-mirror evidence only; §11.4

2.2 What is actually proved

The following are theorems with reproduced or reconstructed proofs. They are safe to use as lemmas; their exact statements and hypotheses are in Chapter 4 and their proofs in Chapters 5–8.

1. **Canonical gap equivalence** (§3.6). For an upper root $r = x + h_r$: $r = p_0 \iff [x, r] \cap \mathbb{P} = \emptyset \iff h_r < G_r$. This is the whole content of “ p_0 is special” at the level of the boundary, and it is categorical, not quantitative.
2. **Strict non-reuse and the reflection gate** (§3.6). Every prime $\ell \geq h$ divides at most one element of the strict reflected block $\mathfrak{B}(p_0)$; every upper mirror c_j is composite. The gate gives compositeness only — *not* activation ($N = 242$).
3. **Genealogical defect rank** (§6.1). $\kappa_{\text{gen}}(p_k) = k$ on the non-diagonal domain, the defect ledger is a lossless code of local prime history, and p_0 is the unique reflection-complete upper root.
4. **Boundary filtration** (§6.2). The reflected blocks of consecutive upper roots nest with a shift, and the gcd-coupling matrix of level k embeds as a submatrix of level $k + 1$.
5. **Phase trichotomy** (§6.3). $m_0 - p_- = G - 2h$ splits the canonical cell into Type I / II / III with exact consequences for immediate success.
6. **Shadow arithmetic** (§5.5). The prime-gap shadow identity $r \mid N - p \iff r \mid \delta - p$, retention–escape $\text{PF}(N - q) \cap Q = \text{PF}(d_q)$, minimum-factor escape, shadow cycle localisation.
7. **Path invariants** (§5.6). The full path Bézout invariant $K_t p_t - C_t x = (-1)^t h$, the winding normal form, the $\leq 2\tau(h)$ reduced-state bound, no-bad-return-to- p_0 , and the global two-sheet collision budget.
8. **Valuation layer** (§5.7). $T_S = U_S J_S$ and the layered Euler defect $\eta(S) + \nu(S) \geq |S| + |\text{Src}(S)|$, canonically $\eta(R) + \nu(R) \geq |R| + 1$.
9. **Post-root factorization ledger** (§7.1, §7.2). $D_q = \gcd(m_0, N - q) = \gcd(B_q, |2h - q|)$, the FRESH/SMOOTH split, and the incidence identities IL1–IL6.
10. **CRD fixed point** (§3.10). $B_N^{\text{low}} = \bigcap_i B_i$ and p_0 fails $\iff N - p_0$ is composite and $\text{PF}(N - p_0) \subseteq B_N^{\text{low}}$.
11. **C0a** (§8.3). The small-gap canonical obligation, via the residual argument.
12. **Exact product locks** (§5.3). The multi-source lock $T_S \equiv (-1)^{|S|} \prod_{\text{Src}} d \pmod{N}$, the rooted full-front lock $T_R \equiv (-1)^{|R|} p_0$, the source-overlap identity, the orthogonality of disjoint child-closed worlds, and the full- Q ledger and production bounds.
13. **Residue no-gos** (§5.4). The residue-only route saturates for every $q \geq 5$; modulus 3 is the sole exception; and topology alone provably cannot supply incremental J .
14. **The rooted first-front layer** (§7.3–§7.5). Conservation and split identities (source status **PROVED – PENDING RED TEAM**), and — fully **PROVED** — the actual prime-power collision budget, the exact weighted cut identity, joint D_q, D_r rigidity, Exact-D pair compensation with its primitive normal form, and the two canonical gap filters.
15. **Scope-corrected partial transcript programming** (§10.2). Prescribed finite divisibilities/valuations can be programmed, but this does *not* control every unprescribed prime factor of a complete BFS row; the old blanket FBTU/finite-local wall is not a theorem.
16. **Closed-support counting and height** (Chapter 10). A fixed prime alphabet supports only finitely many closed active BAD worlds; Matveev gives an unrestricted closed-support height

inequality, an effective bounded-core threshold, and pointwise growth of the largest core label.

17. **Shallow asymptotic structure** (Chapter 10). In the $h = 1$ family, complete radius-two BADness holds for almost every prime start, simultaneously for a growing same- N prefix of nearby roots; short successful paths obey simultaneous label- and cofactor-product barriers.
18. **Exact $h = 1$ conjugacy and canonical embedding** (§10.5). For $N = 2(P - 1)$ the complete non-direct canonical BFS from P is exactly the BFS from root 2 after the initial-root replacement. Any finite hereditary first-support graph/valuation obstruction seen at an arbitrary upper BAD root can be embedded, at the same gap, into infinitely many canonical worlds; therefore such an invariant cannot explain universal p_0 reliability.
19. **Low-gap complete-support restrictions** (§10.6). If $0 < h < q = \min \text{PF}(N - r)$ and no GOOD vertex appears through depth 2, at least three distinct nonroot labels occur by depth 3. Reciprocal returns satisfy $b \geq 3q + 2h$; monomial returns force an exponent drop; the remaining three-label obstruction is reduced to a closed three-prime world or an explicit prime-power Diophantine system.
20. **Low-gap singleton strengthening from run D** (§11.2). The number $2 \cdot 3^u - 1$ is never a proper integer power; under the explicit complete-support hypotheses of the low-gap singleton theorem, either a GOOD vertex is reached by depth 2 or at least four nonroot labels occur by depth 3. In the $q = 3, h = 1$ canonical power family this gives the corresponding unconditional four-label alternative. A monomial return now has order at least 3 and hence $4 \leq v \leq u - 1$.
21. **All-upper-root basin and rank/mirror layer** (§11.3–§11.4). For any upper root r with composite mirror $m = N - r > 1$,

$$r \text{ fails} \iff \text{PF}(m) \subseteq B_N.$$

Every odd $m \geq 3$ occurs as the exact mirror of every fixed rank k for some even N , while $m = 1$ occurs at every $k \geq 1$ but never at p_0 . The boundary case $N = 4, p_0 = 2$ adds the isolated canonical mirror $m = 2$. For fixed (k, m) the realisation fibre is finite with an explicit Ramanujan-prime bound.

22. **Same- N finite-view matching and fixed-support separation** (§11.5). For every fixed K and every finite set of odd prime labels with prescribed exact valuations, infinitely many common N realise those valuations simultaneously in all first $K + 1$ complements. On the other hand, for every fixed finite prime alphabet S , two complete S -smooth complements in the early-rank window must be separated by

$$\gg_S N^{19/40} (\log N)^{-A(S)};$$

hence a prefix through $N^{19/40-\varepsilon}$ eventually contains at most one such complement.

23. **The canonical threshold engine** (§9.2). The container identity, its slack bound, the charge-free overlap normal form, and the trichotomy of surviving routes remain historically correct, but the binding frontier is now §11.6.

2.3 What is only computational

Everything in Chapter 12 and the finite parts of Chapter 11 is **VERIFIED COMPUTATION** or **EMPIRICAL PATTERN** and never more unless an explicit proved finite search bound upgrades the domain. The load-bearing numbers now include: the exhaustive canonical scan to $N \leq 10^7$ with zero canonical failures; the six reconstructed EAC/BAD-basin hosts 128, 1718, 1862, 1928, 2200, 6142; exactly 301 failing upper roots on those hosts; zero composite failures through 10^7 for ranks 10, 14, 26, 28, 30, 62, 63, 71, 74; and the run-G *global finite-fibre* census of every odd composite mirror $m \leq 3053$ at $k \in \{0, 10, 14, 26, 28, 30, 62, 63, 71, 74\}$: 22130 realisations and zero failures. The last statement is all- N only because the finite-fibre theorem proves that every such realisation lies below the certified endpoint, not because a large arbitrary cutoff was scanned. External verification of Goldbach itself reaches $4 \cdot 10^{18}$ [12]; the project has never verified anything at that scale and must not cite it as its own.

2.4 The current frontier in one paragraph

Current (binding 2026-09-07H). The principal unresolved object is the incidence between an upper-root selector and the *moving complete BAD basin* B_N . On the composite-complement domain the exact criterion is

$$p_k(N) \text{ fails} \iff \text{PF}(N - p_k(N)) \subseteq B_N.$$

Run G proves that no fixed rank excludes any odd mirror class, and run H proves that any fixed finite prime-label/valuation view can be made identical across $p_0, \dots, p_K$ at the same N infinitely often. Fixed finite alphabets are also too rigid in the opposite direction: complete S -smooth early complements become quantitatively separated. Therefore the next theorem must see the *full N -dependent support* and its relation to the root position; taking $S = B_N$ separately for each N is not licensed by any fixed- S theorem. See §11.6.

What changed after 07C. The old $h = 1$ closed-support question and the low-gap Diophantine branches remain valid subproblems, but they are no longer the binding global description of the project. The rank/mirror surjectivity theorem shows that first-row arithmetic by itself cannot protect a fixed rank; same- (k, m) opposite-verdict witnesses show that even the exact root row is not the verdict variable. The ambient landscape is indispensable.

Historical v4.79–07C frontier. The container/population/residue programme and the 07C local three-label questions remain in the theorem corpus. They should be used only when they interact with the current moving- B_N target rather than restarted as stand-alone local searches.

2.5 The architecture wall

SCOPE-CORRECTED — what the finite-programming wall actually says. [2026-09-05 hostile audit; see §10.2]

Finite Affine Canonical Programmability survives: compatible prescribed congruence, valuation and compositeness constraints on finitely many affine forms can be realised in infinitely many canonical $h = 1$ worlds. What does *not* survive is the quantifier jump from selected prescribed edges to a complete BAD BFS neighbourhood. Unspecified cofactors may introduce additional children, including GOOD children. The exact $N = 92$ witness contains an arbitrarily reusable selected BAD cycle while the complete canonical BFS succeeds at depth 1.

Consequently:

1. selected finite BAD transcripts are not contradiction-generating by themselves;
2. a theorem about *complete* bounded-depth frontiers is **not** ruled out by FACP/CRT programming;
3. any proposed complete-failure theorem must explicitly control every residual cofactor in every expanded row;
4. the later hereditary first-support embedding theorem supplies a separate rigorous no-go, but only for its stated hereditary graph/valuation class and not for complete support or eventual reachability.

The old blanket “Finite Bad-Tree Universality” and “Finite-local universality wall” are retained historically only in their scope-corrected form.

2.6 What a new model should not do

Short list; the full list is Chapter 18 and the log itself.

1. Do not restart the local certificate programme in a new coordinate system. Five successor theories already did; all five hit the same wall.

2. Do not treat canonical failure as Goldbach failure. Almost every scope error in the log is a version of this.
3. Do not promote a finite scan. The scan to 10^7 says nothing about 10^8 .
4. Do not re-derive proved theorems. Use them by ID.
5. Do not bridge two different functionals with an inequality that holds “morally”. That is precisely what $P1^{br}$ and K1 need and precisely what has never been produced.

Chapter 3

Canonical Notation and Definitions

One symbol, one meaning, for the whole file. If a future revision needs a new object, give it a new symbol; do not overload $h, m, Q, R, T, U, J, \eta, \nu, \gamma, \kappa$. Several failure-log entries are nothing but collisions of notation (FAIL-NOTATION-KAPPA-001, FAIL-SYMBOL-R-COLLISION-001).

3.1 Symbol registry

Symbol	Meaning (fixed throughout)
N	the even target, $N \geq 6$; always $N = 2x$
x	the midpoint $N/2$
$\mathbb{P}$	the primes
p_0	the canonical selector $\min\{p \in \mathbb{P} : p \geq x\}$
p_k	the k -th upper root: $p_0 < p_1 < p_2 < \dots$, all primes $\geq x$
r	a generic upper root (any prime $\geq x$), <i>not necessarily</i> p_0
h, h_r	the offset $p_0 - x$, resp. $r - x$
m_0, m_r	the complement $N - p_0 = x - h$, resp. $N - r = x - h_r$
δ	$2h$; note $N = 2m_0 + \delta$
p_-	the prime predecessor of p_0
G, G_r	the gap $p_0 - p_-$, resp. $r - r^-$
θ	the phase $h/G \in (0, 1)$
Q	$\text{PF}(m_0)$, the canonical prime set
$\text{PF}(n)$	the set of distinct prime factors of n ; $\text{PF}(1) = \emptyset$
$P^+(n), P^-(n)$	largest, resp. smallest, prime factor of n
$\Omega(n)$	number of prime factors of n with multiplicity
$\tau(n)$	number of divisors of n
$\mathcal{A}_N$	the active primes: $\{p \in \mathbb{P} : p < N, p \nmid N\}$
Good_N	$\{p \in \mathcal{A}_N : N - p \in \mathbb{P}\}$ — the Goldbach witnesses
Bad_N	$\mathcal{A}_N \setminus \text{Good}_N$
$\mathfrak{B}(r)$	the strict reflected block of r
c_j, a_j	upper mirror $r - 2j$, lower mirror $N - c_j = m_r + 2j$
t_r	$\lfloor (h_r - 1)/2 \rfloor$, the block half-length
$\kappa_{\text{gen}}(r)$	genealogical defect rank of the upper root r
ρ_0	bidirectional prime rank, $\#\{s \geq 1 : h < D_s < 2h\}$

Symbol	Meaning (fixed throughout)
$R, R_N(r)$	the stopped failed world of the root r
$\text{Src}(S)$	the sources of S : vertices of S with no parent in S
Reach	reachability closure under $p \rightarrow q$
B_i, B_N^{low}	the CRD filtration and its fixed point
G_{esc}	escape capacity (number of escape edges)
D_q, M_q, B_q	$\gcd(m_0, N - q), (N - q)/D_q, m_0/q$
T_S, U_S, J_S	total valuation mass, unit part, layer part on S
η, ν	Euler defect terms: excess valuation, resp. layer count
γ	the coupling/capacity functional in P1^{br} and K1
K_t, C_t, W_t	path Bézout and winding coefficients

Warning 3.1 (Reserved letters). R is *always* the failed world, never a ring, a remainder, or a generic set. κ with no subscript is never used; only κ_{gen} and $\kappa^{\pm}$ appear. Q is $\text{PF}(m_0)$ and nothing else — the smooth domain in the incidence ledger is written Q_{smooth} .

3.2 The active prime graph

Definition 3.2 (Activity, edges, GOOD/BAD). Fix an even $N \geq 6$.

- p is *active* iff $p \in \mathbb{P}$, $p < N$ and $p \nmid N$. Write $\mathcal{A}_N$ for the set of active primes.
- The *row* of p is the integer $N - p$ together with its factorization.
- There is a directed edge $p \rightarrow q$ iff $q \mid N - p$ and $q \nmid N$, i.e. $q \in \text{PF}(N - p) \cap \mathcal{A}_N$. Write $\Gamma_N(p) = \text{PF}(N - p) \cap \mathcal{A}_N$ for the children of p .
- p is *GOOD* iff $N - p \in \mathbb{P}$ (equivalently $(p, N - p)$ is a Goldbach pair), and *BAD* otherwise.

The condition $q \nmid N$ is not cosmetic: without it the graph acquires absorbing cycles through the divisors of N and every statement about sources fails. [CORE, ARS-ACTIVITY]

Remark 3.3 (Unit complement). $N - p = 1$, i.e. $p = N - 1$, is neither GOOD nor a normal BAD row: its row has empty factorization. It is excluded from every count in this file. Failing to exclude it is FAIL-UNIT-COMPLEMENT-001.

Definition 3.4 (Sources). For $S \subseteq \mathcal{A}_N$, $\text{Src}(S) = \{p \in S : \nexists q \in S, q \rightarrow p\}$. A *closed* set is one with $\Gamma_N(S) \subseteq S$.

3.3 The stopped traversal

Definition 3.5 (Stopped traversal / factor-flow). Given N and an active root r , run breadth-first search from r in the graph of Definition 3.2 with one modification: a popped GOOD vertex is *recorded and not expanded*, and a GOOD child is recorded as an *escape edge* and not enqueued. The traversal returns SUCCESS at depth d if a GOOD vertex is first recorded at depth d , and FAILURE otherwise. The set of BAD vertices actually expanded is the *stopped failed world* $R_N(r)$.

If $2r = N$ (the diagonal case, $x \in \mathbb{P}$) the traversal returns SUCCESS at depth 0 with $R = \emptyset$.

Definition 3.6 (Two modes). Two implementations appear in the corpus and are both correct.

census stops at the first popped GOOD vertex. This reproduces the historical node and depth statistics.

world continues to exhaustion, building the full $R_N(r)$ and the full escape-edge set.

They agree on the verdict and on the success depth; they differ only in how much of the world is materialised. [CORE, ARS-TRAVERSAL-MODES]

Lemma 3.7 (Mode equivalence). **PROVED**. *For every N and every active r , *census* and *world* return the same verdict, and when the verdict is *SUCCESS* the same depth.*

Proof. BFS pops vertices in non-decreasing depth order, and neither mode expands a GOOD vertex, so the multiset of vertices popped strictly before the first GOOD pop is identical in the two modes. The first GOOD pop is therefore the same vertex at the same depth, and the verdict is FAILURE in one mode iff no GOOD vertex is ever popped, iff the same in the other. $\square$

3.4 Reference implementation

The following is the canonical executable semantics of Definitions 3.2–3.6. It is self-contained (standard library only), deterministic (fixed Pollard-rho seed), and is the file `p0_algorithm.py` of the project repository with its comments translated into English; the executable code is unchanged. A future model that is unsure what a definition means should run this rather than guess.

Listing 3.1: `p0_algorithm.py` — canonical selector and stopped traversal, runnable as given.

```

1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  p0_algorithm.py -- minimal, fully self-contained implementation of the
5  canonical selector p_0 and of the historical stopped traversal.
6
7  No imports outside the standard library. Copy and run:
8
9      python3 p0_algorithm.py 1928 3 # N = 1928, roots p_0, p_1, p_2
10     python3 p0_algorithm.py # demonstration on the six EAC
11 """
12 import sys
13 from collections import deque
14 from math import gcd
15 from random import Random
16
17 # --- 1. primality: deterministic Miller-Rabin for n < 3.3 * 10^24 -----
18 BASES = (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37)
19
20
21 def is_prime(n):
22     if n < 2:
23         return False
24     for p in BASES:
25         if n % p == 0:
26             return n == p
27     d, s = n - 1, 0
28     while d % 2 == 0:
29         d //= 2
30         s += 1
31     for a in BASES:
32         x = pow(a, d, n)
33         if x in (1, n - 1):
34             continue
35         for _ in range(s - 1):
36             x = x * x % n
37             if x == n - 1:
38                 break
39         else:
40             return False
41     return True
42
43
44 def next_prime(n):
45     m = n + 1
46     while not is_prime(m):
47         m += 1
48     return m
49
50
51 def prev_prime(n):

```

```

52 m = n - 1
53 while m >= 2 and not is_prime(m):
54     m -= 1
55 return m if m >= 2 else None
56
57
58 # --- 2. factorization: trial division + Pollard rho (deterministic seed) ---
59 def _rho(n, rng):
60     if n % 2 == 0:
61         return 2
62     while True:
63         c, y = rng.randrange(1, n), rng.randrange(0, n)
64         g, r, q, x, ys = 1, 1, 1, 0, 0
65         while g == 1:
66             x = y
67             for _ in range(r):
68                 y = (y * y + c) % n
69             k = 0
70             while k < r and g == 1:
71                 ys = y
72                 for _ in range(min(128, r - k)):
73                     y = (y * y + c) % n
74                     q = q * abs(x - y) % n
75                     g, k = gcd(q, n), k + 128
76             r *= 2
77         if g == n:
78             g = 1
79             while g == 1:
80                 ys = (ys * ys + c) % n
81                 g = gcd(abs(x - ys), n)
82         if g != n:
83             return g
84
85
86 def factorize(n):
87     """Returns {p: v_p(n)}; factorize(1) = {}."""
88     f, rng = {}, Random(20260904)
89     for p in range(2, 100000):
90         if p * p > n:
91             break
92         while n % p == 0:
93             f[p] = f.get(p, 0) + 1
94             n //= p
95     stack = [n] if n > 1 else []
96     while stack:
97         m = stack.pop()
98         if m == 1:
99             continue
100         if is_prime(m):
101             f[m] = f.get(m, 0) + 1
102             continue
103         d = _rho(m, rng)
104         stack += [d, m // d]
105     return dict(sorted(f.items()))
106
107
108 # --- 3. canonical selector and the successive upper roots -----
109 def p0(N):
110     """p_0(N) = min{ p prime : p >= N/2 }. """
111     if N % 2 or N <= 2:
112         raise ValueError("N must be even and greater than 2")
113     x = N // 2
114     return x if is_prime(x) else next_prime(x)
115
116
117 def upper_roots(N, count):
118     """[p_0, p_1, ..., p_{count-1}] -- consecutive primes >= N/2. """
119     out, p = [], p0(N)
120     while len(out) < count and p < N:
121         out.append(p)
122         p = next_prime(p)
123     return out
124
125
126 # --- 4. graph: edge, GOOD vertex, activity -----

```

```

127 def is_active(N, p):
128     """p active for N <=> p prime, p < N, p does not divide N."""
129     return is_prime(p) and p < N and N % p != 0
130
131
132 def is_good(N, p):
133     """GOOD <=> N - p prime, i.e. N = p + (N-p) is a Goldbach pair."""
134     return N - p >= 2 and is_prime(N - p)
135
136
137 def children(N, p):
138     """Active children of the row N - p: {q prime : q | N-p, q does not divide N}."""
139     return sorted(q for q in factorize(N - p) if N % q != 0)
140
141
142 # --- 5. HISTORICAL STOPPED TRAVERSAL / FACTOR-FLOW -----
143 def stopped_traversal(N, root):
144     """Full BFS with stopping at GOOD vertices.
145
146     Returns a dict:
147     verdict 'SUCCESS' or 'FAILURE' of the root,
148     depth depth of the first GOOD (None on failure),
149     witness Goldbach pair (p, N-p) found by the traversal,
150     R the stopped bad world (BAD vertices that were expanded),
151     escapes escape edges (p, q): p in R, q GOOD,
152     rows factorizations of N - p for p in R.
153     """
154     x = N // 2
155     if 2 * root == N: # diagonal case
156         return dict(verdict="SUCCESS", depth=0, witness=(root, root),
157                     R=set(), escapes=set(), rows={}, diagonal=True)
158     if not is_active(N, root):
159         raise ValueError("root must be an active prime")
160
161     queue, queued = deque([(root, 0)]), {root}
162     R, rows, escapes = set(), {}, set()
163     verdict, depth, witness = "FAILURE", None, None
164
165     while queue:
166         p, d = queue.popleft()
167         if p in R:
168             continue
169         if is_good(N, p): # GOOD: record, do NOT expand
170             if witness is None:
171                 verdict, depth, witness = "SUCCESS", d, (p, N - p)
172             continue
173         R.add(p) # BAD: expand the whole row
174         rows[p] = factorize(N - p)
175         for q in sorted(q for q in rows[p] if N % q != 0):
176             if is_good(N, q):
177                 escapes.add((p, q))
178                 if witness is None:
179                     verdict, depth, witness = "SUCCESS", d + 1, (q, N - q)
180                 continue # a GOOD child is not enqueued
181             if q not in R and q not in queued:
182                 queued.add(q)
183                 queue.append((q, d + 1))
184
185     return dict(verdict=verdict, depth=depth, witness=witness, R=R,
186                 escapes=escapes, rows=rows, diagonal=False)
187
188
189 # --- 6. report -----
190 def show(N, k=1):
191     x, r0 = N // 2, p0(N)
192     pm = prev_prime(r0)
193     h, m = r0 - x, N - r0
194     fm = factorize(m)
195     print(f"N = {N}, x = N/2 = {x}")
196     print(f"p_0 = {r0}, h = {h}, m_0 = {m} = "
197           + (" * ".join(f"{p}^{e}" if e > 1 else str(p) for p, e in fm.items())
198             or "1"))
199     print(f"previous prime = {pm}, G = {r0 - pm}, h < G : {h < r0 - pm}")
200     print(f"delta = 2h = {2*h}, Q = PF(m_0) = {sorted(fm)}")
201     for i, r in enumerate(upper_roots(N, k)):

```

```

202     t = stopped_traversal(N, r)
203     hr = r - x
204     tt = (hr - 1) // 2
205     kap = sum(1 for j in range(1, tt + 1) if is_prime(r - 2 * j))
206     print(f" p_{i} = {r:<10d} {t['verdict']:<8s} depth={t['depth']}"
207           f" |R|={len(t['R']):<4d} escapes={len(t['escapes']):<3d}"
208           f" kappa_gen={kap} witness={t['witness']}")
209
210
211 if __name__ == "__main__":
212     if len(sys.argv) > 1:
213         show(int(sys.argv[1]), int(sys.argv[2]) if len(sys.argv) > 2 else 1)
214     else:
215         for N in (128, 1718, 1862, 1928, 2200, 6142, 11822, 14198):
216             show(N, 3)
217             print()

```

Remark 3.8 (Reading the code as a definition). Three lines carry all the semantics that prose gets wrong. `if is_good(N, p): ...continue` is the stopping rule (a GOOD vertex is never expanded). `escapes.add((p, q)); ...continue` is the escape rule (a GOOD child is recorded, not enqueued). `if N % q != 0 in children` is activity. Every historical variant that differs from the corpus differs in exactly one of these three lines.

3.5 Historical variants and how they differ

Variant	Difference	Consequence
Canonical (audited)	As Definition 3.5	The reference. All statistics in this file use it.
<i>Expanding GOOD</i>	Expands GOOD vertices too	Depth statistics change; the world is no longer the failed world; several early tables in the corpus were produced this way and do not match later ones. FAIL-TRAVERSAL-EXPANDGOOD-001
<i>Ignoring activity</i>	Allows $q \mid N$ as a child	Creates absorbing cycles; “source” becomes ill-defined; kills the Euler defect bound.
<i>DFS instead of BFS</i>	Depth-first order	Verdict unchanged, depth statistic meaningless. Depth is only defined relative to BFS.
<i>Unit row admitted</i>	$p = N - 1$ counted	Off-by-one in every census; see Remark 3.3.
<i>Non-canonical root</i>	Starts at p_k , $k \geq 1$, or below x	Legitimate, but its results belong to scope class B (root-generic) and must never be quoted as canonical. FAIL-F1-DOMAIN-MIX-001

3.6 The canonical selector and midpoint geometry

Definition 3.9 (Normal form). Write

$$\begin{aligned}
 N &= 2x, & p_0 &= x + h, & m_0 &:= N - p_0 = x - h, & \delta &:= 2h, \\
 p_- &:= \max\{p \in \mathbb{P} : p < p_0\}, & G &:= p_0 - p_-, & Q &:= \text{PF}(m_0), & \theta &:= \frac{h}{G}.
 \end{aligned}$$

If $h = 0$ then $x \in \mathbb{P}$ and $p_0 = x$: the *diagonal* case. Unless stated otherwise the *non-diagonal* case is assumed: x composite, $h \geq 1$. The exact identity

$$N = 2m_0 + \delta$$

holds always. The same symbols with subscript r denote the analogous quantities for an arbitrary upper root $r = x + h_r \geq x$.

Theorem 3.10 (Canonical gap equivalence). **PROVED**. *Let $r = x + h_r \geq x$ be prime, r^- its predecessor and $G_r = r - r^-$. Then*

$$r = p_0 \iff [x, r) \cap \mathbb{P} = \emptyset \iff h_r < G_r.$$

Proof. If $r = p_0$ then $r^- < x$, so $G_r = r - r^- > r - x = h_r$. Otherwise some prime lies in $[x, r)$, hence $r^- \geq p_0 \geq x$ and $G_r \leq h_r$. [CORE, C-GAP-EQUIVALENCE] $\square$

Warning 3.11 (How *not* to read $h < G$). The inequality $h < G$ looks like a statement about “a large prime gap before p_0 ”. That reading is wrong and was a recurring source of error. Theorem 3.10 is *categorical*, not quantitative: no earlier upper prime lies inside the reflected interval. The correct formulation [CORE, V71.43] is

p_0 is a defect-free boundary condition selected by midpoint minimality.

Equivalently [CORE, V71.42]: p_0 is the unique upper root whose predecessor lies geometrically *outside* its reflected block. Nothing about the *size* of G is asserted, and no bound of Baker–Harman–Pintz type is used or needed here.

Definition 3.12 (Strict reflected block). For an upper root $r = x + h_r$ with $h_r \geq 3$ put

$$t_r = \left\lfloor \frac{h_r - 1}{2} \right\rfloor, \quad \mathcal{J}_{h_r} = \{1, \dots, t_r\}, \quad c_j = r - 2j, \quad a_j = N - c_j = m_r + 2j,$$

$$\mathfrak{B}(r) = \{a_1, \dots, a_{t_r}\} \cup \{c_1, \dots, c_{t_r}\}.$$

Then $m_r < a_j < x < c_j < r$ and $a_j + c_j = N$. The endpoints $u = 0$ and $u = h_r$, i.e. m_r and r themselves, are *not* in $\mathfrak{B}(r)$. Call c_j an *upper mirror* and a_j a *lower mirror*.

Lemma 3.13 (Strict non-reuse of large primes). **PROVED**. *If an odd prime ℓ divides two distinct elements of $\mathfrak{B}(r)$ then $\ell < h_r$. Equivalently, every prime $\ell \geq h_r$ occurs in exactly one element of $\mathfrak{B}(r)$.*

Proof. For $i \neq j$, $|a_i - a_j| = |c_i - c_j| = 2|i - j|$, and for any i, j , $c_j - a_i = 2(h_r - i - j)$. Since $i + j \leq 2t_r < h_r$ the last quantity is positive, so all three differences are nonzero. In each case half the difference is an integer in $\{1, \dots, h_r - 1\}$; an odd prime dividing twice that half divides the half itself, hence is smaller than h_r . [CORE, C-STRICT-NONREUSE] $\square$

Warning 3.14 (No widening of the block). Lemma 3.13 may *not* be applied to the widened block $\{m_r + 2u : 0 \leq u \leq h_r\}$. The endpoint pair $(u, v) = (0, h_r)$ has difference exactly $2h_r$ and is not covered by the proof; it would require a separate canonical input $\gcd(h, m_0) = 1$ and is nowhere used that way below. [FAIL, FAIL-BROADBLOCK-ENDPOINT]

Lemma 3.15 (Reflection gate). **PROVED**. *Let $r = p_0$ and $0 < 2j < h$. Then $c_j \in (x, p_0)$ and c_j is composite. Its full factorization is a graph row with parent a_j if and only if a_j is an active prime; in that case $\Gamma_N(\{a_j\}) = \text{PF}(c_j)$.*

Proof. Compositeness of c_j is immediate from $[x, p_0) \cap \mathbb{P} = \emptyset$; the rest is the definition of an edge. [CORE, ARS-REFLECTION-GATE] $\square$

Warning 3.16 (The gate gives compositeness, not activation). **KILLED** is every strengthening of the gate to activation. The gate gives *only* compositeness of each c_j : it does not give primality of a_j , does not give incidence coupling, does not give basin coupling and does not give capacity. Exact counterexample: $N = 242$, $x = 121$, $p_0 = 127$, $p_- = 113$, so $h = 6 < G = 14$; both reflected values $a_1 = 117 = 3^2 \cdot 13$ and $a_2 = 119 = 7 \cdot 17$ are composite, so *not a single active reflected row exists*. [FAIL, FAIL-REFLECTION-ACTIVATION-001]

This is not an accident: $N = 242$ is in phase Type I, and a guaranteed active lower mirror exists exactly in phase Type III (§6.3).

Scope note 3.17 (Diagonal correction to the genealogical layer). **SCOPE-CORRECTED**. The corpus states the genealogy and reflection-completeness uniqueness theorems for “every prime $r \geq x$ ” [CORE, V71.21, V71.31], while the proof uses the strict inequality $p_i > x$. The difference is real.

If $x = N/2$ is *prime* then $p_0 = x$, $h = 0$, and the hole corresponding to p_0 sits at index $h_k/2 > t_k$, i.e. *outside* the strict reflected block. Then $\kappa_{\text{gen}}(p_k) = k - 1$ for $k \geq 1$, and the root p_1 is also reflection-complete — violating literal uniqueness.

VERIFIED COMPUTATION: the smallest witness is $N = 14$, $x = 7 \in \mathbb{P}$, $p_0 = 7$, $p_1 = 11$, block $\{c_1 = 9 = 3^2\}$, $\kappa_{\text{gen}}(p_1) = 0$ although $p_1 \neq p_0$. For $N \leq 6 \cdot 10^4$, $k \leq 5$, *every* violation has x prime, and for x composite there is not a single violation.

What survives. Theorem 3.10 is unaffected in all cases. Only the equivalence with “every $r - 2j$ is composite” fails, because the strict reflected block covers the *open* interval (x, r) while the canonical condition concerns the *closed* interval $[x, r)$. Correct repair: add the hypothesis “ x composite”, or replace $[x, r)$ by (x, r) .

Why this does not disturb the rest of the file. All failure analysis assumes $h \geq 1$: for x prime the canonical root wins immediately and there is no closed world. The domain of every p_0 -specific theorem is therefore non-diagonal by definition.

3.7 The valuation layer

Definition 3.18 (Valuation mass, unit part, layer part). For a finite set S of active rows put, for each active prime q ,

$$d_S(q) = \sum_{p \in S} v_q(N - p), \quad \kappa_S(q) = \#\{p \in S : q \mid N - p\},$$

and define, the products running over $q \in \Gamma_N(S)$,

$$T_S = \prod_q q^{d_S(q)-1}, \quad U_S = \prod_{p \in S} \frac{N - p}{\text{rad}(N - p)}, \quad J_S = \prod_q q^{\kappa_S(q)-1}.$$

U_S is the *unit part* (excess valuation, i.e. repeated primes inside one row), J_S the *layer part* (support reuse, i.e. the same prime appearing in several rows).

Theorem 3.19 (Exact minimal-collision factorization). **PROVED**. $T_S = U_S J_S$.

Proof. Fix $q \in \Gamma_N(S)$ with positive valuations $e_1, \dots, e_k$ ($k = \kappa_S(q)$, $\sum e_i = d_S(q)$). Its exponent in T_S is $\sum e_i - 1$, in U_S it is $\sum e_i - k$, and in J_S it is $k - 1$; these add up. [CORE, G-TUJ] $\square$

Remark 3.20 (Reading). The radical defect of the world splits *exactly* into excess valuation and support reuse. This is completely generic (scope class B): it uses nothing about p_0 .

The Euler-defect terms η and ν built on top of this, and the inequality $\eta(S) + \nu(S) \geq |S| + |\text{Src}(S)|$ that the corpus records, are discussed with their exact status in §5.7. Read that section before using them: the normalisation matters and is easy to get wrong.

Warning 3.21 (η is not a capacity). η measures valuation leaving the set. It is *not* the functional γ appearing in $P1^{br}$ and K1, and the repeated attempt to identify them is the single most common dead end in the log (FAIL-ETA-GAMMA-IDENTIFY-001 and its successors).

3.8 Exceptional autonomous components

Definition 3.22 (EAC). An *exceptional autonomous component* for N is a nonempty set $C \subseteq \text{Bad}_N$ that is closed under Γ_N ($\Gamma_N(C) \subseteq C$) and contains no GOOD vertex. Equivalently: a nonempty closed bad set. Božek [8] introduced this notion for the Goldbach factorization graph.

Warning 3.23 (Orientation). Two orientations of the Goldbach graph exist in the literature and in this project. The corpus uses $p \rightarrow q$ iff $q \mid N - p$ (“the row of p points to its prime factors”). Reversing the arrows turns closed sets into reachable sets and *silently changes what “autonomous” means*. Several early comparisons with [8] were wrong for this reason (FAIL-EAC-ORIENTATION-001). Always state the orientation.

Remark 3.24 (EAC and canonical failure). The canonical root p_0 fails exactly when it lies in an EAC. This is the qualitative form of the fixed-point characterisation in §3.10; the quantitative form is the statement that the relevant EAC is B_N^{low} restricted to the reachable set of p_0 .

3.9 The canonical failed world

Definition 3.25 (Stopped failed world). $R_N(r)$ is the set of BAD vertices expanded by the stopped traversal from r (Definition 3.5). An *escape edge* is a pair (v, q) with $v \in R_N(r)$, $q \in \Gamma_N(v)$ and q GOOD. The traversal fails iff there is no escape edge and r itself is BAD.

Lemma 3.26 (Closure and unique source). **PROVED**. *If the traversal from r fails then $R_N(r)$ is closed ($\Gamma_N(R_N(r)) \subseteq R_N(r)$), consists entirely of BAD vertices, and $\text{Src}(R_N(r)) = \{r\}$ provided no $v \in R_N(r)$ has $r \in \Gamma_N(v)$. For $r = p_0$ the proviso holds unconditionally (no-bad-return-to- p_0 , §5.6), so*

$$\text{Src}(R_N(p_0)) = \{p_0\}.$$

Proof. Closure and badness are immediate from the traversal rules (a BAD vertex is expanded and all its active children enqueued; failure means no child is GOOD). r is a source by construction. Any other source would be a vertex enqueued with no parent in the world, impossible since every enqueued vertex other than r was enqueued from its parent. The canonical case uses Theorem 5.33. [CORE, ARS-CLOSED-WORLD] $\square$

Scope note 3.27 (Failure of the canonical traversal is not Goldbach failure). $R_N(p_0) \neq \emptyset$ and no escape means only that *this traversal from this root* found nothing. Good_N may still be large. The six values of N with $B_N^{\text{low}} \neq \emptyset$ all satisfy Goldbach comfortably. Every theorem whose hypothesis is $\text{Good}_N = \emptyset$ is a *class-D* statement (Goldbach-counterexample-conditioned) and may never be invoked under canonical failure alone. FAIL-UPPERHALF-SCOPE-001

3.10 The CRD fixed point B_N^{low}

Definition 3.28 (CRD filtration). Put

$$B_0 = \{p \in \mathcal{A}_N : p < N/3, N - p \text{ composite}\}, \quad B_{i+1} = \{p \in B_i : \text{PF}(N - p) \subseteq B_i\},$$

and $B_N^{\text{low}} = \bigcap_{i \geq 0} B_i$. The sequence is decreasing and finite, so the intersection is reached in finitely many steps.

Theorem 3.29 (Fixed-point characterisation of canonical failure). **PROVED**. B_N^{low} is the largest Γ_N -closed subset of $\{p < N/3 : N - p \text{ composite}\}$, and

$$p_0 \text{ fails} \iff N - p_0 \text{ is composite and } \text{PF}(N - p_0) \subseteq B_N^{\text{low}}.$$

Proof. B_{i+1} removes from B_i exactly the elements with a child outside B_i , so B_N^{low} is closed and contains every closed subset. For the equivalence: if p_0 fails then $R_N(p_0) \setminus \{p_0\}$ is closed, bad, and contained in the $N/3$ range (a child $q \mid N - p$ with $q \geq N/3$ would force $N - p \in \{q, 2q\}$, both excluded for a BAD row with $p > N/3$), hence contained in B_N^{low} ; and $\text{PF}(N - p_0) \subseteq R_N(p_0)$. Conversely if $\text{PF}(N - p_0) \subseteq B_N^{\text{low}}$ then the traversal never leaves B_N^{low} after the first step and B_N^{low} contains no GOOD vertex. [PCRD, CRD fixed point] $\square$

Remark 3.30 (Why the $N/3$ threshold). For $p > N/3$ and $N - p$ composite, every prime factor q of $N - p$ satisfies $q \leq (N - p)/2 < N/3$. So the interesting closed sets live below $N/3$ and the threshold costs nothing. Using $N/2$ instead re-admits p_0 itself and makes the filtration non-monotone in N ; this is FAIL-BLOW-THRESHOLD-001.

Scope note 3.31 (B_N^{low} is a computation, not a mechanism). Theorem 3.29 converts “does the canonical traversal fail” into “is a certain explicitly computable set nonempty”. That is an exact reformulation and it made the $6 \cdot 10^4$ census possible. It is *not* progress towards a proof: deciding $B_N^{\text{low}} = \emptyset$ for all N is at least as hard as the original question, and the finite-local universality wall (§2.5) says no bounded certificate decides it.

Chapter 4

Theorem Registry

How to use this chapter. Every named result of the project has one row. Before using a result anywhere, read its row: the *status* tells you whether it is a theorem, the *dependencies* tell you what breaks if a dependency is later corrected, and the *location* points at the proof or at the failure-log entry that killed it.

Post-07C note. Results from runs 07D–07H are registered in the binding delta registry §11.8; consult it together with this chapter. The older rows below are not silently rewritten.

Rules. (i) A row is never edited to a higher status. If you prove something that was **CONJECTURE**, add a new row and mark the old one **SUPERSEDED**. (ii) A **VERIFIED COMPUTATION** row always carries its range. A range is part of the statement. (iii) A **KILLED** row is not a lemma. It may not be used “in the direction that still works” without a new row stating precisely that direction. (iv) Dependencies are transitive; the registry lists only immediate ones.

4.1 Foundational and boundary layer

ID	Statement / short name	Status	Dependencies	Location
C-GAP-EQ	Canonical gap equivalence: $r = p_0 \iff [x, r) \cap \mathbb{P} = \emptyset \iff h_r < G_r$	PROVED	—	Thm. 3.10
C-BOUNDARY-READING	$\mathbb{M}$ is a defect-free boundary condition selected by midpoint minimality; <i>not</i> a large-gap statement	PROVED	C-GAP-EQ	Warn. 3.11
C-STRICT-NONREUSE	Every prime $\ell \geq h_r$ divides at most one element of $\mathfrak{B}(r)$	PROVED	Def. 3.12	Lem. 3.13
C-BROADBLOCK-FAIL	The widened block $\{m_r + 2u : 0 \leq u \leq h_r\}$ does <i>not</i> satisfy strict non-reuse (endpoint pair)	KILLED	C-STRICT-NONREUSE	Warn. 3.14
ARS-REFLECTION-GATE	Every upper mirror c_j of p_0 is composite; its row has parent a_j iff a_j is an active prime	PROVED	C-GAP-EQ	Lem. 3.15
C-GATE-ACTIVATION	The gate also gives activation of some a_j	KILLED	—	Warn. 3.16; $N = 242$

ID	Statement / short name	Status	Dependencies	Location
C-DIAGONAL-SCOPE	Genealogy and uniqueness hold on the non-diagonal domain (x composite); the diagonal case is a genuine exception	SCOPE-CORRECTED	V71.21, V71.31	Scope 3.17; $N = 14$
C-NORMALFORM	$N = 2m_0 + \delta$, $\delta = 2h$, $\theta = h/G$	AUDITED IDENTITY		Def. 3.9

4.2 Graph, traversal and world layer

ID	Statement / short name	Status	Dependencies	Location
ARS-ACTIVITY	Activity ($q \nmid N$) is required for the source theory; without it the graph has absorbing cycles	PROVED	—	Def. 3.2
ARS-TRAVERSAL-MONOTONIC	mesus and world agree on verdict and success depth	PROVED	Def. 3.6	Lem. 3.7
ARS-CLOSED-WORLD	On failure, $R_N(r)$ is closed and all-BAD; $\text{Src}(R_N(p_0)) = \{p_0\}$	PROVED	ARS-NORETURN	Lem. 3.26
ARS-NORETURN	No-bad-return-to- p_0 : no $v \in R_N(p_0)$ has $p_0 \in \Gamma_N(v)$	PROVED	C-GAP-EQ	Thm. 5.33
CRD-FIXPOINT	B_N^{low} is the largest closed bad set below $N/3$; p_0 fails iff $N - p_0$ composite and $\text{PF}(N - p_0) \subseteq B_N^{\text{low}}$	PROVED	ARS-CLOSED-WORLD	Thm. 3.29
CRD-THRESHOLD	The $N/3$ threshold is the correct one; $N/2$ breaks monotonicity	PROVED	CRD-FIXPOINT	Rem. 3.30
EAC-DEF	EAC = nonempty closed bad set; p_0 fails iff p_0 lies in an EAC	AUDITED IDENTITY	CRD-FIXPOINT	§3.8
EAC-ORIENT	The orientation convention is $p \rightarrow q \iff q \mid N - p$; reversing it changes “autonomous”	PROVED	—	Warn. 3.23
C-SCOPE-D	Canonical failure does <i>not</i> imply $\text{Good}_N = \emptyset$	PROVED	—	Scope 3.27

4.3 Genealogy, filtration and phase

ID	Statement / short name	Status	Dependencies	Location
V71.21	Genealogical defect rank: $\kappa_{\text{gen}}(p_k) = k$	PROVED	C-DIAGONAL-SCOPE	Thm. 6.2
V71.31	p_0 is the unique reflection-complete upper root ($\kappa_{\text{gen}} = 0$)	PROVED	V71.21	Cor. 6.5
C-LEDGER-LOSSLESS	The defect ledger is a lossless code of local prime history: hole positions recover the prime gaps	PROVED	V71.21	Thm. 6.3
C-LEDGER-EXPLAINS	The lossless ledger <i>explains</i> why p_0 succeeds	KILLED	—	Warn. 6.7

ID	Statement / short name	Status	Dependencies	Location
C0b-RD	The canonical ledger partitions as $\mathcal{J}_h = \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}$ with no fifth class	PROVED	V71.21	Thm. 8.6
C-FILTRATION	Boundary filtration: $c_{i+s}^{(k+1)} = c_i^{(k)}$, $a_{j+s}^{(k+1)} = a_j^{(k)}$, $t_{k+1} = t_k + s$, $s = (p_{k+1} - p_k)/2$	PROVED	Def. 3.12	Thm. 6.8
C-GCD-EMBED	The gcd-coupling matrix $M_{ij}^{(k)} = \gcd(c_i, a_j)$ embeds as a submatrix of level $k + 1$	PROVED	C-FILTRATION	Cor. 6.9
C-PHASE	Phase trichotomy from $m_0 - p_- = G - 2h$: Type I ($2h < G$), Type II ($2h = G$), Type III ($2h > G$)	PROVED	C-GAP-EQ	Thm. 6.12
C-PHASE-I	Type I $\Rightarrow m_0$ composite $\Rightarrow$ immediate success impossible	PROVED	C-PHASE	Thm. 6.12
C-PHASE-II	Type II $\Rightarrow m_0 = p_- \Rightarrow$ immediate success automatic	PROVED	C-PHASE	Thm. 6.12
C-PHASE-III	Type III $\Rightarrow p_- = a_{j^*}$ with $j^* = h - G/2$	PROVED	C-PHASE	Thm. 6.12
C-PHASE-CELL	Cell decomposition $1 + (G/2 - 1) + 1 + (G/2 - 1) = G$; θ symmetric under $\theta \mapsto 1 - \theta$	AUDITED IDENTITY	C-PHASE	§6.3
C-RHO	$\rho_0 = \#\{s \geq 1 : h < D_s < 2h\}$, $D_s = p_0 - p_{-s}$; and $\rho_0 \geq 1 \iff$ Type III	PROVED	C-PHASE	Thm. 6.17
C-RHO-USERFORM	The originally proposed form of ρ_0 (without the lower bound $D_s > h$)	KILLED	C-RHO	Warn. 6.18
C-SIGNATURE	Bidirectional signature $(\kappa^+, \kappa^-) = (\kappa_{\text{gen}}, \rho_0)$	AUDITED IDENTITY	V71.21, C-RHO	§6.4

4.4 Shadow, path and valuation layers

ID	Statement / short name	Status	Dependencies	Location
S-SHADOW-ID	Prime-gap shadow identity: $r \mid N - p \iff r \mid \delta - p$ for $r \mid m_0$	PROVED	C-NORMALFORM	Thm. 5.22
S-RETENTION	Retention-escape: $\text{PF}(N - q) \cap Q = \text{PF}(d_q)$ with $d_q = \gcd(m_0, \delta - q)$	PROVED	S-SHADOW-ID	Thm. 5.24
S-MINFACOR	Minimum-factor escape: if $P^-(N - q) \notin Q$ the traversal escapes Q at q	PROVED	S-RETENTION	Cor. 5.25
S-CYCLE-LOC	Shadow cycle localisation: a cycle inside Q forces all its vertices into an explicit residue window	PROVED	S-SHADOW-ID	Thm. 5.27
S-TWOCYCLE	Shadow two-cycle equation $\delta = q + s + tqs$	PROVED	S-CYCLE-LOC	Prop. 5.28

ID	Statement / short name	Status	Dependencies	Location
S-PPWITNESS	Prime-power witness: a prime power row forces an explicit escape or an explicit divisibility	PROVED	S-RETENTION	Prop. 5.29
P-BEZOUT	Full path Bézout invariant $K_t p_t - C_t x = (-1)^t h$	PROVED	—	Thm. 5.30
P-WINDING	Winding normal form $K_t p_t = W_t N + (-1)^t p_0$	PROVED	P-BEZOUT	Cor. 5.31
P-STATES	At most $2\tau(h)$ reduced states per vertex	PROVED	P-WINDING	Thm. 5.32
P-COLLISION	Global two-sheet collision budget: at most two collision endpoints	PROVED	P-STATES	Thm. 5.34
P-COLLISION-CONTRA	The collision budget yields a contradiction from canonical failure	KILLED	P-COLLISION	§5.6
V-TUJ	$T_S = U_S J_S$ (excess valuation $\times$ support reuse)	PROVED	Def. 3.18	Thm. 3.19
V-EULER	Layered Euler defect $\eta(S) + \nu(S) \geq S + \text{Src}(S) $; canonically $\eta(R) + \nu(R) \geq R + 1$ — <i>normalisation of η, ν not reconstructed here</i>	CONJECTURE	V-TUJ, ARS-CLOSED-WORLD	§5.7
V-EULER-A	Active-mass form $\sum_q d_R(q) \geq R + 1$	VERIFIED COMPUTATION	V-TUJ	§5.7
V-EULER-B	Trivial form $\sum_{p \in R} (\Omega(N-p) - 1) + \nu(R) \geq R + 1$	PROVED	—	§5.7
V-ETA-GAMMA	η is the capacity functional γ	KILLED	—	FAIL-ETA-GAMMA-IDENTIFY

4.5 Post-root factorization and incidence

ID	Statement / short name	Status	Dependencies	Location
PR-DQ	$D_q = \gcd(m_0, N - q) = \gcd(B_q, 2h - q)$ for $q \mid m_0$	PROVED	S-SHADOW-ID	Thm. 7.1
PR-MQ	$M_q = (N - q)/D_q$ is coprime to Q	PROVED	PR-DQ	Cor. 7.3
PR-FRESH	FRESH/SMOOTH split: q is FRESH iff $P^+(N - q) \geq B_q$	AUDITED IDENTICAL	PR-MQ	Def. 7.4
IL1-IL6	Incidence ledger $\prod_q M_q = U_F (J_F B_{\text{re}}) B_{\text{abs}}$ on $S = Q_{\text{smooth}}$	AUDITED IDENTICAL	PR-MQ	Thm. 7.6
IL-DOMAIN	IL1–IL6 hold on Q_{smooth} only; the FRESH part is excluded	SCOPE-CORRECTED	IL1-IL6	Warn. 7.7
PR-BABS	$B_{\text{abs}}, B_{\text{re}}, B_{\text{abs,int}}$: definitions and the exact relation between them	AUDITED IDENTICAL	IL1-IL6	§7.2
PR-HIGHGAP	High-gap amplification $h^2 > 4m_0$	SCOPE-CORRECTED	PR-DQ	Warn. 7.5

4.6 CRD obligations and open targets

ID	Statement / short name	Status	Dependencies	Location
C0a	Small-gap canonical obligation (residual argument)	PROVED	C-GAP-EQ, C0b-RD	§8.3
TI	Injective branch obligation of C0b	OPEN	C0b-RD	§8.5
TD	Root-anchored dominance obligation of C0b	OPEN	C0b-RD	§8.5
TS	Stability branch obligation of C0b	OPEN	C0b-RD	§8.5
TF	Fresh-layer branch obligation of C0b	OPEN	C0b-RD	§8.5
TD-DENSITY	Density form of TD	SUPERSEDED	TD	§8.5
P1BR	Branchwise production upgrade: realised production bounded below by the <i>same</i> functional as capacity	OPEN	TD	§9.4
L1	Actual layer realisation	OPEN	V-EULER	§9.5
E1	Escape or coupling	OPEN	S-RETENTION	§9.6
W1	Simultaneous clean witnesses	OPEN	L1	§9.7
K1	Source-sensitive capacity $K_{\times}(R, \mathcal{P}) + K(\mathcal{P}) \leq \mathcal{C}_N(R)$	OPEN	P1BR	§9.8
X1	Canonical failure exclusion	OPEN	K1	§9.9
RPD	Repeated prime-power redundancy: $D_{\text{high}}(H_Q) \leq O(\log \log N) \eta_{\text{high}} + o(M)$	OPEN	V-EULER	§9.10
C-SAMEFUNC-BRIDGE	Global same-functional margin bridge	OPEN	P1BR, K1	§9.11
C-GENEALFREE-DYN	Genealogy-free boundary-to-dynamic escape for p_0	OPEN	V71.31	§9.11
C-POP1-SEP	One-defect dynamic separation between p_0 and p_1	OPEN	V71.21	§9.11

4.7 Computational rows

ID	Statement / short name	Status	Dependencies	Location
EXP1-SCAN	No canonical failure for even $6 \leq N \leq 10^7$	VERIFIED COMPUTATION		§12.2
EXP7-BLOW	$B_N^{\text{low}} \neq \emptyset$ for exactly six $N \leq 6 \cdot 10^4$: 128, 1718, 1862, 1928, 2200, 6142	VERIFIED COMPUTATION	CRD-FOXP	§6.6
EXP3-CENSUS	301 failing non-canonical upper roots for $N \leq 2 \cdot 10^4$, all at those six N ; 2261 unit rows excluded	VERIFIED COMPUTATION	Rem 103	§12.3
EXP4-MARGIN	Ten $N \leq 2 \cdot 10^5$ with escape margin $G_{\text{esc}} = 1$; per-dyadic-block minimum rises $1 \rightarrow 2 \rightarrow 3$	VERIFIED COMPUTATION	Def 105	§6.7
EXP5-PHASE	Phase-type frequencies for $N \leq 2 \cdot 10^5$	VERIFIED COMPUTATION	CF-PHASE	§6.3

ID	Statement / short name	Status	Dependencies	Location
EXP6-FILT	Filtration embedding verified for $N \leq 2 \cdot 10^4$, $k \leq 5$	VERIFIED COMPUTATION	COFACTOR	§6.2
CERT-34	All 34 hostile certificates of the corpus reproduced (regressions 11822, 3542948; counterexamples 242, 128, 322, 858, 4736, 6142, 14198)	VERIFIED COMPUTATION		§12.4
EMP-DEPTH	Success depth of the canonical traversal is almost always ≤ 2	EMPIRICAL PATTERN		§12.5
EMP-POFIRST	p_0 is not measurably better than p_1 at escaping	EMPIRICAL PATTERN	EXP3-CENSUS	§6.5

4.8 Post-v4.81 continuation registry (2026-09-05–2026-09-07C)

These rows are imported from the final 6420-line checkpoint. Their source statuses are preserved; no novelty claim is added. Exact statements, proofs, hostile checks, experiments and later corrections are preserved verbatim in Appendix B. The 2026-09-07C tail is binding when an earlier frontier summary conflicts with it.

ID	Short name	Status	Location
AUDIT-PREFIX-TRANSFER-20260905	Audit Prefix Transfer	PROVED	Chapter 10
CRT-TRANSCRIPT-EXACT-SCOPE-20260905	Crt Transcript Exact Scope	PROVED	Chapter 10
COPRIME-SUPPORTED-AFFINE-COUNT-20260905	Coprime Supported Affine Count	PROVED	Chapter 10
COMPLETE-H1-TWO-ROW-FINITENESS-20260905	Complete H1 Two Row Finiteness	PROVED	Chapter 10
CLOSED-ALPHABET-COUNT-20260905	Closed Alphabet Count	PROVED	Chapter 10
SMALL-BAD-CHILDREN-PREFIX-20260905	Small Bad Children Prefix	PROVED	Chapter 10
UNRESTRICTED-CLOSED-SUPPORT-HEIGHT-20260905	Unrestricted Closed Support Height	PROVED WITH REPAIR	Chapter 10
EFFECTIVE-BOUNDED-CORE-THRESHOLD-20260905	Effective Bounded Core Threshold	PROVED	Chapter 10
POINTWISE-CORE-LABEL-GROWTH-20260905	Pointwise Core Label Growth	PROVED	Chapter 10
H1-TWO-GENERATION-ENVELOPE-ESCAPE-20260905	H1 Two Generation Envelope Escape	PROVED	Chapter 10
H1-ALL-BRANCH-INJECTIVE-ESCAPE-20260905	H1 All Branch Injective Escape	PROVED	Chapter 10
TWO-SMALLEST-ROWS-HEIGHT-20260906	Two Smallest Rows Height	PROVED	Chapter 10
UNIFORM-FEW-FORMS-SIEVE-CONSUMER-20260906	Uniform Few Forms Sieve Consumer	PROVED	Chapter 10
COMPLETE-H1-FIRST-FRONT-RARE-SUCCESS-20260906	Complete H1 First Front Rare Success	PROVED	Chapter 10
SAME-N-GROWING-ROOTS-COMPLETE-FIRST-FRONT-20260906	Same N Growing Roots Complete First Front	PROVED	Chapter 10
SAME-N-NEARBY-ROOT-FIRST-FRONT-SIEVE-20260906	Same N Nearby Root First Front Sieve	PROVED	Chapter 10
COMPLETE-FIRST-FRONT-NO-GO-GROWING-SAME-N-PREFIX-20260906	Complete First Front No Go Growing Same N Prefix	PROVED	Chapter 10
COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS-20260906	Complete H1 Radius Two Rare Success	PROVED	Chapter 10
MULTIGENERATION-SUCCESS-PATH-PRODUCT-BARRIER-20260906	Multigeneration Success Path Product Barrier	PROVED	Chapter 10
GOLDBACH-TRUE-WITH-COMPLETE-SHALLOW-BAD-20260906	Goldbach True With Complete Shallow Bad	PROVED	Chapter 10
AUDIT-RADIUS-TWO-INPUT-20260906B	Audit Radius Two Input	AUDITED IDENTITY	Chapter 10
WEIGHTED-COMPATIBLE-GCD-AVERAGE-20260906B	Weighted Compatible Gcd Average	PROVED	Chapter 10
DISTINCT-PRIME-FORMS-DEGENERACY-FILTER-20260906B	Distinct Prime Forms Degeneracy Filter	PROVED	Chapter 10
SAME-N-RADIUS-TWO-PAIR-COUNT-20260906B	Same N Radius Two Pair Count	PROVED	Chapter 10
SAME-N-GROWING-ROOTS-COMPLETE-RADIUS-TWO-20260906B	Same N Growing Roots Complete Radius Two	PROVED	Chapter 10
SAME-N-RADIUS-TWO-ADVERSARIAL-SCAN-20260906B	Same N Radius Two Adversarial Scan	VERIFIED COMPUTATION	Chapter 10
COFACTOR-CHAIN-EXACT-MODULUS-20260906B	Cofactor Chain Exact Modulus	PROVED	Chapter 10
ORDERED-COFACTOR-HARMONIC-SIMPLEX-20260906B	Ordered Cofactor Harmonic Simplex	PROVED	Chapter 10

ID	Short name	Status	Location
GROWING-FORMS-UNIFORM-SIEVE-CONSUMER-20260906B	Growing Forms Uniform Sieve Consumer	PROVED	Chapter 10
MULTIGENERATION-COFACTOR-PRODUCT-BARRIER-20260906B	Multigeneration Cofactor Product Barrier	PROVED	Chapter 10
MULTIGENERATION-LABEL-PRODUCT-PAIR-COUNT-20260906B	Multigeneration Label Product Pair Count	PROVED	Chapter 10
SAME-N-GROWING-DEPTH-DUAL-PRODUCT-BARRIERS-20260906B	Same N Growing Depth Dual Product Barrier	PROVED	Chapter 10
MULTIGENERATION-COFACTOR-ADVERSARIAL-SCAN-20260906B	Multigeneration Cofactor Adversarial Scan	VERIFIED COMPUTATION	Chapter 10
SAME-N-SUPPORT-WITNESS-RANK-CORRECTION-20260907	Same N Support Witness Rank Correction	SCOPE-CORRECTED	Chapter 10
EXACT-SCAN-WORK-CONVENTION-AUDIT-20260907	Exact Scan Work Convention Audit	SCOPE-CORRECTED	Chapter 10
PRIME-LIFT-FULL-BFS-CONJUGACY-20260907	Prime Lift Full Bfs Conjugacy	PROVED	Chapter 10
LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE-20260907	Low Gap Complete Three Label Alternative	PROVED	Chapter 10
PRIME-LIFT-AND-LOW-GAP-EXHAUSTIVE-AUDIT-20260907	Prime Lift And Low Gap Exhaustive Audit	VERIFIED COMPUTATION	Chapter 10
CANONICAL-FIRST-SUPPORT-CYCLE-REALIZATION-20260907	Canonical First Support Cycle Realization	PROVED	Chapter 10
SAME-N-FIRST-SUPPORT-CYCLE-NON-DOMINANCE-20260907	Same N First Support Cycle Non Dominance	VERIFIED COMPUTATION	Chapter 10
FIXED-GAP-CANONICAL-VALUATION-EMBEDDING-20260907	Fixed Gap Canonical Valuation Embedding	PROVED	Chapter 10
HEREDITARY-FIRST-SUPPORT-P0-NO-GO-20260907	Hereditary First Support P0 No Go	PROVED	Chapter 10
CANONICAL-VALUATION-EMBEDDING-EXACT-CERTIFICATE-20260907	Canonical Valuation Embedding Exact Certificate	VERIFIED COMPUTATION	Chapter 10
LOW-GAP-SHARP-TWO-STEP-RETURN-BARRIER-20260907C	Low Gap Sharp Two Step Return Barrier	PROVED	Chapter 10
LOW-GAP-SQUARE-ROW-NO-RETURN-20260907C	Low Gap Square Row No Return	PROVED	Chapter 10
LOW-GAP-MONOMIAL-RETURN-EXPONENT-DROP-20260907C	Low Gap Monomial Return Exponent Drop	PROVED	Chapter 10
H1-SINGLETON-SQUARE-ROW-RESIDUE-FILTER-20260907G	H1 Singleton Square Row Residue Filter	PROVED	Chapter 10
CLOSED-SUPPORT-MULTIPLICATIVE-SUBGROUP-TEST-20260907G	Closed Support Multiplicative Subgroup Test	PROVED	Chapter 10
LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT-20260907G	Low Gap Three Label Exact Obstruction Split	PROVED	Chapter 10

4.9 Killed and superseded programmes

ID	Statement / short name	Status	Dependencies	Location
P0-ENGINE	Old finite-selected-transcript p_0 proof architecture	SUPERSEDED	FACP, AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905	§2.5
FACP	Finite Affine Canonical Programmability: every prime is $p_0(2(P-1))$, so no finite affine canonical certificate is universal	PROVED	—	§8.2
FBTU	Partial BAD-transcript programmability; blanket complete-BFS reading withdrawn	SCOPE-CORRECTED	CRT-TRANSCRIPT-EXACT-SCOPE-20260905	§8.2
FLUW	Blanket finite-local complete-BFS universality wall	SCOPE-CORRECTED	AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905	§8.2
CPCT	Canonical Prime Cell Theorem	PROVED	FACP	§8.2
CMFT	Canonical Multiplicative Flow Theory	KILLED	FLUW	§13.2
CGST	Canonical Graph Spectral Theory	KILLED	FLUW	§13.2
CSD	Canonical Shadow Descent	KILLED	FLUW	§13.2
CAPT	Canonical Affine Programmability Transfer	KILLED	FACP	§13.2
RSD	Root Structure Descent	KILLED	FLUW	§13.2
ROUGH-P2	Global rough- P_2 matrix	KILLED	parity	§13.4
H1-FP	H1 root-factor descent with shift-2 orthogonality	KILLED	Pillai/Bennett	§13.2

ID	Statement / short name	Status	Dependencies	Location
TWOCORE	Two-core wall $a^\alpha - a = b^\beta - b$	PROVED	[81]	§13.2

4.10 Imported results: exact product locks and no-gos (CRD Core)

ID	Statement / short name	Status	Dependencies	Location
G-MULTISOURCE-LOCK	$T_S \equiv (-1)^{ S } \prod_{\text{Src}} d \pmod{N}$ for child-closed S	PROVED	V-TUJ	Thm. 5.7
G-ROOT-LOCK	$T_R \equiv (-1)^{ R } p_0 \pmod{N}$; $T_R = (2a + 1)x + (-1)^{ R }h$, $a \geq 3$; $\max(U_R, J_R) > \sqrt{N}$	PROVED	G-MULTISOURCE-LOCK, ARS-CLOSED-WORLD	Thm. 5.8
G-CHILD-CLOSED-OVERLAP-IDENTITY	Overlap identity (OV1), (OV2)	PROVED	V-TUJ	Thm. 5.9
G-DISJOINT-CHILD-LOCKED-ORPHANS	Disjoint orphans are arithmetically orthogonal (OB1)–(OB3)	PROVED	ARS-ACTIVITY, G-COLLISION-CUT	Thm. 5.10
G-FULLQ-LEDGER	$\prod_{q \in Q_r} M_q = U_F(J_F B_{\text{re}}) B_{\text{abs}}$ on the whole Q_r under failure	PROVED	ARS-CLOSED-WORLD, V-TUJ	Thm. 5.11
G-FULLQ-PRODUCTION	$ Q_r > (5/3)^{ Q_r } \text{rad}(m_r) > 1$	PROVED	—	Thm. 5.13
G-COLLISION-CUT	$K(S) = \sum \log \gcd$; $C_{\text{cut}} = (S - 1)L - 2K$; every $\gcd \mid s - s' $	PROVED	—	Thm. 5.14
G-CLEAN-POLYLOG-CLEAN-SELECTION	Clean selection: arithmetic parent selection; support load $\leq \rho(\mathcal{L})$	PROVED	Siegel–Walfisz, lower sieve, C-GAP-EQ	Thm. 5.15
W1-MISATTRIBUTION	Earlier listing of G-CLEAN-POLYLOG-PARENT-SELECTION as W1’s open target	SCOPE-CORRECTED	G-CLEAN-POLYLOG-PARENT-SELECTION	W1-18
C-RESIDUE-ONLY-RETURN	$(\text{TURN}(\text{mod } 3))^3 = G$ for $q \geq 5$: no residue-only forced return	PROVED	—	Thm. 5.17
C-MOD3-TAGGED-FORCED-RETURN	$\text{MOD} = \text{RETURN}(\text{mod } 3)$ forces a sibling $r \equiv 2 \pmod{3}$ with $r \rightarrow 3$	PROVED	—	Thm. 5.18
C-TOPOLOGY-NOGO	Topology alone cannot force $\Delta \log J \gg W(E)$	PROVED	—	Thm. 5.20

4.11 Imported results: the rooted first-front layer (Biblia v7.3–v7.5)

ID	Statement / short name	Status	Dependencies	Location
B-FRESH-PACKING	(FP): $\prod_{q \in E} s_q \geq m_0^{ E } / \prod q$	PROVED – PENDING	INCORRECT TEAM	Prop. 7.10
B-SMOOTH-TAX	(ST): $B_q^{e_q} > q$, $e_q = j_q + u_q$	PROVED – PENDING	INCORRECT TEAM	Prop. 7.11
B-SMOOTH-EXCESS	$\Omega(U_R) + \Omega(J_R) - 1 \geq \sum e_q$	PROVED – PENDING	INCORRECT TEAM	Cor. 7.12

ID	Statement / short name	Status	Dependencies	Location
B-FFC	(FFC), (FFC- Ω): root first-front conservation	PROVED – PENDING RED TEAM	—	Thm. 7.14
B-FFRE	(FFRE): $\prod \text{rad}(N - q) = J_{S_1} R_{\partial}(S_1)$	PROVED – PENDING RED TEAM	B-FFC	Cor. 7.15
B-UJD-SPLIT	(U-split), (JR-split): exact first-front $U/J/\partial$ split	PROVED – PENDING RED TEAM	B-FFRE	Thm. 7.16
B-GCD-PACK	$\nu_{\ell} \leq \min(Q , 1 + \lfloor m_0/\ell \rfloor)$	PROVED – PENDING RED TEAM	—	Lem. 7.18
B-POWERFUL-TAIL	Powerful-integer density is too weak for the adaptive first front	KILLED	—	Scope 7.19
B-RSCP	Root Smoothness Conservation Problem	OPEN	B-UJD-SPLIT, B-SMOOTH-TAX	Target 7.20
B-PP-LEDGER	(PP-I), (PP-D), (PP): lossless prime-power root ledger	PROVED – PENDING RED TEAM	PR-DQ	Thm. 7.25
B-POSTROOT-DEMAND	$D_q = \gcd(B_q, 2h - q)$; (Demand); (Post-root tax)	PROVED – PENDING RED TEAM	B-SHADOW-TAX	Thm. 7.26
B-ROOT-VAL-RIGID	$v_r(N - q) > a \Rightarrow v_r(2h - q) = a$	PROVED – PENDING RED TEAM	B-SHADOW-TAX	Lem. 7.27
B-SHARP-PP-PACK	$\ell^k \mid n_q, n_r \Rightarrow \ell^k \mid q - r$; the diameter bound on $\nu_{\ell,k}$	PROVED – PENDING RED TEAM	—	Lem. 7.28
B-VPC	(VPC): actual prime-power collision budget	PROVED	PR-MQ	Thm. 7.30
B-SHARED-MASS	$L_{\text{shared}} \leq 2 \log \Delta_{\text{odd}}(S)$	PROVED	B-VPC	Cor. 7.31
B-PAIR-FLOOR	$L_{\text{pair}}(q, r) \geq \log 5$	PROVED	PR-FRESH	Lem. 7.32
B-WEIGHTED-CUT	(Cut): $\sum L_{\text{pair}} = \sum_{\lambda} \mu_{\lambda}(s - \mu_{\lambda})w_{\lambda}$; $C_{\text{cut}} \geq \binom{s}{2} \log 5$	PROVED	B-PAIR-FLOOR	Thm. 7.33
B-LOW-MULT	Low-multiplicity layer lower bound $W_{\leq u}, W_1$	PROVED	B-WEIGHTED-CUT, B-VPC	Cor. 7.34
B-EXACTD-OUTSIDE	$t < r \Rightarrow M_q M_r > \delta^2$; failure regime (ED-res)	PROVED	PR-DQ	Thm. 7.36
B-GAP-FILTER-1	$9p_0 \geq 18G + 5G^2 \Rightarrow M_q M_r > \delta^2$	PROVED	B-EXACTD-OUTSIDE, C-GAP-EQ	Cor. 7.37
B-JOINT-D	(Joint-D): $D_q D_r \mid m_0 \gcd(c, r - q)$	PROVED	PR-DQ	Thm. 7.38
B-PNF	(PNF): primitive residual normal form	PROVED	B-EXACTD-OUTSIDE	Thm. 7.39
B-HIGH-GAP	(HG-r), (HG): $h^2 > 4m_0$ under pair failure	PROVED	B-PNF	Thm. 7.40
B-GAP-FILTER-2	$p_0 < 2G + G^2/4$	PROVED	B-HIGH-GAP	Cor. 7.42
B-K1-GAP	$r > 4^{d+1} q u^2$; $p_0 < 2G + G^2/12$	PROVED	B-HIGH-GAP	Thm. 7.43
B-K1-QUOTIENT	$k = 1$ quotient normal form in (A, B)	PROVED	B-PNF	Thm. 7.44
B-K1-COVERAGE	Structured composite coverage; $\gcd(B, qu - 1) = 1$	PROVED	B-K1-QUOTIENT	Prop. 7.46
B-K1-BARRIER	Canonical $k = 1$ gap/covering barrier	OPEN	B-K1-COVERAGE	Target 7.47
B-MULTIPLEX-RIGID	$D \geq \delta m^{\kappa_{\text{col}}}$	PROVED – PENDING RED TEAM	—	Thm. 7.49
B-COMP-BARRIER	$m < (N/6)^{\chi/\eta}$; $\eta \geq t\chi \Rightarrow m < (N/6)^{1/t}$	PROVED – PENDING RED TEAM	B-MULTIPLEX-RIGID	Cor. 7.50
B-SPARSE-CORE-LOCK	$X - 1 \leq U_C J_C \leq U_C (D/2)^{\chi_{\tau}}$	PROVED – PENDING RED TEAM	B-MULTIPLEX-RIGID	Cor. 7.51

ID	Statement / short name	Status	Dependencies	Location
B-SUPPORT-EXPLOSION	$ OS \geq \lceil \log A_S / \log B_S \rceil$	PROVED – PENDING RED TEAM		Thm. 7.53

4.12 Imported results: the canonical threshold programme (v4.79–v4.81)

ID	Statement / short name	Status	Dependencies	Location
V79-NAGURA	$p_0 < 3N/5, m > 2N/5$ for $N \geq 50$	PROVED	[1], C-GAP-EQ	Prop. 9.1
V79-FLOOR	(V79.4): $U_R^2 J_R B_{\text{abs}} > m(5/3)^{ Q } > 2N/3$	PROVED	G-FULLQ-PRODUCT	Thm. 9.2
V79-COEFF-DICH	$F_Q^{\text{coeff}} = 1 \iff \text{primitive};$ $F_Q^{\text{coeff}} > 1 \Rightarrow T_R > \frac{5}{2}N$	PROVED	G-ROOT-LOCK	Thm. 9.3
V80-LOADONE	Primitive $\Rightarrow D_q = 1; \prod_{q \in Q} (N - q) > \frac{25}{36}N^2$	PROVED	V79-COEFF-DICH	Thm. 9.5
C-PO-PRIMITIVE-OWERPRESERVING-QUADRATIC-UJD-CHAF	OWERPRESERVING-QUADRATIC-UJD-CHAF	PROVED	V80-LOADONE	Thm. 9.6
C-PO-PRIMITIVE-FIRSTFRONT-ULTIMARCON-PRODUCTION	FIRSTFRONT-ULTIMARCON-PRODUCTION	PROVED	V79-FLOOR	§9.2.1
C-PO-PRIMITIVE-BASES-Escape-or-must-charge	BASES-Escape-or-must-charge charge base absorption	SUPERSEDED	C-PO-PRIMITIVE-OWERPRESERVING-QUADRATIC-UJD-CHAF	§9.2.1
V81-CONTAINER	(V81.1): $U_R J_R D_{\lambda_N}$ is the row product of the world	PROVED	V80-LOADONE	Thm. 9.7
V81-SLACK	(V81.2): container slack $> 5N/6$ for $N \geq 500$	PROVED	V81-CONTAINER	Thm. 9.8
V81-SIZE-DEATH	The archimedean size route is closed	PROVED	V81-SLACK	Cor. 9.9
V81-OVERLAP	(V81.3): reflected/selected overlaps are charge-free	PROVED	ARS-REFLECTION-CATE	Thm. 9.10
C-PO-PRIMITIVE-FIRSTFRONT-REFLECTION-DISJOINT-CHAF	FIRSTFRONT-REFLECTION-DISJOINT-CHAF alternatives (1) and (2)	KILLED	V81-SLACK, V81-OVERLAP	§9.2.2
V81-C1-SHADOW	(V81.11): primitive load-one $\iff \gcd(m, 2h - u) = 1$ on $R \setminus \{p_0\}$	PROVED	S-SHADOW-ID	Thm. 9.12
V81-C2-RESIDUAL	R' child-closed, nonempty, $\min R' < \sqrt{N}$	PROVED	ARS-CLOSED-WORLD	Thm. 9.14
V81-C3-FACTOR	$T_R \Sigma' = (\prod_{q \in Q} (N - q)) T_{R'}$	PROVED	V81-C2-RESIDUAL	Thm. 9.15
V81-C4-RESLOCK	(V81.13): selected residue lock	PROVED	G-MULTISOURCE-LOCK	Thm. 9.16
V81-C5-ALPHABET	Adaptive fringe cofactor alphabet $\mathcal{L}_N \subseteq [\lambda, 2\lambda)$	PROVED	V80-LOADONE	Thm. 9.17
V81-IMMUNITY	Product-comparison immunity of the minimal survivor	PROVED	V81-SLACK	Thm. 9.19
V81-TRICHOTOMY	Only routes (A) population, (B) residue, (C) GOOD remain	PROVED	V81-IMMUNITY	Thm. 9.20
C-PO-PRIMITIVE-LOWCORE-SMOOTH-POPULATION-CAPACITY	LOWCORE-SMOOTH-POPULATION-CAPACITY $\Psi^*(N, \lambda_N)$	OPEN	V81-TRICHOTOMY, V81-C5-ALPHABET	§9.2.4
C-PO-PRIMITIVE-SELECTED-RESIDUE-LOCK-CAPACITY	SELECTED-RESIDUE-LOCK-CAPACITY ation of $\mathcal{C}_Q \bmod N$	OPEN	V81-C4-RESLOCK, G-ROOT-LOCK	§9.2.4

ID	Statement / short name	Status	Dependencies	Location
C-P0-PRIMITIVE-SELECTED-UP-CAPACITOR-preserving selected charge		OPEN	C-P0-PRIMITIVE-SWIRL	SWIRL-PRESERVING-QUADRA
G-CONDITIONAL-EFFECTIVE-PAIR-NEFF	Conditional effective $N_0^{p_0}$	PROVED (conditional)	V79-FLOOR	§9.2.1
FAIL-SUNIT-COUNT-BOUNDS-SHELOWCORR+DO1	Bounds-SHELOWCORR+DO1 bounds $ C_\lambda $	KILLED	[20]	§9.2.4

4.13 Imported results: surviving lemmas of killed programmes

ID	Statement / short name	Status	Parent (killed)	Location
PCRD-CMFT-1	Lossless anchor filtration; $\mathcal{H}_{j+1} = \mathcal{H}_j \sqcup \mathcal{A}_j$	PROVED	CMFT	Thm. 13.2
PCRD-CMFT-2	Tail hard-edge universality: local tail geometry does not distinguish p_0	PROVED	CMFT	Thm. 13.3
PCRD-CMFT-3	Canonical least-factor split	PROVED	CMFT	Thm. 13.4
PCRD-CMFT-4	All-root collision identity	PROVED	CMFT	Thm. 13.5
PCRD-CMFT-5	Prime-pressure failure ceiling: $\mathcal{P}_N > \sqrt{x} \log x \Rightarrow$ Goldbach	PROVED	CMFT	Thm. 13.6
PCRD-CGST-1	Canonical straddle theorem	PROVED	CGST	Thm. 13.7
PCRD-CGST-2	Arbitrarily deep finite reflection obstruction	PROVED	CGST	Thm. 13.8
PCRD-CGST-3	Diagonal collapse: the cell dimension is false	PROVED	CGST	Thm. 13.9
PCRD-CSD-1	Canonical conductor-gap compatibility	PROVED	CSD	Thm. 13.11
PCRD-CAPT-1	Fixed- (r, s) compression to two linear prime forms	PROVED	CAPT	Thm. 13.12
PCRD-RSD-1	Exact rough signed identity	PROVED	RSD	Thm. 13.14
PCRD-RSD-2	Distribution-roughness phase law $\theta > 2/(k+1)$	PROVED	RSD	Thm. 13.15
PCRD-M-1	$K_{3,3}$ explosion at the natural scales	CONJECTURE (conditional)	ROUGH2	Thm. 13.17
PCRD-M-2	Boundary blindness of generic operator norms	PROVED	ROUGH2	Thm. 13.18
PCRD-M-3	$h = 1$ matrix invisibility	PROVED	ROUGH2	Thm. 13.19
PCRD-M-4	Fixed-modulus spectrum recovers only the singular series	PROVED	ROUGH2	Thm. 13.20
PCRD-P0-3	Gap scalar freedom: $h, G, h/G$ have no closing regime	PROVED	P0-ENGINE	Thm. 13.22
PCRD-H1-1	H1 root-factor descent; $\text{PF}(N - q_*) \cap \text{PF}(m_0) = \emptyset$	PROVED	H1-FP	Thm. 13.24
PCRD-H1-2	H1 shift-2 orthogonality $\gcd(N - u, u + 2) = 1$	PROVED	H1-FP	Thm. 13.25

ID	Statement / short name	Status	Parent (killed)	Location
PCRD-H1-3	3-core normal form; no 3-core found in finite boxes	VERIFIED COMPUTATION	UTAPP	Rem. 13.26
MATCHED-EUCLID-GMATCH	Matched Euclidean gluing lemma (named missing theorem)	OPEN	—	§16.5

Completeness statement. This registry is *not* an exhaustive index of the sources. The CRD Theory Core alone contains 801 live titled results and the Biblia 673; what is imported here is the load-bearing subset — exact identities, no-go theorems, frontier reductions, and the lemmas that survive their killed parent programmes. If a needed result is not here, it may still exist in the sources; search [CORE] by **THEOREM-ID** and [BIB] by theorem title before concluding that something is unproved.

Chapter 5

Core Results (the Biblia layer)

This chapter holds the exact machinery that the project actually owns: shadow arithmetic, path invariants, and the valuation ledger. Almost all of it is *root-generic* — it works from any upper root and never uses minimality. Reading it with that in mind is the fastest cure for the most common error in the project’s history, which is to quote a generic lemma as a canonical one.

5.1 Scope classes A / B / C / D

Definition 5.1 (The four scope classes). A result T is

- class A — genuine p_0 -specific if there is an *explicit proof step* that fails when p_0 is replaced by another upper root, *and* the content of the conclusion actually changes;
- class B — root-generic if every step goes through verbatim for $r = x + h_r$ with h_r, m_r in place of h, m_0 ;
- class C — failure-conditioned if its hypothesis is that the traversal from the root fails;
- class D — Goldbach-counterexample-conditioned if its hypothesis is $\text{Good}_N = \emptyset$.

Separately there is the *vacuity-canonical* case: the construction goes through for p_k on those coordinates where it is defined at all, and canonicity only guarantees that it is defined everywhere. This is **not** class A. The corpus repeatedly confused the two.

Object / result	Class	What happens under $p_0 \rightarrow p_k$
Relation $p \rightarrow q$, $\mathcal{A}_N$, valuations, components	B	nothing; this is prior art §16.2
$T_S = U_S J_S$	B	nothing
High-root compression, unique source	B + C	nothing; $r > N/3$ and $N - r$ composite suffice
$q \leq m_r/3 < N/6$	B + C	nothing
$\gcd(h_r, m_r) = 1$, q^e congruences	B	nothing; root algebra
Shadow identity, retention, D_q , M_q	B	nothing FAIL-MR2-CANONICAL-ONLY-001
Bézout / winding / state count	B	nothing; $h \mapsto h_r$
Shadow cycles, prime-power witness	B + C	nothing
Strict non-reuse of the block	B	nothing; the proof uses index differences only
No-bad-return, two-sheet budget	C	nothing beyond the failure hypothesis

Object / result	Class	What happens under $p_0 \rightarrow p_k$
Fixed point B_N^{low} and the failure characterisation	C	nothing; the object depends on N alone
Reflection gate (<i>activation</i>)	vacuity	for p_k exactly k coordinates are prime and there is nothing there for the gate to open
Exact bundle lift	vacuity	the clauses go through verbatim on every row with composite c'_j ; 25.2% of rows are lost in the recorded scan [CORE, P0-CANONICITY-ABLATION]
Zero hole count as <i>mass</i>	KILLED	loss is $O_k(\log N)$; not a mechanism
$h < G$ / empty interval	A	everything; this is the definition and the only canonical input
$\mathcal{P}_{\text{def}} = \emptyset$	A	exactly k prime atoms appear
$\kappa_{\text{gen}} = 0$	A	the rank rises to k ; the equivalence is exact in both directions
Reflection-completeness	A	uniqueness is lost
Ledger completeness $\mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}$	A	a fifth class $\mathcal{P}_{\text{def}}$ appears
Lossless code of prime history	A	p_k encodes k earlier roots; p_0 encodes none
Phase trichotomy and $j^* = h - G/2$	A	the notion “predecessor outside the block” exists only for the canonical root
$\rho_0 \geq 1 \iff \theta > 1/2$	A	for p_k the predecessor is p_{k-1} , so $D_1 \leq h_k$ and the whole characterisation collapses
Exact-D filter $p_0 < 2G + G^2/4$	A + C	without $h < G$ the filter does not exist; non-canonical countermodels exist
Exact-D model elimination	A + C	the models return FAIL-EXACTD-NONCANONICAL-001
Upper-half compression	D	requires $\text{Good}_N = \emptyset$ FAIL-UPPERHALF-SCOPE-001
Symmetric fracture, forbidden off-set band	D	require the minimal-counterexample hypothesis

The separation result in three sentences.

1. Almost the whole local arithmetic of p_0 — shadow, D_q , M_q , Bézout, winding, cycles, the U/J ledger — is *generic upper-root algebra*. Minimality is used nowhere in it.
2. The only genuine canonical input is $h < G$, equivalently emptiness of $[x, p_0) \cap \mathbb{P}$. It enters proofs in exactly one place: it makes *every* c_j composite.
3. The consequences of that single input — $\kappa_{\text{gen}} = 0$, reflection-completeness, ledger completeness, the phase trichotomy, the ρ_0 formula — are the only known properties of p_0 that have an *exact converse*. Every other candidate either has no converse (least-upper-parent) or was refuted (mass, density, local algebra, statistics).

5.2 The generic layer

Theorem 5.2 (High-root compression and unique source). **PROVED**, class $B + C$. Let $r > N/3$ be an active odd prime, $N - r$ composite, and suppose the traversal from r fails. Then every non-root state lies below $N/3$ and

$$\Gamma_N(R_N(r)) = R_N(r) \setminus \{r\}.$$

Proof. For a bad row, $N - p$ is odd composite, so every prime factor q has an odd cofactor ≥ 3 ,

whence $q \leq (N - p)/3 < N/3$. If r had a parent $p \in R$ then $N - p = kr$ with k odd and $k < 3$, so $k = 1$ and $p = N - r$ would be prime, contradicting compositeness of the complement. [CORE, G-HIGH-ROOT-SOURCE; canonically ARS-UPPER-EXCLUSION] $\square$

Reading. The bad world has one source and the whole upper half apart from the root is external to it. **This is not a property of p_0** — it is a property of being a high root with composite complement. **VERIFIED COMPUTATION** on 301 real closed worlds ($N \leq 2 \cdot 10^4$), zero violations.

Theorem 5.3 (Canonical prime-free geometry). **PROVED**. Hypotheses: *canonical failure, non-diagonal case*. Conclusion:

$$[x, p_0) \cap \mathbb{P} = \emptyset; \quad m_0 \text{ odd composite, } m_0 \geq 9; \quad \gcd(m_0, N) = \gcd(h, m_0) = 1;$$

$$q \in Q \Rightarrow q \leq \frac{m_0}{3} < \frac{N}{6}; \quad q^e \parallel m_0 \Rightarrow x \equiv h, \quad p_0 \equiv 2h = \delta \pmod{q^e}.$$

Classes: *emptiness of the interval* — A ; $\gcd(h, m_0) = 1$ and the q^e congruences — B (identical for any upper root with h_r, m_r); the inequality $q \leq m_r/3$ — $B + C$.

Proof. Emptiness is minimality. m_0 is odd as the complement of an odd prime, and composite, else the root is GOOD. Next $\gcd(m_0, N) = \gcd(p_0, N)$, and $p_0 \mid N$ with $p_0 \geq N/2$ would force $N = 2p_0$, the diagonal case. If $d \mid h$ and $d \mid m_0 = x - h$ then $d \mid x$, hence $d \mid x + h = p_0$; from $0 < d \leq h < p_0$ and primality of p_0 , $d = 1$. The inequality: the cofactor m_0/q is odd and ≥ 3 . The congruences are $m_0 = x - h$ and $p_0 = x + h$. [BIB, normal form; CORE, ARS-CANON-GEOMETRY] $\square$

VERIFIED COMPUTATION: 4915 canonical bad roots for $N \leq 1.5 \cdot 10^4$, zero violations.

Scope note 5.4 (Inequality repair). **SCOPE-CORRECTED**. Some earlier notes wrote $q < m_0/3$. The correct inequality is $q \leq m_0/3$; equality occurs e.g. for $N = 32$, $p_0 = 17$, $m_0 = 15$, $q = 5$. The consequence $q < N/6$ remains strict. [BIB, minor inequality repair]

Theorem 5.5 (Least upper parent). **PROVED**, class A . If $r = p_0$ then for every prime power $q^e \mid N - r$

$$r = \min(\mathcal{P}_N(q^e) \cap [x, N)), \quad \mathcal{P}_N(q^e) = \{p \in \mathcal{A}_N : q^e \mid N - p\}.$$

[CORE, ARS-LEAST-UPPER-PARENT]

Warning 5.6 (No converse). **KILLED** as a selector. In the recorded scan of 301 failing roots, eleven non-canonical roots satisfy the same first-front minimality, among them (128, 73), (128, 79), (1718, 877), (1862, 941), (1928, 971), (6142, 3089); exactly one has the stronger world-level property, (1928, 971). So *first-front parent minimality does not by itself explain p_0* . [CORE, V71.41]

5.3 Exact product locks and world identities

These are the exact multiplicative identities of the CRD Core. All are **PROVED** with short proofs, all are safe to use as lemmas, and none of them was in earlier compressions of this project. They are the algebraic backbone that every capacity argument silently uses.

Theorem 5.7 (Multi-source product lock, G-MULTISOURCE-LOCK). **PROVED**, class B . If a finite $S \subseteq \mathcal{A}_N$ is child-closed and $D_S^{\text{src}} = S \setminus \Gamma_N(S)$, then

$$T_S \equiv (-1)^{|S|} \prod_{d \in D_S^{\text{src}}} d \pmod{N}.$$

Proof. $T_S = \prod_{s \in S} (N - s) / \prod_{q \in \Gamma_N(S)} q$. Modulo N replace each $N - s$ by $-s$ and cancel the invertible active labels of $\Gamma_N(S)$; exactly the sources survive. [CORE, G-MULTISOURCE-LOCK] $\square$

Limitation. A rooted one-source lock cannot upper-bound an enlarged ambient multi-source system; adjoining external parents changes the source term.

Theorem 5.8 (Rooted full-front lock, G-ROOT-LOCK). **PROVED**, class C. If canonical p_0 fails, $R = R_N(p_0)$, $r = |R|$, then

$$T_R = U_R J_R \equiv (-1)^r p_0 \pmod{N}, \quad \text{and for an integer } a \geq 3, \quad T_R = (2a + 1)x + (-1)^r h.$$

In particular $\max(U_R, J_R) > \sqrt{N}$.

Proof sketch. Unique-source (Lemma 3.26) gives $T_R = m_0 \prod_{p \in R \setminus \{p_0\}} (N - p)/p$, and the congruence follows from Theorem 5.7. Every non-root row lies below x , so each ratio exceeds one. Writing $q = P^-(m_0)$, $B = m_0/q$: if $m_0 \neq 9$ then $B \geq 5$, so $T_R \geq B(N - q) \geq 5N - m_0$, exceeding every affine value with $a \leq 2$. If $m_0 = 9$ then $3 \nmid N$, row 3 is bad, and Theorem 5.2 places its non-root child s below $N/3$, giving $T_R \geq 9 \cdot \frac{N-3}{3} \cdot \frac{N-s}{s} > 3N$ for $N \geq 6$, while every affine value with $a \leq 2$ is at most $5x + h < 3N$. [CORE, G-ROOT-LOCK] $\square$

Reading. This is the sharpest *unconditional* statement the project has about a canonical failed world: its total valuation mass is pinned to p_0 modulo N and forced into an explicit affine form. It is the container of route (B) in the surviving-route trichotomy (§9.2).

Theorem 5.9 (Exact source-overlap identity, G-CHILD-CLOSED-OVERLAP-IDENTITY). **PROVED**, class B. Let $A, B \subseteq \mathcal{A}_N$ be finite child-closed sets with $C = A \cap B \neq \emptyset$, $U = A \cup B$, and write $\Sigma(S) = \prod_{d \in D_S^{\text{src}}} d$. Then

$$T_A T_B \Sigma(U) \Sigma(C) = T_U T_C \Sigma(A) \Sigma(B). \quad (\text{OV1})$$

If moreover $D_U^{\text{src}} = D_A^{\text{src}} \uplus D_B^{\text{src}}$, then

$$T_A T_B \Sigma(C) = T_U T_C. \quad (\text{OV2})$$

Proof. For finite child-closed S , $\Gamma_N(S) = S \setminus D_S^{\text{src}}$, so $T_S = \Sigma(S) \prod_{s \in S} (N - s)/s$. The multiplicative row ratio satisfies exact inclusion–exclusion over A, B, U, C ; substituting gives (OV1), and the disjoint-source hypothesis cancels $\Sigma(U) = \Sigma(A) \Sigma(B)$. [CORE, G-CHILD-CLOSED-OVERLAP-IDENTITY] $\square$

Theorem 5.10 (Disjoint child-closed worlds are arithmetically orthogonal, G-DISJOINT-CHILD-CLOSED-ORTHOGONALITY). **PROVED**, class B. Let $S_1, \dots, S_b \subseteq \mathcal{A}_N$ be finite, pairwise disjoint and child-closed, $U = \bigcup_i S_i$. Then for $i \neq j$,

$$\gcd(N - u, N - v) = 1 \quad (u \in S_i, v \in S_j), \quad (\text{OB1})$$

the supports $\Gamma_N(S_i)$ are pairwise disjoint, and

$$T_U = \prod_i T_{S_i}, \quad K(U) = \sum_i K(S_i), \quad \text{Src}(U) = \bigsqcup_i \text{Src}(S_i), \quad (\text{OB2})$$

$$\eta(U) = \sum_i \eta(S_i), \quad \nu(U) = \sum_i \nu(S_i). \quad (\text{OB3})$$

Proof. A prime q dividing both $N - u$ and $N - v$ is active (Definition 3.2), so child-closure puts it in $S_i \cap S_j$ — impossible. Hence no prime-power layer occurs in two sets, and the primewise definitions of T , K , η , ν are additive; absence of cross edges splits the source sets. [CORE, G-DISJOINT-CHILD-CLOSED-ORTHOGONALITY] $\square$

Reading. Disjoint failed packets carry *no* shared arithmetic whatsoever. This is why “combine several failed worlds and count” arguments gain nothing: the combination is a product, not an interaction.

Theorem 5.11 (Full- Q failure ledger, G-FULLQ-LEDGER). **PROVED**, class C. If the stopped traversal from $r > N/2$ fails and $N - r$ is composite, then with $S = Q_r$

$$\prod_{q \in Q_r} M_q = U_F(J_F B_{\text{re}}) B_{\text{abs}}, \quad U_F \mid U_R, \quad J_F B_{\text{re}} \mid J_R.$$

Every residual prime lies in R , and the same formulas hold for any subfamily of Q_r .

Proof. Failure puts every first-front row q in R , so $\kappa_R(\ell) \geq m_\ell$, and closure puts every residual factor in R . For $\ell \notin Q_r$ split $E_\ell = (E_\ell - m_\ell) + (m_\ell - 1) + 1$: the final copy is reused when $\kappa_R > m_\ell$ and absorbed otherwise. For $\ell \in Q_r$ the root row already supplies the base copy and all residual copies enter U_F . The same primewise comparison gives the two divisibilities. [CORE, G-FULLQ-LEDGER] $\square$

Warning 5.12 (This is the correct domain of the incidence ledger). Theorem 5.11 *supersedes* the reading of IL1–IL6 as a Q_{smooth} -only statement: on the failed branch the identity holds on the whole of Q_r . The Q_{smooth} version (Theorem 7.6) is the restriction. The unconditioned extension to arbitrary Q_0 still fails at $N = 20$ (FAIL-Q0-LEDGER-001), because a successful first-front row can give $\kappa_R < m_\ell$.

Theorem 5.13 (Full- Q production, G-FULLQ-PRODUCTION). **PROVED**, class B. For an active prime $r > N/2$ with odd composite $m_r = N - r$,

$$\mathfrak{F}_{Q_r} := \prod_{q \in Q_r} M_q > \left(\frac{5}{3}\right)^{|Q_r|} \text{rad}(m_r) > 1.$$

Proof. For $q \mid m_r$, activity gives $q \nmid N$, hence $q \nmid D_q$ and $D_q \leq m_r/q$. An odd composite has $q \leq m_r/3$, and $m_r < N/2$, so $M_q \geq q(N - q)/m_r > \frac{5}{3}q$. Multiply over distinct $q \in Q_r$. [CORE, G-FULLQ-PRODUCTION] $\square$

Limitation. The unit complement has $Q = \emptyset$ and is excluded (Remark 3.3, FAIL-UNIT-COMPLEMENT-001).

Theorem 5.14 (Exact collision and cut identities, G-COLLISION-CUT). **PROVED**, class B. For odd row values Y_s ($s \in S$), with $L(S) = \sum_{s \in S} \log Y_s$ the total row mass, $K(S)$ the collision mass and $C_{\text{cut}}(S)$ the cut functional,

$$K(S) = \sum_{s < s'} \log \gcd(Y_s, Y_{s'}), \quad C_{\text{cut}}(S) = (|S| - 1)L(S) - 2K(S).$$

If $Y_s = N - s$ then every such gcd divides $|s - s'|$.

Proof. Use $v_\ell(\gcd(Y_s, Y_{s'})) = \sum_{k \geq 1} \mathbf{1}_{\ell^k \mid Y_s} \mathbf{1}_{\ell^k \mid Y_{s'}}$ and sum; the cut identity is $(n-1)\mu - 2\binom{\mu}{2} = \mu(n-\mu)$. A common divisor of $N - s$ and $N - s'$ divides $s - s'$. [CORE, G-COLLISION-CUT] $\square$

Limitation. A singleton layer contributes nothing ($\binom{1}{2} = 0$); the identity is generic and cannot by itself supply a canonical absorption trichotomy (FAIL-GENERIC-ABSORPTION-001).

Theorem 5.15 (Clean polylogarithmic parent selection, G-CLEAN-POLYLOG-PARENT-SELECTION). **PROVED**. Fix $A > 0$. For all sufficiently large even N , let canonical p_0 fail, $R = R_N(p_0)$, and let $\mathcal{L} \subseteq \{(\ell, k) : \ell \in R, \ell^k \leq (\log N)^A\}$ be finite and nonempty, with $\rho_\ell = \#\{k : (\ell, k) \in \mathcal{L}\}$ and $\rho(\mathcal{L}) = \max_\ell \rho_\ell$. Then there exist distinct upper-half external active primes $p_{\ell,k} > p_0$, one per $(\ell, k) \in \mathcal{L}$, with

$v_\ell(N - p_{\ell,k}) = k$ and the clean-support property: if a prime r divides $N - p_{\ell,k}$ and $N - p_{\ell',k'}$ for two distinct selected parents, then

$$r = \ell = \ell'.$$

Consequently $\max_r \#\{p \in \mathcal{P} : r \mid N - p\} \leq \rho(\mathcal{L})$, and if the layers have pairwise distinct base primes the selected complements are pairwise coprime and the global support load is one.

Hypotheses beyond canonical failure: Siegel–Walfisz, a standard lower-bound sieve, Mertens, Theorem 5.2 and Theorem 3.10. [CORE, G-CLEAN-POLYLOG-PARENT-SELECTION]

Warning 5.16 (Correction to an earlier reading of W1). **SCOPE-CORRECTED**. Earlier compressions of this project listed G-CLEAN-POLYLOG-PARENT-SELECTION as the *open target* of W1. It is not: it is a **PROVED** theorem of the CRD Core. What remains open in W1 is the passage from the *polylogarithmic* layer family $\ell^k \leq (\log N)^A$ to the *actual* layers of the world — i.e. W1 is blocked on L1 (§9.5), not on the selection lemma. See §9.7 as corrected.

5.4 Residue no-go theorems

Theorem 5.17 (Residue-only return no-go for $q \geq 5$, C-RESIDUE-ONLY-RETURN-NOGO). **PROVED**. Let $q \geq 5$ be prime, $G = (\mathbb{Z}/q\mathbb{Z})^\times$, $a = N \bmod q$. Then

$$(G \setminus \{a\})^2 = G.$$

Hence every possible tagged quotient residue is a product of two sibling residues neither congruent to $N \bmod q$. Consequently the single residue $b \bmod q$, even together with $\Omega(b) \leq 3$, cannot force a direct sibling return to q . [CORE, C-RESIDUE-ONLY-RETURN-NOGO]

Theorem 5.18 (Exceptional forced return at modulus 3, C-MOD3-TAGGED-FORCED-RETURN). **PROVED**. Assume $N \equiv 2 \pmod{3}$ and $N - p = 3^e b$ with $3 \nmid b$, $b \equiv 2 \pmod{3}$. Then some sibling prime $r \mid b$ satisfies $r \equiv 2 \pmod{3}$, hence $3 \mid N - r$ and $r \rightarrow 3$. [CORE, C-MOD3-TAGGED-FORCED-RETURN]

Warning 5.19 (Modulus 3 is an exception, not a mechanism). Theorems 5.17 and 5.18 together say exactly this: the pure local-residue route *saturates* for every $q \geq 5$, and 3 is the single exception. Any programme that hopes to iterate the mod-3 forced return into a general mechanism is dead before it starts; this is one of the convergent no-gos that killed CRD1 (§8.1).

Theorem 5.20 (Topology alone cannot convert external seed weight into ambient incremental J , C-TOPOLOGY-NOGO). **PROVED**. No theorem using only finite child closure, source counts, support-graph topology, edge multiplicities and the reverse-extension ledger can force $\Delta \log J \gg W(E)$ for arbitrarily large external source weights $W(E)$.

Proof. A finite realisation permits a family with one small receiving target q and arbitrarily many external components feeding it. Topology and attachment count grow linearly, while the horizontal weight at the receiving label grows only like $e \log q$; external source labels may carry arbitrarily larger evaluation weights without altering the recorded topology. The missing ambient factor must therefore come from arithmetic/order/root information, not from the listed ledgers. [CORE, C-TOPOLOGY-NOGO] $\square$

Reading. This is the precise reason the “ η is the capacity” family of attempts is hopeless (Warning 3.21): the quantity that would have to do the work is provably not a function of the topology.

Remark 5.21 (First-occurrence subtlety). Fresh support is *not* incremental J : if $\kappa_R(\ell) = 0$ then the first occurrence of ℓ belongs to the fresh-support factor $F_R(A)$, and only a *later* occurrence contributes a full $\log \ell$ to J_S/J_R . Confusing the two is a recurring source of phantom capacity.

5.5 Shadow and retention arithmetic

Theorem 5.22 (Prime-gap shadow identity). **PROVED**, class B. For $r \in Q$ and any prime p ,

$$r \mid (N - p) \iff r \mid (\delta - p),$$

and the same equivalence holds modulo r^e when $r^e \mid m_0$.

Proof. $N - p = 2m_0 + \delta - p \equiv \delta - p \pmod{r^e}$. [BIB, thm:shadow; CORE, ARS-SHADOW] $\square$

Reading. The whole part of the support that returns to the first alphabet Q is governed by the single number $\delta = 2h$. This is the first point at which the gap parameter enters the multiplicative dynamics. *Limitation:* the theorem does not replace the full factorization of $N - p$; a factor outside Q may lead to success, to a new transient component, or to another sink.

Warning 5.23 (This is *not* a p_0 selector). **KILLED** as a canonical statement. For *any* upper root $r = x + h_r$ with $m_r = x - h_r$ we have $r = m_r + 2h_r$, so for every odd $q \mid m_r$, $q \mid (r - 2j) \iff q \mid (h_r - j)$. Minimality is not used. [FAIL, FAIL-MR2-CANONICAL-ONLY-001]

Theorem 5.24 (Retention-escape). **PROVED**, class B. For $q \in Q$ and $d_q = \gcd(m_0, |\delta - q|)$,

$$\text{PF}(N - q) \cap Q = \text{PF}(d_q).$$

In particular $d_q = 1$ means every active child of the row $N - q$ is new relative to Q .

Proof. For $r \in Q$: $r \mid N - q \iff r \mid \delta - q \iff r \mid \gcd(m_0, |\delta - q|)$; collecting over all r gives the claim. [BIB, thm:retention] $\square$

VERIFIED COMPUTATION, $N \leq 10^4$, zero violations.

Corollary 5.25 (Minimum-factor escape). **PROVED**, class B. Let $a = P^-(m_0)$. If $a > h$ then $d_a = 1$.

Proof. Every prime divisor of d_a lies in Q , hence is $\geq a$; but $a > h$ gives $0 < |2h - a| < a$, and $2h = a$ is impossible by parity. [BIB, cor:min-factor-escape] $\square$

Warning 5.26. A child outside Q does *not* mean success; it means only that the first alphabet has been left.

Theorem 5.27 (Shadow cycle localisation and normal form). **PROVED**, class B + C. On Q consider the auxiliary edge $q \rightsquigarrow s$ whenever $s \mid \delta - q$. Then every directed cycle lies in $(0, \delta)$; a simple cycle of length ≥ 3 has all vertices $< \delta/3 = 2h/3$; and a two-cycle $\{q, s\}$ satisfies

$$\delta = q + s + tqs, \quad t \geq 0 \text{ even},$$

with $t \geq 2$ inside a larger terminal core; hence a terminal component $C \subseteq Q$ with $|C| \geq 3$ satisfies $C \subset (0, 2h/3)$. [BIB, lem:qcycle-below-delta, lem:shadow-two-cycle, cor:qcore-compression]

Reading. The scale $\delta = 2h$ controls the *size* of the whole recursive part of the first front — the cleanest example of a mechanism in which the gap steers multiplicatively. But δ is a function of the root, not of its minimality.

Proposition 5.28 (Shadow two-cycle). **PROVED**, class B. Contained in Theorem 5.27: a two-cycle $\{q, s\}$ in the shadow graph is exactly a solution of $\delta = q + s + tqs$ with $t \geq 0$ even.

Proposition 5.29 (Prime-power witness). **PROVED**, class C. Let $q \in Q$ with $\text{PF}(N - q) \subseteq Q$. Then there is $r \in Q$ with

$$v_r(m_0) = v_r(\delta - q) > 0.$$

Proof. Put $s = \delta - q$, so $N - q = 2m_0 + s$. If for every $r \mid N - q$ the valuations $v_r(m_0)$ and $v_r(s)$ differed, then for odd r we would have $v_r(N - q) = \min\{v_r(m_0), v_r(s)\} \leq v_r(s)$; since all factors of $N - q$ lie in Q , we would get $N - q \mid |s|$, impossible because $N - q > |\delta - q|$. [BIB, thm:witness] $\square$

5.6 Path invariants and winding

Theorem 5.30 (Full path Bézout invariant). **PROVED**, class B. For a path $p_0 \rightarrow p_1 \rightarrow \dots \rightarrow p_t$ with $N - p_i = k_i p_{i+1}$, and $K_0 = C_0 = 1$, $K_{t+1} = K_t k_t$, $C_{t+1} = 2K_t - C_t$:

$$K_t p_t - C_t x = (-1)^t h, \quad \gcd(K_t, C_t) = \gcd(K_t, h). \quad (\text{PBI})$$

Proof. For $t = 0$ the left side is $p_0 - x = h$. Induction: from $p_t = 2x - k_t p_{t+1}$ we get $K_t k_t p_{t+1} - (2K_t - C_t)x = (-1)^{t+1} h$. Each k_i divides $N - p_i$ and is coprime to N , so $\gcd(K_t, x) = 1$; the second identity follows. [BIB, thm:bezout-path; CORE, ARS-BEZOUT] $\square$

Reading. Every path from the root remembers h as an exact Bézout invariant. This is the strongest sense in which “gap geometry” propagates dynamically — but h is replaced by h_r for any root, so the class is B. **VERIFIED COMPUTATION** on real paths, $N \leq 5 \cdot 10^3$, zero violations.

Corollary 5.31 (Irreducible winding normal form). **PROVED**, class B. With $W_t = (C_t - (-1)^t)/2$,

$$K_t p_t = W_t N + (-1)^t p_0, \quad \gcd(W_t, K_t) = 1, \quad (\text{WN})$$

where $W_0 = 0$, $W_{t+1} = K_t - W_t$. On a bad path $W_t > 0$, $W_t \equiv t \pmod{2}$, and for $t \geq 2$, $K_t p_t > 2N$. [BIB, thm:winding-normal-form]

Reading. The root enters the equation multiplicatively and modulo N , not merely as a starting point.

Theorem 5.32 (Finitely many states per vertex). **PROVED**, class B. After reduction by $d = \gcd(K_t, h)$, with $B = K_t/d$, $A = C_t/d$, $H = h/d$, the state has primitive form $Bp - Ax = \pm H$ with $\gcd(A, B) = \gcd(B, H) = 1$; for fixed p , fixed divisor $H \mid h$, fixed sign and $0 < B < x$ the number B is determined uniquely modulo x , so a single vertex carries at most $2\tau(h)$ reduced states. [BIB, cor:path-state-count]

Reading. The divisor count of the gap bounds the richness of the path state — the only place where the size of h (rather than its relation to G) enters.

Theorem 5.33 (No-bad-return-to- p_0). **PROVED**, class C. Under failure no bad path contains an edge $u \rightarrow p_0$. Consequently a non-root endpoint of a bad path satisfies $r < N/3$.

Proof. Under failure $p_0 > N/2$; if $p_0 \mid N - u$ then from $0 < N - u < N < 2p_0$ we get $N - u = p_0$, so $N - u$ is prime and u is GOOD. [BIB, lem:v71-no-return-p0; I160] $\square$

VERIFIED COMPUTATION: all canonical worlds $N \leq 10^4$, zero violations.

Theorem 5.34 (Global two-sheet collision budget). **PROVED**, class C. In the whole bad world reachable from the canonical p_0 there are at most two distinct endpoints r reached by two distinct bad paths whose label products $K_1, K_2 < N$ have opposite winding parities.

Sketch. The two-sheets base gives $K_1 + K_2 = N$ and $m_0 = q_1 q_2$; comparing two collisions gives $m_0 \mid K - K'$, and oddness of the products gives $2m_0 \mid K - K'$. After cutting off small N (Nagura) we have $m_0 > 2N/5$, so $2m_0 > 4N/5$: three distinct numbers in one class modulo $2m_0$ cannot lie in $(0, N)$. For fixed K the winding form gives $Kr \equiv \pm p_0 \pmod{N}$, and the endpoint satisfies $r < N/3$ (Theorem 5.33), so each K encodes at most one endpoint. [BIB, thm:v71-global-two-sheet; l160] $\square$

Reading. The first *global* result over the whole p_0 world, independent of component size and topology.
Limitation: it is not an escape theorem and does not exclude a closed bad world. The attempt to turn it into a contradiction is **KILLED**.

5.7 The Euler-defect layer

Read this box before using η or ν . The corpus records a “layered Euler defect” in the form

$$\eta(S) + \nu(S) \geq |S| + |\text{Src}(S)|, \quad \text{canonically} \quad \eta(R) + \nu(R) \geq |R| + 1,$$

but the material available to this file does not fix the normalisation of η and ν unambiguously. Two natural readings were tested here on all canonical failed worlds with $50 \leq N \leq 6 \cdot 10^4$ (21 093 worlds), and one obvious reading is *false*. Do not use the inequality without saying which normalisation you mean.

Definition 5.35 (Candidate normalisations). With $d_S(q)$, $\kappa_S(q)$ as in Definition 3.18, and $\nu(S)$ the number of BFS layers of S :

(N1) row-excess form $\eta_1(S) = \sum_{p \in S} (\Omega(N - p) - |\text{PF}(N - p) \cap S|)$.

(N2) valuation/reuse split $\eta_2(S) = \sum_q (d_S(q) - \kappa_S(q))$, $\nu_2(S) = \sum_q (\kappa_S(q) - 1)$, over $q \in \Gamma_N(S)$; these are the logarithmic masses of U_S and J_S counted in prime factors.

(N3) active-mass form $\sum_q d_S(q)$, the total active valuation mass of the world.

(N4) trivial form $\eta_4(S) = \sum_{p \in S} (\Omega(N - p) - 1)$ with ν the layer count.

VERIFIED COMPUTATION — canonical failed worlds, $50 \leq N \leq 6 \cdot 10^4$, 21 093 worlds, all with $|\text{Src}| = 1$:

- | | | |
|------|-------------------------------------|---|
| (N1) | $\eta_1(R) + \nu(R) \geq R + 1$ | 6 044 violations ; smallest $N = 556$ ($ R = 17$, $\eta_1 = 10$, $\nu = 6$) |
| (N2) | $\eta_2(R) + \nu_2(R) \geq R + 1$ | 3 804 violations ; smallest $N = 50$ |
| (N3) | $\sum_q d_R(q) \geq R + 1$ | 0 violations |
| (N4) | $\eta_4(R) + \nu(R) \geq R + 1$ | 0 violations |

Proposition 5.36 (The trivial form is a theorem). **PROVED**. For a canonical failed world R , $\eta_4(R) + \nu(R) \geq |R| + 1$.

Proof. Every $p \in R$ is BAD, so $N - p$ is composite and $\Omega(N - p) \geq 2$, whence $\eta_4(R) \geq |R|$; and $\nu(R) \geq 1$. $\square$

Remark 5.37 (What this means for the programme). (N4) is a theorem but carries no information — it is $\Omega \geq 2$ in disguise. (N3) is the only nontrivial normalisation that survived; it says the world carries at least $|R| + 1$ units of active valuation mass, i.e. strictly more mass than vertices. It is here **VERIFIED COMPUTATION** only, and a proof is a reasonable small target (see §15.1). (N1) and (N2) are **KILLED** as stated. Any argument in the corpus that used the Euler defect in the form (N1) or (N2) must be re-derived.

Warning 5.38 (Do not conflate the layers). $T_S = U_S J_S$ (Theorem 3.19) is exact and safe. The Euler defect built on it is not. Warning 3.21 remains in force: none of these quantities is the capacity functional γ of $P1^{\text{br}}/K1$.

Chapter 6

The Canonical p_0 Programme

This chapter is where p_0 genuinely differs from every other upper root, and also where the project has most often overclaimed. Sections 6.1–6.4 are the positive results; §6.5 is the mandatory negative section and is, for a future model, the most valuable part of the chapter.

6.1 Genealogical defect rank

Definition 6.1 (Prime-defect set and genealogical rank). For the upper root $p_k = x + h_k$ (the k -th prime $\geq x$) put

$$\mathcal{P}_{\text{def}}(p_k) = \{j \in [1, t_k] : c_j^{(k)} = p_k - 2j \in \mathbb{P}\}, \quad \kappa_{\text{gen}}(p_k) = |\mathcal{P}_{\text{def}}(p_k)|.$$

Theorem 6.2 (Genealogy of prime holes). **PROVED**, class A. Domain hypothesis: *non-diagonal case* (x composite; see Scope note 3.17). For every $k \geq 0$,

$$\{c_j^{(k)} : j \in \mathcal{P}_{\text{def}}(p_k)\} = \{p_0, \dots, p_{k-1}\}, \quad \kappa_{\text{gen}}(p_k) = k,$$

the unique hole carrying p_i sits at index

$$j_i^{(k)} = \frac{p_k - p_i}{2}$$

and its lower mirror is

$$a_{j_i^{(k)}}^{(k)} = N - c_{j_i^{(k)}}^{(k)} = N - p_i.$$

Proof. Every prime reflected value lies in (x, p_k) , and the primes in that interval are exactly $p_0, \dots, p_{k-1}$. Conversely, for $i < k$ the number $p_k - p_i$ is positive and even, and $p_i > x$ gives $0 < p_k - p_i < h_k$, so $1 \leq (p_k - p_i)/2 \leq \lfloor (h_k - 1)/2 \rfloor = t_k$; at that index $c_j^{(k)} = p_k - (p_k - p_i) = p_i$. Uniqueness follows from injectivity of $j \mapsto p_k - 2j$. The mirror is $a_j = N - c_j$. [CORE, G-UPPERROOT-PRIMEHOLE-GENEALOGY, V71.21–V71.24] $\square$

VERIFIED COMPUTATION: all $N \leq 10^3$ with composite x and $k \leq 5$, zero violations; the diagonal case is isolated separately (Scope note 3.17).

Theorem 6.3 (The defect ledger is a lossless code of prime history). **PROVED**, class A. Knowing p_k and the index set $\mathcal{P}_{\text{def}}(p_k)$ recovers all earlier upper primes,

$$p_i = p_k - 2j_i^{(k)},$$

and the differences of consecutive hole positions recover the consecutive prime gaps:

$$j_i^{(k)} - j_{i+1}^{(k)} = \frac{p_{i+1} - p_i}{2} = \frac{g_{i+1}}{2}, \quad 0 \leq i < k-1.$$

Proof. The first identity is a rewriting of $j_i^{(k)} = (p_k - p_i)/2$; the second is its difference. $\square$

Reading. $\mathcal{P}_{\text{def}}(p_k)$ is not a “counter of missing rows”. It is a *complete, lossless record of the local history of the primes above the midpoint*, gaps included. In exactly that sense p_k “remembers” all its predecessors, and p_0 remembers nobody.

Theorem 6.4 (Defect-rank-zero characterisation). **PROVED**, class A (non-diagonal). For a prime $r = x + h_r \geq x$ the following are equivalent:

- (i) $r = p_0$;
- (ii) $[x, r) \cap \mathbb{P} = \emptyset$;
- (iii) $h_r < G_r$;
- (iv) $\mathcal{P}_{\text{def}}(r) = \emptyset$;
- (v) $\kappa_{\text{gen}}(r) = 0$;
- (vi) every strictly reflected value $r - 2j$, $1 \leq j \leq \lfloor (h_r - 1)/2 \rfloor$, is composite.

[CORE, C-P0-DEFECTRANK-ZERO-EQUIVALENCE]

Proof. (i)–(iii) is Theorem 3.10. Items (ii), (iv), (vi) are equivalent because the strictly reflected values enumerate exactly the numbers of the same parity as r lying *strictly* between x and r — this is where non-diagonality enters. (v) is the cardinality form of (iv). $\square$

Corollary 6.5 (Uniqueness of the reflection-complete root). **PROVED**, class A (non-diagonal). Call an upper root r reflection-complete if every strictly reflected value $r - 2j$ is composite. Then

$$r \text{ is reflection-complete} \iff r = p_0.$$

[CORE, C-P0-UNIQUE-REFLECTION-COMPLETE-ROOT]

VERIFIED COMPUTATION: $N \leq 10^3$, zero violations for composite x ; the diagonal exception is located exactly ($N = 14$, $r = 11$).

Theorem 6.6 (Reflection ledger completeness). **PROVED**, class A. Among all upper prime roots, p_0 is the only one whose strictly reflected ledger is exhausted by four multiplicative-dynamical classes with no prime-defect class:

$$\mathcal{J}_h = \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F} \quad \text{versus} \quad \mathcal{J}_{h_k} = \mathcal{P}_{\text{def}} \uplus \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}, \quad |\mathcal{P}_{\text{def}}| = k.$$

[CORE, C-P0-REFLECTION-LEDGER-COMPLETENESS, V71.32–V71.34]

Warning 6.7 (What these theorems do *not* say). **Do not sell this as an automatic explanation of the success/failure anomaly.** Theorems 6.2–6.6 are an *exact structural exceptionality*, but:

- there is no proved implication $\kappa_{\text{gen}} = 0 \Rightarrow$ traversal success; the target C-P0-GENEALOGYFREE-BOUNDARY-TO-DYNAMIC is **OPEN**;
- the property may be *partly equivalent to the ordering of the primes itself*: it says that p_0 is the first element of the sequence $p_0 < p_1 < \dots$, translated into the language of the reflected block. Theorem 6.3 cuts both ways here: since the defect ledger is a *lossless code* of prime history, the statement “ $\kappa_{\text{gen}}(r) = 0$ ” carries exactly as much information as the statement “ r is the first element of that sequence” — no more, no less;
- the logarithmic mass of the holes is negligible: passing from p_0 to p_k the loss is $\sum_{i < k} \log p_i = O_k(\log N)$, which formally *kills* the mass interpretation (§6.5, FAIL-PO-ZEROHOLE-ESSENTIAL-001).

Honest formulation: *the type of relation available on the boundary coordinate changes, not the amount of information.* A prime hole $c_j = p_i$ has no nontrivial factorization $c_j = s\ell$ with $s, \ell > 1$, so on that coordinate there is no multiplicative carrier of the kind that modern factor-flow machinery requires [CORE, V71.44]. That is *eligibility*, not automatic activation — Warning 3.16 remains binding.

6.2 Boundary filtration

Theorem 6.8 (Embedding of reflected blocks). **PROVED**, class B (the filtration) + A (its initial object). Let $p_k < p_{k+1}$ be consecutive upper primes with $p_{k+1} - p_k = 2s$. Then

$$t_{k+1} = t_k + s, \quad c_{i+s}^{(k+1)} = c_i^{(k)}, \quad a_{j+s}^{(k+1)} = a_j^{(k)} \quad (1 \leq i, j \leq t_k),$$

so $\mathfrak{B}(p_k) \subseteq \mathfrak{B}(p_{k+1})$ and the new coordinates are exactly the indices $1, \dots, s$ of the block of p_{k+1} . In particular there is a filtration

$$\mathfrak{B}_0 \subset \mathfrak{B}_1 \subset \mathfrak{B}_2 \subset \dots, \quad \mathfrak{B}_k := \mathfrak{B}(p_k),$$

whose unique initial object is p_0 .

Proof. Since $h_{k+1} = h_k + 2s$, $t_{k+1} = \lfloor (h_k + 2s - 1)/2 \rfloor = \lfloor (h_k - 1)/2 \rfloor + s = t_k + s$. Further $c_{i+s}^{(k+1)} = p_{k+1} - 2(i + s) = p_k + 2s - 2i - 2s = p_k - 2i = c_i^{(k)}$, and since $a_j = N - c_j$ in both blocks the same index shift gives $a_{j+s}^{(k+1)} = a_j^{(k)}$. Index ranges agree: $i \leq t_k$ gives $i + s \leq t_{k+1}$. $\square$

VERIFIED COMPUTATION: 731 446 checked indices ($N \leq 2 \cdot 10^4$, $k \leq 6$), zero violations; EXP6.

Corollary 6.9 (Embedding of the coupling matrix). **PROVED**, class B. With $M_{ij}^{(k)} = \gcd(c_i^{(k)}, a_j^{(k)})$, $1 \leq i, j \leq t_k$,

$$M_{ij}^{(k)} = M_{i+s, j+s}^{(k+1)},$$

i.e. $M^{(k)}$ is exactly the submatrix of $M^{(k+1)}$ cut out by shifting both indices by s . The entire coupling geometry of the earlier root is inherited unchanged.

Proof. Immediate from Theorem 6.8: both gcd arguments are identical. $\square$

Corollary 6.10 (New strip and new hole). **PROVED**, class A. The new coordinates of the block of p_{k+1} are exactly $j = 1, \dots, s$ with $s = (p_{k+1} - p_k)/2$; among them exactly one, $j = s$, is a prime hole and carries the root pair $(p_k, N - p_k)$. The other $s - 1$ new coordinates are composite.

Proof. New indices are $1 \leq j \leq s$, since indices $> s$ come from the embedding. For $j < s$ the value $c_j^{(k+1)} = p_{k+1} - 2j$ lies strictly between p_k and p_{k+1} , hence is composite; for $j = s$ it equals p_k . The lower mirror is $N - p_k$. $\square$

Reading the filtration.

- p_0 is the rank-zero initial object: $\mathfrak{B}_0$ has no hole.
- p_1 contains the *whole* boundary of p_0 plus one new strip of width $s = (p_1 - p_0)/2$ containing exactly one new hole.
- Every further root inherits the entire earlier geometry (coupling matrix included) and adds exactly one new genealogical hole.

This is interesting independently of Goldbach: the reflected boundaries of the upper primes form a natural ascending chain in which the defect rank is the height.

6.3 The canonical phase diagram $\theta = h/G$

Theorem 6.11 (Canonical prime cell). **PROVED**, class A. Let P be an odd prime, P_- its predecessor and $G = P - P_-$. Then for every $0 \leq h < G$,

$$p_0(N_h) = P, \quad N_h := 2(P - h).$$

In particular every odd prime is a canonical root: for $N = 2(P - 1)$ we get $p_0(N) = P$ and $h = 1$.

Proof. $x = P - h > P - G = P_-$, so $[x, P)$ contains no prime and P is the least prime $\geq x$. [PCRD, Canonical Prime Cell Theorem] $\square$

VERIFIED COMPUTATION: full cells for $P \in \{67, 127, 1009, 104729\}$, agreement at every h . Reading. The label “ p_0 ” does *not* single out any class of primes. All canonical information sits in the pair (P, h) , and the natural parameter is $\theta = h/G \in [0, 1)$. This theorem is also the seed of the architecture wall (§8.2).

Theorem 6.12 (Gap-position identity and trichotomy). **PROVED**, class A. For canonical $p_0 = x + h$ with predecessor p_- and $G = p_0 - p_-$,

$$m_0 - p_- = G - 2h,$$

and in the non-diagonal case ($h \geq 1$, so G is even) exactly one of the following holds.

Type I ($2h < G$, i.e. $\theta < 1/2$). Then $p_- < m_0 < x < p_0$, and since p_- and p_0 are consecutive primes and $m_0 \neq p_-$, the number m_0 is composite. Hence the canonical root is BAD and immediate success is impossible.

Type II ($2h = G$, i.e. $\theta = 1/2$). Then $m_0 = p_-$, so m_0 is prime and $N = p_0 + p_-$ is a Goldbach representation: immediate success is automatic.

Type III ($2h > G$, i.e. $\theta > 1/2$). Then $m_0 < p_- < x < p_0$ and the previous prime appears as a lower mirror of the reflected block at index

$$j^* = h - \frac{G}{2} \in \{1, \dots, t\}, \quad a_{j^*} = p_-.$$

Proof. $m_0 - p_- = (x - h) - (x + h - G) = G - 2h$; its sign gives the trichotomy. Type I: $G - 2h > 0$ gives $m_0 > p_-$; also $m_0 < x < p_0$, and (p_-, p_0) contains no prime, so m_0 is composite (for N large enough that $m_0 > 1$). Type II: $G = 2h$ gives $m_0 = p_-$ directly. Type III: setting $a_j = m_0 + 2j = p_-$ gives $2j = p_- - m_0 = 2h - G$, i.e. $j^* = h - G/2$, an integer because G is even. Range: $j^* \geq 1 \iff 2h \geq G + 2$, which is Type III; and $j^* \leq t = \lfloor (h - 1)/2 \rfloor$ follows from $2j^* = 2h - G \leq h - 1$, i.e. $h \leq G - 1$, which always holds since $h < G$ (Theorem 3.10). $\square$

VERIFIED COMPUTATION: all even $N \leq 2 \cdot 10^5$: zero Type I cases with m_0 prime; zero Type II cases with $m_0 \neq p_-$; zero index violations in Type III.

Corollary 6.13 (Cell decomposition). **PROVED**, class A. The cell of a prime P with even gap G splits into exactly

$$\underbrace{1}_{\text{diagonal } (h=0)} + \underbrace{\left(\frac{G}{2} - 1\right)}_{\text{Type I}} + \underbrace{1}_{\text{Type II } (h=G/2)} + \underbrace{\left(\frac{G}{2} - 1\right)}_{\text{Type III}} = G$$

cases. In particular the number of diagonal cases equals the number of Type II cases, and the distribution of θ is symmetric under $\theta \mapsto 1 - \theta$.

Proof. h runs over $0, \dots, G - 1$; $h = 0$ is diagonal, $h = G/2$ is Type II, and the rest split symmetrically about $G/2$ via the bijection $h \mapsto G - h$. $\square$

EXP5 — phase census. **VERIFIED COMPUTATION**, all even $N \leq 2 \cdot 10^5$:

	diagonal	Type I	Type II	Type III
count	9 591	40 407	9 591	40 409
with immediate success	9 591	0	9 591	7 422

The second row is exactly the content of Theorem 6.12: in Type I immediate success is *impossible*, in Type II and the diagonal case it is *automatic*, and in Type III it happens (here in 18.4% of cases). The difference of 2 between Type I and Type III counts comes from the incomplete cell at the end of the range.

Corollary 6.14 (Reflection-gate phase transition). **PROVED**, class A. A *guaranteed* active prime lower-mirror coordinate in the strictly reflected block exists exactly in phase Type III:

$$\theta > \frac{1}{2} \implies a_{j^*} = p_- \in \mathbb{P}, \quad j^* = h - \frac{G}{2}.$$

If in addition $p_- \nmid N$ (**VERIFIED COMPUTATION**: true in all 40 409 Type III cases with $N \leq 2 \cdot 10^5$) then a_{j^*} is an active row and the reflection gate (Lemma 3.15) gives the edge $a_{j^*} \rightarrow \text{PF}(c_{j^*})$. In Types I and II no such guarantee exists.

Remark 6.15 (This explains the counterexample $N = 242$). Warning 3.16 cites $N = 242$ as proof that an empty interval does *not* activate any reflected row. The phase diagram shows this is not an accident: $N = 242$ has $h = 6$, $G = 14$, so $\theta = 3/7 < 1/2$ — Type I, where the guaranteed-mirror theorem does not apply. By contrast $N = 240$ (same $p_0 = 127$, $h = 7$) is Type II and the canonical root wins immediately, and $N = 124$ ($h = 5$, $G = 6$, $\theta = 5/6$) is Type III with two prime lower mirrors, 59 and $61 = p_-$.

6.4 Bidirectional prime rank ρ_0

Definition 6.16 (Lower rank). For canonical $p_0 = P$ let $P_{-1} > P_{-2} > \dots$ be the consecutive earlier primes and $D_s = P - P_{-s}$ (so $D_1 = G$). Put

$$\rho_0 := \#\{j \in [1, t] : a_j = m_0 + 2j \in \mathbb{P}\},$$

the number of *prime lower mirrors* in the strictly reflected block.

Theorem 6.17 (Lower-rank formula). **PROVED**, class A. For $h \geq 3$,

$$\rho_0 = \#\{s \geq 1 : h < D_s < 2h\}.$$

Proof. A lower mirror $a_j = m_0 + 2j = P - 2h + 2j$ equals $P_{-s} = P - D_s$ iff $2j = 2h - D_s$, i.e. $j = h - D_s/2$ (D_s is even for $P > 3$). The condition $j \geq 1$ gives $D_s \leq 2h - 2$, and $j \leq t = \lfloor (h-1)/2 \rfloor$ gives $2h - D_s \leq h - 1$, i.e. $D_s \geq h + 1$. Since D_s is even, $D_s \leq 2h - 2 \iff D_s < 2h$. The map $s \mapsto j$ is injective because D_s is strictly increasing. $\square$

Warning 6.18 (Correction to the originally proposed formula). **KILLED** as stated. The version proposed in the input material had only the upper bound, $\rho_0 = \#\{s \geq 1 : D_s < 2h\}$. It is missing the lower condition $D_s > h$, which is exactly what confines the mirror to the *strictly* reflected block (index $j \leq t$, not $j \leq h - 1$). Without it the formula also counts mirrors lying outside the strict block. **VERIFIED COMPUTATION**: all even $N \leq 2 \cdot 10^5$ with $h \geq 3$ (72 451 cases): the corrected formula agrees with direct counting in *every* case, zero violations.

Corollary 6.19 (The lower rank detects the phase exactly). **PROVED**, class A.

$$\rho_0 \geq 1 \iff G < 2h \iff \theta > \frac{1}{2} \iff \text{Type III}.$$

Proof. ($\Leftarrow$) In Type III, $h < G = D_1 < 2h$, so $s = 1$ satisfies the condition of Theorem 6.17. ($\Rightarrow$) D_s is strictly increasing, so if $D_1 = G \geq 2h$ then $D_s \geq 2h$ for all s and no s qualifies. $\square$

VERIFIED COMPUTATION: $N \leq 2 \cdot 10^5$: the number of cases with $\rho_0 \geq 1$ is 40 409 — *exactly* the number of Type III cases. Distribution of ρ_0 over the 72 451 cases with $h \geq 3$:

ρ_0	0	1	2	3	4	5	6	7	8	9
$\#N$	32 042	24 288	10 200	3 936	1 331	404	170	57	16	7

Definition 6.20 (Bidirectional signature).

$$(\kappa^+, \kappa^-) := (\kappa_{\text{gen}}, \rho_0).$$

The upper coordinate counts primes *above* the midpoint recorded in the reflected blocks, the lower one counts primes *below*.

Why the signature is a better descriptor than κ_{gen} alone. For the canonical root $\kappa^+ = 0$ always, so κ^+ separates no two canonical worlds — it is a constant. But $\kappa^- = \rho_0$ is non-constant and, by Corollary 6.19, encodes exactly the phase:

$$\kappa^- = 0 \iff \theta \leq \frac{1}{2}, \quad \kappa^- \geq 1 \iff \theta > \frac{1}{2}.$$

The signature $(0, \rho_0)$ is therefore the first quantity in this file that is simultaneously (i) an invariant of the boundary geometry, (ii) p_0 -specific in its first coordinate, and (iii) *nontrivially varying* between canonical worlds.

6.5 What does *not* make p_0 exceptional

Mandatory section. For a future model this is the most valuable part of the chapter: “this does not work, and we know why” is worth as much here as a theorem.

6.5.1 Refutations bearing directly on p_0 -exceptionality

Mass of the reflected boundary. **KILLED**. Hypothesis: the absence of prime holes in the canonical reflected block is itself a significant mass mechanism. Counterexample: for p_1 the block has exactly *one* hole, for p_k exactly k ; iterating the pointwise short-interval bound gives $p_k - x = O_k(x^{0.525})$, so removing the holes costs only $\sum_{i < k} \log p_i = O_k(\log N)$ against a full block mass of order $t_k \log(x/h_k)$. Hence hole mass/block mass $\rightarrow 0$. [FAIL, FAIL-P0-ZEROHOLE-ESSENTIAL-001; CORE, V71.35]

p_0 as an information selector. **SUPERSEDED** (not refuted, but **not established**). The ablation P0-CANONICITY-ABLATION classifies every clause of the bundle-lift construction as “fixed N ” or “arbitrary reflected pair $a + c = N$ ”. Replacing p_0 by p_1 leaves the whole construction *word for word* on every row with composite c'_j . Corpus verdict: p_0 is *quantitatively useful and structurally inessential* for that construction; a non-vacuity guarantee is not a principle of information selection. [FAIL, FAIL-P0-INFORMATION-SELECTOR-001]

Shadow / MR2 congruences. **KILLED** as a canonical statement (Warning 5.23).

First-front parent minimality. **KILLED** as a selector (Warning 5.6).

Activation of reflection by the prime-free boundary. **KILLED** (Warning 3.16, $N = 242$). Structurally explained by the phase diagram (Remark 6.15).

Statistical exceptionalism after controlling for row size. **KILLED**. The median rank percentile of p_0 within fixed N , over all reported metrics, is 0.20982; row-size-aware ranks do not support a single universal claim that p_0 dominates on every axis. Comparing canonical and matched non-canonical B_{abs} on 119 reproducible pairs gives a median difference of $4.6127 \cdot 10^{-5}$ — no clean causal separation. [D4, questions 5, 12] — *reported computational result, not reproduced in this document*.

Minimality does not determine local geometry. **KILLED**. At fixed $P = 927\,961$ ($N = 1\,855\,760$) a single change of h gives $\Delta \log B_{\text{abs}} = 14.434$, $\Delta G_{\text{esc}} = -230$, $\Delta |W| = -240$; full sweeps at fixed p_0 find 23 canonical transitions $B_{\text{abs}} \rightarrow B_{\text{re}}$ and 15 in the reverse direction. [D4, questions 6, 17] — *reported, not reproduced here*.

Largest first-front factor as a heuristic. **KILLED**. The variant “follow the largest factor of m_0 ” fails in two steps where the smallest factor wins (Example 6.21). [PCRD, FAIL-P0-LARGEST-FACTOR-TWOSTEP]

Example 6.21 ($N = 124$). **VERIFIED COMPUTATION**. $x = 62$, $p_0 = 67$, $h = 5$, $G = 6$ (Type III), $m_0 = 57 = 3 \cdot 19$. First-front rows:

$$124 - 3 = 121 = 11^2, \quad 124 - 19 = 105 = 3 \cdot 5 \cdot 7.$$

Following the *largest* factor 19, the children are $\{3, 5, 7\}$ and none is GOOD ($124 - 5 = 119 = 7 \cdot 17$, $124 - 7 = 117 = 3^2 \cdot 13$). Following the *smallest* factor 3, the only child is 11, and $124 - 11 = 113$ is prime. The full traversal succeeds at depth 2 with two escape edges, $3 \rightarrow 11$ and $5 \rightarrow 17$.

6.5.2 The ablation $p_0 \rightarrow p_1$

The ablation removes *only* minimality and keeps everything else. This table is the central empirical result about p_0 .

EXP8 — observables at p_0 and at p_1 . **VERIFIED COMPUTATION**. All even $50 \leq N \leq 2 \cdot 10^5$; 99 976 cases per root; world mode.

observable	p_0	p_1	change
root FAILURES	0	2	$0 \rightarrow 2$
success depth, mean	1.3501	1.5234	+12.8%
world size $ R $, mean	52.71	59.38	+12.7%
margin G_{esc} , mean	24.51	26.96	+10.0%
$\Omega(m)$, mean	2.373	2.521	+6.2%
shadow retention, mean	0.5545	0.5128	−7.5%
$P^+(m)/m$, mean	0.31455	0.23966	−23.8%
κ_{gen} , mean	0.0000	0.9041	$0 \rightarrow \approx 1$
κ_{gen} , maximum	0	1	categorical

Definitions. $\Omega(m)$ — number of prime factors of the complement with multiplicity; shadow retention — $\sum_{q \in Q} |\text{PF}(N - q) \cap Q|$ normalised; $P^+(m)/m$ — normalised largest prime factor; κ_{gen} — genealogical rank (Definition 6.1).

Remark 6.22 (Where 0.9041 comes from, rather than 1). The mean $\kappa_{\text{gen}}(p_1)$ falls short of 1 by exactly $9\,591/99\,976 = 9.59\%$ — exactly the fraction occupied by the diagonal cases (Scope note 3.17 and Corollary 6.13). This is an independent confirmation of the domain repair: for prime x the hole of p_0 falls outside the strict block.

Distribution of Ω (a negative result). [VERIFIED COMPUTATION](#), even $N \leq 2 \cdot 10^5$:

Ω	1	2	3	4	5	6	7	8
$m_0 = N - p_0$	26 585	32 687	24 592	10 967	3 699	1 085	272	71
$m_1 = N - p_1$	18 534	36 385	27 198	12 249	4 048	1 161	307	78

Outside the bucket $\Omega = 1$ (which corresponds to immediate success, hence to a *definition*, not to structure) the two distributions are nearly identical. The local factorization of the canonical complement does *not* look unusual. This is one of the main negative results of the project.

Conclusion of the ablation. All aggregate observables — depth, world size, margin, Ω , retention, normalised largest factor — are *nearly interchangeable* between p_0 and p_1 . The only quantity that changes categorically is the defect rank. This is exactly the content of [CORE, V71.37]: *p_0 -specificity is invisible to many aggregate observables and visible in the exact structure of relations.*

6.5.3 Is the exceptionality local, multiplicative, dynamic, or just the empty interval?

1. **Not local-arithmetic.** Exact-D countermodels satisfy every local condition — normal form, smoothness, pair failure, endpoint primality — and die *only* on canonicity. Smallest recorded model: $q = 3$, $r = 5\,413\,103$, $m_0 = 81\,196\,545$, $N = 297\,720\,668$, endpoint $p = 216\,524\,123$ prime, but $x + 9$ is prime and lies much earlier. [FAIL, FAIL-EXACTD-NONCANONICAL-001]
2. **Not purely multiplicative.** D_q , M_q and the whole prime-power ledger are generic; p_0 enters them only through the value $\delta = 2h$.
3. **Not (yet) dynamic.** There is no proved passage from boundary geometry to traversal behaviour; the target C-P0-GENEALOGYFREE-BOUNDARY-TO-DYNAMIC-ESCAPE is **OPEN**.
4. **But not “just” an empty interval either.** The empty interval is *equivalent* to something substantially richer: the boundary is multiplicatively complete, the ledger has no fifth class, the predecessor lies outside the block, and ρ_0 detects the phase. It is a property of the *type of relation* available on each coordinate, not of the length of an interval.
5. **The reservation in the other direction.** Theorem 6.3 shows the defect ledger is a lossless code of prime history. So “ $\kappa_{\text{gen}}(r) = 0$ ” may be *informationally equivalent* to “ r is the first element of the sequence of upper primes”. We do not claim that the exceptionality of p_0 goes beyond the ordering of the primes; we claim it is an *exact* translation of that ordering into the multiplicative structure of the boundary.

The honest one-sentence summary.

The exceptionality of p_0 is an *exact boundary-closure property*, not a density effect; it is invisible to aggregate observables and visible in the structure of relations; and no theorem is known leading from it to dynamic behaviour.

6.6 The six special N

EXP7 — nonempty fixed points. **VERIFIED COMPUTATION**. For all even $N \leq 6 \cdot 10^4$ the set B_N^{low} is nonempty for *exactly* six values:

N	$ B_N^{\text{low}} $	B_N^{low} (initial segment)
128	10	$\{3, 5, 7, 11, 13, 17, 23, 29, 37, 41\}$
1718	55	$\{3, 5, 7, 11, 13, 17, 29, 31, 37, 41, 43, \dots, 571\}$
1862	55	$\{3, 5, 11, 13, 17, 29, 41, 43, 47, 53, 59, \dots, 619\}$
1928	41	$\{3, 5, 7, 11, 13, 17, 29, 43, 53, 71, 73, \dots, 641\}$
2200	2	$\{3, 13\}$
6142	66	$\{3, 5, 7, 13, 17, 19, 31, 43, 67, 71, 73, \dots, 2011\}$

This replicates exactly the list reported in [PCRD] for the same range.

EXP3 — full census of failing upper roots. **VERIFIED COMPUTATION**. Searching *all* upper roots $p \in (N/2, N)$ with composite complement, for all even $N \leq 2 \cdot 10^4$:

- exactly **301** roots whose traversal fails;
- they sit at exactly *six* values of N , distributed as 128:8, 1718:68, 1862:63, 1928:57, 2200:4, 6142:101;
- **not one** of those 301 roots is canonical;
- unit rows $N - p = 1$ excluded separately: 2261.

The number 301 is the one the corpus carries as “301 composite-complement failed upper roots” FAIL-CENSUS-DOMAIN-001; the computation above fixes its exact domain and structure, which the corpus did not record. The canonical root wins in *all six* cases: $p_0 = 67, 859, 937, 967, 1103, 3079$ at depths 0, 0, 2, 2, 0, 2. These six N are exactly the six numbers with an exceptional autonomous component reported up to 10^8 by Božek [8].

Remark 6.23 (The two censuses agree although they measure different things). EXP3 asks whether some failing upper root exists; EXP7 asks whether a closed bad set exists. The lists coincide on $N \leq 2 \cdot 10^4$. That is expected in one direction (a failing root forces a closed bad set) but the converse is not a theorem, and the agreement is data, not proof.

Example 6.24 (Failure structure at $N = 128$). **VERIFIED COMPUTATION**. For $N = 128$ ($x = 64$, $p_0 = 67$):

k	p_k	$N - p_k$	verdict
0	67	61 (prime)	SUCCESS, $d = 0$
1	71	$57 = 3 \cdot 19$	SUCCESS, $d = 1$ ($19 \rightarrow 109 \in \mathbb{P}$)
2	73	$55 = 5 \cdot 11$	FAILURE
3	79	$49 = 7^2$	FAILURE
4	83	$45 = 3^2 \cdot 5$	FAILURE
5	89	$39 = 3 \cdot 13$	FAILURE
6	97	31 (prime)	SUCCESS, $d = 0$
7	101	$27 = 3^3$	FAILURE
8	103	$25 = 5^2$	FAILURE
9	107	$21 = 3 \cdot 7$	FAILURE
10	109	19 (prime)	SUCCESS, $d = 0$
11	113	$15 = 3 \cdot 5$	FAILURE

All eight failures share the *same* bad world $\{3, 5, 7, 11, 13, 23, 29, 41\}$. The set of failing roots is neither an initial nor a final segment: $p_1 = 71$ wins, $p_2 = 73$ loses. *Minimality alone does not order*

the outcome — the first signal that “ p_0 is simply the smallest” is not an explanation.

Remark 6.25 (The witness $N = 2200$ and the two-core wall). **VERIFIED COMPUTATION**. The core $B_N^{\text{low}} = \{3, 13\}$ at $N = 2200$ is an exact witness that a source-free set is *not* by itself contradictory:

$$2200 - 3 = 2197 = 13^3, \quad 2200 - 13 = 2187 = 3^7, \quad 3^7 - 3 = 13^3 - 13 = 2184.$$

At the same time $p_0(2200) = 1103$ and $m_0 = 1097$ is prime, so the canonical root wins immediately and never enters that core. Excluding such cores in general leads to a Pillai/Bennett-type equation $a^\alpha - a = b^\beta - b$, i.e. to an external Diophantine wall. [PCRD, 2-core wall]

6.7 Escape margin

Success depth turned out to be a *bad* measure of difficulty. The right measure is the margin: how many ways out exist at all.

Definition 6.26 (Escape capacity). For even N with BAD p_0 , let $W = R_N(p_0)$ and

$$G_{\text{esc}}(N) = \#\{(v, q) : v \in W, q \mid N - v, q \in \mathcal{A}_N, N - q \in \mathbb{P}\}.$$

Then Goldbach failure $\Rightarrow G_{\text{esc}}(N) = 0$, and $G_{\text{esc}}(N) > 0 \Rightarrow$ Goldbach for N . The converse implication is *not* proved. [BIB, escape capacity; repair relative to v5.8]

EXP4 — escape margin. **VERIFIED COMPUTATION**. All even $N \leq 2 \cdot 10^5$:

- 73 394 numbers have BAD p_0 ; $G_{\text{esc}} = 0$ *never* occurs;
- $G_{\text{esc}} = 1$ for 52 numbers, of which exactly **10** have $|W| \geq 3$:

$$368, 632, 1258, 1598, 1862, 1928, 3692, 4582, 6142, 14198,$$

with world sizes 12, 14, 25, 22, 23, 16, 21, 22, 12, 64 — *exactly* the list and sizes of [BIB, ten numbers with margin one], obtained there by a scan to $3 \cdot 10^6$;

- distribution of small margins at $|W| \geq 3$: $G_{\text{esc}} = 1$: 10, 2:38, 3:122, 4:477, 5:571, 6:656;
- largest canonical bad world: $|W| = 204$ at $N = 152\,288$.

The minimum margin grows — answer to an open point of the corpus.

VERIFIED COMPUTATION. The corpus left as a task the question “does the $G_{\text{esc}} = 1$ list really end at 14198” [BIB, next move]. Minimum G_{esc} over dyadic blocks $[2^{k-1}, 2^k)$, with $|W| \geq 3$ and $N \leq 2 \cdot 10^5$:

k	7	8	9	10	11	12	13	14	15	16	17	18
$\min G_{\text{esc}}$	2	2	1	1	1	1	1	2	2	3	3	3

In this range the $G_{\text{esc}} = 1$ list at $|W| \geq 3$ does end at $14198 \in [2^{13}, 2^{14})$, and the minimum margin in higher blocks rises to 2 and 3. This is a *signal*, not a proof: the range is finite and an order of magnitude smaller than the scan in [BIB].

Channel substitution. **EMPIRICAL PATTERN**, not a theorem. For $N = 14198$ a world of 64 bad vertices hangs on a *single* edge $71 \rightarrow 277$. Preserving the congruence behind that edge at $N \pm 554$ loses the canonical p_0 ; preserving p_0 in the nearest tested perturbations destroys that edge but creates 10–26 replacement edges. Across the whole campaign 459 compensation candidates were recorded with *zero* true orphans [D4, q. 18; ORPH, q. 9–10]. Reading: closing one channel opens another. Neighbouring sweeps do not isolate causality, so this remains a pattern, not a law.

Chapter 7

Factorization and Reachability Machinery

7.1 Post-root factorization

Theorem 7.1 (The post-root gcd). **PROVED**, class B. Let $q \in Q$ (so q is an odd prime dividing m_0) and put $B_q = m_0/q$. Then

$$D_q := \gcd(m_0, N - q) = \gcd(m_0, |\delta - q|) = \gcd(B_q, |2h - q|), \quad q \nmid D_q.$$

Proof. The first equality is the shadow identity (Theorem 5.22) applied to $N - q = 2m_0 + (\delta - q)$. For the rest: $\gcd(h, m_0) = 1$ (Theorem 5.3) and $q \mid m_0$ give $q \nmid h$, and q is odd, so $q \nmid \delta = 2h$; hence $q \nmid (\delta - q)$ and $q \nmid D_q$. Writing $m_0 = q^a m'$ with $q \nmid m'$ we get $\gcd(m_0, \delta - q) = \gcd(m', \delta - q) = \gcd(q^{a-1} m', \delta - q) = \gcd(B_q, \delta - q)$, and $\gcd(\cdot, y) = \gcd(\cdot, |y|)$. [LOG; FND] $\square$

Remark 7.2 (The identity is root-generic). Every step uses only $m_r = x - h_r$ and $\gcd(h_r, m_r) = 1$, so Theorem 7.1 holds verbatim for any upper root. It is class B and must not be quoted as canonical.

Corollary 7.3 (The cofactor is Q -free). **PROVED**, class B. With $M_q = (N - q)/D_q$ we have $\text{PF}(M_q) \cap Q = \emptyset$: all of the first alphabet that survives in the row $N - q$ is concentrated in D_q .

Proof. By Theorem 5.24, $\text{PF}(N - q) \cap Q = \text{PF}(d_q)$ and $d_q \mid D_q$; dividing out D_q removes every prime of Q from the row. $\square$

Definition 7.4 (FRESH / SMOOTH split). $q \in Q$ is *FRESH* if $P^+(N - q) \geq B_q$, and *SMOOTH* otherwise. Write Q_{smooth} for the smooth part. The split is the domain boundary of the incidence ledger below.

Warning 7.5 (High-gap amplification is conditional). **SCOPE-CORRECTED**. The inequality $h^2 > 4m_0$ (“high-gap amplification”) is *not* a consequence of CRD failure. Its actual hypothesis is failure of the Exact-D pair, and the corpus states it under that hypothesis [CORE, ARS-EXACTD]. Using it under canonical failure alone is an error; this is the **SCOPE-CORRECTED** noted against [BIB].

7.2 The incidence ledger

Theorem 7.6 (Incidence ledger IL1–IL6). **AUDITED IDENTITY** on the domain $S = Q_{\text{smooth}}$. With U_F , J_F the unit and layer parts of the ledger restricted to S , and B_{re} , B_{abs} the reused and absorbed base parts,

$$\prod_{q \in S} M_q = U_F (J_F B_{\text{re}}) B_{\text{abs}}.$$

Two independent reconstruction routes, 1 000 worlds, 0 errors [ILA].

Warning 7.7 (Domain). **SCOPE-CORRECTED**. IL1–IL6 hold on $S = Q_{\text{smooth}}$ only. The FRESH part is excluded. Quoting the identity on all of Q is a recorded error.

Definition 7.8 (Base atoms). B_{abs} is the absorbed base (atoms consumed inside the world), B_{re} the reused base (atoms appearing in more than one row), and $B_{\text{abs,int}}$ the internal absorbed base. The corpus proves $B_{\text{abs,int}} = B_{\text{abs}}$ without vacuity [CORE, ARS-BABS-INTERNAL].

Killed here. *Unconditional CAB*: refuted by $N = 322$ ($U_R = J_R = 1$, $B_{\text{abs}} = 11 \cdot 29$) and, in the internal version, by $N = 858$ FAIL-CAB-001, FAIL-CAB-INTERNAL-001. $\mathcal{C}_S < \mathcal{D}_S$ from the first front: refuted by $N = 4736$ ($\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$) FAIL-POTENTIAL-CAPACITY-001. *Failure always produces nontrivial B_{abs}* : refuted FAIL-BABS-PRODUCTION-001. *Four-channel ledger on all of Q_0* : refuted by $N = 20$ (a GOOD row outside R gives $\kappa_R < m_\ell$) [ILA]. *Constant $C = 1$ in charging*: 53 violations, maximum 9.42243 FAIL-CHILD-C1-001. *Naive support expansion of all sources*: Hall compression $456 \rightarrow 17$ at $N = 51\,032$ FAIL-HALL-SUPPORT-001.

Escape-or-reuse (**OPEN** as a theorem, **VERIFIED COMPUTATION** as data). Every internal base atom $\ell \in B_{\text{abs,int}}$ has either a GOOD edge reachable inside one of its first-front branches, or an active parent outside R . Both alternatives are necessary: $(N, \ell, p) = (6142, 877, 3511)$ shows the second is needed. Bounded charging multiplicity: observed maximum 1, boundedness unproved. Data: 61 452 instances, 0 orphans; independent red team 60 413 atoms, 0 orphans [ORPH, RT]. Known scope corrections around it: external reuse is *not* local $((N, \ell) = (1\,825\,372, 52\,153))$, distance 53 750, FAIL-LOCAL-REUSE-001; the external parent is *not* necessarily a success root $((6142, 877, 3511))$: 3511 is active, above p_0 , and its own traversal fails, FAIL-EXTERNAL-SUCCESS-001; escape need not be directly adjacent ($N = 14198$: atoms 19, 167 at distances 8 and 7, FAIL-DIRECT-ADJACENCY-001).

7.3 The rooted first-front layer

This section and the two that follow are the Biblia's v7.3–v7.5 rooted first-front theory. It is the most quantitatively developed part of the whole corpus and it is where the exact ledgers live. Read the statuses carefully: the *conservation and split identities* carry the source's **PROVED – PENDING RED TEAM** label, while the *collision budget and Exact-D geometry* of §7.4–§7.5 are fully **PROVED**.

Throughout, $q \in Q = \text{PF}(m_0)$, and

$$n_q := N - q, \quad B_q := \frac{m_0}{q}, \quad D_q := \gcd(m_0, n_q), \quad I_q := \text{rad } D_q, \quad M_q := \frac{n_q}{D_q},$$

$$U_q^{\text{row}} := \frac{n_q}{\text{rad}(n_q)}, \quad E_q := \frac{\text{rad}(n_q)}{I_q}, \quad j_q := \omega(n_q) - 1, \quad u_q := \Omega(n_q) - \omega(n_q),$$

so that $e_q := \Omega(n_q) - 1 = j_q + u_q$ splits the row cost exactly into a *collision/branching channel* j_q and a *squarefull channel* u_q .

7.3.1 FRESH / SMOOTH dichotomy

Definition 7.9 (FRESH and SMOOTH, restated). For $q \in Q$: the row is **FRESH** if $P^+(n_q) \geq B_q$, in which case it has a subthreshold prime child $s_q \geq B_q$; it is **SMOOTH** if $P^+(n_q) < B_q$, in which case every child of the row exceeds the threshold already at the second step.

Proposition 7.10 (Fresh prime packing, B-FRESH-PACKING). **PROVED – PENDING RED TEAM**. For distinct first-front primes q_i choose FRESH children s_i with two-step cost $< N$. The paths have the same parity, so the two-sheet budget (Theorem 5.34) forbids their merging; hence the s_i are pairwise distinct and for every $E \subseteq Q_{\text{fresh}}$

$$\prod_{q \in E} s_q \geq \frac{m_0^{|E|}}{\prod_{q \in E} q}. \quad (\text{FP})$$

[BIB, prop:v73-fresh-packing]

Proposition 7.11 (Smoothness Complexity Tax, B-SMOOTH-TAX). **PROVED – PENDING RED TEAM**. On the SMOOTH branch, with $e_q = \Omega(n_q) - 1$,

$$B_q^{e_q} > q. \quad (\text{ST})$$

Moreover $e_q = (\omega(n_q) - 1) + (\Omega(n_q) - \omega(n_q)) = j_q + u_q$, so the cost splits exactly into the collision/branching channel J and the squarefull channel U . [BIB, prop:v73-smooth-tax]

Corollary 7.12 (Root smoothness excess tax). **PROVED – PENDING RED TEAM**. For the full closed world,

$$\Omega(U_R) + \Omega(J_R) - 1 \geq \sum_{q \in Q_{\text{smooth}}} e_q > \sum_{q \in Q_{\text{smooth}}} \frac{\log q}{\log(m_0/q)}.$$

Warning 7.13 (The two-channel wall, and what it forbids). The external argument “many SMOOTH rows $\Rightarrow$ many powerful integers” is **KILLED**. A SMOOTH row can pay almost its whole cost through support *diversity* — the J channel — with no high prime multiplicity at all. This is exactly why Proposition 7.11 must keep both j_q and u_q . The correct picture is two-channel:

$$\text{SMOOTH} \longrightarrow \begin{cases} j_q \text{ large} & \longrightarrow J/R_{\partial} \text{ load} \\ u_q \text{ large} & \longrightarrow U \text{ load} \end{cases}$$

and what is needed is a theorem bounding the *joint* capacity of both channels while keeping the root geometry of p_0 . That is the *Root Smoothness Conservation Problem* (§7.3.4).

7.3.2 Root First-Front Conservation

Put $S_1 = \{p_0\} \cup Q$ and $b_1 := |\Gamma_N(S_1) \setminus S_1|$. The root row contains every $q \in Q$, so p_0 is the unique source of S_1 .

Theorem 7.14 (Root First-Front Conservation, B-FFC). **PROVED – PENDING RED TEAM**. The fundamental factor-flow decomposition gives

$$m_0 \frac{\prod_{q \in Q} (N - q)}{\prod_{q \in Q} q} = U_{S_1} J_{S_1} R_{\partial}(S_1). \quad (\text{FFC})$$

Applying Ω ,

$$\Omega(U_{S_1}) + \Omega(J_{S_1}) + |\Gamma(S_1) \setminus S_1| = \Omega(m_0) + \sum_{q \in Q} (\Omega(N - q) - 1). \quad (\text{FFC-}\Omega)$$

[BIB, thm:v73-ff-conservation]

Corollary 7.15 (Exact recycle–escape identity, B-FFRE). **PROVED – PENDING RED TEAM**. Since $U_{S_1} = \frac{m_0}{\text{rad}(m_0)} \prod_{q \in Q} \frac{N - q}{\text{rad}(N - q)}$ and $\prod_{q \in Q} q = \text{rad}(m_0)$, the squarefull factors cancel and

$$\prod_{q \in Q} \text{rad}(N - q) = J_{S_1} R_{\partial}(S_1). \quad (\text{FFRE})$$

The first front therefore has an exact decomposition of its support into recycle/collision and true boundary escape. [BIB, cor:v73-recycle-escape]

7.3.3 The exact first-front split ledger

Theorem 7.16 (Exact First-Front $U/J/\partial$ Split, B-UJD-SPLIT). **PROVED – PENDING RED TEAM**. With $S_1 = \{p_0\} \cup Q$ and $b_1 = |\Gamma(S_1) \setminus S_1|$,

$$\boxed{\Omega(U_{S_1}) = \Omega(m_0) - |Q| + \sum_{q \in Q} u_q,} \quad (\text{U-split})$$

$$\boxed{\Omega(J_{S_1}) + b_1 = |Q| + \sum_{q \in Q} j_q.} \quad (\text{JR-split})$$

Summing recovers Root First-Front Conservation: $\Omega(U_{S_1}) + \Omega(J_{S_1}) + b_1 = \Omega(m_0) + \sum_{q \in Q} e_q$.

Proof. By definition $U_{S_1} = \frac{m_0}{\text{rad}(m_0)} \prod_{q \in Q} \frac{N-q}{\text{rad}(N-q)}$; applying Ω and $\omega(m_0) = |Q|$ gives (U-split). From Corollary 7.15, $\prod_{q \in Q} \text{rad}(N-q) = J_{S_1} R_\partial(S_1)$, and the boundary radical is squarefree with b_1 prime atoms, so $\sum_{q \in Q} \omega(N-q) = \Omega(J_{S_1}) + b_1$; since $\omega(N-q) = 1 + j_q$ this is (JR-split). [BIB, thm:v74-exact-split] $\square$

Corollary 7.17 (Smoothness tax in the separated channels). **PROVED – PENDING RED TEAM**. For $q \in Q_{\text{smooth}}$ one has canonically $B_q^{e_q} > q$ with $e_q = j_q + u_q$. Hence every SMOOTH row must pay its required cost through exactly one of three routes: larger u_q feeding U , larger j_q feeding J or the boundary support, or a mixture splitting the cost between the channels. **There is no “free” SMOOTH variant** — but at the present stage there is still no global upper capacity bound producing a contradiction.

Lemma 7.18 (First-front GCD packing, B-GCD-PACK). **PROVED – PENDING RED TEAM**. For a prime ℓ put $\nu_\ell := \#\{q \in Q : \ell \mid N-q\}$. All q counted by ν_ℓ lie in one class modulo ℓ , so

$$\boxed{\nu_\ell \leq \min\left(|Q|, 1 + \left\lfloor \frac{m_0}{\ell} \right\rfloor\right).}$$

[BIB, lem:v74-gcd-pack]

Limitation. The lemma is correct but is *not* a knockout: it bounds incidence multiplicity, not total capacity.

Scope note 7.19 (Why the classical powerful-tail bound is too weak). **KILLED** as a route. The global sparsity of powerful integers gives $O(N^{(1+\alpha)/2})$ exceptions up to N , whereas the first front requires control of only $|Q| = \omega(m_0) \leq \log_2 m_0$ special points. Global density of powerful integers is therefore far too weak to force even one non-powerful row in the *adaptive* set $\{N-q : q \mid m_0\}$. [BIB, v7.4 powerful-tail remark]

7.3.4 The Root Smoothness Conservation Problem

Open target 7.20 (Root Smoothness Conservation Problem, B-RSCP). **OPEN**. For a hypothetical failure, find a uniform, deterministic capacity theorem linking

$$B_q^{j_q+u_q} > q \quad (q \in Q_{\text{smooth}})$$

to the exact ledger

$$\Omega(U_{S_1}) = \Omega(m_0) - |Q| + \sum u_q, \quad \Omega(J_{S_1}) + b_1 = |Q| + \sum j_q,$$

together with source attenuation and terminal absorption taxes, and leading to a contradiction or a forced escape. [BIB, def:v74-rscp]

Warning 7.21. This target is the Biblia’s own statement of the *same* obstruction that Chapter 9 records as the same-functional bridge (§9.11). It is not an independent problem: solving one solves the other. What the Biblia adds is the exact two-channel bookkeeping the bridge has to respect.

7.4 The lossless prime-power root ledger

Proposition 7.22 (Root-recycle-free row identity). **PROVED – PENDING RED TEAM**. For every $q \in Q$,

$$I_q = \text{rad gcd}(B_q, |2h - q|), \quad U_q^{\text{row}} E_q = \frac{N - q}{I_q}.$$

In particular the first copy of every prime recycled from the root alphabet is removed exactly from the weighted tax. [BIB, prop:v75-root-recycle-free]

Proposition 7.23 (Full first-front recycle split). **PROVED – PENDING RED TEAM**. For the full first front $S_1 = \{p_0\} \cup Q$ the recycling factor splits as

$$J_{S_1} = J^{\text{rec}} J_Q^{\text{ext}}, \quad J^{\text{rec}} = \prod_{q \in Q} I_q,$$

and after removing the root recycle the remaining lower demand is

$$U_{S_1} J_Q^{\text{ext}} R_{\partial}(S_1) > \left(\frac{9}{5}\right)^{|Q|} m_0.$$

This is a lower-capacity statement; it does not yet give an upper bound closing SMOOTH.

Warning 7.24 (Two sharp interpretive countermodels). SMOOTH does *not* force $E_q > 1$: one can have $E_q = 1$ with the whole cost staying in the valuations. And the same external prime may occur in several rows. Externalisation must therefore never be identified with pairwise-disjoint support, nor with automatic escape.

Let $\emptyset \neq S \subseteq Q_{\text{smooth}}$, $\nu_{\ell,k}(S) := \#\{q \in S : \ell^k \mid n_q\}$ and $a_r := v_r(m_0)$ for $r \in Q$.

Theorem 7.25 (Lossless Prime-Power Root Ledger, B-PP-LEDGER). **PROVED – PENDING RED TEAM**. Three exact decompositions hold:

$$\log \prod_{q \in S} \frac{n_q}{I_q} = \sum_{\ell \notin Q} \sum_{k \geq 1} \nu_{\ell,k} \log \ell + \sum_{r \in Q} \sum_{k \geq 2} \nu_{r,k} \log r, \quad (\text{PP-I})$$

$$\log \prod_{q \in S} \frac{D_q}{I_q} = \sum_{r \in Q} \sum_{k=2}^{a_r} \nu_{r,k} \log r, \quad (\text{PP-D})$$

$$\log \prod_{q \in S} \frac{n_q}{D_q} = \sum_{\ell \notin Q} \sum_{k \geq 1} \nu_{\ell,k} \log \ell + \sum_{r \in Q} \sum_{k \geq a_r + 1} \nu_{r,k} \log r. \quad (\text{PP})$$

Proof. For each prime ℓ the layer-cake identity $\sum_{q \in S} \nu_{\ell}(n_q) = \sum_{k \geq 1} \nu_{\ell,k}(S)$ holds. Dividing by I_q subtracts one copy of every root prime present in the row, so the root sum starts at $k = 2$. Dividing by the full D_q subtracts $\min\{a_r, \nu_r(n_q)\}$, leaving exactly the layers $k \geq a_r + 1$. External primes do not divide m_0 , so their layers start at $k = 1$. [BIB, thm:v75-pp-root-ledger] $\square$

Reading. This is the exact sense in which the root-supported valuation mass can be subtracted *without losing demand*: (PP-D) isolates precisely what the root alphabet already paid for.

Theorem 7.26 (Exact post-root demand). **PROVED – PENDING RED TEAM**. For $q \in Q$,

$$D_q = \text{gcd}(B_q, |2h - q|) \quad \text{and} \quad \frac{n_q}{D_q} \geq \frac{N - q}{\min\{B_q, |2h - q|\}}. \quad (\text{Demand})$$

Setting $w_q := \sum_{\ell \notin Q} \nu_{\ell}(n_q) + \sum_{r \in Q} (v_r(n_q) - a_r)_+$, every SMOOTH row satisfies

$$B_q^{w_q} > \frac{n_q}{D_q} \geq \frac{N - q}{\min\{B_q, |2h - q|\}}. \quad (\text{Post-root tax})$$

Proof. Since $q \mid m_0$ and $\gcd(h, m_0) = 1$ we get $q \nmid 2h - q$, so $\gcd(m_0, n_q) = \gcd(m_0, 2h - q) = \gcd(B_q, |2h - q|)$. Hence $D_q \leq B_q$ and $D_q \leq |2h - q|$, which is (Demand). After dividing by D_q all remaining prime factors of a SMOOTH row are still smaller than B_q , and their number with multiplicity is exactly w_q . [BIB, thm:v75-post-root-demand] $\square$

Lemma 7.27 (Root valuation rigidity). **PROVED – PENDING RED TEAM**. Let $r^a \parallel m_0$. Then

$$\boxed{v_r(N - q) > a \implies v_r(|2h - q|) = a,} \quad \text{and hence} \quad \boxed{r^a > |2h - q| \implies v_r(N - q) \leq a.}$$

Proof. Write $m_0 = r^a M$ with $r \nmid M$ and $b = v_r(|2h - q|)$. From $n_q = 2m_0 + (2h - q)$,

$$v_r(n_q) = \begin{cases} b, & b < a, \\ a, & b > a, \\ a + v_r(2M + \frac{2h-q}{r^a}), & b = a. \end{cases}$$

Excess above a can occur only in the case $b = a$; and if $r^a > |2h - q|$ then $v_r(|2h - q|) = a$ is impossible. [BIB, lem:v75-root-valuation-rigidity] $\square$

Lemma 7.28 (Sharp Prime-Power Packing, B-SHARP-PP-PACK). **PROVED – PENDING RED TEAM**. If $\ell^k \mid n_q$ and $\ell^k \mid n_r$ then

$$\boxed{\ell^k \mid q - r.}$$

All incident roots therefore lie in one class modulo ℓ^k , and for any eligible set $\mathcal{E}_{\ell,k}$ containing all actually incident roots,

$$\boxed{\nu_{\ell,k} \leq \min(|\mathcal{E}_{\ell,k}|, 1 + \left\lfloor \frac{\text{diam } \mathcal{E}_{\ell,k}}{\ell^k} \right\rfloor).}$$

In particular $\ell^k > \text{diam } \mathcal{E}_{\ell,k}$ forces $\nu_{\ell,k} \leq 1$. [BIB, lem:v75-pp-pack]

Warning 7.29 (Potential capacity is not actual capacity). **KILLED**. One may *not* assume $\mathcal{C}_S < \mathcal{D}_S$ uniformly on the basis of the first front alone. The canonical model

$$N = 4736, \quad p_0 = 2371, \quad h = 3, \quad m_0 = 2365 = 5 \cdot 11 \cdot 43, \quad Q = S = \{5, 11, 43\}$$

has all three rows SMOOTH,

$$N - 5 = 3 \cdot 19 \cdot 83, \quad N - 11 = 3^3 \cdot 5^2 \cdot 7, \quad N - 43 = 13 \cdot 19^2,$$

and for the potential-capacity definition used, $\mathcal{C}_S \approx 22.60 > 20.16 \approx \mathcal{D}_S$. This is not a closed bad world, so it does *not* refute a future theorem with a closure hypothesis; it does refute deriving the needed inequality from local eligibility geometry alone. **FAIL-POTENTIAL-CAPACITY-001**

7.4.1 Actual prime-power collision budget

For residual rows $M_q = (N - q)/D_q$ set $\mu_{\ell,k}(S) := \#\{q \in S : \ell^k \mid M_q\}$ and $\Delta_{\text{odd}}(S) := \prod_{q < r, q, r \in S} |q - r|_{\text{odd}}$.

Theorem 7.30 (Actual Prime-Power Collision Budget, B-VPC). **PROVED**. Exactly,

$$\boxed{\sum_{\ell} \sum_{k \geq 1} \binom{\mu_{\ell,k}}{2} \log \ell = \sum_{q < r} \log \gcd(M_q, M_r) \leq \log \Delta_{\text{odd}}(S).} \quad (\text{VPC})$$

Proof. For a fixed pair $q < r$, $\sum_{\ell,k} \mathbf{1}_{\ell^k \mid M_q, M_r} \log \ell = \log \gcd(M_q, M_r)$. Summing over pairs, the number of pairs sharing layer (ℓ, k) is $\binom{\mu_{\ell,k}}{2}$. Moreover $\gcd(M_q, M_r) \mid \gcd(N - q, N - r) \mid r - q$, and the M_q are odd, so the common divisor divides the odd part $|r - q|_{\text{odd}}$. Multiplying over pairs gives the last inequality. [BIB, thm:v75-actual-pp-collision] $\square$

Corollary 7.31 (Actual shared-mass bound). **PROVED**. With $L_{\text{shared}} := \sum_{\mu_{\ell,k} \geq 2} \mu_{\ell,k} \log \ell$,

$$L_{\text{shared}} \leq 2 \log \Delta_{\text{odd}}(S).$$

Proof. For $\mu \geq 2$ one has $\mu \leq 2 \binom{\mu}{2}$; sum over the layers that actually occur and apply Theorem 7.30. $\square$

For a pair $q < r$ put $g_{q,r} := \gcd(M_q, M_r)$ and $L_{\text{pair}}(q, r) := \log \frac{M_q M_r}{g_{q,r}^2}$.

Lemma 7.32 (Pair isolation floor). **PROVED**. For two distinct SMOOTH roots, $L_{\text{pair}}(q, r) \geq \log 5$.

Proof. $g_{q,r}$ is odd and divides $|r - q|_{\text{odd}} < r/2$. For the larger SMOOTH root r , $M_r > \frac{9}{5}r$, so $M_r/g_{q,r} > 18/5$. That quotient is a positive odd integer, hence at least 5; and $M_q/g_{q,r} \geq 1$. [BIB, lem:v75-pair-log5] $\square$

Theorem 7.33 (Exact weighted cut identity, B-WEIGHTED-CUT). **PROVED**. Let $s = |S|$ and let the layer $\lambda = (\ell, k)$ carry weight $w_\lambda = \log \ell$ and multiplicity μ_λ . Then

$$\sum_{q < r} L_{\text{pair}}(q, r) = \sum_{\lambda} \mu_\lambda (s - \mu_\lambda) w_\lambda, \quad (\text{Cut})$$

and consequently $C_{\text{cut}} := \sum_{\lambda} \mu_\lambda (s - \mu_\lambda) w_\lambda \geq \binom{s}{2} \log 5$.

Proof. Layer λ contributes w_λ to $L_{\text{pair}}(q, r)$ exactly when it belongs to precisely one of the two rows; the number of such unordered pairs is $\mu_\lambda (s - \mu_\lambda)$. Summing over layers gives the identity, and Lemma 7.32 gives the lower bound. [BIB, thm:v75-weighted-cut] $\square$

Corollary 7.34 (Low-multiplicity layer lower bound). **PROVED**. For $1 \leq u \leq s - 1$ put $W_{\leq u} := \sum_{1 \leq \mu_\lambda \leq u} w_\lambda$, $R_u := \frac{2(s-u-1)}{u}$ and $H_u := \frac{s-1}{u} \lfloor \frac{(u+1)^2}{4} \rfloor$. Then

$$W_{\leq u} \geq \frac{[\binom{s}{2} \log 5 - R_u \log \Delta_{\text{odd}}(S)]_+}{H_u},$$

and in particular

$$W_1 \geq \frac{[\binom{s}{2} \log 5 - 2(s-2) \log \Delta_{\text{odd}}(S)]_+}{s-1}.$$

Proof. Subtract a multiple of the collision budget from (Cut). For $\mu \geq u + 1$, $\frac{\mu(s-\mu)}{\binom{\mu}{2}} = \frac{2(s-\mu)}{\mu-1} \leq R_u$; for $1 \leq \mu \leq u$ the coefficient $\mu(s-\mu) - R_u \binom{\mu}{2} = \frac{s-1}{u} \mu(u+1-\mu)$ does not exceed H_u . Apply $K \leq \log \Delta_{\text{odd}}(S)$ and take positive parts. $\square$

Warning 7.35 (The sharp abstract obstruction). Cut geometry alone does *not* force singletons once the right-hand side of Corollary 7.34 vanishes. Abstract systems in which every layer has multiplicity exactly 2 (or generally $u + 1$) realise the obstruction. Globalisation needs congruences, source structure or absorption data in addition.

7.5 Exact-D pair compensation and canonical pair geometry

This section reduces a hypothetical Exact-D failure to an explicit rigid regime, then to an explicit normal form, and finally to a canonical gap filter. Everything here is **PROVED**. The

branch $k = 1$ ends at a stated **OPEN** barrier. Notation: $t := 2h$, $m_0 = qrc$ with $q < r$ distinct in Q_{smooth} , and $\delta := |r - q|_{\text{odd}}$.

Theorem 7.36 (Exact-D pair compensation outside the regime $r < t$, B-EXACTD-OUTSIDE). **PROVED**.

If $t < r$ then $M_q M_r > \delta^2$; more precisely $\frac{M_q M_r}{\delta^2} > \frac{36q}{5}$. Consequently every potential Exact-D failure satisfies

$$q < r < t, \quad r \mid D_q, \quad t \equiv q \pmod{r}. \quad (\text{ED-res})$$

Proof. Suppose $M_q M_r \leq \delta^2$. Since r is SMOOTH, $M_r > \frac{9}{5}r$, so

$$D_q = \frac{N - q}{M_q} > \frac{9r(N - q)}{5\delta^2} > \frac{9qr^2c}{5\delta^2} > \frac{36}{5}qc, \quad (\text{ED1})$$

using $\delta < r/2$. If $t < r$ then $|t - q| < r$; from $D_q \mid rc$, $D_q \mid |t - q|$ and $\gcd(t, m_0) = 1$, D_q cannot contain r , so $D_q \mid c$, contradicting (ED1). Hence failure forces $r < t$ (equality is excluded by parity). By (ED1), $D_q > c$, so every divisor $D_q \mid rc$ must contain r , i.e. $r \mid D_q \mid (t - q)$. [BIB, thm:v75-exactd-pair-outside] $\square$

Corollary 7.37 (First canonical gap filter). **PROVED**. Writing $G = p_0 - p_-$ (so canonicity gives $h < G$, hence $t < 2G$), every Exact-D pair failure satisfies

$$p_0 < 2G + \frac{5(2G - q)\delta^2}{9r} < 2G + \frac{5}{9}G^2,$$

and in particular

$$9p_0 \geq 18G + 5G^2 \implies M_q M_r > \delta^2 \quad \text{for all SMOOTH pairs.}$$

Theorem 7.38 (Joint D_q, D_r rigidity, B-JOINT-D). **PROVED**. If $s^a \parallel m_0$ and $\alpha = v_s(D_q)$, $\beta = v_s(D_r)$, then

$$\alpha + \beta \leq a + \min\{a, v_s(r - q)\},$$

and globally

$$D_q D_r \mid m_0 \gcd(m_0, r - q) = m_0 \gcd(c, r - q). \quad (\text{Joint-D})$$

Proof. $\max(\alpha, \beta) \leq a$, while $s^{\min(\alpha, \beta)} \mid \gcd(D_q, D_r) \mid \gcd(N - q, N - r) \mid r - q$. Adding exponents gives the first bound; the primewise product gives (Joint-D). Since neither q nor r divides $r - q$, the common part $\gcd(m_0, r - q)$ comes from c . [BIB, thm:v75-joint-D] $\square$

Theorem 7.39 (Primitive residual normal form, B-PNF). **PROVED**. Assume the regime (ED-res) and write $t - q = rk$, $m_0 = qrc$ with k positive and odd; put $g := \gcd(c, k)$, $c = gu$, $k = gv$, $\gcd(u, v) = 1$. Then exactly

$$\begin{array}{lll} t = q + rgv, & m_0 = qrgu, & D_q = rg, \\ N - q = rg(2qu + v), & M_q = 2qu + v, & p_0 = q + rg(qu + v), \\ h = \frac{q+rgv}{2}, & D_r = \gcd(qgu, q + (gv - 1)r), & M_r = \frac{2qrgu + q + (gv - 1)r}{D_r}. \end{array} \quad (\text{PNF})$$

The formulas remain correct when q or r occur to higher powers in m_0 ; the extra copies are contained in $c = gu$.

Proof. The key identity is $D_q = \gcd(rc, rk) = r \gcd(c, k) = rg$; the rest follows by substituting $m_0 = qrgu$ and $t = q + rgv$ into $N = 2m_0 + t$ and $p_0 = m_0 + t$. For D_r use $D_r = \gcd(B_r, t - r) = \gcd(qgu, q + (gv - 1)r)$. Since $\gcd(t, m_0) = 1$ we have $q \nmid gv$. [BIB, thm:v75-primitive-normal] $\square$

Theorem 7.40 (High-gap amplification, B-HIGH-GAP). **PROVED** under the *Exact-D pair-failure hypothesis*. With $d := v_2(r - q) \geq 1$, every *Exact-D pair failure* in primitive normal form satisfies

$$r > 2^{2d+1}(2qu + v) > 2^{2d+2}qu \quad (\text{HG-r})$$

and

$$\frac{h^2}{m_0} > 4^d gv^2 + 2^{2d-1} \frac{gv^3}{qu} + \frac{v}{2u} > 4^d gv^2 \geq 4, \quad (\text{HG})$$

so in particular $h^2 > 4m_0$.

Proof. Put $A = 2qu + v$. From $D_r \leq B_r = qgu$ we get $M_r \geq (2 + \frac{qv-1}{qgu})r + \frac{1}{gu} > 2r$. Failure $M_q M_r \leq (r - q)^2 / 4^d$ then gives $2 \cdot 4^d Ar < r^2$, which is (HG-r). Moreover $\frac{h^2}{m_0} = \frac{rgv^2}{4qu} + \frac{v}{2u} + \frac{q}{4rgu}$; substituting $r > 2^{2d+1}(2qu + v)$ into the first term gives (HG). [BIB, thm:v75-high-gap] $\square$

Warning 7.41 (Hypothesis, restated). **Warning 7.5** applies verbatim: the hypothesis is *Exact-D pair failure*, not CRD failure. Using $h^2 > 4m_0$ under canonical failure alone is an error.

Corollary 7.42 (Quadratic canonical gap filter). **PROVED**. With $\Lambda := 4^d gv^2 + 2^{2d-1} \frac{gv^3}{qu} + \frac{v}{2u}$ and $h < G$,

$$p_0 < 2G + \frac{G^2}{\Lambda} < 2G + \frac{G^2}{4^d gv^2} \leq 2G + \frac{G^2}{4}.$$

Finite check of the quadratic filter. **VERIFIED COMPUTATION** as reported by the source. An exact check to $p_0 \leq 10^9$ found no canonical failure satisfying the filter $p_0 < 2G + G^2/4$; of the candidates only $p_0 = 11$ passed, and it admits no permissible pair of distinct roots. This is a finite verification, not a global proof. — *reported computational result, not reproduced in this document.*

7.5.1 The branch $k = 1$

For $k = 1$ one has $g = v = 1$, so $t = q + r$, $m_0 = qru$ with $u \geq 3$ odd, and exactly $D_q = r$, $M_q = 2qu + 1$, $D_r = q$, $M_r = 2ru + 1$.

Theorem 7.43 ($k = 1$ high-gap strengthening). **PROVED**. For $k = 1$ every pair failure satisfies

$$r > 4^{d+1}qu^2, \quad \frac{h^2}{m_0} > 4^d u, \quad \text{hence} \quad p_0 < 2G + \frac{G^2}{4^d u} \leq 2G + \frac{G^2}{12}.$$

The only non-automatic SMOOTH condition on the larger row is $P^+(2ru + 1) < qu$.

Proof. Failure gives $(2qu + 1)(2ru + 1) \leq (r - q)^2 / 4^d < r^2 / 4^d$; the lower bounds on both factors give $r > 4^{d+1}qu^2$. Then $\frac{h^2}{m_0} = \frac{(q+r)^2}{4qru} > \frac{r}{4qu} > 4^d u$, and the filter for p_0 follows from $h < G$. The condition $u \geq 3$ is necessary: at $u = 1$ the factor $r = D_q = B_q$ breaks the strict SMOOTH threshold. [BIB, thm:v75-k1-gap] $\square$

Theorem 7.44 ($k = 1$ quotient normal form). **PROVED**. With $A := 2qu + 1$, $B := 2ru + 1$ one has exactly

$$q = \frac{A-1}{2u}, \quad r = \frac{B-1}{2u}, \quad A \equiv B \equiv 3 \pmod{4},$$

$$4um_0 = (A-1)(B-1), \quad 4uh = A + B - 2, \quad 4ux = AB - 1, \quad 4up_0 = AB + A + B - 3,$$

$$up_0 + 1 = \frac{(A+1)(B+1)}{4} = (qu+1)(ru+1).$$

Moreover $d = v_2(B-A) - 1$ and $\delta = \frac{B-A}{2^{d+1}u}$; pair failure is equivalent to

$$4^{d+1}u^2AB \leq (B-A)^2,$$

SMOOTH of the larger row reads $P^+(B) < \frac{A-1}{2}$, and canonicity reads

$$\left[\frac{AB-1}{4u}, \frac{AB+A+B-3}{4u} \right) \cap \mathbb{P} = \emptyset$$

with prime endpoint $p_0 = \frac{AB+A+B-3}{4u}$. [BIB, thm:v75-k1-quotient]

Remark 7.45 (The continued-fraction relation, and why it is not a descent). With $a := qu+1 = \frac{A+1}{2}$ and $b := ru+1 = \frac{B+1}{2}$ one has $ab - up_0 = 1$, and the Euclidean algorithm gives explicitly

$$\frac{a}{p_0} = [0; r, u, q], \quad \frac{u}{b} = [0; r, u].$$

The modular-inverse relation is therefore an exact relation between consecutive convergents — but by itself it supplies no descent. Compare the *Matched Euclidean Gluing Lemma* recorded as missing in §16.5.

Proposition 7.46 (Structured composite coverage in the prime-free interval). **PROVED**. For odd s with $\frac{r-q}{2q} < s < \frac{r}{q}$, the number $z = r - sq$ satisfies $0 < z < h$ and

$$p_0 - z = q(ru + s + 1).$$

Moreover for every prime $\ell \mid B$, $2up_0 \equiv qu - 1 \pmod{\ell}$, hence $\gcd(B, qu - 1) = 1$. Each such ℓ covers exactly one residue class among the shifts $p_0 - 2j$, and under pair failure the number of its hits in $0 < 2j < h$ is at least $4^d u$. [BIB, prop:v75-k1-coverage]

Open target 7.47 (Canonical $k = 1$ gap/covering barrier). **OPEN**. The canonical branch $k = 1$ has *not* been eliminated. The present apparatus produces a gap filter, a smooth-shift condition and a covering requirement, but no contradiction. This is the sharpest surviving remnant of the Exact-D programme and is a self-contained problem: exhibit a canonical $k = 1$ Exact-D pair failure, or prove none exists. [BIB, Canonical $k = 1$ Gap/Covering Barrier]

7.6 Sparse geometric core locks (FFS layer)

The factor-flow-system (FFS) layer of the Biblia proves that *collision compression must be paid for with geometry*. These results are stated for a finite factor-closed core C of a prime affine FFS; the Goldbach specialisations take $\delta = 2$ because all active primes are odd. Statuses are the source's own: **tagN** = source claim “new, proved” and **tagG** = Goldbach specialisation, both imported as **PROVED – PENDING RED TEAM**.

Definition 7.48 (Collision multiplexing ratio). For a finite factor-closed core C let $\eta(C) = \Omega(J_C) > 0$, let $\chi(C)$ be the minimal number of distinct parent-pairs in a collision witness forest, and put

$$\kappa_{\text{col}}(C) := \frac{\eta(C)}{\chi(C)} \geq 1.$$

$\kappa_{\text{col}} = 1$ means no multiplexing: each unit of collision excess needs its own pair. A value above 1 means some witness pairs carry several distinct prime labels at once.

Theorem 7.49 (Multiplex collision rigidity, B-MULTIPLEX-RIGID). **PROVED – PENDING RED TEAM**.
 With $m = \min C$, $D = \max C - \min C$, $\eta = \Omega(J_C) > 0$ and $\chi = \chi(C)$,

$$D \geq \delta m^{\eta/\chi} = \delta m^{\kappa_{\text{col}}(C)}.$$

Proof. Closure puts every child in C , so every prime label occurring in J_C is at least m and $J_C \geq m^\eta$. Choose a collision witness forest realising χ ; lattice collision divisibility bounds each of its χ aggregated labels by D/δ , and the product of the aggregated labels is exactly J_C , so $J_C \leq (D/\delta)^\chi$. Combine and take χ -th roots. [BIB, Multiplex Collision Rigidity] $\square$

Reading. If many independent collision-excess labels are packed into few parent-pairs, those pairs must be extremely far apart: at $\kappa_{\text{col}} = 2$ the diameter grows at least quadratically in the minimal atom, at 3 at least cubically. Topological excess is converted into a power lower bound for the diameter.

Corollary 7.50 (Goldbach collision-compression barrier). **PROVED – PENDING RED TEAM**. Let C be a terminal bad SCC of the graph for even N that is not a pure support cycle, $m = \min C$, $D = \max C - \min C$. Then $D \geq 2m^{\eta/\chi}$, and since $\max C < N/3$,

$$m < \left(\frac{N}{6}\right)^{\chi/\eta}.$$

In particular for every integer $t \geq 2$: $\eta \geq t\chi \Rightarrow m < (N/6)^{1/t}$; equivalently a hypothetical core with $m > N^{1/t}$ must satisfy $\eta < t\chi$.

Proof. All active primes are odd, so $\delta = 2$; apply Theorem 7.49 and $\max C < N/3$ (Theorem 5.2). $\square$

Reading. This converts a structural question into a very concrete one: instead of killing all large cores directly, it suffices to exhibit triple collision multiplexing in the relevant class. It does not solve Goldbach — χ may be close to η , recovering only a linear diameter barrier.

Corollary 7.51 (Sparse Geometric Core Lock, B-SPARSE-CORE-LOCK). **PROVED – PENDING RED TEAM**.
 Under the same hypotheses, for every collision witness forest with χ_τ distinct parent-pairs,

$$N - 1 \leq U_C J_C \leq U_C \left(\frac{D}{2}\right)^{\chi_\tau}, \quad \text{hence} \quad D \geq 2 \left(\frac{N - 1}{U_C}\right)^{1/\chi_\tau}.$$

Proof. For a terminal bad SCC the unit-slope product lock gives $T_C = U_C J_C \geq N - 1$, and lattice collision divisibility with $\delta = 2$ gives $J_C \leq (D/2)^{\chi_\tau}$. [BIB, Sparse Geometric Core Lock] $\square$

Remark 7.52. Unlike the older bound $U_C G_C \geq N - 1$, the right-hand side here uses only the sparse family of parent-pairs needed to realise the collision fibres. It is a geometric version of exact- J , not a product over all pairwise gcds of rows.

Theorem 7.53 (Quantitative support explosion). **PROVED – PENDING RED TEAM**. Let S be a finite subsystem of an exact prime affine FFS and $K_\tau(S)$ any collision witness product. Put

$$\mathcal{M}(S) = P_{\text{src}}(S) \frac{\prod_{p \in S} R(p)}{\prod_{p \in S} p}, \quad B_S = \max_{p \in S} R(p).$$

If $A_S := \mathcal{M}(S)/(U_S K_\tau(S)) > 1$ then the number of distinct boundary atoms satisfies

$$|\partial S| \geq \left\lceil \frac{\log A_S}{\log B_S} \right\rceil,$$

and more generally $\mathcal{M}(S) > U_S K_\tau(S) B_S^r \Rightarrow |\partial S| \geq r + 1$.

Proof. The geometric escape criterion gives $R_\partial(S) \geq A_S$. Every boundary atom divides some $R(p)$, hence is at most B_S , and R_∂ is a product of $|\partial S|$ distinct primes, so $R_\partial(S) \leq B_S^{|\partial S|}$. Take logarithms and use integrality. [BIB, Quantitative Support Explosion] $\square$

Reading. The escape criterion is not merely “ $R_\partial > 1$ ”; the same computation can force *many* new boundary atoms. Together with Corollary 7.51 the FFS layer therefore has two independent elimination mechanisms, one for existence of escape and one for its multiplicity.

Chapter 8

Canonical Root Dynamics (CRD)

8.1 Epistemic convention

CRD is a *reformulation*, not a mechanism. Its content is:

1. the failed world is a closed bad set with a unique source (Lemma 3.26);
 2. the union of all closed bad sets below $N/3$ is computable by a decreasing filtration (Definition 3.28);
 3. canonical failure is exactly membership of $\text{PF}(N - p_0)$ in that fixed point (Theorem 3.29).
- Anything beyond that — capacity arguments, container arguments, spectral arguments — belongs to the successor theories of §13.2 and is **KILLED**. When reading older CRD material, keep this boundary in mind: the three items above survived every audit; the rest did not.

8.2 The architecture wall

Theorem 8.1 (Finite Affine Canonical Programmability). **PROVED** in [PCRD] (proof not reproduced here), class A as a statement about canonicity. For any finite family of affine forms $L_j(P) = a_jP + b_j$ and any prescribed finite valuation and compositeness conditions, under the standard coprimality conditions there exist — by the Chinese remainder theorem and Dirichlet’s theorem — infinitely many primes P realising the whole pattern. Setting $N = 2(P - 1)$ gives canonical $p_0 = P$, $h = 1$.

Theorem 8.2 (Partial BAD-transcript programmability; scope-corrected). **SCOPE-CORRECTED**. Compatible prescribed finite BAD edges and edge valuations can be programmed in canonical $h = 1$ worlds by CRT/Dirichlet methods, as formalised by CRT-TRANSCRIPT-EXACT-SCOPE-20260905. The theorem controls the selected requirements only. It does not assert that every unprescribed prime factor of an expanded row is BAD, nor that a complete BFS neighbourhood of a prescribed depth is BAD. The earlier blanket reading of FBTU is withdrawn.

Corollary 8.3 (Partial-transcript no-go; not a complete-BFS wall). **SCOPE-CORRECTED**. A contradiction cannot follow merely from the existence of a finite compatible selected divisibility/valuation transcript, because such a transcript may be realised in canonical worlds with extra cofactors. This corollary does **not** rule out a theorem controlling a complete bounded-depth BFS frontier. The exact $N = 92$ certificate in §10.2 shows why the quantifiers differ: a selected BAD cycle coexists with a depth-1 GOOD child.

Remark 8.4 (Link to the phase diagram). All $h = 1$ worlds with $G \geq 4$ lie in phase Type I (Theorem 6.12), hence have m_0 composite and $\rho_0 = 0$. The whole $h = 1$ programme of [PCRD] therefore lives *inside a single phase* — the one in which the reflection gate gives no guarantee. Two

reported witnesses, verified here:

N	p_0	G	m_0	type, depth
103 344	51 673	14	$51\,671 = 163 \cdot 317$	Type I, $d = 4$
364 844	182 423	6	$182\,421 = 3^2 \cdot 20\,269$	Type I, $d = 6$

VERIFIED COMPUTATION.

8.3 C0a — the small-gap canonical obligation

Theorem 8.5 (Small-gap residual closure, C0a). **PROVED**, class C . Hypotheses: *canonical failure*, $h \leq 2$. Then $h \in \{1, 2\}$, m_0 is odd composite, $Q \neq \emptyset$, and

$$\max\{U_R, J_R, B_{\text{abs}}\} > \left[\left(\frac{5}{3} \right)^{|Q|} \text{rad}(m_0) \right]^{1/3} > 1.$$

[CORE, C0a-RESIDUAL-CLOSURE]

Reading. For $h \leq 2$ the reflected block is empty, so boundary geometry gives nothing; the theorem replaces it by a nonzero obligation on the row side. *Limitation:* it yields neither a path to GOOD nor a capacity contradiction.

8.4 C0b — the exact canonical reflection-defect ledger

Theorem 8.6 (C0b-RD). **PROVED**, class $A + C$. Hypotheses: *canonical failure*, $N \geq 50$, $h \geq 3$. With $t = \lfloor (h-1)/2 \rfloor$ and

$$\begin{aligned} \mathcal{I} &= \{j : \text{PF}(c_j) \cap R \neq \emptyset, a_j \in \mathcal{A}_N\}, & \mathcal{D} &= \{j : \text{PF}(c_j) \cap R \neq \emptyset, a_j \notin \mathcal{A}_N\}, \\ \mathcal{S} &= \{j : \text{PF}(c_j) \cap R = \emptyset, P^+(c_j) < h\}, & \mathcal{F} &= \{j : \text{PF}(c_j) \cap R = \emptyset, P^+(c_j) \geq h\}, \end{aligned}$$

one has

$$\mathcal{J}_h = \mathcal{I} \uplus \mathcal{D} \uplus \mathcal{S} \uplus \mathcal{F}$$

and for $j \in \mathcal{F}$ the number $\ell_j = P^+(c_j)$ is active, lies outside R , and occurs in exactly one element of $\mathfrak{B}(p_0)$. Moreover, if $\mathcal{I} = \emptyset$ and $k = \lceil t/3 \rceil$, then at least one of

$$\sum_{j \in \mathcal{D}} \Omega(a_j) \geq 2k, \quad \sum_{j \in \mathcal{S}} \Omega(c_j) > k \frac{\log x}{\log h}, \quad \prod_{\ell \in \Lambda_{\mathcal{F}}} \ell \geq h^k$$

holds. With $Q_{\leq} = \{q \in Q : q \leq t+1\} \neq \emptyset$ and $j(q) \equiv h \pmod{q}$, $1 \leq j(q) \leq q-1$, one has $q \mid c_{j(q)}$, $j(q) \in \mathcal{I} \cup \mathcal{D}$ and $B_{\leq} < h^{\lfloor \mathcal{J}_{\leq} \rfloor}$; and if $Q_{\leq} = \emptyset$ then $\mathfrak{F}_Q > (5(t+1)/3)^{|Q|}$. [CORE, C0b-RD]

Reading. This is the strongest *quantitative* obligation that canonical boundary geometry issues at all: every coordinate of the reflected block must land in one of the four classes. *Known limitation:* C0b-RD supplies neither a path to GOOD nor a capacity bound.

Scope note 8.7 (Why the hypothesis $Q_{\leq} \neq \emptyset$ is needed). **SCOPE-CORRECTED**. In version v1 this condition was omitted; with $Q_{\leq} = \emptyset$ the inequality would read $1 < 1$. The configuration is not rare: among canonical instances with $h \geq 3$ and $N < 2 \cdot 10^5$, 47 551 of 72 446 have $Q_{\leq} = \emptyset$; the smallest case is $N = 50$, $p_0 = 29$, $h = 4$, $t = 1$, $m_0 = 21$, $Q = \{3, 7\}$. [CORE, C0b-RD audit repair] — *reported computational result, not reproduced in this document*.

8.5 The C0b branch targets: TI, TD, TS, TF

COb-RD says one of four classes must carry the obligation. The four branch targets are the statements “if the obligation lands in class X , then the dynamics must realise it”. All four are **OPEN**. They are the entry point to the frontier; full frontier entries with obstructions and success criteria are in Chapter 9.

ID	Branch	Obligation to be discharged
TI	$\mathcal{I}$ (incident, active mirror)	Turn an active reflected row into realised production: the mirror a_j is an active parent of c_j , so it <i>could</i> feed the world; TI asks that it must feed it <i>measurably</i> .
TD	$\mathcal{D}$ (incident, inactive mirror)	The root-anchored dominance obligation. The mirror is not active, so the mass must be recovered from $\Omega(a_j)$; TD asks for a lower bound on realised production anchored at the root. This is the smallest missing bridge (§9.11).
TS	$\mathcal{S}$ (h -smooth mirror)	The coordinate is smooth, so its factors are small and lie in the low range; TS asks that smooth coordinates cannot all be absorbed.
TF	$\mathcal{F}$ (fresh large factor)	$\ell_j = P^+(c_j) \geq h$ is active and outside R ; TF asks that a fresh external atom forces an escape or a coupling.
TD-DENSITY	density form of TD	SUPERSEDED : replaced by the root-anchored form after the density version was found to require a bound of the same strength as its conclusion.

Scope note 8.8 (TF is local, and locality is not enough). **SCOPE-CORRECTED**. “A local $\mathcal{F}$ certificate together with $h < G$ forces routing” is false: at $N = 128$, $\ell = 13$ the certificate holds, but p_0 wins and $R = \emptyset$. FAIL-TF-LOCAL-SCOPE-001

Chapter 9

Current Proof Frontier

This is the only chapter where new work should start. Everything before it is either settled or dead. Each target below carries seven fields: exact statement, why it matters, what is already proved, the exact remaining obligation, known obstructions, what would count as success, and the failure-log entries that constrain it.

Before reading further, internalise the shape of the problem. The chain is

$$\underbrace{\text{C0a}}_{\text{PROVED}} \Rightarrow \underbrace{\text{C0b-RD}}_{\text{PROVED}} \Rightarrow \underbrace{\{\text{TI, TD, TS, TF}\}}_{\text{OPEN}} \Rightarrow \underbrace{\text{P1}^{\text{br}}}_{\text{OPEN}} \Rightarrow \underbrace{\{\text{L1, E1, W1, K1}\}}_{\text{OPEN}} \Rightarrow \underbrace{\text{X1}}_{\text{OPEN}} .$$

The chain does *not* end at Goldbach even if every link is proved: it ends at “the canonical traversal succeeds”, which is strictly stronger than Goldbach for each N but is not implied by it. The architecture wall (§8.2) says the chain cannot be closed by finite local data alone, so at least one link must import global input.

Read §9.2 first. That chain is the historical frontier. The project’s *current* frontier is the canonical threshold programme of v4.79–v4.81 (§9.2), whose target is deliberately weaker than X1 — an asymptotic $N_0^{p_0}$ rather than all N — and which has already converted two of its own demanded outputs into proved impossibilities. The two live targets are named there. The older chain is retained because its statements are correct and its obstructions are real, and because C0b-RD still feeds it.

9.1 The exact Goldbach boundary

What is proved. For every even N in the non-diagonal case: the canonical root is well defined and unique; if it fails, the failed world is closed, all-BAD, has p_0 as its unique source, and lies below $N/3$; the reflected boundary is defect-free and its ledger has exactly four classes; and the row side satisfies C0a when $h \leq 2$ and C0b-RD when $h \geq 3$.

What would imply Goldbach. If for every N the canonical traversal succeeded, then in particular $\text{Good}_N \neq \emptyset$ for every N , which is binary Goldbach. So a proof of universal canonical success is *sufficient*.

What does not follow. Goldbach does *not* imply universal canonical success; the six N of §6.6 have $B_N^{\text{low}} \neq \emptyset$ while satisfying Goldbach comfortably, and 301 non-canonical upper roots fail while their N satisfy Goldbach. Every statement conditioned on $\text{Good}_N = \emptyset$ is class D and is unusable under canonical failure alone. **FAIL-UPPERHALF-SCOPE-001**

The current smallest missing bridge. The root-anchored TD obligation feeding P1^{br} . Everything upstream is proved; everything downstream is blocked on it or on the same defect in

another guise.

9.2 The canonical threshold programme (post-v4.79/4.80/4.81)

This is the project's actual current frontier, and it is newer than the TI/TD/TS/T-F/P1BR/K1/X1 chain of §9.3–§9.9. The older chain is not wrong; it is simply not where the work is. A model starting fresh should read this section first.

The target here is deliberately weaker than X1 and therefore more attackable:

$$\exists N_0^{p_0} < \infty \quad \forall \text{ even } N \geq N_0^{p_0} : \quad p_0(N) \text{ succeeds.}$$

The verdict after v4.81 is: *yes in architecture, not in closure*. Two of the four demanded outputs of the previous frontier were *proved impossible*; exactly three routes remain, and the preferred one is named.

9.2.1 The v4.79/v4.80 engine

Throughout: $N = 2x$, $p_0 = x + h$, $m = N - p_0$, $Q = \text{PF}(m)$, $R = R_N(p_0)$, canonical failure, $N \geq 50$.

Proposition 9.1 (Nagura window). **PROVED**. For $N \geq 50$, Nagura's explicit interval theorem together with canonical minimality gives

$$p_0 < \frac{3N}{5}, \quad m > \frac{2N}{5}.$$

[CORE, v4.79 (V79.2); Nagura [1]]

Reading. This is why the two-sheet argument (Theorem 5.34) may use $m_0 > 2N/5$; it is the only place an external prime-counting input enters the canonical layer, and it is completely explicit.

Theorem 9.2 (Root-row vertical compensation and weighted residual floor). **PROVED**. The root row itself contributes $V_0 := m/\text{rad}(m)$ to the canonical vertical defect U_R . Combined with full- Q residual production (Theorem 5.13) and the rooted residual-absorption split,

$$U_R^2 J_R B_{\text{abs}} > m \left(\frac{5}{3}\right)^{|Q|} > \frac{2N}{3}. \quad (\text{V79.4})$$

[CORE, vNEXT4.79]

Theorem 9.3 (First-front coefficient dichotomy). **PROVED**. With the exact first-front coefficient defect

$$F_Q^{\text{coeff}} = \frac{m}{\text{rad}(m)} \prod_{\substack{q|m \\ \kappa_R(q) \geq 2}} q,$$

one has the sharp dichotomy

$$F_Q^{\text{coeff}} = 1 \iff m \text{ squarefree and } \kappa_R(q) = 1 \ (q \mid m),$$

while

$$F_Q^{\text{coeff}} > 1 \implies T_R > \frac{5}{2}N,$$

which forces a half-power U/J channel. Nonprimitive first-front failure is therefore already coefficient-heavy. [CORE, vNEXT4.79–4.80]

Definition 9.4 (Primitive branch). The canonical failed world is *primitive* when $F_Q^{\text{coeff}} = 1$, i.e. m is squarefree and every first-front label has load one. Otherwise it is *nonprimitive*.

Theorem 9.5 (Primitive load-one and selected first-front production). **PROVED**. *On the primitive branch the load-one condition is exactly*

$$D_q = \gcd(m, N - q) = 1 \quad (q \in Q),$$

so the selected first-front product is exactly

$$U_F(J_F B_{\text{re}}) B_{\text{abs}} = \prod_{q \in Q} (N - q).$$

With the adaptive least non-dividing prime $\lambda_N = \min\{\ell \in \mathbb{P} : \ell \geq 3, \ell \nmid N\}$ one has $q < N/(2\lambda_N)$ for every $q \in Q$, hence

$$\prod_{q \in Q} (N - q) > \left(1 - \frac{1}{2\lambda_N}\right)^{|Q|} N^{|Q|} \geq \frac{25}{36} N^2.$$

The primitive selected production therefore has leading logarithmic coefficient $|Q| \geq 2$, not merely one. [CORE, vNEXT4.80]

Theorem 9.6 (Owner-preserving quadratic UJD charge, C-PO-PRIMITIVE-OWNERPRESERVING-QUADRATIC-UJD-CHARGE). **PROVED**. *The adaptive high range forms a unique-parent acyclic forest. Each absorbed high atom is replaced by a selected terminal descendant whose support lies in $C_{\lambda_N} \setminus Q$; low absorbed support and selected terminal support are disjoint, so vertical, horizontal and denominator support can each be single-charged. The resulting owner-preserving selected divisor $\mathcal{C}_Q$ satisfies*

$$\mathcal{C}_Q \mid U_R J_R D_{\lambda_N}.$$

[CORE, vNEXT4.80]

Consequences recorded by v4.80. C-PO-PRIMITIVE-BASEABS-ESCAPE-OR-REUSE-CHARGE is **SUPERSEDED**: no external-parent escape/reuse theorem is needed to charge primitive base absorption. C-PO-PRIMITIVE-FIRSTFRONT-UNITMARGIN-PRODUCTION ($U_F(J_F B_{\text{re}}) B_{\text{abs}} > \frac{10N}{9}$, **PROVED** in v4.79) is *retained* but superseded as the *binding primitive floor*. C-PO-PRIMITIVE-SELECTEDFAMILY-CAPACITY is refined to C-PO-PRIMITIVE-SELECTED-UJD-CAPACITY, which remains **OPEN**. G-CONDITIONAL-EFFECTIVE-PO-NO is **PROVED** only as a *conditional theorem*: no numerical $N_0^{p_0}$, no universal p_0 -success theorem and no Goldbach proof is claimed.

9.2.2 The v4.81 container theorem and what it kills

The v4.80 preferred frontier C-PO-PRIMITIVE-FIRSTFRONT-REFLECTION-DISJOINT-CAPACITY asked for one of four outputs: (1) reflected production disjoint from $\mathcal{C}_Q$ at fixed positive logarithmic scale; (2) same-label overlap forcing an additional exact J/U layer; (3) a GOOD event; (4) strict scale descent. v4.81 proves that **(1) and (2) cannot exist**, by exact identity rather than heuristic.

Theorem 9.7 (Exact rooted container identity). **PROVED**. *On the primitive canonical failed branch,*

$$U_R J_R D_{\lambda_N} = \frac{\prod_{u \in R} (N - u)}{m \prod_{r \in H_\lambda} r}. \quad (\text{V81.1})$$

So the “container” of the v4.80 selected charge is, up to an explicit denominator, the row product of the world itself. [CRD v4.81, Thm. container]

Theorem 9.8 (Container slack for $\mathcal{C}_Q$). **PROVED**. For every $N \geq 500$,

$$\frac{U_R J_R D_{\lambda_N}}{\mathcal{C}_Q} = \frac{\prod_{r \in C_{\lambda} \setminus Q} (N-r) \prod_{r \in H_{\lambda} \setminus Z} (N-r)}{\prod_{r \in H_{\lambda} \setminus A_H} r} > \frac{5N}{6}. \quad (\text{V81.2})$$

The selected charge therefore never approaches its container: the gap is at least one full logarithmic unit, and grows by one further unit for every low non-front state. [CRD v4.81, Thm. slackCQ]

Corollary 9.9 (Death of the size route). **PROVED**. Two consequences close the archimedean size route completely.

1. Trivially: a divisor never exceeds what it divides. Once $\mathcal{C}_Q \mid U_R J_R D_{\lambda_N}$ is a theorem (Theorem 9.6), no lower bound for $\mathcal{C}_Q$ can contradict any true upper capacity for the container.
2. Quantitatively: any adjoined production P with $\mathcal{C}_Q P \mid U_R J_R D_{\lambda_N}$ is priced exactly by (V81.2) — it must supply one factor $\geq \frac{5}{6}N$ for every low non-front state before it can begin to bite.

Hence “new selected storage disjoint from $\mathcal{C}_Q$ at fixed positive logarithmic scale” — the frontier’s own demanded output (1) — is insufficient by construction, not merely unproved.

Theorem 9.10 (Exact overlap normal form: overlaps are charge-free). **PROVED**. If a prime ℓ divides a selected row $N - u$ ($u \in Q \cup Z$) and a strict reflected carrier $c_j = p_0 - 2j$, then the entire content of that coincidence is

$$\ell \mid a_j - u, \quad 0 < a_j - u < x, \quad (\text{V81.3})$$

which is equivalent to the already-counted divisibility $\ell \mid N - u$ together with the residue condition $2j \equiv p_0 \pmod{\ell}$. Because $a_j \notin R$, the coincidence creates no rooted parent, no κ_R increment, no J -layer and no U -layer.

Corollary 9.11. Alternative (2) of the v4.80 frontier is empty.

Reading. This is the sharpened, final form of the reflection-gate warning: not only does the prime-free boundary fail to *activate* a reflected row (Warning 3.16, $N = 242$), an overlap between the reflected block and the selected family carries *no charge at all*.

9.2.3 New positive structure surviving the attack

All of the following is owner-preserving and reconstructible from the rooted CRD definitions.

Theorem 9.12 (C1: gap-scale shadow normal form). **PROVED**. $D_q = \gcd(m, 2h - q)$, and primitive load-one is equivalent to

$$\gcd(m, 2h - u) = 1 \quad (u \in R \setminus \{p_0\}). \quad (\text{V81.11})$$

This puts the first front and the reflected block on a single scale $\delta = 2h$.

Remark 9.13. This is the Prime-Gap Shadow Identity (Theorem 5.22) re-proved from the v4.80 definitions and used for the first time in the v4.79/v4.80 chain. The v4.81 salvage audit records it as the *one genuinely missing tool* the older corpus contributed to the current frontier.

Theorem 9.14 (C2: primitive residual world). **PROVED**. $R' := R \setminus (\{p_0\} \cup Q)$ is child-closed, nonempty, satisfies $\text{Src}(R') \subseteq \Gamma_N(Q)$ and $\min R' < \sqrt{N}$.

Theorem 9.15 (C3: owner-preserving capacity factorisation). **PROVED**. With $\Sigma' = \prod_{d \in \text{Src}(R')} d$,

$$T_R \Sigma' = \left(\prod_{q \in Q} (N - q) \right) T_{R'}.$$

Theorem 9.16 (C4: selected residue lock). **PROVED**.

$$\mathcal{C}_Q B_H \equiv (-1)^{|Q|+|Z|} m \prod_{z \in Z} z \pmod{N}. \quad (\text{V81.13})$$

Theorem 9.17 (C5: adaptive fringe cofactor alphabet). **PROVED**. *Every edge into H_λ has cofactor a prime in $[\lambda, 2\lambda]$ coprime to N , and that prime is itself a low state. The alphabet is the full prime range $\mathcal{L}_N \subseteq [\lambda, 2\lambda]$.*

Warning 9.18. This is *not* the rejected two-letter claim $\{\lambda, 2\lambda-1\}$, which is **KILLED** as FAIL-FRIDGE-ALPHABET-TRANSI. Theorem 9.17 is precisely the “what survives” clause of that failure entry.

9.2.4 The minimal hostile survivor and the surviving routes

Theorem 9.19 (Product-comparison immunity). **PROVED**. *A primitive canonical failed world survives every archimedean product comparison between a single-charged selected family and the rooted $U/J/D$ container. The reason is now exactly quantified: a selected family can beat its container only if it leaves fewer than $|H_\lambda|$ rows unselected, whereas $\mathcal{C}_Q$ leaves at least $|C_\lambda \setminus Q| + |H_\lambda| - k$.*

Theorem 9.20 (Trichotomy of surviving routes). **PROVED**. *Any proof that the minimal hostile survivor class is empty must proceed through at least one of*

(A) population capacity $\vee$ **(B) residue/congruence obstruction** $\vee$ **(C) a GOOD witness.**

- (A)** an upper bound for $|C_\lambda|$ (or $|R|$) contradicting a lower bound for the same quantity. This is the only archimedean route left, because by (V81.1) every product container scales with $|R|$.
- (B)** an obstruction inside a container of bounded size. The only such containers in the corpus are the residue class modulo N (Theorem 9.16, Theorem 5.8) and the gap-scale divisors $h - j < h$ of $\text{COB-RD}(3)$.
- (C)** an actual reachable $g \in R$ with $N - g$ prime.

Proof. Immediate from Theorem 9.19 together with the observation that every capacity object in the corpus (T_R , U_R , J_R , D_λ , the FC and LCAP envelopes) is a product over rows or labels of R , hence has logarithm $\Theta(|R| \log N)$, whereas any selected subfamily has logarithm at most $|S| \log N$ with $S \subseteq R$. $\square$

C-P0-PRIMITIVE-LOWCORE-SMOOTHROW-POPULATION-CAPACITY — the current preferred target

Status. **OPEN**.

Statement. Primitive canonical failed world, $N \geq 500$, with the partition $R = \{p_0\} \uplus C_\lambda \uplus H_\lambda$. By Theorem 9.17 exactly $|H_\lambda|$ rows carry a high factor, each of the form $L\ell$ with $L \in \mathcal{L}_N$. Hence

exactly $|R| - |H_\lambda|$ distinct integers $N - u$ ($u \in R$) are C_λ -smooth and lie in $[m, N)$,

all of them $> (1 - \frac{1}{2\lambda})N$ except the root row $m > 2N/5$. Wanted: an effective upper bound

$$|C_\lambda| \leq \Psi^*(N, \lambda_N)$$

for the number of integers in the window $((1 - \frac{1}{2\lambda})N, N)$ all of whose prime factors lie in a prescribed set of $|C_\lambda|$ primes below $N/(2\lambda)$, contradicting the matching lower bound $|C_\lambda| \geq |R| - |H_\lambda| - 1$ forced by child-closure.

Why it matters. By Theorem 9.20 route (A) is the only archimedean route left; by Theorem 9.17 the high band is completely described, so the whole unknown is the low core; and by (V81.1) a bound on $|C_\lambda|$ is exactly what converts every existing product capacity into a finite quantity.

Already proved. The container identity (V81.1), the slack bound (V81.2), the fringe alphabet, the shadow normal form (V81.11), the residual world R' , and the capacity factorisation.

Honest difficulty statement. This is an analytic question about S -smooth integers in a short interval with prescribed *sparse* support. $\Psi(x, y)$ technology does not apply directly, because the support set is not an initial segment of the primes.

Known obstruction. Evertse's S -unit bound is *vacuous* here: the equation $u + w = N$ with $u \in S$, w S -smooth, $S = R \setminus \{p_0\}$ has $|R| - 1$ solutions, while the bound is $3 \cdot 7^{2|S'|+3}$ with $|S'| \leq |S| + \omega(N)$ — exponential in the very quantity to be bounded. Recorded as FAIL-SUNIT-COUNT-BOUNDS-THE-LOWCORE-001.

Success criterion. The effective bound Ψ^* above, or a proof that no such bound can exist at this scale.

C-P0-PRIMITIVE-SELECTED-RESIDUE-LOCK-CAPACITY — the secondary target

Status. **OPEN**.

Statement. Produce a second, *independent* determination of $\mathcal{C}_Q \bmod N$ (or of $T_{R'} \bmod N$) from the reflection-complete boundary, incompatible with (V81.13) above a computable threshold.

Input. (V81.13), G-ROOT-LOCK (Theorem 5.8), the capacity factorisation (Theorem 9.15), and the gap-scale normal form (V81.11). Reflected carriers entering the exact lock are the natural source of a second determination.

Guard — read before attempting. A congruence that is a formal consequence of the *same* factorisation is *vacuous*. The check performed in Theorem 9.16 shows that combining (V81.12), (V81.13) and G-ROOT-LOCK reproduces only $m \equiv -p_0$. Any candidate must import an *independent* evaluation.

Why it is route (B). It lives inside a container of bounded size (the residue class modulo N), which is the only kind of container Theorem 9.20 leaves for a non-archimedean argument.

The nonprimitive branch. Separately alive but without a consumer. The coefficient lock gives $T_R > \frac{5}{2}N$; a large F_Q^{coeff} yields a superlinear amplifier, while cheap nonprimitive defect bases localise below a small power and into a Baker-accessible reduction. *These are reductions, not a capacity theorem*, and the same container degeneracy (Corollary 9.9) forbids the scalar route there too. The exact Baker requirement recorded in v4.81 is *not* met by the available effective machinery.

Status line, verbatim from the source. Binary Goldbach remains **OPEN**; universal canonical p_0 -success remains **OPEN**; $N_0^{p_0}$ is **not proved**, neither effectively nor ineffectively; no status of G-CONDITIONAL-EFFECTIVE-P0-NO is changed.

9.2.5 Updated proof DAG

$$\begin{array}{l}
p_0(N) \text{ fails, } \quad N \geq 500 \\
\Downarrow \\
F_Q^{\text{coeff}} > 1 \vee F_Q^{\text{coeff}} = 1 \\
\Downarrow \\
\left\{ \begin{array}{l} \textbf{nonprimitive: } T_R > \frac{5}{2}N, \text{ split at } N^{3/40} \quad (\text{no consumer; the scalar route is forbidden}) \\ \textbf{primitive: } m \text{ squarefree, } \kappa_R(q) = 1 \end{array} \right. \\
\Downarrow \quad (\text{primitive}) \\
D_q = 1 \iff \gcd(m, 2h - q) = 1 \iff \gcd(m, 2h - u) = 1 \quad \forall u \in R \setminus \{p_0\} \quad (\text{V81.11}) \\
\Downarrow \\
U_F(J_F B_{\text{re}}) B_{\text{abs}} = \prod_{q \in Q} (N - q) > \left(\frac{5N}{6}\right)^{|Q|} \\
\Downarrow \\
\mathcal{C}_Q \mid U_R J_R D_{\lambda_N} \quad \text{and} \quad \frac{U_R J_R D_{\lambda_N}}{\mathcal{C}_Q} > \frac{5N}{6} \quad (\text{V81.1, V81.2}) \\
\Downarrow \\
\text{archimedean product comparison is immune} \implies \text{routes (A), (B), (C) only}
\end{array}$$

9.3 The C0b branch targets

TI — injective branch obligation

Statement. If the C0b-RD obligation is carried by the class $\mathcal{I}$ (incident coordinates with an *active* lower mirror), then the map $j \mapsto (a_j, \text{PF}(c_j))$ realises production in the world: the active mirrors inject into distinct parent–child incidences, and the number of such incidences is bounded below by $|\mathcal{I}|$ up to an explicit constant.

Why it matters. $\mathcal{I}$ is the only class whose coordinates are already *active graph rows*. If the obligation lands there and TI holds, the boundary obligation converts directly into world structure with no further input. It is the cheapest of the four branches.

Already proved. The reflection gate (Lemma 3.15): for $j \in \mathcal{I}$ the edge $a_j \rightarrow \text{PF}(c_j)$ exists and $\Gamma_N(\{a_j\}) = \text{PF}(c_j)$. Strict non-reuse (Lemma 3.13): every prime $\ell \geq h$ occurs in at most one block element, so distinct $j \in \mathcal{I}$ with a large factor give distinct incidences.

Remaining obligation. Injectivity for the *small* factors, i.e. for primes $\ell < h$, where strict non-reuse gives nothing. Equivalently: bound $\#\{j \in \mathcal{I} : \text{PF}(c_j) \subseteq [2, h)\}$ from above by a fraction of $|\mathcal{I}|$.

Obstructions. (i) The class $\mathcal{I}$ can be empty — $N = 242$ has no active reflected row at all (Warning 3.16); TI is then vacuous and the obligation moves to $\mathcal{D}, \mathcal{S}, \mathcal{F}$. (ii) The guaranteed active mirror exists only in phase Type III (Corollary 6.14), so TI can only be the operative branch on $\theta > 1/2$. (iii) Small-factor collisions are exactly what the widened block would have controlled, and the widened block is refuted (Warning 3.14).

Success criterion. An unconditional inequality of the form $\#\{\text{distinct incidences from } \mathcal{I}\} \geq c|\mathcal{I}|$ with $c > 0$ absolute, valid in all phases where $\mathcal{I} \neq \emptyset$, and *stated in the same functional* that P1^{br} consumes.

Constraining log entries. FAIL-REFLECTION-ACTIVATION-001, FAIL-BROADBLOCK-ENDPOINT, FAIL-TF-LOCAL-SCOPE-001.

TD — root-anchored dominance obligation

Statement. If the obligation is carried by $\mathcal{D}$ (incident coordinates whose lower mirror is *inactive*), then the recovered mass $\sum_{j \in \mathcal{D}} \Omega(a_j) \geq 2\lceil t/3 \rceil$ of C0b-RD translates into a lower bound on *realised production in the world*, anchored at the root, measured by the same functional that bounds capacity.

Why it matters. This is **the smallest missing bridge in the whole project**. C0b-RD is proved and hands over a genuine quantitative obligation; TD is the first step that turns a boundary count into a dynamical count. Every downstream target (P1^{br}, L1, W1, K1, X1) either depends on TD or fails for the same reason TD fails.

Already proved. C0b-RD (Theorem 8.6) including the $\mathcal{D}$ -branch inequality $\sum_{j \in \mathcal{D}} \Omega(a_j) \geq 2k$ under $\mathcal{I} = \emptyset$; the closed-world and unique-source structure (Lemma 3.26); no-bad-return (Theorem 5.33); the exact factorization $T_S = U_S J_S$ (Theorem 3.19).

Remaining obligation. Produce a lower bound

$$(\text{realised production of the world}) \geq \gamma(\text{obligation mass of } \mathcal{D})$$

in which γ is *the same* functional as the one appearing on the capacity side of P1^{br} and K1. Not a comparable functional, not a functional that dominates it on average: the same one.

Obstructions. (i) a_j inactive means a_j is composite or divides N ; the mass $\Omega(a_j)$ is therefore not attached to any graph vertex, and there is no proved route from it to an incidence. (ii) The density formulation TD-DENSITY is **SUPERSEDED** precisely because it needed a bound of the same strength as its own conclusion. (iii) The scope-corrected partial-transcript wall (Corollary 8.3) forbids inferring the bound from selected programmed edges alone; a theorem controlling the complete frontier is not excluded. (iv) Every recorded attempt to substitute a different but “morally equivalent” functional is in the log as a scope error.

Success criterion. A proof of the displayed inequality with an explicit γ , plus a verification that the γ used is literally the one in §9.4. A proof with a different functional is *not* partial progress; it is a new open problem.

Constraining log entries. FAIL-ETA-GAMMA-IDENTIFY-001, FAIL-POTENTIAL-CAPACITY-001, FAIL-CAB-001, FAIL-CAB-INTERNAL-001, FAIL-BABS-PRODUCTION-001.

TS — smooth branch obligation

Statement. If the obligation is carried by $\mathcal{S}$ (coordinates with $P^+(c_j) < h$ and $\text{PF}(c_j) \cap R = \emptyset$), then the smooth mass $\sum_{j \in \mathcal{S}} \Omega(c_j) > k \log x / \log h$ cannot be entirely absorbed inside the world: some smooth coordinate must contribute an escape or a coupling.

Why it matters. $\mathcal{S}$ is the branch where the mass bound is largest (it grows like $\log x / \log h$ per unit), so if the obligation lands there the quantitative margin is greatest. It is the branch most likely to be provable by elementary means.

Already proved. The $\mathcal{S}$ -branch inequality of C0b-RD; minimum-factor escape (Corollary 5.25); shadow cycle localisation (Theorem 5.27), which confines a terminal core inside Q to $(0, 2h/3)$ — exactly the range in which smooth coordinates live.

Remaining obligation. Convert “ c_j is h -smooth and its factors avoid R ” into a statement about the world. The factors of c_j are active primes outside R ; they are candidate escape targets, but their rows have not been shown to reach GOOD.

Obstructions. (i) A factor outside Q or outside R does not mean success — only that the current alphabet was left (warning after Corollary 5.25). (ii) Smoothness bounds *numbers*, not *vertices*; the passage from $\Omega(c_j)$ to distinct active vertices needs the same injectivity that TI lacks.

Success criterion. An unconditional statement: if $\mathcal{I} = \emptyset$ and the $\mathcal{S}$ inequality is the operative one, then $G_{\text{esc}}(N) > 0$. That alone would be a real theorem, since it would give Goldbach for every N in that regime.

Constraining log entries. FAIL-HALL-SUPPORT-001, FAIL-DIRECT-ADJACENCY-001.

TF — fresh branch obligation

Statement. If the obligation is carried by $\mathcal{F}$ (coordinates with $\ell_j = P^+(c_j) \geq h$, active and outside R), then the product bound $\prod_{\ell \in \Lambda_{\mathcal{F}}} \ell \geq h^k$ forces an escape edge or an external coupling.

Why it matters. $\mathcal{F}$ is the only branch that hands over *named active primes outside the world*. Those are exactly the objects an escape argument needs.

Already proved. The $\mathcal{F}$ -branch product inequality; that each ℓ_j is active, outside R , and occurs in exactly one block element (Theorem 8.6 combined with Lemma 3.13).

Remaining obligation. Show that an active prime outside R occurring as a large factor of a reflected coordinate must be, or must lead to, a GOOD vertex.

Obstructions. (i) **Locality is not enough:** at $N = 128$, $\ell = 13$, the local $\mathcal{F}$ certificate holds but the canonical root wins outright and $R = \emptyset$, so the certificate constrains nothing. FAIL-TF-LOCAL-SCOPE-001 (ii) An external parent need not be a success root: (6142, 877, 3511) has 3511 active and above p_0 , yet its own traversal fails. FAIL-EXTERNAL-SUCCESS-001 (iii) External reuse is not local: $(N, \ell) = (1\,825\,372, 52\,153)$ at distance 53 750. FAIL-LOCAL-REUSE-001

Success criterion. A theorem of the form: under canonical failure with $\mathcal{I} = \mathcal{D} = \mathcal{S} = \emptyset$, some $\ell \in \Lambda_{\mathcal{F}}$ has $N - \ell$ prime, or has an active parent outside R whose row meets Good_N . The second alternative must be included — it is provably necessary.

Constraining log entries. FAIL-TF-LOCAL-SCOPE-001, FAIL-EXTERNAL-SUCCESS-001, FAIL-LOCAL-REUSE-001.

9.4 P1^{br} — branchwise production upgrade

P1BR

Statement. Let γ be the capacity functional of the canonical world. P1^{br} asserts that the production realised branchwise — summed over the branches of the first front, each measured on its own subtree — admits a lower bound in γ that exceeds the capacity bound in the same γ :

$$\sum_{\text{branches } \beta} \gamma(\text{production}(\beta)) > \gamma(\mathcal{C}_N(R)).$$

Why it matters. P1^{br} is the node where the boundary obligations become a contradiction. Everything downstream (L1, E1, W1, K1, X1) consumes its conclusion.

Already proved. The branchwise decomposition itself is legitimate: the world is closed with a unique source, so the first front partitions it into branches. The exact ledger $T_S = U_S J_S$ holds on every branch separately. The capacity side $\mathcal{C}_N(R)$ is well defined.

Remaining obligation. A *positive external packet* feeding the *same* P1 capacity — the corpus target C-POSITIVE-EXTERNAL-PACKET-TO-SAME-P1-CAPACITY. Concretely: exhibit a quantity that (a) is provably positive under canonical failure, (b) is external to the world (so it is not already counted in $\mathcal{C}_N(R)$), and (c) is measured in γ rather than in a functional that merely correlates with γ .

Obstructions. (i) The status of P1^{br} was overridden several times in the corpus ([CORE, vNEXT4.25–4.33]) as successive candidate packets were found to be either non-positive or already internal. (ii) $\mathcal{C}_S < \mathcal{D}_S$ from the first front is **KILLED** by $N = 4736$ ($\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$). FAIL-POTENTIAL-CAPACITY-001 (iii) Unconditional CAB is **KILLED** by $N = 322$ and $N = 858$. (iv) The naive support expansion of all sources is **KILLED** by Hall compression $456 \rightarrow 17$ at $N = 51\,032$. FAIL-HALL-SUPPORT-001

Success criterion. A packet satisfying (a), (b), (c) above, with the positivity proved and the functional identity checked symbol by symbol. Any proof that silently changes γ between the

production and capacity sides is invalid; that substitution is the single most repeated error in the log.

Constraining log entries. FAIL-POTENTIAL-CAPACITY-001, FAIL-CAB-001, FAIL-CAB-INTERNAL-001, FAIL-HALL-SUPPORT-001, FAIL-ETA-GAMMA-IDENTIFY-001, FAIL-CHILD-C1-001.

9.5 L1 — actual layer realisation

L1

Statement. The layer part J_R of the canonical world is realised by *actual* distinct layers of the internal filtration, i.e. the support reuse counted by J_R corresponds to genuinely distinct BFS levels rather than to repeated incidences inside one level.

Why it matters. Every capacity argument that uses J_R implicitly assumes this. Without L1, J_R is a bookkeeping quantity with no dynamical meaning, and the Euler-defect route collapses — as the computation in §5.7 shows it does under the naive normalisations.

Already proved. $T_S = U_S J_S$ exactly (Theorem 3.19); the world is closed with a unique source, so the BFS filtration is well defined and canonical.

Remaining obligation. A lower bound on the number of layers in terms of $\log J_R$, or a counterexample. Note that the two false normalisations (N1) and (N2) of §5.7 are exactly failed attempts at L1 in disguise.

Obstructions. The measured data go the wrong way: 3 804 of 21 093 canonical worlds violate the (N2) inequality, so support reuse is systematically *larger* than the layer count can absorb.

Success criterion. Either a proof of $\nu(R) \geq c \log J_R$ for some $c > 0$, or an explicit family showing $\nu(R) = O(1)$ with J_R unbounded (which would kill L1 and, with it, every capacity route that passes through J). The second is probably easier and is listed as an experiment in Chapter 18.

Constraining log entries. FAIL-ETA-GAMMA-IDENTIFY-001 and the computational refutations of (N1), (N2) recorded in §5.7.

9.6 E1 — escape or coupling

E1

Statement. Under canonical failure, every internal base atom $\ell \in B_{\text{abs,int}}$ either has a GOOD edge reachable inside one of its first-front branches, or has an active parent outside R . (This is the *escape-or-reuse* alternative promoted to a theorem.)

Why it matters. E1 is the only frontier target with overwhelming computational support and a completely explicit statement: 61 452 instances with 0 orphans, plus an independent red team over 60 413 atoms with 0 orphans. If any target is true, this one is.

Already proved. Retention–escape (Theorem 5.24); minimum-factor escape (Corollary 5.25); the prime-power witness (Proposition 5.29); $B_{\text{abs,int}} = B_{\text{abs}}$ without vacuity [CORE, ARS-BABS-INTERNAL].

Remaining obligation. The disjunction itself, unconditionally. Both alternatives are provably necessary: (6142, 877, 3511) shows the second cannot be dropped.

Obstructions. (i) External reuse is not local (FAIL-LOCAL-REUSE-001), so the “active parent outside R ” cannot be found by bounded search. (ii) The external parent need not itself be a success root (FAIL-EXTERNAL-SUCCESS-001). (iii) Escape need not be directly adjacent (FAIL-DIRECT-ADJACENCY-001: at $N = 14198$ the atoms 19 and 167 escape at distances 8 and 7).

(iv) Charging multiplicity is observed to be 1 but is *not* proved bounded, and the constant- $C = 1$ charging scheme is **KILLED** (53 violations, maximum 9.42243, FAIL-CHILD-C1-001).

Success criterion. A proof of the disjunction, or a bounded-charging lemma (multiplicity $\leq C$ for an absolute C) which the corpus identifies as the missing ingredient.

Constraining log entries. FAIL-LOCAL-REUSE-001, FAIL-EXTERNAL-SUCCESS-001, FAIL-DIRECT-ADJACENCY-001, FAIL-CHILD-C1-001.

9.7 W1 — simultaneous clean witnesses

W1 — corrected

Statement. Several clean witnesses can be selected simultaneously at the *actual* layer scale of the world: a selection of parents, one per branch, simultaneously clean (no shared support, no double charging), covering the layers that really occur in R rather than only the polylogarithmic ones.

Correction to an earlier reading. **SCOPE-CORRECTED**. Earlier compressions listed G-CLEAN-POLYLOG-PARENT-SELECTION as W1's open target. It is *not* open: it is a **PROVED** theorem of the CRD Core (Theorem 5.15), giving exactly such a clean selection for every finite layer family with $\ell^k \leq (\log N)^A$, with global support load at most $\rho(\mathcal{L})$ — and load one when the base primes are distinct. See Warning 5.16.

Why it matters. Capacity arguments need many witnesses at once, and the selection problem is *solved* in the polylogarithmic range. What is missing is the range, not the selection.

Already proved. Theorem 5.15 (the selection itself); strict non-reuse (Lemma 3.13) for primes $\ell \geq h$; the global two-sheet collision budget (Theorem 5.34).

Remaining obligation. Extend the selection from $\ell^k \leq (\log N)^A$ to the actual layers of R . That is precisely the content of L1 (§9.5): **W1 is blocked on L1, not on a selection lemma**. If L1 is refuted — and §5.7 suggests it may be — then W1 is not merely open but misconceived, and the correct object is the polylogarithmic family alone.

Obstructions. Hall-type compression destroys naive selections outside the proved range: at $N = 51\,032$ the naive support expansion of all sources compresses $456 \rightarrow 17$ (FAIL-HALL-SUPPORT-001). This is not a technical nuisance — it says the bipartite incidence structure has small vertex covers, which is exactly what a simultaneous selection must avoid, and it is why Theorem 5.15 needs Siegel–Walfisz and a lower-bound sieve rather than a combinatorial argument.

Success criterion. Either a clean selection at the actual layer scale, or a proof that the polylogarithmic range is the maximum possible — which would close W1 negatively and remove it from the frontier.

Constraining log entries. FAIL-HALL-SUPPORT-001, FAIL-CHILD-C1-001.

9.8 K1 — source-sensitive capacity

K1

Statement.

$$K_{\times}(R, \mathcal{P}) + K(\mathcal{P}) \leq \mathcal{C}_N(R)$$

where $K_{\times}$ is the cross term between the world R and the selected parent set $\mathcal{P}$, $K(\mathcal{P})$ the internal term of $\mathcal{P}$, and $\mathcal{C}_N(R)$ the capacity of the world — all three measured in the *same* functional as $P1^{\text{br}}$.

Why it matters. K1 is the quantitative closure: with $P1^{\text{br}}$ above and K1 below, the two bounds cross and canonical failure is excluded.

Already proved. The capacity side $\mathcal{C}_N(R)$ is well defined and computable; the world's ledger identities (IL1–IL6) hold on Q_{smooth} .

Remaining obligation. *Exactly the same defect as TD.* K1 needs the cross term and the capacity to be expressed in one functional. Every recorded attempt produced two functionals with a comparison that holds on average and fails pointwise. The corpus's vertical-origin overrides ([CORE, vNEXT4.8–4.13]) end at the target **C-NATURAL-SCALE-WEIGHTED-CONSUMER**: a consumer of the packet weighted by a natural scale, which has never been produced.

Obstructions. (i) The same-functional obstruction, shared with TD and P1^{br} — these three are not independent problems. (ii) $N = 4736$ kills the first-front form of the capacity inequality. (iii) $N = 322$ and $N = 858$ kill the unconditional CAB form.

Success criterion. The displayed inequality with all three terms written in one explicit functional and proved unconditionally under canonical failure. Equivalently: solve TD and K1 together, since a solution to one in the correct functional very likely gives the other.

Constraining log entries. FAIL-POTENTIAL-CAPACITY-001, FAIL-CAB-001, FAIL-CAB-INTERNAL-001, FAIL-ETA-GAMMA-IDENTIFY-001.

9.9 X1 — canonical failure exclusion

X1

Statement. For every even N in the non-diagonal case, the canonical traversal from p_0 terminates in SUCCESS. Equivalently: $\text{PF}(N - p_0) \not\subseteq B_N^{\text{low}}$ whenever $N - p_0$ is composite (Theorem 3.29).

Why it matters. X1 is the terminal node. It implies binary Goldbach. It is the central hypothesis of the whole project, written here as (**H**- p_0).

Already proved. Nothing towards X1 itself. The fixed-point reformulation (Theorem 3.29) is exact but not progress.

Remaining obligation. Everything above, plus the passage from “the capacity inequality fails” to “the traversal succeeds”.

Obstructions. The architecture wall (§8.2) — no finite local certificate can do it; the parity barrier (§13.4) — no purely sieve-theoretic route reaches a prime complement; and the fact that X1 is *strictly stronger* than Goldbach for each N , so any proof strategy that would also prove Goldbach must be at least as hard as Goldbach.

Success criterion. A proof. Nothing weaker counts; in particular no finite scan, however large, changes the status of X1.

Computational status. **VERIFIED COMPUTATION**: no canonical failure for even $N \leq 10^7$ (this project, exhaustive). Independent Goldbach verification reaches $4 \cdot 10^{18}$ [12] but says nothing about X1.

9.10 RPD — repeated prime-power redundancy

RPD

Statement. The repeated external prime-power layer redundancy is bounded:

$$D_{\text{high}}(H_Q) \leq C \log \log N \cdot \sum_{\text{layers}} (\kappa - 1)_+ + o(M),$$

equivalently $D_{\text{high}}(H_Q) \leq O(\log \log N) \cdot \eta_{\text{high}} + o(M)$, where M is the total mass and η_{high} the high-range excess valuation.

Why it matters. If RPD is true then $\Delta_J \gg M$: the layer defect dominates the mass, which is exactly the quantitative separation that the capacity route needs. RPD is the one open target that would supply the *margin* rather than merely the structure.

Already proved. $T_S = U_S J_S$; the high/low decomposition of the valuation mass; the prime-power witness (Proposition 5.29).

Remaining obligation. The $\log \log N$ bound. The natural approach is a divisor-type bound on how often a fixed prime power can recur across layers, which is where $\log \log N$ would come from.

Obstructions. (i) The bound is asserted for the *external* redundancy; the internal part is already controlled and is not the difficulty. (ii) The $o(M)$ term hides the interaction between layers, and no proof technique in the corpus separates layers without losing a factor of the size that RPD is supposed to save.

Success criterion. A proof of the displayed inequality with an explicit C , or a family of N showing D_{high} can exceed $\eta_{\text{high}} \log \log N$ by an unbounded factor.

Provenance note. RPD is recorded in the corpus as a v4.45 open target inside a commented archive block of the CRD failure log; it is reproduced here because it is the only surviving open target that carries an explicit quantitative margin. Its exact original phrasing should be checked against the source before it is used as a hypothesis.

9.11 Open targets specific to p_0 alone

These are stated *without* reference to a Goldbach proof and each is a self-contained problem.

C-SAMEFUNC-BRIDGE — global same-functional margin bridge

Statement. There is a single functional γ in which both the production lower bound (TD, $P1^{\text{br}}$) and the capacity upper bound (K1) can be stated, with a positive margin between them.

Why it matters. This is the abstract form of the obstruction shared by TD, $P1^{\text{br}}$ and K1. Solving it solves all three; failing to solve it blocks all three. Anyone attacking one of them individually should first decide whether they are really attacking this.

Already proved. Nothing. Several *pairs* of functionals are known in which one side is provable; no pair is known in which both are.

Remaining obligation / obstruction / success criterion. As for TD; see §9.3.

Constraining log entries. FAIL-ETA-GAMMA-IDENTIFY-001 and every entry filed under capacity.

C-GENEALFREE-DYN — boundary to dynamic escape

Statement. Does the absence of any genealogical boundary defect ($\kappa_{\text{gen}} = 0$) force the whole C0b-RD obligation into *actual* dynamical channels?

Why it matters. This is the single missing implication between Chapter 6 (exact boundary structure) and any dynamical statement. The entire “why p_0 works” question reduces to it.

Already proved. The boundary side completely (Theorems 6.2–6.6); the obligation side completely (Theorem 8.6). The implication between them: nothing.

Minimum acceptable form. A theorem whose conclusion *actually fails*, or acquires an explicit leakage term, when the defect rank is one. Anything that also holds at rank one is not progress — it is a class-B result in disguise.

Obstruction. Warning 6.7(2): the statement $\kappa_{\text{gen}}(r) = 0$ may be informationally equivalent to “ r is the first upper prime”. If so, no theorem can extract more from it than from the ordering of the primes, and the target should be restated.

Constraining log entries.

FAIL-P0-ZEROHOLE-ESSENTIAL-001,

FAIL-P0-INFORMATION-SELECTOR-001.

C-POP1-SEP — one-defect dynamic separation

Statement. Prove a dynamical difference between p_0 and p_1 that is caused by the single embedded channel of the earlier root, not by aggregate size.

Why it matters. This is the sharpest available test of C-GENEALFREE-DYN and the one with the best data.

Already proved. The categorical separation on the boundary (κ_{gen} : 0 versus 0.9041 mean, maximum 1) and the aggregate near-identity of every other observable (§6.5.2).

Remaining obligation. Isolate causality. The aggregate observables differ by 6–24%, which is consistent with pure size effects; the defect rank differs categorically but has negligible logarithmic mass.

Obstruction. FAIL-P0-ZEROHOLE-ESSENTIAL-001: the mass of the hole is $O(\log N)$, so any argument that routes the separation through mass is dead before it starts.

Success criterion. A statement true for p_0 and false for p_1 whose proof uses the hole *structurally* (as absence of a multiplicative carrier, [CORE, V71.44]) and not quantitatively.

9.12 Smaller open questions about p_0

Each of the following is a self-contained problem that does not mention Goldbach.

Q1. Does p_0 have dynamical consequences? **OPEN**. The signature $(\kappa^+, \kappa^-) = (0, \rho_0)$ is the first boundary invariant that varies nontrivially between canonical worlds. Does $\rho_0 \geq 1$ (Type III, where a guaranteed active reflected row exists) correlate with, or imply, anything about the traversal? Preliminary data: immediate success occurs in 18.4% of Type III cases and is *impossible* in Type I — but that is a consequence of the definition, not of dynamics.

Q2. Does the escape margin grow? **OPEN**. Is $G_{\text{esc}}(N) \rightarrow \infty$? Data (§6.7): the per-dyadic-block minimum runs 1, 1, 1, 1, 1, 1, 2, 2, 3, 3, 3 up to $2 \cdot 10^5$. A theorem “ $G_{\text{esc}}(N) \geq f(N)$ for increasing f ” would be strictly stronger than the existence of a Goldbach pair and is, as far as is known, untouched.

Q3. Is there a missing state variable? **OPEN**. Four pairs of structural twins survived enrichment by all global-source information; they are separated only by the *internal witness signature* (localisation, nearest escape distance in the same branch, external-reuse flag) [F3]. Is there a scalar invariant, or is the system state intrinsically a profile?

Q4. A second canonical anchor? **OPEN**. Let $p^* = \max\{p \in \mathbb{P} : p < N\}$ and $g = N - p^*$. Then $g \leq N - p \leq m_0$ for upper sources, $p^* - p_0 = m_0 - g$, and $\gcd(m_0, g) = \gcd(m_0, p^* - p_0) = \gcd(g, p^* - p_0)$ **PROVED** [BIB]. Does p^* have its own rigidity? Evidence against: p^* finances on average only 25.07% of the B_{abs} atoms against 81.44% for the whole ensemble of upper sources; priority assessed as low [D4, q. 11, 21].

Q5. Is p_0 exceptional outside this one graph? **OPEN**. An analogous selector appears in the Erdős–Straus model and in the Schinzel model [BIB]. Does “the least element above the midpoint, chosen unconditionally” have the same reflection-completeness property there? The definition of the reflected block transfers verbatim, so the question is well posed.

Q6. Diagonal closure. **OPEN** (Scope note 3.17). What is the right formulation of the genealogical layer covering $x \in \mathbb{P}$? Replacing $[x, r)$ by (x, r) makes the theorem unconditionally true, but then p_0 is no longer uniquely characterised. Is there a simultaneous formulation?

Q7. Invariants of the boundary filtration. **OPEN**. The filtration $\mathfrak{B}_0 \subset \mathfrak{B}_1 \subset \dots$ (Theorem 6.8) with the coupling-matrix embedding (Corollary 6.9) is a purely combinatorial structure independent of Goldbach. Does the sequence $(M^{(k)})_k$ have invariants — rank, spectrum, zero pattern — that do not reduce to the gap sequence $g_1, g_2, \dots$?

Q8. Prove the active-mass Euler bound. **OPEN**. Normalisation (N3) of §5.7, $\sum_q d_R(q) \geq |R| + 1$, held in all 21 093 canonical worlds tested. It is the only nontrivial surviving form; a proof would be small but real, and would repair the part of the corpus that used the false normalisations.

Chapter 10

Post-v4.81 Research Continuation: 5–7 September 2026

Historical binding waypoint through 2026-09-07C. This chapter integrates the lossless 2026-09-05–06B archive and the 2026-09-07C checkpoint. Its theorems and scope corrections remain part of the corpus, but its frontier is superseded by the binding 2026-09-07H update in Chapter 11. Binary Goldbach and universal canonical p_0 -reachability remain **OPEN**. The exact 6420-line raw checkpoint is preserved in Appendix B.

10.1 Merge provenance and source dominance

The merge used three supplied sources: the v1.0 master, the standalone lossless checkpoint archive through 2026-09-06B, and the final 2026-09-07C checkpoint. The archive’s 5108-line raw payload is an *exact prefix* of the final 6420-line checkpoint, so duplicating both raw transcripts would add no information. The final checkpoint is therefore the canonical raw payload; the earlier archive contributes its provenance discipline and strict prefix genealogy.

Master source — 13,744 lines

goldbach_master_for_future_ai(5)(1).tex

SHA-256: 685dd184 103a54d7 0d32d9f5 0b440520 9c287c39 c1a41288 8e4f717a 47fcb197

Lossless archive — 5,346 lines

goldbach_checkpoint_archive_2026-09-05_to_2026-09-06B(1).tex

SHA-256: d8a4cfe0 d9db56eb e7c4bf45 eed1becc 01f5cad6 1d8784e9 51a0661b 96ad2d54

Final research checkpoint — 6,420 lines

goldbach_research_checkpoint_2026-09-07(3).txt

SHA-256: 49d1ce74 1f76bec6 297e46c2 546c562c 7089e5b9 e0be31c2 a3d18e67 1a814392

10.2 Binding FBTU scope correction

The audit AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905 is binding. Prescribed CRT constraints force selected edges; they do not force the entire factorisation of an expanded row. At

$$N = 92, \quad p_0 = 47, \quad 92 - 47 = 3^2 \cdot 5,$$

the selected walk $47 \rightarrow 5 \rightarrow 29 \rightarrow 7$ and the BAD cycle $5 \rightarrow 29 \rightarrow 7 \rightarrow 5$ are real, while $92 - 3 = 89$ is prime, so the complete BFS succeeds at depth 1. This kills the blanket implication “programmable selected BAD transcript $\Rightarrow$ programmable complete BAD neighbourhood”. The repaired theorem CRT-TRANSCRIPT-EXACT-SCOPE- 20260905 survives.

10.3 Closed-support counting, height and forced envelope escape

The September 5 continuation produced a coherent fixed-/bounded-support layer. For a fixed prime alphabet S of size s , coprime S -smooth solutions of a fixed affine equation are bounded by 2^{9s+16} via the verified Beukers–Schlickewei input. This yields both COMPLETE-H1-TWO-ROW-FINITENESS-20260905 and CLOSED-ALPHABET-COUNT-20260905: one fixed finite alphabet cannot support infinitely many complete closed active BAD worlds.

The Matveev route was then repaired to remove the old branching/pure-row hypothesis. If C is any nonempty closed active BAD set, $s = |C|$ and $M = \max C$, then

$$\log \frac{N-M}{M} < 1.4 \cdot 30^{s+3} s^{9/2} \prod_{q \in C} \log q \left(1 + \log \frac{\log N}{\log 3} \right).$$

Consequences retained as proved are the explicit bounded-core threshold and the pointwise lower growth

$$M \geq (1 - o(1)) \frac{(\log \log N)(\log \log \log N)}{\log \log \log \log N}$$

along any sequence carrying a closed active BAD set. These are necessary conditions, not contradictions: moving support remains possible.

In the canonical $h = 1$ family, bounded first support also forces complete-row escape from a fixed envelope: beyond the explicit threshold $F_2(y)$ every first child row has a prime factor $> y$, and the large factors arising from distinct first children are pairwise disjoint. The inference that this automatically iterates was explicitly killed by ENVELOPE-ESCAPE-AUTOMATIC-ITERATION-20260905.

10.4 Shallow asymptotics and same- N no-gos

The September 6/06B continuation establishes several asymptotic traversal constraints, all with novelty still unaudited externally.

1. COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS-20260906: for almost every prime $P \in [X, 2X]$, the canonical $h = 1$ start has complete BADness through radius two; the exceptional relative proportion has an explicit decaying bound.
2. SAME-N-GROWING-ROOTS-COMPLETE-RADIUS-TWO-20260906B: the same complete radius-two BADness holds simultaneously for a growing prefix of nearby same- N roots. Thus shallow BADness is not a p_0 -only signature.
3. SAME-N-GROWING-DEPTH-DUAL-PRODUCT-BARRIER-20260906B: for fixed $c + \kappa < 1$, almost every prime start and every same- N root $k \leq (\log X)^\kappa$ has no short successful simple path whose label product or cofactor product is $\leq X^{1-\varepsilon}$, up to depth $c \log \log X / \log \log \log X$.
4. Exact non-dominance survives hostile checking: at $N = 2180$, p_1 is directly GOOD while p_0 is BAD through radius two; at $N = 2024$, p_0 succeeds earlier while p_1 is BAD through radius two.

The product barriers quantify over *successful paths*. They are vacuous inside a completely closed failed core, so the attempted inference from those barriers to closed-core height was correctly killed by SUCCESS-PRODUCT-BARRIER-TO-CLOSED-CORE-HEIGHT-20260906B.

10.5 Exact $h = 1$ conjugacy, cycle realisation and canonical embedding

Prime-lift full-BFS conjugacy. Let $N = dx$, let $m = x - 1 \geq 2$, and suppose $R = (d - 1)x + 1$ is prime. In the composite- m case the complete BFS from the inactive prime divisor d and from

R have identical ordered transcripts after replacing the initial root. For $d = 2$ this gives the exact all-generation normal form

$$N = 2(P - 1), \quad P = p_0(N) \quad \implies \quad \text{BFS from } P \equiv \text{BFS from } 2$$

for every non-direct $h = 1$ case. The tempting general implication “prime lift $\Rightarrow$ success” is false: $(N, d, R) = (1862, 7, 1597)$ gives a complete failed 22-test conjugate pair.

Canonical first-support cycles. For every prescribed ordered list of distinct odd primes there is a fixed gap h and infinitely many canonical worlds whose complete first support contains a directed BAD cycle through those primes. Extra factors and GOOD branches are uncontrolled. Hence canonical first-support acyclicity and any uniform bound on cycle length are false outside the special $h = 1$ situation.

Fixed-gap canonical valuation embedding. Take any actual upper BAD root (N_0, R_0) with composite first complement and gap h_0 . Its complete induced first-support graph on $S = \text{PF}(N_0 - R_0)$, including root valuations and all internal row valuations, embeds exactly into infinitely many canonical BAD-root first-support graphs with the same numerical gap h_0 . Additional support outside S is forced and remains uncontrolled. Therefore every hereditary property of this labelled induced graph that holds for all canonical BAD roots must also hold for every upper BAD root. This is the rigorous replacement for the old overbroad finite-local wall: finite hereditary first-support graph/valuation invariants cannot universally separate p_0 , while complete support and later reachability remain outside the theorem.

10.6 Low-gap complete-support restrictions

Let $r = N/2 + h$ be an upper prime root, $m = N - r$ composite and $q = \min \text{PF}(m)$ with $0 < h < q$.

1. If no GOOD vertex occurs through depth 2, then at least three distinct nonroot primes have been discovered by depth 3 (LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE-20260907).
2. If q participates in a reciprocal BAD two-cycle $q \rightarrow b \rightarrow q$, then

$$b = 2h + (2j + 1)q \quad (j \geq 1), \quad b \geq 3q + 2h,$$

and this numerical bound is sharp for both canonical and noncanonical roots.

3. If the complete row is a prime square, $N - q = b^2$, then $q \nmid N - b$ and $(b - 1)^2 < N - b < b^2$. In the $h = 1$ singleton-square subcase, necessarily u is odd, $u \geq 3$, and $q \equiv 23 \pmod{24}$ when $m = q^u$.
4. If $m = q^u$, $N - q = b^v$ is a complete prime-power row and the reciprocal return occurs, then

$$3 \leq v \leq u - 1.$$

Thus $u = 2, 3$ exclude this immediate monomial return.

The attempted universal FIVE-label strengthening was refuted exactly at $N = 44670$, root rank 12, where the depth-three nonroot labels are precisely $\{149, 211, 23, 1933\}$. The proved lower bound THREE survives. The final exact reduction LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT-20260907C says that a genuine three-label case must be either (A) a complete closed active BAD world on exactly three primes, or (B) an explicit prime-power system

$$m = q^u, \quad N - q = b^v, \quad N - b = q^a c^w,$$

with the restrictions recorded in the raw theorem. In the singleton square branch the remaining equations can be written

$$2q^u + 2h = q + b^2 = b + c^w, \quad w \geq 3 \text{ odd};$$

for $h = 1$, also u is odd and $q \equiv 23 \pmod{24}$.

10.7 Negative structural diagnostics

The multiplicative subgroup generated modulo N by a closed BAD support necessarily contains -1 , and contains any upper root whose complete first support lies in that closed set. As a general explanation of p_0 this route fails: at five of the six historical hosts the maximal closed low BAD support generates the entire unit group. At $N = 2200$ the generated subgroup is proper, but membership is still not sufficient for failure. This kills simple multiplicative-character separation as a general p_0 mechanism while leaving more structured valuation/additive residue arguments open.

10.8 07C frontier waypoint (superseded by 07H)

Global exact question: the $h = 1$ closed-support obstruction

Can complete factor search from 2 fail when

$$N = 2(P - 1), \quad P \geq 5 \text{ prime}, \quad P - 2 \text{ composite?}$$

Equivalently, can there exist a finite set C of odd active primes below $2(P - 1)/3$ such that $\text{PF}(P - 2) \subseteq C$ and, for every $q \in C$, $2(P - 1) - q$ is composite with its *entire* prime support contained in C ? By the full-BFS conjugacy this is exactly the unresolved non-direct canonical $h = 1$ failure question. Selected cycles, finite internal valuations and support-height lower bounds do not answer it.

Local exact question: three-label obstruction

Can either branch of LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT-20260907C actually occur? The singleton-square branch reduces to the displayed exponential/prime-power system above. A proof of impossibility would strengthen the current THREE-label theorem to FOUR in the stated low-gap domain; an exact example would kill that strengthening cleanly.

Asymptotic radius-three question

The previous 2026-09-06B frontier remains open: control the canonical depth-3 successful-path system in the region where *both* the label product and the cofactor product exceed $X^{9/10}$. A suitable $o(X/\log X)$ count would give almost-everywhere complete radius-three BADness in the $h = 1$ family, not a Goldbach proof.

10.9 p_0 status at the 07C waypoint

The empirical eventual reliability of p_0 remains unexplained. Several simple candidate explanations are now rigorously unavailable: shallow BADness and product barriers are shared by nearby roots; finite first-support acyclicity can reverse between p_0 and p_1 ; arbitrary first-support BAD cycles occur in canonical worlds; and hereditary first-support label/valuation invariants cannot separate all canonical BAD roots from all upper BAD roots. What may still distinguish p_0 must use information outside that hereditary local class — complete support, leakage/external cofactors, absolute root location/size, or truly multigeneration reachability.

Chapter 11

Rank/Mirror and Same- N

Continuation: 7 September 2026 D–H

Binding override after 2026-09-07H. This chapter integrates the standalone rank-spectrum checkpoint (07E), the rank-mirror checkpoint (07F), and the independently audited 07G/07H continuation. When E/F and G disagree, G is binding; in particular the canonical mirror range has the isolated boundary value $m = 2$ at $N = 4, p_0 = 2$, and the Scott–Bozek two-vertex classification is not used as a proved dependency beyond the parity-safe scope already recorded. Novelty of the new applications is *not externally audited*. Binary Goldbach, universal p_0 success, and universal composite-success of the currently surviving noncanonical ranks all remain **OPEN**.

11.1 Source dominance and merge provenance

The source order for this update is

$$07E, 07F \longrightarrow 07G \text{ audit/correction} \longrightarrow 07H \text{ continuation.}$$

The first 6420 lines of the supplied `goldbach_research_checkpoint_2026-09-07(9).txt` are byte-for-byte identical, line-for-line, to the raw checkpoint already preserved in Appendix B. Therefore this version preserves the later source losslessly by appending only the continuation beginning at source line 6421, rather than duplicating the identical prefix.

Source	Lines	SHA-256
<code>rank_spectrum_checkpoint_2026-09-07E(1).txt</code>	2180	950209fe93257b6ec4acde7750e9f0957c42f765039a3e8a3b9a
<code>rank_mirror_checkpoint_2026-09-07F(1).txt</code>	106	5b71165858ef6d393b26eb6780de177311f9e8f508c37a7bf2c8
<code>goldbach_research_checkpoint_2026-09-07(9).txt</code>	7027	9db047c79d70c66fbfa4e364f0d9a1dfc3ec73fe104dc2dfc027
07D–H continuation delta, source lines 6421–7687	1267	36b78243f3f4bd7adf0078a64aa0093ae0ca6133136a061ca05e

The exact E and F sources appear in Appendices C and D; the exact nonduplicated D–H continuation appears in Appendix E.

11.2 Run D: low-gap complete-support strengthening

Run D extends the 07C low-gap branch but does not become the global frontier.

Theorem 11.1 (No proper power in $2 \cdot 3^u - 1$). **PROVED**. *For every integer $u \geq 1$, the integer $2 \cdot 3^u - 1$ is not a proper integer power A^k with $A \geq 2$ and $k \geq 2$.*

The proof is the elementary factorisation argument recorded in the run-D source and in the delta registry §11.8. It excludes proper powers, not compositeness.

Theorem 11.2 (Low-gap singleton upward four-label alternative). **PROVED**. *Let*

$$r = q^u + 2h \text{ prime}, \quad u \geq 2, \quad q \text{ odd prime}, \quad 0 < h < q, \quad N = 2q^u + 2h.$$

Assume $N - q$ is composite, has at least two distinct prime factors, and every prime factor of $N - q$ exceeds q . Then either a GOOD vertex is reached by depth 2 from r , or at least four distinct nonroot prime labels occur by depth 3 in the complete stopped search.

The complete-factor hypothesis is essential. In the special canonical family $q = 3, h = 1$, Theorem 11.1 supplies it after the depth-1 GOOD case is removed. Hence for $P = 3^u + 2$ prime and $N = 2(P - 1)$, the canonical search either succeeds by depth 2 or has at least four nonroot labels by depth 3. The explicit source witness $u = 14, P = 4782971, N = 9565940$ shows the BAD-through-depth-2 branch is nonvacuous.

The companion run-D monomial-return theorem proves that a monomial return

$$N - q = b^v, \quad q \mid N - b$$

in the same low-gap setup satisfies

$$d = \text{ord}_q(b) = \text{ord}_q(2h) \geq 3, \quad d \mid (v - 1), \quad d \mid (q - 1),$$

and consequently

$$4 \leq v \leq u - 1, \quad u \geq 5.$$

Later D carryover in the 07G source records stronger $h = 1$ filters for special parity/exponent branches; these are preserved verbatim in the raw delta and are not promoted here into a global closure theorem.

11.3 All-upper-root fixed point and exact rank spectrum

For this chapter write

$$B_N := B_N^{\text{low}},$$

the largest active closed BAD subset below $N/3$ from Definition 3.28. The canonical fixed-point theorem extends without difficulty to every upper root.

Theorem 11.3 (All-upper-root fixed point). **PROVED**. *Let N be even and let $r \geq N/2$ be prime with composite mirror $m = N - r > 1$. Then*

$$\boxed{r \text{ fails} \iff \text{PF}(m) \subseteq B_N}.$$

Indeed every child of an odd composite row lies below $N/3$; a failing reachable set therefore feeds a closed low BAD set, while closure of B_N gives the converse.

If $p_k(N)$ is the zero-based k th prime at or above $N/2$ and $m = N - p_k(N)$, then

$$k = \pi(N - m) - \pi((N/2)^-) - 1,$$

and on the composite-mirror domain

$$\boxed{p_k(N) \text{ fails} \iff \text{rad}(m) \mid R_N}, \quad R_N := \prod_{q \in B_N} q.$$

Thus the failure-rank spectrum is exactly the prime-counting image of the B_N -smooth, prime-backed mirrors. This is the correct exact replacement for a hoped-for rank-only congruence law.

11.3.1 Same (k, m) , opposite verdicts

The ambient dependence is not cosmetic. Exact certificates give:

$$(k, m) = (7, 27) : \quad N = 116 \text{ success}, \quad N = 128 \text{ failure},$$

$$(k, m) = (20, 799) : \quad N = 1862 \text{ failure}, \quad N = 1886 \text{ success},$$

and

$$(k, m) = (48, 729) : \quad N = 2200 \text{ failure}, \quad N = 2180, 2210 \text{ success}.$$

Therefore no verdict function of (k, m) alone exists, even when the complete integer factorisation of m is supplied. The indispensable variable is the ambient N -dependent basin.

11.3.2 Finite rank spectrum on the six reconstructed hosts

The independent 07G reconstruction found 301 failing upper roots on

$$128, 1718, 1862, 1928, 2200, 6142,$$

with 214 distinct composite mirrors and 188 distinct failed ranks. Within $0 \leq k \leq 118$, the ranks avoiding failure on all six host spectra are exactly

$$\boxed{\{0, 10, 14, 26, 28, 30, 62, 63, 71, 74\}}.$$

This is a finite-host statement. It is not a theorem that the nine noncanonical indices are universally composite-safe.

11.4 Fixed-rank mirror universality and effective finite fibres

Theorem 11.4 (Fixed-rank mirror surjectivity, audited). **PROVED**. *Fix an odd integer $m \geq 3$ and $k \geq 0$. There exists an odd prime $P > m$ such that, for $N = P + m$,*

$$P = p_k(N), \quad N - p_k(N) = m.$$

For $m = 1$ the same holds exactly for $k \geq 1$; $k = 0$ is impossible.

Define

$$\kappa_m(P) = \#\{q \in \mathbb{P} : (P + m)/2 \leq q < P\}.$$

Along consecutive primes above m , κ_m starts at 0, has upward jumps of size at most 1, and tends to infinity by the prime number theorem. Hence it assumes every nonnegative integer. For $m = 1$ it starts at 1.

Scope note 11.5 (Audited boundary correction). If all even $N \geq 4$ are admitted, then $N = 4, p_0 = 2$ gives mirror 2. Thus

$$\{N - p_0(N) : p_0(N) < N\} = \{2\} \cup \{3, 5, 7, \dots\},$$

whereas for every fixed $k \geq 1$ the range is $\{1, 3, 5, 7, \dots\}$. The odd-only formula for p_0 is correct after restricting to odd roots or $N \geq 6$. All composite-mirror conclusions are unchanged.

Mirror surjectivity immediately rules out every proposed protection that depends only on the value, support, radical, Ω , largest factor, or complete first-row factorisation of $N - p_k$ at a fixed rank.

The same theorem has a complementary finiteness statement.

Theorem 11.6 (Effective finite rank–mirror fibres). **PROVED**. *Let $m \geq 1$ be odd, $P > m$ an odd prime, $N = P + m$, and $P = p_k(N)$. Put*

$$c(m) = \left\lceil \frac{m-1}{4} \right\rceil, \quad n = k + c(m) + 2.$$

Then, writing R_n for the n th Ramanujan prime,

$$P < R_n < 4n \log(4n), \quad N < m + 4n \log(4n).$$

Hence each fixed (k, m) fibre is finite and effectively enumerable.

The external numerical bound is Sondow’s $R_n < 4n \log(4n)$ [14]; the rank/mirror reduction is elementary. For $m = 1$ it follows that every $k \geq 1$ has a nonempty but finite set of unit-row inputs, while p_0 has none.

Theorem 11.7 (Upper-corner last-realisation principle). **PROVED**. *Fix odd $M \geq 3$ and $K \geq 0$. Among all odd $1 \leq m \leq M$ and ranks $k \leq K$, the largest realised $N = P + m$ is attained at the upper corner $m = M, k = K$.*

For $M = 3053, K = 74$ the exact endpoint is

$$P = 4271, \quad N = 7324.$$

Consequently increasing an arbitrary N cutoff cannot add new cases for these already bounded mirrors and ranks.

11.4.1 Globally exhausted bounded-mirror experiments

The finite-fibre theorem upgrades two previously cutoff-labelled scans into complete all- N statements for their fixed domains.

First, the 214 dangerous mirrors extracted from the six reconstructed failure hosts, transplanted into

$$k \in \{10, 14, 26, 28, 30, 62, 63, 71, 74\},$$

have exactly 4035 realisations over *all* N ; none fails, while 147 full worlds have exactly one GOOD escape edge. The control ranks 7, 8, 9, 11, 20, 48 recover genuine failures, so the harness is not vacuous.

Second, enumerating *every* odd composite mirror $m \leq 3053$ at

$$k \in \{0, 10, 14, 26, 28, 30, 62, 63, 71, 74\}$$

gives exactly 22130 realised composite cases over all N , with zero failures. This is a rigorous exhaustive statement for a bounded mirror box, not evidence that larger mirrors are harmless.

Warning 11.8 (Do not promote the survivor ranks). The set above was discovered inside the same finite failure landscape. Neither the 4035 transplant exhaustion nor the 22130 bounded-mirror census proves universal composite-success of any noncanonical rank. A new failure may require a new mirror integer and a new moving basin.

11.5 Run H: same- N finite views versus complete support

Run H attacks a stronger possibility: perhaps p_0 can be recognised by finite local data *at the same ambient* N . The answer is negative for any fixed finite valuation projection.

Theorem 11.9 (Same- N prefix valuation realisation). **PROVED**. Fix $K \geq 0$, a finite set T of odd primes, and arbitrary prescribed nonnegative integers e_ℓ for $\ell \in T$. There are infinitely many even N such that, for the zero-based primes p_j at or above $N/2$,

$$p_0 = N/2 + 1, \quad v_\ell(N - p_j) = e_\ell \quad (\ell \in T, 0 \leq j \leq K).$$

All $K + 1$ complements may additionally be required composite. Along an infinite subfamily there are fixed offsets

$$0 = d_0 < d_1 < \cdots < d_K, \quad p_j = p_0 + d_j.$$

The construction chooses one reduced congruence class by CRT and invokes Freiberg's theorem on bounded-diameter strings of consecutive primes in a prescribed reduced residue class [13]. It controls the requested valuations but deliberately leaves the remaining cofactors unbounded.

Corollary 11.10 (Fixed-label projection no-go). For any fixed finite label set T and fixed K , infinitely many same- N worlds have identical root-to- T multiplicities and identical retained rows/GOOD-BAD flags on T after relabelling the root. Therefore a fixed prime-label projection cannot identify p_0 among a fixed early prefix.

This is *not* a theorem about a complete bounded-depth ball: new prime labels can already appear on the first full BFS layer and tend to infinity.

The complementary theorem controls complete factorizations in a genuinely fixed alphabet.

Theorem 11.11 (Early-rank fixed-support separation). **PROVED**. Let S be a fixed nonempty finite set of odd primes, $s = |S|$, and put

$$A(S) = 1.4 \cdot 30^{s+3} s^{9/2} \prod_{q \in S} \log q, \quad c(S) = \frac{(\log 3/e)^{A(S)}}{6}.$$

For all sufficiently large even N , if

$$N/2 \leq p_i < p_j \leq 2N/3$$

and both $N - p_i$ and $N - p_j$ are completely S -smooth, then

$$j - i > c(S) \frac{N^{19/40}}{(\log N)^{A(S)}}.$$

Consequently, for every fixed $0 < \varepsilon < 19/40$, sufficiently large N have at most one S -smooth complement among

$$N - p_0, \dots, N - p_{\lfloor N^{19/40-\varepsilon} \rfloor}.$$

The proof combines the already audited rational Matveev estimate with the Baker–Harman–Pintz exponent 0.525 [15]; the exponent $19/40 = 1 - 0.525$ is exactly the resulting rank scale. No BAD closure is assumed.

Corollary 11.12 (Finite-view matching with cofactor escape). Fix a finite valuation prescription and $K \geq 1$. The same- N theorem matches that finite view across $p_0, \dots, p_K$ infinitely often. Yet for any additional fixed finite prime alphabet S , after finitely many exceptions at least K of those $K + 1$ complete complements contain a prime outside S .

The quantifier is crucial:

$$\boxed{\text{fixed } S \neq S = B_N \text{ chosen separately for each } N.}$$

No theorem in this chapter permits the latter substitution.

A useful elementary overlap consumer also survives: if upper roots $r < s$ share a prime factor q in their complements, then

$$2q \mid (s - r), \quad h_s \geq h_r + 2q.$$

Thus sufficiently low-gap later roots have initial support disjoint from every earlier upper root. Certified failed examples show that such disjointness still does not force success.

11.6 Binding current frontier after 07H

Primary target: selector versus moving basin

Find a necessary condition on a complete closed BAD basin B_N that interacts with the *actual position* of the first upper prime (or, more generally, with the rank-realisation curve). For a fixed rank k the exact failure question is whether there exist a composite m and prime $P = N - m = p_k(N)$ such that

$$\text{rad}(m) \mid R_N, \quad R_N = \prod_{q \in B_N} q.$$

For $k = 20$ such an intersection exists at $(N, m) = (1862, 799)$. For $k = 10, 14, 26, 28, 30, 62, 63, 71, 74$ none is known.

Uniformise the fixed- S geometry or prove why it cannot be

The fixed-support Matveev/BHP theorem is pointwise strong but cannot accept $S = B_N$ when B_N changes with N . A useful advance must control the size, height, growth, or internal spacing of the *moving* alphabet well enough that the fixed- S constants do not become vacuous, or replace them by an inequality uniform in an admissible family of basins.

Adversarial alternative: construct a genuinely new failure host

The bounded mirror fibres through $m \leq 3053$ are exhausted. More scanning of the same mirrors cannot add information. A computational attack should instead search for a new complete closed BAD basin / EAC host and then inspect its entire upper-root rank spectrum. A new host would immediately stress the current survivor ranks with genuinely new ambient data.

The 07D low-gap Diophantine branch remains a legitimate secondary target, but the H results show that repeated fitting of fixed residues, fixed valuations, or fixed alphabets cannot by itself explain p_0 .

11.7 p_0 status after 07H

The project now has a sharper separation between what is genuinely canonical and what is shared by nearby ranks.

- p_0 remains the least upper prime, the reflection-complete boundary root, and the unique fixed rank with no unit-row input.
- It is *not* characterised by the possible odd values or complete factorisations of its first mirror: every fixed rank realises every odd $m \geq 3$ somewhere.
- It is *not* characterised by any prescribed finite list of small prime valuations, even when comparison is made at the same N : such data can be synchronised across any fixed prefix.
- Fixed finite full-support alphabets are themselves too rigid to describe many early roots simultaneously for large N .
- What remains potentially special is therefore global: how the least-upper-prime choice sits relative to the full, moving basin B_N and how its complete cofactors enter or escape that basin over multiple generations.

OPEN — **no promoted conclusion.** No result through 07H proves Goldbach, universal canonical success, a global classification of EAC hosts, or universal composite-success of any noncanonical rank. The absence of a novelty claim is deliberate: several new statements are applications of classical PNT, Ramanujan-prime, Matveev, prime-gap, and consecutive-prime

inputs and have not been externally checked for priority.

11.8 Delta theorem/audit registry

The identifiers below are deliberately set as breakable typewriter text rather than squeezed into the older compact registry table.

TWICE-POWER-OF-THREE-MINUS-ONE-NOT-PERFECT-POWER-20260907D **PROVED**
 $2 \cdot 3^u - 1$ is never a proper integer power.

LOW-GAP-SINGLETON-UPWARD-FOUR-LABEL-ALTERNATIVE-20260907D **PROVED**
 Complete-support low-gap success-or-four alternative.

H1-POWER-OF-THREE-COMPLETE-FOUR-LABEL-ALTERNATIVE-20260907D **PROVED**
 Canonical $q = 3, h = 1$ four-label alternative.

LOW-GAP-MONOMIAL-RETURN-ORDER-AT-LEAST-THREE-20260907D **PROVED**
 Return order at least 3, hence $4 \leq v \leq u - 1$.

07E all-upper-root fixed point / exact rank formula **PROVED**

An upper root fails exactly when its mirror support lies in B_N ; the failure-rank spectrum is the corresponding prime-counting image.

RANK-MIRROR-SURJECTIVITY-20260907G **PROVED WITH REPAIR**

Every odd mirror occurs at every fixed rank; the audit adds the canonical boundary mirror $m = 2$ at $N = 4$ and narrows the scope.

07F same- (k, m) opposite-verdict certificates **VERIFIED COMPUTATION**

Exact rank and exact first row do not determine the eventual verdict.

07F basin-coprime gap and fixed-gap S -unit consumers **PROVED**

Support-overlap and finite-pair restrictions for simultaneous failures.

EFFECTIVE-FINITE-RANK-MIRROR-FIBRES-20260907G **PROVED**

Explicit Ramanujan-prime upper bound for each fixed (k, m) fibre.

DANGEROUS-MIRROR-TRANSPLANTS-GLOBALLY-EXHAUSTED-20260907G **PROVED** / **VERIFIED COMPUTATION**

The 4035 dangerous-mirror transplant realisations are the complete global finite set for those mirrors and ranks; none fails.

COMPLETE-BOUNDED-MIRROR-SAFETY-CENSUS-20260907G **VERIFIED COMPUTATION**

Exactly 22130 all- N realisations for odd composite $m \leq 3053$ at the ten selected ranks, with zero failures.

LAST-RANK-MIRROR-REALIZATION-AT-THE-UPPER-CORNER-20260907G **PROVED**

The last realisation in a bounded (m, k) rectangle occurs at its upper corner; $(M, K) = (3053, 74)$ gives $N = 7324$.

SCOTT-BOZEK-PARITY-GAP-REAPPEARS-IN-SOL-CONVERSATION-20260907G **SCOPE-CORRECTED**

The unqualified twin-core classification is not accepted as a complete dependency; the rank results do not use it.

SAME-N-PREFIX-VALUATION-REALIZATION-20260907H **PROVED**

Exact prescribed finite valuations can be synchronised at the same N across $p_0, \dots, p_K$.

EARLY-RANK-FIXED-SUPPORT-SEPARATION-20260907H **PROVED**

Two complete S -smooth early-rank mirrors must be quantitatively separated.

FINITE-VIEW-MATCHING-WITH-COFACTOR-ESCAPE-20260907H **PROVED**

A fixed finite local view can match across roots while complete cofactors necessarily escape every fixed larger finite alphabet except for at most one early member.

Chapter 12

Computational Evidence

Everything in this chapter is **VERIFIED COMPUTATION** or **EMPIRICAL PATTERN**. **Nothing here is evidence for a universal statement.** A range is part of a claim: quoting a number without its range is an error, and quoting a number without its *domain* (which roots, which complements, whether unit rows are included) is the error that produced the “301” confusion (FAIL-CENSUS-DOMAIN-001).

Every number below is regenerated by the single script in Appendix A, which has no external dependencies. The block IDs EXP1–EXP8 are the script’s own.

12.1 Provenance and reproduction

ID	What it computes	Range	Key result
EXP1	exhaustive canonical scan, census mode	even $N \leq 10^7$	0 failures; mean inclusive work $W = 7.0527$ tested vertices
EXP2	verdict of roots $p_0, \dots, p_6$	even $N \leq 2 \cdot 10^5$	0 for $k = 0$; 2, 4, 5, 3, 2, 1 for $k \geq 1$
EXP3	census of all failing upper roots	even $N \leq 2 \cdot 10^4$	301 roots, six N , 0 canonical
EXP4	escape margin G_{esc}	even $N \leq 2 \cdot 10^5$	52 numbers with margin 1; 10 with $ W \geq 3$
EXP5	phase trichotomy and ρ_0	even $N \leq 2 \cdot 10^5$	0 violations; $\rho_0 \geq 1 \iff \text{Type III}$
EXP6	embedding of reflected blocks	$N \leq 2 \cdot 10^4, k \leq 6$	731 446 indices, 0 violations
EXP7	fixed point B_N^{low}	even $N \leq 6 \cdot 10^4$	nonempty for exactly six N
EXP8	observables p_0 vs p_1	even $N \leq 2 \cdot 10^5$	aggregates $\pm 6\text{--}24\%$; $\kappa_{\text{gen}}: 0 \rightarrow 0.9041$
EULER	Euler-defect normalisations (N1)–(N4)	even $50 \leq N \leq 6 \cdot 10^4$	(N1) 6 044 and (N2) 3 804 violations; (N3), (N4) none
CERT	the 34 hostile certificates of the corpus	fixed witnesses	34/34 pass

Exact invocation reproducing the numbers in this file:

```
python3 p0_experiments.py -scan 10000000 -ablation 200000 -census 20000
                        -margin 200000 -phase 200000
                        -filtration 20000 -fixedpoint 60000
                        -observables 200000 -json data_p0.json
```

EXP1 at `-scan 10000000` takes about 75 s and about 50 MB; EXP7 at `-fixedpoint 60000` about 70 s. The defaults (`-scan 1000000 -fixedpoint 20000`) finish in a few minutes and reproduce everything except the 10^7 scan records.

12.2 The canonical scan

Scope note 12.1 (Work convention audit, 2026-09-07). The binding convention is inclusive work W : every dequeued/tested vertex, including the successful endpoint, is counted. The regenerated scan gives mean $W = 7.052697210539442$, maximum $W = 151$ first at $N = 3\,542\,948$, and maximum success depth 10 first at $N = 11\,822$. The older pseudocode variable “processed” counted only expanded composite rows E ; on success $W = E + 1$. This is a source/counting repair only and changes no success verdict.

EXP1. VERIFIED COMPUTATION. All even $4 \leq N \leq 10^7$, canonical root, **census** mode:

cases	4 999 999
failures of p_0	0
mean number of tested vertices W (successful endpoint included)	7.0527
record vertex count	151 ($N = 3\,542\,948$)
record success depth	10 ($N = 11\,822$)

The same scan restricted to $N \leq 10^6$ gives 499 999 cases, 0 failures, mean 5.8044, record 133 ($N = 976\,904$), record depth 10 ($N = 11\,822$) — reproducing the table of [BIB] to the last digit. The depth record 10 is attained only by $N = 11\,822$ and remains stable through the whole 10^7 range.

Distribution of the success depth $d(N)$ over all 4 999 999 cases:

d	0	1	2	3	4	5	6	7	8	9	10
$\#N$	983 722	1 426 153	1 500 435	802 099	240 796	42 059	4 436	276	18	4	1

$d = 0$ means immediate success ($N - p_0$ prime, or the diagonal case).

Warning 12.2 (What the scan does not say). 0 failures up to 10^7 is consistent with a failure at $10^7 + 2$. The scan is **VERIFIED COMPUTATION** and stays **VERIFIED COMPUTATION**. In particular it is *not* evidence for X1 (§9.9); it is the statement that X1 has no small counterexample.

12.3 Root ablation and the census

EXP2 — verdict of the root p_k . VERIFIED COMPUTATION. All even $N \leq 2 \cdot 10^5$, roots $p_0, \dots, p_6$; failures on a *composite* row and degenerate failures on the unit row $N - p_k = 1$

counted separately:

	k	0	1	2	3	4	5	6
failures, composite row		0	2	4	5	3	2	1
failures, unit row		0	4	2	3	3	2	2

Smallest failures on a composite row: $k = 1$: $(N, p_1) = (1862, 941)$; $k = 2, \dots, 5$: $(128, 73), (128, 79), (128, 83), (128, 89)$; $k = 6$: $(1862, 977)$.

The full census EXP3 and the six special N are in §6.6; the escape margin EXP4 is in §6.7; the phase census EXP5 in §6.3; the filtration check EXP6 in §6.2; the fixed point EXP7 in §6.6; the ablation EXP8 in §6.5.2. They are not repeated here.

12.4 Hostile certificates

CERT — 34/34 pass. **VERIFIED COMPUTATION**. Every finite certificate cited anywhere in the corpus was re-run against the reference implementation of §3.4. Groups:

A1 regression on the record cases $N = 11\,822$ (depth 10) and $N = 3\,542\,948$ (151 vertices).

B1 gap equivalence (Theorem 3.10) on all even $N \leq 2 \cdot 10^5$.

C1–C5 genealogy, reflection completeness, filtration embedding, and the diagonal calibration ($N = 14$, $r = 11$).

D1–D5 canonical geometry, shadow identity, Bézout invariant, no-bad-return, unique source.

E1–E14 the hostile counterexamples: $N = 242$ (reflection activation), $N = 128$ (TF local scope), $N = 322$ and $N = 858$ (CAB), $N = 4736$ (potential capacity), $N = 6142$ (external success), $N = 14198$ (direct adjacency), $N = 20$ (four-channel ledger), $N = 1718$ (generic absorption), $N = 51\,032$ (Hall compression), Exact-D countermodels, the $K_{3,3}$ grid at $N = 10^6$, the $N = 2200$ two-core, and the $h = 1$ witnesses 103 344 and 364 844.

F1–F5 ablation and the 301 census with its exact domain.

All 34 pass. Where a corpus claim and a certificate disagreed, the disagreement is recorded as a scope correction in §13.1, not silently resolved.

12.5 Other observables

Depth. **EMPIRICAL PATTERN**. The success depth of the canonical traversal is ≤ 2 for 3 910 310 of 4 999 999 cases (78.21%) and ≤ 4 for 4 953 205 cases (99.06%). The tail is extremely thin: a single case reaches $d = 10$. No mechanism is known that would force a bounded depth, and the depth is *not* the right measure of difficulty — see §6.7.

p_0 **is not measurably better at escaping.** **EMPIRICAL PATTERN**. Beyond the aggregate table of §6.5.2: within the 301 failing upper roots of EXP3, the canonical root wins at all six N , but the set of failing roots is neither an initial nor a final segment of $p_1, p_2, \dots$ (Example 6.24: p_1 wins, p_2 fails). Ordering by height does not order the outcome.

Two censuses agree. **EMPIRICAL PATTERN**. The six N with $B_N^{\text{low}} \neq \emptyset$ (EXP7, $N \leq 6 \cdot 10^4$) coincide with the six N carrying failing upper roots (EXP3, $N \leq 2 \cdot 10^4$) and with the six numbers with an exceptional autonomous component reported to 10^8 by Božek [8]. The three lists measure different things; their coincidence is data, not a theorem, and it has not been

checked beyond $6 \cdot 10^4$ in this project.

Chapter 13

Failed and Killed Approaches

This chapter is the curated narrative; Chapter 14 is the complete raw record. Read this one first: it groups 239 individual entries into the dozen mistakes that actually recur. A model that internalises this chapter will not need the raw log except to check a specific ID.

The five recurring shapes of error.

1. **Domain mixing** — quoting a result outside the class (A/B/C/D) on which it was proved. Most common single failure.
2. **Functional substitution** — proving a bound in one functional and consuming it in another. Kills TD, P1^{br}, K1.
3. **Vacuity mistaken for canonicity** — a construction that “only works for p_0 ” because for p_k it is undefined on k coordinates.
4. **Finite certificate** — deriving a contradiction from a bounded canonical transcript. Provably impossible (§8.2).
5. **Promotion** — turning a scan with zero counterexamples into a theorem, or an exact identity into a mechanism.

13.1 Curated ledger of negative results

The table is complete with respect to all sources. Rows are grouped by theme; each entry carries an exact counterexample or an exact reason.

Claim / route	Status	Exact reason
Exceptionality of p_0		
Zero hole count as a mass mechanism	KILLED	loss $O_k(\log N)$ against block mass $t_k \log(x/h_k)$ FAIL-P0-ZEROHOLE-ESSENTIAL-001
p_0 as information selector	SUPERSEDED	every clause survives ablation; not refuted, but not established FAIL-P0-INFORMATION-SELECTOR-001
MR2 as a canonical identity	KILLED	$q \mid (r - 2j) \iff q \mid (h_r - j)$ for any upper root FAIL-MR2-CANONICAL-ONLY-001
Least-upper-parent as selector	KILLED	eleven non-canonical roots share the property [CORE, V71.41]
Prime-free boundary activates a reflected row	KILLED	$N = 242$: $a_1 = 117$, $a_2 = 119$ both composite FAIL-REFLECTION-ACTIVATION-001
Statistical dominance of p_0	KILLED	median rank percentile 0.20982 [D4, q. 12]

Claim / route	Status	Exact reason
Minimality determines local geometry	KILLED	23+15 transitions $B_{\text{abs}} \leftrightarrow B_{\text{re}}$ at fixed p_0 [D4, q. 17]
Largest-factor heuristic	KILLED	$N = 124$ [PCRD, FAIL-P0-LARGEST-FACTOR-TWOSTEP]
Finite canonical data give a contradiction	KILLED	affine programmability and finite bad-tree universality [PCRD]
Mechanisms around p_0		
Literal label conservation along a path	KILLED	$v_\ell(N - \ell) = v_\ell(N) = 0$ [LOG]
Absorption trichotomy from factor-flow alone	KILLED	$N = 1718$, $\ell = 59$: both payments have zero valuation FAIL-GENERIC-ABSORPTION-001
Four-channel ledger on all of Q_0	KILLED	$N = 20$: a GOOD row outside R gives $\kappa_R < m_\ell$ [ILA]
Unconditional CAB	KILLED	$N = 322$: $U_R = J_R = 1$, $B_{\text{abs}} = 11 \cdot 29$; internal version $N = 858$ FAIL-CAB-001, FAIL-CAB-INTERNAL-001
$\mathcal{C}_S < \mathcal{D}_S$ from the first front	KILLED	$N = 4736$: $\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$ FAIL-POTENTIAL-CAPACITY-001
Escape must be directly adjacent	KILLED	$N = 14198$: atoms 19,167 at distances 8 and 7 FAIL-DIRECT-ADJACENCY-001
External reuse is local	KILLED	$(N, \ell) = (1825372, 52153)$, distance 53750 FAIL-LOCAL-REUSE-001
External parent is a success root	KILLED	(6142, 877, 3511): 3511 active, above p_0 , own traversal fails FAIL-EXTERNAL-SUCCESS-001
Constant $C = 1$ in charging	KILLED	53 violations, maximum 9.42243 FAIL-CHILD-C1-001
Naive support expansion of all sources	KILLED	Hall compression $456 \rightarrow 17$ at $N = 51032$ FAIL-HALL-SUPPORT-001
Exact-D local algebra replaces minimality	KILLED	non-canonical models die only on $h < G$ FAIL-EXACTD-NONCANONICAL-001
Failure always produces nontrivial B_{abs}	KILLED	FAIL-BABS-PRODUCTION-001
Widened block in the non-reuse lemma	KILLED	the endpoint pair has difference $2h$ FAIL-BROADBLOCK-ENDPOINT
Euler defect in normalisation (N1) or (N2)	KILLED	6044 resp. 3804 violations for $N \leq 6 \cdot 10^4$ (§5.7)
Scope corrections		
Upper-half compression under p_0 failure	SCOPE-CORRECTED	the theorem assumes $\text{Good}_N = \emptyset$ FAIL-UPPERHALF-SCOPE-001
Local $\mathcal{F}$ certificate + $h < G$ forces routing	SCOPE-CORRECTED	$N = 128$, $\ell = 13$: certificate holds, p_0 wins, $R = \emptyset$ FAIL-TF-LOCAL-SCOPE-001
Source-free kernel = EAC = terminal component	SCOPE-CORRECTED	three different constructions FAIL-KERNEL-EAC-001
Parent lattice over all primes	SCOPE-CORRECTED	the correct object is the intersection with $\mathcal{A}_N$ FAIL-PARENT-LATTICE-DOMAIN-001

Claim / route	Status	Exact reason
“301” counts all failing roots	SCOPE-CORRECTED	the census covers composite complements only FAIL-CENSUS-DOMAIN-001
Genealogy and reflection-completeness without hypothesis	SCOPE-CORRECTED	diagonal case, $N = 14$ (Scope note 3.17)
Formula $\rho_0 = \#\{D_s < 2h\}$	SCOPE-CORRECTED	missing lower condition $D_s > h$ (Warning 6.18)
High-gap amplification as unconditional	SCOPE-CORRECTED	hypothesis is <i>pair failure</i> , not CRD failure (Warning 7.5)
$q < m_0/3$	SCOPE-CORRECTED	conject is $q \leq m_0/3$; $N = 32$ attains equality (Scope note 5.4)
IL1–IL6 on all of Q	SCOPE-CORRECTED	domain is Q_{smooth} (Warning 7.7)

13.1.1 What survived hostile testing

- Incidence ledger IL1–IL6 on the domain $S = Q_{\text{smooth}}$: two independent reconstruction routes, 1 000 worlds, 0 errors [ILA] — **AUDITED IDENTITY**.
- The escape-or-reuse alternative: 61 452 instances, 0 orphans; independent red team 60 413 atoms, 0 orphans [ORPH, RT] — **VERIFIED COMPUTATION**, as a theorem **OPEN**.
- The whole genealogical layer (§6.1) and the filtration (§6.2) — **PROVED**.
- The phase diagram and the bidirectional rank (§6.3, §6.4) — **PROVED**.
- Global two-sheet collision budget, no-bad-return-to- p_0 , the B_N^{low} characterisation — **PROVED**.
- All 34 hostile certificates of the corpus were reproduced here and pass (§12.4).

13.2 Post-CRD successor theories

[PCRD] documents a sequence of theories built after CRD1, each intended to preserve information lost by its predecessor. All reached a wall. The table summarises them; their proofs are *not* reproduced here.

Theory	Fundamental object		What it was to recover	What stopped it
CRD1	support/valuation dynamics from p_0		reachability, shadow, capacity	the “same container” argument, generic smooth counting, residue saturation, exact self-consistency of the fixed point
CMFT	lossless factorization (δ, q, a, b)	mirror-tensor	parent-valuation-cofactor correlation	the least-factor tree <i>is</i> the Buchstab/sieve decomposition; the cofactor star at fixed q is an affine bijection
CGST	canonical cell N_h and the convolution $a_P * a_P$		exact additive pairing + phase	the factorization layer collapses diagonally in (h, t) ; exception propagation reduces to prime-pair geometry
CSD	exceptional spectral configurations + the gap		a pointwise inverse anomaly	finite conductor patterns are consistent with an arbitrarily long <i>fixed</i> canonical gap

Theory	Fundamental object	What it was to recover	What stopped it
CAPT	Li–Liu almost-prime witnesses	a massive real family of sources	at fixed (r, s) the system reduces to two linear prime forms; $(3, 3)$ is the natural degeneration
RSD	rough reservoir + Liouville sign + P_3 correction	explicit exposure of parity	either level of distribution $> 2/3$, or a pointwise binary prime–Liouville correlation
rough- P_2 matrix	$A_N(q, r) = \mathbf{1}_{\mathbb{P}}(N - qr)$	global higher-order correlation	boundary blindness, local spectrum = singular series, co-degrees = prime-tuple problem, $K_{3,3}$ grids exist arithmetically
H1-FP	root-factor descent at $h = 1$ with shift-2 orthogonality	elementary closure of the 2-core	reduces to a Pillai/Bennett equation $a^\alpha - a = b^\beta - b$

Example 13.1 (A $K_{3,3}$ grid at $N = 10^6$). **VERIFIED COMPUTATION**. For $q \in \{61, 67, 73\}$ and $r \in \{113, 179, 599\}$ all nine numbers $N - qr$ are prime, and all six factor vertices are BAD. A full rank-two bipartite rough- P_2 grid is therefore arithmetically real and is *not* a contradiction. [PCRD, hostile exact survivor]

Updated verdict on the p_0 programme. The *old local p_0 -exceptionality proof architecture* is **SUPERSEDED**, but the stronger historical claim that finite complete-BFS neighbourhoods are universally programmable was scope-corrected in September 2026. Consequently universal p_0 reachability and the $h = 1$ complete-support problem remain live open questions. What is ruled out rigorously is narrower: selected partial transcripts by themselves, and hereditary first-support graph/valuation separators, cannot explain universal p_0 reliability.

What remains alive: p_0 as a selector, the exact full-BFS/root-2 conjugacy at $h = 1$, complete-support/moving-support analysis, non-hereditary selector invariants, and the full reachability problem.

13.3 Surviving lemmas of the killed programmes

A killed theory is not a theory whose theorems are false. Each successor programme of §13.2 died at an *architecture* wall, and each left behind correct, reusable lemmas. Compression is exactly where those get thrown out with the dead programme, so they are collected here with their statuses intact. *Every item below is **PROVED** in [PCRD] unless marked otherwise; the proofs are not reproduced here.* Using one of them is legitimate; restarting its parent programme is not.

13.3.1 From CMFT (Canonical Mirror Factorization Theory)

CMFT’s objects: for $N = 2x$, $\mathcal{U}_N = \{p \in \mathbb{P} : x \leq p < N\}$, $\mathcal{M}_N = \{m \leq x : N - m \in \mathbb{P}\}$, so that Goldbach is exactly $\mathcal{M}_N \cap \mathbb{P} \neq \emptyset$; the canonical root has mirror anchor $m_0 = N - p_0 = \max \mathcal{M}_N$; and the CMFT atom is $\mathfrak{c} = (\delta, q, a, b)$ with $m_0 - \delta = q^a b$ and $p_0 + \delta + q^a b = N$.

Theorem 13.2 (Lossless anchor filtration, PCRD-CMFT-1). **PROVED**. For upper primes $p_0 < p_1 < \dots$ with $m_j = N - p_j$, the full left history of the anchor p_j grows by adjoining the entire hyperedge row

m_j :

$$\mathcal{H}_{j+1} = \mathcal{H}_j \sqcup \mathcal{A}_j, \quad L_{j+1}(\ell, k) - L_j(\ell, k) = \mathbf{1}_{\ell^k | m_j}.$$

The ablation $p_0 \rightarrow p_1$ loses exactly the full factorization of m_0 .

Why it is worth keeping. This is the precise *information-theoretic* form of the ablation result of §6.5.2: it says what is lost, not merely that something is.

Theorem 13.3 (Tail hard-edge universality, PCRD-CMFT-2). **PROVED** / *no-go*. For any later anchor p_j , after truncating the earlier prefix its complement m_j becomes the maximal element of the tail mirror field and satisfies identical local gcd, offset and bilinear identities. Hence

$$\text{the local geometry of the tail does not distinguish } p_0.$$

The genuinely unique information carried by p_0 is only the zero lost prefix.

Reading. This is the sharpest available formulation of what §6.5 establishes empirically, and it is a theorem. Combined with Warning 6.7(2) it is the strongest reason to doubt that any purely local property of p_0 can do dynamical work.

Theorem 13.4 (Canonical least-factor split, PCRD-CMFT-3). **PROVED**. For a composite mirror row m choose uniquely $q = P^-(m)$, $b = m/q$. If $q > x^{1/3}$ then b must be prime, hence

$$m \text{ composite} \implies P^-(m) \leq x^{1/3} \quad \vee \quad m = qr \text{ a balanced semiprime.}$$

Why it is worth keeping. It repairs the divisor-multiplicity bias of any raw bilinear form over rows: without it, divisor-rich rows are counted $\tau(m)$ times.

Theorem 13.5 (All-root collision identity, PCRD-CMFT-4). **PROVED**. For a composite row family S ,

$$\sum_{\ell} \sum_{k \geq 1} \binom{\nu_{\ell, k}(S)}{2} \log \ell = \sum_{i < j} \log \gcd(m_i, m_j).$$

A pigeonhole over least factors yields a positive collision floor.

Limitation, recorded by the source. The floor has *no consumer*: nothing in the corpus converts it into a contradiction. Compare Theorem 7.30, which is the residual-row analogue and does have one.

Theorem 13.6 (Prime-pressure failure ceiling, PCRD-CMFT-5). **PROVED**. With $\mathcal{P}_N = \sum_{m \in \mathcal{M}_N} \Lambda(m)$, a hypothetical Goldbach failure eliminates every $k = 1$ contribution, so

$$\mathcal{P}_N \leq E(x) := \sum_{r^k \leq x, k \geq 2} \log r \leq \sqrt{x} \log x, \quad \text{whence} \quad \mathcal{P}_N > \sqrt{x} \log x \implies \text{Goldbach.}$$

Reading. A clean quantitative reformulation of Goldbach for a single N , entirely elementary. It did not supply the matching lower bound, which is why CMFT died — but the reformulation itself is correct and reusable, and it is a class-D-free statement (it does not assume $\text{Good}_N = \emptyset$; it concludes Goldbach).

The CMFT wall. Raw bilinear mass counted divisor-rich rows repeatedly. After the least-factor repair the machinery turned out to be *exactly* the Buchstab/sieve decomposition, and the fixed- q cofactor star $p \leftrightarrow b = (N - p)/q$ is an affine bijection, not an independent Type-II correlation. Giant stars have the natural scale of primes in arithmetic progressions and can be far larger than the “forced” $x^{2/3}$ with no anomaly at all.

13.3.2 From CGST (Canonical Gap-Spectral Theory)

CGST's reformulation: with $A_P = \{u \in \mathbb{Z} : P + 2u \in \mathbb{P}\}$ and $N_h = 2(P - h)$,

$$N_h \text{ is Goldbach} \iff -h \in A_P + A_P.$$

Theorem 13.7 (Canonical straddle theorem, PCRD-CGST-1). **PROVED**. *If no direct pair through P exists, every Goldbach representation has the form*

$$P + 2t + (P - 2(h + t)) = 2(P - h), \quad t \geq 1,$$

and the lower prime must lie below the previous prime P_- .

Reading. This is the additive counterpart of the phase trichotomy (Theorem 6.12) and explains why Type I forces the search strictly past p_- .

Theorem 13.8 (Arbitrarily deep finite reflection obstruction, PCRD-CGST-2). **PROVED**. *For any fixed T , CRT and Dirichlet construct infinitely many canonical $h = 1$ worlds in which $P - 2, P - 4, \dots, P - 2(T + 1)$ are all composite. Hence no bounded-depth reflection search can be a universal closer.*

Reading. The reflection-specific instance of the architecture wall (§8.2). If a proposed argument inspects the reflected block to any fixed depth, this theorem kills it immediately.

Theorem 13.9 (Diagonal collapse, PCRD-CGST-3). **PROVED**. *In cell coordinates the reflection row is $C_P(h, t) = P - 2(h + t)$, so the entire valuation/factor tensor on the (h, t) plane depends only on $h + t$. The extra “cell dimension” is a false degree of freedom for the factorization layer.*

Warning 13.10. Any construction that treats (h, t) as two independent parameters at the factorization layer is already dead. The exception-amplification wall follows: two cells h and $h + s$ share exact rows only when the upper-prime offsets form a prime pair of difference $2s$, and same-row factor certificates satisfy $\gcd(C_P(h, t), C_P(h + s, t)) = \gcd(C_P(h, t), s)$, which is 1 for $s = 1$. Factorization-driven propagation of one Goldbach exception to neighbouring N is therefore far too weak.

13.3.3 From CSD (Canonical Spectral Defect)

Theorem 13.11 (Canonical conductor-gap compatibility, PCRD-CSD-1). **PROVED**. *For any fixed modulus pattern M and any fixed length L , CRT and Dirichlet construct infinitely many canonical $h = 1$ worlds with $M \mid N$ and a prime-free interval of length at least L immediately before p_0 . Hence any finite conductor pattern is compatible with an arbitrarily long fixed canonical gap.*

Reading. The bridge “exceptional characters $\Rightarrow$ canonical gap contradiction” does not exist, and this theorem says why in one line. Keep it: it is the cheapest available refutation of an entire family of circle-method hybrids.

13.3.4 From CAPT (Canonical Almost-Prime Transfer)

Theorem 13.12 (Fixed- (r, s) compression, PCRD-CAPT-1). **PROVED**. *After freezing (r, s) in the Li-Liu representation $N = p + rq$ with $N - q = sb$, one has $q = N - sb$ and*

$$p = rsb - (r - 1)N,$$

so the system becomes two linear prime forms in the single variable b .

Remark 13.13 (The $(3, 3)$ degeneration). For $(r, s) = (3, 3)$: $q = N - 3b$, $p = 9b - 2N$ — an ordinary two-linear-prime pattern that occurs abundantly in real canonical worlds. Second-layer compression is therefore not a contradiction. Large-factor consequences of Li-Liu that remain correct: under Goldbach failure $r \neq 1$, and from $rq < N$ with $r \leq q^{9/10}$ one gets $r < N^{9/19}$ and $q > N^{10/19 - o(1)}$ for almost all of the relevant mass.

13.3.5 From RSD (Rough Signed Decomposition)

Theorem 13.14 (Exact rough signed identity, PCRD-RSD-1). **PROVED**. *On the rough support $(\rho_z(n) = \mathbf{1}_{(n, P(z))=1}$, $z > X^{1/4}$, so $\Omega(n) \leq 3$),*

$$\mathbf{1}_{\mathbb{P}}(n)\rho_z(n) = \rho_z(n) \left[\frac{1 - \lambda(n)}{2} - \mathbf{1}_{\Omega(n)=3} \right],$$

and for shifted-prime weights

$$G_z(N) = \frac{1}{2}R_z(N) - \frac{1}{2}L_z(N) - T_{3,z}(N),$$

so that under Goldbach failure $L_z(N) = R_z(N) - 2T_{3,z}(N)$.

Reading. This is the cleanest *exact* statement in the corpus of where the parity obstruction sits for this problem. It is worth keeping precisely because it is an identity, not an estimate.

Theorem 13.15 (Distribution-roughness phase law, PCRD-RSD-2). **PROVED**. *If the usable distribution level is $D = X^\theta$ and one sieves to $z = X^\beta$, the lower linear sieve needs $\beta < \theta/2$, while forcing $\Omega(n) \leq k$ by roughness needs $\beta > 1/(k+1)$. The two are compatible iff*

$$\theta > \frac{2}{k+1}.$$

For $k = 2$ the natural threshold is $\theta > 2/3$.

Reading. A quantitative triage rule: any proposal in this family can be checked against it in one line. Pascadi's $5/8 - o(1)$ [95] for special factorable weights is below $2/3$, so branch A is closed at current technology.

Remark 13.16 (The parity endpoint, exactly). At $z = X^{5/16-\varepsilon}$ the surviving P_3 sits in a narrow balanced box that switching/modern Chen sieves control well, but after reduction to P_1/P_2 the exact identity

$$P_1(N) = \frac{D_{1,2}(N) - \Delta_2(N)}{2}, \quad \Delta_2 = P_2 - P_1$$

remains, and Goldbach failure means full saturation $\Delta_2 = D_{1,2}$. Any theorem giving “even a little desaturation” is *already equivalent* to finding a prime complement on that family. This is the precise sense in which the parity barrier is the endpoint (§13.4).

13.3.6 From the global rough- P_2 matrix route

With $A_N(q, r) = \mathbf{1}_{\mathbb{P}}(N - qr)$:

Theorem 13.17 ($K_{3,3}$ explosion, PCRD-M-1). **CONJECTURE** (conditional **PROVED**). *At rough- P_2 mass of order $X/\log^2 X$, Kővári-Sós-Turán forces a $K_{3,3}$ at the natural scales $|Q| \sim X^{1/2}$, $|R| \sim X^{11/16}$. $K_{4,4}$ is not forced by that density scale.*

Theorem 13.18 (Boundary blindness, PCRD-M-2). **PROVED**. *Extending the bulk matrix by the boundary row/column $r = 1$ encodes the Goldbach entries. Zeroing the whole boundary axis is a perturbation of rank at most 2, whose operator norm is asymptotically small against the bulk scale. A generic operator norm or energy can therefore almost fail to notice the loss of every Goldbach entry.*

Theorem 13.19 ($h = 1$ matrix invisibility, PCRD-M-3). **PROVED**. *For $N = 2(P - 1)$ we have $m_0 = P - 2 = X - 1$, so every odd product $qr \leq X$ automatically satisfies $qr \leq X - 1 = m_0$. The canonical hard edge removes no rough- P_2 matrix cell: p_0 disappears from the domain geometry entirely.*

Theorem 13.20 (Fixed-modulus spectrum, PCRD-M-4). **PROVED**. *The local admissibility matrix modulo a prime $\ell \nmid N$ has the form $J - P_N$; its singular values are explicit and recover only local density / singular-series data. The fixed-modulus spectrum does not see Goldbach failure.*

Remark 13.21 (The codegree wall). The pointwise codegree $C_N(q_1, q_2) = \sum_r A_N(q_1, r) A_N(q_2, r)$ counts simultaneous primality of the three linear forms $r, N - q_1 r, N - q_2 r$, i.e. it enters the prime-tuple problem; average codegrees are boundary-blind by Theorem 13.18. Only a specially constructed bulk-to-boundary inverse theorem would remain, and no candidate is known that does not reduce to an earlier binary-correlation wall.

13.3.7 From the p_0 kill programme and the H1-FP revival

Theorem 13.22 (Gap scalar freedom, PCRD-P0-3). **PROVED** (from known facts). *For a fixed prime P with gap $G = P - P_-$, all $N_h = 2(P - h)$, $0 \leq h < G$, have canonical root P . With unbounded prime gaps the ratio h/G can approximate all of $[0, 1]$; with bounded-gap results there are infinitely many small G . Hence the scalars $h, G, h/G$ have no universal closing regime.*

Warning 13.23. This is the exact statement that kills every attempt to close the problem by a size condition on $\theta = h/G$ (§6.3). The phase diagram is a correct classification; it is not a filter.

Theorem 13.24 (H1 root-factor descent, PCRD-H1-1). **PROVED**. *If $q, r \mid m_0 = P - 2$ then*

$$r \mid N - q \iff r \mid q - 2.$$

The root-factor subsystem is therefore a strictly decreasing DAG, and the least root factor q_ satisfies*

$$\text{PF}(N - q_*) \cap \text{PF}(m_0) = \emptyset,$$

so failure must immediately produce external factors.

Theorem 13.25 (H1 shift-2 orthogonality, PCRD-H1-2). **PROVED**. *For $N = 2(P - 1)$ with P prime and $u < N/3$ prime,*

$$\gcd(N - u, u + 2) = 1.$$

Every terminal fixed core must therefore satisfy full shift-2 coprimality on all of its hyperedges.

Remark 13.26 (The 3-core normal form). The two smallest rows of a terminal core have disjoint factor supports. For $K = \{a < b < c\}$ this leads to one of two coupled forms, e.g.

$$N - b = a^\alpha, \quad N - a = b^\beta c^\gamma, \quad N - c = a^u b^v.$$

A hostile reverse search over large finite boxes found no 3-core; the status is **VERIFIED COMPUTATION**, not a theorem.

Why the H1-FP revival was not enough. Closing $N = 2(P - 1) \Rightarrow B_N^{\text{low}} = \emptyset$ requires classifying complete self-consistent shifted S -unit systems and the primality of the linear form $N/2 + 1$. Already the minimal case $|K| = 2$ touches an unresolved Pillai/Bennett-type uniformity ($a^\alpha - a = b^\beta - b$; Remark 6.25). p_0 supplies no mechanism of its own for that problem.

13.3.8 Live CRD results retained through every pivot

[PCRD] records the following as explicitly retained when CRD1 was frozen. They are the intersection of everything that survived; all appear elsewhere in this file and are listed together here so a future model can see the retained core at a glance.

- the fixed-point characterisation B_N^{low} (Theorem 3.29);
- the C0b decomposition and its correct gap-scale congruence consequences (Theorem 8.6);
- T2 first-front residue closure;
- T3 mod-3 exclusion/rigidity (Theorems 5.17, 5.18);
- the source-free core K with its shifted S -unit equations (Remark 6.25, Remark 13.26);
- the exact container/product identities (§5.3);
- the correct prime-power collision geometry (§7.4);
- the empirical six-EAC corpus (§6.6).

What killed CRD1 — the convergence, not a single false lemma.

1. Selected-divisor capacity routes were *same-container* arguments: one cannot beat a container with one of its own divisors. (This is exactly what Corollary 9.9 later made quantitative.)
2. Generic smooth-population bounds lost self-consistency: the factor alphabet is simultaneously the set of state labels.
3. The pure local-residue route saturates for $q \geq 5$; modulus 3 is an exception, not a mechanism (Theorem 5.17).
4. Reflection is not a universal closer: $h = 1$ gives an *empty* strict reflected block.
5. Nonprimitive large product/reuse is compatible with a failed world and gives no capacity contradiction.
6. A source-free core is not itself contradictory: $N = 2200$, $K = \{3, 13\}$ is an exact witness.
7. After removing the intermediate ledgers the frontier returned to the exact self-consistency $\text{PF}(m_0) \subseteq B_N^{\text{low}}$ with no independent intermediate parameter.

13.4 The parity barrier and why it is the real wall

Two distinct obstructions are present and must not be conflated.

The partial-transcript wall (§8.2) is internal and scope-corrected: compatible selected finite divisibility/valuation patterns can be realised canonically, so those selected patterns alone cannot supply a contradiction. It does *not* rule out a theorem about a complete bounded-depth frontier. The parity barrier is external and classical: sieve methods cannot distinguish numbers with an even number of prime factors from those with an odd number, so no purely sieve-theoretic argument produces a *prime* complement.

Thus selected-edge programming cannot replace complete-support control, while pure sieve input cannot by itself force primality. The current moving-support frontier may require a combination of exact complete-row arithmetic and analytic input; the old slogan “not locally” was too broad.

RSD and parity. RSD attempted to expose parity explicitly through a rough reservoir with a Liouville sign and a P_3 correction. It requires either level of distribution $> 2/3$ — beyond Pascadi’s $5/8$ [95] — or a pointwise binary correlation between primes and the Liouville function, which is at least as hard as the problems calibrated by Mangerel [97]. **KILLED**.

The weaker P_2 desaturation target. Also **KILLED**: at $\Omega \leq 2$ a sign- (-1) witness *is* a prime complement, so the “weaker” target is the original problem.

Chen-type input. R. Li’s bound $D_{1,2}(N) \geq 1.9728 C(N)N/\log^2 N$ [96] and Li–Liu’s $(1 + 1.9)$ [17] supply almost-prime representations in quantity, but not the last bit distinguishing P_1 from P_2 . That last bit is the parity barrier.

Friedlander–Iwaniec. The asymptotic sieve for primes [40] is the template for what the missing input looks like: a bilinear form estimate. No bilinear input of the required shape has been produced in this project, and the rough- P_2 matrix was the attempt to manufacture one.

Warning 13.27 (Do not try to route around parity by enlarging the graph). Several killed routes are variations on “add more structure to the graph until the parity obstruction disappears”. The obstruction is not a property of the graph; it is a property of the sieve information being used.

Adding vertices, weights or higher-order correlations to a construction whose only input is sieve-type data cannot remove it.

Chapter 14

Full Failure Log

This chapter is reproduced in full and unabridged. It contains the 239 legacy entries of the project failure log plus seven post-v4.81 binding additions. The exact source text for the new additions is also preserved in Appendix B. No entry was removed for appearing unimportant; the record’s value lies precisely in its completeness.

How to use it.

- The 239 legacy entries remain alphabetically ordered by FAIL-ID; the seven post-v4.81 additions are appended in their research chronology and are also listed in the index.
- The index in §14.1 gives ID, status and page.
- Every entry is labelled `fail:FAIL-ID`, so `\ref{fail:FAIL-XXX-001}` resolves.
- Cross-references that pointed into the original document and cannot be resolved here appear as `[target]` rather than being silently dropped.
- Legacy status macros of the source were mapped onto the status vocabulary of §1; the mapping is stated below.

Legacy status mapping. `R, RI, RM` → **KILLED**; `SC` → **SCOPE-CORRECTED**; `SU, RET` → **SUPERSEDED**; `SR, C` → **VERIFIED COMPUTATION**; `P` → **PROVED**; `PA` → **PROVED WITH REPAIR**; `H` → **EMPIRICAL PATTERN**; `O` → **OPEN**. The mapping is status-preserving in the sense that no entry was moved to a stronger status.

14.1 Index of failure-log entries

ID	Status	Str.
AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905	SCOPE-CORR.	219
ENVELOPE-ESCAPE-AUTOMATIC-ITERATION-20260906	KILLED.	219
AUTOMATIC-THIRD-GENERATION-SIEVE-SWITCH-20260906	KILLED.	219
SUCCESS-PRODUCT-BARRIER-TO-CLOSED-CORE-HEIGHT-20260906B	KILLED.	219
PRIME-LIFT-PRIMALITY-FORCES-SUCCESS-20260907	KILLED.	219
LOW-GAP-FIVE-LABEL-OVERGENERALIZATION-20260907C	SCOPE-CORR.	219
SUBGROUP-SEPARATION-OF-P0-AS-GENERAL-EXPLANATION-20260907D	KILLED.	219
FAIL-ABOVE-BV-MASS-INVISIBLE-001	SCOPE-CORR.	118
FAIL-ABSORBED-FRIDGE-RESIDUAL-HAS-ARBITRARY-001	KILLED.	118
FAIL-AP2-HIDDEN-OMEGA-DEPENDENCY-001	SCOPE-CORR.	119
FAIL-BABS-PRODUCTION-001	SUPERSEDED.	119

ID	Status	Str.
FAIL-BAKER-CORE-FAN-DIRECTLY-GIVES-LEVELC-SCOPE-CORR.001	SCOPE-CORR.	120
FAIL-BOUNDED-ARITY-WEIGHT-001	KILLED (infer.).	120
FAIL-BUNDLE-ENLARGES-CAPACITY-001	KILLED.	121
FAIL-BV-BOUNDARY-DESTROYS-MASS-001	KILLED.	122
FAIL-BV-INACCESSIBILITY-HIDES-MIRROR-MASS-001	SUPERSEDED.	122
FAIL-C-SYMBOL-COLLISION-001	SCOPE-CORR.	122
FAIL-CAB-001	KILLED without a failure condition.	123
FAIL-CAB-INTERNAL-001	KILLED without a failure condition.	124
FAIL-CANONICAL-BT-CAPACITY-TRANSPORT-TO-EXTENSION-CORR	SCOPE-CORR (domain failure).	124
FAIL-CANONICAL-EXACT-LAYER-TRANSFER-MISSING-SCOPE-CORR.	SCOPE-CORR.	125
FAIL-CASCADE-DIRECTION-001	KILLED as stated and as proved.	125
FAIL-CENSUS-DOMAIN-001	SCOPE-CORR.	126
FAIL-CHILD-C1-001	KILLED by source-recorded computation.	126
FAIL-CLEAN-ROOTED-CAP-P1-001	KILLED (infer.).	127
FAIL-COFACTOR-ENTROPY-ALONE-CLOSES-LOADONEKILLED	KILLED (infer.).	128
FAIL-COFACTOR-SIBLING-PRIME-UNSCOPED-001	KILLED (infer.).	128
FAIL-CRD2-NEEDED-FOR-FIRST-OCCURRENCE-001	KILLED as a present engineering requir	129
FAIL-CRD2-PREMATURE-001	DESIGN PROHIBITION / FRONTIER CERTIFIC	129
FAIL-CRITICAL-VERTICAL-CARRIES-TLOGN-001	KILLED.	129
FAIL-CRITICAL-VERTICAL-GENERIC-001	KILLED as a description.	130
FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001	KILLED.	130
FAIL-CUBEROOT-PIGEONHOLE-IS-BOUNDARY-CASE-SCOPE-CORR.	SCOPE-CORR.	131
FAIL-CYCLE-COFACTOR-LOCK-ALONE-CLOSES-DEEPCYCLES-001	KILLED as a sufficient uniform closure	131
FAIL-DC-CAPACITY-COMPRESSSION-001	SCOPE-CORR.	131
FAIL-DEADSMOOTH-CAN-CARRY-POLYNOMIAL-HIGHKILL-001	KILLED.	131
FAIL-DEADSMOOTH-PILLAI-001	SCOPE-CORR. The search result stands a	132
FAIL-DEADSMOOTH-UNBOUNDED-001	KILLED.	132
FAIL-DEEP-AFFINE-PATH-RESIDUE-IS-CAPACITY-KILLED	KILLED for the capacity inference.	133
FAIL-DEEP-FIBRE-NEEDS-FIRSTENTRY-OWNER-ESCAPED	KILLED as a proof-design necessity.	134
FAIL-DEEP-FIBRE-TO-SUBPOLY-BY-ANCESTRY-ALONEKILLED	KILLED as a proof mechanism from one r	134
FAIL-DEEP-ONE-AP-PACKING-CLOSES-001	KILLED as a sufficient closing mechani	134
FAIL-DEEP-PURE-QMONOMIAL-CAN-CYCLE-001	KILLED.	134
FAIL-DEEP-SELFRENEWAL-IS-INDEPENDENT-TERMIKILLED	KILLED as a proof-design description.	135

ID	Status	Str.
FAIL-DEEP-U-TO-U-RETURN-AS-PRIMARY-FRONTIER-001	SUPERSEDED as a primary frontier.	135
FAIL-DESCENT-ONE-STATE-001	SUPERSEDED.	135
FAIL-DIRECT-ADJACENCY-001	KILLED for direct adjacency.	136
FAIL-ETA-EQUIVALENT-TO-WEIGHTED-J-001	KILLED (infer.).	136
FAIL-EXACTD-NONCANONICAL-001	KILLED as a substitute for minimality;	137
FAIL-EXACTD-RESIDUAL-SURVIVES-SUBROOT-001	KILLED.	137
FAIL-EXTERNAL-ETA-IDENTIFICATION-001	KILLED.	138
FAIL-EXTERNAL-FIXED-POLYLOG-SQUAREFREE-PENDING-001	CLOSED PENDING INDEPENDENT AUDIT .	138
FAIL-EXTERNAL-MERGER-FIRST-SPLIT-001	SUPERSEDED as proof organisation.	139
FAIL-EXTERNAL-MULTISOURCE-CRT-IS-THE-NEXT-CAPACITY-001	CAPACITY (mech.) relative to the current	139
FAIL-EXTERNAL-SOURCE-MASS-IS-INDEPENDENT-ROUTE-001	SUPERSEDED as a primary factorisation	139
FAIL-EXTERNAL-SUCCESS-001	KILLED.	140
FAIL-EXTERNAL-UT-IDENTITY-IS-UPPER-CAPACITY-001	KILLED (infer.).	140
FAIL-F1-DOMAIN-MIX-001	SCOPE-CORR.	141
FAIL-FC8-FLOOR-FOR-RHS-001	KILLED (infer.).	141
FAIL-FC8-SELF-CONTRADICTION-001	KILLED (infer.). (FC8) itself is PROVE	142
FAIL-FC8-SLACK-IS-VERTICAL-MASS-001	SCOPE-CORR.	142
FAIL-FIRSTENTRY-SQUARE-ONLY-HAS-Q-SPACING-001	KILLED.	143
FAIL-FIRSTFRONT-CONTAINMENT-CLAIMED-NEW-001	KILLED (infer.).	143
FAIL-FIRSTFRONT-RESIDUAL-SCALE-IS-CONTROLLED-ONLY-BY-Q-001	KILLED (infer.).	143
FAIL-FIRSTFRONT-SCALAR-ROUTE-001	KILLED (mech.) with the scope below.	144
FAIL-FIXED-FINITE-MIRROR-HUB-001	KILLED.	144
FAIL-FIXED-K-UNIFORM-SELBERG-FREE-001	KILLED (infer.).	144
FAIL-FRESH-BRANCH-PRIMACY-001	SCOPE-CORR.	145
FAIL-FRIDGE-ALPHABET-TRANSFER-001	SCOPE-CORR.	145
FAIL-FRIDGE-BOUNDED-MULTIPLICITY-001	KILLED (mech.).	146
FAIL-FRIDGE-BRANCH-DOUBLE-CHARGE-001	SCOPE-CORR.	146
FAIL-FRIDGE-CONGRUENCE-CONTRADICTION-001	SCOPE-CORR.	147
FAIL-GAMMA-BOUND-ROUTES-001	KILLED for all four, as routes; the un	147
FAIL-GAMMA-POLYLOG-ASSUMED-UNIVERSAL-001	KILLED (infer.).	147
FAIL-GENERIC-ABSORPTION-001	KILLED on the generic/non-canonical dom	148
FAIL-GENERIC-AVERAGE-DEPTH-0-LOGLN-001	SUPERSEDED.	148
FAIL-GENERIC-PRIME-ROW-RPD-001	KILLED as a generic theorem.	149
FAIL-GREEDY-AVOIDANCE-NAIVE-001	KILLED as an argument. The conclusion	149

ID	Status	Str.
FAIL-HALL-SUPPORT-001	KILLED (mech.).	150
FAIL-HIGH-LOADONE-COFACTOR-CAN-HAVE-ARBITRARY-001	KILLED.	151
FAIL-HIGH-SEMIPRIME-SIBLING-REUSE-IS-CHEAP-001	KILLED.	151
FAIL-HIGH-VERTICAL-IS-INDEPENDENT-EXTERNAL-001	KILLED.	151
FAIL-HIGHFLOOR-DOMAIN-VACUITY-001	SCOPE-CORR. The theorem is PROVED; its	152
FAIL-IMPORT-V32-POSITIVE-K-TO-DEEP-LOW-OWN-001	KILLED.	152
FAIL-INCREMENTAL-J-IS-BOUNDED-BY-OLD-J-001	KILLED.	153
FAIL-INDIVIDUAL-MESOSCALE-BV-001	KILLED (infer.).	153
FAIL-INVERSE-SIBLING-INJECTIVE-001	KILLED (infer.).	154
FAIL-KERNEL-EAC-001	SCOPE-CORR.	154
FAIL-KPOWER-SPECTRUM-IS-OPTIMAL-DJ-BOUND-001	SUPERSEDED.	155
FAIL-LARGE-FRESH-SUPPORT-MAY-BE-MOSTLY-REPRESENTED-001	KILLED.	155
FAIL-LARGE-MONOMIAL-PACKET-CAN-OCCUR-IN-BOTH-001	KILLED.	156
FAIL-LARGEBASE-DEEP-VERTICAL-ESCAPE-001	KILLED as a live frontier on $q > N^{1/3}$	156
FAIL-LAYERED-GAMMALOGN-IS-CONTRADICTION-001	KILLED (infer.).	156
FAIL-LEDGER-INDEPENDENCE-LI5-SCOPE-001	SCOPE-CORR. POSITIVE-ROUND-LEDGER-IN	G- 157
FAIL-LEGACY-CONTINUUM-001	KILLED.	157
FAIL-LEGACY-ENCODING-001	KILLED.	158
FAIL-LEGACY-PIGEONHOLE-001	KILLED as a mechanism.	158
FAIL-LEGACY-WAVES-001	KILLED.	159
FAIL-LF2-AS-UPPER-BOUND-001	KILLED.	159
FAIL-LINEAR-SMOOTH-MIRROR-PACKET-CAN-STAY-KILLED-SUPPORTED-001	KILLED supported only on the recorded	160
FAIL-LITERAL-LINEAGE-001	KILLED.	160
FAIL-LOCAL-REUSE-001	KILLED by source-recorded computation.	161
FAIL-LOSSLESS-INTERFACE-001	KILLED for lossless transfer, under ev	161
FAIL-LOWBASE-POWERFUL-ROWS-AS-INDEPENDENT-001	KILLED (mech.) as an independent prima	162
FAIL-LOWCORE-CARDINALITY-FC8-WEIGHT-001	KILLED (infer.).	162
FAIL-LOWCORE-CHILDCLOSED-001	KILLED (infer.).	163
FAIL-LOWPOLYLOG-MULTIPLICITY-IS-PARENT-EXISTENCE-001	KILLED. Not a description of the current	164
FAIL-MANY-FIRSTFRONT-ROOTS-CAN-ALL-LIE-ABOUT-001	KILLED for $h > N^{1/3}$.	164
FAIL-MATCHING-CAPACITY-001	KILLED (infer.) — the matching theor	164
FAIL-MERGER-COUNT-IMPLIES-AMBIENT-J-001	KILLED (infer.).	165
FAIL-MERGER-WEIGHT-TO-J-TOPOLOGY-001	KILLED (infer.).	165
FAIL-MIRROR-FIXED-POLYLOG-RESERVOIR-001	SCOPE-CORR.	166
FAIL-MIRROR-MASS-ONE-SCALE-001	KILLED.	166
FAIL-MIRROR-MULTIPLICITY-BY-REPEATED-BASE-001	KILLED (infer.).	166

ID	Status	Str.
FAIL-MONOMIAL-MOD3-NOVELTY-vNEXT4.11	SCOPE-CORR (provenance correction).	167
FAIL-MONOMIAL-NEEDS-SPARSITY-001	KILLED (mech.).	167
FAIL-MONOMIAL-NODESCENT-CHEAP-001	KILLED.	167
FAIL-MONOMIAL-PATH-LENGTH-TWO-IS-ONLY-EMPIRICAL-001	KILLED.	168
FAIL-MONOMIAL-VERTICAL-SELF-SUFFICIENT-001	KILLED.	168
FAIL-MR2-CANONICAL-ONLY-001	KILLED as a scope claim.	168
FAIL-N1/4-IS-MULTIPLICITY-REALISATION-CEILING-001	KILLED.	168
FAIL-N1/6-DESIGNATED-BASE-BARRIER-001	KILLED (infer.).	169
FAIL-NAIVE-MU-LE-3ETA-001	KILLED.	169
FAIL-NATURAL-DENOMINATOR-IS-ONLY-BASELINE-001	KILLED.	169
FAIL-NESTED-PRIMEPOWER-PACKING-GIVES-CUMULATIVE-001	KILLED as a description.	170
FAIL-NONMONOMIAL-VERTICAL-IS-INDEPENDENT-001	KILLED as a description.	170
FAIL-ONE-REPRESENTATIVE-CONSUMES-MIRROR-MULTIPLICATION-001	KILLED (infer.).	170
FAIL-ORTHOGONAL-BRIDGE-001	KILLED.	171
FAIL-ORTHOGONALITY-AS-HYPOTHESIS-001	SCOPE-CORR.	172
FAIL-OWNED-RETURN-FIRST-EDGE-IS-FREE-SOURCE-001	KILLED in the v18 owner-completed doma	172
FAIL-OWNER-SEALED-RETURN-CAN-STAY-SQUAREFREE-001	KILLED.	172
FAIL-OWNERLESS-FIRST-OCCURRENCE-HAS-NO-ADDRESS-001	KILLED as a structure-001	173
FAIL-PO-INFORMATION-SELECTOR-001	KILLED (mech.). Not refuted; not estab	173
FAIL-PO-ZEROHOLE-ESSENTIAL-001	KILLED as a mechanism for the active/o	174
FAIL-P4-IS-THE-BEST-BV-ARITY-001	KILLED.	174
FAIL-PAIR-LOADED-DESCENT-OPEN-001	CLOSED AS A POLYLOG- () SCALE-REENTRY	175
FAIL-PAIRWISE-K-IS-THE-ONLY-POSSIBLE-X1-CANONICAL-001	KILLED.	175
FAIL-PARENT-LATTICE-DOMAIN-001	SCOPE-CORR.	175
FAIL-PATH-PRODUCT-AGGREGATION-ERASES-DEEP-VALUATION-001	KILLED for the inference [remote refl	176
FAIL-PER-ROW-RADC-COMPRESSION-001	KILLED as a failure certificate.	176
FAIL-PLAIN-AP-COUNT-KILLS-VERTICAL-001	SCOPE-CORR.	176
FAIL-POLYLOG-IS-NEEDED-FOR-DETERMINISTIC-SCALE-001	SCOPE-CORR.	177
FAIL-POSITIVE-ROUND-BLOCKS-REPEATED-FAN-PROCESSED-001	KILLED.	177
FAIL-POSITIVE-ROUND-CONSEQUENCE-001	KILLED (infer.). The depletion apparat	177
FAIL-POSITIVE-ROUND-EXHAUSTION-001	KILLED, quantitatively.	178
FAIL-POSITIVE-ROUND-LEDGER-CONSEQUENCE-001	KILLED for ledger-only derivations. No	179
FAIL-POSITIVE-SEMIPRIME-ROUND-IS-NEXT-PRIME-001	SCOPE-CORR.	179
FAIL-POTENTIAL-CAPACITY-001	KILLED.	180
FAIL-POWER-THRESHOLD-TO-POLYLOG-001	SCOPE-CORR.	181
FAIL-PRODUCT-BUDGET-ALONE-CANNOT-PRESERVE-CHARACTERISTICS-001	KILLED (infer.).	181
FAIL-PROFILE-CARRIER-001	SUPERSEDED.	181

ID	Status	Str.
FAIL-PURE-PATH-ONLY-RETAINS-MOD-Q-001	KILLED.	182
FAIL-PURE-Q-GRAPH-CAN-BRANCH-001	KILLED.	182
FAIL-Q0-LEDGER-001	KILLED.	183
FAIL-Q0-ONLY-OWNER-SEALED-RETURN-IS-CHEAP-KILLED.	KILLED.	183
FAIL-QFREE-FIRST-COPY-CAN-HIDE-BELOW-CARDINALITY-001	KILLED.	184
FAIL-QUOTIENT-SIEVE-POLYNOMIAL-RANGE-001	KILLED (mech.).	184
FAIL-RDS-GEOMETRY-DENSITY-001	KILLED (infer.).	184
FAIL-REFLECTION-ACTIVATION-001	KILLED.	185
FAIL-REPEATED-BASE-MUST-BE-FIXED-POLYLOG-001	KILLED.	186
FAIL-REPEATED-BASE-QUADRATIC-COLLISION-UNASSUMED-SCOPE-CORR.	SCOPE-CORR.	186
FAIL-REPEATED-HUB-QUADRATIC-COST-EQUALS-ACTUAL-EDRAGE-001	KILLED.	186
FAIL-RETURN-COEFFICIENT-K-IS-THE-INDEPENDENT-STRUCTURE-001	SUPERSEDED.	187
FAIL-RETURN-COFACTOR-IS-FREE-001	KILLED as a description.	187
FAIL-RETURNED-SEED-FIRST-OCCURRENCE-IS-J-001	KILLED.	187
FAIL-RETURNING-BASIN-IS-ABSOLUTELY-COSTFREE-SCOPE-CORR.	SCOPE-CORR.	187
FAIL-REUSE-DOUBLE-CHARGE-001	KILLED (infer.).	188
FAIL-ROOT-GENERATION-AC-ALONE-CLOSES-HQ-RP-SCOPE-CORR.	SCOPE-CORR.	188
FAIL-ROOTED-AMBIENT-001	KILLED.	188
FAIL-ROUTING-INDIFFERENCE-SCOPE-001	SCOPE-CORR. The theorem survives; its	189
FAIL-ROUTING-LOSES-SEED-WEIGHT-001	KILLED.	190
FAIL-ROUTING-RESCUES-P1BR-001	KILLED (mech.). The routing statement	190
FAIL-SATURATION-DOUBLE-CHARGE-001	KILLED (infer.).	191
FAIL-SEALED-MULTISOURCE-LOCK-001	KILLED (mech.).	191
FAIL-SELECTOR-CLOSES-CASCADE-001	KILLED (infer.). The selector is PROVE	192
FAIL-SHORTEN-PARENT-INTERVAL-TO-GET-P3-001	KILLED (mech.) relative to that stated	193
FAIL-SIMPLE-STATE-001	SUPERSEDED.	193
FAIL-SINGLE-BASE-CONCENTRATION-NEEDED-FOR-WHITEL-MIGRATION-001	KILLED.	194
FAIL-SINGLETON-STAR-LOCAL-WINDOW-001	KILLED (infer.).	194
FAIL-SIZE-COMPARISON-AGAINST-RECORDED-CAPACITY-SCOPE-CORR.	SCOPE-CORR.	195
FAIL-SMALL-GAMMA-MAKES-CANONICAL-BLOCK-INVERTIBLE-001	KILLED.	195
FAIL-STAR-SELF-REENTRY-001	KILLED.	195
FAIL-STRIP-COMMON-BASE-THEN-IMPORT-CLEAN-RUND-001	KILLED.	196
FAIL-TAGGED-P3-RESIDUE-FORCES-RETURN-001	KILLED (infer.) for every prime q_5 .	196
FAIL-TD-NAIVE-001	KILLED.	196
FAIL-TD-SMOOTH-INACTIVE-001	KILLED as a proof route resting on the	197
FAIL-TF-LOCAL-SCOPE-001	SCOPE-CORR.	199
FAIL-TWOVARIABLE-FULL-LEVEL-001	KILLED (infer.).	199
FAIL-UNIQUE-FRESH-TARGETS-MAY-FUNNEL-INTO-KH-LEVEL-001	KILLED. The high layer extracted in	200
FAIL-UNIQUE-MISSING-DATUM-001	SCOPE-CORR.	200

ID	Status	Str.
FAIL-UNIT-COMPLEMENT-001	KILLED.	201
FAIL-UNIVERSAL-BRIDGE-IMPOSSIBILITY-001	SCOPE-CORR. The mathe- matics is correct	201
FAIL-UPPERHALF-SCOPE-001	SCOPE-CORR.	202
FAIL-V13-EXTERNAL-VALUATION-AS-SEED-WEIGHTS-001	SCOPE-CORR.	202
FAIL-V18-SOURCE-STAR-COUNTEREXAMPLE-SPECIFIC-001	KILLED (infer.).	203
FAIL-V410-ANONYMOUS-STAR-COFACTOR-001	SUPERSEDED on the monomial branch.	203
FAIL-V47-DEEP-RPD-GLOBAL-GENERATION-LOCALISATION-001	SCOPE-CORR.	204
FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001	SCOPE-CORR.	204
FAIL-V48-HIGHTARGET-LOADONE-NECESSARY-001	SCOPE-CORR.	204
FAIL-V48-LOCAL-J-AS-AMBIENT-M-SCALE-001	SCOPE-CORR.	205
FAIL-V48-LOWCHILD-VERBATIM-DOMAIN-TRANSFER-001	KILLED as a direct domain application.	205
FAIL-V48-MONOMIAL-NOVELTY-001	SCOPE-CORR (provenance correction).	205
FAIL-V48-MULTIPARENT-SEPARATE-BRANCH-NECESSARY-001	SUPERSEDED as a neces- sary proof split.	205
FAIL-V48-POSITIVE-DEEP-SEMIPRIME-CLOSES-P1BR-001	KILLED for immediate clo- sure.	206
FAIL-V48-VARIABLE-M-SELFREENTRY-NEW-001	SUPERSEDED.	206
FAIL-V49-CROSSGEN-GCD-IS-P1BR-BRIDGE-001	KILLED as a sufficient gen- eral bridge.	206
FAIL-V49-DEEP-SEMIPRIME-AUTOMATIC-POSITIVE-SCOPE-CORR-001	SCOPE-CORR.	207
FAIL-V49-DIRECT-REFLECTED-FACTOR-INHERITANCE-001	KILLED/SCOPE-CORR.	207
FAIL-V49-LATER-DESCENDANTS-DESTROY-ALL-REFLECTED-ARITHMETIC-001	KILLED	207
FAIL-V49-RELATIVE-T-OR-FRESH-MASS-ALONE-CLASHES-001	KILLED as an immediate contradiction r	207
FAIL-V50-ANCESTRY-DIAGONAL-TRANSPORTS-DIRECT-001	SCOPE-CORR	208
FAIL-V50-BOUNDED-BRANCHING-CLOSES-SINGLESEED-001	KILLED as a weighted- capacity inferenc	208
FAIL-V50-REFLECTED-PROVENANCE-USELESS-AFTER-EDITION-001	KILLED on the return- heavy cheap- (J)	208
FAIL-V50-RETURNED-SEED-IMPLIES-CYCLE-001	KILLED.	208
FAIL-V50-SOURCEHEAVY-REFLECTED-SEEDS-CLOSE-001	KILLED.	209
FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIED-POWER-001	KILLED	209
FAIL-V51-ANCESTRY-LOCALISATION-FAILED-001	KILLED.	209
FAIL-V51-CYCLE-LOCK-FALSE-001	KILLED as a description of the theorem	210
FAIL-V51-UNWEIGHTED-REFLECTED-MATCHING-CARD-DEEP-ORBS-001	SCOPE-CORR	210
FAIL-VERTICAL-DESCENT-AUTOMATIC-BV-001	KILLED (infer.).	210
FAIL-VERTICAL-FC8-SEPARATION-001	SCOPE-CORR / KILLED (mech.).	211
FAIL-VERTICAL-HIGHGAP-UNBOUNDED-CAPACITY-001	KILLED (mech.).	211
FAIL-VERTICAL-SQRTLOG-WALL-001	SUPERSEDED within that conditional bra	211
FAIL-VERTICAL-TO-ETA-LOSES-LOG-001	KILLED (mech.).	212

ID	Status	Str.
FAIL-VERTICAL-UNLOCALISED-001	KILLED as a description.	212
FAIL-VNEXT433-UNIFORM-FLOOR-GAP-001	KILLED as a failure certificate; it mu	213
FAIL-WEIGHTED-REUSE-AMPLIFICATION-001	SCOPE-CORR.	213
FAIL-WHOLE-BLOCK-EXTERNAL-SET-LOSES-REFLECTION-MAINTENANCE-001	KILLED.	213
FAIL-WORLD-SIZE-NAIVE-001	KILLED as stated.	214
FAIL-v40-HIGH-FLOOR-N-OVER-LOG2-IS-SHARP-001	SUPERSEDED.	214
FAIL-v41-MONOMIAL-LENGTH-TWO-NOVELTY-001	SCOPE-CORR.	215
FAIL-BOUNDED-COFACTOR-ROWSLOTS-ARE-UNCONSTRAINED-001	KILLED.	215
FAIL-BOUNDED-COFACTOR-SUM-MUST-PAY-ONE-PER-KILLED.	KILLED.	215
FAIL-CHEAP-RADIC-OSCILLATION-HAS-A-HIDDEN-KILLED-Sink-counting diagnosis.	KILLED.	215
FAIL-COFACTOR-RENEWAL-IMPLIES-DIVERGENCE-001	KILLED.	216
FAIL-FIRST-COPY-HIGH-ROW-POPULATION-IS-BY-KILLED-CONTRADICTION-001	KILLED.	216
FAIL-FIXED-COFACTOR-PRIMALITY-GIVES-NO-MORE-KHANS-PACING-001	KILLED.	216
FAIL-HIGH-BASE-R-GREATER-Y-IMPLIES-RADIC-MONOTONICITY-001	KILLED.	216
FAIL-LARGE-COFACTOR-IS-AN-ANONYMOUS-COEFFICIENT-001	KILLED.	217
FAIL-LARGE-COFACTOR-RENEWAL-IS-DIVERGENCE-001	KILLED.	217
FAIL-PRODUCT-MODULUS-COMPRESSION-ALONE-CLOSES-DEEP-001	KILLED.	217
FAIL-RADIC-HALVING-IS-AUTOMATIC-DYNAMICAL-SCOPE-CORR.	SCOPE-CORR.	217
FAIL-RADIC-LEVEL-IS-MONOTONE-POTENTIAL-001	KILLED.	217
FAIL-RADIC-LEVEL-RETURN-ITSELF-PAYS-J-001	KILLED.	218
FAIL-SAME-EXACT-LEVEL-IS-A-NEW-EXTERNAL-RESTARTED as a packet-design inference.	KILLED.	218
FAIL-SELBERG-LOGSAVING-ALONE-CLOSES-DEEP-001	KILLED.	218

14.2 Entries

FAIL-ABOVE-BV-MASS-INVISIBLE-001

FAIL-ID:

FAIL-ABOVE-BV-MASS-INVISIBLE-001.

Date / iteration:

vNEXT4.19.

Old description:

Mirror mass above one scalar BV boundary is unavailable to clean selection.

Status:

SCOPE-CORRECTED.

Replacement:

Track both designated-base scale Q and roughness scale z . With polylogarithmic z , the clean designated-base range reaches $Q_* = N^{1/2}/\text{polylog } N$. Significant mass beyond Q_* forces a polynomial actual-state packet rather than disappearing.

FAIL-ABSORBED-FRIDGE-RESIDUAL-HAS-ARBITRARY-COFACTOR-001

FAIL-ID:

FAIL-ABSORBED-FRIDGE-RESIDUAL-HAS-ARBITRARY-COFACTOR-001.

Status:**KILLED**.**Rejected statement:**

A first-front residual prime $\ell \in (N/6, N/3)$ may occur with an arbitrary cofactor in $N - q$.

Certificate:

since $q < N/6$, $(N - q)/\ell$ is an odd integer strictly between 1 and 6, hence $N - q = 3\ell$ or $N - q = 5\ell$. The root-supported gcd then divides 3 or 5.

FAIL-AP2-HIDDEN-OMEGA-DEPENDENCY-001**FAIL-ID:**

FAIL-AP2-HIDDEN-OMEGA-DEPENDENCY-001.

Date / iteration:

vNEXT4.5 audit of vNEXT4.2.

Claim:

The recorded AP1/AP2 consequence follows merely from “standard Mertens/PNT estimates”.

Status:**SCOPE-CORRECTED**.**Exact repair:**

The support-capacity inequality contains $\omega_h(x) \log c_{\max}/t$, so one also needs

$$\omega_h(x) \leq \omega(x) = O\left(\frac{\log x}{\log \log x}\right).$$

With that explicit dependency, the quantified population conclusion survives.

Do not repeat:

Do not suppress the $\omega(x)$ dependency when invoking this population lemma.

FAIL-BABS-PRODUCTION-001**FAIL-ID:**

FAIL-BABS-PRODUCTION-001.

Date / iteration:

Canonical bridge audit §4, 31 August 2026.

Claim:

Every failed world produces a nontrivial absorbed base product $B_{\text{abs}} > 1$, which is therefore the universal carrier of failure mass.

Original motivation:

B_{abs} is the natural object of ambient promotion and was observed nontrivial in most instances.

Status:**SUPERSEDED**.**Minimal / strong counterexample:**

$(N, r) = (128, 79)$.

Exact certificate:

Full- Q production is nontrivial for this root while the recorded B_{abs} equals 1. Production and absorption are different channels, and only the former is always nonvacuous on the composite-complement domain.

Failed hypothesis or mechanism:

One channel of a four-channel ledger was promoted to the whole ledger.

Why the argument failed:

B_{abs} collects exactly the residual primes with $\kappa_R(\ell) = m_\ell$; nothing forces that set to be nonempty.

What survives:

ARS-BABS-INTERNAL, that $\mathcal{B}_{\text{abs,int}} = \mathcal{B}_{\text{abs}}$ under failure — which explicitly yields *no* nonvacuity

conclusion — and B_{abs} as one channel and the proper object of ambient promotion.

Safe weaker version:

Carry failure mass on $\mathfrak{F}_Q = \prod_{q \in Q_r} M_q$, not on B_{abs} .

Current replacement:

Theory Core, G-FULLQ-PRODUCTION, which gives $\mathfrak{F}_{Q_r} > (5/3)^{|Q_r|} \text{rad}(m_r) > 1$ unconditionally on the composite-complement domain.

Do not repeat:

Do not make one ledger channel universal.

Related current target:

$P1^{\text{br}}$, K1.

FAIL-BAKER-CORE-FAN-DIRECTLY-GIVES-LEVELC-RPD-WITNESS-001

FAIL-ID:

FAIL-BAKER-CORE-FAN-DIRECTLY-GIVES-LEVELC-RPD-WITNESS-001.

Status:

SCOPE-CORRECTED.

Correction:

G-BAKER-SPARSE-EXACT-LAYER-FAN packages Baker's analytic theorem together with extra CRD domain assumptions and an upper-half conclusion. The v4.46 witness uses only the analytic sparse-moduli input and repeats the endpoint subtraction on a different fixed proportional interval.

What survives:

the Level-C prime-row no-go is a legitimate new arithmetic consequence, but its provenance must be stated correctly.

FAIL-BOUNDED-ARITY-WEIGHT-001

FAIL-ID:

FAIL-BOUNDED-ARITY-WEIGHT-001.

Date / iteration:

Bombieri-Vinogradov four-ary continuation, 1 September 2026.

Claim:

Once every selected parent has at most four unintended children, the large-state packet E is automatically a lower-order cost, or the clean parent family closes P1BR by cardinality alone.

Status:

KILLED.

Minimal / strong counterexample:

Structural; one unintended prime factor is already enough.

Exact certificate:

A squarefree selected row may have

$$N - p = \ell r, \quad r > N^{1/4-\varepsilon},$$

with $r \in R$. The four-ary conclusion permits this configuration, and the only universal weight estimate it supplies is

$$\log r = \log(N - p) - \log \ell.$$

Thus even $\omega(E_\ell) = 1$ permits the unintended label to carry the entire non-base cofactor mass. More generally

$$\log E \leq L(\mathcal{P}) - \log \prod \ell,$$

with equality when every unintended child lies in R .

Failed hypothesis or mechanism:

An unweighted degree bound was used as a weighted logarithmic capacity bound.

Why the argument failed:

The number of factors and the sum of their logarithms are different functionals. Roughness makes each surviving factor larger, not cheaper.

What survives:

The four-ary theorem turns E into a bounded-degree dynamic packet. The exact clean source-surplus identity then cancels the base copy of every internal unintended factor and exposes its true cost as $(\kappa_R(r) - 1) \log r$.

Safe weaker version:

Use bounded arity only for dynamic branching and population statements. For P1BR prove an external-mass-versus-rooted-reuse inequality with logarithmic weights.

Current replacement:

Theory Core v1.6, G-CLEAN-MULTIPARENT-SURPLUS-IDENTITY and G-BV-FOUR-ARY-DISPOSITION.

Do not repeat:

$\omega(E) = O(|A_0|)$ does not imply $\log E = o(L(\mathcal{P}))$.

Related current target:

P1BR, K1, large-layer disposition.

FAIL-BUNDLE-ENLARGES-CAPACITY-001**FAIL-ID:**

FAIL-BUNDLE-ENLARGES-CAPACITY-001.

Date / iteration:

vNEXT4.1, Phase 7.

Claim:

That preserving a canonical bundle D in the lifted row forces a canonical contribution — to the denominator of (RC4'), or to a state-difference packing budget — and so extracts more from R than an arbitrary external row does.

Original motivation:

The lifted row shares exact prime powers with a canonical reflected row, and empirically bundle rows are much more often nonpositive than single-base rows on the same reflected row (87% against 72% in COMP-VNEXT41-DISPOSITION). Nonpositive rounds are the ones with a recorded consumer, so this looked exploitable.

Status:

KILLED.

Exact certificate:

Theory Core G-EXTERNAL-INTERFACE-CAPACITY. For any globally coprime family $\mathcal{P}$ of external rows and any failed world R , the integers $\gcd(N - p, N - u)$, $p \in \mathcal{P}$, are pairwise coprime divisors of $N - u$, so $\sum_p \sum_u \log \gcd(N - p, N - u) \leq \sum_u \log(N - u) < |R| \log N$. The bound has *no* hypothesis on how $\mathcal{P}$ was chosen. Consequently at most $2|R| < 6.6\gamma \log N$ members with $N - p > \sqrt{N}$ can be nonpositive, whatever the provenance; the empirical bias towards nonpositivity is therefore not exploitable beyond that ceiling.

Failed hypothesis or mechanism:

Confusing a richer *description* of a row with a larger *charge* on R . The charge is a gcd against the fixed integers $N - u$, and those integers have finitely many divisors.

What survives:

The bundle lift, the exact valuations and the mirror relation, all **PROVED**; and the ceiling (IC3), which is new and which also strengthens the vNEXT4 statement (MD1) by dropping the designated-base hypothesis.

Do not repeat:

Before claiming a selection strategy extracts more from R , check it against (IC1), which is provenance-blind.

Related current target:

P1BR-ADDITIVE-COUPLING.

FAIL-BV-BOUNDARY-DESTROYS-MASS-001**FAIL-ID:**

FAIL-BV-BOUNDARY-DESTROYS-MASS-001.

Date / iteration:

vNEXT4.15.

Claim:

Low-child states above the current BV-accessible range lose the natural reflected mass because they cannot be cleanly selected.

Status:**KILLED**.**Exact correction:**

Above $Y_{BV} = N^{1/(6+2\vartheta)}(\log N)^{-O(1)}$, each distinct state already carries $(1/6 - o(1)) \log N$ weight. A proportional family therefore retains the original $t \log \Lambda$ scale as weighted state mass.

Do not repeat:

The analytic boundary changes the type of object, not its total natural logarithmic weight.

FAIL-BV-INACCESSIBILITY-HIDES-MIRROR-MASS-001**FAIL-ID:**

FAIL-BV-INACCESSIBILITY-HIDES-MIRROR-MASS-001.

Date / iteration:

vNEXT4.17; interpretation superseded by vNEXT4.19.

Old description:

Above the then-used same-scale BV threshold the mirror packet becomes analytically invisible.

Status:**SUPERSEDED**.**Replacement:**

Already vNEXT4.17 converted significant above-threshold mass to distinct active states. vNEXT4.19 further shows that the relevant selector geometry has two independent scales Q, z , so the designated-base range extends almost to $\sqrt{N}$ when z is polylogarithmic.

FAIL-C-SYMBOL-COLLISION-001**FAIL-ID:**

FAIL-C-SYMBOL-COLLISION-001.

Date / iteration:

vNEXT2 independent audit.

Claim:

The symbol c may be read uniformly across the Theory Core.

Original motivation:

Both quantities are small integers attached to a packet and were introduced in different rounds.

Status:**SCOPE-CORRECTED**.**Minimal / strong counterexample:**

Internal. In (LE1)–(LE6), $c = |D_S^{\text{src}}|$ is the *source count*; in (FC7)–(FC10), $c = |C|$ is the *low-core cardinality*. For a canonical R the first is 1 and the second is typically large, so any statement that combines an LE identity with an FC bound while reading c uniformly is wrong by a factor that grows with the world.

Exact certificate:

(LE4) reads $\eta(S) + \nu(S) \geq |S| + c$ with $c = 1$ for a canonical rooted world, whereas (FC8) reads $J_R < N^{c+\tau}(25/18)^\tau / \prod_{q \in C \setminus Q} q$ with $c = |C|$. Substituting one into the other is a category error.

Failed hypothesis or mechanism:

Symbol reuse across rounds without a concordance entry.

What survives:

Both families of statements, unchanged, on their own readings.

Safe weaker version:

Use $\gamma = |C|$ for the low core and reserve c for the source count, as fixed by Convention [conv:c-collision] in the Theory Core. (FC7)–(FC10) are not restated; their c is γ .

Current replacement:

Theory Core, Convention [conv:c-collision].

Do not repeat:

Do not introduce a symbol already bound in an earlier round without a concordance line.

Related current target:

(FC8), K1.

FAIL-CAB-001**FAIL-ID:**

FAIL-CAB-001.

Date / iteration:

Post-v7.5 CAB audit, 30 August 2026.

Claim:

For every canonical stopped world, $\log B_{\text{abs}} \leq A(\log U_R + \log J_R)$ for a fixed finite A .

Original motivation:

Every absorbed base atom was expected to be financed by power excess or support reuse.

Status:

KILLED without a failure condition.

Minimal / strong counterexample:

$N = 322, p_0 = 163$.

Exact certificate:

$N - p_0 = 159 = 3 \cdot 53$, and $322 - 3 = 319 = 11 \cdot 29$. The audited stopped world has $U_R = J_R = 1$ but $B_{\text{abs}} = 11 \cdot 29 > 1$. The other first-front state 53 is GOOD because $322 - 53 = 269 \in \mathbb{P}$.

Failed hypothesis or mechanism:

A successful-world local ledger was treated as a failure-world capacity law.

Why the argument failed:

Absorption can occur on the bad branch while another first-front branch already succeeds, leaving zero proposed denominator.

What survives:

This witness does not refute a theorem explicitly conditioned on a closed canonical failure.

Safe weaker version:

Any CAB repair must state the failure domain and a nonzero source term; no such repair is currently proved.

Current replacement:

C0b-RD and the branchwise production program.

Do not repeat:

Never cite $N = 322$ as refuting a failure-conditioned theorem, and never cite CAB unqualified.

Related current target:

$P1^{\text{br}}, K1$.

FAIL-CAB-INTERNAL-001

FAIL-ID:

FAIL-CAB-INTERNAL-001.

Date / iteration:

Post-v7.5 CAB audit, 30 August 2026.

Claim:

CAB becomes valid in every canonical stopped world after discarding boundary atoms and retaining only internal absorbed atoms.

Original motivation:

The $N = 322$ defect looked purely external.

Status:

KILLED without a failure condition.

Minimal / strong counterexample:

$N = 858, p_0 = 431, \ell = 23$.

Exact certificate:

$N - p_0 = 427 = 7 \cdot 61$, $858 - 7 = 851 = 23 \cdot 37$, and the audited stopped world has $U_R = J_R = 1$. The atom 23 is internal and absorbed; 37 is a boundary atom. The state 61 is GOOD because $858 - 61 = 797 \in \mathbb{P}$.

Failed hypothesis or mechanism:

Boundary deletion was assumed to remove the zero-payment phenomenon.

Why the argument failed:

The same phenomenon occurs at an internal atom on a bad branch of a globally successful rooted traversal.

What survives:

Again, this successful-world witness does not refute a statement explicitly conditioned on canonical failure.

Safe weaker version:

None is currently proved; require a genuine failure-specific source mechanism.

Current replacement:

C0b-RD and targets $L1, E1, K1$.

Do not repeat:

“Internal” does not mean “already paid”.

Related current target:

$K1$.

FAIL-CANONICAL-BT-CAPACITY-TRANSPORT-TO-EXTERNAL-HQ-001

FAIL-ID:

FAIL-CANONICAL-BT-CAPACITY-TRANSPORT-TO-EXTERNAL-HQ-001.

Status:

SCOPE-CORRECTED (domain failure).

Killed transfer:

apply the canonical fringe Brun–Titchmarsh prime-power packing theorem unchanged to all of $H_Q \setminus R$.

Certificate:

its decisive population cap is tied to canonical fringe paths / the γ -controlled canonical domain. External descendants do not inherit that cap.

What survives:

the theorem applies on its canonical domain and the overlap $H_Q \cap R$ is negligible in the v4.45 high-band accounting.

Do not repeat:

replacing γ by $|H_Q|$ is a new crude estimate, not a legal transport of the canonical theorem.

FAIL-CANONICAL-EXACT-LAYER-TRANSFER-MISSING-001

FAIL-ID:

FAIL-CANONICAL-EXACT-LAYER-TRANSFER-MISSING-001.

Status:

SCOPE-CORRECTED.

Killed formulation:

the v18–v32 exact-layer/star machinery is external and cannot be applied to an exact layer already contained in the canonical world R .

Certificate:

v4.26 already contains C-CANONICAL-POLYLOG-STAR-ABSORPTION; exact-layer parent slots, quotient inflation and cofactor conservation are generic exact-layer results. Moreover v4.35 shows that an already extracted polynomial canonical vertical layer has only the full-scale receiving channels J_R , U_R^{nr} and D_C .

What was actually missing at v4.35:

a termination/payment theorem for repeated q -free $U \rightarrow U$ migration, or a legal generation/owner transfer to the v28–v32 package. v4.36 then supersedes this dynamic issue as the primary vertical frontier by giving a static upper capacity for the whole U -mass.

FAIL-CASCADE-DIRECTION-001

FAIL-ID:

FAIL-CASCADE-DIRECTION-001.

Date / iteration:

vNEXT3.1 correction round.

Claim:

G-CASCADE-GAMMA-TERMINUS: the (FC10) cascade closes precisely on $\gamma \leq (\log N)^A$, proved by bounding $\sum_i M_i \log z_i \ll (\log N)^A \log N$.

Original motivation:

The selector supplies at most one clean row per designated base and the bases are polylogarithmically bounded, so the total production looked easy to estimate.

Status:

KILLED as stated and as proved.

Minimal / strong counterexample:

The proof itself. (DP5) states that *simultaneous nonpositivity implies* $\sum_i M_i \log z_i < \log J_R$; to conclude that some round is positive one must therefore exhibit a **lower** bound $\sum_i M_i \log z_i \geq \log J_R$. The vNEXT3 proof supplied an **upper** bound $\sum_i M_i \log z_i \ll (\log N)^A \log N$ and used it in that place. An upper bound on a quantity can never certify that the quantity is large; the argument establishes nothing, in either direction.

Failed hypothesis or mechanism:

The two roles of $\sum_i M_i \log z_i$ — what (DP5) concludes about it, and what must be supplied to invoke (DP5) contrapositively — were interchanged.

Why the argument failed:

Counting the *number* of available bases bounds production from above. Forcing a positive round needs the *logarithmic mass* of the bases actually used, bounded from below.

What survives:

The intended conclusion, correctly proved, with the quantifiers made explicit and the domain moved to the global-failure DAG: see G-SELECTABLE-CLEAN-LOG-MASS and G-GAMMA-POLYLOG-FORCES-POSITIVE-ROUND in the Theory Core. The correct relation between the exponents is $A > B + 1$, with B fixed first.

Safe weaker version:

None; the statement must be reproved from a lower bound, which is what the replacement does.

Current replacement:

Theory Core, Theorem [thm:clean-log-mass] and Theorem [thm:gamma-forces-positive].

Do not repeat:

Before invoking a contrapositive, check which side of the inequality must be supplied and which side is concluded.

Related current target:

γ , P1BR-POSITIVE-ROUND-CONSEQUENCE.

FAIL-CENSUS-DOMAIN-001**FAIL-ID:**

FAIL-CENSUS-DOMAIN-001.

Date / iteration:

Canonical bridge audit §12, item 2.

Claim:

The recorded figure “301” counts all foundational failed upper roots in the scanned range.

Original motivation:

The label was abbreviated in later reports.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

The unit-complement row, already registered as FAIL-UNIT-COMPLEMENT-001.

Exact certificate:

The scan enumerated *composite-complement* failed upper roots. A row with $N - r = 1$ also fails, has no children, and is outside that census by construction; $(N, r) = (20, 19)$ is the smallest instance.

Failed hypothesis or mechanism:

A domain-qualified count was quoted without its qualifier.

Why the argument failed:

Every theorem keyed to that census inherits the composite-complement hypothesis.

What survives:

The number itself, with its domain label attached, as a regression fixture. Status **VERIFIED COMPUTATION**.

Safe weaker version:

Always write “301 composite-complement failed upper roots in the recorded scan”.

Current replacement:

Benchmark suite below, where the domain label is preserved.

Do not repeat:

Never quote a census figure without its domain.

Related current target:

C0a and every generic failed-root lemma.

FAIL-CHILD-C1-001**FAIL-ID:**

FAIL-CHILD-C1-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

The selected child-prime normalization has universal charging constant $C = 1$.

Original motivation:

Every selected parent financed at most one audited base atom in the focused incidence table.

Status:

KILLED by source-recorded computation.

Minimal / strong counterexample:

The campaign records 53 violations; the empirical maximum is 9.42243 at $N = 406996$.

Exact certificate:

The numerical ratio and the location of its maximum are preserved from the final report. The individual raw child-mass table was unavailable in this extraction, so this certificate was not independently recomputed.

Failed hypothesis or mechanism:

A one-to-one incidence observation was used as a weighted logarithmic inequality.

Why the argument failed:

Multiplicity of financed atoms and size of their weights are different quantities.

What survives:

A more generous partner-prime normalization had empirical maximum 1, but remains computational only.

Safe weaker version:

Treat every finite observed constant as a conjectural target until an exact source-sensitive bound is proved.

Current replacement:

Theory Core target $K1$.

Do not repeat:

Never promote the best observed normalisation to a universal constant.

Related current target:

$K1$.

FAIL-CLEAN-ROOTED-CAP-P1-001**FAIL-ID:**

FAIL-CLEAN-ROOTED-CAP-P1-001.

Date / iteration:

Clean-support-load continuation, 1 September 2026.

Claim:

Once one selects pairwise-coprime upper-half external parents, the rooted capacity $\mathcal{C}_N(R) = L(R)$ supplied by the clean base-layer theorem can be beaten by the cofactor production of those same parents, closing P1BR.

Status:

KILLED.

Minimal / strong counterexample:

Structural; no finite witness is needed.

Exact certificate:

In a canonical failed world, every non-root state $r \in R \setminus \{p_0\}$ lies below $N/3$, hence $N - r > 2N/3$. Every selected upper-half parent satisfies $p > p_0$, hence $N - p < m = N - p_0 < N/2$. With at most one selected parent per distinct internal base label, $|\mathcal{P}| \leq |R| - 1$, so

$$L(\mathcal{P}) < (|R| - 1) \log m < \log m + (|R| - 1) \log(2N/3) < L(R).$$

Thus the exact rooted upper bound $K_\times + K(\mathcal{P}) \leq L(R)$ is too large for this naive production choice to exceed.

Failed hypothesis or mechanism:

Solving the collision-selection problem was mistaken for solving the production-versus-capacity comparison.

Why the argument failed:

The clean theorem makes external complements small and noncolliding, but the failed-world rows are individually larger: non-root rows exceed $2N/3$, whereas selected upper-half-parent complements are below $m < N/2$. The order geometry points in the wrong direction for $L(\mathcal{P}) > L(R)$.

What survives:

G-CLEAN-POLYLOG-PARENT-SELECTION and G-CLEAN-BASE-ROOTED-CAPACITY remain exact **PROVED** statements. They close the W1/K1 collision side on the fixed-polylog base-layer subdomain.

Safe weaker version:

Use the clean selection to isolate the remaining cost: the base-copy surcharge from unintended labels of R . A future P1BR route must control that smaller support set, or use a different common production/capacity functional; it may not simply take $L(R)$.

Do not repeat:

“Pairwise coprime parents” is a capacity statement, not a production surplus.

Related current target:

P1BR, W1, K1.

FAIL-COFACTOR-ENTROPY-ALONE-CLOSES-LOADONE-001**FAIL-ID:**

FAIL-COFACTOR-ENTROPY-ALONE-CLOSES-LOADONE-001.

Status:

KILLED.

Killed inference:

once cross-row cofactor radical reuse is identified with incremental J , a large load-one packet automatically has ambient $\Omega(M \log N)$ cofactor entropy.

Certificate:

a common tiny factor such as 3 across M rows carries only $(M - 1) \log 3 = O(M)$ entropy while designated high labels have $\Theta(M \log N)$ mass.

What survives:

cofactor entropy is exact horizontal storage, but the correct next reduction is the cube-root row trichotomy / boundary argument.

FAIL-COFACTOR-SIBLING-PRIME-UNSCOPED-001**FAIL-ID:**

FAIL-COFACTOR-SIBLING-PRIME-UNSCOPED-001.

Date / iteration:

v1.22 cofactor-window continuation, 1 September 2026.

Claim:

For every internal returned fresh seed, the cofactor $r_j = (N - w_j)/\ell_j$ is prime.

Status:

KILLED.

Why the inference fails:

v1.21 proves only that all prime factors of r_j lie in U and are at least $s = \min U$. Primality follows in v1.22 only when $r_j < s^2$, supplied by the bounded window $r_j < 2\Lambda h < h^2 \leq s^2$ under no descent.

What survives:

On $b_j \leq \Lambda h$ with $s \geq h > 2\Lambda$, the return cofactor is prime and yields the sibling packet of TF-COFACTOR-WINDOW-SIBLING-RENEWAL.

Do not repeat:

Do not replace cofactor roughness by primality outside the explicit short-window inequality.

Related current target:

TF, above-BV return transport.

FAIL-CRD2-NEEDED-FOR-FIRST-OCCURRENCE-001

FAIL-ID:

FAIL-CRD2-NEEDED-FOR-FIRST-OCCURRENCE-001.

Status:

KILLED as a present engineering requirement.

Killed inference:

first/second occurrence, reflected provenance, parenthood, cofactor data or no-double-charge ownership currently require a new fundamental dynamic state variable.

Certificate:

the v4.39 audit finds all theorem-relevant objects reconstructible from full CRD1 together with derived subsets/matchings. No pair with equal full CRD1 state and different theorem-relevant deterministic outcome has been exhibited.

What survives:

introduce CRD2 only if a future proof exhibits non-reconstructible data, genuine history dependence, same-state/different- outcome behaviour, or irreducible persistent ownership. Until then use CRD1-R / CRD1.5 as a derived relation-preserving interface.

FAIL-CRD2-PREMATURE-001

FAIL-ID:

FAIL-CRD2-PREMATURE-001.

Date / iteration:

vNEXT4.3–vNEXT4.4.

Claim:

Because the $J, U, \eta, \kappa, \Delta$ ledgers forget parent or bundle provenance, CRD1 must be replaced immediately by a richer fundamental CRD2 state space.

Status:

DESIGN PROHIBITION / FRONTIER CERTIFICATE; not a claim that CRD2 will never be needed.

Exact certificate:

The full ambient object

$$\mathfrak{C}_N = (N, \mathcal{A}_N, p_0, \nu_N), \quad \nu_N(p, q) = v_q(N - p),$$

already reconstructs active supports/valuations, parent sets, sibling and factorisation hyper-edges, exact cofactors $(N - p)/q^e$, inactive residuals, and reflected provenance (j, a_j, c_j) . The proven vNEXT4 independence certificates concern compressed ledgers, not equality of the full CRD1 state.

What survives:

Use a relation-preserving derived view (CRD1-R/CRD1.5) when a theorem must keep provenance alive.

Migration trigger:

CRD2 becomes justified only after a theorem-relevant non-reconstructible datum, genuine history dependence, a full-state information-loss pair, or an irreducible ownership state is exhibited.

Related current target:

C-VERTICAL-LOWCORE-CAPACITY; CRD2 is dormant.

FAIL-CRITICAL-VERTICAL-CARRIES-TLOGN-001

FAIL-ID:

FAIL-CRITICAL-VERTICAL-CARRIES-TLOGN-001.

Date / iteration:

vNEXT4.7.

Claim:

On $h \geq \sqrt{N/3}$, the block-derived vertical family can still absorb a positive proportion of a $t \log N$ -scale obligation.

Status:

KILLED.

Exact certificate:

The monomial collapse puts every vertical base in $(\sqrt{2N/3}, \sqrt{N})$, so standard prime counting gives

$$|\mathcal{V}| \ll \sqrt{N}/\log N = O(t/\log N),$$

and exactly one vertical layer per base gives

$$\log U_{\mathcal{V}} \ll \sqrt{N} = O(t) = o(t \log N).$$

What survives:

The critical square-root branch itself is not closed, because production of enough large block factors degenerates there. What is closed is the possibility that an already-produced critical block family can hide its logarithmic cost in vertical layers.

Related current target:

C-MONOMIAL-REFLECTED-PAIR-SPARSITY.

FAIL-CRITICAL-VERTICAL-GENERIC-001**FAIL-ID:**

FAIL-CRITICAL-VERTICAL-GENERIC-001.

Date / iteration:

vNEXT4.7.

Claim:

Near the critical square-root gap, the surviving vertical branch still has arbitrary prime-power/cofactor structure.

Status:

KILLED as a description.

Exact certificate:

On

$$h \geq \sqrt{N/3},$$

every block-derived vertical event is exactly

$$N - w = q^2.$$

What survives:

A specific monomial reflected/quadratic prime-pair correlation, not generic U_R -capacity.

FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001**FAIL-ID:**

FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001.

Status:

KILLED.

Killed inference:

a linear aggregate defect on bases $q \leq N^{1/3}$ automatically yields one base of polynomial multiplicity.

Certificate:

there are $\asymp N^{1/3}/\log N$ available prime bases; for a packet of size $M \asymp N^{1/3}$, plain pigeonhole forces only $\gg \log N$ average multiplicity.

Do not repeat:

do not replace the current low-layer consumer by an unweighted single-base pigeonhole.

FAIL-CUBEROOT-PIGEONHOLE-IS-BOUNDARY-CASE-001**FAIL-ID:**

FAIL-CUBEROOT-PIGEONHOLE-IS-BOUNDARY-CASE-001.

Status:

SCOPE-CORRECTED.

Earlier failure:

FAIL-CUBEROOT-DEFECT-PIGEONHOLE-GIVES-POLYNOMIAL-STAR-001 correctly kills raw cube-root pigeonhole as a concentration theorem.

New explanation:

this is exactly the boundary value $m = 2$ of **G-POWER-THRESHOLD-DEFECT-COMPRESSION**: the threshold $N^{1/(m+1)}$ is then $N^{1/3}$ and the guaranteed star exponent $1/3 - 1/(m+1)$ is zero. Every fixed $m \geq 3$ gives a nontrivial polynomial star.

What survives:

do not infer polynomial concentration from cube-root pigeonhole alone; first split the alphabet at a lower power threshold.

FAIL-CYCLE-COFACTOR-LOCK-ALONE-CLOSES-DEEP-CYCLES-001**FAIL-ID:**

FAIL-CYCLE-COFACTOR-LOCK-ALONE-CLOSES-DEEP-CYCLES-001.

Source:

explicit v4.53 Failure Log addition.

Status:

KILLED as a sufficient uniform closure mechanism.

Reason:

the exact relation

$$Q^L D = mN + (-1)^L$$

is useful, but in the small- Q regime it does not by itself bound the population of non-pure cycles at the required polynomial scale.

FAIL-DC-CAPACITY-COMPRESSION-001**FAIL-ID:**

FAIL-DC-CAPACITY-COMPRESSION-001.

Status:

SCOPE-CORRECTED.

Killed form:

within the Q -indexed scalar route of **C-Q-INDEXED-SCALAR-PRODUCTION-SUBCRITICALITY**, increasing D_C cannot lower FC8 below the hard $|Q|$ -scale floor needed by that same production.

What survives:

once a proposed production violates the one-logarithmic-unit per- Q -index hypothesis, the exact term $-\log D_C$ in A_R matters again. Do not promote this entry to a universal ban on denominator compression.

FAIL-DEADSMOOTH-CAN-CARRY-POLYNOMIAL-HIGHGAP-PACKET-001**FAIL-ID:**

FAIL-DEADSMOOTH-CAN-CARRY-POLYNOMIAL-HIGHGAP-PACKET-001.

Status:

KILLED.

Killed inference:

the dead-smooth carrier or reflected classes may carry a fixed polynomial fraction of a high-gap reflected packet.

Certificate:

v4.40 counts integers supported on the fixed alphabet $\text{PF}(x)$ by exponent vectors and obtains only $x^{o(1)}$ possibilities below $2x$. Hence $|\text{DS}|, |\text{RDS}| = N^{o(1)} = o(t)$ on $h > N^{1/3}$.

Compatibility:

FAIL-DEADSMOOTH-UNBOUNDED-001 remains valid. It rejected a finite/uniform inference from the old local ceiling. The new statement is asymptotic fixed-support counting and does not assert that two dead-smooth rows are impossible or that the class is absolutely bounded.

FAIL-DEADSMOOTH-PILLAI-001**FAIL-ID:**

FAIL-DEADSMOOTH-PILLAI-001.

Date / iteration:

Phase-3 continuation, 31 August 2026.

Claim:

No canonical N carries two simultaneous dead-smooth carriers, since none was found below $3 \cdot 10^5$.

Original motivation:

The search returned nothing at all, and the configuration looks arithmetically absurd.

Status:

SCOPE-CORRECTED. The *search result* stands as COMP-DEADSMOOTH-CENSUS; the *exclusion* does not.

Minimal / strong counterexample:

None is needed: the claim has no proof.

Exact certificate:

By TD-DEADSMOOTH-NORMALFORM(5), two simultaneous dead-smooth carriers with $\omega(a_j) = 1$ are exactly a Pillai coincidence $|q^e - q'^{e'}| < h$ with $q^e, q'^{e'} > 4x/5$, $q, q' < h$ and $qq' \mid x$. Excluding that unconditionally requires effective linear forms in logarithms. The corpus contains no such argument.

Failed hypothesis or mechanism:

An empty finite search was read as an exclusion theorem.

Why the argument failed:

This is PROHIB-STATUS-LADDER: a finite computation, however clean, is not a universal statement.

What survives:

The exact reduction to a Pillai equation, which is a genuine normal form and the right object for a future attack.

Safe weaker version:

“No such pair occurs below $3 \cdot 10^5$ ”, labelled **VERIFIED COMPUTATION**.

Current replacement:

Theory Core, TD-DEADSMOOTH-NORMALFORM(5) and COMP-DEADSMOOTH-CENSUS.

Do not repeat:

Never convert an empty search into an exclusion.

Related current target:

TD-SMALLSMOOTH.

FAIL-DEADSMOOTH-UNBOUNDED-001**FAIL-ID:**

FAIL-DEADSMOOTH-UNBOUNDED-001.

Date / iteration:

Phase-3 continuation, 31 August 2026.

Claim:

The exceptional dead-smooth class is negligible for every N , so the side condition $\lceil t/3 \rceil > \sigma_N^+(h)$ of TD-COB-CLOSURE can be discharged and removed.

Original motivation:

Dead-smooth carriers are extraordinarily rare — 19 among 514 447 tested, none doubled — which invites the conclusion that the class is uniformly negligible.

Status:

KILLED.

Minimal / strong counterexample:

$N = 60\,060$, $p_0 = 30\,047$, $h = 17$, $t = 8$.

Exact certificate:

$x = 30\,030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ is a primorial, so $\mathcal{P}_N(h) = \{3, 5, 7, 11, 13\}$ — every odd prime below h divides x . The ceiling then degenerates: $\sigma_N^+(h) = 6.32$ while $\lceil t/3 \rceil = 3$. The side condition fails.

Failed hypothesis or mechanism:

Rarity of the class in a sample was confused with a bound valid for every N . The ceiling depends on how many small primes divide x , and x may legitimately be a primorial.

Why the argument failed:

σ^{dead} and σ^+ are small *because* x usually has few small prime factors, not because the dead-smooth normal form is intrinsically bounded. No theorem forbids $x = \prod_{\ell \leq h} \ell$.

What survives:

TD-DEADSMOOTH-CEILING and TD-COB-CLOSURE exactly as stated, with the side condition retained as a hypothesis. It is an arithmetic condition on N alone, checkable in $O(\pi(h) \log x)$ operations, and it held in every tested canonical instance with $t \geq 16$.

Safe weaker version:

State the closure with the side condition, and treat the small-smooth- x regime as a separate open case.

Current replacement:

Theory Core, TD-COB-CLOSURE and the retained target TD-DENSITY.

Do not repeat:

Do not discharge a hypothesis because a sample makes it look automatic; check the extremal arithmetic first.

Related current target:

TD-DENSITY, $P1^{\text{br}}$.

FAIL-DEEP-AFFINE-PATH-RESIDUE-IS-CAPACITY-001**FAIL-ID:**

FAIL-DEEP-AFFINE-PATH-RESIDUE-IS-CAPACITY-001.

Alias:

v4.50 R50-1.

Status:

KILLED for the capacity inference.

Exact theorem retained:

for any actual path $v_0 \rightsquigarrow p$,

$$Q \mid N - p \implies \Xi(\mathcal{P}) \equiv v_0 N^{-1} \pmod{Q}.$$

Why rejected as capacity:

this residue is forced by the endpoint through the exact affine path identity and imposes no independent restriction on intermediate vertices or cofactors.

Replacement theorem:

G-DEEP-PATH-RESIDUE-ENDPOINT-TAUTOLOGY.

FAIL-DEEP-FIBRE-NEEDS-FIRSTENTRY-OWNER-ESCAPE-001**FAIL-ID:**

FAIL-DEEP-FIBRE-NEEDS-FIRSTENTRY-OWNER-ESCAPE-001.

Source:

explicit v4.54 Failure Log addition and correction to v4.53.

Status:**KILLED** as a proof-design necessity.**Reason:**

every pure path terminates at a non-pure row, its length is $N^{o(1)}$, and the refined fibre theorem already forces polynomial boundary defect/off-layer support without first passing through ancestry predecessors.

Supersession:

C-DEEP-FIRST-ENTRY-OWNER-CAPACITY is **SUPERSEDED** as the binding frontier. The v4.53 first-entry matching remains a valid secondary local diagnostic.

FAIL-DEEP-FIBRE-TO-SUBPOLY-BY-ANCESTRY-ALONE-001**FAIL-ID:**

FAIL-DEEP-FIBRE-TO-SUBPOLY-BY-ANCESTRY-ALONE-001.

Source:

explicit v4.52 Failure Log addition.

Status:

KILLED as a proof mechanism from one reflected root, common-sink topology, total path cofactor products, and bounded branching alone.

Reason:

the current corpus yields renewal alternatives rather than a subpolynomial fibre bound.

Scope guard:

this does not refute a stronger arithmetic fibre theorem using edge-local valuation or additional selected-family structure.

FAIL-DEEP-ONE-AP-PACKING-CLOSES-001**FAIL-ID:**

FAIL-DEEP-ONE-AP-PACKING-CLOSES-001.

Status:**KILLED** as a sufficient closing mechanism.**Killed route:**

use only the fact that a selected row/index lies in one residue class modulo the deep modulus Q .

Reason:

the resulting ceiling of shape $1 + N/(6Q)$ is too weak when $Q = N^{o(1)}$.

FAIL-DEEP-PURE-QMONOMIAL-CAN-CYCLE-001**FAIL-ID:**

FAIL-DEEP-PURE-QMONOMIAL-CAN-CYCLE-001.

Source:

explicit v4.53 Failure Log addition.

Status:**KILLED**.

Killed possibility:

the pure subgraph defined by $N - p = Qq$ can support a nontrivial active cycle.

Reason:

the affine fixed-point identity of v4.53 forbids every nontrivial active cycle in the pure $d = 1$ subgraph.

Scope guard:

non-pure cycles with variable multipliers d_i are not ruled out by this statement; they satisfy the separate multiplier law.

FAIL-DEEP-SELFRENEWAL-IS-INDEPENDENT-TERMINAL-001**FAIL-ID:**

FAIL-DEEP-SELFRENEWAL-IS-INDEPENDENT-TERMINAL-001.

Source:

explicit v4.53 Failure Log addition and v4.53 frontier correction.

Status:

KILLED as a proof-design description.

Killed description:

a cheap polynomial deep fibre may require an independent global theorem controlling exact-layer self-renewal as a terminal phenomenon.

Reason:

the v4.53 stability theorem forces the cheap fibre into short pure affine paths whose starts expose boundary owners / off-layer support.

Supersession:

C-DEEP-SELF-RENEWAL-QCOFACTOR-CAPACITY is **SUPERSEDED** as the standalone binding frontier. This does not say all renewed external population is contradictory.

FAIL-DEEP-U-TO-U-RETURN-AS-PRIMARY-FRONTIER-001**FAIL-ID:**

FAIL-DEEP-U-TO-U-RETURN-AS-PRIMARY-FRONTIER-001.

Status:

SUPERSEDED as a primary frontier.

Superseded description:

v4.35 temporarily treated repeated migration of deep vertical mass to new q -free bases as the smallest obstruction.

Certificate:

v4.36 proves, pending audit, the static global capacity

$$\log U_R^{\text{nr}} \ll \sqrt{\frac{N\gamma}{\log(2 + \gamma)}},$$

without proving dynamic termination.

What survives:

nested peeling remains useful for structural analysis of near-extremal packets, but it is not necessary merely to prevent unbounded vertical storage.

FAIL-DESCENT-ONE-STATE-001**FAIL-ID:**

FAIL-DESCENT-ONE-STATE-001.

Date / iteration:

vNEXT4.15.

Old description:

Passing immediately to the full descendant closure reduces the descent branch to the existence

of one state $r \leq h$.

Status:

SUPERSEDED.

Replacement:

G-FIRST-ROW-HIGHLOW-DISPOSITION retains the selected first parent rows and yields full logarithmic defect, weighted state expansion, explicit horizontal reuse, a fixed-polylog population, or a quantitative reciprocal band.

FAIL-DIRECT-ADJACENCY-001

FAIL-ID:

FAIL-DIRECT-ADJACENCY-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

Every internally absorbed atom is incident, or uniformly adjacent, to a GOOD escape edge in its first-front branch.

Original motivation:

Most observed witnesses were shallow.

Status:

KILLED for direct adjacency.

Minimal / strong counterexample:

$N = 14198$, internal atoms 19 and 167.

Exact certificate:

The audited sets are $B_{\text{abs,int}} = \{19, 167\}$ and $B_{\text{re}} = \{17, 83, 149\}$. Both internal atoms use the same merged GOOD edge $71 \rightarrow 277$, reached only at branch distances 8 and 7, respectively. Atom 19 has 93 external parents and atom 167 has 12; neither is directly incident to that GOOD edge.

Failed hypothesis or mechanism:

Typical shallow depth was promoted to a local theorem.

Why the argument failed:

Branches can merge long before their eventual escape.

What survives:

A nonlocal branch-escape disjunct remains viable.

Safe weaker version:

Reachability of a GOOD edge, with no fixed adjacency bound.

Current replacement:

Theory Core target $E1$.

Do not repeat:

Do not charge only the nearest one or two edges.

Related current target:

$E1, W1$.

FAIL-ETA-EQUIVALENT-TO-WEIGHTED-J-001

FAIL-ID:

FAIL-ETA-EQUIVALENT-TO-WEIGHTED-J-001.

Status:

KILLED.

Rejected inference:

an unweighted lower bound on $\eta(R) = \sum_q (\kappa_R(q) - 1)$ is equivalent to the corresponding lower bound on $\log J_R$.

Reason:

exactly

$$\log J_R = \sum_q (\kappa_R(q) - 1) \log q,$$

while only $\log J_R \geq \eta(R) \log 3$ follows from the unweighted count.

FAIL-EXACTD-NONCANONICAL-001**FAIL-ID:**

FAIL-EXACTD-NONCANONICAL-001.

Date / iteration:

Exact-D audit; reaffirmed in the canonical bridge audit §11.2.

Claim:

The Exact-D local row algebra — pair compensation, smoothness, primitive normal form, prime endpoint — is enough to isolate the canonical root, so canonicity need not be used as boundary transport.

Original motivation:

The local conditions are sharp and eliminate most models.

Status:

KILLED as a substitute for minimality; **SCOPE-CORRECTED** as machinery.

Minimal / strong counterexample:

The recorded noncanonical normal forms, including $(q, r, m_0) = (3, 5\,413\,103, 81\,196\,545)$ — note $81\,196\,545 = 3 \cdot 5 \cdot 5\,413\,103$ — together with the two recorded $k = 1$ examples.

Exact certificate:

These models satisfy the local D_q, D_r conditions, smoothness, pair failure and a prime endpoint, and are eliminated *only* by $h < G$. Removing $h < G$ restores them.

Failed hypothesis or mechanism:

Row algebra was asked to do the work of the empty prime interval.

Why the argument failed:

Local normal forms constrain a row; they do not constrain the position of the root relative to x .

What survives:

ARS-EXACTD: pair compensation outside $q < r < 2h$, the primitive normal form, and $h^2 > 4m$ in the residual regime — all under an Exact-D *pair failure* hypothesis, not under CRD failure alone.

Safe weaker version:

Use Exact-D only where the pair-failure hypothesis is explicit, and state $h < G$ as a separate input.

Current replacement:

Theory Core, ARS-EXACTD and the ablation discussion of $h < G$.

Do not repeat:

Do not attempt to repair Exact-D by improving a constant; the corpus already localises the missing step as interval-to-ledger transport.

Related current target:

C0a, T_D , $P1^{\text{br}}$.

FAIL-EXACTD-RESIDUAL-SURVIVES-SUBROOT-001**FAIL-ID:**

FAIL-EXACTD-RESIDUAL-SURVIVES-SUBROOT-001.

Date / iteration:

vNEXT4.20.

Branch:

Retain Exact-D Pair Compensation failure as an obstruction when $h \leq \sqrt{x}$.

Status:

KILLED.

Certificate:

Existing Exact-D high-gap amplification says pair failure forces $h^2 > 4m$. But $h \leq \sqrt{x}$ gives $h^2 \leq x < 4m$ for sufficiently large x . Thus all first-front SMOOTH pairs satisfy Exact-D compensation on this domain.

Scope guard:

This closes only the pair-failure branch; it does not by itself strengthen the global weighted-cut/K1 bound.

FAIL-EXTERNAL-ETA-IDENTIFICATION-001**FAIL-ID:**

FAIL-EXTERNAL-ETA-IDENTIFICATION-001.

Date / iteration:

vNEXT2 independent audit, Phase 3.

Claim:

The high-floor reuse $\eta(U) \geq \lceil n/3 \rceil + c$ of an external orthogonal packet can be charged to the canonical capacity ledger.

Original motivation:

η is the same functional in both places, and a large value looks like a large obligation wherever it occurs.

Status:

KILLED.

Minimal / strong counterexample:

Structural, and it is exactly the orthogonality that was thought to help.

Exact certificate:

The packet is orthogonal, $K_{\times}(R, U) = 0$. That identity says precisely that U contributes nothing to the rooted collision ledger. Any functional built from cross-collision is therefore identically zero on this configuration, so no such functional can carry $\eta(U)$ into $\mathcal{C}_N(R)$.

Failed hypothesis or mechanism:

Orthogonality was read as a licence to add the two ledgers; it is the obstruction to doing so.

What survives:

TF-HIGH-FLOOR-HORIZONTAL-RIGIDITY itself, now **PROVED** with both of its skeleton gaps closed.

Safe weaker version:

Report $\eta(U)$ as a property of U only.

Current replacement:

Theory Core, Proposition [prop:missing-bridge], which states the exact minimal lemma required: a *source-side* statement that re-closing $R \cup U$ changes the multisource product by a factor whose logarithm is bounded below in terms of $\eta(U)$.

Do not repeat:

Never add an orthogonal packet's ledger to the rooted one; the zero cross-term is the reason it cannot be done.

Related current target:

TF, K1, W1.

FAIL-EXTERNAL-FIXED-POLYLOG-SQUAREFREE-PENDING-001**FAIL-ID:**

FAIL-EXTERNAL-FIXED-POLYLOG-SQUAREFREE-PENDING-001.

Date / iteration:

vNEXT4.15.

Old obstruction:

The fixed-polylog external BAD clean bridge lacked an explicit squarefree-parent theorem.

Status:

CLOSED PENDING INDEPENDENT AUDIT.

Repair:

Large-square thinning applies to the external BAD parent fans as well, yielding squarefree, pairwise-coprime complements with exact designated valuation one and no unintended fixed-polylog factors.

FAIL-EXTERNAL-MERGER-FIRST-SPLIT-001**FAIL-ID:**

FAIL-EXTERNAL-MERGER-FIRST-SPLIT-001.

Date / iteration:

vNEXT4.14.

Strategy:

Partition external seeds into merger/disjoint subfamilies before analysing source retention.

Status:**SUPERSEDED** as proof organisation.**Replacement:**

Union all failed seed basins first. The union is child-closed below $N/3$, source retention applies to the whole family, and one overlap ledger covers either orthogonality or nonempty intersection with R .

FAIL-EXTERNAL-MULTISOURCE-CRT-IS-THE-NEXT-CAPACITY-BRIDGE-001**FAIL-ID:**

FAIL-EXTERNAL-MULTISOURCE-CRT-IS-THE-NEXT-CAPACITY-BRIDGE-001.

Status:**KILLED** relative to the current data.**Rejected strategic description:**

The next optimal step is to force a contradiction from source-front CRT compatibility of the external v18 star.

Reason:

the source-front lock constrains one affine coefficient but does not upper-bound it, and the non- q_0 moduli are largely pairwise coprime, so CRT compatibility is not itself a capacity obstruction.

Current replacement:

attack the canonical root-tied target **C-FIRSTFRONT-RESIDUAL-TO-LOWCORE-CAPACITY**; retain K1-INC and owner-anchored incremental capacity as secondary global routes.

FAIL-EXTERNAL-SOURCE-MASS-IS-INDEPENDENT-RADICAL-RESERVOIR-001**FAIL-ID:**

FAIL-EXTERNAL-SOURCE-MASS-IS-INDEPENDENT-RADICAL-RESERVOIR-001.

Status:**SUPERSEDED** as a primary factorisation branch.**Superseded description:**

external seed/source weight must remain as a third independent radical-storage reservoir in addition to fresh support and incremental J .

Certificate:

v4.39 proves the exact extension identity

$$\prod_{a \in A} \text{rad}(N - a) = F_R(A) \frac{J_S}{J_R}.$$

All new complement radical mass is therefore stored in one baseline copy of fresh support plus rooted/repeated copies in incremental J .

What survives:

Σ_{ext} remains useful for source-sensitive congruence/product arguments, but it is not needed as an independent primary factorisation leaf.

FAIL-EXTERNAL-SUCCESS-001**FAIL-ID:**

FAIL-EXTERNAL-SUCCESS-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

If an internal atom has an active parent outside the current bad world, then that parent supplies a successful rooted escape.

Original motivation:

Leaving the current closed set was conflated with entering the global success basin.

Status:

KILLED.

Minimal / strong counterexample:

$(N, \ell, p) = (6142, 877, 3511)$.

Exact certificate:

$3511 \in \mathcal{A}_{6142}$, it lies outside the audited p_0 -world, and $6142 - 3511 = 2631 = 3 \cdot 877$; hence it is an external parent of 877. The independently checked stopped traversal rooted at 3511 fails.

Failed hypothesis or mechanism:

“External to R ” was replaced by “outside the global bad basin.”

Why the argument failed:

A different failed basin can feed the same atom.

What survives:

The parent is an exact ambient incidence and can be promoted into an enlarged valuation ledger.

Safe weaker version:

External active parent *or* internal branch escape, with no success claim about the parent.

Current replacement:

Theory Core, G-AMBIENT-PROMOTION and targets $E1, W1, K1$.

Do not repeat:

Existence of a parent is not success of that parent.

Related current target:

$E1, W1$.

FAIL-EXTERNAL-UT-IDENTITY-IS-UPPER-CAPACITY-001**FAIL-ID:**

FAIL-EXTERNAL-UT-IDENTITY-IS-UPPER-CAPACITY-001.

Status:

KILLED.

Killed inference:

the exact relative identity $T_{S^*}/T_R = (U_{S^*}/U_R)(J_{S^*}/J_R)$ and source lock automatically bound

the external vertical factor from above.

Certificate:

increasing an exact valuation from 1 to e enlarges Δ_U without changing the support graph, source set, or source-lock residue class. The known canonical upper capacity is for canonical U_R^{nr} , not arbitrary external incremental U .

What survives:

the identity is a strong exact payment ledger, but a new receiving-capacity theorem would be required to close it.

FAIL-F1-DOMAIN-MIX-001

FAIL-ID:

FAIL-F1-DOMAIN-MIX-001.

Date / iteration:

Interrupted T_D round, dry run, 31 August 2026.

Claim:

A falsifier run over mixed controls can refute a candidate statement whose domain is “closed failed world”.

Original motivation:

A single sweep over many roots was cheaper than three separate domains.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

The dry run itself.

Exact certificate:

The recorded log states that the candidate F1 had domain “closed failed world” while the code mixed in successful-envelope controls; the domain filter and the counter for fresh states belonging to R were corrected, and three separate domains were preserved thereafter. This is a **VERIFIED COMPUTATION** record of a tooling defect, not a mathematical failure of F1.

Failed hypothesis or mechanism:

Domain bookkeeping inside the falsifier, not the statement under test.

Why the argument failed:

A counterexample from the wrong domain is not a counterexample.

What survives:

The candidate statement, untouched, still awaiting a correctly scoped test.

Safe weaker version:

Every falsifier must declare and enforce its domain before it may report a kill.

Current replacement:

The benchmark suite below, in which every row carries its domain and evidence class.

Do not repeat:

Never let a falsifier report a kill it obtained outside the declared domain.

Related current target:

TF, T_D .

FAIL-FC8-FLOOR-FOR-RHS-001

FAIL-ID:

FAIL-FC8-FLOOR-FOR-RHS-001.

Status:

KILLED.

Rejected inference:

proving $\log J_R > (|Q| + \tau) \log N$ automatically contradicts FC8.

Reason:

$(|Q| + \tau) \log N$ is only a lower floor for

$$A_R = (\gamma + \tau) \log N + \tau \log \frac{25}{18} - \sum_{q \in C \setminus Q} \log q.$$

The exact target must use the exact right-hand side, not a lower bound for it.

FAIL-FC8-SELF-CONTRADICTION-001**FAIL-ID:**

FAIL-FC8-SELF-CONTRADICTION-001.

Date / iteration:

vNEXT2 independent audit, Phase 2.

Claim:

The low-core capacity bound (FC8), being an upper bound on J_R under canonical failure, can be contradicted by the forced lower bounds already available, closing the $\mathcal{D}$ /low-core front.

Original motivation:

(FC8) is the first genuine upper capacity for J_R , and upper capacities are what a contradiction architecture wants.

Status:

KILLED. (FC8) itself is **PROVED**; the inference from it is not.

Minimal / strong counterexample:

Structural; no finite witness needed.

Exact certificate:

The only unconditional lower bound on J_R available from canonical failure is $J_R \geq 3^{\eta(R)}$ by (LE3). A contradiction with (FC8) therefore requires $\eta(R) > (\gamma + \tau) \log_3 N + \tau \log_3(25/18) - \sum_{C \setminus Q} \log_3 q$, that is $\eta(R) \gtrsim \gamma \log N$. Nothing in the corpus forces a reuse count carrying a factor $\log N$; the strongest available floor is $\eta(R) \geq |Q| + |H|$ (Theorem [thm:lowcore-floor]), which is bounded by $|R|$.

Failed hypothesis or mechanism:

An upper capacity was treated as self-contradictory without a matching lower bound of the same order.

Why the argument failed:

The two sides differ by a factor $\log N$ per low-core element; that gap is not an accident of effort but of what the available identities can produce.

What survives:

(FC8) as an upper capacity usable through the external comparison (FC10), exactly as its own capacity paragraph states.

Safe weaker version:

Report (FC8) only in the (FC10) form.

Current replacement:

Theory Core, Proposition [prop:fc8-barrier].

Do not repeat:

Do not attempt another self-contained contradiction from (FC8); the barrier is at the level of available bounds.

Related current target:

K1, P1BR.

FAIL-FC8-SLACK-IS-VERTICAL-MASS-001**FAIL-ID:**

FAIL-FC8-SLACK-IS-VERTICAL-MASS-001.

Status:**SCOPE-CORRECTED**.**Killed inference:**a large $\Delta_{\text{FC8}} = B_R - \log(D_C J_R)$ by itself forces a comparably large $\log U_R^{\text{nr}}$.**Certificate:**

v4.35 gives the exact identity

$$\Delta_{\text{FC8}} = \log U_R^{\text{nr}} + \Delta_{\text{full}}, \quad \Delta_{\text{full}} > 0.$$

The positive full-layer row/path envelope slack is an independent part of the FC8 slack.

What survives:LCAP1/LCAP4 may still require vertical mass after the size of $D_C J_R$ has been specified.**FAIL-FIRSTENTRY-SQUARE-ONLY-HAS-Q-SPACING-001****FAIL-ID:**

FAIL-FIRSTENTRY-SQUARE-ONLY-HAS-Q-SPACING-001.

Source:

explicit v4.54 Failure Log addition.

Status:**KILLED**.**Killed description:**the square first-entry packet can only be spaced using one residue class modulo Q .**Correction:**

a pure-path start satisfies

$$p_i \equiv N(1 - Q) \pmod{Q^2},$$

so distinct square first-entry targets have spacing at least $2Q^2$.**FAIL-FIRSTFRONT-CONTAINMENT-CLAIMED-NEW-001****FAIL-ID:**

FAIL-FIRSTFRONT-CONTAINMENT-CLAIMED-NEW-001.

Status:**KILLED**.**Rejected description:** $Q \subseteq C$ was missing from the previous Core.**Certificate:**

it is explicitly used in the independently audited FC7/FC8 proof and audit. v4.34 only extracted it for a new scale comparison.

FAIL-FIRSTFRONT-RESIDUAL-SCALE-IS-CONTROLLED-ONLY-BY-Q-001**FAIL-ID:**

FAIL-FIRSTFRONT-RESIDUAL-SCALE-IS-CONTROLLED-ONLY-BY-Q-001.

Status:**KILLED** on $h \leq \sqrt{x}$.**Rejected statement:**The only general lower scale for $M_q = (N - q)/\gcd(m, N - q)$ is the old $M_q > (5/3)q$ bound.**Certificate:**canonical shadow geometry gives $D_q = \gcd(m, |2h - q|)$ and uniformly $M_q > (5/6)h$; when $q \leq 4h$, $M_q > (5/6)\Lambda h$.

FAIL-FIRSTFRONT-SCALAR-ROUTE-001

FAIL-ID:

FAIL-FIRSTFRONT-SCALAR-ROUTE-001.

Status:

KILLED with the scope below.

Killed inference:

a production indexed by $S \subseteq Q$, contributing less than one ambient logarithmic unit per index, can close the current FC8 capacity without multiplicity amplification or a new structural upper bound.

Certificate:

$Q \subseteq C$ is already proved and the exact FC8 right-hand side A_R satisfies $A_R > |Q| \log N$, whereas such a production is $< |Q| \log N$.

What survives:

first-front shadow geometry, H_3/H_5 structure, multiplicity-amplified arguments, independent capacity compression, analytic input, and non-scalar contradictions are not killed.

FAIL-FIXED-FINITE-MIRROR-HUB-001

FAIL-ID:

FAIL-FIXED-FINITE-MIRROR-HUB-001.

Date / iteration:

vNEXT4.16.

Escape:

Store all active smooth-mirror valuation mass on an N -independent finite family of small primes.

Status:

KILLED.

Exact certificate:

The mirror support capacity

$$M_{\text{act}}^- \leq t \sum_{r \in P_{\text{mir}}} \frac{\log r}{r-1} + |P_{\text{mir}}| \log N$$

is only $O(t) + O(\log N)$ for fixed finite support, while the natural smooth branch requires $M_{\text{act}}^- \gg t \log \Lambda$ with $\Lambda \rightarrow \infty$.

Scope guard:

This does not say the direct least-prime children of the original parent rows cannot all equal 3; it says the mirror side must then create additional growing active support.

FAIL-FIXED-K-UNIFORM-SELBERG-FREE-001

FAIL-ID:

FAIL-FIXED-K-UNIFORM-SELBERG-FREE-001.

Date / iteration:

vNEXT4.12.

Claim:

A standard fixed- k Selberg sieve immediately gives

$$\#\{q : q, N - kq^2 \text{ prime}\} \ll \sqrt{N/k} / \log^2 N$$

with one absolute constant uniformly over all polylogarithmic odd k .

Status:

KILLED.

Exact obstruction:

The fixed- k local root count contains the quadratic character (kN/p) ; uniform singular-series

control is an additional theorem obligation.

Replacement:

Sum over k first and use the hybrid dimension-one linear sieve in the coefficient variable.

FAIL-FRESH-BRANCH-PRIMACY-001

FAIL-ID:

FAIL-FRESH-BRANCH-PRIMACY-001.

Date / iteration:

Phase-3 continuation, 31 August 2026.

Claim:

The external fresh branch $\mathcal{D}_{\text{fr}}^{\text{ext}}$ is where the $\mathcal{D}$ obligation is strongest, so cut statements should be developed there first and restricted to it.

Original motivation:

External carriers are candidate *new sources* and therefore the ones that bear on $P1^{\text{br}}$; that relevance was transferred to the cut itself.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

Structural, by TD-LOCALIZED-CUT.

Exact certificate:

For $\tau \geq 2$ distinct active primes bounded by y , $C_{\text{cut}} > \tau(\tau - 1) \log((N - y)/y)$. A fresh carrier gives only $y = N/6$ and hence the constant $\log 5$; a live smooth carrier gives $y < h$ and hence $\log((N - h)/h)$, larger by orders of magnitude once $h = o(N)$. The fresh branch is therefore the *weakest* of the three for cut purposes.

Failed hypothesis or mechanism:

Relevance to $P1^{\text{br}}$ was conflated with strength of the cut.

Why the argument failed:

Two different functionals were being compared. Only external carriers can be new sources; but the cut constant is governed by the *localisation* of the primes, and small primes localise better.

What survives:

Both facts, kept apart. Only the external branch bears on $P1^{\text{br}}$; the live-smooth branch gives the largest cut defect.

Safe weaker version:

State the cut with the explicit bound y and read the constant off it, rather than fixing $\log 5$.

Current replacement:

Theory Core, TD-LOCALIZED-CUT, with TD-FRESH-CUT and TD-FIVEFOLD-CUT as its $y = N/6$ cases.

Do not repeat:

Do not restrict a theorem to the sub-branch that happens to be relevant downstream; state it where its proof lives.

Related current target:

T_D , $P1^{\text{br}}$.

FAIL-FRIDGE-ALPHABET-TRANSFER-001

FAIL-ID:

FAIL-FRIDGE-ALPHABET-TRANSFER-001.

Date / iteration:

vNEXT2 independent audit, Phase 4.

Claim:

For $\lambda_N > 3$ the upper fringe is charged by the alphabet $\{\lambda, 2\lambda - 1\}$, by analogy with $\{3, 5\}$.

Original motivation:

$\{3, 5\} = \{\lambda, 2\lambda - 1\}$ at $\lambda = 3$, and (AN7) features $2\lambda - 1$.

Status:

SCOPE-CORRECTED.

Exact certificate:

The admissible coefficients are determined by (AN5), $N(1 - 1/\lambda)/q < b < N/q$, together with b odd, $(b, N) = 1$, $b \geq \lambda$. For q ranging over $(N/(2\lambda), N/\lambda)$ this interval has length $N/(\lambda q) \in (1, 2)$ only near the top of the band; lower in the band it admits several odd values, so the alphabet is a range, not a pair. The two-letter alphabet at $\lambda = 3$ is an accident of $2 < b < 6$ admitting exactly $\{3, 5\}$.

Failed hypothesis or mechanism:

A coincidence at one value of λ was read as a pattern.

What survives:

(AN5) itself, which is the correct general statement and the right starting point: compute the alphabet from the interval before proposing any potential.

Safe weaker version:

For general λ , state the admissible coefficient set as the odd integers coprime to N in the (AN5) interval.

Current replacement:

Theory Core, Phase-4 note in Section [sec:vnext2].

Do not repeat:

Do not propose a monotone potential of $D(q) = 4q - N$ type for $\lambda > 3$ before the alphabet is computed; D was built from the coefficient 3.

Related current target:

$\lambda_N > 3$ fringe capacity.

FAIL-FRIDGE-BOUNDED-MULTIPLICITY-001**FAIL-ID:**

FAIL-FRIDGE-BOUNDED-MULTIPLICITY-001.

Status:

KILLED.

Certificate:

the first-front-to-fringe map is already injective and the H_3, H_5 bands are disjoint. Thus there is no hidden large fibre to overflow.

What survives:

an independent theorem proving that the relevant incidences cannot exist at all, or cannot coexist with canonical high-gap geometry, remains possible.

FAIL-FRIDGE-BRANCH-DOUBLE-CHARGE-001**FAIL-ID:**

FAIL-FRIDGE-BRANCH-DOUBLE-CHARGE-001.

Status:

SCOPE-CORRECTED.

Killed inference:

count the same fringe entrance/path population τ as a new independent scalar payment after FC8/LCAP4 already price those rows.

What survives:

exact $3/5$ arithmetic may still yield a forbidden configuration, a new nonlocal relation, or GOOD. Structural use is not double charging.

FAIL-FRIDGE-CONGRUENCE-CONTRADICTION-001

FAIL-ID:

FAIL-FRIDGE-CONGRUENCE-CONTRADICTION-001.

Status:

SCOPE-CORRECTED.

Certificate:

the finite exact example $N = 4298$ in the revised report shows that the bare congruence pattern $q \mid m$, $N - q = b\ell$ with $b \in \{3, 5\}$ is arithmetically realisable with several incidences.

Scope:

that example is outside the canonical high-gap/sub-root failed domain. It kills a contradiction from the bare congruences alone, not a future theorem using the missing canonical/high-gap or primality input.

FAIL-GAMMA-BOUND-ROUTES-001

FAIL-ID:

FAIL-GAMMA-BOUND-ROUTES-001.

Date / iteration:

vNEXT2 independent audit, Phase 2.

Claim:

The adaptive indegree bound, the denominator of (FC8), a distinct resource per low-core element, or the layered Euler defect bounds $\gamma = |C|$.

Original motivation:

Each route looks like it converts local structure into a global count.

Status:

KILLED for all four, as routes; the underlying statements are untouched.

Exact certificate:

(i) Summing (AN2) against Theorem [thm:lowcore-floor] gives $\gamma + |H| \leq \sum_{q \in C} \lceil N/(6q) \rceil$, which every summand already satisfies term by term. (ii) Nothing distinguishes $C \setminus Q$ from an arbitrary set of distinct odd primes, so only the primorial bound applies, and Corollary [cor:primorial-ceiling] then gives $\gamma \log \gamma \lesssim 2\gamma \log N$, i.e. $\gamma \lesssim N^2$. (iii) A single row can host $\log_3 N$ low prime factors, so the parent count bounds γ only by $|R| \log_3 N$. (iv) The layered Euler route produces Theorem [thm:lowcore-floor] itself, which bounds $\eta(R)$ below and the low-core primorial mass above, and γ in neither direction.

What survives:

Theorem [thm:lowcore-floor] and Corollary [cor:primorial-ceiling], both new and both genuine.

Do not repeat:

Do not retry these four; a bound on γ needs a mechanism that is not a summation of an existing local bound.

Related current target:

(FC8), K1.

FAIL-GAMMA-POLYLOG-ASSUMED-UNIVERSAL-001

FAIL-ID:

FAIL-GAMMA-POLYLOG-ASSUMED-UNIVERSAL-001.

Status:

KILLED.

Rejected inference:

$\gamma = |C| = O(\log N / \log \log N)$, and hence $t \gg \gamma$, holds throughout the relevant canonical domain.

Reason:

the standard divisor-count bound concerns $\omega(m)$ and hence $|Q|$, not γ . Existing polylogarithmic statements for γ are hypotheses on subdomains, not universal conclusions.

Do not repeat:

do not compare reflected population t with γ without a proved theorem giving that comparison.

FAIL-GENERIC-ABSORPTION-001**FAIL-ID:**

FAIL-GENERIC-ABSORPTION-001.

Date / iteration:

Post-v7.5 generic absorption audit.

Claim:

Every residual label in a closed world is paid by within-row power, cross-row reuse, or the proposed generic absorption channel.

Original motivation:

$T = UJ$ suggested that no residual singleton could remain unpaid.

Status:

KILLED on the generic/noncanonical domain.

Minimal / strong counterexample:

$N = 1718, r = 1109$, not the canonical root (the canonical selector is the diagonal 859).

Exact certificate:

$N - r = 609 = 3 \cdot 7 \cdot 29$, and

$$1718 - 7 = 1711 = 29 \cdot 59, \quad D_7 = 29, \quad M_7 = 59.$$

The chosen closed recurrent core contains 59, only the row 7 contains the factor 59, and $1718 - 59 = 1659 = 3 \cdot 7 \cdot 79$. Thus the audited 59-valuations of both the within-row and cross-row payment terms are zero.

Failed hypothesis or mechanism:

A generic identity was asked to produce canonical source information.

Why the argument failed:

A singleton residual can enter the state set without being paid by the proposed local channels.

What survives:

$T = UJ$ itself and any future statement that uses genuine canonical minimality essentially.

Safe weaker version:

Keep the generic identity; add a separately proved canonical transport statement before asserting absorption.

Current replacement:

Theory Core, C0b-RD and branch targets.

Do not repeat:

Do not derive a canonical trichotomy from Factor-Flow alone.

Related current target:

$T_D, T_F, P1^{\text{br}}$.

FAIL-GENERIC-AVERAGE-DEPTH-O-LOGLN-001**FAIL-ID:**

FAIL-GENERIC-AVERAGE-DEPTH-O-LOGLN-001.

Status:

SUPERSEDED.

Old diagnostic:

size, exact valuation, parent-slot and scalar Brun-Titchmarsh admit Level-B integer packets with full-scale vertical depth but too little horizontal multiplicity for an $O(\log \log N)$ average-depth bound.

Superseded by:

FAIL-GENERIC-PRIME-ROW-RPD-001, which realises the same obstruction on actual prime

rows.

FAIL-GENERIC-PRIME-ROW-RPD-001

FAIL-ID:

FAIL-GENERIC-PRIME-ROW-RPD-001.

Status:**KILLED** as a generic theorem.**Killed statement:**for every packet X of genuine prime rows below $N/3$,

$$\sum_{Y < r \leq N^{1/3}} \sum_{k \geq 2} (n_k^X(r) - 1)_+ \ll (\log \log N) \sum_{Y < r \leq N^{1/3}} (\kappa_X(r) - 1)_+ + o(|X|).$$

Certificate:

using the analytic sparse-moduli theorem underlying Baker's exact-layer fan, choose a mesoscale prime $r > Y$ and an exact depth $e \asymp (\log \log N)^3$. A fixed proportional interval below $N/3$ contains $N^{1-o(1)}$ primes in the required exact residue class. After Brun–Titchmarsh/Mertens control of unintended band supports one can select $h = N^{1/3+o(1)}$ prime rows with

$$R_X \gg h(\log \log N)^3, \quad \eta(X) = O(h \log \log N).$$

Scope guard:

this is Level C only. It does not show that the rows are actual members of $H_Q \setminus R$ or descendants of the designated v4.41 matching.

Do not repeat:

generic primality, spacing, nested packing and standard BT cannot establish the Level-D RPD theorem.

FAIL-GREEDY-AVOIDANCE-NAIVE-001

FAIL-ID:

FAIL-GREEDY-AVOIDANCE-NAIVE-001.

Date / iteration:

vNEXT3 hostile proof of the external selector.

Claim:

In the external selector one may choose complements greedily, avoiding the primes already used, and the surviving proportion of the fan is $\prod_{r \in \mathcal{E}} (1 - \frac{1}{r-1}) \gg 1/\log \log N$.

Original motivation:

$|\mathcal{E}| = O((\log N)^{A+1})$, so the avoided set is small compared with the fan, and the product looks Mertens-like.

Status:

KILLED as an argument. The *conclusion* of the selector is true — see G-BAD-BASE-SQUAREFREE-POLYLOG-EXTENSION-PROVED — but not by this route.

Minimal / strong counterexample:

Structural, two independent failures.

Exact certificate:

(i) Without a prior roughness condition, $\mathcal{E}$ may contain small primes. If $3 \in \mathcal{E}$ then $\sum_{r \in \mathcal{E}} 1/(r-1) \geq \frac{1}{2}$, and the Brun–Titchmarsh union bound over $\mathcal{E}$ already exceeds the whole fan, so the union bound gives nothing. (ii) To use $\prod_{r \in \mathcal{E}} (1 - \frac{1}{r-1})$ as a density one needs a *sieve* over $\mathcal{E}$, not a union bound. The combined modulus is $\prod_{r \in \mathcal{E}} r = \exp(\theta(y))$ with $y \approx (\log N)^{A+2}$, that is $\exp((\log N)^{A+2})$, which vastly exceeds N ; Siegel–Walfisz does not apply there, and a sieve at that level would require a level-of-distribution input such as Bombieri–Vinogradov, which was never declared.

Failed hypothesis or mechanism:

A product of local densities was used where only a union bound was available, and the modulus range needed for the alternative was never checked.

What survives:

Everything, after reordering. Imposing w -roughness *first* — no prime below $w = (\log N)^{A+2}$ divides a complement except its own designated base — forces $\mathcal{E}$ to consist only of primes $\geq w$, whence $\sum_{r \in \mathcal{E}} 1/(r-1) \leq |\mathcal{E}|/(w-1) \ll 1/\log N$ and a plain union bound suffices. The sieve then used has polylogarithmic sifting limit and is truncated at a polylogarithmic level, so every modulus stays inside the Siegel–Walfisz range and no level-of-distribution theorem is imported.

Attribution correction:

the $1/\log \log N$ loss is real, but it belongs to the roughness sieve, where it is $\asymp 1/\log w$. The coprimality step costs only $1 + O(\log w/\log N)$. vNEXT2 attributed the loss to the wrong step.

Safe weaker version:

State the selection as: sieve for roughness, then delete squares, then union-bound the coprimality classes.

Current replacement:

Theory Core, Theorem [thm:selector].

Do not repeat:

Do not multiply local densities across conditions that are only accessible by a union bound, and always check that the combined modulus stays in the range of the distribution theorem being used.

Related current target:

(FC10), K1.

FAIL-HALL-SUPPORT-001**FAIL-ID:**

FAIL-HALL-SUPPORT-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

A Hall expansion over the support counts of all sources yields the required capacity bound.

Original motivation:

Every source contributes support, so a global counting argument looked available.

Status:

KILLED.

Minimal / strong counterexample:

The recorded compression figure: 456 sources compress to 17 factor atoms. Status **VERIFIED COMPUTATION**; the raw table was not reloaded in this extraction.

Exact certificate:

With that degree of compression the naive expansion has no room to run: the support side is far smaller than the source side, so no Hall-type surplus exists in the required direction.

Failed hypothesis or mechanism:

Sources and atoms were counted as if they were in near-bijection.

Why the argument failed:

Compression is exactly the regime in which a counting surplus disappears.

What survives:

The precise asymptotic polylog Hall matching theorem is *not* refuted by this: G-POLYLOG-MATCHING runs on a fixed finite family of polylog prime-power layers with an enormous per-layer fan, not on an all-source support count.

Safe weaker version:

Restrict any Hall argument to the polylog layer family where the fan is proved.

Current replacement:

Theory Core, G-POLYLOG-PARENTS, G-POLYLOG-MATCHING, target W1.

Do not repeat:

Do not run a counting argument across a compression.

Related current target:

W1, K1.

FAIL-HIGH-LOADONE-COFACTOR-CAN-HAVE-ARBITRARY-ARITY-001**FAIL-ID:**

FAIL-HIGH-LOADONE-COFACTOR-CAN-HAVE-ARBITRARY-ARITY-001.

Status:

KILLED.

Killed inference:

after selecting a unique-parent target $q > N^{1/3}$, the remaining complement may still contain an uncontrolled high- arity cofactor supported entirely above the cube root.

Certificate:

v4.41 proves exact valuation depth $e \in \{1, 2\}$ and the row trichotomy: a low child $r \leq N^{1/3}$, the square q^2 , or the squarefree semiprime qr with both factors above the cube root.

What survives:

low boundary layers, monomial squares and high semiprime dynamics.

FAIL-HIGH-SEMIPRIME-SIBLING-REUSE-IS-CHEAP-001**FAIL-ID:**

FAIL-HIGH-SEMIPRIME-SIBLING-REUSE-IS-CHEAP-001.

Status:

KILLED.

Killed inference:

a linear high-semiprime packet may heavily reuse its second factors while keeping incremental J negligible.

Certificate:

every repeated sibling $r > N^{1/3}$ contributes more than $(\log N)/3$ per excess row occurrence. If that ambient payment is absent, a $> 3/4$ subfamily has distinct siblings and therefore pairwise-coprime squarefree two-prime complements.

What survives:

the clean positive-sign branch itself was not a contradiction; v4.42 later supersedes it as the primary node.

FAIL-HIGH-VERTICAL-IS-INDEPENDENT-EXTERNAL-CHANNEL-001**FAIL-ID:**

FAIL-HIGH-VERTICAL-IS-INDEPENDENT-EXTERNAL-CHANNEL-001.

Status:

KILLED.

Killed inference:

above the exact cube-root floor, a large $\nu_{>N^{1/3}}$ term may remain a third independent storage channel alongside horizontal reuse and low boundary descent.

Certificate:

every high squared prime either forms a monomial row or has a cofactor $< N^{1/3}$ and therefore already crosses the low boundary. Together with monomial-component capacity and floor-

truncated conservation,

$$\eta_{>N^{1/3}}(H) + 2D_{\leq N^{1/3}}(H) \geq (|H| - 1)/2 + c_{N^{1/3}}.$$

What survives:

a genuine aggregate low-layer defect on bases $q \leq N^{1/3}$.

FAIL-HIGHFLOOR-DOMAIN-VACUITY-001

FAIL-ID:

FAIL-HIGHFLOOR-DOMAIN-VACUITY-001.

Date / iteration:

vNEXT2 independent audit, Phase 1E.

Claim:

TF-HIGH-FLOOR-HORIZONTAL-RIGIDITY applies to packets that actually occur.

Original motivation:

The theorem is stated unconditionally on its hypothesis and its proof is complete.

Status:

SCOPE-CORRECTED. The theorem is **PROVED**; its *antecedent* is unverified.

Exact certificate:

An exhaustive search over even $N < 3 \cdot 10^4$ and all admissible seeds found *no* bad child-closed packet with $\min U > N^{1/3}$ and $U \subset (0, N/3)$. Status of that search: **VERIFIED COMPUTATION**.

Failed hypothesis or mechanism:

None mathematically; the risk is rhetorical — quoting $\eta(U) \geq \lceil n/3 \rceil + c$ as though it were about an exhibited object.

What survives:

The theorem in full, with both skeleton gaps now proved: σ has no cycles (a 1-cycle forces $N = u(u+1)$, hence $u \mid N$ and u inactive; a 2-cycle forces $(v-u)(v+u+1) = 0$; longer cycles are excluded because $f(x) = \sqrt{N-x}$ is strictly decreasing), and there are no three consecutive square rows (the contraction constant of f puts u_2, u_3 within 0.111 of the fixed point for $N \geq 50$, forcing $u_2 = u_3$ and hence a 1-cycle).

Safe weaker version:

Always state it as a conditional.

Do not repeat:

An empty finite search is not evidence that the hypothesis is unsatisfiable, and a complete proof is not an exhibited instance.

Related current target:

TF, K1.

FAIL-IMPORT-V32-POSITIVE-K-TO-DEEP-LOW-OWNER-001

FAIL-ID:

FAIL-IMPORT-V32-POSITIVE-K-TO-DEEP-LOW-OWNER-001.

Source:

v4.53 scope correction relative to v4.32.

Status:

KILLED.

Killed import:

transfer the old lower bound

$$k_i > \frac{N}{6s_i}$$

from the upper-half owner geometry directly to the present low descendant basin.

Reason:

the algebraic owner-return normal form transfers, but the old sign and positive lower bound

used an upper-half owner which is absent here.

What survives:

the signed deep owner-return normal form and the same-layer second-parent inverse bound remain valid in their stated domains.

FAIL-INCREMENTAL-J-IS-BOUNDED-BY-OLD-J-001

FAIL-ID:

FAIL-INCREMENTAL-J-IS-BOUNDED-BY-OLD-J-001.

Status:

KILLED.

Rejected statement:

$J_{R \cup P} / J_R \leq J_R$ for arbitrary external extensions.

Certificate:

vNEXT4.31 gives the exact primewise increment and permits new support multiplicity to create genuinely new J storage.

Do not repeat:

K1-INC itself is an open upper-capacity theorem, not a consequence of the definition of J .

FAIL-INDIVIDUAL-MESOSCALE-BV-001

FAIL-ID:

FAIL-INDIVIDUAL-MESOSCALE-BV-001.

Date / iteration:

v1.10 scale-wall continuation, 1 September 2026.

Claim:

The v1.6 “sub-sum of ordinary Bombieri–Vinogradov” argument can be transferred unchanged from a polylogarithmic base modulus to one fixed base $q = N^\beta$, $\beta > 0$, and still prove a positive lower- sieve fan.

Original motivation:

Extend the clean-parent theorem directly from polylogarithmic labels to polynomially sized states.

Status:

KILLED.

Exact certificate:

For one polynomial base the expected prime/sieve main term is of size at most

$$\asymp \frac{N}{q \log N \log z} = N^{1-\beta} \text{polylog}(N)^{-1},$$

whereas the ordinary Bombieri–Vinogradov all-moduli error available from simply taking a sub-sum is

$$O_B\left(\frac{N}{(\log N)^B}\right).$$

For every fixed B and every fixed $\beta > 0$, the latter is larger by a factor $N^\beta / \text{polylog}(N)$. Therefore the v1.6 error comparison does not prove positivity for an individual mesoscopic base.

Failed hypothesis or mechanism:

A global mean error was treated as though it scaled with the individual base modulus.

What survives:

Aggregate use over many bases does work at the unsieved parent-incidence level. G-BV-HARMONIC-PARENT-MASS proves that fixed positive reciprocal mass below the BV level yields $\gg N / \log N$ parent incidences and $\gg N / \log^2 N$ distinct external parents. TD-HARMONIC-SUPPORT-DISPOSITION supplies the reciprocal-mass input on the D branch.

Safe weaker version:

Use the existing individual selector only while the prescribed base modulus is polylogarithmic, or prove a genuinely aggregate many-base theorem with its own remainder accounting.

Current replacement:

v1.10 adaptive lift remains the individual polylog tool. v1.11 adds genuinely aggregate clean extraction: **G-BV-HARMONIC-CLEAN-EXTRACTION** selects a clean load-one subfamily from mesoscopic reciprocal mass, and **G-BV-POLYNOMIAL-HARMONIC-CLEAN-EXTRACTION** gives a polynomially large clean family on fixed power bands. Neither theorem supplies a fan for every prescribed mesoscopic base.

Do not repeat:

Do not cite ordinary BV's total error as an individual polynomial-modulus lower bound.

Related current target:

TD scale bridge, W1, P1BR/K1.

FAIL-INVERSE-SIBLING-INJECTIVE-001**FAIL-ID:**

FAIL-INVERSE-SIBLING-INJECTIVE-001.

Date / iteration:

v1.23 inverse-sibling continuation, 1 September 2026.

Claim:

Distinct returned fresh seeds ℓ necessarily produce distinct prime siblings s satisfying $\ell s < N$.

Status:

KILLED.

Why the inference fails:

A single packet state can divide the return cofactors of several distinct parent rows. The inverse product inequality contains no injectivity information.

What survives:

After first passing to distinct return parents, repeated choice of the same sibling s gives genuine rooted reuse inside U and is charged by

$$\eta(U) \geq |P| - |S|.$$

Thus collisions are an explicit alternative rather than an ignored loss.

Current replacement:

Theory Core, **TF-HIGH-RETURN-INVERSE-PACKET**.

Do not repeat:

Never convert the inverse-scale map directly into a cardinality lower bound without the $\eta(U)$ collision term.

Related current target:

TF inverse-scale analytic re-entry.

FAIL-KERNEL-EAC-001**FAIL-ID:**

FAIL-KERNEL-EAC-001.

Date / iteration:

v0.1 audit, notation and object review.

Claim:

The iterated source-free core $C_{sf} = \bigcap_{j \geq 0} \Gamma_N^j(R)$ is an EAC, equivalently a terminal strongly connected component.

Original motivation:

Both objects are “what is left at the bottom”.

Status:**SCOPE-CORRECTED**.**Minimal / strong counterexample:**

Definitional; the two constructions are not the same operation.

Exact certificate:

C_{sf} is defined by iterated image and need not be strongly connected, nor a single component.

A terminal SCC is defined by the condensation having no outgoing edge. **G-FAIL-CLOSURE** produces a terminal SCC; it does not produce C_{sf} , and neither statement implies the other.

Failed hypothesis or mechanism:

Two distinct objects shared an informal name.

Why the argument failed:

A theorem about one was used as a theorem about the other.

What survives:

Both objects, kept separate, as in the Theory Core notation registry.

Safe weaker version:

Name the construction whenever either object is used.

Current replacement:

Theory Core, **G-FAIL-CLOSURE** and the notation entry for C_{sf} .

Do not repeat:

Existence of a terminal bad SCC somewhere does not show that p_0 reaches it.

Related current target:

X1.

FAIL-KPOWER-SPECTRUM-IS-OPTIMAL-DJ-BOUND-001**FAIL-ID:**

FAIL-KPOWER-SPECTRUM-IS-OPTIMAL-DJ-BOUND-001.

Status:**SUPERSEDED**.**Superseded description:**

the coefficient $(k\alpha - 1)/(k - 1)^2$ from v4.37 is the primary canonical D/J lower bound.

Certificate:

v4.38 constructs exactly γ fully- C -supported rows and uses the uniform radical-fibre count to obtain

$$\log(D_C J_R) \geq \gamma \log \gamma - o(\gamma \log N).$$

For $\gamma = N^{\alpha+o(1)}$ the coefficient is $\alpha - o(1)$, and at the cube-root splice it is $1/3 - o(1)$ instead of $1/24 - o(1)$.

What survives:

the k -power theorem remains valid and useful for explicit powerful-divisor structure.

FAIL-LARGE-FRESH-SUPPORT-MAY-BE-MOSTLY-REPEATED-001**FAIL-ID:**

FAIL-LARGE-FRESH-SUPPORT-MAY-BE-MOSTLY-REPEATED-001.

Status:**KILLED**.**Killed inference:**

a large fresh-support factor may consist almost entirely of fresh labels repeated on many extension rows while incremental J stays small.

Certificate:

for fresh labels with $\kappa_A(q) \geq 2$, at least one full copy of $\log q$ already belongs to incremental J . Hence

$$W_1(A) \geq \log F_R(A) - \log(J_S/J_R),$$

and the relative radical floor gives

$$\log(J_S/J_R) \geq \mathcal{R}(A)/3 \quad \vee \quad W_1(A) \geq \mathcal{R}(A)/3.$$

What survives:

the cheap branch is not arbitrary fresh support but the exact unique-parent layer.

FAIL-LARGE-MONOMIAL-PACKET-CAN-OCCUR-IN-BOTH-MOD3-CLASSES-001

FAIL-ID:

FAIL-LARGE-MONOMIAL-PACKET-CAN-OCCUR-IN-BOTH-MOD3-CLASSES-001.

Status:

KILLED.

Killed inference:

for $3 \nmid N$, a polynomial family $N - w = q^2$ with $q > N^{1/3}$ may occur for either nonzero residue of N modulo 3.

Certificate:

if $N \equiv 1 \pmod{3}$, then $w = N - q^2 \equiv 0 \pmod{3}$; prime w must equal 3, so at most one such edge exists.

What survives:

a polynomial monomial packet can only remain when $N \equiv 2 \pmod{3}$.

FAIL-LARGEBASE-DEEP-VERTICAL-ESCAPE-001

FAIL-ID:

FAIL-LARGEBASE-DEEP-VERTICAL-ESCAPE-001.

Status:

KILLED as a live frontier on $q > N^{1/3}$.

Killed description:

genuinely independent cubic-or-deeper vertical storage may persist on bases above $N^{1/3}$.

Certificate:

$q^3 < N$ is necessary for depth at least three. Above the cube-root scale vertical depth is at most two, and the one depth-two copy is termwise accompanied by the same-base radical support already counted by FC7.

What survives:

cubic-and-deeper layers on $q < N^{1/3}$.

FAIL-LAYERED-GAMMALOGN-IS-CONTRADICTION-001

FAIL-ID:

FAIL-LAYERED-GAMMALOGN-IS-CONTRADICTION-001.

Date / iteration:

vNEXT4.5.

Claim:

Once $\log D + \log J + \log U = \Omega(\gamma \log N)$, FC8 or Goldbach follows automatically.

Status:

KILLED.

Exact certificate:

The same functional has the natural sandwich

$$\log W_C^{\text{nr}} = \gamma \log N + O(\gamma + |H|).$$

The lower scale and upper capacity match in the leading term.

Failed mechanism:

A conservation identity was mistaken for an independent overload inequality.

What survives:

An additive/canonical theorem may still force an incompatible $D/J/U$ partition.

Do not repeat:

Do not count the exact row mass twice as two independent obligations.

FAIL-LEDGER-INDEPENDENCE-LI5-SCOPE-001**FAIL-ID:**

FAIL-LEDGER-INDEPENDENCE-LI5-SCOPE-001.

Date / iteration:

vNEXT4.1, Phase 0B.

Claim:

vNEXT4 introduced (LI5) under the heading “that trivial bound is the only one available”, suggesting η_{supp} is the unique ledger-derivable lower bound on $\eta(S)$.

Original motivation:

The witness family drives $\eta(R_n)$ to infinity at constant round datum, which does refute every *growing* bound.

Status:

SCOPE-CORRECTED. G-POSITIVE-ROUND-LEDGER-INDEPENDENCE is preserved in full.

Exact certificate:

The $n = 1$ member has $\eta(R_1) = 2$. Hence any F with $F \leq \eta$ satisfies only $F(\mathfrak{d}) \leq 2$ at that datum; constants up to 2 are consistent with the family. So the family excludes unbounded growth in Δ , not the existence of some bounded lower bound.

What survives:

(LI5'): no universally valid lower bound on $\eta(S)$ expressible solely as a function of the round datum can grow unboundedly with Δ ; in particular none supplies the $\gamma \log N$ -scale bound the (LE3)+(FC8) route consumes. This is the only form used downstream, and it is what makes the route unavailable.

Do not repeat:

Distinguish “no bound of the required size” from “no bound”. Only the former was proved.

Related current target:

P1BR-ADDITIVE-COUPLING.

FAIL-LEGACY-CONTINUUM-001**FAIL-ID:**

FAIL-LEGACY-CONTINUUM-001.

Date / iteration:

v0.1 audit of legacy candidates.

Claim:

As N grows the discrete CRD structure becomes effectively continuous, and that transition is itself usable.

Original motivation:

Asymptotic densities do appear.

Status:

KILLED.

Exact certificate:

No density or covering statement is attached to the phrase. It must be replaced by a precise density or covering theorem before anything can be deduced.

Failed hypothesis or mechanism:

An informal limit replaced a quantified statement.

What survives:

Genuine asymptotics, such as G-POLYLOG-PARENTS, stated with explicit uniformity and an explicitly INEFFECTIVE threshold.

Current replacement:

Theory Core, the polylog theorems.

Do not repeat:

Do not use a limit as a lemma.

Related current target:

W1.

FAIL-LEGACY-ENCODING-001**FAIL-ID:**

FAIL-LEGACY-ENCODING-001.

Date / iteration:

v0.1 audit of legacy candidates.

Claim:

A binary or positional encoding of the states supplies an established CRD invariant.

Original motivation:

Encodings are cheap to define.

Status:

KILLED.

Exact certificate:

No retained theorem uses such an encoding, and none of the exact identities — $T = UJ$, the source lock, the collision and cut identities — is expressible through one.

Failed hypothesis or mechanism:

A representation was mistaken for a structure.

What survives:

Nothing in this direction.

Current replacement:

Theory Core, GEN-SBR and the specialisation dictionaries, which is where representation-level statements belong.

Do not repeat:

A change of notation is not a theorem.

Related current target:

none.

FAIL-LEGACY-PIGEONHOLE-001**FAIL-ID:**

FAIL-LEGACY-PIGEONHOLE-001.

Date / iteration:

v0.1 audit of legacy candidates.

Claim:

A parametric symmetric pigeonhole argument supplies the weighted CRD capacity bound.

Original motivation:

The finite lemma is genuinely true.

Status:

KILLED as a mechanism.

Minimal / strong counterexample:

Structural.

Exact certificate:

The bare finite lemma is true and supplies no weighted logarithmic inequality; capacity in CRD is a statement about $\sum \mu_{\ell,k} \log \ell$, which the lemma never reaches.

Failed hypothesis or mechanism:

A cardinality lemma was used where a weighted one is required.

Why the argument failed:

Counting objects is not bounding their logarithmic mass.

What survives:

The finite lemma itself, as an unrelated true statement.

Safe weaker version:

None in this direction.

Current replacement:

Theory Core, targets $P1^{\text{br}}$ and K1.

Do not repeat:

Do not offer an unweighted lemma as a weighted bound.

Related current target:

$P1^{\text{br}}$, K1.

FAIL-LEGACY-WAVES-001**FAIL-ID:**

FAIL-LEGACY-WAVES-001.

Date / iteration:

v0.1 audit of legacy candidates.

Claim:

Describing residue behaviour as interfering sinusoidal waves supplies a rigorous CRD invariant.

Original motivation:

The picture is suggestive.

Status:

KILLED.

Exact certificate:

Only the underlying residue lattice has proved content; no retained theorem uses the wave description, and no invariant is derived from it anywhere in the corpus.

Failed hypothesis or mechanism:

Imagery in place of a statement.

What survives:

The residue lattice, used directly.

Current replacement:

Theory Core, prime-power layer formalism.

Do not repeat:

Do not carry a metaphor into a proof.

Related current target:

none.

FAIL-LF2-AS-UPPER-BOUND-001**FAIL-ID:**

FAIL-LF2-AS-UPPER-BOUND-001.

Date / iteration:

vNEXT3.1 correction round.

Claim:

In the (FC8) frontier certificate, “(LF2) bounds $\eta(R)$ above by $|R|$ ”.

Original motivation:

$|Q| + |H| \leq |R|$ is true, and the certificate needed $\eta(R)$ to be small.

Status:

KILLED.

Exact certificate:

(LF2) reads $\eta(R) \geq |Q| + |H|$. It is a *lower* bound. The inequality $|Q| + |H| \leq |R|$ says only that this recorded lower bound is itself at most $|R|$; it places no ceiling on $\eta(R)$, which by

(LE2) can exceed $|R|$ whenever rows carry repeated support.

Failed hypothesis or mechanism:

A bound on the right-hand side of a lower-bound inequality was read as a bound on the left-hand side.

What survives:

The certificate, correctly stated: to contradict (FC8) through (LE3) one needs a *lower* bound on $\eta(R)$ of order $\gamma \log N$, and the recorded (LF2) does not carry a factor $\log N$.

Current replacement:

Theory Core, Proposition [prop:fc8-corrected], status FRONTIER CERTIFICATE / RELATIVE TO RECORDED BOUNDS.

Do not repeat:

Never infer a ceiling from a floor.

Related current target:

K1, γ .

FAIL-LINEAR-SMOOTH-MIRROR-PACKET-CAN-STAY-POLYLOG-SUPPORTED-001

FAIL-ID:

FAIL-LINEAR-SMOOTH-MIRROR-PACKET-CAN-STAY-POLYLOG-SUPPORTED-001.

Status:

KILLED asymptotically on the recorded high-gap/BHP domain.

Killed inference:

a linear smooth-mirror family may carry the required active mass on only polylogarithmically many active support primes.

Certificate:

existing mirror active-mass plus support capacity gives, for $|J_{\text{sm}}| \geq \rho t$,

$$|P_{\text{mir}}| \geq N^{0.2375\rho - o(1)}$$

unless there is already a linear active-state population.

What survives:

this is asymptotic and does not replace finite small-gap case analysis.

FAIL-LITERAL-LINEAGE-001

FAIL-ID:

FAIL-LITERAL-LINEAGE-001.

Date / iteration:

Post-v7.5 layer-lineage audit.

Claim:

Once a prime label ℓ is produced, its valuation can be followed literally through the row indexed by ℓ .

Original motivation:

Residual factors were pictured as conserved tokens moving down a path.

Status:

KILLED.

Minimal / strong counterexample:

Universal for active ℓ .

Exact certificate:

Since $\ell \nmid N$, $N - \ell \equiv N \not\equiv 0 \pmod{\ell}$; hence

$$v_{\ell}(N - \ell) = 0.$$

Failed hypothesis or mechanism:

Row labels and factor labels were identified.

Why the argument failed:

The dynamics changes the row index; it does not conserve a prime token along a path.

What survives:

Global valuation accounting $T = UJ$, collision layers, and residual products.

Safe weaker version:

Track incidence across a finite set of rows, not literal pathwise conservation.

Current replacement:

Theory Core, G-TUJ and the prime-power layer formalism.

Do not repeat:

Never write a lineage law without an exact valuation identity at every transition.

Related current target:

$E1, K1$.

FAIL-LOCAL-REUSE-001**FAIL-ID:**

FAIL-LOCAL-REUSE-001.

Date / iteration:

Escape-or-Reuse campaign, 30 August 2026.

Claim:

An upper-source reuse parent occurs within five source indices.

Original motivation:

A local window would make simultaneous witness selection inexpensive.

Status:

KILLED by source-recorded computation.

Minimal / strong counterexample:

The strongest recorded witness is $(N, \ell) = (1825372, 52153)$ with source-index distance 53750.

Exact certificate:

Only 0.0265727 of defined recorded witnesses lay within five indices; the cited distance is the recorded maximum. The raw row list was not reloaded in this extraction.

Failed hypothesis or mechanism:

Source order was treated as a metric controlling arithmetic incidence.

Why the argument failed:

Parent lattices are residue classes, not short intervals in the adaptive source enumeration.

What survives:

Polylogarithmic parent abundance and finite matching for prescribed prime powers.

Safe weaker version:

Use the full active parent lattice or prove a new quantitative selector theorem.

Current replacement:

Theory Core, G-POLYLOG-PARENTS, G-POLYLOG-MATCHING, and $W1$.

Do not repeat:

Do not impose an unproved absolute locality window.

Related current target:

$W1, K1$.

FAIL-LOSSLESS-INTERFACE-001**FAIL-ID:**

FAIL-LOSSLESS-INTERFACE-001.

Date / iteration:

vNEXT4.1, Phase 4.

Claim:

That a canonical row c_j can be transferred to a global prime selection with its information preserved — the round's primary success criterion, third arrow.

Original motivation:

The bundle query does preserve strictly more than the old single-base query: exact valuations, the cofactor class and the mirror relation $D \mid p - a_j$. It was natural to hope the preserved part could be made all of A_j .

Status:

KILLED for lossless transfer, under every distribution theorem presently available. The bundle lift itself is **PROVED**.

Exact certificate:

Theory Core PO-QUERY-CAPACITY: if every member of a residue class $n \equiv a \pmod{M}$ satisfies $v_q(N - n) = e$, then $q^{e+1} \mid M$. Taking $n = a$ and $n = a - M$ forces $q^e \mid M$; and if $v_q(M) = e$ exactly then one k in each complete residue system mod q gives $q^{e+1} \mid N - a + kM$. Hence $D \text{ rad } D \mid M$ and $\log D \leq \frac{E}{E+1} \log M$. With Siegel–Walfisz, $\log M \leq K \log \log N$, so the preserved fraction is $< K \log \log N / \log(N/2) \rightarrow 0$. Under GRH the fraction is at most $\frac{1}{2}$ (at most $\frac{1}{4}$ for squarefree bundles), so GRH does not make it lossless either. Losslessness needs a level of distribution $\theta \geq 1$ for an *individually specified* modulus of size $N^{1-o(1)}$ — stronger than GRH, and not the average-level statement of Elliott–Halberstam.

Failed hypothesis or mechanism:

Treating “retain more information” as free. Retention is paid for in the modulus, at a rate fixed by (QC1), and the modulus is capped by the available distribution theorem.

What survives:

The exact capacity: $\log D \leq \frac{E}{E+1} K \log \log N$, and everything the bundle does preserve, which is genuinely more than the old query. See PO-EXACT-BUNDLE-LIFT and PO-ADAPTIVE-BUNDLE-DICHOTOMY.

Do not repeat:

Do not propose an interface that preserves a positive proportion of a canonical row without first naming the level-of-distribution theorem that pays for it, and do not import Bombieri–Vinogradov silently — this corpus does not use it anywhere.

Related current target:

P1BR-ADDITIVE-COUPPLING, PO-BUNDLE-ROUGH-HIGHPOWER-BRANCH.

FAIL-LOWBASE-POWERFUL-ROWS-AS-INDEPENDENT-VERTICAL-WALL-001**FAIL-ID:**

FAIL-LOWBASE-POWERFUL-ROWS-AS-INDEPENDENT-VERTICAL-WALL-001.

Status:

KILLED as an independent primary frontier.

Killed description:

a polynomial-size canonical low core may keep an asymptotically negligible $D_C J_R$ while financing essentially all compulsory mass through diffuse low-base powerful rows.

Certificate:

v4.37 proves, pending audit, that for $\gamma = N^{\alpha+o(1)}$ and every fixed $\alpha > 0$,

$$\log(D_C J_R) = \Omega_\alpha(\gamma \log N).$$

Equivalently, near-pure vertical escape forces $\gamma = N^{o(1)}$.

What survives:

a subpolynomial low core or a non-negligible D/J payment; rowwise powerful-divisor structure remains useful downstream.

FAIL-LOWCORE-CARDINALITY-FC8-WEIGHT-001**FAIL-ID:**

FAIL-LOWCORE-CARDINALITY-FC8-WEIGHT-001.

Date / iteration:

vNEXT4.4.

Claim:

Once the reflected block forces large $\gamma = |C|$, the layered Euler theorem automatically converts this into enough rooted reuse to contradict FC8.

Status:**KILLED**.**Exact certificate:**

Canonical layered Euler gives only

$$\eta(R) + \nu(R) \geq |R| + 1 \geq \gamma + 2,$$

hence only

$$\max\{\log J_R, \log U_R\} \geq \frac{\gamma + 2}{2} \log 3.$$

This is $O(\gamma)$ guaranteed log mass. The recorded FC8 route through $J_R \geq 3^{\eta(R)}$ needs a lower bound at scale $\gamma \log N$ unless a separate large denominator is supplied. Moreover the population may be paid vertically through prime-power layers U_R ; the generic failed-world prime-power examples show that such vertical storage is a real mechanism.

Failed hypothesis or mechanism:

Cardinality was confused with weighted prime-base mass, and horizontal reuse was treated as if it were the only defect channel.

What survives:

vNEXT4.4 produces a genuine FC8 denominator on the sub-square-root large-factor branch, and the exact horizontal/vertical split remains available. A theorem forcing large bases or upper-bounding cheap vertical storage would reopen the route.

Do not repeat:

A new lower bound on γ , $\eta + \nu$, or an unweighted layer count is not a consumer unless its logarithmic bases are controlled.

Related current target:

C-VERTICAL-LOWCORE-CAPACITY.

FAIL-LOWCORE-CHILDCLOSED-001**FAIL-ID:**

FAIL-LOWCORE-CHILDCLOSED-001.

Date / iteration:

v1.23 low-core continuation, 1 September 2026.

Claim:

Since every reused rooted label is below $N/6$, the set $C = R \cap (0, N/6)$ may be treated as a child-closed failed subsystem.

Status:**KILLED**.**Why the inference fails:**

The theorem controls the location of *repeated targets*, not all children of low rows. A low row $u < N/6$ has $N - u > 5N/6$, and a prime factor of this complement can lie in $(N/6, N/3)$. Thus an edge may leave C for the upper fringe.

What survives:

Every repeated target and the complete factor J_R are supported in C ; every high state has a unique-parent forest structure and returns to C when parent ancestry is followed backwards.

Current replacement:

Theory Core, G-CANONICAL-REUSE-LOWCORE and G-CANONICAL-UPPER-FRIDGE-FOREST.

Do not repeat:

Do not apply G-MULTISOURCE-LOCK, low-packet product locks, or child-closed component arguments directly to C without separately proving closure for the subset in question.

Related current target:

K1 low-core upper capacity.

FAIL-LOWPOLYLOG-MULTIPLICITY-IS-PARENT-EXISTENCE-001**FAIL-ID:**

FAIL-LOWPOLYLOG-MULTIPLICITY-IS-PARENT-EXISTENCE-001.

Date / iteration:

vNEXT4.20.

Old description:

The low-polylog $t \log \Lambda$ mirror multiplicity cannot be consumed because there are not enough actual prime parent rows preserving the required layers.

Status:

KILLED as a description of the current obstruction.

Certificate:

After an $o(t \log \Lambda)$ high-power tail is removed, all retained polylog prime-power blocks can be simultaneously matched to distinct external BAD primes satisfying $p_{j,q} \equiv c_j \pmod{q^{e+1}}$.

Replacement:

The live obstruction is joint unintended support load / single charging, not parent abundance.

FAIL-MANY-FIRSTFRONT-ROOTS-CAN-ALL-LIE-ABOVE-4H-001**FAIL-ID:**

FAIL-MANY-FIRSTFRONT-ROOTS-CAN-ALL-LIE-ABOVE-4H-001.

Status:

KILLED for $h > N^{1/3}$.

Rejected statement:

A large fraction of Q may lie above $4h$ without additional restriction.

Certificate:

three such distinct primes would divide $m < N$ but have product $> (4h)^3 > 64N$.

Current safe statement:

$\#\{q \in Q : q > 4h\} \leq 2$.

FAIL-MATCHING-CAPACITY-001**FAIL-ID:**

FAIL-MATCHING-CAPACITY-001.

Date / iteration:

Canonical bridge audit §5.3, 31 August 2026.

Claim:

An injective, load-one assignment of external parents to layers bounds the ambient collision cost, closing the capacity target.

Original motivation:

Load one removes repeated use of a single parent, which was mistaken for removing the cost.

Status:

KILLED — the matching theorem itself is **PROVED**; the inference drawn from it is not.

Minimal / strong counterexample:

Structural; no finite witness is needed.

Exact certificate:

Injectivity leaves two costs untouched. One selected row $N - p$ may meet many atoms of R , inflating $K_{\times}(R, \mathcal{P})$; and two distinct selected rows may share unintended new support,

inflating $K(\mathcal{P})$. Greedy selection of pairwise-coprime complements would have to avoid, for each previously chosen p' , every class $p \equiv N \pmod{t}$ with $t \mid N - p'$; Siegel–Walfisz for a single polylog modulus gives no uniform positive count under a growing adaptive system of such prohibitions, and some t may be large.

Failed hypothesis or mechanism:

Existence and load were confused with cost.

Why the argument failed:

Abundance is a statement about one modulus at a time; capacity is a statement about a whole selected family at once.

What survives:

G-POLYLOG-PARENTS and G-POLYLOG-MATCHING, now for all polylog prime-power layers, both with an explicitly INEFFECTIVE threshold.

Safe weaker version:

Report load one and the two uncontrolled terms separately; never report their conjunction as capacity.

Current replacement:

Theory Core, target K1, which is exactly the missing bound $K_{\times}(R, \mathcal{P}) + K(\mathcal{P}) \leq \mathcal{C}_N(R)$.

Do not repeat:

This is the precise point at which abundance stops becoming clean capacity; do not cross it silently.

Related current target:

W1, K1.

FAIL-MERGER-COUNT-IMPLIES-AMBIENT-J-001

FAIL-ID:

FAIL-MERGER-COUNT-IMPLIES-AMBIENT-J-001.

Status:

KILLED.

Killed inference:

e merger/attachment events automatically force $\Delta \log J = \Omega(e \log N)$.

Certificate:

v4.38 proves a real attachment charge, but if all events reuse a fixed small label q , its weight is only $O(e \log q)$. The ELCFS finite-realisation theorem shows that source counts/topology/reverse-generation bookkeeping alone cannot manufacture the missing ambient $\log N$ factor.

What survives:

attachment forces an unweighted horizontal defect and CRD-specific arithmetic may still retain scale.

FAIL-MERGER-WEIGHT-TO-J-TPOLOGY-001

FAIL-ID:

FAIL-MERGER-WEIGHT-TO-J-TPOLOGY-001.

Status:

KILLED.

Killed sufficient strategy:

prove $\Delta \log J \gg W(E)$ using only child closure, source counts, graph topology, support multiplicities and the v28 reverse-extension ledger.

Certificate:

ELCFS finite realisation permits arbitrarily large source weights attached through a fixed low target. The topology grows while the horizontal receiving weight remains only $e \log q$.

What survives:

CRD arithmetic absent from ELCFS—the map $N - p$, difference divisibility, reflected prove-

nance, exact cofactors and canonical root extremality—may still supply the missing scale.

FAIL-MIRROR-FIXED-POLYLOG-RESERVOIR-001

FAIL-ID:

FAIL-MIRROR-FIXED-POLYLOG-RESERVOIR-001.

Date / iteration:

vNEXT4.17, sharpened by vNEXT4.19.

Escape:

Store all natural mirror mass below $(\log N)^D$ for super-polylogarithmic Λ .

Status:

SCOPE-CORRECTED.

Certificate:

For fixed D ,

$$M_{\leq (\log N)^D}^- = O_D(t \log \log N).$$

Thus this reservoir can remain competitive only when $\log \Lambda = O(\log \log N)$. That low-polylog balance is the current open frontier rather than a killed escape.

FAIL-MIRROR-MASS-ONE-SCALE-001

FAIL-ID:

FAIL-MIRROR-MASS-ONE-SCALE-001.

Date / iteration:

vNEXT4.17.

Escape:

Store $\gg t \log \Lambda$ active mirror mass in one dyadic base band.

Status:

KILLED.

Certificate:

Uniformly $M_{\bar{Y}}^- \ll t$. Natural mass therefore occupies $\Omega(\log \Lambda)$ scales.

FAIL-MIRROR-MULTIPLICITY-BY-REPEATED-BASE-001

FAIL-ID:

FAIL-MIRROR-MULTIPLICITY-BY-REPEATED-BASE-001.

Date / iteration:

vNEXT4.17.

Claim:

Represent reflected multiplicity m_r^- by the same number of clean selected parent copies carrying designated base r in one family.

Status:

KILLED.

Exact obstruction:

The clean multiparent identity contains

$$-\binom{\rho_r}{2} \log r,$$

a quadratic selected-family self-collision cost.

Safe replacement:

Convert aggregate reflected mass first to distinct-base reciprocal mass/population, or use separate rounds with a proved global single-charge mechanism.

FAIL-MONOMIAL-MOD3-NOVELTY-vNEXT4.11

FAIL-ID:

FAIL-MONOMIAL-MOD3-NOVELTY-vNEXT4.11.

Date / iteration:

vNEXT4.11.

Claim:

The mod-3 collapse used in the repeated- b classification is new in vNEXT4.11.

Status:

SCOPE-CORRECTED (provenance correction).

Exact certificate:

Its source is the earlier **C-MONOMIAL-MOD3-COLLAPSE** theorem from vNEXT4.7.

What survives:

The repeated-entry-cofactor classification remains valid subject to its own audit; only the novelty/provenance claim is corrected.

FAIL-MONOMIAL-NEEDS-SPARSITY-001

FAIL-ID:

FAIL-MONOMIAL-NEEDS-SPARSITY-001.

Date / iteration:

vNEXT4.8.

Claim:

The prime-square branch can only be handled by proving that q and $N - q^2$ are sparse.

Status:

KILLED.

Exact certificate:

Monomial square edges form components with at most two edges. A large monomial family therefore forces many component starts; every nonroot start has a nonmonomial entry edge, and on $h > N^{1/3}$ those starts produce a linear packet of actual entry-parent rows. Existing rooted transport then routes the packet into horizontal reuse, low-scale states, or reciprocal mass.

What survives:

Prime-square sparsity remains an optional analytic strengthening, not the necessary binding step.

FAIL-MONOMIAL-NODESCENT-CHEAP-001

FAIL-ID:

FAIL-MONOMIAL-NODESCENT-CHEAP-001.

Date / iteration:

vNEXT4.8.

Claim:

Above $N^{1/3}$, a large monomial entry packet whose descendant closure stays above h can remain supported only by cheap vertical layers.

Status:

KILLED.

Exact certificate:

If P is the selected entry-parent packet and its descendant closure U satisfies $\min U > h > N^{1/3}$, then

$$\eta(U) \geq |P|, \quad \log J_R > |P| \log h.$$

What survives:

The downstream question is how to consume this horizontal payment inside K1.

FAIL-MONOMIAL-PATH-LENGTH-TWO-IS-ONLY-EMPIRICAL-001

FAIL-ID:

FAIL-MONOMIAL-PATH-LENGTH-TWO-IS-ONLY-EMPIRICAL-001.

Status:

KILLED.

Killed inference:

the observed maximum monomial path length two is only a finite-census phenomenon.

Certificate:

the pre-v4.41 Core theorem already proves path length at most two. The interval $(\sqrt{N - \sqrt{N}}, \sqrt{N})$ additionally yields uniqueness of a two-edge component.

FAIL-MONOMIAL-VERTICAL-SELF-SUFFICIENT-001

FAIL-ID:

FAIL-MONOMIAL-VERTICAL-SELF-SUFFICIENT-001.

Date / iteration:

vNEXT4.8.

Claim:

A large monomial vertical family may account for the whole canonical layered defect by itself.

Status:

KILLED.

Exact certificate:

$$\eta(R) + \nu_{\neq \square}(R) \geq \left\lceil \frac{M_{\square}}{2} \right\rceil + 1.$$

Every monomial family forces a companion horizontal or nonmonomial vertical defect.

FAIL-MR2-CANONICAL-ONLY-001

FAIL-ID:

FAIL-MR2-CANONICAL-ONLY-001.

Date / iteration:

vNEXT4.2.

Claim:

For $q \mid m$, the equivalence $q \mid c_j \iff q \mid h - j$ is a genuinely canonical p_0 -only relation.

Status:

KILLED as a scope claim.

Exact certificate:

For an arbitrary upper prime root $r = x + h_r$, $m_r = x - h_r$, one has $r = m_r + 2h_r$. Hence for every odd $q \mid m_r$,

$$q \mid (r - 2j) \iff q \mid h_r - j.$$

No minimality of p_0 is used.

What survives:

The relation itself remains valid and useful; only its classification changes from “canonical selector information” to generic upper-root geometry.

Do not repeat:

Reserve “genuinely canonical” for conclusions using minimality/prime-freeness, e.g. $h < G$, not identities valid for every upper root.

FAIL-N1/4-IS-MULTIPLICITY-REALISATION-CEILING-001

FAIL-ID:

FAIL-N1/4-IS-MULTIPLICITY-REALISATION-CEILING-001.

Date / iteration:

vNEXT4.20.

Killed description: $N^{1/4-\varepsilon}$ is an intrinsic ceiling on simultaneous realisation of the low-polylog mirror multiplicity.**Status:****KILLED**.**Correction:**

That scale belongs to the clean rough-support/lower-sieve architecture. Without demanding global complement coprimality, the full low-polylog block reservoir, of size up to $N^{1/2-o(1)}$, has distinct prime parent representatives.

FAIL-N1/6-DESIGNATED-BASE-BARRIER-001**FAIL-ID:**

FAIL-N1/6-DESIGNATED-BASE-BARRIER-001.

Date / iteration:

vNEXT4.19 correction of vNEXT4.15–vNEXT4.17.

Killed claim:Variable-mass clean selection cannot act on designated bases larger than $N^{1/6-o(1)}$.**Status:****KILLED**.**Exact correction:**

The selector requires only

$$Qz^{2+\vartheta} \ll N^{1/2}/\text{polylog } N.$$

With $z = (\log N)^D$, designated bases may reach

$$Q = N^{1/2}/\text{polylog } N.$$

Call $N^{1/6}$ only the same-scale specialization $z \asymp Q$.**FAIL-NAIVE-MU-LE-3ETA-001****FAIL-ID:**

FAIL-NAIVE-MU-LE-3ETA-001.

Status:**KILLED**.**Killed inference:**for $q > N^{1/4}$, valuation depth at most three gives $\mu_H(q) \leq 3(\kappa_H(q) - 1)$.**Certificate:**if $\kappa_H(q) = 1$ and $d_H(q) = 3$, then $\mu_H(q) = 2$ while $\eta_q = 0$.**Safe replacement:** $\mu_H(q) \leq 3\eta_q + 2$; for general depth m , $\mu_H(q) \leq m\eta_q + (m - 1)$.**Do not repeat:**

never discard the occupied-base additive term in a power-threshold argument.

FAIL-NATURAL-DENOMINATOR-IS-ONLY-BASELINE-001**FAIL-ID:**

FAIL-NATURAL-DENOMINATOR-IS-ONLY-BASELINE-001.

Date / iteration:

vNEXT4.14.

Claim:

A high reflected denominator family may carry only its first copies in D_C , with no further canonical consequence.

Status:**KILLED**.**Exact certificate:**

If its selected parent descendants do not reach $\leq h$, then

$$\log J_R > \frac{\log h}{3 \log N} L(E),$$

so a fixed positive fraction of the same logarithmic family weight is paid horizontally.

FAIL-NESTED-PRIMEPOWER-PACKING-GIVES-CUMULATIVE-SAVING-001**FAIL-ID:**

FAIL-NESTED-PRIMEPOWER-PACKING-GIVES-CUMULATIVE-SAVING-001.

Status:**KILLED**.**Killed inference:**

simultaneous use of the nested spacing bounds for $r^2, r^3, \dots, r^e$ must create an extra cumulative saving beyond the single-layer parent-slot envelope.

Certificate:

if κ rows lie in one exact layer $v_r(N-p) = e$, then $n_k(r) = \kappa$ for every $1 \leq k \leq e$. Consequently all nested layers are realised by the same packet and

$$\sum_{k \geq 2} (n_k(r) - 1)_+ = (e - 1)(\kappa - 1).$$

The support graph sees only $\kappa - 1$ repeated r -edges.

Do not repeat:

nested packing alone cannot turn valuation depth into an $O(1)$ or $O(\log \log N)$ bounded-multiplicity graph charge.

FAIL-NONMONOMIAL-VERTICAL-IS-INDEPENDENT-001**FAIL-ID:**

FAIL-NONMONOMIAL-VERTICAL-IS-INDEPENDENT-001.

Date / iteration:

vNEXT4.7.

Claim:

For $h > N^{1/3}$, a nonmonomial event $N - w = kq^2$, $k > 1$, may remain an independent vertical obstruction.

Status:**KILLED** as a description.**Exact certificate:**

Every prime $r \mid k$ is rooted and satisfies

$$r < N/h^2 < h.$$

For a family of such events, repeated siblings force horizontal/collision mass, while distinct siblings create an explicit lower-scale rooted packet.

What survives:

A downstream consumer/upper-capacity problem for the exported sibling/collision alternatives.

FAIL-ONE-REPRESENTATIVE-CONSUMES-MIRROR-MULTIPLICITY-001**FAIL-ID:**

FAIL-ONE-REPRESENTATIVE-CONSUMES-MIRROR-MULTIPLICITY-001.

Date / iteration:

vNEXT4.16.

Claim:

The existing clean selector closes the smooth-mirror harmonic branch merely by selecting one representative incidence for each active base.

Status:**KILLED**.**Exact obstruction:**

Reflected packing permits about t/r incidences at a small base r . Present clean selectors retain one incidence per base, or repeated multiplicity only for a fixed-polylog total multiset.

Replacement target:

Preserve a positive fraction of the weighted multiplicity before applying the signed K1 ledger.

FAIL-ORTHOGONAL-BRIDGE-001**FAIL-ID:**

FAIL-ORTHOGONAL-BRIDGE-001.

Date / iteration:

vNEXT3, Phases 2–3.

Claim:

Although cross-collision is dead under $K_{\times}(R, U) = 0$, a source- or product-side functional can still carry $\eta(U)$ from an external orthogonal packet into a capacity determined by R .

Original motivation:

The source product of **G-MULTISOURCE-LOCK** does change when a packet is adjoined, so it looked like a live channel.

Status:**KILLED**.**Minimal / strong counterexample:**

Structural, by exact decoupling.

Exact certificate:

For disjoint bad child-closed R, U , **G-ORTHOGONAL-DECOUPLING** gives $\Gamma_N(S) = \Gamma_N(R) \dot{\cup} \Gamma_N(U)$, $\text{Src}(S) = \text{Src}(R) \dot{\cup} \text{Src}(U)$, $T_S = T_R T_U$, $J_S = J_R J_U$, $U_S = U_R U_U$, $\eta(S) = \eta(R) + \eta(U)$, $\nu(S) = \nu(R) + \nu(U)$, and the multisource lock for S factors as the product of the two locks. Every recorded functional is therefore a disjoint-union homomorphism, and the recorded state of R is *identical* for every admissible U . The target inequality $\log(T_{R \cup U}/T_R) \geq c_0 \eta(U)$ does hold, with $c_0 = \log 3$, and is worthless: its left side equals $\log T_U$.

Failed hypothesis or mechanism:

The source product does change, but it changes by exactly U 's own source product; nothing is shared.

Why the argument failed:

$K_{\times}(R, U) = 0$ is not an extra hypothesis to be worked around — it is *automatic* for disjoint child-closed packets, and it is a symptom of total decoupling rather than of one blocked channel.

What survives:

G-BRIDGE-ROUTING-EQUIVALENCE: a bridge can exist only if $U \cap R \neq \emptyset$, which is precisely the TD/TF routing target. The two fronts are one.

Safe weaker version:

Report $\eta(U)$ as a property of U alone.

Current replacement:

Theory Core, Theorem [thm:decoupling], Corollary [cor:worthless-coupling], Theorem [thm:bridge-routing].

Do not repeat:

Do not seek any functional coupling two disjoint child-closed packets; all recorded functionals

provably factor.

Related current target:

TD, TF, K1.

FAIL-ORTHOGONALITY-AS-HYPOTHESIS-001

FAIL-ID:

FAIL-ORTHOGONALITY-AS-HYPOTHESIS-001.

Date / iteration:

vNEXT3, Phase 2.

Claim:

$K_{\times}(R, U) = 0$ is a special favourable hypothesis on the external packet, to be secured by construction.

Original motivation:

It is written as a hypothesis throughout the external-packet material.

Status:

SCOPE-CORRECTED.

Exact certificate:

(OD4). If R and U are disjoint and both child-closed, then $\Gamma_N(R) \subseteq R$ and $\Gamma_N(U) \subseteq U$ are disjoint, so no prime divides a complement on both sides and every cross gcd is 1. Orthogonality is a consequence of disjointness plus child-closure, not an independent assumption.

What survives:

Every statement conditioned on orthogonality, now with the hypothesis recognised as redundant given the other two.

Do not repeat:

Do not spend effort “arranging” orthogonality, and do not read it as evidence of a favourable configuration; it signals decoupling.

Related current target:

TF, K1.

FAIL-OWNED-RETURN-FIRST-EDGE-IS-FREE-SOURCE-LOSS-001

FAIL-ID:

FAIL-OWNED-RETURN-FIRST-EDGE-IS-FREE-SOURCE-LOSS-001.

Status:

KILLED on the v18 owner-completed domain.

Rejected statement:

The first later return edge into an external rough seed can always be absorbed solely by destroying that seed as a source.

Certificate:

the owner row p_i already supplies $p_i \rightarrow r$; a later $u \rightarrow r$ is a second occurrence and creates a full $+\log r$ incremental J charge.

Scope guard:

This does not invalidate the generic vNEXT4.28 ledger; it strengthens it on the retained-owner domain.

FAIL-OWNER-SEALED-RETURN-CAN-STAY-SQUAREFREE-001

FAIL-ID:

FAIL-OWNER-SEALED-RETURN-CAN-STAY-SQUAREFREE-001.

Status:

KILLED.

Rejected statement:

A return row using only the owner support $\{q_0\} \cup \text{PF}(b_i)$ may be a cheap squarefree recycle.

Certificate:

its complement is larger than the squarefree owner complement while its radical divides that owner complement; verticality is forced.

Do not repeat:

Owner support preservation is a verticality certificate, not a squarefree escape.

FAIL-OWNERLESS-FIRST-OCCURRENCE-HAS-NO-ADDITIVE-STRUCTURE-001**FAIL-ID:**

FAIL-OWNERLESS-FIRST-OCCURRENCE-HAS-NO-ADDITIVE-STRUCTURE-001.

Status:

KILLED.

Killed description:

a fresh returned reflected factor with no prior owner is arithmetically anonymous apart from its support incidence.

Certificate:

for $c_j = sq$ and a low active parent $N - w = bq$, v4.39 gives

$$\frac{10}{9}s < b < 2s, \quad a_j - w = (b - s)q,$$

with $b - s$ positive even and

$$a_j - w > N/18.$$

What survives:

this additive relation still needs a capacity / second- occurrence consumer.

FAIL-P0-INFORMATION-SELECTOR-001**FAIL-ID:**

FAIL-P0-INFORMATION-SELECTOR-001.

Date / iteration:

vNEXT4.1, Phase 9.

Claim:

The central hypothesis of the round: that p_0 identifies the particular additive–multiplicative correlations a global prime-distribution theorem should preserve.

Original motivation:

The canonical reflected block is the only place in CRD where a *composite* row of size $> N/2$ with a canonical mirror is produced deterministically.

Status:

KILLED. Not refuted; **not established**.

Exact certificate:

Theory Core P0-CANONICITY-ABLATION. Every clause of P0-EXACT-BUNDLE-LIFT classifies as (AB-A) “fixed N only” or (AB-B) “an arbitrary reflected pair $a + c = N$ ”. Replacing p_0 by any upper-half prime p_1 leaves (BL3)–(BL6) and (BL5) word for word intact on every row whose c'_j is composite. Canonicity is used in exactly one place: the prime-free interval $[x, p_0)$ makes *every* c_j composite, so branch (MC1) is available on the whole block. Measured on the 301 recorded noncanonical failed upper roots, 32,766 of 130,090 reflected rows — 25.2% — have c'_j prime and are unbundleable (their single block costs $c_j'^2 > N$).

Failed hypothesis or mechanism:

A non-vacuity guarantee is not an information-selection principle. Canonicity guarantees the construction is always available; it does not make the construction, or its conclusion, different.

What survives:

One genuinely canonical statement, (MR2): for $q^e \parallel m$, $q \mid c_j \iff q \mid h - j$, from ARS-CANON-GEOMETRY. No consequence was built on it in this round.

Do not repeat:

Do not record a P0-INFORMATION-SELECTOR-CERTIFICATE without an ablation in which the conclusion actually fails after the canonical datum is deleted. None was found.

Related current target:

P1BR-ADDITIVE-COUPPLING, P0-BUNDLE-ROUGH-HIGHPOWER-BRANCH.

FAIL-P0-ZEROHOLE-ESSENTIAL-001**FAIL-ID:**

FAIL-P0-ZEROHOLE-ESSENTIAL-001.

Date / iteration:

vNEXT4.2.

Claim:

The fact that the canonical reflected block has no prime holes is, by itself, the essential p_0 -specific additive–multiplicative glue.

Status:

KILLED as a mechanism for the active/overflow mass argument.

Exact certificate:

If p_1 is the immediate next prime after canonical $p_0 > x$, the strict p_1 -reflected block has exactly one prime hole, namely p_0 , at index $(p_1 - p_0)/2$. More generally the fixed p_k block has exactly the k earlier primes as holes. Iterating the recorded pointwise short-interval bound a fixed number of times gives $p_k - x = O_k(x^{0.525})$, so deleting the holes costs only $O_k(\log x)$ in the block active-mass identity when the width tends to infinity.

Why the argument failed:

The earlier 25.2% statistic concerned a broad family of noncanonical upper roots; it was not the immediate $p_0 \rightarrow p_1$ ablation. The correct immediate ablation is a one-hole perturbation, not a positive-density destruction of the block.

What survives:

p_0 still gives the exact zero-hole formulation and may be essential through stronger canonical information such as $h < G$, Exact-D, or another conclusion which actually fails already at p_1 .

Do not repeat:

Do not infer a P0-INFORMATION-SELECTOR-CERTIFICATE from the absence of prime holes alone.

Related current target:

C-CRITICAL-SQRT-GAP-BRANCH.

FAIL-P4-IS-THE-BEST-BV-ARITY-001**FAIL-ID:**

FAIL-P4-IS-THE-BEST-BV-ARITY-001.

Date / iteration:

vNEXT4.21.

Killed claim:

Ordinary full-interval Bombieri–Vinogradov can support only the old P_4 rough cofactor output.

Status:

KILLED.

Replacement:

The explicit Richert-weighted dimension-one sieve in vNEXT4.21 gives a tagged squarefree P_3 fan with rough floor $N^{6/25-o(1)}$, pending independent analytic audit of the weighted-sieve interface.

FAIL-PAIR-LOADED-DESCENT-OPEN-001

FAIL-ID:

FAIL-PAIR-LOADED-DESCENT-OPEN-001.

Date / iteration:

vNEXT4.15.

Old obstruction:

Rows $N - w = Bq_1q_2$ with $q_i \geq h$ and $B > 1$ might create an uncontrolled low residual packet.

Status:

CLOSED AS A POLYLOG- Λ SCALE-REENTRY PROBLEM.

Exact certificate:

If $\Lambda \leq (\log N)^A$, then $B < 2\Lambda$; residual prime factors are fixed-polylog BAD states. A large family therefore produces reuse or enters the repaired rooted/external clean ledger.

FAIL-PAIRWISE-K-IS-THE-ONLY-POSSIBLE-X1-CAPACITY-001

FAIL-ID:

FAIL-PAIRWISE-K-IS-THE-ONLY-POSSIBLE-X1-CAPACITY-001.

Status:

KILLED.

Rejected statement:

The final X1 stitching must upper-bound the full pairwise functional $K_{\times}(R, P) + K(P)$.

Certificate:

vNEXT4.31 proves the exact promotion hierarchy

$$L(A) \leq \log \frac{J_{R \cup P}}{J_R} \leq K_{\times}(R, P) + K(P).$$

Thus the weaker target K1-INC, bounding only the actual incremental J by the same P1BR functional, is logically sufficient.

Do not repeat:

Do not identify “K1 is sufficient” with “K1 is logically necessary.”

FAIL-PARENT-LATTICE-DOMAIN-001

FAIL-ID:

FAIL-PARENT-LATTICE-DOMAIN-001.

Date / iteration:

v0.1 audit, canonical-ambient split correction.

Claim:

The parent lattice of a layer may be treated as the full residue class $p \equiv N \pmod{\ell^k}$ over all primes.

Original motivation:

The analytic count is naturally stated over all primes in an interval.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

Definitional.

Exact certificate:

The correct object is $\mathcal{P}_N(\ell^k) = \{p \in \mathcal{A}_N : \ell^k \mid N - p\}$, the intersection of the congruence class with the *finite active prime universe* $\mathcal{A}_N$. Primes dividing N , and primes at least N , are not states.

Failed hypothesis or mechanism:

An ambient analytic set was used as a set of CRD states.

Why the argument failed:

Counting in the class is legitimate; treating every counted prime as a usable parent is not.

What survives:

G-POLYLOG-PARENTS, which restricts to $p \in (N/2, N)$ and proves activity and externality explicitly.

Safe weaker version:

Intersect with $\mathcal{A}_N$ at the moment the class is introduced.

Current replacement:

Theory Core, notation entry $\mathcal{P}_N(\ell^k)$ and G-POLYLOG-PARENTS.

Do not repeat:

A residue class is not a state set.

Related current target:

W1, K1.

FAIL-PATH-PRODUCT-AGGREGATION-ERASES-DEEP-VALUATION-001**FAIL-ID:**

FAIL-PATH-PRODUCT-AGGREGATION-ERASES-DEEP-VALUATION-001.

Source:

explicit proposed Failure Log item in v4.51; also v4.51 R51-2.

Status:

KILLED for the inference

remote reflected root+aggregate affine path product $\implies$ control of the terminal exponent E .

Certificate:

after appending the last edge $p \rightarrow r$, the aggregate affine identity recombines the terminal row into the single integer $N - p$; modulo N , the factor r^{E-1} in the total path product is cancellable because $r \nmid N$.

Scope guard:

this is not a no-go for CRD1-R. The full valuation tensor still knows $v_r(N - p) = E$. The loss occurs only after compressing the terminal edge into the aggregate path product.

Replacement:

retain the final row/layer datum (p, r, E) edge-locally.

FAIL-PER-ROW-RADC-COMPRESSION-001**FAIL-ID:**

FAIL-PER-ROW-RADC-COMPRESSION-001.

Status:

KILLED as a failure certificate.

Reason:

LCAP4 controls the full product $D_C J_R U_R^{\text{nr}}$, not the radical factor $D_C J_R$ alone. Large mass may be stored in U_R^{nr} . The revised report had conflated the three-channel product with one channel.

FAIL-PLAIN-AP-COUNT-KILLS-VERTICAL-001**FAIL-ID:**

FAIL-PLAIN-AP-COUNT-KILLS-VERTICAL-001.

Status:

SCOPE-CORRECTED.

Killed sufficient strategy:

use Brun–Titchmarsh or Bombieri–Vinogradov pointwise to rule out one dynamically selected canonical exact layer or to infer non-reachability in R .

Safe surviving statement:

ambient AP estimates do not see canonical reachability and need not kill singleton or exceptional layers.

v4.36 correction:

Brun–Titchmarsh becomes effective after summing over *all* repeated prime-power layers together with the canonical γ -row budget and the weighted layer count $A_2(Y) \ll \sqrt{Y}$. Thus the old no-go was correct pointwise and too broad as an aggregate prohibition.

FAIL-POLYLOG-IS-NEEDED-FOR-DETERMINISTIC-EXACT-STAR-EXPORT-001**FAIL-ID:**

FAIL-POLYLOG-IS-NEEDED-FOR-DETERMINISTIC-EXACT-STAR-EXPORT-001.

Status:

SCOPE-CORRECTED.

Scope correction:

the polylogarithmic restriction belongs to some analytic star-selection / finite-alphabet stages of the old v4.26 apparatus. If a fixed active base already has polynomial row multiplicity, exact-exponent pigeonhole loses only $N^{o(1)}$, and parent-slot / quotient inflation gives a q -free residual export of $\Omega(n \log N)$ for arbitrary base size.

What survives:

the hard part is producing polynomial concentration from the dispersed cube-root defect, not consuming such concentration after it is known.

FAIL-POSITIVE-ROUND-BLOCKS-REPEATED-FAN-PROGRESS-001**FAIL-ID:**

FAIL-POSITIVE-ROUND-BLOCKS-REPEATED-FAN-PROGRESS-001.

Date / iteration:

vNEXT4.18.

Claim:

Because a positive signed singleton round is a dead end, the repeated-base fan gives no progress unless all rounds are nonpositive.

Status:

KILLED.

Correction:

Independently of signs, every selected row supplies a distinct rough active child above $N^{1/4-\varepsilon}$. The polynomial state/source packet therefore survives the positive-round wall, although it still is not an upper-capacity contradiction.

FAIL-POSITIVE-ROUND-CONSEQUENCE-001**FAIL-ID:**

FAIL-POSITIVE-ROUND-CONSEQUENCE-001.

Date / iteration:

vNEXT3.1, Phase-3 semantic audit.

Claim:

Forcing a positive signed round is progress toward a contradiction, so the (FC10) route closes once γ is controlled.

Original motivation:

The (DP1)–(DP5) depletion apparatus is elaborate and its conclusion is that nonpositive rounds cannot continue; the natural reading is that the alternative must be favourable.

Status:

KILLED. The depletion apparatus is untouched; the inference from its escape branch is not supported.

Exact certificate:

By (EB6), $\Delta > 0$ says exactly $\sum_{r \in B_0} \log r > W_R(B_-)$: the fresh support mass of the selected rows exceeds the rooted reuse mass they touch. A mechanical audit of every DOMAIN line and every proof in the Theory Core found three theorems assuming *nonpositive* rounds — G-ROW-SATURATION-ROOTED-DEFECT, G-EXTERNAL-BAD-SUPPORT-SATURATION-DISPOSITION, G-GLOBALLY-CLEAN-EXTERNAL-SUPPORT-BUDGET — and *none* assuming a positive round. Nothing derives a GOOD state, a capacity violation, or any canonical obligation from $\Delta > 0$.

Failed hypothesis or mechanism:

The escape branch of a dichotomy was treated as a conclusion. Structurally $\Delta > 0$ is *growth* of the enlarged world, not contradiction; and growth cannot be iterated to exhaustion, because designated bases live below $(\log N)^A$ while $|\mathcal{A}_N| \sim N/\log N$.

What survives:

Everything about nonpositive rounds, and G-GAMMA-POLYLOG-FORCES-POSITIVE-ROUND, which correctly delivers the positive round. What is corrected is only what that delivery is worth.

Safe weaker version:

“ γ polylog forces a positive signed round.” Never “... forces a contradiction”.

Current replacement:

Theory Core, Proposition [prop:positive-dead-end] and the new open target P1BR-POSITIVE-ROUND-CONSEQUENCE.

Do not repeat:

Do not announce that a wall has moved because an object is now forced to exist; check first that something consumes that object.

Related current target:

P1BR-POSITIVE-ROUND-CONSEQUENCE, P1BR, K1.

FAIL-POSITIVE-ROUND-EXHAUSTION-001**FAIL-ID:**

FAIL-POSITIVE-ROUND-EXHAUSTION-001.

Date / iteration:

vNEXT4, Phase 9.

Claim:

That repeated positive rounds must eventually run out of fresh primes, giving a contradiction by exhaustion.

Original motivation:

Nonpositive rounds deplete the integer potential $\mathfrak{J}_k$ of (DP4); by symmetry one hopes positive rounds deplete something too.

Status:

KILLED, quantitatively.

Exact certificate:

All fresh support of an external upper-half round lies below $N/6$ by (RC2). The total logarithmic demand of the whole selector output at exponent A is at most $(\log N)^{A+1}$; the supply in that band is $\vartheta(N/6) \geq N/12$. The ratio tends to 0 like $(\log N)^{A+1}/N$. Theory Core, G-AMBIENT-SUPPLY-NONDEPLETION.

Failed hypothesis or mechanism:

$\Delta > 0$ is a statement of growth. Growth has no terminal state when the reservoir is superpolynomially larger than the demand.

What survives:

The depletion statement for *nonpositive* rounds, (DP1)–(DP4), which is unaffected and remains the only monotone integer potential in the DAG.

Do not repeat:

Do not argue by exhaustion without comparing demand to ϑ of the relevant band.

Related current target:

P1BR-ADDITIVE-COUPLING.

FAIL-POSITIVE-ROUND-LEDGER-CONSEQUENCE-001**FAIL-ID:**

FAIL-POSITIVE-ROUND-LEDGER-CONSEQUENCE-001.

Date / iteration:

vNEXT4, Phase 7.

Claim:

That the failure recorded at FAIL-POSITIVE-ROUND-CONSEQUENCE-001 was a gap in the present list of theorems, so that a consequence of $\Delta > 0$ could be found by writing more theorems in the same language.

Original motivation:

vNEXT3.1 established option F by a mechanical audit of the file's headers. An audit of a text is silent about what could be added to the text.

Status:

KILLED for ledger-only derivations. **Not** a refutation of P1BR-POSITIVE-ROUND-CONSEQUENCE itself.

Exact certificate:

Define the round datum $\mathfrak{d}(S, \mathcal{P})$ as the rows, their supports, the heights on those supports and κ_S restricted to those supports. Every quantity in (EB6)–(EB10), (DP1)–(DP5) and (PD1) is a function of $\mathfrak{d}$. The family $\mathfrak{S}_n$ of Theory Core G-POSITIVE-ROUND-LEDGER-INDEPENDENCE has constant $\mathfrak{d}$, constant $\Delta > 0$, and $\eta(R_n) = n + 1 \rightarrow \infty$; it is child-closed with a unique source, has no unit rows, and is routing-compatible. Its member $n = 1$ has $\eta(R_1) = 2$ while Δ can be made arbitrarily large by raising the two fresh heights. Hence the only lower bound on η expressible in $\mathfrak{d}$ is the trivial $\eta_{\text{supp}} = \sum_{r \in B_-} (\kappa(r) - 1)$, which is 0 whenever $B_- = \emptyset$; in particular no such bound grows with Δ . The witnesses exist by GEN-FINITE-REALISATION.

Failed hypothesis or mechanism:

The (LE3)+(FC8) route consumes a *lower* bound on $\eta(R)$ of order $\gamma \log N$. A nonpositive round supplies one by (EB8). A positive round supplies an *upper* bound on the rooted mass of its own support and constrains $\Gamma_N(R)$ nowhere else — which is exactly where the witnesses place all their incidence.

What survives:

The asymmetry is proved, not assumed: for a nonpositive round the same datum does bound $\eta(S)$ below. The obstruction is the sign, not the bookkeeping.

Safe weaker version:

A consequence may still be derivable using an axiom that the witness family violates. Three are identified and recorded as live: (L1) $\mathcal{D}(p) = N - p$; (L2) $\gcd(N - p, N - p') \mid p' - p$; (L3) band localisation plus prime counting. All three are additive.

Do not repeat:

Loss-control rule (LR1): a claimed consequence of a positive round must name a quantity that is not a function of the round datum.

Related current target:

P1BR-ADDITIVE-COUPLING.

FAIL-POSITIVE-SEMIPRIME-ROUND-IS-NEXT-PRIMARY-001**FAIL-ID:**

FAIL-POSITIVE-SEMIPRIME-ROUND-IS-NEXT-PRIMARY-001.

Status:

SUPERSEDED.

Superseded description:

after v4.41, the binding branch is to convert a positive clean semiprime round directly through P1BR additive coupling.

Certificate:

v4.42 continues through the actual designated BAD states q_i and their stopped descendant closure. It forces ambient incremental J or a linear cube-root boundary load before any positive-sign consumer is needed.

What survives:

positive sign remains a valid certificate where proved, and P1BR-ADDITIVE-COUPLING remains open in its own right.

FAIL-POTENTIAL-CAPACITY-001**FAIL-ID:**

FAIL-POTENTIAL-CAPACITY-001.

Date / iteration:

v7.5 first-front audit; retained in v0.1.

Claim:

A first-front-only potential capacity $\mathcal{C}_S < \mathcal{D}_S$ holds canonically and supplies the needed contradiction.

Original motivation:

Pairwise row isolation seemed to make the available factor capacity smaller than root demand.

Status:

KILLED.

Minimal / strong counterexample:

$N = 4736, p_0 = 2371, h = 3, m = 2365 = 5 \cdot 11 \cdot 43$.

Exact certificate:

The three first-front rows are

$$4736 - 5 = 4731 = 3 \cdot 19 \cdot 83, \quad 4736 - 11 = 4725 = 3^3 \cdot 5^2 \cdot 7,$$

$$4736 - 43 = 4693 = 13 \cdot 19^2.$$

With the source definitions, the audited values are $\mathcal{C}_S \approx 22.60 > \mathcal{D}_S \approx 20.16$.

Failed hypothesis or mechanism:

Potential parents were counted as if their availability already encoded closure and actual incidence.

Why the argument failed:

Pair isolation controls collisions, not which rows are dynamically realised or simultaneously usable.

What survives:

Exact collision identities for actual rows.

Safe weaker version:

Introduce closure/activation first, then charge only realised rows.

Current replacement:

Theory Core, G-COLLISION-CUT, G-AMBIENT-PROMOTION, and target $P1^{\text{br}}$.

Do not repeat:

Potential incidence is not actual incidence.

Related current target:

$L1, W1, K1$.

FAIL-POWER-THRESHOLD-TO-POLYLOG-001

FAIL-ID:

FAIL-POWER-THRESHOLD-TO-POLYLOG-001.

Status:

SCOPE-CORRECTED.

Original killed formulation:

power-threshold compression can never force a polylogarithmic / near-logarithmic star, because reaching that scale requires $m \rightarrow \infty$.

Correction:

variable m is legal with its explicit occupied-base error. Taking $m \asymp \log N / \log \log N$ can force $q_0 \leq (\log N)^{1+o(1)}$ with multiplicity $N^{1/3-o(1)}$ on the low branch.

What the no-go really says:

one threshold cannot simultaneously reach that star scale and retain the old $M \log N$ -scale generic high- J lower bound. The high branch falls to $M(\log \log N)^2 / \log N$.

Do not repeat:

do not interpret the v4.43 fixed- m calculation as a ban on growing m .

FAIL-PRODUCT-BUDGET-ALONE-CANNOT-PRESERVE-NEARCRITICAL-STAR

FAIL-ID:

FAIL-PRODUCT-BUDGET-ALONE-CANNOT-PRESERVE-NEARCRITICAL-STAR.

Status:

KILLED.

Killed mechanism:

preserve $N^{1/3-o(1)}$ rows under repeated exact-layer peeling using only distinct peeled bases, growth of the common modulus, parent-slot, and separate valuation-layer counts.

Sharp cumulative-loss bound:

writing $X = \log N$, $B_i = \log Q_i$ and $\ell_i = \log q_i$,

$$\mathcal{L}_t := \sum_{i < t} \log E_i \leq \sum_{i < t} \log \frac{X - B_i}{\ell_i} \leq \log \frac{Q_t}{Q_{\text{in}}} + (1 + o(1)) \frac{\log N}{(\log \log N)^2}.$$

The coefficient 1 before $\log(Q_t/Q_{\text{in}})$ is asymptotically sharp from the inspected scalar data alone.

Certificate:

a primorial hostile sequence with consecutive small primes up to $a \log N$, $1/3 < a < 2/3$, satisfies simultaneously

$$Q_t = N^{a+o(1)}, \quad \prod_i E_i = N^{a+o(1)}.$$

Thus the formal cardinality can collapse before the modulus reaches the $N^{2/3}$ parent-slot scale.

Sharp scope guard:

the no-go ignores global compatibility of all valuations in the same integer $N - p$ and therefore does not apply to simultaneous joint-valuation compression.

Current replacement:

G-SUBCRITICAL-ALPHABET-SIMULTANEOUS-VALUATION-FREEZE.

FAIL-PROFILE-CARRIER-001

FAIL-ID:

FAIL-PROFILE-CARRIER-001.

Date / iteration:

Interrupted T_D round, 31 August 2026.

Claim:

The cardinality profile of the four C0b classes separates canonical from noncanonical roots and can therefore carry a theorem.

Original motivation:

Hard canonical controls were reported as D-heavy and $\mathcal{F}$ as associated with easy or direct-GOOD controls.

Status:

SUPERSEDED.

Minimal / strong counterexample:

The recorded profiles themselves.

Exact certificate:

The recovered log states that raw profiles did not separate the relevant outcomes and that the reported associations were explicitly not promoted to theorems. The associated 97.81% D-fresh rate is **VERIFIED COMPUTATION**, with no surviving artifact.

Failed hypothesis or mechanism:

A correlation was treated as a mechanism.

Why the argument failed:

Cardinalities discard the arithmetic that produces the classes.

What survives:

The *negative* conclusion only: cardinality-only theorem design is rejected as a current route.

Safe weaker version:

Select a branch by the theorem it produces, not by the correlation it exhibits.

Current replacement:

Theory Core, TD-CARRIER-SPLIT. T_D is selected because it factorises, not because it correlates.

Do not repeat:

Never promote a computational rate to a mechanism.

Related current target:

T_D .

FAIL-PURE-PATH-ONLY-RETAINS-MOD-Q-001**FAIL-ID:**

FAIL-PURE-PATH-ONLY-RETAINS-MOD-Q-001.

Source:

explicit v4.54 Failure Log addition.

Status:

KILLED.

Killed description:

repeated pure edges preserve only the original endpoint congruence modulo Q .

Correction:

the fixed cofactor Q on every pure edge lifts the path-start congruence to modulus Q^{L+1} .

Scope guard:

this does not contradict the earlier aggregate affine-path no-go, which allowed arbitrary path cofactors and therefore could lose the deep exponent.

FAIL-PURE-Q-GRAPH-CAN-BRANCH-001**FAIL-ID:**

FAIL-PURE-Q-GRAPH-CAN-BRANCH-001.

Source:

explicit v4.54 Failure Log addition.

Status:

KILLED.

Killed possibility:

a pure target q can have two distinct pure parents.

Reason:

a pure parent is uniquely determined by

$$p = N - Qq.$$

Thus the pure graph is injective.

FAIL-Q0-LEDGER-001**FAIL-ID:**

FAIL-Q0-LEDGER-001.

Date / iteration:

Post-v7.5 incidence-ledger audit, 30 August 2026.

Claim:

The four channels proved for the smooth first-front domain cover every $q \in Q_0 = \text{PF}(N - p_0)$ without an under-parented channel.

Original motivation:

The same residual split appeared syntactically available for all root factors.

Status:

KILLED.

Minimal / strong counterexample:

$N = 20, p_0 = 11, Q_0 = \{3\}$.

Exact certificate:

$20 - p_0 = 9, 20 - 3 = 17 \in \mathbb{P}$, so 3 is GOOD. Under stopped traversal $R = \{11\}$. Moreover $D_3 = \gcd(9, 17) = 1, M_3 = 17$, while the required 17-incidence in R is zero although the root demand is one.

Failed hypothesis or mechanism:

A successful first-front state was treated as though it were an expanded bad-world row.

Why the argument failed:

Stopping at GOOD deletes precisely the row needed by the proposed channel accounting.

What survives:

The exact ledger on its verified smooth/failed domain; the arithmetic identity $D_q M_q = N - q$.

Safe weaker version:

Retain the original domain or add an explicit under-parented/success channel.

Current replacement:

Theory Core, Full-Q Failure Ledger and C0b-RD, both with explicit stopped-world hypotheses.

Do not repeat:

Never enlarge a ledger domain merely because its formulas remain syntactically defined.

Related current target:

T_I, T_D, T_S, T_F .

FAIL-Q0-ONLY-OWNER-SEALED-RETURN-IS-CHEAP-001**FAIL-ID:**

FAIL-Q0-ONLY-OWNER-SEALED-RETURN-IS-CHEAP-001.

Status:

KILLED.

Rejected statement:

If owner-sealed verticality occurs only at the small common hub q_0 , the branch is cheap.

Certificate:

polynomial q_0^2 occupancy enters the already retained vNEXT4.26 exact-layer star peeling and exports full q_0 -free logarithmic mass.

FAIL-QFREE-FIRST-COPY-CAN-HIDE-BELOW-CARDINALITY-001

FAIL-ID:

FAIL-QFREE-FIRST-COPY-CAN-HIDE-BELOW-CARDINALITY-001.

Source:

explicit v4.52 Failure Log addition.

Status:

KILLED.

Killed possibility:

on a polynomial exact fibre with cheap J/U_{res} , all unique external q -free first-copy support can remain at primes $\leq n$.

Reason:

the unique external q -free support has $\Omega(n \log N)$ weight, while all primes at most n carry only $O(n)$ total Chebyshev weight.

Replacement:

a constant-scale portion escapes above n , yielding C-DEEP-CARDINALITY-SCALE-SIBLING-RENEWAL.

FAIL-QUOTIENT-SIEVE-POLYNOMIAL-RANGE-001

FAIL-ID:

FAIL-QUOTIENT-SIEVE-POLYNOMIAL-RANGE-001.

Date / iteration:

vNEXT4.6.

Claim:

The fixed- (s, b) two-linear-form upper sieve can be union-bounded without loss over all $s \leq N^\delta$.

Status:

KILLED.

Exact certificate:

For a given s , there are $O(s)$ possible parent coefficients b , while the reflected progression has only $O(t/s)$ indices. These factors cancel, leaving roughly one $t/\log^2 t$ cost per quotient band. The present union bound therefore gives no polynomial range.

What survives:

Fixed and sufficiently slowly growing S satisfying (QS4).

Do not repeat:

Do not spend another round merely optimising constants in QS3 in hopes of reaching N^δ .

FAIL-RDS-GEOMETRY-DENSITY-001

FAIL-ID:

FAIL-RDS-GEOMETRY-DENSITY-001.

Date / iteration:

v1.13 support-free continuation, 1 September 2026.

Claim killed:

From canonical reflected geometry and $\text{rad}(c_j) \mid d_j$ alone, one may infer a universal bound $|\text{RDS}| \leq t/8$.

Status:

KILLED.

Exact certificate:

Take

$$N = 4368, \quad x = 2184, \quad p_0 = 2203, \quad h = 19, \quad \text{prevprime}(p_0) = 2179, \quad G = 24, \quad t = 9.$$

Thus $h < G$, so the root is canonical. The strict reflected block contains

$$j = 3 : d_j = 13, \quad c_j = 2197 = 13^3,$$

and

$$j = 8 : d_j = 3, \quad c_j = 2187 = 3^7.$$

Hence both indices satisfy $\text{rad}(c_j) \mid d_j$, so $|\text{RDS}| \geq 2 > 9/8 = t/8$.

Scope warning:

This is a successful-world certificate: $m = N - p_0 = 2165 = 5 \cdot 433$ and the first-front state 5 is GOOD because $N - 5 = 4363$ is prime. Therefore the witness kills only a *geometry-only* density inference; it does not refute a stronger theorem conditioned on a genuinely stopped canonical failed world.

What survives:

TS-REFLECTED-DEADSMOOTH-CEILING is exact and remains PROVED. Any stronger failure-conditioned density bound requires additional dynamic input, not the reflected geometry alone.

Do not repeat:

Do not infer a small universal RDS density from powerful-number rarity or from finite empirical scarcity without using a proved failure-conditioned mechanism.

FAIL-REFLECTION-ACTIVATION-001

FAIL-ID:

FAIL-REFLECTION-ACTIVATION-001.

Date / iteration:

Canonical bridge audit, 31 August 2026.

Claim:

Canonical $h < G$ alone forces at least one reflected index j with $a_j = N - c_j$ an active prime, hence an actual reflected CRD row.

Original motivation:

The Reflection Gate Lemma makes every c_j composite, and compositeness was read as if it also produced a parent row.

Status:

KILLED.

Minimal / strong counterexample:

$N = 242$, $x = 121$, $p_0 = 127$, previous prime 113, so $h = 6 < G = 14$.

Exact certificate:

The two internal offsets give $(c_1, a_1) = (125, 117)$ and $(c_2, a_2) = (123, 119)$, with $125 = 5^3$, $117 = 3^2 \cdot 13$, $123 = 3 \cdot 41$, $119 = 7 \cdot 17$. Both reflections are composite, so no reflected row exists at all. Verified exactly.

Failed hypothesis or mechanism:

The gate was treated as a state activation statement.

Why the argument failed:

ARS-REFLECTION-GATE is an *if and only if*: the factorisation of c_j is a CRD row precisely when a_j is an active prime, and nothing in $h < G$ supplies that.

What survives:

Compositeness of every c_j , and the classification COb-RD which does not require any a_j to be prime.

Safe weaker version:

State separately *boundary* $\rightarrow$ *active reflected state* as an open implication.

Current replacement:

Theory Core, ARS-REFLECTION-GATE; target BIA remains **OPEN**.

Do not repeat:

Do not hide a missing activation step under the word “reflection”.

Related current target:

T_I , $P1^{\text{br}}$.

Domain note:

$N = 242$ is not a failed canonical instance, so it does not refute BIA itself; it refutes any proof of BIA that uses only selector geometry before invoking failure.

FAIL-REPEATED-BASE-MUST-BE-FIXED-POLYLOG-001**FAIL-ID:**

FAIL-REPEATED-BASE-MUST-BE-FIXED-POLYLOG-001.

Date / iteration:

vNEXT4.18.

Restriction:

The fixed-polylog multiset limit of the four-ary multiparent selector is a fundamental limit on repeated rows from one base.

Status:

KILLED.

Correction:

A greedy rough-support matching argument selects

$$\gg N^{1/4-\varepsilon}/(\log N)^2$$

rows from one already-constructed base fan while keeping unintended rough supports pairwise disjoint.

FAIL-REPEATED-BASE-QUADRATIC-COLLISION-UNAVOIDABLE-001**FAIL-ID:**

FAIL-REPEATED-BASE-QUADRATIC-COLLISION-UNAVOIDABLE-001.

Date / iteration:

vNEXT4.18.

Claim:

Repeating a designated base necessarily incurs $\binom{\rho_q}{2} \log q$ collision even when rows are separated.

Status:

SCOPE-CORRECTED.

Correction:

The quadratic term is unavoidable inside one multiparent round. Separate singleton rounds with planned designated overlap and disjoint unintended rooted payment supports admit exact single charging.

FAIL-REPEATED-HUB-QUADRATIC-COST-EQUALS-ACTUAL-STORAGE-001**FAIL-ID:**

FAIL-REPEATED-HUB-QUADRATIC-COST-EQUALS-ACTUAL-STORAGE-001.

Status:

KILLED.

Rejected statement:

A fresh repeated hub of multiplicity M creates $\binom{M}{2} \log q$ units of horizontal factor-flow storage.

Certificate:

actual incremental J at that layer is $(M-1) \log q$; the exact difference $((M-1)(M-2)/2) \log q$ is pairwise curvature.

Do not repeat:

Pairwise-gcd statistics and incremental storage answer different questions.

FAIL-RETURN-COEFFICIENT-K-IS-THE-INDEPENDENT-BINDING-WALL-001

FAIL-ID:

FAIL-RETURN-COEFFICIENT-K-IS-THE-INDEPENDENT-BINDING-WALL-001.

Status:

SUPERSEDED.

Superseded description:

Analyse $k_i = (p_i - u_i)/r_i$ in isolation.

Replacement:

Use the full owner-return relation $a_i = q_0 c_i + k_i$ and compare the support of a_i with the owner support. Preserved support forces verticality; changed support gives a foreign sibling and hence reuse or renewed state population.

FAIL-RETURN-COFACTOR-IS-FREE-001

FAIL-ID:

FAIL-RETURN-COFACTOR-IS-FREE-001.

Date / iteration:

vNEXT4.6.

Claim:

In $N - w = bq$, the cofactor b can be discarded as an unstructured residual.

Status:

KILLED as a description.

Exact certificate:

For a block-derived low-core target,

$$\frac{10}{9}s < b < 2s, \quad (b, N) = 1, \quad \text{PF}(b) \subset R.$$

Hence b is a rooted cofactor packet of comparable scale and its support must be retained.

Do not repeat:

Do not project a quotient-return event to the base q alone.

FAIL-RETURNED-SEED-FIRST-OCCURRENCE-IS-J-001

FAIL-ID:

FAIL-RETURNED-SEED-FIRST-OCCURRENCE-IS-J-001.

Status:

KILLED.

Killed inference:

if $\ell \notin R$ is returned by an extension row, that occurrence immediately contributes $\log \ell$ to $\log(J_{R \cup A}/J_R)$.

Certificate:

when $\kappa_R(\ell) = 0$, the first occurrence is the baseline copy in the exact fresh factor $F_R(A)$. Only a second occurrence contributes full-weight incremental J .

What survives:

owner-anchored labels or later returns do enter incremental J exactly as recorded by the v32 second-occurrence theorem.

FAIL-RETURNING-BASIN-IS-ABSOLUTELY-COSTFREE-001

FAIL-ID:

FAIL-RETURNING-BASIN-IS-ABSOLUTELY-COSTFREE-001.

Status:

SCOPE-CORRECTED.

Correction:

A returning basin may leave a functional evaluated on the fixed set R unchanged, but relative

to the enlarged child-closed world it has the exact incremental TUJ/source-lock cost. A nonempty source-free BAD extension satisfies $T_S/T_R \geq N - 1$.

Do not repeat:

Distinguish fixed- R invisibility from zero incremental cost.

FAIL-REUSE-DOUBLE-CHARGE-001

FAIL-ID:

FAIL-REUSE-DOUBLE-CHARGE-001.

Date / iteration:

v1.21 depletion audit, 1 September 2026.

Claim:

If one nonpositive clean round forces a rooted reuse payment $J_* \mid J_R$, then running clean selection again automatically forces a new, disjoint payment and therefore decreases the remaining rooted budget.

Status:

KILLED.

Exact logical obstruction:

The second “round” may select the identical family or may use complements sharing exactly the same rooted negative labels. Then the same set B_- and the same factor J_* are being observed again. Nothing implies $J_*^2 \mid J_R$ or any decrease of a residual quotient.

What survives:

If complete complement supports are pairwise disjoint across rounds, then the negative rooted label sets are disjoint and the product of the corresponding J_i divides J_R .

Current replacement:

Theory Core, G-MULTIROUND-EXTERNAL-SINGLE-CHARGE and
G-BAD-BASE-FORBIDDEN-SUPPORT-CLEAN-CONTINUATION.

Do not repeat:

Never infer depletion from repeated invocation of a selector alone. First prove cross-round support freshness or enter the explicit support-saturation alternative.

Related current target:

K1, P1BR, TF renewal termination.

FAIL-ROOT-GENERATION-AC-ALONE-CLOSES-HQ-RPD-001

FAIL-ID:

FAIL-ROOT-GENERATION-AC-ALONE-CLOSES-HQ-RPD-001.

Status:

SCOPE-CORRECTED.

Killed overreach:

a distribution estimate for $m_k(r) = \#\{i : r^k \mid N - q_i\}$ on the initial v4.41 targets automatically proves RPD for H_Q .

Certificate:

$H_Q = S_Q \cap (N^{1/3}, N/3)$ contains later high descendants of the stopped basins as well as the initial q_i .

What survives:

an initial-target estimate is a valid closure input only after an ancestry-localisation theorem shows that every uncharged full-scale deep obstruction can be moved to an equivalent designated generation.

FAIL-ROOTED-AMBIENT-001

FAIL-ID:

FAIL-ROOTED-AMBIENT-001.

Date / iteration:

Canonical bridge audit, 31 August 2026.

Claim:

After adjoining a selected set $\mathcal{P}$ of external active parents, the old rooted lock still controls the enlarged system, so T_R, U_R, J_R bound the promoted valuation mass.

Original motivation:

The rooted congruence was the strongest available global identity and was reused outside its domain.

Status:

KILLED.

Minimal / strong counterexample:

Structural, by G-MULTISOURCE-LOCK.

Exact certificate:

For child-closed S , $T_S \equiv (-1)^{|S|} \prod_{d \in D_S^{\text{src}}} d \pmod{N}$. Adjoining $\mathcal{P}$ and re-closing changes the source set from $\{p_0\}$ to a set containing $\mathcal{P}$, so the right-hand side changes. A large ambient J is consistent with a large source product.

Failed hypothesis or mechanism:

A one-source identity was applied to a multi-source system.

Why the argument failed:

The lock is an identity *about the sources*, not a bound on the collision channel.

What survives:

The exact multi-source congruence itself, and the promotion inequality $\log(J_{R \cup \mathcal{P}}/J_R) \leq K_{\times}(R, \mathcal{P}) + K(\mathcal{P})$.

Safe weaker version:

Any future capacity theorem must book the new sources explicitly.

Current replacement:

Theory Core, G-MULTISOURCE-LOCK, G-AMBIENT-PROMOTION, target K1.

Do not repeat:

Never carry a rooted bound across a change of source set.

Related current target:

K1, W1.

FAIL-ROUTING-INDIFFERENCE-SCOPE-001**FAIL-ID:**

FAIL-ROUTING-INDIFFERENCE-SCOPE-001.

Date / iteration:

vNEXT4.1, Phase 0A.

Claim:

vNEXT4 (RI1), TD-UNIVERSAL-ROUTING $\not\Rightarrow$ P1BR.

Original motivation:

Under routing every fresh seed returns, and the return branch was shown to move no recorded quantity of R .

Status:

SCOPE-CORRECTED. The theorem survives; its reading does not.

Exact certificate:

The proof enumerates the *present* functionals $\kappa_R, \Gamma_N(R), \eta(R), J_R, U_R, T_R, \gamma, C, H, \tau, Q$ and observes that each is a function of the set R , which does not change when a closure returns to it. That establishes a non-implication *through those functionals*. It says nothing about a theorem that uses the arithmetic of the return path — the actual integers $N - p$ along it, the congruences placing a returning seed inside R , or $\gcd(N - p, N - u) \mid p - u$. Quantifying over all future theorems was never proved and cannot be proved this way.

What survives:

Theory Core, G-ROUTING-INDIFFERENCE-SCOPED: routing alone together with the currently recorded CRD ledgers does not close P1BR.

Safe weaker version:

the scoped form above, which is what is used.

Do not repeat:

Loss-control rule (LR5) is amended: routing may not be cited as a *sufficient* route to P1BR through the recorded ledgers; it remains admissible as an *ingredient* of an additive argument.

Related current target:

P1BR-ADDITIVE-COUPLING, TD-UNIVERSAL-ROUTING.

FAIL-ROUTING-LOSES-SEED-WEIGHT-001**FAIL-ID:**

FAIL-ROUTING-LOSES-SEED-WEIGHT-001.

Status:

KILLED.

Killed description:

after basin merging/returns, the original logarithmic weight of the external seed packet may disappear from all exact ledgers.

Certificate:

v4.38 proves

$$\prod_{\ell \in E} \ell \mid \Sigma_{\text{ext}}(S) F_R(A),$$

so

$$W(E) \leq \log \Sigma_{\text{ext}}(S) + \log F_R(A).$$

For $|E| \gg N^{1/3}$ one of the two reservoirs has ambient polynomial weight.

What survives:

neither reservoir yet has the required rooted upper consumer.

FAIL-ROUTING-RESCUES-P1BR-001**FAIL-ID:**

FAIL-ROUTING-RESCUES-P1BR-001.

Date / iteration:

vNEXT4, Phase 6.

Claim:

That proving TD-UNIVERSAL-ROUTING — every BAD child-closed closure meets R — would remove the obstruction at P1BR-POSITIVE-ROUND-CONSEQUENCE, since it removes the disjoint-packet configuration “without requiring any new theory”.

Original motivation:

vNEXT3.1 recorded, correctly, that routing removes the disjoint-packet discussion, the recorded-ledger bridge theorem and the CRD-2 question. It was natural to expect that it also unblocked the positive round.

Status:

KILLED. The routing statement itself is untouched and remains **OPEN**; only its claimed relevance to P1BR is rejected.

Exact certificate:

Under routing, $B_{\text{ext}} = \emptyset$ and every fresh seed of a positive round returns. A returning seed r produces the enlarged world $R' = R \cup R_N(r)$, but κ_R , $\Gamma_N(R)$, $\eta(R)$, J_R , U_R , T_R , γ , $|R|$, C , H , τ , Q are functions of the set R alone and do not move; and the capacity bounds (FC7)–(FC10) do not transfer to R' , whose source set is not a single upper-half root with composite complement. See Theory Core, G-RETURN-BRANCH-NO-CANONICAL-CHARGE

and G-ROUTING-INDIFFERENCE, (RI1).

Second, independent certificate:

The witness family of G-POSITIVE-ROUND-LEDGER-INDEPENDENCE is routing-compatible — every fresh state's closure meets R_n — and still has $\eta(R_n)$ unbounded at fixed round datum.

Failed hypothesis or mechanism:

Confusing “the seed reaches R ” with “ R is charged”. Reaching a fixed set does not alter any functional of that set.

What survives:

Routing remains the right question for the bridge, the disjoint-packet configuration and the CRD-2 question, and the restricted form TD-POSITIVE-ROUND-ROUTING is recorded in the Core.

Do not repeat:

Loss-control rule (LR5): TD-UNIVERSAL-ROUTING may not be cited as a *sufficient* route to P1BR through the recorded ledgers. Scope-corrected at vNEXT4.1: see FAIL-ROUTING-INDIFFERENCE-SCOPE-001; routing remains admissible as an ingredient of an additive argument.

Related current target:

P1BR-ADDITIVE-COUPPING, TD-UNIVERSAL-ROUTING.

FAIL-SATURATION-DOUBLE-CHARGE-001

FAIL-ID:

FAIL-SATURATION-DOUBLE-CHARGE-001.

Date / iteration:

v1.22 support-saturation continuation, 1 September 2026.

Claim:

If a current external BAD packet substantially overlaps the support of an earlier nonpositive globally clean family, then the pivot payment (SP3) is an additional disjoint factor which can be multiplied into the old v1.21 depletion product.

Status:

KILLED.

Why the inference fails:

The pivot uses a subfamily of the *same old rows*. Its negative rooted support can therefore be a subset of the already charged payment set of those rows. Relabelling changes the certificate used to lower-bound the payment, not the underlying layer-profile balance.

What survives:

The overlap gives the quantitative localisation (SP1)–(SP5), and may reveal a positive signed subfamily. It becomes a new depletion charge only after a separate proof of support disjointness from all previous payment sets.

Current replacement:

Theory Core, G-SATURATED-BASE-RELABEL-PIVOT together with
G-MULTIROUND-EXTERNAL-SINGLE-CHARGE.

Do not repeat:

Never add a pivot-derived rooted factor to $\prod_i J_i$ merely because the designated bases are new.

Related current target:

K1, globally clean renewal termination.

FAIL-SEALED-MULTISOURCE-LOCK-001

FAIL-ID:

FAIL-SEALED-MULTISOURCE-LOCK-001.

Date / iteration:

v1.6 sealed-source continuation, 1 September 2026.

Claim:

If many external upper-half bad parents have all children inside the canonical failed world R , adjoining them and applying **G-MULTISOURCE-LOCK** yields a new modular obstruction stronger than the rooted lock.

Original motivation:

The sealed branch creates genuine new sources, and the multi-source congruence explicitly records the source product.

Status:

KILLED.

Minimal / strong counterexample:

Structural; the proposed new congruence reduces identically to the old one.

Exact certificate:

Let $\mathcal{P}$ be finite, upper-half, external and bad with $\Gamma_N(\mathcal{P}) \subseteq R$, and put $S = R \cup \mathcal{P}$. Then

$$\text{Src}(S) = \{p_0\} \dot{\cup} \mathcal{P}, \quad T_S = T_R \prod_{p \in \mathcal{P}} (N - p).$$

The multi-source lock reads

$$T_R \prod_{p \in \mathcal{P}} (N - p) \equiv (-1)^{|R|+|\mathcal{P}|} p_0 \prod_{p \in \mathcal{P}} p \pmod{N}.$$

Using $N - p \equiv -p \pmod{N}$ and cancelling the active labels p , which are units modulo N , leaves exactly

$$T_R \equiv (-1)^{|R|} p_0 \pmod{N},$$

the original **G-ROOT-LOCK** congruence.

Failed hypothesis or mechanism:

A larger source set was assumed to create new modular information merely because the source product changed.

Why the argument failed:

The row factors contributed by the new sealed sources are congruent to the negatives of those same source labels, so their entire modular contribution cancels.

What survives:

G-SEALED-SOURCE-LOCK-REDUCTION is an exact **PROVED** structural statement. The sealed branch remains potentially useful through population size, valuation/reuse overload, or deeper topology, but not through the source-lock congruence alone.

Safe weaker version:

Use the sealed-source lift as bookkeeping only; seek a noncongruential invariant for the new sources.

Current replacement:

Theory Core v1.6, **G-SEALED-SOURCE-LOCK-REDUCTION**; see also the four-ary sign-overload disposition theorem in the Core.

Do not repeat:

Do not count new sealed sources as new modular constraints without simplifying the multi-source congruence first.

Related current target:

sealed-source branch, P1BR, K1.

FAIL-SELECTOR-CLOSES-CASCADE-001

FAIL-ID:

FAIL-SELECTOR-CLOSES-CASCADE-001.

Date / iteration:

vNEXT3, Phase 4.

Claim:

Once G-BAD-BASE-SQUAREFREE-POLYLOG-EXTENSION is proved, the (FC10) clean-round route yields a contradiction.

Original motivation:

The selector was the last analytic blocker on that route.

Status:

KILLED . The selector is **PROVED** ; the inference is not.

Exact certificate:

The selector supplies at most one clean row per designated base, and the bases lie in $[3, (\log N)^A]$, so $\sum_i M_i \log z_i \ll (\log N)^A \log N$. (FC10) forces a positive signed round only once $\sum_i M_i \log z_i \geq \gamma \log(25N^2/18)$, i.e. once $\gamma \lesssim (\log N)^A$. The cascade therefore closes exactly on the subdomain $\gamma \leq (\log N)^A$ and on no larger one by this route.

Failed hypothesis or mechanism:

Removing an analytic obstruction was read as closing the chain; the counting obstruction downstream is untouched.

What survives:

G-CASCADE-GAMMA-TERMINUS, which states the subdomain exactly, and the selector itself.

Safe weaker version:

“Contradiction on $\gamma \leq (\log N)^A$.”

Current replacement:

Theory Core, the retired G-CASCADE-GAMMA-TERMINUS, whose proof body is kept at Subsection [fail:cascade-body] of this document and no longer in the Core.

Do not repeat:

Do not announce a cascade closure without counting the number of rounds the selector actually supplies against the capacity it must exhaust.

Related current target:

γ , K1, X1.

FAIL-SHORTEN-PARENT-INTERVAL-TO-GET-P3-001**FAIL-ID:**

FAIL-SHORTEN-PARENT-INTERVAL-TO-GET-P3-001.

Date / iteration:

vNEXT4.21.

Strategy:

Shorten the parent interval so the cofactor is smaller and use classical short-interval BV plus an unweighted lower linear sieve to force P_3 .

Status:

KILLED relative to that stated analytic architecture.

Reason:

The loss in available distribution level dominates the gain in cofactor length. The successful vNEXT4.21 P_3 theorem instead uses weighted-sieve information at the full-interval BV level.

Scope guard:

This is not a universal impossibility result for stronger short-interval/switching technology.

FAIL-SIMPLE-STATE-001**FAIL-ID:**

FAIL-SIMPLE-STATE-001.

Date / iteration:

v0.1 foundations review.

Claim:

The simple directed support graph on $\mathcal{A}_N$ is a faithful presentation of the CRD dynamics.

Original motivation:

Reachability questions are graph questions.

Status:

SUPERSEDED.

Minimal / strong counterexample:

Structural.

Exact certificate:

The support graph loses the valuations $\nu_N(p, q) = v_q(N - p)$, hence the prime-power layers $\mu_{\ell, k}$, hence the collision and cut functionals; and it carries no record of ambient external parents.

G-TUJ, G-FULLQ-LEDGER, G-COLLISION-CUT and G-AMBIENT-PROMOTION are all invisible to it.

Failed hypothesis or mechanism:

Edge existence was taken for the whole structure.

Why the argument failed:

$T = UJ$, residual production, prime-power collision and ambient promotion all depend on valuations, not on the mere existence of an edge.

What survives:

Graph-level facts: closure, terminal SCC, source deletion.

Safe weaker version:

Use the support graph only for reachability statements.

Current replacement:

Theory Core, the rooted ambient valuation-incidence object $\mathfrak{C}_N = (N, \mathcal{A}_N, p_0, \nu_N)$.

Do not repeat:

Do not state a valuation theorem in graph language.

Related current target:

L1, K1.

FAIL-SINGLE-BASE-CONCENTRATION-NEEDED-FOR-VERTICAL-MIGRATION-001**FAIL-ID:**

FAIL-SINGLE-BASE-CONCENTRATION-NEEDED-FOR-VERTICAL-MIGRATION-001.

Status:

KILLED.

Killed inference:

a polynomial exact star on one base must first be extracted before diffuse vertical storage can be forced into another CRD ledger.

Certificate:

v4.37 k -power divisor packing sums over all powerful parts simultaneously and forces a D/J lower bound without a common base.

What survives:

exact-star peeling is still useful when concentration is actually present.

FAIL-SINGLETON-STAR-LOCAL-WINDOW-001**FAIL-ID:**

FAIL-SINGLETON-STAR-LOCAL-WINDOW-001.

Date / iteration:

vNEXT4.16.

Claim:

A fixed small prime r cannot divide parent cofactors for many high reflected targets because the legal coefficient interval is too short.

Status:

KILLED.

Exact obstruction:

For $c_j = sq$, the admissible cofactor interval has length $> 5s/9$. If $s > (18/5)r$, it already contains an odd multiple of r .

What remains:

Primality/rootedness, mirror feedback, or a signed multiplicity capacity must be used.

FAIL-SIZE-COMPARISON-AGAINST-RECORDED-CAPACITY-001**FAIL-ID:**

FAIL-SIZE-COMPARISON-AGAINST-RECORDED-CAPACITY-001.

Status:

SCOPE-CORRECTED.

Killed inference:

derive $X \geq Y$ using only the same local size inequalities $x_i \leq y_i$ whose product already proves $X \leq Y$.

Reason:

this merely asks the local envelope to contradict itself.

What survives:

an independent global, canonical, high-gap, analytic or other nonlocal theorem may imply $X \geq Y$ on a special failed domain; together with the envelope this would prove that the domain is empty. Therefore the strong revised2 phrase “no global lower bound can ever contradict the envelope” is not an active prohibition.

FAIL-SMALL-GAMMA-MAKES-CANONICAL-BLOCK-INVISIBLE-001**FAIL-ID:**

FAIL-SMALL-GAMMA-MAKES-CANONICAL-BLOCK-INVISIBLE-001.

Status:

KILLED.

Killed inference:

$\gamma < N^{1/3}/\log N$ makes the high-gap reflected block too small to yield a polynomial CRD object.

Certificate:

the whole-block disposition with any fixed $1/3 < \delta < 0.475$ forces

$$E \subset (\mathcal{A}_N \setminus R) \cap (0, N/5), \quad |E| \gg N^{1/3}.$$

What survives:

the packet is external and requires a source/fresh consumer; it is not automatically absorbed by R .

FAIL-STAR-SELF-REENTRY-001**FAIL-ID:**

FAIL-STAR-SELF-REENTRY-001.

Status:

KILLED.

Killed inference:

the star $N^{1/3-1/(m+1)-o(1)}$ produced at fixed m can be fed back into the same power-threshold theorem to obtain a terminating descent.

Certificate:

the theorem requires a packet carrying defect $\gg N^{1/3}$, while the fixed- m star has an exponent deficit $1/(m+1) > 0$.

Scope guard:

v4.44 does not use such self-reentry; it chooses growing m before concentration and then uses

simultaneous joint-profile freezing.

FAIL-STRIP-COMMON-BASE-THEN-IMPORT-CLEAN-ROUND-001

FAIL-ID:

FAIL-STRIP-COMMON-BASE-THEN-IMPORT-CLEAN-ROUND-001.

Source:

v4.52 scope correction and explicit Failure Log addition.

Status:

KILLED.

Killed inference:

remove the common r^e and then automatically apply G-EXTERNAL-BASE-SIGNED-REUSE as though the residuals were pairwise coprime and squarefree.

Reason:

the full complements are not pairwise coprime, and the stripped residuals b_p need not be pairwise coprime or squarefree.

Replacement:

use the exact r -stripped surplus ledger and split into paid residual collision/verticality or owned unique external sibling renewal. The stronger pairwise-coprime interface estimate remains conditional.

FAIL-TAGGED-P3-RESIDUE-FORCES-RETURN-001

FAIL-ID:

FAIL-TAGGED-P3-RESIDUE-FORCES-RETURN-001.

Date / iteration:

vNEXT4.21.

Killed inference:

Knowing $b \bmod q$ together with $\Omega(b) \leq 3$ forces some sibling $r \mid b$ to satisfy $r \equiv N \pmod{q}$.

Status:

KILLED for every prime $q \geq 5$.

Exact certificate:

With $G = (\mathbb{Z}/q\mathbb{Z})^\times$ and $a = N \bmod q$,

$$(G \setminus \{a\})^2 = G.$$

Thus two non-return sibling residues already realise every possible product residue.

Exceptional case:

$q = 3$ is different. If $N \equiv 2 \pmod{3}$ and $b \equiv 2 \pmod{3}$, at least one sibling returns to 3.

FAIL-TD-NAIVE-001

FAIL-ID:

FAIL-TD-NAIVE-001.

Date / iteration:

Interrupted T_D branch round, 31 August 2026.

Claim:

Every composite D -carrier a_j has an active prime factor, so every $\mathcal{D}$ index produces an actual CRD state.

Original motivation:

Fresh carriers do produce active states, and the smooth case was assumed to differ only quantitatively.

Status:

KILLED.

Minimal / strong counterexample:

$N = 1862$, $r = 1733$, $j = 2$.

Exact certificate:

$m = N - r = 129$ and $a_j = m + 2j = 133 = 7 \cdot 19$, while $7 \mid 1862$ and $19 \mid 1862$. Both prime factors of the carrier divide N , so neither is active and a_j contributes no CRD row. The reflected partner is $c_j = 1729 = 7 \cdot 13 \cdot 19$. Verified exactly.

Failed hypothesis or mechanism:

Compositeness of the carrier was confused with activity of its factors.

Why the argument failed:

Activity is a condition on N , not on a_j ; a carrier can be built entirely from primes dividing N .

What survives:

The smooth/fresh split of TD-CARRIER-SPLIT covers this instance instead of hiding it: a *fresh* carrier, that is one with $P^+(a_j) \geq h$, does give an active prime state $s_j < N/6$ by strict reflected-block nonreuse. The exception lives in $\mathcal{D}_{\text{sm}}$.

Safe weaker version:

Assert activity only for $j \in \mathcal{D}_{\text{fr}}$.

Current replacement:

Theory Core, TD-CARRIER-SPLIT and C-STRICT-NONREUSE.

Do not repeat:

Never promote “composite” to “has an active factor”.

Related current target:

T_D , $P1^{\text{br}}$.

Domain note:

$r = 1733$ is a *noncanonical* upper root; the canonical selector for $N = 1862$ is $p_0 = 937$. The certificate is an exact arithmetic fact and is domain-independent, but it must not be quoted as a canonical-world witness.

FAIL-TD-SMOOTH-INACTIVE-001**FAIL-ID:**

FAIL-TD-SMOOTH-INACTIVE-001.

Date / iteration:

Phase-3 T_D second-layer attack, 31 August 2026.

Claim:

The canonical geometry alone — p_0 canonical, $h < G$, $a_j = m + 2j$ odd composite, c_j composite — forces a smooth reflected carrier to transport some cost: an active state, a prime-power layer, or a source term.

Original motivation:

TD-CARRIER-SPLIT gives $\Omega(a_j) > \log m / \log h$, a large factorisation mass, and mass was assumed to be transportable.

Status:

KILLED as a proof route resting on the canonical geometry. It is *not* a refutation of any statement conditioned on canonical failure; see the domain note below.

Minimal / strong counterexample:

$N = 168$, $x = 84$, $p_0 = 89$, $h = 5 < G = 6$, $m = 79$, $t = 2$, $j = 1$. Smallest instance with *composite* complement, hence with a genuine first front: $N = 2684$, $p_0 = 1361$, $h = 19 < G = 34$, $m = 1323 = 3^3 \cdot 7^2$, $j = 4$, $a_4 = 1331 = 11^3$, $c_4 = 1353 = 3 \cdot 11 \cdot 41$, $11 \mid 2684$.

Exact certificate:

$a_1 = m + 2 = 81 = 3^4$ and $c_1 = N - a_1 = 87 = 3 \cdot 29$. Since $3 \mid 168$, the only prime of a_1 divides N , so $3 \notin \mathcal{A}_N$. The carrier therefore contributes no active state, no prime-power layer of the CRD system, and no source. Its whole factorisation mass $\Omega(a_1) = 4$ is inert. Verified exactly.

Domain note — read before transferring this witness:

all six known instances are *successful* worlds. At $N = 168$ and $N = 492$ the complement m is prime, so p_0 is GOOD at depth zero and $R = \emptyset$; at $N = 2684, 4380, 4386, 6260$ the complement is composite but the traversal succeeds deeper. In none of them is R a failed world, so in none of them is the index j shown to lie in $\mathcal{D}$. By PROHIB-DOMAIN-TRANSFER this witness therefore kills only proofs that use the canonical geometry alone; the failure-conditioned form of the claim is untouched and remains open. This is the same shape of kill as FAIL-TF-LOCAL-SCOPE-001.

Further instances:

$(N, p_0, j, a_j) = (492, 251, 1, 3^5), (4380, 2203, 5, 3^7), (4386, 2203, 2, 3^7), (6260, 3137, 1, 5^5)$. Over canonical $N < 1.2 \cdot 10^5$, 22.24% of composite reflected carriers have at least one prime factor dividing N .

Failed hypothesis or mechanism:

Ω -mass of a_j was identified with CRD mass. The two live in different objects: a_j is not a row, and its prime factors need not be states.

Why the argument failed:

Activity is a condition relative to N . A carrier assembled from primes dividing N is invisible to the CRD system.

Structural note:

such a carrier is necessarily *smooth*. If $\ell \mid a_j$ and $\ell \mid N$ then $\ell \mid N - a_j = c_j$, so ℓ divides two distinct members of the strict reflected block and C-STRICT-NONREUSE forces $\ell < h$. The fresh branch is therefore immune, which is exactly the content of TD-CARRIER-SPLIT: only $\mathcal{D}_{\text{sm}}$ can be inert.

What survives:

The smooth branch is bounded in *count* rather than transported in *mass*. TD-SMOOTH-CEILING gives $|\mathcal{D}_{\text{sm}}| \leq \sigma_N(h)$ unconditionally, and that ceiling is what the second-layer trichotomy consumes.

Safe weaker version:

Never assert that a smooth carrier produces a state. Use only its cardinality, via the ceiling.

Current replacement:

Theory Core, TD-SMOOTH-CEILING and TD-SECOND-LAYER.

Do not repeat:

Do not attempt to transport smooth carrier mass to actual layers. That direction is closed by this certificate; the productive direction is the ceiling.

Relation to FAIL-TD-NAIVE-001:

complementary, not stronger. FAIL-TD-NAIVE-001 uses a noncanonical root, $(N, r) = (1862, 1733)$ with canonical selector 937, and kills “every composite carrier has an active factor”. This entry uses canonical roots with $h < G$ and kills the same claim for any proof resting on that geometry, while strengthening the configuration from “some prime factor is inactive” to “the carrier is wholly inert”. Neither entry reaches the canonical *failed* domain, because no canonical failed world is known.

Related current target:

$T_D, P1^{\text{br}}$.

Exact classification (Phase-3 continuation):

the failure mode is now completely understood. By TD-CARRIER-DICHOTOMY, a carrier produces no active state *precisely* when $\text{rad}(a_j) \mid h - 2j$. All six instances above are of that kind, and each realises the normal form $x = q^e + d$: $84 = 3^4 + 3$, $246 = 3^5 + 3$, $1342 = 11^3 + 11$, $2190 = 3^7 + 3$, $2193 = 3^7 + 6$, $3130 = 5^5 + 5$. The class is bounded by TD-DEADSMOOTH-CEILING and is empty whenever no prime below h divides x — 48.0% of canonical instances with $N < 3 \cdot 10^5$. Across that whole range only 19 dead-smooth carriers exist among 514447, and no N carries two.

FAIL-TF-LOCAL-SCOPE-001

FAIL-ID:

FAIL-TF-LOCAL-SCOPE-001.

Date / iteration:

Interrupted T_D branch round, hostile scope witness, 31 August 2026.

Claim:

A fresh reflected state ℓ_j , certified locally as $\mathcal{F}$ -class together with $h < G$, routes to Good_N or into R .

Original motivation:

The recorded census showed $\mathcal{F}$ to be empirically rigid, and the local certificate looked self-sufficient.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

$N = 128$, $p_0 = 67$, $j = 1$, $c_1 = 65$, $\ell = 13$.

Exact certificate:

$x = 64$, $p_0 = 67$, previous prime 61, so $h = 3 < G = 6$; $t = 1$ and $c_1 = 65 = 5 \cdot 13$ with $P^+(c_1) = 13 \geq h$, so the local $\mathcal{F}$ certificate holds. The state 13 is active and not GOOD, since $128 - 13 = 115 = 5 \cdot 23$, and its stopped traversal fails with the separate closed world $\{3, 5, 7, 11, 13, 23, 29, 41\}$. Verified exactly.

Failed hypothesis or mechanism:

The certificate was evaluated without the hypothesis that defines R .

Why the argument failed:

Here $128 - 67 = 61$ is prime, so p_0 *succeeds* at depth zero and $R = R_N(p_0) = \emptyset$. The condition $\text{PF}(c_j) \cap R = \emptyset$ is satisfied vacuously. The instance is a *successful world*, and R is not even defined as a failed world.

What survives:

The canonical-failure-conditioned form of TF is untouched. What is killed is every proof of it that uses only the local $\mathcal{F}$ certificate together with $h < G$.

Safe weaker version:

Always carry the failure hypothesis into the $\mathcal{F}$ certificate; never evaluate $\mathcal{I}, \mathcal{D}, \mathcal{S}, \mathcal{F}$ at a root that succeeds.

Current replacement:

Theory Core, target TF.

Do not repeat:

A vacuous R is not an empty support intersection in the intended sense.

Related current target:

TF, T_D .

FAIL-TWOVARIABLE-FULL-LEVEL-001

FAIL-ID:

FAIL-TWOVARIABLE-FULL-LEVEL-001.

Date / iteration:

vNEXT4.12.

Claim:

The exact local density $(2p - 1)/p^2 = 2/p - 1/p^2$ for the polynomial pair $q(N - kq^2)$ permits an elementary sieve level equal to a fixed power of N .

Status:

KILLED.

Exact certificate:

The coefficient coordinate has length only

$$K = N/h^2.$$

The elementary remainder term therefore restricts the second logarithmic saving to $\log K$, not $\log N$.

Do not repeat:

A local dimension-two identity is not a distribution level theorem.

FAIL-UNIQUE-FRESH-TARGETS-MAY-FUNNEL-INTO-O(n)-PARENTS-001**FAIL-ID:**

FAIL-UNIQUE-FRESH-TARGETS-MAY-FUNNEL-INTO-O(n)-PARENTS-001.

Status:

KILLED on the high layer extracted in v4.40.

Killed inference:

even after extracting linearly many fresh load-one targets, they may all be fed by $o(n)$ parent rows.

Certificate:

after removing targets below $n/(\log N)^2$, one parent can feed at most three surviving targets because their product divides one complement $< N$. Therefore the parent population is at least $(1/27 - o(1))n$, and a greedy matching gives at least $(1/63 - o(1))n$ vertex-disjoint parent-target edges.

What survives:

the cofactor supports of those matched rows remain uncontrolled.

FAIL-UNIQUE-MISSING-DATUM-001**FAIL-ID:**

FAIL-UNIQUE-MISSING-DATUM-001.

Date / iteration:

vNEXT3.1, Phase-5 reclassification.

Claim:

The vNEXT3 information-loss certificate named “a share of the ambient valuation tensor” as *the exact datum missing* for a coupling.

Original motivation:

It is the most natural non-additive quantity in sight.

Status:

SCOPE-CORRECTED.

Exact certificate:

What is proved is that all recorded packet functionals factor on disjoint child-closed unions, hence cannot express competition for an R -only capacity. What is *not* proved is that ambient tensor occupancy is the unique repair, that no other non-additive functional works, or that a CRD-2 is necessary. No uniqueness argument was given, and none is available.

What survives:

The diagnosis, reclassified as **FRONTIER CERTIFICATE / INFORMATION-LOSS DIAGNOSIS**, together with four candidates recorded as **CONJECTURAL THEORY-DESIGN DIRECTIONS**: ambient tensor occupancy, cofactor provenance, parent-pair correlations, non-additive resource ownership.

Do not repeat:

Do not call a design hypothesis an “exact missing datum” without a uniqueness proof.

Related current target:

TD-UNIVERSAL-ROUTING, CRD-2 question.

FAIL-UNIT-COMPLEMENT-001

FAIL-ID:

FAIL-UNIT-COMPLEMENT-001.

Date / iteration:

Domain audit, frozen 31 August 2026.

Claim:

Every failed active root has composite complement.

Original motivation:

Failure has no prime-complement success, so the complement was informally treated as composite.

Status:

KILLED.

Minimal / strong counterexample:

$N = 20, r = 19$.

Exact certificate:

$19 \in \mathcal{A}_{20}$, $20 - 19 = 1$, and $1 \notin \mathbb{P}$. The row has no prime children, hence the stopped traversal fails, although the complement is not composite.

Failed hypothesis or mechanism:

The implication “nonprime $\Rightarrow$ composite” silently assumed the complement exceeds one.

Why the argument failed:

The unit row was omitted from the domain.

What survives:

If $N - r > 1$ and r is not GOOD, then $N - r$ is composite.

Safe weaker version:

State the composite-complement hypothesis explicitly whenever $Q = \text{PF}(N - r)$ or a root ledger is used.

Current replacement:

Theory Core, rooted algorithm and Theorems G-FULLQ-LEDGER/G-FULLQ-PRODUCTION on their stated domains.

Do not repeat:

Do not infer compositeness from failure before checking the unit row.

Related current target:

C0a and every generic failed-root lemma.

FAIL-UNIVERSAL-BRIDGE-IMPOSSIBILITY-001

FAIL-ID:

FAIL-UNIVERSAL-BRIDGE-IMPOSSIBILITY-001.

Date / iteration:

vNEXT3.1, Phase-4 scope correction.

Claim:

G-BRIDGE-ROUTING-EQUIVALENCE of vNEXT3, read as: no coupling of disjoint packets is possible at all.

Original motivation:

(OD1)–(OD4) really do make every recorded functional additive, which felt exhaustive.

Status:

SCOPE-CORRECTED. The mathematics is correct; the quantifier was too large.

Exact certificate:

(OD1)–(OD4) quantify over the recorded ledger $T, J, U_\bullet, \eta, \nu, K, \text{Src}$. They say nothing about an arbitrary functional $\Psi_N(R, U)$, which need not be a disjoint-union homomorphism. Candidates not excluded: ambient valuation-tensor occupancy, parent or cofactor correlations, residue geometry, shared ambient prime supply, non-additive resource ownership.

What survives:

G-RECORDED-LEDGER-BRIDGE-ROUTING: no bridge built *solely from the recorded disjoint-union-homomorphic ledgers* can charge $\eta(U)$ against an R -only capacity, and any such bridge requires $U \cap R \neq \emptyset$.

Do not repeat:

Do not promote “every functional we have recorded” to “every functional”.

Related current target:

TD/TF routing, TD-UNIVERSAL-ROUTING.

FAIL-UPPERHALF-SCOPE-001**FAIL-ID:**

FAIL-UPPERHALF-SCOPE-001.

Date / iteration:

v0.1 hostile mathematical audit §3.

Claim:

Upper-half source compression applies to a failed canonical world, so every active upper-half prime is a source of the bad graph.

Original motivation:

Both settings are called “failure”.

Status:

SCOPE-CORRECTED.

Minimal / strong counterexample:

Hypothesis mismatch; no finite witness is required.

Exact certificate:

The source theorem assumes $\text{Good}_N = \emptyset$, that is, that N is a Goldbach counterexample. Only then do stopped and raw reachability coincide and every active $N/2 < p < N$ become a source. Canonical p_0 -failure asserts nothing about other roots: the recorded corpus contains many failed worlds for arbitrary roots at values of N that have Goldbach representations.

Failed hypothesis or mechanism:

The global hypothesis was imported into the local one.

Why the argument failed:

Using a theorem whose hypothesis is global Goldbach failure in order to contradict p_0 -failure assumes what is to be proved for every other root.

What survives:

ARS-UPPERHALF-COMPRESSION on its own domain, and the direct-Goldbach-failure DAG in which it is legitimate.

Safe weaker version:

Keep two separate DAGs — universal p_0 -success, and direct contradiction under global failure — and never move a statement between them without annotation.

Current replacement:

Theory Core, ARS-UPPERHALF-COMPRESSION with its prohibition, and the exact Goldbach boundary section.

Do not repeat:

PROVED $\neq$ COMPUTATION $\neq$ OPEN $\neq$ REJECTED, and failure of the canonical traversal is not global Goldbach failure.

Related current target:

X1.

FAIL-V13-EXTERNAL-VALUATION-AS-SEED-WEIGHT-001**FAIL-ID:**

FAIL-V13-EXTERNAL-VALUATION-AS-SEED-WEIGHT-001.

Date / iteration:

vNEXT4.14 correction of vNEXT4.13.

Claim:

The full weighted mass $W_{\text{ext}} = \sum v_q(c_j) \log q$ can be passed unchanged into a family consumer that retains one seed per prime label.

Status:

SCOPE-CORRECTED.

Exact correction:

On $h > N^{1/3}$, high reflected valuations are at most two, so for the distinct external labels

$$W(E_{\text{ext}}) \geq W_{\text{ext}}/2.$$

What survives:

The natural-scale external branch survives with an absolute factor-two loss.

FAIL-V18-SOURCE-STAR-COUNTEREXAMPLE-SPECIFIC-001**FAIL-ID:**

FAIL-V18-SOURCE-STAR-COUNTEREXAMPLE-SPECIFIC-001.

Date / iteration:

vNEXT4.19 status repair of vNEXT4.18.

Killed inference:

A polynomial BAD source packet by itself is evidence specific to a Goldbach counterexample and should replace the mirror route as the primary proof target.

Status:

KILLED.

Reason:

The source-star selector is essentially a generic reduced residue-class/sieve construction once an active BAD base exists. Large source products are compatible with multisource defect identities and are lower obligations, not upper collision/source capacities.

What survives:

G-POLYNOMIAL-REPEATED-BASE-ROUGH-FAN and G-PLANNED-BASE-OVERLAP-SINGLE-CHARGE remain PROVED-PENDING-INDEPENDENT-AUDIT; C-POLYNOMIAL-SOURCE-STAR-CAPACITY becomes a secondary structural frontier.

FAIL-V410-ANONYMOUS-STAR-COFACTOR-001**FAIL-ID:**

FAIL-V410-ANONYMOUS-STAR-COFACTOR-001.

Date / iteration:

vNEXT4.11 correction of the vNEXT4.10 frontier.

Claim/strategy:

For a large inverse-sibling star, treat $(N - v)/\ell$ as an anonymous large quotient and split it into generic rough/smooth cases.

Status:

SUPERSEDED on the monomial branch.

Exact certificate:

Retained provenance gives the two-level factorisation

$$N - v = bw, \quad N - w = q^2, \quad w > \sqrt{N},$$

with the whole entry cofactor b already small. Repeated values of b produce the exact entry comb, including $\text{rad}(b)^{L-1} \mid J_R$.

Replacement:

Use the whole-entry-cofactor atom/comb, not an anonymous quotient split.

FAIL-V47-DEEP-RPD-GLOBAL-GENERATION-LOCALISATION-001

FAIL-ID:

FAIL-V47-DEEP-RPD-GLOBAL-GENERATION-LOCALISATION-001.

Source target:

C-DEEP-RPD-ANCESTRY-LOCALISATION.

Status:

SCOPE-CORRECTED .

Overstrong formulation:

a full uncharged deep-RPD obstruction can be pigeonholed into one common global generation.

Why not proved:

generation-edge amortisation charges cross-generation edges through horizontal surplus or source destruction but does not bound the total number of generations strongly enough for that pigeonhole.

Safe v4.47 replacement:

polynomial exact deep layer, followed by original v4.41 submatching, large incremental J , or a local immediate-ancestry restart.

v4.48 refinement:

the primary replacement is further sharpened to C-DEEP-RPD-DESIGNATED-OWNER-LOCALISATION; total unique parenthood is not required.

FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001

FAIL-ID:

FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001.

Alias:

v4.48 R48-4.

Status:

SCOPE-CORRECTED .

Killed automatic inference:

if an ancestor was born as a factor of a reflected carrier, a later descendant automatically divides that same carrier.

Certificate:

divisibility by the original carrier is not preserved by a generic edge $v_{j-1} = N - c_j v_j$.

What survives:

the v4.47 reflected congruences remain valid on the explicit intersection domain where a target really has both tags.

Replacement:

path-labelled affine ancestry transport, G-CRD-PATH-AFFINE-COFACTOR-NORMAL-FORM and G-DEEP-ANCESTRAL-AFFINE-CONGRUENCE.

FAIL-V48-HIGHTARGET-LOADONE-NECESSARY-001

FAIL-ID:

FAIL-V48-HIGHTARGET-LOADONE-NECESSARY-001.

Alias:

v4.48 R48-3.

Status:

SCOPE-CORRECTED .

Old scope:

treat the cube-root high-target row trichotomy as requiring global load one of the designated target.

Correction:

inspection of the proof uses only an actual BAD edge $u \rightarrow p$, $p > N^{1/3}$, and factorisation of $N - u$. The accepted replacement is G-CUBEROOT-HIGH-TARGET-OWNER-TRICHOTOMY.

FAIL-V48-LOCAL-J-AS-AMBIENT-M-SCALE-001

FAIL-ID:

FAIL-V48-LOCAL-J-AS-AMBIENT-M-SCALE-001.

Alias:

v4.48 R48-1.

Status:

SCOPE-CORRECTED.

Killed promotion:

from $|X| = N^{1/3-o(1)}$ and $M \gg N^{1/3}$, promote $\Omega(|X| \log N)$ automatically to an ambient/full- M -scale payment.

Correction:

retain only the explicit subpolynomial comparison delivered by the pigeonhole losses, recorded in v4.48 as $|X| \geq MN^{-o(1)}$. The local J -payment is valid; the stronger scale label was not justified.

FAIL-V48-LOWCHILD-VERBATIM-DOMAIN-TRANSFER-001

FAIL-ID:

FAIL-V48-LOWCHILD-VERBATIM-DOMAIN-TRANSFER-001.

Alias:

v4.48 R48-5.

Status:

KILLED as a direct domain application.

Killed transfer:

apply C-LOWCHILD-J-OR-SMALLHUB unchanged to the deep designated-owner family.

Domain mismatch:

the old theorem requires a family of size $\gg N^{1/3}$, whereas the v4.48 owner family is guaranteed only at $MN^{-o(1)} = N^{1/3-o(1)}$.

Replacement:

G-DESIGNATED-LOWCHILD-SCALE-DISPOSITION and the open consumer
C-DEEP-LOWCHILD-RENEWAL-CAPACITY.

FAIL-V48-MONOMIAL-NOVELTY-001

FAIL-ID:

FAIL-V48-MONOMIAL-NOVELTY-001.

Date / iteration:

vNEXT4.9 audit of vNEXT4.8.

Claim:

C-MONOMIAL-TWO-EDGE-RIGIDITY introduced a new square-map theorem.

Status:

SCOPE-CORRECTED (provenance correction).

Exact certificate:

The injectivity/no-cycle/no-three-consecutive-square argument is already contained in the **PROVED** theorem TF-HIGH-FLOOR-HORIZONTAL-RIGIDITY.

What survives:

The extracted standalone formulation and its use of component starts in vNEXT4.8 remain valid.

FAIL-V48-MULTIPARENT-SEPARATE-BRANCH-NECESSARY-001

FAIL-ID:

FAIL-V48-MULTIPARENT-SEPARATE-BRANCH-NECESSARY-001.

Alias:

v4.48 R48-2.

Status:

SUPERSEDED as a necessary proof split.

Old design:

separate deep targets by total indegree and cash out the multi-parent part before constructing a restart matching.

Replacement:

choose one actual parent per target. Because one parent row cannot feed three distinct targets above $N^{1/3}$, the chosen-edge graph has bounded degree and yields a vertex-disjoint designated-owner matching directly.

FAIL-V48-POSITIVE-DEEP-SEMIPRIME-CLOSES-P1BR-001**FAIL-ID:**

FAIL-V48-POSITIVE-DEEP-SEMIPRIME-CLOSES-P1BR-001.

Alias:

v4.48 R48-7.

Status:

KILLED for immediate closure.

Killed inference:

once the deep semiprime owner family enters a positive clean signed round, P1BR follows.

Reason:

the established positive-round independence theorem remains binding.

New datum retained:

every designated target additionally satisfies the common deep-layer relation $Q \mid N - p_i$, which was not part of the old round datum.

Live replacement:

P1BR-DEEP-DESIGNATED-TARGET-COUPLING.

FAIL-V48-VARIABLE-M-SELFREENTRY-NEW-001**FAIL-ID:**

FAIL-V48-VARIABLE-M-SELFREENTRY-NEW-001.

Alias:

v4.48 R48-6.

Status:

SUPERSEDED.

Superseded idea:

restart the v4.43 fixed- m recurrence with growing $m \sim \log \log N$ as a new proof route.

Reason:

v4.44 already performed the variable-threshold correction and replaced the fixed- m recurrence by simultaneous valuation freezing, including the near-critical polylog star boundary.

FAIL-V49-CROSSGEN-GCD-IS-P1BR-BRIDGE-001**FAIL-ID:**

FAIL-V49-CROSSGEN-GCD-IS-P1BR-BRIDGE-001.

Status:

KILLED as a sufficient general bridge.

Certificate:

the cheap semiprime orthogonal-broom branch admits a large selected subfamily with the relevant owner-support / later-target gcds equal to 1.

Consequence:

the common deep modulus does not automatically couple the old owner round to the descendant target rows through cross-generation gcd.

FAIL-V49-DEEP-SEMPIRIME-AUTOMATIC-POSITIVE-ROUND-CLOSURE-001

FAIL-ID:

FAIL-V49-DEEP-SEMPIRIME-AUTOMATIC-POSITIVE-ROUND-CLOSURE-001.

Status:

SCOPE-CORRECTED.

Killed promotion:

treat the localised deep semiprime family as automatically in the former ambient positive-round closure after the $N^{-o(1)}$ localisation loss.

Reason:

the scale/domain transfer is not automatic, and positive signed production remains insufficient for P1BR without an independent additive/capacity coupling.

Replacement:

v4.49 orthogonal-broom analysis and the later local reflected-anchor frontier.

FAIL-V49-DIRECT-REFLECTED-FACTOR-INHERITANCE-001

FAIL-ID:

FAIL-V49-DIRECT-REFLECTED-FACTOR-INHERITANCE-001.

Status:

KILLED / **SCOPE-CORRECTED**.

Killed inference:

from $\ell \rightsquigarrow p$ and $\ell \mid c_j^{\text{ref}}$, infer $p \mid c_j^{\text{ref}}$.

Reason:

direct divisor incidence is not hereditary along a generic CRD path.

Replacement:

the exact path-labelled identity **G-REFLECTED-ANCESTRY-EXACT-IDENTITY** and branch application **C-DEEP-ANCESTRAL-REFLECTED-TAG-TRANSPORT**.

FAIL-V49-LATER-DESCENDANTS-DESTROY-ALL-REFLECTED-ARITHMETIC-001

FAIL-ID:

FAIL-V49-LATER-DESCENDANTS-DESTROY-ALL-REFLECTED-ARITHMETIC-001.

Status:

KILLED.

Killed description:

once direct factor inheritance is lost, reflected arithmetic is lost entirely.

Counterdatum:

exact ancestry retains

$$N(2s\Xi - 1) - 2d_j = 2(-1)^t sC_t(N - p).$$

Scope guard from v4.50:

the exact identity is useful data, but its bare reduction modulo the terminal deep modulus is automatic and is not by itself capacity.

FAIL-V49-RELATIVE-T-OR-FRESH-MASS-ALONE-CLOSES-001

FAIL-ID:

FAIL-V49-RELATIVE-T-OR-FRESH-MASS-ALONE-CLOSES-001.

Status:

KILLED as an immediate contradiction route.

Reason:

the exact relative $T/U/J/F$ identities give lower reservoirs but no independent upper capacity for the external extension.

What survives:

the identities remain valid accounting tools.

FAIL-V50-ANCESTRY-DIAGONAL-TRANSPORTS-DIRECT-CAPACITY-001

FAIL-ID:

FAIL-V50-ANCESTRY-DIAGONAL-TRANSPORTS-DIRECT-CAPACITY-001.

Alias:

v4.50 R50-2.

Status:

SCOPE-CORRECTED.

What remains proved:

the exact identity

$$N(2s\Xi - 1) - 2d_j = 2(-1)^t sC_t(N - p).$$

What is removed:

the claim that reducing this identity modulo $Q \mid N - p$ creates a new reflected path-profile capacity condition.

Current distinction:

direct reflected incidence is a capacity datum; bare remote reflected ancestry is not, at residue level.

FAIL-V50-BOUNDED-BRANCHING-CLOSES-SINGLESEED-001

FAIL-ID:

FAIL-V50-BOUNDED-BRANCHING-CLOSES-SINGLESEED-001.

Alias:

v4.50 R50-5; consistent with FAIL-BOUNDED-ARITY-WEIGHT-001.

Status:

KILLED as a weighted-capacity inference.

Safe use:

branching may be combined with exact cofactor/path arithmetic or another single-charged quantity, but it is not a capacity theorem by itself.

FAIL-V50-REFLECTED-PROVENANCE-USELESS-AFTER-SELECTION-001

FAIL-ID:

FAIL-V50-REFLECTED-PROVENANCE-USELESS-AFTER-SELECTION-001.

Alias:

v4.50 R50-3.

Status:

KILLED on the return-heavy cheap- J branch.

Certificate:

C-RETURNHEAVY-REFLECTED-CUBEROOT-LOADONE-MATCHING recovers a positive cube-root-scale vertex-disjoint matching whose targets retain direct reflected tags with distinct reflected indices and quotient-return normal forms.

FAIL-V50-RETURNED-SEED-IMPLIES-CYCLE-001

FAIL-ID:

FAIL-V50-RETURNED-SEED-IMPLIES-CYCLE-001.

Alias:

v4.50 R50-4.

Status:

KILLED.

Killed inference:

if a reflected seed has some parent in the failed extension, then the parent lies in the same seed basin and closes a directed cycle.

Reason:

source retention supplies a parent somewhere in the basin union; it may belong to a different seed basin.

What survives:

when a genuine same-basin return or actual cycle is proved, G-CRD-CYCLE-COFACTOR-LOCK applies.

FAIL-V50-SOURCEHEAVY-REFLECTED-SEEDS-CLOSE-001**FAIL-ID:**

FAIL-V50-SOURCEHEAVY-REFLECTED-SEEDS-CLOSE-001.

Alias:

v4.50 R50-6.

Status:

KILLED.

What is proved:

a source-heavy packet gives a large external source product / T_U -type lower defect.

What is missing:

an independent source-sensitive upper capacity.

Consequence:

full C-REFLECTED-FIRST-OCCURRENCE-CONVERSION remains open; only the return-heavy half is proved pending audit.

FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIES-DEEP-POWER-001**FAIL-ID:**

FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIES-DEEP-POWER-001.

Source:

v4.51 R51-3.

Status:

KILLED.

Killed inference:

use the alternate-path cofactor congruence to obtain extra divisibility from the common factor r^{E-1} .

Reason:

$r \nmid N$, so the relevant factor is invertible modulo N and cancels.

What survives:

G-ALTERNATE-PATH-COFACTOR-LOCK remains **PROVED**.

FAIL-V51-ANCESTRY-LOCALISATION-FAILED-001**FAIL-ID:**

FAIL-V51-ANCESTRY-LOCALISATION-FAILED-001.

Source:

v4.51 R51-5.

Status:

KILLED.

Killed description:

the inability of aggregate path products to see E means ancestry localisation itself failed.

Correction:

ancestry localisation remains useful and v4.51 strengthens it through G-DEEP-ANCHOR-FIBRE-COUNTING-TRADEOFF and C-DEEP-LINKED-REFLECTED-ANCHOR-RECOVERY. What fails is only an overly coarse path-product consumer.

FAIL-V51-CYCLE-LOCK-FALSE-001**FAIL-ID:**

FAIL-V51-CYCLE-LOCK-FALSE-001.

Source:

v4.51 R51-4.

Status:

KILLED as a description of the theorem.

Correction:

G-CRD-CYCLE-COFACTOR-LOCK remains **PROVED** and is subsumed as a corollary of the stronger G-ALTERNATE-PATH-COFACTOR-LOCK.

FAIL-V51-UNWEIGHTED-REFLECTED-MATCHING-CARRIES-DEEP-MASS-001**FAIL-ID:**

FAIL-V51-UNWEIGHTED-REFLECTED-MATCHING-CARRIES-DEEP-MASS-001.

Source:

v4.51 R51-1 / scope correction to v4.50.

Status:

SCOPE-CORRECTED.

Killed proof-DAG edge:

full cube-root reflected matching from v4.50 $\implies$ its descendant closure carries the current common-deep ob

Reason:

the deep obstruction was extracted from the descendant closure of the v4.41/v4.42 high matching, while the v4.50 reflected matching was re-extracted from the original external seed packet. A large deep obstruction may be concentrated in basins omitted by that unweighted matching.

What survives:

the v4.50 reflected matching theorem itself remains **PROVED-PENDING-INDEPENDENT-AUDIT**; v4.51 repairs the transfer by selecting reflected anchors after the actual deep rows are known.

FAIL-VERTICAL-DESCENT-AUTOMATIC-BV-001**FAIL-ID:**

FAIL-VERTICAL-DESCENT-AUTOMATIC-BV-001.

Date / iteration:

vNEXT4.9.

Claim:

A dyadic band produced at $Y \geq h > N^{1/3}$ automatically lies in the range of G-DYADIC-RECIPROCAL-REENTRY.

Status:

KILLED.

Exact certificate:

The current analytic theorem requires

$$2Y^{3+\vartheta} \leq \frac{N^{1/2}}{(\log N)^{C_0}},$$

so its upper scale is only roughly $N^{1/(6+2\vartheta)}$, modulo logarithms. The window between that scale and the cubic range is not covered.

Do not repeat:

Keep deterministic inverse-scale transport and analytic BV re-entry as separate arrows in the proof DAG.

FAIL-VERTICAL-FC8-SEPARATION-001

FAIL-ID:

FAIL-VERTICAL-FC8-SEPARATION-001.

Date / iteration:

vNEXT4.5.

Claim:

The nonroot vertical factor U_R is an essentially independent store which the exact FC8 low-core ledger does not see.

Status:

SCOPE-CORRECTED / **KILLED**.

Exact certificate:

In the proved $3 \nmid N$ fringe domain,

$$D_C J_R U_R^{\text{nr}} = W_C^{\text{nr}}.$$

Thus denominator, horizontal reuse, and nonroot vertical layers are exact factors of one conserved low-core mass. The root-only term $U_0 = m/\text{rad}(m)$ is separate but has $\log U_0 < \log N$.

What survives:

The partition among D, J, U can still be the obstruction; what is killed is treating U^{nr} as outside the same functional.

Do not repeat:

Preserve the full C -part of each row before projecting to J and U .

Related current target:

C-MONOMIAL-REFLECTED-PAIR-SPARSITY.

FAIL-VERTICAL-HIGHGAP-UNBOUNDED-CAPACITY-001

FAIL-ID:

FAIL-VERTICAL-HIGHGAP-UNBOUNDED-CAPACITY-001.

Date / iteration:

vNEXT4.11.

Claim:

High-base block-derived vertical mass has no independent upper capacity and can only be controlled through J_R , $K(R)$, or a γ -dependent ledger.

Status:

KILLED.

Exact certificate:

For every $Y > N^{1/3}$, exact depth two and the square-parent slot interval give

$$m_q \leq 1 + \frac{N}{6q^2}, \quad V_{\geq Y}^{\text{nr}} \ll \frac{N}{Y}.$$

In particular $V_{\text{blk}} \ll N/h$ on the block-derived high-gap branch.

What survives:

This is an upper capacity on the vertical channel. It is not by itself a Goldbach contradiction and does not prove that the whole reflected obligation must enter that channel.

FAIL-VERTICAL-SQRTLOG-WALL-001

FAIL-ID:

FAIL-VERTICAL-SQRTLOG-WALL-001.

Date / iteration:

vNEXT4.12.

Claim:

Under the fixed-proportion vertical-heavy hypothesis, the best available confinement is $h \ll \sqrt{N/\log N}$.

Status:

SUPERSEDED within that conditional branch.

Replacement:

The hybrid linear sieve gives

$$h \ll_{\delta} \sqrt{\frac{N}{\log N \log \log N}}$$

when $V_{\text{blk}} \geq \delta t \log h$.

Scope guard:

vNEXT4.13 shows that this remains a conditional vertical-heavy subroutine; it is not an unconditional ceiling on the whole canonical branch.

FAIL-VERTICAL-TO-ETA-LOSES-LOG-001**FAIL-ID:**

FAIL-VERTICAL-TO-ETA-LOSES-LOG-001.

Date / iteration:

vNEXT4.9.

Claim:

Routing a linear vertical family only to $\eta(R) \gg h$ irretrievably loses the $\log h$ factor and therefore cannot preserve a $t \log h$ -scale obligation.

Status:

KILLED.

Exact certificate:

The distinct parent rows are forced through a much smaller low-base sibling support. Cauchy/pigeonhole concentration yields

$$K_{<h}^{(1)}(R) \gg h \log h$$

for a linear family of distinct vertical bases, and the weighted theorem gives

$$V_{\text{blk}} \gg t \log h \implies \log J_R \gg V_{\text{blk}} \vee K_{<h}^{(1)}(R) \gg t \log h.$$

What survives:

Large $K(R)$ is still not a contradiction; the missing ingredient is a source-sensitive consumer/upper capacity.

FAIL-VERTICAL-UNLOCALISED-001**FAIL-ID:**

FAIL-VERTICAL-UNLOCALISED-001.

Date / iteration:

vNEXT4.6.

Claim:

Vertical storage may occur anywhere among block-derived rooted low-core factors.

Status:

KILLED as a description.

Exact certificate:

If $c_j = sq$ and $N - w = bq$, then

$$v_q(N - w) \geq 2 \iff q \mid b.$$

Since $b < 2s$ and $q > N/(2s)$, this forces

$$s > \sqrt{N}/2.$$

What survives:

The true vertical zone is the large-quotient region; below it every block-derived rooted factor is D/J-only.

FAIL-VNEXT433-UNIFORM-FLOOR-GAP-001

FAIL-ID:

FAIL-VNEXT433-UNIFORM-FLOOR-GAP-001.

Status:

KILLED as a failure certificate; it must not be used.

Incorrect claim:

the v4.33 uniform residual floor was allegedly unproved for $q > 4h$.

Repair:

the already proved G-FULLQ-PRODUCTION gives $M_q > (5/3)q$, hence $q > 4h$ gives $M_q > 20h/3$. For $q \leq 4h$, the shadow bound works after explicitly excluding $q = 2h$ by parity. Thus the full sub-root floor survives.

Do not repeat:

before recording a gap for a quantity, search the whole Core for an older bound on that same quantity, not only the newest report block.

FAIL-WEIGHTED-REUSE-AMPLIFICATION-001

FAIL-ID:

FAIL-WEIGHTED-REUSE-AMPLIFICATION-001.

Status:

SCOPE-CORRECTED.

Generic witness:

for $N = 2200$, root 1471 and $R = \{3, 13, 1471\}$, the complements are $3^6, 13^3, 3^7$ and $\log J_R = \log 3$; the vertical channel carries nearly all prime-power mass. This refutes a near-unit $\gamma \log N$ lower bound on $\log J_R$ over the full generic FC7/FC8 failed-root domain.

Critical scope:

the root is not canonical and lies outside the near-square-root subdomain. The witness does *not* refute a theorem in which independent canonical/high-gap information forces an incompatible lower bound; such a theorem would prove that its failed domain is empty.

FAIL-WHOLE-BLOCK-EXTERNAL-SET-LOSES-REFLECTED-PROVENANCE-001

FAIL-ID:

FAIL-WHOLE-BLOCK-EXTERNAL-SET-LOSES-REFLECTED-PROVENANCE-001.

Status:

KILLED.

Killed inference:

projecting the reflected block to a coarse set of external prime factors destroys the generating reflected indices so severely that no polynomial-weight tagged family can be recovered.

Certificate:

v4.39 assigns each prime to one reflected index, notes that one index carries $< \log N$ assigned weight, and recovers an injectively tagged subfamily preserving an $\alpha - o(1)$ fraction of the weight for a $N^{\alpha-o(1)}$ packet.

What survives:

the recovered packet may still lack an immediate second occurrence / upper consumer.

FAIL-WORLD-SIZE-NAIVE-001

FAIL-ID:

FAIL-WORLD-SIZE-NAIVE-001.

Date / iteration:

vNEXT4, Phase 1b.

Claim:

For a failed world R with composite complement, $|R| \leq \eta(R) + 1$, on the ground that every row satisfies $\omega(N - u) \geq 2$ and that $\sum_{u \in R} \omega(N - u) = \eta(R) + |R| - 1$.

Original motivation:

Every state of a failed world is BAD, so its complement is composite; a composite number “has at least two prime factors”.

Status:

KILLED as stated.

Minimal / strong counterexample:

$N = 2200$, root $r_* = 1471$,

$$R = \{3, 13, 1471\}, \quad N - 1471 = 729 = 3^6, \quad N - 3 = 2197 = 13^3, \quad N - 13 = 2187 = 3^7.$$

Here $\Gamma_N(R) = \{3, 13\} = R \setminus \{r_*\}$, $\kappa_R(3) = 2$, $\kappa_R(13) = 1$, so $\eta(R) = 1$ while $|R| = 3 > \eta(R) + 1 = 2$.

Failed hypothesis or mechanism:

A composite number has at least two prime factors *counted with multiplicity*; it need not have two *distinct* ones. Every row of this world is a prime power, so $\omega = 1$ on all three.

Why the argument failed:

κ is layer-one incidence and therefore sees ω , not Ω . The step that requires $\omega \geq 2$ is the only place where badness was used, and badness gives $\Omega \geq 2$.

What survives:

The corrected count $|R| \leq \eta(R) + 1 + 2a_0 + a_1$ with $a_0 \leq 1$ and $a_1 \leq \gamma(\lfloor \log_2 N \rfloor - 1)$, hence $|R| < 4\gamma \log N$ under the (FC9) hypotheses: Theory Core, G-CANONICAL-WORLD-SIZE-CAP, (WS1)–(WS3). Zero violations over the recorded 301 worlds.

Safe weaker version:

The corrected form above; nothing weaker is needed.

Do not repeat:

Whenever a count over the rows of a failed world uses $\omega(N - u) \geq 2$, the prime-power rows must be excluded explicitly. Recorded as loss-control rule (LR6).

Related current target:

P1BR-ADDITIVE-COUPLING, γ .

FAIL-v40-HIGH-FLOOR-N-OVER-LOG2-IS-SHARP-001

FAIL-ID:

FAIL-v40-HIGH-FLOOR-N-OVER-LOG2-IS-SHARP-001.

Status:

SUPERSEDED.

Superseded description:

the useful unique-parent target floor is only $q > n/(\log N)^2 = N^{1/3-o(1)}$.

Certificate:

the total log mass of all primes $q \leq N^{1/3}$ is $O(N^{1/3}) = o(n \log N)$ on $n \gg N^{1/3}$. v4.41 therefore moves essentially all unique-parent mass to the strict floor $q > N^{1/3}$.

What survives:

the v4.40 matching is valid but its constants/floor are superseded by the v4.41 cube-root matching.

FAIL-v41-MONOMIAL-LENGTH-TWO-NOVELTY-001

FAIL-ID:

FAIL-v41-MONOMIAL-LENGTH-TWO-NOVELTY-001.

Status:

SCOPE-CORRECTED .

Scope correction:

v4.41 correctly proves/uses the interval mechanism, but the statement that every monomial component has at most two edges was already **PROVED** in the Theory Core as C-MONOMIAL-TWO-EDGE-RIGIDITY.

New part retained:

v4.42 records the stronger fact that for fixed N at most one monomial component can contain two edges.

FAIL-BOUNDED-COFACTOR-ROWSLOTS-ARE-UNCONSTRAINED-001

FAIL-ID:

FAIL-BOUNDED-COFACTOR-ROWSLOTS-ARE-UNCONSTRAINED-001.

Status:

KILLED .

Killed inference:

fixing the integer cofactor c leaves no useful deterministic capacity on the owned high row-slot packet.

Replacement:

fixed c forces

$$2QFc \mid p_i - p_j,$$

and summing over $c \leq C$ gives the v4.57 harmonic row-slot capacity.

FAIL-BOUNDED-COFACTOR-SUM-MUST-PAY-ONE-PER-C-001

FAIL-ID:

FAIL-BOUNDED-COFACTOR-SUM-MUST-PAY-ONE-PER-C-001.

Status:

KILLED .

Killed inference:

summing fixed- c Selberg bounds over a polynomial cofactor range necessarily creates an $O(C)$ loss comparable to the whole packet.

Replacement:

choose

$$C_\varepsilon = MN^{-1/(\log \log N)^\varepsilon} = o(M).$$

Then the total $O(1)$ -per- c remainder is $o(M)$, while discarded cofactors still carry full logarithmic mass.

FAIL-CHEAP-RADIC-OSCILLATION-HAS-A-HIDDEN-ROUTING-SINK-001

FAIL-ID:

FAIL-CHEAP-RADIC-OSCILLATION-HAS-A-HIDDEN-ROUTING-SINK-001.

Status:

KILLED as a routing diagnosis.

Reason:

exact target-row entry accounting shows that cheap oscillation can survive only by consuming fresh target labels. Repeated targets are paid; the unresolved issue is population capacity, not missing routing bookkeeping.

FAIL-COFACTOR-RENEWAL-IMPLIES-DIVERGENCE-001

FAIL-ID:

FAIL-COFACTOR-RENEWAL-IMPLIES-DIVERGENCE-001.

Status:

KILLED.

Killed inference:

a renewed polynomial owned support population obtained from cofactor factorisation is already a contradiction.

Reason:

the standing renewal-is-not-divergence restriction remains binding; there is no proved global upper capacity for arbitrary fresh external population.

FAIL-FIRST-COPY-HIGH-ROW-POPULATION-IS-BY-ITSELF-A-CONTRADICTION-001

FAIL-ID:

FAIL-FIRST-COPY-HIGH-ROW-POPULATION-IS-BY-ITSELF-A-CONTRADICTION-001.

Status:

KILLED.

Killed inference:

a polynomial family of distinct external high target rows, each used once, is already an upper-capacity contradiction.

Reason:

this is the standing “renewal is not divergence” restriction in the present deep-owned domain. A fresh external population is an obligation until a CRD-specific capacity theorem consumes it.

FAIL-FIXED-COFACTOR-PRIMALITY-GIVES-NO-MORE-THAN-SPACING-001

FAIL-ID:

FAIL-FIXED-COFACTOR-PRIMALITY-GIVES-NO-MORE-THAN-SPACING-001.

Status:

KILLED.

Killed inference:

after v4.57 fixed- c spacing, primality of both endpoints cannot yield further capacity.

Replacement:

after parity parametrisation the endpoint primes are two explicit non-proportional affine forms with determinant $-2FN$, so the already audited classical dimension-two Selberg upper-bound sieve gives logarithmic-square compression.

FAIL-HIGH-BASE-R-GREATER-Y-IMPLIES-RADIC-MONOTONICITY-001

FAIL-ID:

FAIL-HIGH-BASE-R-GREATER-Y-IMPLIES-RADIC-MONOTONICITY-001.

Status:

KILLED.

Killed inference:

the large-base hypothesis $r > Y$ by itself forces monotone r -adic dynamics.

Certificate:

the explicit v4.55 high-base diagnostic witness with nonmonotone level pattern. It does not satisfy the full deep exponent hypothesis and therefore does not rule out stronger theorems using deep/reflected provenance.

FAIL-LARGE-COFACTOR-IS-AN-ANONYMOUS-COEFFICIENT-SINK-001

FAIL-ID:

FAIL-LARGE-COFACTOR-IS-AN-ANONYMOUS-COEFFICIENT-SINK-001.

Status:

KILLED.

Killed inference:

large c_i may absorb the row-slot obligation without producing a trackable CRD resource.

Replacement:

large cofactors carry explicit logarithmic mass with exact support/horizontal/vertical factorisation; this yields genuine residual J/U or owned fresh-support renewal.

FAIL-LARGE-COFACTOR-RENEWAL-IS-DIVERGENCE-001

FAIL-ID:

FAIL-LARGE-COFACTOR-RENEWAL-IS-DIVERGENCE-001.

Status:

KILLED.

Killed inference:

the v4.58 full-scale large-cofactor renewal branch alone gives a global contradiction.

Reason:

renewed support is a real relation-preserving CRD obligation, but no proved capacity theorem yet bounds its total fresh external population.

FAIL-PRODUCT-MODULUS-COMPRESSION-ALONE-CLOSES-DEEP-001

FAIL-ID:

FAIL-PRODUCT-MODULUS-COMPRESSION-ALONE-CLOSES-DEEP-001.

Status:

KILLED.

Killed inference:

the v4.57 product-modulus upper bound already contradicts the deep lower bound on $Q = r^e$.

Reason:

the deep hypothesis permits $Q = N^{o(1)}$, so even a bound of scale $QF \leq N^{1-\alpha+o(1)}$ need not contradict the branch.

FAIL-RADIC-HALVING-IS-AUTOMATIC-DYNAMICAL-DESCENT-001

FAIL-ID:

FAIL-RADIC-HALVING-IS-AUTOMATIC-DYNAMICAL-DESCENT-001.

Status:

SCOPE-CORRECTED.

Overstrong inference:

organising threshold classes at $e, \lceil e/2 \rceil, \lceil e/4 \rceil, \dots$ automatically produces strict forward descent at every owned transition.

Correction:

threshold layers may remain useful organisationally, but an edge may jump upward to a larger exact valuation. Any finite-return argument must use an independent payment/capacity statement.

FAIL-RADIC-LEVEL-IS-MONOTONE-POTENTIAL-001

FAIL-ID:

FAIL-RADIC-LEVEL-IS-MONOTONE-POTENTIAL-001.

Status:

KILLED.

Killed inference:

for an owned transition $p \rightarrow s$, the quantity $v_r(N - p)$ must decrease along the forward dynamics.

Certificate:

the exact v4.55 three-cycle template

$$N = r^{a+b+c} + 1, \quad N - p_0 = r^a p_1, \quad N - p_1 = r^b p_2, \quad N - p_2 = r^c p_0,$$

which realises arbitrary cyclic valuation pattern $a \rightarrow b \rightarrow c \rightarrow a$ whenever the states are active primes.

Safe replacement:

retain the exact adjacent-row transition identities and price actual repeated target rows rather than valuation-level recurrence.

FAIL-RADIC-LEVEL-RETURN-ITSELF-PAYS-J-001**FAIL-ID:**

FAIL-RADIC-LEVEL-RETURN-ITSELF-PAYS-J-001.

Status:

KILLED.

Killed inference:

a return to a previously used value of $v_r(N - s)$ must create horizontal reuse.

Certificate:

horizontal reuse is indexed by target prime labels; the v4.55 cycle realises level recurrence through distinct target rows.

Safe replacement:

after the first copy at one target row, every later occurrence at that same target row is fully priced by incremental J .

FAIL-SAME-EXACT-LEVEL-IS-A-NEW-EXTERNAL-RESTART-001**FAIL-ID:**

FAIL-SAME-EXACT-LEVEL-IS-A-NEW-EXTERNAL-RESTART-001.

Status:

KILLED as a packet-design inference.

Killed inference:

an owned edge landing on another high row with the same exact valuation e should be treated as a new external restart branch.

Replacement:

saturate the exact level in the stopped reflected-seed basin first; then every high exact- e target already belongs to the saturated layer.

FAIL-SELBERG-LOGSAVING-ALONE-CLOSES-DEEP-001**FAIL-ID:**

FAIL-SELBERG-LOGSAVING-ALONE-CLOSES-DEEP-001.

Status:

KILLED.

Killed inference:

the v4.58 log-saved bound

$$QF \ll_{\varepsilon} \frac{N(\log \log N)^{1+\varepsilon}}{M \log N}$$

is already incompatible with the current deep modulus.

Reason:

Q may still be only $N^{o(1)}$; the logarithmic saving is real but not a contradiction.

14.3 Post-v4.81 failure-log additions (2026-09-05–2026-09-07C)

AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905

Status. **SCOPE-CORRECTED**. The supplied CRT proof programs selected BAD edges/valuations but does not control every unprescribed factor of every expanded row. The blanket bounded-depth complete-BFS no-go is therefore not established. Exact witness: $N = 92$, $p_0 = 47$ has a selected BAD cycle $5 \rightarrow 29 \rightarrow 7 \rightarrow 5$ but succeeds at depth 1 via 3 because $92 - 3 = 89$ is prime. **Survives:** compatible partial-transcript programming.

ENVELOPE-ESCAPE-AUTOMATIC-ITERATION-20260905

Status. **KILLED** as an inference. The proved $h = 1$ envelope-escape theorem can force new primes $> y$ at depth 2, but after the envelope grows the threshold $F_2(y)$ changes and need not remain below the same N . Automatic repeated escape is not justified. **Survives:** the one-step and all-first-branch escape theorems under their explicit size hypotheses.

AUTOMATIC-THIRD-GENERATION-SIEVE-SWITCH-20260906

Status. **KILLED** as an inference. Radius-two upper-bound sieve estimates do not automatically iterate to a complete third generation: the number and arithmetic shape of residual forms change, and complete cofactors must be retained. No proved radius-two theorem is refuted.

SUCCESS-PRODUCT-BARRIER-TO-CLOSED-CORE-HEIGHT-20260906B

Status. **KILLED** as an inference. Label/cofactor product barriers quantify over successful paths. A completely closed BAD core contains no such path, so the successful-path theorem is vacuous there and cannot be combined with the closed-support height bound to obtain a contradiction.

PRIME-LIFT-PRIMALITY-FORCES-SUCCESS-20260907

Status. **KILLED**. Prime lift primality alone does not force BFS success. Exact counterexample: $(N, d, R) = (1862, 7, 1597)$, where the full BFS from inactive root 7 and from prime lift 1597 are conjugate and both fail with a complete closed active support certificate. **Survives:** PRIME-LIFT-FULL-BFS-CONJUGACY, and the special unresolved $d = 2/h = 1$ case.

LOW-GAP-FIVE-LABEL-OVERGENERALIZATION-20260907C

Status. **KILLED** by exact counterexample. At $N = 44670$, $r = 22469 = p_{12}$, $m = 149^2$, $h = 134 < 149$, the complete BAD frontiers through depth 2 discover exactly four nonroot labels by depth 3: $\{149, 211, 23, 1933\}$. Both depth-3 endpoints are GOOD. The proved universal lower bound THREE survives; FOUR remains open in the stated low-gap domain.

SUBGROUP-SEPARATION-OF-P0-AS-GENERAL-EXPLANATION-20260907C

Status. **KILLED** as a general explanation. For five of the six historical hosts, the subgroup of $(\mathbb{Z}/N\mathbb{Z})^\times$ generated by the maximal closed low BAD support is the entire unit group, so no character trivial on that support can separate p_0 . At $N = 2200$ the subgroup is proper, but membership is still not sufficient for failure. The elementary necessary subgroup lemma survives as an exclusion test when a verified full BAD envelope is available.

Chapter 15

Research Agenda

15.1 How to choose what to work on

Three rules for triage, in order of importance.

1. **Check complete support first.** A finite *selected* transcript can be programmed, but the old blanket theorem about complete bounded-depth BFS was scope-corrected (Corollary 8.3). If an argument constrains only chosen divisibilities while ignoring residual cofactors, it cannot prove full failure or success. A complete-frontier theorem is still admissible, but it must control every factor of every expanded row.
2. **Check the functional.** If the argument proves a lower bound in one functional and consumes it in another, it is the P1^{br}/K1 error. Write down both functionals explicitly *before* starting.
3. **Check the scope class.** If the argument uses a class-B lemma to prove a class-A statement, it proves nothing about p_0 . If it uses a class-D hypothesis under canonical failure, it is invalid.

2026-09-07C agenda override. The tier table below is retained as the historical v4.81 agenda. For new work, start instead with the three targets in §11.6: (1) the exact $h = 1$ full-BFS/root-2 closed-support question, (2) the low-gap three-label Diophantine obstruction split, and (3) the radius-three large-label/large-cofactor asymptotic region. The moving closed-support problem is the global wall.

15.1.1 Tiered targets

Tier	Target	Why it is in this tier
1	Prove or refute (N3): $\sum_q d_R(q) \geq R + 1$ (§5.7, Q8)	Small, self-contained, elementary in appearance, and it repairs the part of the corpus that used the false Euler normalisations. Zero counterexamples in 21 093 worlds. A model with a few hours should attempt this first.
1	Refute L1 by exhibiting $\nu(R) = O(1)$ with J_R unbounded (§9.5)	A <i>negative</i> result that would remove a whole family of capacity arguments. The data already point this way. Cheap to test computationally.
1	Extend the B_N^{low} census beyond $6 \cdot 10^4$	The coincidence of the three six-element lists (§12.5) is unexplained. A seventh N would be the single most informative new datum in the project.

Tier	Target	Why it is in this tier
2	TS: smooth branch to escape (§9.3)	The branch with the largest quantitative margin and the most elementary statement. Success would give Goldbach in a specific regime, which is a real theorem.
2	E1: escape-or-reuse, or the bounded-charging lemma (§9.6)	Overwhelming data, explicit statement, and the missing ingredient is named (bounded charging multiplicity). The most likely target to actually fall.
2	Invariants of the boundary filtration (Q7)	Purely combinatorial, independent of Goldbach, publishable on its own terms if anything nontrivial exists.
3	TD / C-SAMEFUNC-BRIDGE (§9.3)	The real bridge. Hard, and previous attempts all failed the same way. Attempt only with the same-functional constraint written down first.
3	RPD (§9.10)	The only open target carrying an explicit quantitative margin. Its original phrasing should be re-checked against the source before use.
4	x1 / (H- p_0)	Strictly stronger than Goldbach. Do not attack directly.
4	Anything requiring level of distribution $> 2/3$ or a binary prime–Liouville correlation	These are the external walls (§13.4). Working on them is working on a different, famous problem.

15.1.2 Experiments that would change the picture

1. B_N^{low} **census to 10^6** . Currently $6 \cdot 10^4$. Cost: roughly quadratic in the limit as implemented; a smarter implementation should use the fact that B_0 can be built once per N from a shared sieve. A seventh nonempty B_N^{low} , or its absence at 10^6 , is decisive information about whether the six are an artefact of smallness.
2. **Full upper-root census beyond $2 \cdot 10^4$** . The 301 number is the project’s most-cited statistic and it is known only to $2 \cdot 10^4$. Extending it to 10^5 would test whether “failing roots cluster at $B_N^{\text{low}} \neq \emptyset$ numbers” persists.
3. J_R **versus $\nu(R)$ scatter**. Directly tests L1 and is a few lines on top of the existing world routine.
4. G_{esc} **to 10^6** . Tests Q2 (does the margin grow?). The current dyadic minima 1, 1, 1, 1, 1, 2, 2, 3, 3, 3 are suggestive but the range is small.
5. **Phase-conditioned traversal statistics**. Does anything about the traversal, beyond immediate success, depend on θ ? Currently unknown, and ρ_0 is the only boundary invariant that varies between canonical worlds (Q1).

Chapter 16

Literature and Source Registry

16.1 Internal sources

Tag	Source	Role
[BIB]	the Goldbach Biblia (v7.5 and its post-v7.5 technical log)	Shadow, retention, path/winding invariants, prime-power witness, escape capacity, the 10^6 scan tables, the ten margin-one numbers.
[CORE]	CRD Core	Canonical geometry, C-GAP-EQUIVALENCE, C-STRICT-NONREUSE, the reflection gate, C0a, C0b-RD, the genealogical layer (V71.21–V71.44), the open-target registry, P0-CANONICITY-ABLATION.
[PCRD]	GOLDBACH_POST_CRD_RESEARCH_TALENTS	THIS FILE DOES NOT TAKE PRECEDENCE ON CONFLICTS. B_N^{low} fixed point and characterisation, the six N to $6 \cdot 10^4$, the successor-theory walls, finite affine canonical programmability, finite bad-tree universality, the canonical prime cell theorem, the H1 descent and the 2-core wall, the final verdict, its own theorem registry and 30-row log.
[FAIL]	the project failure log	All 239 live entries, reproduced in Chapter 14.
[LOG]	post-v7.5 technical log	D_q , M_q , the prime-power ledger, primitive residual normal form.
[ILA]	incidence-ledger audit	IL1–IL6, two reconstruction routes, 1 000 worlds.
[ORPH], [RT]	orphan census and red team	Escape-or-reuse data: 61 452 and 60 413 instances, 0 orphans.
[D4], [F3], [FRO], [FND], [V481], [NOV]	question-answer campaigns and the novelty note	Statistical ablations, structural twins, the rooted container identity, the prior-art boundary. Several of their numbers are <i>reported, not reproduced here</i> and are marked as such wherever they appear.

Warning 16.1 (Precedence). Where sources conflict, [PCRD] takes precedence, then [CORE], then [BIB]. Where this file’s own computation conflicts with any of them, this file states both and marks the discrepancy — it does not silently overwrite the source. The recorded discrepancies are: the Euler-defect normalisations (§5.7), the diagonal scope of the genealogical layer (Scope note 3.17), the lower bound in the ρ_0 formula (Warning 6.18), and the hypothesis of high-gap amplification (Warning 7.5).

16.2 Prior art and the novelty boundary

Warning 16.2. No claim of novelty is made anywhere in this file. Establishing novelty would require a separate literature review that has not been done. Any label of the form “derived here” means *derived from the corpus’s own definitions in this document*, not “new in number theory”.

The bibliography is a search record, not a citation list. It reproduces the Biblia’s full list of 92 sources plus three later additions. Many entries are *not* cited anywhere in the running text: they are there so that a future model can see what has already been looked at and does not repeat the search. An entry’s presence means “this was examined”; the disposition tables of §16.3–§16.4.1 say with what result. An entry’s absence is the useful signal — it means the project has probably not looked there.

16.2.1 What must *not* be attributed to this project

The relation $p \rightarrow q$ for $q \mid N - p$; the notion of a Goldbach factorization graph; recursive search through complement factors; valuation weights $v_q(N - p)$; reachability, source and sink components, autonomous closed components; the study of Goldbach pairs near $N/2$; elementary parent residue classes and gcd identities. [NOV]

Topic	Source	Boundary
Factorization graph and exceptional autonomous components	Božek [8]	The graph and the terminal pathology are not ours. Orientation matters: a source component in the paper’s orientation corresponds to terminal behaviour after reversal (Warning 3.23).
Recursive factor search from a given root	OEIS A306746	Search from an arbitrary root is prior art; the root condition there is <i>not</i> the unconditional least prime $\geq N/2$.
First upper prime already forming a pair	OEIS A078496	$a(n) = \min\{p > n : p, 2n - p \in \mathbb{P}\}$ — a selector <i>conditioned on the outcome</i> . This is not p_0 .
Nearest symmetric Goldbach pair	OEIS A082467	Goldbach near the midpoint is prior art.
Short intervals containing a prime	Baker–Harman–Pintz [15]	Calibrates $h \ll x^{0.525}$; it is not an escape lemma and is not used as one.
Verification of Goldbach	Oliveira e Silva et al. [12]	Says nothing about (H- p_0). Must never be cited as this project’s own computation.
Linear forms in logarithms	Matveev [81]	Astronomical constants; asymptotic content only. Used only for the 2-core wall.
Almost-prime representations $(1 + 1.9)$	Li–Liu [17]	External input for CAPT; does not give the last bit P_1 versus P_2 .
Exponent of distribution $5/8$	Pascadi [95]	External input for RSD; below the $2/3$ that RSD needs.
Chen-type lower bound	R. Li [96]	$D_{1,2}(N) \geq 1.9728 C(N)N/\log^2 N$.
Asymptotic sieve for primes	Friedlander–Iwaniec [40]	The template for the missing bilinear information.

Topic	Source	Boundary
Goldbach problem for the Liouville function	Mangerel [97]	Calibrates the centred-correlation wall.

16.2.2 The safe formulation

Building on earlier work on Goldbach factorization graphs and on recursive search through complement factors, we study the *unconditional* selector $p_0(N) = \min\{p \in \mathbb{P} : p \geq N/2\}$ as a distinguished initial condition of the full dynamics. To the best of our knowledge we have not found earlier work using this selector — without requiring $N - p_0$ to be prime — as a distinguished root, and systematically comparing its reachable bad world with those of non-canonical roots. *This is a report on the extent of a literature search, not a claim of priority.*

16.3 Register of rejected literature applications

This section exists so that a future model does not re-import a tool the project has already tried and discarded. Each row records a specific *application* of a specific external result, and why it does not do the job here. A **KILLED** row means the application is dead, *not* that the cited theorem is wrong. An **OPEN** row means the tool may still be usable once a stated precondition is met.

16.3.1 First-front applications (Biblia v7.4)

Claim / tool	Status	Reason
$ Q \sim N/\log N$	KILLED	$Q = \text{PF}(m_0)$, so $ Q = \omega(m_0)$; a global density comparison does not apply to the first front.
SMOOTH $\Rightarrow$ huge powerful part	KILLED	A SMOOTH row can be squarefree with many distinct small prime factors (Warning 7.13).
Positive density of squarefree values $\Rightarrow$ positive fraction of Q	KILLED	A result for all arguments, or all primes, does not transfer to the <i>adaptive</i> set of prime divisors of a single m_0 .
Bombieri–Vinogradov on Q	KILLED	The theorem controls primes in arithmetic progressions <i>on average over moduli</i> , not the distribution of the prime divisors of one specific m_0 .
Filaseta–Trifonov as a uniform lower gap between consecutive powerful numbers	KILLED	The short-interval squarefull-counting literature does not supply the universal lower gap barrier that was assumed.
Darmon–Granville $\Rightarrow$ uniform bound for a varying family	KILLED	Coefficients, the difference and the curve itself vary with N ; finiteness for a fixed curve gives no uniformity.
Fixed-polynomial radical theorem with $F(X) = \prod_{q \in Q} (X - q)$	NEEDS VERIFICATION	Degree, height and the whole polynomial vary with N ; an explicit uniform dependence is required.
Pairwise classical <i>abc</i> closes SMOOTH	KILLED	The shift $ q - r $ can be of order N and consumes the radical; without extra global structure the required first-front lower bound does not arise.
Corvaja–Zannier greatest-prime-factor result	OPEN	Qualitatively supports escape, but gives no <i>moving</i> threshold $P^+(N - q) \geq m_0/q$.

Claim / tool	Status	Reason
Evertse–Schlickewei–Schmidt [20]	OPEN	Usable only after freezing a small alphabet/rank inside one counterexample. See also the vacuity result recorded in §9.2.4.
Gallagher’s larger sieve [43]	OPEN	Requires first showing that a large part of Q occupies few classes modulo many primes.
Multiplicity-order literature	OPEN	Known generic / almost-all results give no deterministic lower side for specific ports modulo a specific N .

16.3.2 Iteration-12 audit of 52 primary sources

An honest protocol instead of a fictitious “1000”. The scope and metadata of **52 distinct primary sources** were checked; for 18 of them the main theorems and the proof passages directly relevant to the project were read as well. This is *not* called “reading 1000 papers”: automatically fetching a thousand bibliographic records is not reading and does not raise mathematical credibility. Below, mode T means “theorems and relevant sections”, mode Z means “scope audit from abstract, introduction and statements of results”.

Sources	Mode	Conclusion for the project
Bhowmik–Grimmelt [38]; Li–Liu [17]	T	Goldbach still open. The first localises the barrier in minor arcs and exceptional sets; the second gives $(1 + 1.9)$, not an adaptive traversal path. Contributed calibration of the main gap.
Lichtman [28]; Ford [32]	T	Strong distribution and the anatomy of shifted primes work <i>on average</i> . They do not give a pointwise choice from a reachable parent set.
Grimmelt–Teräväinen [47]; Halupczok [48]	Z	Transference and Goldbach in progressions / short intervals still leave an exceptional set or averaged conditions. Useful as a weighting template, insufficient pointwise.
Friedlander–Iwaniec [40]; Polymath8 [41]	T	The most important analytic input: breaking parity needs Type-II bilinear information, not a linear sieve alone.
Green–Tao [42]; Bienvenu–Shao–Teräväinen [46]	T/Z	Transference needs a pseudorandom majorant and positive relative density for a <i>fixed</i> configuration. Those hypotheses are not available for adaptive reachability.
Ford [33, 49]; Assing–Blomer–Li [30]	T	Pratt-type closures and uniform Titchmarsh problems are the closest structural analogues, but the relation $q \mid N - p$ moves with N .
Matomäki–Radziwiłł [50]	T/Z	“Almost all” short intervals give no certificate for every adaptive set.
Helfgott [18, 19]; Oliveira e Silva et al. [12]; Huang–Li [27]; Li [16]	T/Z	Circle method, finite verification, EH and short intervals are important calibration, but none synchronises a prime with the basin of p_0 .
Evertse–Schlickewei–Schmidt [20]	T/Z	The number of S -unit solutions may depend only on dimension and rank, which is better than fixed- S constants. Application needs genuinely small rank; the known zero-wrap branch is empty.
Vaaler [51]	T/Z	Heights may give a short multiplicative dependence <i>after</i> an independent rank and height bound. They do not handle the present δ -forcing automatically.
Amoroso–Sombra–Zannier [52]	Z	Sparse-polynomial / unlikely-intersection results work for fixed families; Smith normal form is an excellent lattice algorithm but supplies no arithmetic contradiction by itself.

Sources	Mode	Conclusion for the project
Bugeaud [22]; Siksek–Stoll [23]; Bennett–Siksek [21, 5]	Z	Effective once a specific curve or support is frozen; no uniformity for a moving full SCC.
Bellman [53]; Karp [54]	T	The variational calculus of paths reduces to dynamic programming and a cycle criterion. Gave the cycle–potential duality, not prime selection.
Prajna–Jadbabaie [55]	Z	Barrier certificates and occupation measures are the right shape for reachability; on an SCC they need an arithmetic inequality that is missing.
Angeli–Philippe–Athanasopoulos–Jungers [56]	T/Z	A complete Lyapunov function is constant on recurrent classes, which justifies SCC compression. Multi-part versions do not evade the cycle barrier.
Friedman–Tillich [57]; Jiang–Lim–Yao–Ye [58]; De Leenheer [59]; Dodziuk [7]	T/Z	Calculus and Hodge theory on graphs formalise the gradient and the cyclic part. They contributed an exact no-go, not a primality theorem.
Bass [60]; Rangarajan [61]	Z	The graph zeta function and $\text{coker}(I - A)$ repackaging determinant/Smith data. No new information about the rows $N - p$.
Z-transform / Koopman resolvent literature	Z	Appropriate for a <i>fixed</i> operator; the adaptive graph is not an LTI system.
Discrete variational calculus literature	Z	Gives stationarity equations, but for a sum of path costs it is not stronger than Bellman.

16.4 Post-v4.81 external-input audit (September 2026)

The continuation checked only the exact external statements it consumed; it did not perform a full literature-priority audit of the new project theorems.

Input	Source status	Exact use
BS-1996	primary statement verified	Beukers–Schlickewei’s uniform bound for $x + y = 1$ in finitely generated multiplicative groups; used only for solution counts, not heights.
MATVEEV-IN-BMS-9.4	exact product-minus-one form checked	Real/rational Matveev bound as stated in Bugeaud–Mignotte–Siksek, used for closed-support height and explicit envelope escape.
FKL-2010-LEMMA-5.1	exact statement and uniformity checked	Upper-bound sieve for a growing number of primitive linear forms; used for radius-two and multigeneration counting, never as a prime-tuples lower bound.
PINTZ-GOLDBACH-EXCEPTIONAL-SET	theorem statement/hypotheses checked	$E(Y) < Y^{0.72}$ above an ineffective constant, used only to show that ordinary Goldbach truth can coexist with complete shallow BADness on a density-one set of the special starts.

16.4.1 Frontier salvage audit (v4.81)

The v4.81 attack performed a targeted salvage audit of the older Biblia against the *current* frontier. Its verdict in one sentence: *the old corpus contributed exactly one genuinely missing tool — the gap-scale shadow identity — and one correct **shape** for the replacement frontier (the smoothness budget); everything else in the neighbourhood of the current obstruction is either already in the Core, or vacuous on the primitive branch, or domain-mismatched.*

Salvaged and used.

Old item	Restated form	Status and use
Prime-Gap Shadow Identity (Theorem 5.22)	$N = 2m + \delta, \delta = 2h; \text{ for } r \mid m: r \mid N - y \iff r \mid \delta - y$	PROVED , re-proved from v4.80 definitions. Used to obtain the gap-scale form of primitive load-one (V81.11). <i>Not previously used in the v4.79/v4.80 chain.</i>
Retention-Escape (Theorem 5.24)	$\text{PF}(N - q) \cap Q = \text{PF}(d_q), d_q = \gcd(m, \delta - q)$	SCOPE-CORRECTED . Identical to $D_q = \gcd(m, N - q)$, giving $D_q = \gcd(m, 2h - q)$. The old “escape” reading (“all children are new”) must <i>not</i> be upgraded to success.
Gap/first-front disjointness	$\gcd(h, m) = 1, \text{PF}(h) \cap Q = \emptyset$	PROVED , re-proved; used for the block/ Q separation.
Multi-source product lock (Theorem 5.7)	$T_S \equiv (-1)^{ S } \prod_{\text{Src}} d \pmod{N}$	Already in the Core; the <i>new</i> use is the selected-family instance (V81.13) and the residual instance $T_{R'} \equiv (-1)^{ R' } \Sigma'$.
Two-alphabet retention	$\gcd(N - q, mh) = \gcd(N - q, m) \gcd(N - q, h); \gcd(E_q, E_r) \mid q - r$	PROVED (the split). On the primitive branch the two-alphabet retention of every first-front row collapses to the h -alphabet.
Smoothness budget (every row of a child-closed world is C -smooth)	The count of available C -smooth numbers in the row window is a necessary condition	SCOPE-CORRECTED to EMPIRICAL PATTERN : the old text carries it under an equidistribution assumption. Adopted here <i>only as the shape</i> of the new preferred frontier (§9.2.4). The equidistribution assumption is not imported.

Interesting but insufficient.

Old item	Content	Why insufficient at the frontier
Prime-power (Prop. 5.29)	witness $\text{PF}(N - q) \subseteq Q \Rightarrow \exists r \in Q : v_r(m) = v_r(\delta - q)$	On the primitive branch the hypothesis is never satisfied ($\text{PF}(N - q) \cap Q = \emptyset$): vacuous domain exactly where it is needed.
Shadow (Prop. 5.28), $\delta = q + r + tqr$	two-cycle Exact form of a Q -shadow two-cycle	The primitive branch makes the Q -shadow graph <i>empty</i> ; no cycle exists to exploit. Vacuous domain.

Old item	Content	Why insufficient at the frontier
Quotient-rigidity master lemma	$ u - v < 1.1u/m^2$, $\min(u, v) > 0.9m^2$ for row quotients	Proved inside a terminal-SCC band normalisation; its row-window and row-gap hypotheses are unavailable for primitive first-front rows. Domain mismatch.
One-escape residue lock	$g \equiv (-1)^{ W } p_0(U_W J_W)^{-1} \pmod{N}$	Requires a unique good child; in a failed world there is none. The useful residue content is exactly G-MULTISOURCE-LOCK. Retained as motivation for the secondary frontier.
C -smooth budget calibration on the six EAC	Empirical ratios 0.62–0.79	VERIFIED COMPUTATION only; the six EAC are terminal-SCC objects, <i>not</i> canonical failed worlds. Not usable as evidence for the canonical branch.

Rejected or already superseded.

Old item	Disposition
“Terminal bad SCC $\Rightarrow$ Goldbach contradiction” programme	Not revived. A terminal bad SCC is not by itself a Goldbach counterexample (Scope note 3.27). No lemma of that chain is used.
Exponent-determinant collapses	Domain is the two- or three-element terminal core with fixed Ω ; the primitive first front has neither. SUPERSEDED by the adaptive low-core theory.
Fringe alphabet $\{\lambda, 2\lambda - 1\}$	FAIL-FRIDGE-ALPHABET-TRANSFER-001. Theorem 9.17 is <i>not</i> that claim: it derives the alphabet as the full prime range $\mathcal{L}_N \subseteq [\lambda, 2\lambda)$, which is precisely the “what survives” clause of that failure entry.
“Prime-free boundary activates a reflected parent”	FAIL-REFLECTION-ACTIVATION-001 ($N = 242$, $p_0 = 127$) is respected throughout; Theorem 9.10 is its sharpened form.
Evertse S -unit counting as a closure device	Vacuous, as the Biblia itself noted; recorded as FAIL-SUNIT-COUNT-BOUNDS-THE-LOWCORE-001.

16.5 Literature verdicts and named missing lemmas

Iteration-157 verdict. Checked: the Euclidean algorithm for positive 2×2 matrices, the free monoid of positive SL_2 , Conway–Coxeter friezes [88, 89], unit equations [90], graph toric ideals [83, 84], higher Lawrence configurations [91], cyclic congruence systems [92], and prime-escape theorems from subset products [93]. *None* supplies a dynamical transition preserving simultaneously N , the row identities, the full valuations and the self-label roles. The nearest missing theorem is named:

Matched Euclidean Gluing Lemma (OPEN). To the word or matrix of one packet one must assign an *actual* continuation to the next packet in the same system, preserving all roles, in such a way that a Euclidean measure strictly decreases or an escape is forced.

Without that lemma, word length, continued-fraction depth and matrix entry sums are only *static* coordinates. **They must not be used as Lyapunov functions for self-label dynamics.** Compare Remark 7.45, where the same continued-fraction structure appears exactly and still yields no descent.

The one external input whose shape is known. Across every audit the same conclusion recurs: what is missing is *Type-II bilinear information* in the sense of the asymptotic sieve for primes [40, 41]. Every internal construction that appeared to supply it turned out on inspection to be a Buchstab/sieve decomposition or an affine reparametrisation (§13.3.1). A future model should treat “does this really produce bilinear information, or is it an affine relabelling?” as the first question to ask of any new proposal.

Chapter 17

Historical and Superseded Results

Kept because the *history* matters: each item below was once load-bearing, and knowing why it stopped being load-bearing is what prevents its reconstruction under a new name. Nothing in this chapter may be used as a lemma. Where a superseded statement contains a correct sub-lemma, the sub-lemma appears in Chapter 4 under its own ID; the programme does not come back with it.

Item	Status	History and what replaced it
CRD1 as a proof engine	SUPERSEDED	Replaced by the fixed-point reformulation (Theorem 3.29), which is exact but is a computation, not a mechanism. What died: the “same container” argument, generic smooth counting, residue saturation, and the exact self-consistency of the fixed point.
TD-DENSITY	SUPERSEDED	The density formulation of TD required a bound of the same strength as its own conclusion. Replaced by the root-anchored form (§9.3).
p_0 as information selector	SUPERSEDED	Not refuted, but not established: every clause of the bundle-lift construction survives the ablation $p_0 \rightarrow p_1$ on rows with composite c'_j . The corpus verdict is “quantitatively useful, structurally inessential”. FAIL-P0-INFORMATION-SELECTOR-001
Euler defect in normalisations (N1), (N2)	KILLED	Refuted here computationally (§5.7). Any earlier argument using $\eta + \nu \geq S + \text{Src}(S) $ in those normalisations must be re-derived. The surviving forms are (N3) (VERIFIED COMPUTATION) and (N4) (trivial).
Prefix flux $B_k \equiv (-1)^k p_0$	AUDITED IDENTITY	Recorded in [BIB] (iterations 20, 22) with the flag “not auditable from packet”. It has never been used downstream and is kept only so that it is not rediscovered and trusted.
Rooted container identity	AUDITED IDENTITY	[1481]. Closed the size-based route — i.e. its role was to <i>end</i> a line of attack, not to open one.
Four-channel ledger on all of Q_0	KILLED	Replaced by IL1–IL6 on Q_{smooth} after $N = 20$ refuted the wider domain. [ILA]
Unconditional CAB	KILLED	$N = 322$ and $N = 858$. No repaired version exists; the capacity route now goes through K1, which is open.

Item	Status	History and what replaced it
Upper-half source compression	PROVED (class D)	Correct, but its hypothesis is $\text{Good}_N = \emptyset$. It was repeatedly invoked under canonical failure, which is invalid. FAIL-UPPERHALF-SCOPE-001
Symmetric fracture, forbidden offset band	PROVED (class D)	Same situation: correct under the minimal-counterexample hypothesis, unusable otherwise.
$h = 1$ local closure	SCOPE-CORRECTED	The blanket complete-BFS reading of finite bad-tree universality was withdrawn. The exact $h = 1$ full-BFS problem is live and is conjugate to root 2; selected transcript programming survives only with uncontrolled cofactors. [2026-09-05-07C]
CMFT, CGST, CSD, CAPT, RSD, rough- P_2	KILLED	§13.2. Their individual failure reasons remain binding (cofactor leakage, missing complete-support consumer, functional mismatch or parity/prime-correlation barriers); they must not now be justified by the old over-broad complete-BFS finite-local wall.
$q < m_0/3$	SCOPE-CORRECTED	Corrected to $q \leq m_0/3$; $N = 32$ attains equality. The consequence $q < N/6$ is unaffected.
$\rho_0 = \#\{D_s < 2h\}$	SCOPE-CORRECTED	Corrected here by adding the lower condition $D_s > h$ (Warning 6.18).
Genealogy stated for “every prime $r \geq x$ ”	SCOPE-CORRECTED	Corrected here to the non-diagonal domain; smallest witness $N = 14$ (Scope note 3.17).
High-gap amplification $h^2 > 4m_0$	SCOPE-CORRECTED	Hypothesis is Exact-D pair failure, not CRD failure (Warning 7.5).
COB-RD v1 (without $Q_{\leq} \neq \emptyset$)	PROVED WITH REPAIR	The missing hypothesis made the inequality read $1 < 1$ in 47 551 of 72 446 canonical instances with $h \geq 3$ and $N < 2 \cdot 10^5$. Repaired version is Theorem 8.6 (Scope note 8.7).

Chapter 18

Handoff to the Next Model

If you read only one chapter, read this one — then Chapter 2, then the binding update Chapter 11. Historical frontiers remain in Chapters 9 and 10.

18.1 What not to try again

1. **Do not confuse selected finite programming with complete BFS.** Compatible selected divisibility/valuation transcripts can be realised in canonical $h = 1$ worlds (Theorems 8.1, 8.2), but this does not control unprescribed cofactors. Therefore a selected BAD path/cycle is not a complete BAD neighbourhood. The old blanket finite-local wall is scope-corrected, not a prohibition on every bounded-depth complete-frontier theorem.
2. **Do not restart the programme in new coordinates.** Five successor theories have already done exactly that (§13.2). A new coordinate system does not change what information is available.
3. **Do not prove a bound in one functional and use it in another.** This is the single defect blocking TD, $P1^{\text{br}}$ and K1 simultaneously. “Morally the same” and “dominates on average” are not the same functional.
4. **Do not treat canonical failure as Goldbach failure.** Class-D theorems (upper-half compression, symmetric fracture, forbidden offset band) require $\text{Good}_N = \emptyset$ and are unusable under canonical failure alone. FAIL-UPPERHALF-SCOPE-001
5. **Do not quote a count without its domain.** Which roots, which complements, unit rows included or not. FAIL-CENSUS-DOMAIN-001, FAIL-UNIT-COMPLEMENT-001
6. **Do not promote a scan.** 0 failures to 10^7 is not evidence for X1; it is the statement that X1 has no small counterexample.
7. **Do not use the mass of the boundary holes.** $O_k(\log N)$ against a block mass of order $t_k \log(x/h_k)$. Any argument routed through hole mass is dead before it starts. FAIL-P0-ZEROHOLE-ESSENTIAL-001
8. **Do not strengthen the reflection gate to activation.** $N = 242$ is final: both lower mirrors are composite and no active reflected row exists. The phase diagram explains why (Type I). FAIL-REFLECTION-ACTIVATION-001
9. **Do not widen the reflected block.** The endpoint pair has difference exactly $2h$ and is not covered by strict non-reuse. FAIL-BROADBLOCK-ENDPOINT
10. **Do not use the Euler defect in normalisation (N1) or (N2).** Refuted here: 6 044 and 3 804 violations respectively for $N \leq 6 \cdot 10^4$ (§5.7).
11. **Do not attack X1 or Goldbach directly.** X1 is strictly stronger than Goldbach for each N .
12. **Do not route around parity by enlarging the graph.** The obstruction lives in the sieve information, not in the graph (§13.4).
13. **Do not search for a fixed-rank law in the first mirror alone.** Every fixed rank realises every odd mirror $m \geq 3$, and exact same- (k, m) certificates have opposite eventual verdicts. Any

viable rule must use ambient N -dependent information.

14. **Do not use a fixed finite valuation/label projection to isolate p_0 .** Run H synchronises arbitrary prescribed finite valuations across $p_0, \dots, p_K$ at the same N infinitely often.
15. **Do not substitute the moving basin B_N into a fixed- S theorem.** The quantifiers are different; all constants in the Matveev/BHP separation theorem depend on the fixed alphabet.
16. **Do not extend the bounded-mirror transplant scan by increasing only the N cutoff.** For $m \leq 3053, k \leq 74$ the finite-fibre theorem proves the entire realisation box ends at $N = 7324$.
17. **Do not re-prove what is already PROVED.** Use it by ID from Chapter 4 and the delta registry §11.8.
18. **Do not raise a status.** If you prove something, add a row; do not edit an old one.

18.2 What currently looks best

1. **Selector–moving-basin interaction.** This is the binding target. Use

$$p_k(N) \text{ fails} \iff \text{rad}(N - p_k(N)) \mid R_N$$

together with the fact that p_0 is the maximal prime-backed mirror selector. A useful theorem must constrain B_N in a way that depends on this actual location, not on a fixed external alphabet.

2. **Uniform moving-support geometry.** The strongest analytic ingredient now available is the Matveev/BHP separation of two complete S -smooth early mirrors. Search for a bound on the size/height/spacing of B_N that makes this statement uniform over an admissible moving family. A contradiction between a lower growth requirement and an upper rank/mirror-capacity requirement would be genuinely global.
3. **Construct a new complete failure host.** The six known hosts have been mined heavily, and all fixed mirrors $m \leq 3053$ in the selected rank box are exhausted. A new closed BAD basin would provide independent arithmetic data and immediately test whether the apparent survivor ranks are accidental.
4. **Exploit low-gap support disjointness only when coupled to B_N .** Shared child factors force $2q$ -spaced roots; low-gap roots can have pairwise disjoint initial supports. By itself this does not imply success, but a basin-capacity theorem consuming many disjoint supports could.
5. **Keep the run-D Diophantine branch as a secondary exact target.** The $q = 3, h = 1$ three-label branch is killed and monomial returns are more restricted. Further exact closure elimination is useful only if it feeds the moving-basin programme rather than becoming another local invariant.

Warning 18.1 (What does *not* look good, despite appearances). The genealogical layer is the most beautiful part of the project and the most likely to mislead. It is exact, it has genuine converses, and it does not imply anything dynamical — and Warning 6.7(2) raises the real possibility that it cannot, because “ $\kappa_{\text{gen}}(r) = 0$ ” may carry exactly the information “ r is the first upper prime” and nothing more. Before building on it, decide whether your argument would survive that equivalence.

18.3 The most important dependency theorems

If any of the following were found to be wrong, large parts of the file would fall. They are listed in order of how much depends on them.

ID	Name	What depends on it
C-GAP-EQ	Canonical gap equivalence (Thm. 3.10)	Everything canonical. It is the definition of the canonical input and the only class-A hypothesis anywhere.
ARS-CLOSED-WORLD	Closure and unique source (Lem. 3.26)	The whole failure analysis, the fixed point, the Euler layer, every capacity statement.
ARS-NORETURN	No-bad-return-to- p_0 (Thm. 5.33)	Uniqueness of the source, hence ARS-CLOSED-WORLD in the canonical case, hence the two-sheet budget.
CRD-FIXPOINT	Fixed-point characterisation (Thm. 3.29)	The entire computational programme; the six special N ; the tractability of the census.
C0b-RD	Reflection-defect ledger (Thm. 8.6)	All four branch targets TI, TD, TS, TF, hence $P1^{br}$, hence everything downstream. <i>Note its repaired hypothesis</i> (Scope note 8.7).
V71.21	Genealogical rank (Thm. 6.2)	Ledger completeness, uniqueness of the reflection-complete root, the lossless code, the ablation's interpretation. <i>Note its domain</i> (Scope note 3.17).
C-STRICT-NONREUSE	Strict non-reuse (Lem. 3.13)	TI's injectivity for large primes, W1's collision-freeness, the $\mathcal{F}$ branch of C0b-RD.
S-SHADOW-ID	Shadow identity (Thm. 5.22)	Retention, D_q , M_q , cycle localisation, the prime-power witness — the whole multiplicative layer.
V-TUJ	$T_S = U_S J_S$ (Thm. 3.19)	Every ledger and capacity statement. Note that the <i>Euler defect</i> built on it does not inherit its reliability (§5.7).
G-MULTISOURCE-LOCK	Multi-source product lock (Thm. 5.7)	G-ROOT-LOCK, the selected residue lock (V81.13), and route (B) of the surviving-route trichotomy.
V81-CONTAINER	Rooted container identity (Thm. 9.7)	The whole current frontier: the slack bound, the death of the size route, and the trichotomy that names the only remaining routes.
V81-C5-ALPHABET	Adaptive fringe cofactor alphabet (Thm. 9.17)	The preferred target: it is what makes the high band completely known, so that the low core is the entire unknown.
FACP & scope-corrected FBTU	Partial-transcript constraint	FACP and the repaired transcript theorem constrain selected finite-pattern arguments only; they do not rule out complete-frontier theorems. See §10.2.

18.4 The five most valuable next moves

1. **Prove a moving- B_N capacity/height inequality with selector input.** The desired statement should combine the complete closure equations for B_N with the fact that p_0 is the first upper prime and $PF(N - p_0) \subseteq B_N$ under failure.
2. **Try to uniformise H2.** Replace “fixed finite S ” in the early-rank smooth-support separation theorem by a quantitatively controlled family S_N large enough to contain a hypothetical B_N . Record exactly where the Matveev constant destroys the argument if it cannot be done.
3. **Search for a new EAC/BAD-basin host, not for more old-mirror transplants.** If one is found, compute the full upper-root spectrum and ask whether any of 10, 14, 26, 28, 30, 62, 63, 71, 74 finally intersects it.
4. **Develop a simultaneous-failure spacing theorem.** Combine $\gcd(N - r, N - s) = \gcd(N -$

$r, s - r$), the $2q$ shared-factor spacing law, and the fixed-gap S -unit count to bound how many low-rank roots can all be B_N -smooth in one host.

5. **Only then return to the canonical low-gap Diophantine branch.** Use the run-D four-label and order- ≥ 3 theorems as exact local inputs to a complete-basin argument; do not search isolated congruences for their own sake.

Warning 18.2 (Two moves demoted by the v4.81 result). Anything of the shape “find new selected storage disjoint from $\mathcal{C}_Q$ at fixed positive logarithmic scale” is now known to be *insufficient by construction*, not merely unproved (Corollary 9.9); and anything of the shape “exploit an overlap between the reflected block and the selected family” is known to be charge-free (Theorem 9.10). Do not spend a run on either.

Final orientation after 2026-09-07H. Binary Goldbach is **OPEN**. Universal canonical success (**H**- p_0) is **OPEN** and is strictly stronger. The old finite-selected-transcript architecture is **SUPERSEDED**, and runs G–H now rule out a much broader family of fixed first-row and fixed finite-label explanations. The live object is the incidence between the actual root selector and the complete moving BAD basin B_N . Fixed (k, m) fibres, bounded old-mirror scans, and fixed- S arguments have sharply defined limits and must not be extrapolated past them. Future work should begin at §11.6; the raw E/F and D–H sources are preserved losslessly in the appendices so later models can audit every scope decision.

Appendix A

Reproduction Code

A.1 The data generator

The following script reproduces *every* number marked **VERIFIED COMPUTATION** in this file. It has no external dependencies (standard library, Python 3.9+). Blocks EXP1–EXP8 correspond to the identifiers in the provenance table of §12.1. Comments are the project’s own, transliterated to ASCII; the executable code is unchanged.

Listing A.1: `p0_experiments.py` — complete self-contained generator for every **VERIFIED COMPUTATION** number in this file.

```
1 #!/usr/bin/env python3
2 # -*- coding: utf-8 -*-
3 """
4 p0_experiments.py -- JEDEN samowystarczalny skrypt odtwarzajacy KAZDA liczbe
5 oznaczona COMPUTATIONAL w dossier p_0 oraz dane wszystkich rysunkow.
6
7 Bez zaleznosci zewnetrznych (tylko biblioteka standardowa Pythona 3.9+).
8
9     python3 p0_experiments.py --scan 10000000 --census 20000 --phase 200000 \
10         --fixedpoint 60000 --json dane.json
11
12 Domyślne, szybkie ustawienia (kilka minut):
13     python3 p0_experiments.py
14
15 Kazdy blok EXP wypisuje dokładny zakres, który jest częścią twierdzenia
16 obliczeniowego -- zgodnie z reguła "nigdy nie cytuj liczby bez jej domeny".
17 """
18 from __future__ import annotations
19
20 import argparse
21 import json
22 from array import array
23 from collections import Counter, deque
24
25
26 # ===== sito
27 def build(limit):
28     spf = array('i', range(limit + 1))
29     i = 2
30     while i * i <= limit:
31         if spf[i] == i:
32             for j in range(i * i, limit + 1, i):
33                 if spf[j] == j:
34                     spf[j] = i
35             i += 1
36     isp = bytearray(limit + 1)
37     for n in range(2, limit + 1):
38         isp[n] = 1 if spf[n] == n else 0
39     return spf, isp
40
41
42 def pf(n, spf):
```

```

43 out = []
44 while n > 1:
45     q = spf[n]
46     out.append(q)
47     while n % q == 0:
48         n //= q
49 return out
50
51
52 def bigomega(n, spf):
53     c = 0
54     while n > 1:
55         q = spf[n]
56         while n % q == 0:
57             n //= q
58             c += 1
59 return c
60
61
62 class Env:
63     """Sito + selektory kanoniczne + dwa tryby przejścia."""
64
65     def __init__(self, limit):
66         self.L = limit
67         self.spf, self.isp = build(limit)
68
69     def nextp(self, n):
70         m = n + 1
71         while m <= self.L and not self.isp[m]:
72             m += 1
73         return m
74
75     def prevp(self, n):
76         m = n - 1
77         while m >= 2 and not self.isp[m]:
78             m -= 1
79         return m if m >= 2 else -1
80
81     def p0(self, N):
82         x = N // 2
83         return x if self.isp[x] else self.nextp(x)
84
85     def census(self, N, r):
86         """tryb census: (sukces, glebokosc, liczba wierzchołkow)"""
87         if 2 * r == N:
88             return True, 0, 1
89         q = deque([(r, 0)])
90         seen = {r}
91         nodes = 0
92         while q:
93             p, d = q.popleft()
94             nodes += 1
95             n = N - p
96             if n >= 2 and self.isp[n]:
97                 return True, d, nodes
98             while n > 1:
99                 f = self.spf[n]
100                 while n % f == 0:
101                     n //= f
102                 if N % f and f not in seen:
103                     seen.add(f)
104                     q.append((f, d + 1))
105         return False, -1, nodes
106
107     def world(self, N, r):
108         """tryb world: (sukces, glebokosc, |R|, liczba krawedzi GOOD)"""
109         if 2 * r == N or (N - r >= 2 and self.isp[N - r]):
110             return True, 0, 0, 0
111         q = deque([(r, 0)])
112         queued = {r}
113         R, esc, ok, dep = set(), 0, False, None
114         while q:
115             p, d = q.popleft()
116             if p in R:
117                 continue

```

```

118     n = N - p
119     if n >= 2 and self.isp[n]:
120         if not ok:
121             ok, dep = True, d
122         continue
123     R.add(p)
124     while n > 1:
125         f = self.spf[n]
126         while n % f == 0:
127             n //= f
128         if N % f:
129             if N - f >= 2 and self.isp[N - f]:
130                 esc += 1
131                 if not ok:
132                     ok, dep = True, d + 1
133                 elif f not in R and f not in queued:
134                     queued.add(f)
135                     q.append((f, d + 1))
136     return ok, dep, len(R), esc
137
138
139 # ===== EXP1: wyczerpujący skan p_0
140 def exp1_scan(limit):
141     """Kazde parzyste 4<=N<=limit, korzen kanoniczny, tryb census."""
142     e = Env(limit)
143     fails, tot, s = [], 0, 0
144     maxn, maxd, hist = (0, 0), (0, 0), Counter()
145     for N in range(4, limit + 1, 2):
146         r = e.p0(N)
147         ok, d, nodes = e.census(N, r)
148         tot += 1
149         s += nodes
150         if not ok:
151             fails.append((N, r))
152         else:
153             hist[d] += 1
154             if d > maxd[0]:
155                 maxd = (d, N)
156             if nodes > maxn[0]:
157                 maxn = (nodes, N)
158     return {"limit": limit, "cases": tot, "failures": len(fails),
159           "failure_list": fails[:20], "mean_nodes": s / tot,
160           "max_nodes": maxn, "max_depth": maxd,
161           "depth_hist": {str(k): hist[k] for k in sorted(hist)}}
162
163
164 # ===== EXP2: ablacja p_0 -> p_k (werdykt roota)
165 def exp2_ablation(limit, kmax=6):
166     e = Env(limit)
167     comp = [0] * (kmax + 1)
168     unit = [0] * (kmax + 1)
169     first = {}
170     for N in range(4, limit + 1, 2):
171         p = e.p0(N)
172         for k in range(kmax + 1):
173             if p >= N:
174                 break
175             ok, _, _ = e.census(N, p)
176             if not ok:
177                 if N - p == 1:
178                     unit[k] += 1
179                 else:
180                     comp[k] += 1
181                     first.setdefault(k, (N, p))
182         p = e.nextp(p)
183     return {"limit": limit, "kmax": kmax, "composite_complement_failures": comp,
184           "unit_row_failures": unit,
185           "first": {str(k): v for k, v in sorted(first.items())}}
186
187
188 # ===== EXP3: census WSZYSTKICH zawodzacych gornych korzeni
189 def exp3_upper_census(limit):
190     e = Env(limit)
191     rows, unit = [], 0
192     for N in range(4, limit + 1, 2):

```

```

193     r0 = e.p0(N)
194     p = r0
195     while p < N:
196         if e.isp[p] and N % p:
197             if N - p == 1:
198                 unit += 1
199             elif not e.census(N, p)[0]:
200                 rows.append((N, p, p == r0))
201             p += 1
202         while p < N and not e.isp[p]:
203             p += 1
204     Ns = sorted({r[0] for r in rows})
205     return {"limit": limit, "failed_roots": len(rows), "unit_rows": unit,
206           "N_values": Ns, "per_N": {str(n): sum(1 for r in rows if r[0] == n)
207                                     for n in Ns},
208           "canonical_among_them": sum(1 for r in rows if r[2])}
209
210
211 # ===== EXP4: margines ucieczki G_esc
212 def exp4_margin(limit):
213     e = Env(limit)
214     ones, sizes = [], []
215     hist = Counter()
216     maxW = (0, 0)
217     dyad = {}
218     for N in range(4, limit + 1, 2):
219         r = e.p0(N)
220         ok, _, W, esc = e.world(N, r)
221         if W == 0:
222             continue
223         hist[esc] += 1
224         sizes.append(W)
225         if W > maxW[0]:
226             maxW = (W, N)
227         if esc == 1:
228             ones.append((N, r, W))
229         if W >= 3:
230             dyad.setdefault(N.bit_length(), []).append(esc)
231     return {"limit": limit, "bad_p0_worlds": len(sizes),
232           "G_esc_hist_small": {str(k): hist[k] for k in sorted(hist) if k <= 6},
233           "margin_one_all": [o[0] for o in ones],
234           "margin_one_W_ge_3": [(N, r, W) for N, r, W in ones if W >= 3],
235           "max_world": maxW,
236           "dyadic_min_G": {str(k): min(v) for k, v in sorted(dyad.items())}}
237
238
239 # ===== EXP5: trichotomia fazowa i dwukierunkowa ranga rho_0
240 def exp5_phase(limit):
241     e = Env(limit)
242     primes = [n for n in range(2, limit + 1) if e.isp[n]]
243     idx = {p: i for i, p in enumerate(primes)}
244     types = Counter()
245     imm = Counter()
246     viol = Counter()
247     rho_hist = Counter()
248     rho_theta = {}
249     theta_hist = Counter()
250     for N in range(6, limit + 1, 2):
251         x = N // 2
252         P = e.p0(N)
253         h, m0 = P - x, N - P
254         Pm = e.prevp(P)
255         G = P - Pm
256         if m0 - Pm != G - 2 * h:
257             viol["identity"] += 1
258         t = "diag" if h == 0 else ("I" if 2 * h < G else
259                                 ("II" if 2 * h == G else "III"))
260         types[t] += 1
261         theta_hist[int(20 * h / G)] += 1
262         if e.isp[m0] or m0 == 1:
263             imm[t] += 1
264             if t == "I":
265                 viol["typeI_prime_m0"] += 1
266         if t == "II" and m0 != Pm:
267             viol["typeII"] += 1

```



```

268     if t == "III":
269         js, tt = h - G // 2, (h - 1) // 2
270         if not (1 <= js <= tt and m0 + 2 * js == Pm):
271             viol["typeIII_mirror"] += 1
272         if N % Pm == 0:
273             viol["typeIII_inactive"] += 1
274     if h >= 3:
275         tt = (h - 1) // 2
276         direct = sum(1 for j in range(1, tt + 1) if e.isp[m0 + 2 * j])
277         i0, f, r = idx[P], 0, 1
278         while i0 - r >= 0:
279             D = P - primes[i0 - r]
280             if D >= 2 * h:
281                 break
282             if D > h:
283                 f += 1
284             r += 1
285             if direct != f:
286                 viol["rho0"] += 1
287             rho_hist[direct] += 1
288             rho_theta.setdefault(round(h / G, 2), []).append(direct)
289     return {"limit": limit, "type_counts": dict(types),
290           "immediate_success_by_type": dict(imm),
291           "violations": dict(viol),
292           "theta_hist_20": {str(k): theta_hist[k] for k in sorted(theta_hist)},
293           "rho_hist": {str(k): rho_hist[k] for k in sorted(rho_hist)},
294           "rho0_mean_by_theta": {str(k): round(sum(v) / len(v), 4)
295                                for k, v in sorted(rho_theta.items())
296                                if len(v) >= 50}}
297
298
299 # ===== EXP6: filtracja brzegu (embedding)
300 def exp6_filtration(limit, kmax=6):
301     e = Env(limit)
302     checked = viol = 0
303     for N in range(50, limit + 1, 2):
304         x = N // 2
305         roots, p = [], e.p0(N)
306         for _ in range(kmax + 1):
307             if p >= N:
308                 break
309             roots.append(p)
310             p = e.nextp(p)
311         for k in range(len(roots) - 1):
312             pk, pk1 = roots[k], roots[k + 1]
313             s = (pk1 - pk) // 2
314             tk, tk1 = (pk - x - 1) // 2, (pk1 - x - 1) // 2
315             if tk1 != tk + s:
316                 viol += 1
317             for i in range(1, tk + 1):
318                 checked += 1
319                 if pk1 - 2 * (i + s) != pk - 2 * i:
320                     viol += 1
321     return {"limit": limit, "checked": checked, "violations": viol}
322
323
324 # ===== EXP7: fixed point B_N^low
325 def exp7_fixedpoint(limit):
326     e = Env(limit)
327     out = []
328     for N in range(6, limit + 1, 2):
329         cap = N // 3
330         B, rows = set(), {}
331         for p in range(3, cap + 1, 2):
332             if not e.isp[p] or N % p == 0:
333                 continue
334             v = N - p
335             if v < 2 or e.isp[v]:
336                 continue
337             B.add(p)
338             rows[p] = pf(v, e.spf)
339         changed = True
340         while changed:
341             changed = False
342             for p in list(B):

```

```

343         for q in rows[p]:
344             if N % q and q not in B:
345                 B.discard(p)
346                 changed = True
347                 break
348     if B:
349         out.append((N, sorted(B)))
350     return {"limit": limit, "nonempty_count": len(out),
351            "nonempty": [(N, B) for N, B in out]}
352
353
354 # ===== EXP8: ablacja obserwabli p_0 vs p_1
355 def exp8_observables(limit):
356     e = Env(limit)
357     acc = {k: {n: [] for n in ("depth", "world", "esc", "Omega", "ret", "kap")}}
358           | {"fail": 0, "n": 0, "pp": []} for k in (0, 1)}
359     for N in range(50, limit + 1, 2):
360         x = N // 2
361         r0 = e.p0(N)
362         r1 = e.nextp(r0)
363         for k, r in ((0, r0), (1, r1)):
364             if r >= N or N - r <= 1:
365                 continue
366             m = N - r
367             ok, dep, W, esc = e.world(N, r)
368             a = acc[k]
369             a["n"] += 1
370             if not ok:
371                 a["fail"] += 1
372             if dep is not None:
373                 a["depth"].append(dep)
374             a["world"].append(W)
375             a["esc"].append(esc)
376             a["Omega"].append(bigomega(m, e.spf))
377             Q = pf(m, e.spf)
378             Qs = set(Q)
379             a["pp"].append(max(Q) / m)
380             a["ret"].append(sum(1 for q in Q for z in pf(N - q, e.spf) if z in Qs))
381             hh = r - x
382             tt = (hh - 1) // 2
383             a["kap"].append(sum(1 for j in range(1, tt + 1) if e.isp[r - 2 * j]))
384
385     def st(v):
386         v = sorted(v)
387         return {"mean": round(sum(v) / len(v), 4), "median": v[len(v) // 2],
388                "max": v[-1]}
389     return {"limit": limit,
390            **{f"p{k}": {n: st(acc[k][n]) for n in
391                        ("depth", "world", "esc", "Omega", "ret", "kap")}}
392            | {"fail": acc[k]["fail"], "n": acc[k]["n"],
393              "Pplus_over_m_mean": round(sum(acc[k]["pp"]) / len(acc[k]["pp"]), 5),
394              "Omega_hist": {str(a): b for a, b in
395                           sorted(Counter(acc[k]["Omega"]).items())},
396              "depth_hist": {str(a): b for a, b in
397                           sorted(Counter(acc[k]["depth"]).items())}}
398            for k in (0, 1)}
399
400
401 # ===== pojedynczy swiadkowie
402 def witnesses(limit=1_100_000):
403     """Wszyscy pojedynczy swiadkowie cytowani w dossier."""
404     e = Env(limit)
405
406     def fac(n):
407         c = Counter(pf(n, e.spf))
408         d = Counter()
409         m = n
410         for q in c:
411             while m % q == 0:
412                 m //= q
413                 d[q] += 1
414         return dict(sorted(d.items()))
415
416     w = {}
417     for N in (124, 242, 240, 2200, 103344, 364844, 14198, 11822):

```

```

418 P = e.p0(N)
419 Pm = e.prevp(P)
420 h, G, m0 = P - N // 2, P - Pm, N - P
421 ok, dep, W, esc = e.world(N, P)
422 t = (h - 1) // 2
423 w[str(N)] = {"p0": P, "h": h, "G": G, "prev_prime": Pm, "m0": m0,
424             "m0_fac": {str(a): b for a, b in fac(m0).items()},
425             "type": "diag" if h == 0 else
426                     ("I" if 2 * h < G else
427                      ("II" if 2 * h == G else "III")),
428             "kappa_gen": sum(1 for j in range(1, t + 1)
429                               if e.isp[P - 2 * j]),
430             "rho0": sum(1 for j in range(1, t + 1) if e.isp[m0 + 2 * j]),
431             "verdict": "SUCCESS" if ok else "FAILURE",
432             "depth": dep, "world": W, "escapes": esc}
433 w["2200_core"] = {"2200-3": 2200 - 3, "13^3": 13 ** 3,
434                  "2200-13": 2200 - 13, "3^7": 3 ** 7,
435                  "shift": 3 ** 7 - 3 == 13 ** 3 - 13 == 2184}
436 N = 1_000_000
437 Q, R = (61, 67, 73), (113, 179, 599)
438 w["K33"] = {"N": N, "Q": Q, "R": R,
439             "all_entries_prime": all(e.isp[N - q * r] for q in Q for r in R),
440             "all_factor_vertices_bad": all(not e.isp[N - v] for v in Q + R)}
441 w["cells"] = [{"P": P, "G": P - e.prevp(P),
442               "all_h_have_p0_eq_P": all(e.p0(2 * (P - h)) == P
443                                           for h in range(0, P - e.prevp(P)))}
444               for P in (67, 127, 1009, 104729)]
445 return w
446
447
448 # ===== driver
449 def main():
450     ap = argparse.ArgumentParser()
451     ap.add_argument("--scan", type=int, default=1_000_000)
452     ap.add_argument("--ablation", type=int, default=200_000)
453     ap.add_argument("--census", type=int, default=20_000)
454     ap.add_argument("--margin", type=int, default=200_000)
455     ap.add_argument("--phase", type=int, default=200_000)
456     ap.add_argument("--filtration", type=int, default=20_000)
457     ap.add_argument("--fixedpoint", type=int, default=20_000)
458     ap.add_argument("--observables", type=int, default=200_000)
459     ap.add_argument("--json", default="dane_p0.json")
460     a = ap.parse_args()
461     out = {
462         "EXP1_scan": exp1_scan(a.scan),
463         "EXP2_ablation": exp2_ablation(a.ablation),
464         "EXP3_upper_census": exp3_upper_census(a.census),
465         "EXP4_margin": exp4_margin(a.margin),
466         "EXP5_phase": exp5_phase(a.phase),
467         "EXP6_filtration": exp6_filtration(a.filtration),
468         "EXP7_fixedpoint": exp7_fixedpoint(a.fixedpoint),
469         "EXP8_observables": exp8_observables(a.observables),
470         "witnesses": witnesses(),
471     }
472     with open(a.json, "w") as f:
473         json.dump(out, f, indent=1)
474     print(json.dumps({k: (v if k == "witnesses" else
475                          {kk: vv for kk, vv in v.items()
476                            if not isinstance(vv, (list, dict)) or len(str(vv)) < 400})
477                          for k, v in out.items()}}, indent=1)[:6000])
478
479
480 if __name__ == "__main__":
481     main()

```

A.2 The Euler-defect check

This script produced the refutations of normalisations (N1) and (N2) and the zero-violation results for (N3) and (N4) in §5.7. It is new in this file.

Listing A.2: `euler_check.py` — tests the four candidate normalisations of the layered Euler defect on all canonical failed worlds.

```

1  #!/usr/bin/env python3
2  """Test the four candidate normalisations of the layered Euler defect
3  on every canonical failed world with 50 <= N <= LIMIT."""
4  from collections import deque
5
6  LIM = 200001
7  spf = list(range(LIM))
8  i = 2
9  while i * i < LIM:
10     if spf[i] == i:
11         for j in range(i * i, LIM, i):
12             if spf[j] == j:
13                 spf[j] = i
14         i += 1
15
16
17 def fac(n):
18     f = {}
19     while n > 1:
20         p = spf[n]
21         e = 0
22         while n % p == 0:
23             n //= p
24             e += 1
25         f[p] = e
26     return f
27
28
29 def isp(n):
30     return n >= 2 and spf[n] == n
31
32
33 def world(N):
34     """Canonical stopped traversal, world mode.
35     Returns (R, rows, depth, success, root) or None when the root wins
36     immediately (diagonal case or N - p0 prime)."""
37     x = N // 2
38     r = x if isp(x) else None
39     if r is None:
40         p = x + 1
41         while not isp(p):
42             p += 1
43         r = p
44     if 2 * r == N:
45         return None
46     if isp(N - r):
47         return None
48     q = deque([(r, 0)])
49     seen = {r}
50     R, rows, depth = set(), {}, {}
51     ok = False
52     while q:
53         p, d = q.popleft()
54         if p in R:
55             continue
56         if isp(N - p):
57             ok = True
58             continue
59         R.add(p)
60         rows[p] = fac(N - p)
61         depth[p] = d
62         for c in sorted(rows[p]):
63             if N % c == 0:
64                 continue
65             if isp(N - c):
66                 ok = True
67                 continue
68             if c not in R and c not in seen:
69                 seen.add(c)
70                 q.append((c, d + 1))
71     return (R, rows, depth, ok, r)
72
73

```

```

74 bad = [0, 0, 0, 0]
75 first = [None, None, None, None]
76 tested = 0
77 for N in range(50, 60001, 2):
78     w = world(N)
79     if w is None:
80         continue
81     R, rows, depth, ok, r = w
82     if not R:
83         continue
84     tested += 1
85     nu = len(set(depth.values()))
86     # (N1) row-excess form
87     eta1 = sum(sum(rows[p].values()) - len([c for c in rows[p] if c in R])
88               for p in R)
89     # (N2) valuation / reuse split
90     d = {}
91     kap = {}
92     for p in R:
93         for q_, e_ in rows[p].items():
94             if N % q_ == 0:
95                 continue
96             d[q_] = d.get(q_, 0) + e_
97             kap[q_] = kap.get(q_, 0) + 1
98     eta2 = sum(d[q_] - kap[q_] for q_ in d)
99     nu2 = sum(kap[q_] - 1 for q_ in d)
100    # (N3) active-mass form ; (N4) trivial form
101    mass = sum(d.values())
102    eta4 = sum(sum(rows[p].values()) - 1 for p in R)
103    target = len(R) + 1
104    for k, val in enumerate((eta1 + nu, eta2 + nu2, mass, eta4 + nu)):
105        if val < target:
106            bad[k] += 1
107            if first[k] is None:
108                first[k] = (N, len(R), val, target)
109    print("worlds tested:", tested)
110    for k, name in enumerate(("N1", "N2", "N3", "N4")):
111        print(name, "violations:", bad[k], "first:", first[k])

```

A.3 Auxiliary scripts not embedded here

The project repository also contains the following, which are *not* embedded because no number in this file depends on them.

File	Role
p0_algorithm.py	Listing 3.1: the self-contained reference algorithm (embedded, §3.4).
p0_experiments.py	Listing A.1: the data generator (embedded above).
p0_core.py	Extended engine: both traversal modes, canonical geometry, reflected block, genealogical rank, D_q/M_q , FRESH/SMOOTH split, $B_{\text{abs}}/B_{\text{re}}$, CLI.
p0_certificates.py	The 34 hostile certificates of §12.4: regressions 11 822 and 3 542 948, counterexamples 242, 128, 322, 858, 4736, 6142, 14198, 20, 1718, 51 032, the Exact-D models, the $K_{3,3}$ grid and the 2-core.
p0_bigscan.py, p0_phase.py, p0_figures.py, p0_upper_root_census.py, p0_escape_margin.py	Supporting scripts for individual experiments; all of their outputs are reproducible from p0_experiments.py.

Appendix B

Exact Post-v4.81 Raw Checkpoint Transcript

This appendix is the lossless canonical payload of the September 5–7C continuation. It contains all 6420 source lines in original order. The standalone archive’s 5108-line payload is an exact prefix of this transcript, so it is not duplicated separately. Status labels and later-overrides-earlier semantics are source data, not editorial promotion by this merge.

```
1 GOLDBACH RESEARCH CONTINUATION - 2026-09-05
2 Single raw checkpoint. Work in progress; no Goldbach proof asserted.
3 NOVELTY of new statements: NOT YET AUDITED unless explicitly stated otherwise.
4
5 INPUT PROVENANCE
6 Latest checkpoint: upload/astra_checkpoint.txt, mathematical sections 1-10.
7 Recent reports: upload/astra_run_1.md (singleton-excess construction),
8 upload/astra_run_2.md (PR-MQ repair and global counting attempt).
9 Navigation: upload/goldbach_master_for_future_ai(5).tex.
10 Historical sources are being searched selectively, not certified wholesale.
11 Later corrections override earlier PROVED labels.
12
13 RESULT-ID: AUDIT-PREFIX-TRANSFER-20260905
14 STATUS: PROVED (independent proof audit of the supplied checkpoint, not a new claim).
15 EXACT STATEMENT:
16 Let  $3=P_0<P_1<\dots$  be the odd primes,  $X\geq 4$  an integer,  $I$  the first index
17 with  $P_I=X$ ,  $K\geq 0$  an integer,  $b_j=(P_{j-1}-1)/2$ ,  $L=b_{(I+K)}$ .
18 For  $x\geq 4$  let  $i$  satisfy  $P_{(i-1)}<x\leq P_i$ ,  $p_k(2x)=P_{(i+k)}$ ,
19  $m_k(2x)=2x-p_k(2x)$ . For any  $E$  of positive odd integers  $\leq X$ ,  $B=|E|$ ,
20  $T=\sum_{x=4..X} \sum_{k=0..K} 1_E(m_k(2x))$  satisfies
21  $T\leq 6(K+1) B (1+\log^+(L/B))$  when  $B>0$ , and  $T=0$  when  $B=0$ .
22 HYPOTHESES: No failure hypothesis;  $E$  may depend on  $X$  and  $K$ . Nonpositive
23 complements do not belong to  $E$ . The claim counts incidences, not just inputs.
24 PROOF:
25 Fix selected endpoint  $j$ . The disjoint source cells for ranks  $0..K$  unite,
26 after restricting to  $x\geq 4$ , into the portion of
27  $P_{\max}(0,j-K-1)<x\leq P_j$  that actually occurs. Dropping  $x\leq X$  enlarges it.
28 In output coordinate  $t=(m-1)/2$  the containing interval is
29  $J_j=(2a_j-b_j,b_j]$  intersect  $Z$ ,  $a_j=b_{\max}(0,j-K-1)$ .
30 Its core  $C_j=(a_j,b_j]$  has  $|J_j|=2|C_j|$ . Zero-length cores are absent.
31 Each integer belongs to at most  $K+1$  cores, since the cores are unions of
32 at most  $K+1$  consecutive disjoint elementary gaps  $(b_{(i-1)},b_i]$ . All are
33 inside  $\{1,\dots,L\}$ . With  $A=\{(m-1)/2:m\in E\}$ , use the uncentred integer-interval
34 maximal function  $M_A(t)=\sup_{J\text{ containing }t} |A\cap J|/|J|$ .
35 A longest-first disjoint selection of witnessing intervals for  $M_A>u$ ,
36  $0<u\leq 1$ , covers their union after tripling the retained intervals. Integer
37 intervals of cardinality  $n$  can be tripled to cardinality  $3n-2$ ; hence
38  $|M_A>u|\leq 3B/u$ . Only finitely many witnessing intervals exist: their
39 length is  $<B/u$  and each intersects the finite  $A$ .
40 Layer cake gives, on any set  $U$  of cardinality at most  $L$ ,
41  $\sum_U M_A \leq \int_0^1 \min(L, 3B/u) du$ 
42  $\leq 3B(1+\log^+(L/B))$ .
43 For every  $t$  in  $C_j$ ,  $|A\cap J_j|\leq 2|C_j|M_A(t)$ .
44 Average over  $C_j$ , sum  $j$ , and use core multiplicity  $K+1$ . This proves  $T$ .
45 HOSTILE CHECK:
46 The source cell is closed at its upper endpoint, so  $x$  prime is included.
47 Endpoint  $j=0$  never occurs for  $x\geq 4$ . Initial cells are truncated using  $P_0=3$ .
48 Truncating the last cell and excluding nonpositive complements only decrease  $T$ .
49 Grouping by endpoint is essential: summing separate  $k$  bounds would lose
50 another  $K+1$ . No prime-gap distribution enters the finite inequality.
51 DEPENDENCIES: Elementary maximal-interval covering only.
52 IMPLICATION:
53 The supplied density-zero transfer, fixed-rank normal-order deduction, and
54 growing direct-prefix no-go follow, subject to their stated standard
55 number-theoretic inputs. The squarefree-density proof also needs finite
56 residue equidistribution; density-zero transfer alone is insufficient.
57 RELATION TO GOLDBACH: No positive representation count. No inference about
58 factorization-graph traversal from the direct-prefix failure estimate.
59 RELATION TO  $p_0$  EXCEPTIONALITY: Shared by all fixed nearby roots.
60 WHAT REMAINS OPEN: Nonlocal rooted reachability and every-point positivity.
```

61
62 AUDIT NOTE: The supplied Scott-Bozek branch is not used as a classification.
63 No conclusion about its completeness is promoted without rechecking primary
64 sources. The global/rooted failure distinction is maintained throughout.
65
66 FAIL-ID: AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905
67 STATUS: SCOPE-CORRECTED; the blanket bounded-depth BFS no-go is NOT ESTABLISHED
68 by the supplied CRT argument. This is not a proof of a bounded-depth closer.
69 CLAIM / ROUTE:
70 Master theorem FBTU and its following finite-local wall (around lines
71 4320-4333), citing the post-CRD dossier, use programming of finite BAD
72 subtrees to rule out bounded-depth failure-dynamics arguments.
73 The dossier explicitly calls the object a BAD-subtree. It supplies no proof
74 that every unspecified prime factor of every expanded row is also BAD.
75 The p0 dossier marks the imported claims PENDING INDEPENDENT AUDIT.
76 EXACT FAILURE:
77 Prescribed divisibilities force selected edges, not an entire factor row.
78 A factor not named in a CRT transcript may be a GOOD child. A selected BAD
79 path or BAD cycle is an existential statement. Failure of the full BFS
80 through depth d is a universal statement about ALL reachable vertices at
81 depths $\leq d$. CRT compatibility alone does not justify that quantifier change.
82 COUNTEREXAMPLE / OBSTRUCTION:
83 $N=92$, canonical $P=47$, $h=1$ (46 composite, 47 prime).
84 $92-47=45=3^2 \cdot 5$; $92-3=89$ is prime.
85 $92-5=87=3 \cdot 29$; $92-29=63=3^2 \cdot 7$; $92-7=85=5 \cdot 17$.
86 Thus $47 \rightarrow 5 \rightarrow 29 \rightarrow 7$ is a selected all-BAD path and $5 \rightarrow 29 \rightarrow 7 \rightarrow 5$
87 is an all-BAD cycle, while the complete canonical traversal succeeds at
88 depth 1 via 3. Arbitrarily long BAD walks can be unrolled in this one world;
89 they do not give arbitrarily large shortest GOOD depth.
90 This example refutes the implication from a selected BAD transcript to a
91 full BAD traversal, not the existence of prescribed partial trees.
92 WHAT SURVIVES:
93 The finite congruence/valuation programming theorem with its actual
94 compatibility hypothesis. The historical FAIL-P0-FINITE-VALUATION-SIGNATURE
95 and reflection obstruction survive on those restricted domains.
96 DO NOT REPEAT:
97 Do not use FAIL-P0-FINITE-BAD-TRAVERSAL or the master finite-local wall to
98 claim that ALL complete finite BFS neighbourhoods can be reproduced by CRT.
99 Do not reverse this correction into a claimed p0 proof or into a claim that
100 arbitrarily deep complete BAD neighbourhoods do not exist.
101 POSSIBLE REPLACEMENT:
102 An exact theorem about complete neighbourhoods, controlling the residual
103 cofactors, or a correctly restricted theorem about selected edges only.
104
105 RESULT-ID: CRT-TRANSCRIPT-EXACT-SCOPE-20260905
106 STATUS: PROVED (precise repair of FACP, not a novelty claim).
107 EXACT STATEMENT / HYPOTHESES:
108 Fix finitely many integer affine forms $L_j(P)=a_j P+b_j$ and finitely many
109 requirements $v_l(L_j(P))=e$ or $v_l(L_j(P))>e$. Assume all conditions admit
110 one residue class $P \equiv A \pmod M$ with $\gcd(A,M)=1$; for an exact valuation include
111 $l^*(e+1)$ in M . Assume every form required to be composite has a specified
112 prime divisor and is positive and tends to infinity along the progression.
113 Then infinitely many primes P realize these conditions; every required
114 composite form is composite for all sufficiently large such P . These P
115 are canonical roots of $N=2(P-1)$, $h=1$.
116 PROOF:
117 The constraints are unions of residue classes for finite prime-power
118 moduli. Choose the assumed common reduced class $A \pmod M$. Dirichlet's
119 theorem supplies infinitely many primes in it. Exact valuations are decided
120 $\text{mod } l^*(e+1)$, lower valuations $\text{mod } l^e$. A growing positive form divisible
121 by a fixed prime eventually exceeds that prime. Finally $P-1$ is an even
122 composite integer for $P \geq 5$, proving canonicity. No assertion about the
123 unprescribed cofactors occurs in the proof.
124 HOSTILE CHECK:
125 Conditions involving the same prime are NOT independently programmable.
126 For a fixed-label edge $a \rightarrow b$ in an $h=1$ world one needs
127 $2P \equiv a+2 \pmod b$; if b also divides $P-2$, compatibility forces $a \equiv 2 \pmod b$.
128 If b divides $a+2$, the edge may instead force $P \equiv 0 \pmod b$, violating the
129 reduced-class hypothesis. Compatibility must be checked, not waved away.
130 DEPENDENCIES: CRT and Dirichlet's theorem on reduced progressions.
131 RELATION TO GOLDBACH: No global failure hypothesis or conclusion.
132 RELATION TO p0 EXCEPTIONALITY: Shows a precise class of partial local
133 constraints is compatible with p0; does not cover complete BFS closure.
134 WHAT REMAINS OPEN: Uniform control of all unprescribed child factors.
135
136 EXPERIMENT-ID: EXACT-WITNESS-N92
137 STATUS: VERIFIED COMPUTATION, with displayed arithmetic independently sufficient.
138 DOMAIN: one even $N=92$, active vertices 47,3,5,29,7,17; unit rows excluded.
139 METRIC: selected BAD path/cycle versus shortest path to a GOOD vertex.
140 METHOD: trial division through the integer square root, exact arithmetic.
141 RESULT: displayed rows above; additionally $92-17=75=3 \cdot 5^2$.
142 No exhaustive-range or asymptotic claim is attached to this witness.
143
144 EXTERNAL-INPUT: BS-1996
145 STATUS: PRIMARY STATEMENT VERIFIED.
146 F. Beukers and H. P. Schlickewei, The equation $x+y=1$ in finitely generated
147 groups, Acta Arithmetica 78 (1996), 189-199, Theorem 1.1.
148 Author-hosted full text: <https://web.archive.org/web/20070808100600/http://www.science.uu.nl/~beuke106/s-units.pdf>
149 (The available author PDF has a January 8, 2007 typesetting date.)
150 Exact input: in the $\mathbb{Q}$ -closure of a finitely generated subgroup of
151 $(\mathbb{C}^*)^2$ of rank r , $x+y=1$ has at most $2^{(8r+8)}$ solutions. In particular the
152 same bound holds in the subgroup itself. No height bound is asserted.
153 Only this input theorem is external; applications and rank bookkeeping follow.
154
155 RESULT-ID: COPRIME-SUPPORTED-AFFINE-COUNT-20260905
156 STATUS: PROVED.
157 EXACT STATEMENT:
158 Let S be a finite set of rational primes, $s=|S|$, and let A,B,C be fixed
159 nonzero rational numbers. The number of ordered pairs of positive coprime
160 S -smooth integers (u,v) satisfying $Au+Bv=C$ is at most $2^{(9s+16)}$.
161 Here 1 is allowed as an S -smooth integer. This bound is uniform in S,A,B,C .
162 PROOF:
163 Since $\gcd(u,v)=1$, choose a partition $S=S_1 \sqcup S_2$ such that
164 $PF(u)$ is contained in S_1 and $PF(v)$ in S_2 . An unused prime can be assigned
165 to either side. There are at most 2^s partitions covering all possibilities.
166 For a fixed partition take the subgroup of $(\mathbb{Q}^*)^2$ generated by

167 (1,1) for 1 in S₁, (1,1) for 1 in S₂, and the single vector (A/C,B/C).
168 Its rank is at most s+1, independently of all prime divisors of A,B,C.
169 The injective map (u,v)→(Au/C,Bv/C) lands in this group and satisfies
170 x+y=1. BS-1996 gives at most 2²(8s+16) solutions in this partition.
171 Summing over 2^s partitions proves the assertion.
172 HOSTILE CHECK:
173 The coefficient vector adds ONE group generator, not one generator per
174 prime dividing the coefficients. Signs are permitted in Q⁺; the group need
175 not consist of positive vectors. The Q-closure theorem bounds its subgroup.
176 Overcounting unused-prime assignments only enlarges the upper bound.
177 The coprimality hypothesis is essential to the s+1 rank accounting.
178 DEPENDENCIES: BS-1996. No Goldbach or primality-distribution assumption.
179 IMPLICATION: A usable uniform COUNT of coprime supported affine pairs,
180 with a common alphabet; it is not a bound on the size of a solution.
181 RELATION TO GOLDBACH: Auxiliary counting lemma; no positive prime-pair count.
182 RELATION TO p₀ EXCEPTIONALITY: None by itself.
183 WHAT REMAINS OPEN: A consumer forcing more than this many pairs in a shared
184 small alphabet, or controlling moving labels more strongly.
185 NOVELTY: NOT YET AUDITED; an elementary application of an established theorem.
186
187 RESULT-ID: COMPLETE-H1-TWO-ROW-FINITENESS-20260905
188 STATUS: PROVED.
189 EXACT STATEMENT:
190 Fix a finite set S of odd primes, s=|S|. There are at most
191 s * 2²(9s+16) even integers N with P=N/2+1 prime, m=N-P=2>1,
192 PF(m) subset S, and PF(N-q) subset S for q=min PF(m).
193 No assertion of complete traversal failure is assumed; N-q need not be BAD.
194 PROOF:
195 For each such N the selected q belongs to S. Write v=N-q=2m+2-q.
196 Any common prime divisor of m and v divides q-2 and is at least q by
197 minimality of q. Since 0<q-2<q, this is impossible. Hence gcd(m,v)=1.
198 Apply COPRIME-SUPPORTED-AFFINE-COUNT with (u,v)=(m,N-q) and
199 A=2, B=-1, C=q-2. The equation 2m-v=q-2 has nonzero right side.
200 For each of the s choices of q there are at most 2²(9s+16) possible pairs.
201 Each m determines N=2m+2 uniquely. Summing proves the bound.
202 HOSTILE CHECK:
203 Uses the COMPLETE supports of m AND N-q. Specifying only selected divisors
204 of these rows does not satisfy the hypotheses. The theorem allows m prime,
205 so restricting it to BAD roots only reduces the set. N-q is positive for
206 all admitted N, because q<=m<N. q is odd and q-2 is never zero.
207 DEPENDENCIES: Elementary h=1 minimum-factor orthogonality (reproved above)
208 and COPRIME-SUPPORTED-AFFINE-COUNT.
209 IMPLICATION:
210 A fixed finite alphabet cannot carry complete root and next-generation rows
211 for infinitely many canonical h=1 inputs, even when all exponents vary.
212 This gives a quantitative obstruction to the strong/full-support reading of
213 FBTU, beyond the elementary observation that fixing all factors AND all
214 exponents fixes N. Partial CRT programming survives.
215 RELATION TO HISTORICAL FAILURES:
216 The corpus already warns that fixed-S finiteness is not a moving-S proof.
217 We do NOT use this as a Goldbach closer. Its consumer here is a correction
218 to an infinite full-transcript assertion with genuinely FIXED labels.
219 RELATION TO GOLDBACH: Does not rule out a single failed world.
220 RELATION TO p₀ EXCEPTIONALITY: A real arithmetic restriction in the h=1
221 subfamily, but not a distinction between all p₀ and all nearby roots.
222 WHAT REMAINS OPEN: Complete BAD neighbourhoods with moving prime labels;
223 existence of arbitrarily large shortest GOOD depth for canonical roots.
224 NOVELTY: NOT YET AUDITED.
225
226 RESULT-ID: CLOSED-ALPHABET-COUNT-20260905
227 STATUS: PROVED.
228 EXACT STATEMENT:
229 Fix a finite set S of odd primes, s=|S|. Let C_S be the set of even N for
230 which there is a nonempty set C subset S of active primes (p does not divide
231 N), all N-p composite, with PF(N-p) subset C for every p in C.
232 Then |C_S| <= binomial(s,2) * 2²(9s+16), over ALL positive N.
233 PROOF:
234 A singleton C={a} is impossible: closure would give N=a=a^e and hence a|N,
235 contrary to activity. Let a<b be the two smallest elements of C.
236 The two complete rows U=N-a and V=N-b are coprime. Indeed, a common prime
237 1 belongs to C and divides U-V=b-a. It cannot be a because a does not
238 divide U by activity. Every other prime of C is at least b>b-a, also
239 impossible. Thus gcd(U,V)=1. Both rows are S-smooth and satisfy
240 U-V=b-a. For each of binomial(s,2) choices of a,b, apply
241 COPRIME-SUPPORTED-AFFINE-COUNT with A=1, B=-1, C=b-a.
242 Each U determines N=U+a. The union bound proves the claim.
243 HOSTILE CHECK:
244 No SCC minimality is needed; full factor closure and activity are needed.
245 The set C and all its valuations may vary with N. They all lie inside ONE
246 fixed S. Dropping composite-row or prime-root conditions in the counting
247 lemma is harmless because the count becomes larger. Singleton inactivity
248 is why trivial p|N components are excluded.
249 DEPENDENCIES: BS-1996; two-smallest-row coprimality, already in the corpus
250 and reproved here. No Scott/Bozek two-vertex classification is used.
251 IMPLICATION / NEW CONSUMER:
252 With S={odd primes <=y}, s=pi(y)-1, the total number of N admitting ANY
253 closed active BAD set supported below y is at most
254 binomial(s,2)*2²(9s+16), with no upper bound on N required.
255 Consequently, for y(X)=c log X log log X and fixed 0<c<1/(9 log 2),
256 #{N<=X: such a closed set exists with max C<=y(X)}
257 <= X^c (9c log 2+o(1))=o(X).
258 For each X, all moving supports in this event are contained in the SAME
259 alphabet S_y(X), so taking their union is justified. The prime number theorem
260 gives pi(y(X))=(c+o(1))log X. No union over arbitrary unbounded labels is made.
261 RELATION TO HISTORICAL FIXED-S WALL:
262 This does not repeat the invalid within-one-world capacity argument.
263 It counts DISTINCT N across a family sharing an explicitly bounded envelope.
264 It still gives no every-N exclusion and no height bound for exceptional N.
265 RELATION TO GOLDBACH:
266 Applies also to ordinary successful N containing bad components. Under global
267 Goldbach failure a closed BAD set exists, but nothing proved here forces
268 its labels below y(X). Hence this is not a Goldbach proof or a new estimate
269 for the full Goldbach exceptional set.
270 RELATION TO p₀ EXCEPTIONALITY: Global graph restriction; no selector dependence.
271 WHAT REMAINS OPEN: Large or moving support, and canonical access to it.
272 NOVELTY: NOT YET AUDITED; new application within this checkpoint, not a claim

273 that bounded-prime smooth-pair counting is new to number theory.
274
275 RESULT-ID: SMALL-BAD-CHILDREN-PREFIX-20260905
276 STATUS: PROVED.
277 EXACT STATEMENT:
278 Fix $0 < \alpha < 1/2$ and $0 < \epsilon < 1/2$. Put $z = (\log X)^\alpha$,
279 $s = \#\{\text{odd primes } q \leq z\}$, $\mu = \sum_{3 \leq q \leq z} \frac{1}{q}$.
280 Let $K = K(X) > 0$ be an integer with $K+1 = o(\mu)$, equivalently it suffices to
281 assume $K+1 = o(\log \log \log X)$. For all but $o(X)$ integers $4 \leq x \leq X$,
282 every upper root $p_k(2x)$, $0 \leq k \leq K$, is BAD and has at least
283 $(1-\epsilon)\mu$ distinct active BAD children $q \leq z$.
284 In particular the lower bound is at least
285 $(1-2\epsilon)\log \log \log X$ for all sufficiently large X when $0 < \epsilon < 1/2$.
286 This concerns actual vertices on the COMPLETE first frontier, not merely
287 an arbitrarily prescribed partial tree. It does NOT say that all children
288 are BAD or that shortest GOOD depth tends to infinity.
289
290 HYPOTHESES / CONVENTIONS:
291 BAD here means a positive composite complement, excluding unit rows.
292 All quantifiers over k are simultaneous on one set of x of density 1.
293 Logs are natural and X tends to infinity; no useful small- X rate is claimed.
294
295 PROOF:
296 1. Uniform arithmetic progression counts.
297 Let $H = X-3$ and J be the number of source prime cells meeting $[4, X]$, so
298 $J = O(X/\log X)$. For ANY $k > 0$ and ANY odd $d > 1$, as x runs across a source
299 cell, $m_k(2x)$ is an arithmetic progression with difference 2. Therefore
300 $A_d(k) = \#\{4 \leq x \leq X: d \text{ divides } m_k(2x)\} = H/d + O(J)$,
301 with absolute error at most J , uniform in d and k . Negative complements
302 can provisionally be counted here as integers; we remove them in step 4.
303 The final partial cell also has discrepancy at most 1.
304
305 2. Mean and variance on actual root complements.
306 Define $f_k(x) = \sum_{3 \leq q \leq z} \frac{1}{q} \cdot \mathbf{1}_{(q \text{ divides } m_k(2x))}$. Then
307 $\sum_x f_k = H\mu + O(Js)$,
308 $\sum_x f_k^2 = H(\mu^2 + \mu \sum_{1/q \leq 2} \frac{1}{q^2}) + O(Js^2)$.
309 The latter follows by expanding the square and using $A_{-}(qr)$ for distinct
310 q, r . It follows, without assuming independence in x , that
311 $\sum_x (f_k - \mu)^2 \leq H\mu + O(J(s^2 + \mu s))$.
312 Consequently
313 $\#\{x: f_k(x) < (1-\epsilon)\mu\}$
314 $\leq H/(\epsilon^2 \mu)$
315 $+ O(J(s^2 + \mu s)/(\epsilon^2 \mu^2))$.
316 All constants are independent of k . Taking a union over $0 \leq k \leq K$ is allowed.
317 The elementary bound $s \leq z$ and $\mu \leq \sum_{n \leq z} \frac{1}{n} = O(\log z)$ shows that
318 the second error, divided by X and multiplied by $K+1 = o(\mu)$, tends to 0:
319 $J/X = O(1/\log X)$ and $z^2 = (\log X)^{2\alpha}$ with $2\alpha < 1$.
320 The first error tends to 0 because $K+1 = o(\mu)$.
321
322 3. Every sufficiently small prime vertex is BAD on almost all inputs.
323 For each odd prime $q \leq z$, the condition $2x - q$ prime holds for at most $\pi(2X)$
324 values of x . Thus the number of x having ANY GOOD $q \leq z$ is at most
325 $s\pi(2X) = O(Xz/\log X) = o(X)$.
326 This is an ordinary counting upper bound, not an independence heuristic.
327 For $x > \sqrt{x}$ all $N - q > 1$ and exceed q , so rows avoiding the prime case are
328 composite. Also all roots exceed z .
329
330 4. Positivity, activity and root status.
331 The prime number theorem gives at least $c \cdot x/\log x$ primes in $[x, 2x]$ for all
332 sufficiently large x . Uniformly for $\sqrt{x} \leq x \leq X$ this number exceeds $K+1$,
333 since $K+1 = o(\mu) = O(\log z)$. Hence $p_k(2x) < 2x$ and $m_k(2x) > 0$ for all $k \leq K$
334 in this range. Removing $x \leq \sqrt{x}$ costs $o(X)$.
335 Outside the exceptional set in step 2, $f_k > 1$ eventually, so m_k is composite
336 and p_k is not diagonal. Because $x \leq p_k < 2x$, a nondiagonal prime p_k cannot
337 divide $2x$. For $q | m_k$, $q | 2x$ would imply $q | p_k$ and thus $q = p_k$, impossible
338 since $m_k < p_k$. All counted child primes are therefore active. Step 3 makes
339 their complete complements composite. This proves the assertion.
340
341 5. Growth of μ ; no normal-order theorem is needed here.
342 The Euler product over primes $\leq z$ contains all reciprocals $1/n$ for $n \leq z$,
343 so $\prod_{q \leq z} (1 - 1/q)^{-1} \geq \sum_{n \leq z} \frac{1}{n} \geq \log z$.
344 Taking logarithms and using $\sum_{q \leq z} \frac{1}{q} = \mu$ gives
345 $\sum_{q \leq z} \frac{1}{q} \geq \log \log z - O(1)$. Removing $q=2$ changes a constant.
346 Thus $\mu \geq \log \log \log X + \log \alpha - O(1)$, proving the asserted lower bound.
347 The standard sharper Mertens estimate is unnecessary for this lower bound.
348
349 HOSTILE CHECK:
350 The square expansion involves q^2 , never prime powers q^2 for the diagonal
351 term. Divisibility discrepancy is counted by SOURCE cells, whose number is
352 independent of k ; no unjustified prime-residue distribution is assumed.
353 Goodness is tested in $N - q$, not in the root complement. Step 3 is a union
354 bound over all small q , so no conditional-primality assumption is hidden.
355 The full frontier can still have a large GOOD child and succeed at depth 1.
356
357 DEPENDENCIES:
358 Exact source-cell geometry; elementary second moments and Euler product;
359 $\pi(t) = O(t/\log t)$ and the prime number theorem $\pi(t) \sim t/\log t$. The latter is
360 used only for uniform positivity of the first slowly growing prefix.
361
362 ACTUAL TRAVERSAL CONSUMERS:
363 (a) In world/exhaustion mode, which expands all reachable BAD vertices and
364 stops only each GOOD branch, $|R_N(p_k)| \geq 1 + (1-\epsilon)\mu$ simultaneously.
365 Each small BAD child of a BAD root is necessarily in that world even when
366 some other child is GOOD.
367 (b) In the reference algorithm the root's children are tested in increasing
368 prime order. Before finding a GOOD child or exhausting the root row, it
369 must perform at least $(1-\epsilon)\mu$ DISTINCT unsuccessful child-goodness
370 tests, because every child $\leq z$ precedes every possible GOOD child.
371 The same conclusion holds for a lazy FIFO BFS with children enqueued in
372 increasing order, counting tested vertices before first GOOD.
373 (c) This is NOT a lower bound of μ on expanded BAD rows before first escape
374 in an eager-child implementation: it may detect a large GOOD child while
375 still processing the root. The distinction is material.
376
377 RELATION TO GOLDBACH:
378 An algorithmic lower bound on a specific search, not a lack of prime pairs.

379 RELATION TO p0 EXCEPTIONALITY:
380 A nontrivial traversal statistic, but it is shared simultaneously by p0 and
381 a growing prefix of nearby roots. Slow early testing / a growing BAD first
382 front cannot by itself explain unique eventual success of p0.
383 WHAT REMAINS OPEN:
384 Control of LARGE child exits, complete BAD depth, and multigeneration
385 correlations. Nothing here forces either success or closed failure.
386 NOVELTY: NOT YET AUDITED.
387
388 SOURCE-CORRECTION ADDENDUM TO AUDIT-FBTU:
389 The older Biblia independently records the precise obstruction already:
390 iteration 108, immediately before theorem iteration108-support (around
391 lines 28288-28310 of upload/goldbach_biblia(1).tex), reports that CRT retained
392 up to eight prescribed BAD-path edges but additional prime factors let BFS
393 escape. It explicitly distinguishes retaining existing edges from preventing
394 new factors. Therefore the present audit repairs a later scope regression,
395 not the first discovery of cofactor leakage. The small N=92 certificate and
396 the complete-two-row finiteness bound provide additional exact witnesses.
397
398 HISTORICAL LEMMA RECOVERED FOR FURTHER IMPROVEMENT:
399 Biblia theorem iteration108-support (around lines 28317-28373) uses Matveev
400 to obtain a support-height bound, but assumes at least two nonbranching
401 rows, via $|R| \geq b+2$. The text correctly notes this excludes highly branched
402 worlds. The current continuation is checking whether that restriction can
403 be removed by choosing two arbitrary distinct rows and using max C itself
404 as the row-gap bound. This is a genuine changed hypothesis/consumer, not
405 an attempt to reuse generic fixed-S finiteness as a full Goldbach proof.
406
407 EXTERNAL-INPUT: MATVEEV-IN-BMS-9.4
408 STATUS: EXACT STATEMENT AND DERIVATION CHECKED IN PRIMARY RESEARCH SOURCE.
409 Y. Bugeaud, M. Mignotte, S. Siksek, Classical and modular approaches to
410 exponential Diophantine equations I. Fibonacci and Lucas perfect powers,
411 Annals of Mathematics 163 (2006), 969-1018, Theorem 9.4, real-field clause.
412 Author's full text, printed pp.16-17:
413 <https://samirsiksek.github.io/siksek.github.io/papers/fibannalsfinal.pdf>
414 The source derives this product-minus-one form from Matveev's Corollary 2.3.
415 For positive rational $\alpha_1, \dots, \alpha_t$ and integer $b_i, B = \max |b_i| \geq 1$,
416 $\Lambda = \prod \alpha_i^{-b_i} \neq 0$, and
417 $A_i = \max(h(\alpha_i), |\log \alpha_i|, 0.16)$, its $D=1$ real case gives
418 $\log |\Lambda| > -1.4 \cdot 30^{(t+3)} \cdot t^{(9/2)} \cdot (1 + \log B) \cdot \prod A_i$.
419 Here $h(a/b) = \log \max(|a|, |b|)$ for coprime integers $a, b, b > 0$.
420 We use this verified form, not an unverified stronger interpretation.
421
422 RESULT-ID: UNRESTRICTED-CLOSED-SUPPORT-HEIGHT-20260905
423 STATUS: PROVED WITH REPAIR / GENERALIZATION of Biblia iteration108-support.
424 EXACT STATEMENT:
425 Let $N \geq 8$ be even and let C be a nonempty set of active prime labels $\leq N$,
426 with every N -p composite and $PF(N-p)$ subset C for every p in C .
427 Put $s = |C|$, $M = \max C$, and
428 $A(C) = 1.4 \cdot 30^{(s+3)} \cdot s^{(9/2)} \cdot \prod_{q \in C} \log q$.
429 Then $s \geq 2$, all q in C are odd, and
430 $\log((N-M)/M)$
431 $< A(C) \cdot (1 + \log(\log N / \log 3))$. (H)
432 No assumption on the number of nonbranching rows, SCC minimality,
433 canonicity, or global Goldbach failure is made.
434
435 PROOF:
436 Activity excludes 2 because N is even. The singleton case is excluded as
437 in CLOSED-ALPHABET-COUNT. Choose any $a < b$ in C ; one may choose the two smallest.
438 Write $U = N - a$ and $V = N - b$. Closure implies U, V are C -smooth and $U > V > 0$.
439 Let $e_q = v_q(U) - v_q(V)$. Since $U, V \leq N$ and $q \geq 3$,
440 $|e_q| \leq \log N / \log q \leq \log N / \log 3$.
441 Omit zero coefficients if necessary. At least one remains, because $U! = V$.
442 For $\Lambda = U/V = 1 - (b-a)/(N-b) > 0$, use the preceding Matveev form with
443 $\alpha_i = q_i, A_i = \log q_i$ ($h(q_i) = \log q_i > 0.16$), $D=1$.
444 All logarithms of odd primes exceed 1. Thus adding back omitted primes
445 only enlarges the product; the factor $1.4 \cdot 30^{(t+3)} \cdot t^{(9/2)}$ increases with t .
446 The result is
447 $\log \Lambda > -A(C) \cdot (1 + \log(\log N / \log 3))$.
448 On the other hand $b - a < M$ and $N - b \geq N - M$, so
449 $\Lambda < M / (N - M)$.
450 Compare and negate to obtain (H).
451
452 HOSTILE CHECK:
453 Λ is PRODUCT-minus-one, precisely the verified external theorem's form.
454 No confusion with the logarithmic form is needed. The inequality remains
455 valid if $M > N/2$, but its left side then gives no information. This is an
456 explicit limitation, not an omitted hypothesis. No coprimality of U, V is
457 needed here, and no assumption about two small pure-power rows is used.
458 For $M \leq \sqrt{N}$, (H) immediately yields the familiar positive order-log N
459 lower bound on $A(C)$, without any branching restriction.
460
461 DEPENDENCIES: Exact full factor closure; the verified real rational Matveev
462 bound. No Scott classification, prime-gap bound, RH, or abc assumption.
463
464 RESULT-ID: EFFECTIVE-BOUNDED-CORE-THRESHOLD-20260905
465 STATUS: PROVED.
466 EXACT STATEMENT:
467 For real $y \geq 5$, put $s_y = \#\{\text{odd primes } \leq y\}$,
468 $A_y = 1.4 \cdot 30^{(s_y+3)} \cdot s_y^{(9/2)} \cdot \prod_{3 \leq q \leq y, \text{ prime}} \log q$,
469 $D_y = A_y + \log(2y)$,
470 $F(y) = \exp(2 D_y \log(2 D_y))$.
471 If an even N admits ANY nonempty closed active BAD set C with $\max C \leq y$,
472 then $N < F(y)$. Thus $N \geq F(y)$ excludes all such sets, regardless of their
473 branching and cardinality. F is explicit and effectively computable.
474
475 PROOF:
476 When $N \leq 2y$ the bound is immediate because $D_y > \log(2y)$ and $F(y) > 2y$.
477 When $N > 2y$, use (H), $A(C) \leq A_y$, $M \leq y$, and $t = \log N > 1$ to get
478 $t - \log(2y) < A_y \cdot (1 + \log t)$.
479 Indeed $\log((N-M)/M) \geq \log((N-y)/y) > t - \log(2y)$, while $\log 3 > 1$
480 implies $\log(t / \log 3) < \log t$. Consequently $t < D_y \cdot (1 + \log t)$.
481 For any $D \geq 1$, the inequality $t < D(1 + \log t)$, $t > 1$, implies
482 $t < 2D \log(2D)$.
483 To verify it, put $t_0 = 2D \log(2D)$, $u = \log(2D) > 0$. Then
484 $t_0/D - (1 + \log t_0) = u - 1 - \log u \geq 0$.

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485 The function  $t/D-1-\log t$  is strictly increasing for  $t>D$ , and  $t_0>D$ .
486 Thus  $t>t_0$  is impossible. Apply this with  $D=D_y$  and exponentiate.
487
488 HOSTILE CHECK:
489  $A_y$  is a uniform envelope for ALL choices of  $C$  inside primes  $\leq y$ .
490 The bound is enormous and is not a practical census endpoint even for small  $y$ .
491 It bounds  $N$  for a specified maximum label  $y$ ; it is NOT a Goldbach  $N_0$ .
492 It supplies the height control that BS counting alone does not provide.
493
494 RELATION TO HISTORICAL FAILURES:
495 The old Matveev support lemma required  $|R|>b+2$  to obtain two rows with
496 labels  $\leq \sqrt{N}$ . This proof removes that condition: it uses arbitrary
497 distinct rows and max  $C$  as their gap bound. If max  $C$  is already large,
498 the desired small-support exclusion is automatic. The old warning about
499 huge constants and moving/unbounded support still applies.
500
501 RESULT-ID: POINTWISE-CORE-LABEL-GROWTH-20260905
502 STATUS: PROVED (asymptotic consequence, no assertion of novelty).
503 EXACT STATEMENT:
504 Uniformly along any sequence  $N \rightarrow \infty$  possessing a nonempty closed
505 active BAD set  $C$ , its largest prime label  $M$  obeys
506  $M \geq (1-o(1)) \cdot (\log \log N)(\log \log \log N)/(\log \log \log \log N)$ . (LG)
507 The iterated logarithms are used only beyond their positivity thresholds.
508
509
510 PROOF:
511 The effective theorem first implies  $M \rightarrow \infty$  along such a sequence.
512 The prime number theorem gives  $s_y = (1+o(1))y/\log y$ . Therefore
513  $\log A_y$ 
514  $\leq s_y(\log 30 + \log \log y) + O(\log y)$ 
515  $\leq (1+o(1))y \log \log y / \log y$ .
516 Since  $D_y = A_y + \log(2y)$ , the same main upper bound holds for  $\log D_y$ ,
517 and also for  $\log(2 D_y \log(2 D_y))$ , the extra term  $\log \log D_y$  being
518 lower order. Apply  $N < F(M)$ , take log twice, and conclude
519  $\log \log N \leq (1+o(1))M \log \log M / \log M$ .
520 The function  $g(u) = u \log \log u / \log u$  is increasing for sufficiently large  $u$ .
521 Its asymptotic inverse is  $v \log v / \log \log v$ , since direct substitution
522 shows  $g(v \log v / \log \log v) \sim v$ . This proves (LG).
523
524 IMPLICATION / SCOPE FOR THE THREE HEIGHT RESULTS:
525 These constrain every genuine closed BAD set, including one arising from
526 rooted failure in an otherwise Goldbach-successful  $N$ . A full failed rooted
527 world has a terminal SCC; all its vertices are factors of composite odd
528 rows and hence  $< N/3$ . Apply the same results to this terminal core, so the
529 bound need not be vacuously satisfied just by the large starting root.
530 The existence of a terminal SCC follows from finite graph closure; it has
531 at least two vertices by activity. No classification of such SCCs is used.
532 RELATION TO GOLDBACH:
533 Global Goldbach failure forces a closed BAD component, but its labels can
534 satisfy (LG). The result is a necessary condition, not a contradiction.
535 RELATION TO  $p_0$  EXCEPTIONALITY:
536 Shared by all failed traversals. No canonical asymmetry is proved.
537 WHAT REMAINS OPEN:
538 An upper restriction incompatible with this lower growth, or a direct
539 arithmetic escape theorem for the moving support. We have neither.
540 NOVELTY: NOT YET AUDITED. Classical logarithmic-form methods are the source;
541 the advance over the supplied lemma is the removal of its branching hypothesis
542 and an explicit common-envelope threshold with correct scope.
543
544 RESULT-ID: H1-TWO-GENERATION-ENVELOPE-ESCAPE-20260905
545 STATUS: PROVED.
546 EXACT STATEMENT:
547 For  $y \geq 5$  let  $s_y = \#\{\text{odd primes } \leq y\}$  and define
548  $A_y \sim 1.4 \cdot 30^{s_y+4} \cdot (s_y+1)^{(9/2)} \cdot \log 2^*$ 
549  $\text{product}_{(3 \leq l \leq y, \text{ prime})} \log l$ ,
550  $D_y \sim 2 \cdot A_y + \log(2y)$ ,
551  $F_2(y) = \exp(2 \cdot D_y + \log(2 \cdot D_y))$ .
552 Let  $N = 2(P-1)$ ,  $P \geq 5$  prime, with  $m = P-2$  composite and  $PF(m)$  contained in
553  $\{\text{odd primes } \leq y\}$ . Put  $q = \min PF(m)$ . If  $N \geq F_2(y)$ , then
554  $P \sim (N-q) \cdot y$ . (E2)
555 Thus either  $q$  itself is GOOD, giving success at depth 1, or the stopped
556 traversal from canonical  $P$  reaches a prime  $r \cdot y$  at depth 2.
557 No global failure or full rooted failure is assumed.
558
559 PROOF:
560 Suppose instead that both complete rows  $m$  and  $v = N - q$  are supported on odd
561 primes  $\leq y$ .  $q \leq y$  and  $v = 2m + 2 - q$ , so
562  $\text{Lambda} = 2m/v - 1 = (q-2)/v > 0$ .
563 For  $N \geq 2y$ ,  $\text{Lambda} < 2y/N$ . ( $N \geq F_2(y)$  implies  $N \geq 2y$ .)
564 The expression  $2m/v$  has a nonzero exponent 1 on base 2, and its other bases
565 are odd primes  $\leq y$ . Their exponent differences have absolute value at most
566  $\log N / \log 3$ ; all exponents are bounded by  $2 \log N$ .
567 Apply the verified rational real Matveev form to these bases. Base 2 has
568 height  $\log 2 > 0.16$  and is always present. Omitted odd bases may be padded since
569 their logarithms exceed 1. Hence, with  $t = \log N$ ,
570  $\log \text{Lambda} > -A_y \sim (2 + \log t)$ ,
571 where  $1 + \log(2 \log N) < 2 + \log t$  has been used.
572 Combining bounds gives
573  $t < \log(2y) + A_y \sim (2 + \log t) \leq D_y \sim (1 + \log t)$ .
574 The inversion proved above implies  $N < F_2(y)$ , a contradiction. This proves E2.
575 If  $q$  is BAD, every prime  $r|N-q$  is an actual child and is active:  $r|N$  would
576 force  $r|q$ , hence  $r=q$  and  $q|N$ , already impossible because  $q|m$  and
577  $N = 2m + 2$ . Consequently a factor  $r \cdot y$  is reached at depth 2. If  $q$  is GOOD, the
578 stopped traversal stops that branch at  $q$  and has already succeeded.
579
580 HOSTILE CHECK:
581 The large  $r$  is a factor of the COMPLETE row  $N-q$ , not a residual quotient
582 obtained by gcd removal. It is genuinely outside the WHOLE prescribed
583 prime envelope, not just a surplus valuation of an old prime. A GOOD  $q$  is
584 handled before claiming that its children are traversed. A large reached
585 prime  $r$  need not be GOOD; no Goldbach conclusion follows in the BAD- $q$  branch.
586 The finite  $h=1$  transcript can still be prescribed by CRT: this theorem
587 forces some of its previously uncontrolled cofactors to carry larger primes.
588
589 DEPENDENCIES: Matveev-in-BMS-9.4, exact  $h=1$  arithmetic, elementary inversion.
590 RELATION TO HISTORICAL FAILURES:

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591 This is an actual escape consumer, so the fixed-alphabet finiteness warning
592 does not invalidate it. It still cannot iterate with a fixed useful bound,
593 since the new alphabet may grow and its threshold changes. It does not
594 revive the failed collision-to-new-support argument PR-MQ.
595 RELATION TO GOLDBACH: Success OR a provable enlarged reachable alphabet.
596 RELATION TO p0 EXCEPTIONALITY:
597 A concrete reachability theorem in the canonical h=1 subfamily. We have not
598 shown a corresponding property fails at p1 for the same N, nor covered all
599 canonical gaps. It therefore is not an explanation of universal p0 success.
600 WHAT REMAINS OPEN: A mechanism that turns repeated necessary envelope escape
601 into GOOD reachability, with control that survives the changing envelope.
602 NOVELTY: NOT YET AUDITED.
603
604 EXTERNAL CLASSICAL INPUT CHECKS
605 PNT:  $\pi(t) \sim t / \log t$ , Hadamard and de la Vallee Poussin (1896).
606 Statement and a complete proof checked in D. Zagier, Newman's Short Proof
607 of the Prime Number Theorem, American Mathematical Monthly 104 (1997),
608 705-708: https://sites.math.rutgers.edu/~zeilberg/EM18/zagier.pdf
609 The upper bound  $\pi(t) = O(t / \log t)$  and the interval consequence
610  $\pi(2t) - \pi(t) \sim t / \log t$  follow directly. No error term or AP uniformity is used.
611 Dirichlet (1837): if  $\gcd(a, m) = 1$ ,  $m \geq 1$ , infinitely many primes are
612  $a \pmod m$ . The exact statement is independently given in Paul Pollack,
613 Hypothesis H and an impossibility theorem, Rendiconti del Seminario
614 Matematico Universita Politecnico Torino 68 (2010), starting at p.183:
615 https://seminariomatematico.polito.it/rendiconti/68-2/183.pdf
616 No theorem about several simultaneously prime affine forms is used.
617
618 EXPERIMENT-ID: FINITE-CELL-AND-CLOSED-WORLD-STRESS-20260905
619 STATUS: VERIFIED COMPUTATION.
620 DOMAIN:
621 All integer  $4 < x \leq 200000$  for the residue and moment checks; ranks
622  $k=0,1,2,5,20$ ; small-prime cutoffs  $z=7,31,101$  (15 configurations).
623 These  $z$  are diagnostics of the finite identities; they are not an assertion
624 that the asymptotic scale  $z = (\log X)^\alpha$  is numerically informative here.
625 The residue/moment identities include negative complements. The graph
626 consumer excludes nonpositive complements,  $x$  below  $\sqrt{x}$ , and inputs
627 with a GOOD prime  $\leq z$ . Unit rows contribute zero to small factor counts.
628 For graph closure, exactly the six previously supplied  $N$  were reconstructed:
629 128,1718,1862,1928,2200,6142; this is NOT a scan for all  $N \leq 6142$  or beyond.
630 METRICS:
631 Exact residue discrepancy for every  $q$  and pair-product  $q*r$ ; second moment;
632 actual BAD-child count; maximal low closed set by deletion to a fixed point;
633 terminal SCCs; smallest-row gcd; pure-row count; and complete stopped BFS
634 from ranks 0,1,2. Pure row means exactly one distinct prime factor.
635 CHECKS:
636 For all residue counts, the integer certificate  $|d*A_d - H| \leq d*J$  passed.
637 The explicit second-moment upper bound
638  $H * \mu + J * (s^2 + 2 * \mu * s)$  passed in all 15 cases. On every eligible input
639 the actual number of small BAD children equalled the small-divisor count.
640 Every reconstructed terminal component was closed, active, entirely BAD,
641 and its two smallest rows were coprime. All assertions passed.
642 EXTREMAL / RELEVANT WITNESSES:
643  $N=128$ : terminal size=8, pure rows=2, max label=41;  $C=[3, 5, 7, 11, 13, 23, 29, 41]$ .
644  $N=1718$ : terminal size=28, pure rows=1, max label=571;  $C=[3, 5, 7, 11, 13, 17, 29, 31, 37, 41, 43, 53, 59, 67, 79, 89, 127, 137, 149, 181, 211, 239,$ 
645  $\hookrightarrow 241, 383, 523, 563, 569, 571]$ .
646  $N=1862$ : terminal size=21, pure rows=1, max label=619;  $C=[3, 5, 11, 13, 17, 41, 43, 47, 53, 67, 83, 107, 113, 167, 179, 251, 359, 593, 607, 617,$ 
647  $\hookrightarrow 619]$ .
648  $N=1928$ : terminal size=14, pure rows=0, max label=641;  $C=[3, 5, 7, 11, 13, 17, 53, 71, 73, 103, 113, 383, 619, 641]$ .
649  $N=2200$ : terminal size=2, pure rows=2, max label=13;  $C=[3, 13]$ .
650  $N=6142$ : terminal size=10, pure rows=0, max label=877;  $C=[3, 5, 7, 13, 17, 19, 157, 227, 409, 877]$ .
651 The components at  $N=1928$  and 6142 have ZERO pure rows. They show the
652 branching hypothesis removed from the old Matveev lemma is a real domain
653 extension. They satisfy the new necessary height inequality, with enormous
654 slack; they do not refute any global-Goldbach-failure theorem.
655 Maximum finite residue discrepancy among the recorded configurations: 5335.66666667; source-cell upper bound  $J=17983$ .
656 Maximum recorded small BAD-child count: 5.
657 All asymptotic and all- $N$  assertions rely on the written proofs.
658 Floating-point logarithmic comparisons in the code illustrate the size
659 of the Matveev margin and do not constitute a rigorous numerical theorem.
660
661 REPRODUCIBLE CODE (single-file Python, requires NumPy)
662 Save the following code separately only when reproducing the diagnostics.
663 BEGIN DIAGNOSTIC CODE
664 from math import isqrt, gcd, log
665 from collections import deque
666 import json
667 from pathlib import Path
668 import numpy as np
669
670 X=200000
671 limit=2*X+2000
672 isp=np.ones(limit+1,dtype=bool);isp[:2]=False
673 for p in range(2,isqrt(limit)+1):
674     if isp[p]:isp[p*p:] = False
675 ps=np.flatnonzero(isp)
676 spf=np.arange(limit+1,dtype=np.int64)
677 for p in ps[ps<=isqrt(limit)]:
678     sl=spf[int(p)*int(p):int(p)]
679     mask=sl==np.arange(int(p)*int(p),limit+1,int(p))
680     sl[mask]=p
681
682 def fac(n):
683     out={}
684     while n>1:
685         p=int(spf[n]);out[p]=out.get(p,0)+1;n//=p
686     return out
687
688 def world(N,r):
689     if 2*r==N:return {'depth':0,'R':[],'good':[r]}
690     todo=deque([(r,0)]);seen={r};R=[];good=[];first=None
691     while todo:
692         p,d=todo.popleft()
693         if isp[N-p]:
694             good.append(p)
695             if first is None:first=d

```

```

694         continue
695     assert N>p>1
696     R.append(p)
697     for q in fac(N-p):
698         assert N%q!=0
699         if q not in seen:seen.add(q);todo.append((q,d+1))
700     return {'depth':first,'R':sorted(R),'good':sorted(good)}
701
702 def tarjan(adj):
703     index={} ; low={} ; stack=[] ; active=set() ; comps=[]
704     def go(v):
705         index[v]=low[v]=len(index);stack.append(v);active.add(v)
706         for w in adj[v]:
707             if w not in index:go(w);low[v]=min(low[v],low[w])
708             elif w in active:low[v]=min(low[v],index[w])
709         if low[v]==index[v]:
710             comp=[]
711             while True:
712                 w=stack.pop();active.remove(w);comp.append(w)
713                 if w==v:break
714             comps.append(sorted(comp))
715     for v in adj:
716         if v not in index:go(v)
717     return comps
718
719 xs=np.arange(4,X+1,dtype=np.int64);ii=np.searchsorted(ps,xs)
720 H=len(xs);J=len(np.unique(ii));res=[]
721 for k in [0,1,2,5,20]:
722     ms=2*xs-ps[ii+k]
723     for z in [7,31,101]:
724         small=[int(q) for q in ps if 3<=q<=z]
725         s=len(small);mu=sum(1/q for q in small)
726         f=np.zeros(H,dtype=np.int64)
727         maxdisc=0
728         for j,q in enumerate(small):
729             f+=(ms%q==0)
730             ds=[q]*[q*r for r in small[:j]]
731             for d in ds:
732                 count=int((ms%d==0).sum())
733                 # Exact integer discrepancy certificate, no floating point.
734                 assert abs(d*count-H)<=d*J
735                 maxdisc=max(maxdisc,abs(count-H/d))
736             variance=float(((f-mu)**2).sum())
737             variance_bound=H*mu+J*(s*s+2*mu*s)
738             assert variance<=variance_bound+1e-7
739             pos=ms>0
740             smallgood=np.zeros(H,dtype=bool)
741             smallbad_count=np.zeros(H,dtype=np.int64)
742             for q in small:
743                 good=isp[2*xs-q] if 2*xs[0]-q>=0 else np.array([2*int(x)-q>=2 and bool(isp[2*int(x)-q]) for x in xs])
744                 smallgood|=good
745                 smallbad_count+=(pos & (ms%q==0) & (2*xs>2*q) & ~good)
746             eligible=pos & (xs==isqrt(X)+1) & ~smallgood
747             assert np.all(smallbad_count[eligible]==f[eligible])
748             res.append({'k':k,'z':z,'nonpositive':int((-pos).sum()),
749                 'mu':mu,'actual_mean':float(f.mean()),'actual_variance':variance/H,
750                 'max_residue_discrepancy':maxdisc,'source_cell_bound':J,
751                 'any_small_good_inputs':int(smallgood.sum()),
752                 'eligible_graph_bridge_inputs':int(eligible.sum()),
753                 'max_small_bad_children':int(smallbad_count.max())})
754
755 cores=[]
756 for N in [128,1718,1862,1928,2200,6142]:
757     B={int(p) for p in ps if 3<=p<N/3 and N%int(p)!=0 and not isp[N-int(p)]}
758     while True:
759         nxt={p for p in B if set(fac(N-p))<=B}
760         if nxt==B:break
761         B=nxt
762     adj={p:set(fac(N-p)) for p in B}
763     comps=tarjan(adj)
764     terminal=[c for c in comps if all(adj[p]<=set(c) for p in c)]
765     out=[]
766     for C in terminal:
767         a,b=C[:2]
768         assert gcd(N-a,N-b)==1
769         rows={p:fac(N-p) for p in C}
770         assert all(set(row)<=set(C) and sum(row.values())>=2 for row in rows.values())
771         pure=sum(len(row)==1 for row in rows.values())
772         M=max(C);s=len(C)
773         logA=log(1.4)+(s+3)*log(30)+4.5*log(s)+sum(log(log(q)) for q in C)
774         lhs=log((N-M)/M)
775         # Logarithmic comparison avoids overflows and is diagnostic only.
776         assert log(lhs)<logA+log(1+log(log(N)/log(3)))
777         out.append({'C':C,'size':s,'pure_rows':pure,'M':M,'two_min_rows_gcd':gcd(N-a,N-b),
778             'height_lhs':lhs,'log10_A':logA/log(10),'rows':rows})
779     root_index=int(np.searchsorted(ps,N//2))
780     roots=[]
781     for k in range(3):
782         p=int(ps[root_index+k]);w=world(N,p)
783         roots.append({'k':k,'p':p,'m_factors':fac(N-p),'depth':w['depth'],'R_size':len(w['R'])})
784     cores.append({'N':N,'B_size':len(B),'terminal':out,'roots':roots})
785
786 n92=world(92,47)
787 assert n92['depth']==1
788 for a,b in [(47,5),(5,29),(29,7),(7,5)]:
789     assert (92-a)%b==0 and not isp[92-a] and not isp[92-b]
790 output={'domain':{'x_min':4,'x_max':X,'ranks':[0,1,2,5,20],
791     'z_values':[7,31,101],'scope':'exhaustive residue/moment checks at these finite parameters; not an asymptotic census'},
792     'moment_checks':res,'known_closed_worlds':cores,
793     'N92':{'root':47,'shortest_good_depth':n92['depth'],'R':n92['R'],'good':n92['good']},
794     'checks':'all assertions passed'}
795 Path('/workspace/scratch/db0c19a2d4d9/checkpoint_diagnostics.json').write_text(json.dumps(output,indent=2)+'\n')
796 print(json.dumps({'checks':output['checks'],'moment_cases':len(res),
797     'known_worlds':[{'N':c['N'],'B_size':c['B_size'],'terminal':[key:t[key] for key in ['size','pure_rows','M','log10_A']] for t in
    ↪ c['terminal']], 'roots':c['roots']} for c in cores),

```

```

798         'N92':output['N92']],indent=2))
799
800 END DIAGNOSTIC CODE
801
802 COMPLETE DIAGNOSTIC OUTPUT
803 {
804     "domain": {
805         "x_min": 4,
806         "x_max": 200000,
807         "ranks": [
808             0,
809             1,
810             2,
811             5,
812             20
813         ],
814         "z_values": [
815             7,
816             31,
817             101
818         ],
819         "scope": "exhaustive residue/moment checks at these finite parameters; not an asymptotic census"
820     },
821     "moment_checks": [
822         {
823             "k": 0,
824             "z": 7,
825             "nonpositive": 0,
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833             "max_small_bad_children": 3
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835         {
836             "k": 0,
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841             "actual_variance": 0.900648678697622,
842             "max_residue_discrepancy": 5335.66666666666715,
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855             "max_residue_discrepancy": 5335.66666666666715,
856             "source_cell_bound": 17983,
857             "any_small_good_inputs": 197552,
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861         {
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883             "any_small_good_inputs": 171998,
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900         {
901             "k": 2,
902             "z": 7,
903             "nonpositive": 4,

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911     "max_small_bad_children": 3
912 },
913 {
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921     "source_cell_bound": 17983,
922     "any_small_good_inputs": 171998,
923     "eligible_graph_bridge_inputs": 27998,
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938 },
939 {
940     "k": 5,
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949     "eligible_graph_bridge_inputs": 109748,
950     "max_small_bad_children": 3
951 },
952 {
953     "k": 5,
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957     "actual_mean": 1.0651409771146567,
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977 },
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1017 ],
1018 "known_closed_worlds": [
1019   {
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1021     "B_size": 10,
1022     "terminal": [
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1024         "C": [
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1026           5,
1027           7,
1028           11,
1029           13,
1030           23,
1031           29,
1032           41
1033         ],
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1036         "M": 41,
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1119      29,
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1121      37,
1122      41,
1123      43,
1124      53,
1125      59,
1126      67,
1127      79,
1128      89,
1129      127,
1130      137,
1131      149,
1132      181,
1133      211,
1134      239,
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1137      523,
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1139      569,
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1260     },
1261     "569": {
1262         "3": 1,
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1265     "571": {
1266         "31": 1,
1267         "37": 1
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1269 }
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1272 {
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1276         "859": 1
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1280 },
1281 {
1282     "k": 1,
1283     "p": 863,
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1288     },
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1290     "R_size": 29
1291 },
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1300 }
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1302 },
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1306     "terminal": [
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1310                 5,
1311                 11,
1312                 13,
1313                 17,
1314                 41,
1315                 43,
1316                 47,
1317                 53,
1318                 67,
1319                 83,
1320                 107,
1321                 113,
1322                 167,
1323                 179,
1324                 251,
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1326                 593,

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1477         17,
1478         53,
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1481         103,
1482         113,
1483         383,
1484         619,
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1559 ],
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1600                 13
1601             ],
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1660         19,
1661         157,
1662         227,
1663         409,
1664         877
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1718           "13": 1
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1722   ],
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1724     {
1725       "k": 0,
1726       "p": 3079,
1727       "m_factors": {
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1729         "1021": 1
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1737       "m_factors": {
1738         "7": 1,
1739         "19": 1,
1740         "23": 1
1741       },
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1744     },
1745     {
1746       "k": 2,
1747       "p": 3089,
1748       "m_factors": {
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1750         "71": 1
1751       },

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1752     "depth": null,
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1754   }
1755 ]
1756 }
1757 ],
1758 "N92": {
1759   "root": 47,
1760   "shortest_good_depth": 1,
1761   "R": [
1762     5,
1763     7,
1764     17,
1765     29,
1766     47
1767   ],
1768   "good": [
1769     3
1770   ]
1771 },
1772 "checks": "all assertions passed"
1773 }
1774
1775 RESULT-ID: H1-ALL-BRANCH-INJECTIVE-ESCAPE-20260905
1776 STATUS: PROVED (strict strengthening of the preceding two-generation theorem).
1777 EXACT STATEMENT:
1778 Keep the hypotheses and explicit  $F_2(y)$  of H1-TWO-GENERATION-ENVELOPE-ESCAPE:
1779  $N=2(P-1)$ ,  $P$  prime,  $m=P-2$  composite,  $Q=PF(m)$  subset {odd primes  $\leq y$ },
1780 and  $N>=F_2(y)$ . Then EVERY  $q$  in  $Q$  satisfies  $P^+(N-q)>y$ .
1781 Moreover the sets
1782  $L_q=\{r>y: r \text{ prime and } r \text{ divides } N-q\}$ ,  $q$  in  $Q$ ,
1783 are all nonempty and pairwise disjoint.
1784 Consequently either the canonical traversal succeeds at depth  $\leq 1$ , or its
1785 actual depth-2 frontier contains at least  $|Q|$  DISTINCT prime labels  $>y$ .
1786 If any of those new labels is GOOD, the traversal succeeds at depth  $\leq 2$ .
1787
1788 PROOF:
1789 Inspect the previous full proof. Its use of  $q=\min PF(m)$  was unnecessary:
1790 all it needed was  $q$  in  $Q$ ,  $3\leq q\leq y$ , and  $v=N-q=2m+2-q$ .
1791 The same nonzero form  $\Lambda=2m/v-1=(q-2)/v$  has the same uniform bound
1792 for every  $q$  in  $Q$ . Thus every  $L_q$  is nonempty.
1793 If a prime  $r$  belonged to  $L_q$  and  $L_{q'}$  for  $q'\neq q$ , it would divide
1794  $(N-q)-(N-q')=q'-q$ . But  $0<|q'-q|\leq y<r$ , impossible.
1795 If every  $q$  in  $Q$  is BAD, the stopped traversal expands each such child in
1796 world mode, and each selected  $r_q$  in  $L_q$  is reached. Their disjointness
1797 makes them  $|Q|$  distinct new prime vertices, none in  $Q$ . Also each  $N-q$ 
1798 is odd and composite, so each factor  $r_q < N/3 < P$ ; none is the original
1799 root. Their shortest root distance is therefore exactly 2. They are active
1800 by the activity proof in the preceding result. If some  $q$  in  $Q$  is GOOD, success
1801 has already occurred at depth 1. This proves every alternative.
1802
1803 HOSTILE CHECK:
1804 This is NOT a claim that every residual gcd quotient is free of  $Q$ . Old
1805 prime-power excess may coexist with the forced large factor in each row.
1806 The injection uses  $r>y$  and the ACTUAL row difference; it is not Hall
1807 expansion inferred from a generic collision average. In an implementation
1808 that stops as soon as any GOOD is found, the entire frontier need not be
1809 materialized; the alternative is success or the complete stopped graph's
1810 stated expansion. No expansion is asserted at later generations.
1811
1812 DEPENDENCIES:
1813 Exact  $h=1$  relation; verified Matveev bound; elementary row-gap divisibility.
1814 No PR-MQ, Scott classification, global failure, or conjectural prime tuples.
1815 RELATION TO HISTORICAL FAILURES:
1816 The positive consumer escaping the previous collision-budget obstruction is
1817 an INDEPENDENT height bound on complete  $y$ -smooth rows. It is precisely the
1818 piece absent from attempts to infer new support merely from singleton mass.
1819 This does not erase those counterexamples: for each fixed  $y$  the result
1820 applies only beyond the enormous explicit  $F_2(y)$ .
1821 RELATION TO GOLDBACH:
1822 Success OR an injective collection of actual new reachable primes. Still no
1823 control of the complete rows at those new primes, so no proof of termination.
1824 RELATION TO  $p_0$  EXCEPTIONALITY:
1825 A nontrivial dynamical theorem for canonical  $h=1$  roots with bounded initial
1826 prime envelope relative to  $N$ . It is not yet a theorem distinguishing  $p_0$  from
1827  $p_1$ : no same- $N$  failure of this property for  $p_1$  has been established.
1828 WHAT REMAINS OPEN:
1829 Control after this forced exit. The new labels may be as large as  $N/3$ , at
1830 which point  $F_2$  of the enlarged envelope greatly exceeds  $N$ ; automatic
1831 iteration is invalid.
1832 NOVELTY: NOT YET AUDITED.
1833
1834 FINAL HOSTILE AUDIT / LIMITATIONS
1835 1. H1-ALL-BRANCH-INJECTIVE-ESCAPE is a conditional arithmetic theorem with a
1836 very severe  $N$ -versus- $y$  hypothesis. No explicit example with  $N>=F_2(y)$ ,
1837 and no infinite prime family satisfying that small-envelope hypothesis,
1838 is supplied in this run. Its logical consumer is correct; its practical
1839 nonvacuity at that scale and usefulness for a Goldbach architecture are
1840 not established. Do not silently report it as a typical-input statement.
1841 2. The support-height theorem applies to a terminal core without assuming
1842 two pure rows. The finite cores with no pure rows verify this is an actual
1843 extension of the historical domain, but the numeric bound is very weak.
1844 3. BS-1996 counts solutions; it does not bound their heights. The height
1845 bound uses a separate, explicitly verified Matveev input. These dependencies
1846 must not be interchanged.
1847 4. The new bounded-alphabet counts do not bound the union over arbitrary
1848 moving alphabets. Their density corollary explicitly confines all supports
1849 to the primes  $\leq y(X)$  before applying the bound.
1850 5. The small-child theorem produces MANY actual BAD children, not an ALL-BAD
1851 first frontier. Large GOOD children are its exact remaining obstruction.
1852 Runtime consequences specify the sorting and early-stopping conventions.
1853 6. The strongest growing-direct-prefix estimate from the INPUT checkpoint
1854 remains valid after the audit. It controls direct Goldbach pairs only;
1855 no theorem here promotes its failure to graph-traversal failure.
1856 7. FBTU is not declared false as a theorem about admissible partial trees.
1857 The inferred blanket theorem about complete BFS neighbourhoods is not

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1858 established by the supplied proof. Older Biblia iteration 108 already
 1859 documents the residual-factor obstruction. No opposite complete-depth
 1860 theorem has been proved in this continuation.
 1861 8. The Scott-Bozek even-exponent branch was neither eliminated nor used as
 1862 a dependency. No published theorem is declared false in this report.
 1863 9. No source correction, necessary condition, finite computation, or
 1864 density-one statement is called a proof of binary Goldbach.
 1865 10. Proofs and targeted arithmetic checks were performed in this continuation.
 1866 They have not received a separate external adversarial review. No claim
 1867 of literature novelty is made for any new application.
 1868
 1869 FAIL-ID: ENVELOPE-ESCAPE-AUTOMATIC-ITERATION-20260905
 1870 STATUS: KILLED AS AN INFERENCE; the two-generation theorem remains PROVED.
 1871 CLAIM / ROUTE: Iterate H1-ALL-BRANCH-INJECTIVE-ESCAPE until a finite graph
 1872 cannot contain more vertices, thereby obtaining a Goldbach pair.
 1873 WHY IT LOOKED PROMISING: The theorem gives an injective set of actual new
 1874 large primes using an independent height bound, rather than a scalar identity.
 1875 EXACT FAILURE: At the next generation the relation $N=2m+2$ is no longer a
 1876 relation with the NEW row anchor. Also the larger alphabet envelope may be
 1877 of order N , while the premise $N \geq F_2(y_{\text{new}})$ then fails spectacularly.
 1878 The new labels may enter a closed large core. Neither injectivity nor the
 1879 height premise persists automatically.
 1880 COUNTEREXAMPLE / OBSTRUCTION: The explicit changing-envelope dependence of
 1881 F_2 and loss of the $h=1$ row identity are sufficient obstructions; this is
 1882 not a constructed canonical failure.
 1883 WHAT SURVIVES: One fully justified two-generation expansion, and the global
 1884 support-height constraint on any subsequently closed core.
 1885 DO NOT REPEAT: Do not replace a fixed y in the proved premise by the newly
 1886 produced, larger y without verifying the premise again.
 1887 POSSIBLE REPLACEMENT: A different invariant or a height-sensitive argument
 1888 that survives changing alphabets and is forced by actual closed failure.
 1889
 1890 REFERENCE BOOK
 1891 Yann Bugeaud, Linear Forms in Logarithms and Applications, EMS, 2018.
 1892 Publisher: <https://ems.press/content/book-files/22003>
 1893 For ordinary prime-counting and probabilistic number theory, the input
 1894 checkpoint's G. Tenenbaum, Introduction to Analytic and Probabilistic
 1895 Number Theory, 3rd ed., AMS, 2015, remains an appropriate background book.
 1896
 1897 =====
 1898 STRONGEST NEW RESULTS
 1899 =====
 1900 1. UNRESTRICTED-CLOSED-SUPPORT-HEIGHT: removes the historical two-pure-row
 1901 assumption. EFFECTIVE-BOUNDED-CORE-THRESHOLD gives a completely explicit
 1902 $F(y)$ excluding every closed active BAD core supported below y once
 1903 $N \geq F(y)$. POINTWISE-CORE-LABEL-GROWTH gives the resulting iterated-log
 1904 lower bound on the largest core prime, uniformly over arbitrary branching.
 1905 2. H1-ALL-BRANCH-INJECTIVE-ESCAPE: beyond explicit $F_2(y)$, a canonical $h=1$
 1906 root with y -smooth composite complement either succeeds at depth 1 or
 1907 reaches at least $|PF(m)|$ distinct primes $> y$ at depth 2. Severe premise;
 1908 automatic iteration and large-scale nonvacuity are not established.
 1909 3. CLOSED-ALPHABET-COUNT: at most $\binom{s+2}{2} (9s+16)$ distinct even N
 1910 support a closed active BAD set inside any one fixed alphabet of s primes.
 1911 A common slowly growing prime envelope gives a rigorous density corollary.
 1912 4. SMALL-BAD-CHILDREN-PREFIX: simultaneous actual first-front BAD growth for
 1913 a slowly growing prefix of roots, with exact search-cost consumers and
 1914 no inference about shortest GOOD depth. This is shared by nearby roots.
 1915 5. AUDIT-FBTU: corrects the later partial-tree/full-BFS scope regression;
 1916 $N=92$ is a short exact certificate, and complete $h=1$ two-row finiteness
 1917 disproves an infinite fixed-alphabet/full-row interpretation.
 1918 The original prefix-transfer estimate survived the audit; it is inherited
 1919 work, not counted as a new result of this continuation.
 1920
 1921 =====
 1922 CURRENT FRONTIER
 1923 =====
 1924 No proof of binary Goldbach. No unconditional Goldbach threshold N_0 .
 1925 A globally failed N supplies a closed active BAD core. A failed canonical
 1926 traversal also supplies a reachable closed BAD core, but does not imply that
 1927 all other vertices of the graph are BAD. The new results constrain each such
 1928 core's height and support, and exclude a quantitatively specified small-label
 1929 regime. They do not rule out moving supports above that regime.
 1930 The $h=1$ expansion theorem ends at new prime labels, not at GOOD vertices.
 1931 No proved invariant controls their next complete rows strongly enough to
 1932 force another expansion, and no contradictory upper support bound was found.
 1933 A pointwise positive prime-pair count or a contradiction to full composite
 1934 closure remains missing. Existing density-one conclusions cannot remove
 1935 every exceptional N .
 1936
 1937 =====
 1938 p0 STATUS
 1939 =====
 1940 Found a genuine restricted $h=1$ arithmetic/reachability theorem with an
 1941 explicit size hypothesis. Did NOT find a property proved for every canonical
 1942 p_0 whose corresponding claim fails at a nearby root for the same N .
 1943 The density-transfer and small-BAD-child results are shared by nearby roots.
 1944 The claimed broad finite-local no-go needs the scope correction above; partial
 1945 CRT programming is not a theorem about complete bounded-depth failure.
 1946 There is still no convincing mathematical explanation of universal canonical
 1947 traversal success, and that universal success remains unproved.
 1948
 1949 =====
 1950 EXACT NEXT QUESTION
 1951 =====
 1952 For every integer $d \geq 1$, does there exist a prime $P \geq 5$ with $N=2(P-1)$,
 1953 $P-2$ composite, such that EVERY vertex at directed distance at most d from
 1954 canonical P in the factorization graph is BAD (positive composite complement)?
 1955 Equivalently, are the minimum distances from canonical $h=1$ roots to GOOD
 1956 vertices unbounded, allowing infinity for a failed traversal?
 1957 A proof must control the COMPLETE factor frontiers, including every residual
 1958 cofactor; programming a selected BAD path is not an answer. Resolving this
 1959 would settle the strongest presently unsupported bounded-depth no-go in the
 1960 corpus, or provide a uniform finite-depth success theorem on the $h=1$ family.
 1961
 1962 =====
 1963 CONTINUATION 2026-09-06


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1964 =====
1965 Previous sections remain an archive of the 2026-09-05 run; later entries
1966 below override any conflicting frontier statement. The present run starts
1967 from the complete-support / moving-support obstruction. No assertion of a
1968 Goldbach proof or of universal canonical traversal success is assumed.
1969
1970 RESULT-ID: TWO-SMALLEST-ROWS-HEIGHT-20260906
1971 STATUS: PROVED; refinement of UNRESTRICTED-CLOSED-SUPPORT-HEIGHT.
1972 EXACT STATEMENT:
1973 Let  $N > 8$  be even, and  $C$  a nonempty finite set of primes  $p < N$  with  $p$  not
1974 dividing  $N$ ,  $N-p$  composite, and  $PF(N-p)$  subset  $C$ . Write  $a < b$  for the two
1975 smallest labels in  $C$ ,  $U = N-a$ ,  $V = N-b$ ,  $S = PF(UV)$ ,  $t = |S|$ , and
1976  $A(S) = 1.4 \cdot 30^{-(t+3)} \cdot t^{-(9/2)} \cdot \text{product}_{(q \text{ in } S)} \log q$ .
1977 Then:
1978 (i)  $a < b < \sqrt{N}$ , and  $\gcd(U, V) = 1$ ;
1979 (ii)  $\log((N-b)/(b-a))$ 
1980  $< A(S) \cdot (1 + \log(\log N / \log 3))$ ;
1981 (iii)  $\log(\sqrt{N}-1)$ 
1982  $< A(S) \cdot (1 + \log(\log N / \log 3))$ .
1983 In particular the full support of just these two rows, even inside an
1984 arbitrarily branching core with large other labels, must satisfy
1985  $\text{product}_{(q \text{ in } S)} \log q >$ 
1986  $\log(\sqrt{N}-1) /$ 
1987  $[1.4 \cdot 30^{-(t+3)} \cdot t^{-(9/2)} \cdot (1 + \log(\log N / \log 3))]$ .
1988
1989 PROOF:
1990 A singleton  $C = \{a\}$  would give  $N-a = a^e$ , hence  $a|N$ , contradicting activity.
1991 All labels are odd. The least prime divisor  $r$  of the odd composite  $U$  is
1992 at most  $\sqrt{U} < \sqrt{N}$ . Closure gives  $r$  in  $C$ , and activity gives  $r! = a$ ,
1993 because  $a|(N-a)$  would imply  $a|N$ . Hence  $b < r < \sqrt{N}$ , proving (i)'s
1994 size assertion, without any assumption about pure-power rows.
1995 If a prime  $\ell$  divided  $U$  and  $V$ , then  $\ell$  belongs to  $C$  and divides  $b-a$ .
1996 It cannot equal  $a$ , by activity of  $a$ , and every other element of  $C$  is at
1997 least  $b > b-a$ . Thus no such  $\ell$  exists and  $\gcd(U, V) = 1$ .
1998 Now  $U/V = 1 - (b-a)/(N-b) > 0$ . Its prime bases are exactly  $S$ , with nonzero
1999 signed exponents  $e_q = v_q(U) - v_q(V)$  and  $|e_q| \leq \log N / \log 3$ . Apply the
2000 rational real-field product-minus-one Matveev bound recorded above, with
2001  $A_q = \log q$ , to obtain (ii). Finally  $b-a < b < \sqrt{N}$  and
2002  $N-b > N-\sqrt{N}$ , whence  $(N-b)/(b-a) > \sqrt{N}-1$ . This gives (iii).
2003
2004 HOSTILE CHECK:
2005 The fragile elementary step is excluding  $a$  as a factor of its own row;
2006 that requires activity, not merely compositeness. If inactive vertices
2007 were allowed, singleton examples  $N = a^2$  invalidate the claim. The bound
2008 uses the actual union of two row supports, not necessarily all of  $C$ .
2009 Nothing requires that either row be a prime power. The older result that
2010 a terminal BAD SCC has a cycle below  $\sqrt{N}$  is already stated in the
2011 Biblia (late synthesis, near line 49182 in the supplied copy); the small
2012 cycle itself is not claimed as new. Here the consumer is the quantitative
2013 height comparison of the two least rows. This improves the previous
2014  $M/(N-M)$  comparison, but remains only a necessary condition on a failed
2015 world. It does not put an upper bound on the moving support.
2016
2017 DEPENDENCIES:
2018 Full factor closure; activity; Matveev-in-BMS-9.4. The external input is
2019 being rechecked in the author's PDF in this run. No Scott classification.
2020 RELATION TO GOLDBACH: Applies to any closed failed core, whether supplied
2021 by global failure or by one failed traversal. Does not exclude all cores.
2022 RELATION TO p0: Shared by all roots whose traversals fail. No asymmetry.
2023 WHAT REMAINS OPEN: A contradictory upper support bound or another genuine
2024 consumer of the close pair of coprime rows.
2025 NOVELTY: NOT YET AUDITED.
2026
2027 SOURCE-AUDIT 2026-09-06:
2028 The Matveev input was rechecked in Bugeaud--Mignotte--Siksek, Theorem 9.4,
2029 author PDF printed pp.16--17. It is indeed the product-minus-one bound,
2030 with real-field coefficient  $1.4 \cdot 30^{-(n+3)} \cdot n^{4.5} \cdot (1 + \log B)$  for  $D=1$ .
2031 The preceding two-smallest-row refinement uses exactly that statement.
2032
2033 EXTERNAL-INPUT: FKL-2010-LEMMA-5.1
2034 STATUS: EXACT STATEMENT CHECKED in primary research, including uniformity.
2035 Kevin Ford, Sergei V. Konyagin, Florian Luca, Prime chains and Pratt trees,
2036 Geom. Funct. Anal. 20 (2010), 1231--1258, Lemma 5.1, printed pp.12--13.
2037 https://arxiv.org/pdf/0904.0473
2038 For  $k$  primitive integer linear forms  $a_i n + b_i$ ,  $a_i > 0$ , let  $\nu(\ell)$  count
2039 their roots modulo  $\ell$ ,  $B = \sum_{\ell} (k - \nu(\ell)) \cdot \log(\ell) / \ell$ , and
2040  $S = \text{product}_{\ell} (1 - \nu(\ell) / \ell) \cdot (1 - 1/\ell)^{-k}$ .
2041 There is an absolute  $\delta > 0$  such that, if  $x > 10$ ,
2042  $k \leq \delta \cdot \log(x) / \log \log(x)$ ,  $B \leq \delta \cdot \log(x)$ , the number of positive  $n \leq x$ 
2043 for which all forms are prime and exceed  $k$  is at most
2044  $2^k \cdot k! \cdot x \cdot S / (\log x)^k \cdot$ 
2045  $\exp(O((k \cdot B + k^2 \cdot \log \log x) / \log x))$ .
2046 The displayed 0 constant is absolute. We only use  $k=2, 3, 4$ . All pairwise
2047 determinants are checked nonzero below; nonprimitive forms and prime
2048 values  $\leq k$  are removed before application. This is an upper-bound sieve,
2049 not a prime-tuples lower bound.
2050
2051 RESULT-ID: UNIFORM-FEW-FORMS-SIEVE-CONSUMER-20260906
2052 STATUS: PROVED from FKL-2010-LEMMA-5.1.
2053 EXACT STATEMENT:
2054 Fix  $r$  in  $\{2, 3, 4\}$ ,  $\epsilon > 0$ ,  $C > 0$ . Let primitive linear forms
2055  $L_i(n) = a_i n + b_i$ ,  $a_i > 0$ , have nonzero pairwise determinants and
2056  $1 \leq \Delta = \text{abs}(\text{product}_i a_i \cdot \text{product}_{i < j} (a_i \cdot b_j - a_j \cdot b_i)) \leq X^C$ .
2057 Suppose  $X^{\epsilon} \cdot \text{product}_{i \leq r} L_i(n) \leq X^C$ . Uniformly in these forms, the number of
2058 positive  $n \leq z$  with every  $L_i(n)$  prime and  $> r$  is
2059  $O_{(\epsilon, C)}(z^{(\log \log X)^r / (\log z)^r})$ .
2060 If  $\nu(\ell) > 1$  for every prime  $\ell$ , improve the exponent  $r$  on  $\log \log X$ 
2061 to  $r-1$ . The same improvement holds when the only possible exception is
2062 one odd prime  $q \geq 3$ ; its extra local factor is bounded by  $(3/2)^r$ .
2063 All assertions are for sufficiently large  $X$ ; changing constants covers
2064 bounded  $X$  in any use where the logarithms are replaced by positive logs.
2065
2066 PROOF:
2067 For  $\ell$  not dividing  $\Delta$ ,  $\nu(\ell) = r$ , since all leading coefficients are
2068 invertible and the  $r$  roots are distinct. Always  $0 \leq \nu(\ell) \leq r$ . Thus
2069  $B \leq r \cdot \sum_{\ell | \Delta} \log(\ell) / \ell = O_C(\log \log X)$ .

```

2070 Here is an elementary justification of the uniform bound: put
2071 $T = \max(3, \log(3 \cdot \Delta))$. For $\ell \leq T$ the sum is $O(\log T)$, by partial
2072 summation from $\theta(t) = \sum_{\ell \leq t} \log \ell = O(t)$. For $\ell > T$ it is at most
2073 $(\log \Delta) / T \leq 1$. Chebyshev's θ bound is sufficient (and also follows
2074 from the PNT already used in the checkpoint).
2075 Similarly
2076 $\text{product}(\ell | \Delta) (1 - 1/\ell)^{-1} = O(\log \log(3 \cdot \Delta))$.
2077 For completeness, the factors $\ell > T$ have bounded product because their
2078 logarithms are $O(\sum_{\ell | \Delta, \ell > T} 1/\ell) = O(1)$. For $\ell \leq T$, put
2079 $\sigma = 1 + 1/\log T$. Comparing Euler factors at 1 and σ costs $O(1)$ in
2080 logarithm: the first-power difference is at most
2081 $(\sigma - 1) \sum_{\ell \leq T} \log(\ell) / \ell = O(1)$, and higher powers have a
2082 bounded sum. The product at σ is $\leq \zeta(\sigma) \leq 1 + 1/(\sigma - 1)$,
2083 by the integral comparison for $\sum n^{-(\sigma)}$. Thus the product is $O(\log T)$.
2084 For ℓ not dividing Δ , the local singular-series factor is ≤ 1 ,
2085 by $1 - r/\ell \leq (1 - 1/\ell)^r$; primes $\ell \leq r$ outside Δ cannot occur
2086 unless local admissibility fails. At exceptional primes use the bound
2087 $(1 - 1/\ell)^{-r}$. If $\nu(\ell) \geq 1$, use the sharper exponent $-(r - 1)$.
2088 This proves $S = O_C((\log \log X)^r)$, or the stated improvement. If $S = 0$,
2089 a prime $\ell \leq r$ divides at least one form for every n , so no tuple of
2090 prime values $> r$ occurs. Finally $B = O_C(\log \log X)$ and $\log z \geq \epsilon \log X$
2091 verify both FKL size hypotheses for large X ; its exponential factor is
2092 bounded. This proves the claimed uniform bounds.
2093
2094 HOSTILE CHECK:
2095 A fixed-coefficient asymptotic sieve result would be insufficient here;
2096 FKL supplies the needed uniform B condition. Common leading primes may
2097 have $\nu = 0$ even for primitive forms, so the exponent $r - 1$ is NOT used
2098 without the explicitly stated local check. A zero pair determinant
2099 would destroy the r -dimensional sieve gain; such cases are checked
2100 separately in each consumer. This lemma is an application, not a new
2101 claim about the underlying sieve theorem.
2102 DEPENDENCIES: FKL Lemma 5.1; elementary Euler products and Chebyshev bound.
2103 NOVELTY: NOT YET AUDITED.
2104
2105 RESULT-ID: COMPLETE-H1-FIRST-FRONT-RARE-SUCCESS-20260906
2106 STATUS: PROVED.
2107 EXACT STATEMENT:
2108 For X sufficiently large, the number of primes P in $[X, 2X]$ for which
2109 $N = 2(P - 1)$, root P , has a GOOD vertex at directed distance 0 or 1 is
2110 $O(X \cdot (\log \log X)^2 / (\log X)^2)$.
2111 Thus, with relative density one among primes P , P and EVERY prime
2112 factor q of $P - 2$ are BAD for this same N . This is a statement about the
2113 COMPLETE radius-one frontier, including all large prime factors.
2114 It does not say the traversal fails at later generations.
2115
2116 PROOF:
2117 P is canonical because $N/2 = P - 1$ and $P - 1$ is an even integer > 2 .
2118 A GOOD root requires $P - 2$ prime. The two-form sieve for $n, n - 2$ counts
2119 these primes by $O(X / (\log X)^2)$.
2120 Otherwise $m = P - 2$ is an odd composite. If a first-generation vertex q is
2121 GOOD, write $m = a \cdot q$, where q is an odd prime and $a \geq 3$ is odd. Then
2122 $P = a \cdot q + 2$, $N - q = (2a - 1) \cdot q + 2$
2123 are both prime. Count such pairs (a, q) ; overcounting several GOOD
2124 children of the same P is harmless.
2125 For $q \leq \sqrt{X}$, fix q and sieve in $a \leq 2X/q$ the two forms
2126 $q \cdot a + 2$, $2q \cdot a + 2 - q$.
2127 They are primitive: $\gcd(q, 2) = 1$ and $\gcd(2q, 2 - q) = 1$. Their determinant is
2128 $-q \cdot (q + 2)$, nonzero, and $\Delta = 2q^3 \cdot (q + 2) \leq O(X^2)$. Except at $\ell = q$
2129 there is at least one root of their product modulo every prime ℓ
2130 (the first form suffices unless $\ell = q$). The uniform two-form consumer
2131 therefore gives
2132 $O(X \cdot \log \log X / (q \cdot (\log X)^2))$.
2133 Summing over prime $q \leq \sqrt{X}$ costs $O(\log \log X)$, by the Euler-product
2134 bound proved above, giving $O(X \cdot (\log \log X)^2 / (\log X)^2)$.
2135 For $q > \sqrt{X}$, fix odd $a \geq 3$. Now $a \leq 2\sqrt{X}$ and $q \leq 2X/a$. Sieve in q
2136 using the THREE forms
2137 q , $a \cdot q + 2$, $(2a - 1) \cdot q + 2$.
2138 All are primitive. Their pair determinants are 2, 2, $2(1 - a)$, nonzero.
2139 The product Δ is polynomially bounded in X . There is at least one
2140 root modulo every prime, because the first form is q itself. Hence
2141 the count for each a is
2142 $O(X \cdot (\log \log X)^2 / (a \cdot (\log X)^3))$.
2143 Sum over $a \leq 2\sqrt{X}$; the harmonic sum is $O(\log X)$. The resulting bound
2144 is again $O(X \cdot (\log \log X)^2 / (\log X)^2)$. All prime values in the actual
2145 count exceed the sieve's r , since $q > \sqrt{X}$ and $P, N - q$ have size at least
2146 X up to an absolute constant. The sieve variable in both cases has
2147 upper limit $\geq \sqrt{X}$, verifying the uniformity premise. This proves
2148 the asserted count.
2149 PNT gives $\#\{P \text{ in } [X, 2X]\} \sim X / \log X$; dividing proves relative density one.
2150 When the root is BAD, each $q | m$ is active: a prime dividing q and N
2151 would divide P , impossible since $q < P$. Also $q \leq m/3$, so $N - q > 1$. Removing
2152 the prime case makes $N - q$ positive composite, not a unit or inactive row.
2153
2154 HOSTILE CHECK:
2155 The identity in the sieve is $N - q = 2a \cdot q + 2 - q$, not $2a \cdot q + 4 - q$.
2156 The large- q range is NOT discarded as negligible and is NOT treated
2157 with a short progression estimate. Switching to the small cofactor a
2158 creates three distinct prime forms and controls that entire range.
2159 This is the step absent from a small-child-only argument. The result
2160 is relative to the thin family $N/2 + 1$ prime, not density one among all
2161 even N . The union bound counts all first-front GOOD children; no
2162 independence of primality and factorization is assumed.
2163 DEPENDENCIES: The uniform two-/three-form sieve consumer; PNT only for
2164 the relative-density interpretation. No conjectural prime tuples.
2165 RELATION TO GOLDBACH: A rigorous limitation on shallow traversal in an
2166 infinite canonical family. It neither supplies nor denies a Goldbach
2167 representation using deeper reached vertices or other primes.
2168 RELATION TO p0: Actual complete-frontier statement for canonical $h = 1$.
2169 Nearby roots for the SAME N are tested in the next result.
2170 WHAT REMAINS OPEN: Complete radius-two control, and every fixed depth.
2171 NOVELTY: NOT YET AUDITED.
2172
2173 RESULT-ID: SAME-N-GROWING-ROOTS-COMPLETE-FIRST-FRONT-20260906
2174 STATUS: PROVED.
2175 EXACT STATEMENT:

2176 Let P run over primes in $[X, 2X]$, let $N=2(P-1)$, and let $p_k(N)$ be the
 2177 $(k+1)$ -st prime at or above $N/2$; in particular $p_0=P$. Let $K(X) \geq 1$ be
 2178 integer-valued with
 2179 $K = o(\log X / (\log \log X)^3)$.
 2180 For all but $o(X/\log X)$ such primes P , EVERY root p_k , $0 \leq k \leq K$, is BAD
 2181 and EVERY prime divisor q of $N-p_k$ is also BAD. Thus none of these
 2182 roots has a GOOD vertex at distance ≤ 1 in its complete stopped graph.
 2183 This compares all the roots for the SAME N , and includes all their
 2184 large as well as small first-front factors.
 2185 For $1 \leq K \leq \log X / (\log \log X)^3$, a quantitative bound for the exceptional
 2186 proportion (relative to $X/\log X$) is
 2187 $O((\log \log X)^2 / \log X + \sqrt{K} (\log \log X)^3 / \log X)$. (R1)
 2188 For example $K = \text{floor}(\log X / (\log \log X)^5)$ gives proportion $O(1/\log \log X)$.
 2189 These are asymptotic statements; they do not assert this proportion at
 2190 currently accessible numerical sizes.
 2191
 2192 PROOF:
 2193 Write $L = \log X$ and $l = \log \log X$. First isolate p_0 : the preceding theorem
 2194 counts its exceptional P by $O(X^{1/2}/L^2)$.
 2195 For any later prime $Q = p_k$, put $h = Q - P + 1$. Then h is an odd integer ≥ 3 ,
 2196 $P = Q - h + 1$, $N = 2(Q - h)$, $m = N - Q = Q - 2h$.
 2197 We first count all such pairs (P, Q) with $3 \leq h \leq H$, without imposing any
 2198 rank condition on Q . Assume $2 \leq H \leq \sqrt{X}/10$, P in $[X, 2X]$, Q prime.
 2199 Then $Q \leq 3X$ for large X . For EACH h the number of these pairs with a
 2200 GOOD root or a GOOD first-front child is
 2201 $O(X^{1/3}/L^3)$, uniformly in h . (R2)
 2202 We prove (R2) carefully.
 2203 If m is prime, the three forms in Q are
 2204 Q , $Q - h + 1$, $Q - 2h$.
 2205 They are primitive, distinct since $h > 1$, with nonzero shift differences
 2206 $h-1$, $2h$, $h-1$. Their Delta is polynomially bounded in X and $\nu(\text{ell}) \geq 1$.
 2207 The three-form consumer bounds the count by $O(X^{1/2}/L^3)$.
 2208 Otherwise m is odd composite. For a GOOD child q write $m = a \cdot q$, $a \geq 3$ odd.
 2209 The quantities
 2210 $Q = a \cdot q + 2h$,
 2211 $P = a \cdot q + h + 1$,
 2212 $g = N - Q = (2a-1) \cdot q + 2h$
 2213 are prime. Split into $q \leq \sqrt{X}$ and $q > \sqrt{X}$.
 2214 For small prime q , sieve in $a \leq 3X/q$ the THREE forms
 2215 $q \cdot a + 2h$, $q \cdot a + h + 1$, $2q \cdot a + 2h - q$.
 2216 If q divides h or $h+1$, one of Q, P would be divisible by q while larger
 2217 than q , so this tuple contributes nothing. For every remaining q these
 2218 three forms are primitive, including the last since its constant is
 2219 odd and coprime to q . Their pair determinants are
 2220 $q \cdot (1-h)$, $-q \cdot (q+2h)$, $-q \cdot (q+2)$,
 2221 all nonzero. Delta is bounded by X to an absolute constant power.
 2222 Except possibly at $\text{ell} = q$ there is a root modulo every prime. Hence
 2223 the uniform three-form count is $O(X^{1/2}/(q \cdot L^3))$. Summing $1/q$ over
 2224 prime $q \leq \sqrt{X}$ gives $O(1)$, and this range contributes $O(X^{1/3}/L^3)$.
 2225 For large $q > \sqrt{X}$, fix $a \leq 3\sqrt{X}$ and sieve in $q \leq 3X/a$ the FOUR forms
 2226 q , $a \cdot q + 2h$, $a \cdot q + h + 1$, $(2a-1) \cdot q + 2h$.
 2227 If any of the last three forms is not primitive, every value is a
 2228 multiple of a fixed integer $d > 1$ with $d \leq 2h+1 < X$. But its actual prime
 2229 value $(Q, P, \text{ or } g)$ is $> X$ for large X . Hence that a has no solutions.
 2230 For the remaining a all forms are primitive. The determinants between
 2231 the first form and the others are $2h$, $h+1$, $2h$. The other three are
 2232 $a \cdot (1-h)$, $2h \cdot (1-a)$, $h+1-2a$.
 2233 The first two do not vanish. If the last vanished, then $h+1=2a$ and
 2234 $\gcd(a, h+1) = a > 1$, a case just excluded. Thus there is no hidden
 2235 coincidence of two prime forms. Delta is polynomially bounded in X ,
 2236 and $\nu(\text{ell}) \geq 1$ because the first form is the variable q . The four-form
 2237 bound is $O(X^{1/3}/(a \cdot L^4))$. Summing $1/a$ up to $3\sqrt{X}$ costs $O(L)$,
 2238 giving $O(X^{1/3}/L^3)$. All relevant prime values exceed four. Both
 2239 sieve-variable upper limits are $\geq \sqrt{X}$, so the uniform constants
 2240 are absolute, independently of $h \leq H$. This proves (R2).
 2241 Summing (R2) over $h \leq H$ bounds the number of exceptional P with ANY
 2242 such nearby Q by $O(H \cdot X^{1/3}/L^3)$. No extra factor K is required: every
 2243 relevant pair (P, Q) is already in this count.
 2244 It remains to show that the first K successors usually lie within
 2245 $Q - P \leq H - 1$. For the indicated range of K , PNT ensures that $p_K \leq 3X$ for
 2246 every P in $[X, 2X]$ and sufficiently large X . Write each $p_K - P$ as the
 2247 sum of its K consecutive prime gaps. Summing over starting primes P
 2248 in $[X, 2X]$, every gap is used at most K times and all gaps lie between
 2249 primes in $[X, 3X]$. Their total length is $\leq 2X$. Therefore
 2250 $\sum_{P \text{ in } [X, 2X] \text{ prime}} (p_K - P) \leq 2KX$,
 2251 so at most $2KX/(H-1)$ of these P have $p_K - P > H - 1$.
 2252 Combining all three exceptional sets and dividing by X/L yields
 2253 $O(1/2/L + K \cdot L/(H-1) + H \cdot 1/3/L^2)$.
 2254 Take H to be an integer within $O(1)$ of $\sqrt{K \cdot L^3/1^3}$. Uniformly in
 2255 $1 \leq K \leq L/1^3$, H tends to infinity and is $\leq \sqrt{X}/10$ for large X .
 2256 This proves (R1). If $K = o(L/1^3)$, both terms in (R1) tend to zero.
 2257 For every remaining P , all roots have positive odd complements; their
 2258 proper prime factors are active by $\gcd(N, Q) = 1$ and $q < Q$. Indeed $Q > P$
 2259 is close to P , $Q > N/2$ and $Q < N$, so it cannot divide the even number N
 2260 (the only possible quotient would be 1). A root is BAD after excluding
 2261 the prime case, and each child complement exceeds 1 and is composite
 2262 after excluding all the counted GOOD children. This completes the
 2263 full stopped-graph conclusion.
 2264
 2265 HOSTILE CHECK:
 2266 1. The rank condition was used only in the unweighted prime-gap sum.
 2267 No claim about typical polylogarithmic gaps for uniformly sampled
 2268 even N is required. Sampling is over prime P , not over gap-weighted x .
 2269 2. The extra primality $P = Q - h + 1$ is essential. Without it the sieve loses
 2270 one logarithm and summing over h in the required range does not give
 2271 this result. For $h=1$ these forms coincide; p_0 was handled separately
 2272 by the two-/three-form theorem, not by a false four-prime count.
 2273 3. The potentially vanishing determinant $h+1-2a$ forces an imprimitive
 2274 form and has no actual prime solutions in the stated large- X range.
 2275 4. The conclusion is FULL radius-one BAD, not a selected path, not a
 2276 statement that only many children are BAD, and not failure of BFS.
 2277 5. The property is explicitly SHARED by nearby roots at the same N .
 2278 It is not evidence of an advantage peculiar to p_0 . It rules out
 2279 shallow-success explanations only in the precise asymptotic family
 2280 and prefix stated here, leaving all deeper behavior open.
 2281 6. The count remains true if the algorithm has arbitrary child order:

2282 it is a graph-theoretic distance assertion, not a runtime statistic.
2283
2284 DEPENDENCIES: Uniform FKL sieve consumer, elementary harmonic sums,
2285 PNT for the prime-sample density and availability of K primes below 3X.
2286 RELATION TO GOLDBACH: No representation is ruled out outside the tested
2287 one-step factor neighborhoods. Many steps or other roots may succeed.
2288 RELATION TO p0: A rigorous SAME-N asymptotic no-go for distinguishing
2289 p_0 from a growing nearby prefix by existence of a depth-0/1 GOOD node.
2290 WHAT REMAINS OPEN: Control the complete second frontier; prove or refute
2291 unbounded shortest GOOD distance in the h=1 family.
2292 NOVELTY: NOT YET AUDITED.
2293
2294 RESULT-ID: SAME-N-NEARBY-ROOT-FIRST-FRONT-SIEVE-20260906
2295 STATUS: PROVED.
2296 EXACT STATEMENT:
2297 Let X be sufficiently large and $3 \leq H = \sqrt{X}/10$. Count pairs of odd
2298 primes P,Q with $X \leq P \leq 2X$, $P \leq Q \leq 3X$, and $h = Q - P + 1 \leq H$. Set $N = 2(P-1)$.
2299 The number of these pairs for which the stopped traversal from Q has
2300 a GOOD vertex at distance 0 or 1 is
2301 $O(H * X * (\log \log X)^3 / (\log X)^3)$,
2302 with an absolute implied constant. No consecutiveness assumption on
2303 P,Q is needed. All first-front prime factors are covered.
2304
2305 PROOF:
2306 Fix h. Since P,Q are odd and $Q \geq P$, h is odd and $h \geq 3$. The identities are
2307 $P = Q - h + 1$, $N = 2Q - 2h$, $m = N - Q = Q - 2h$.
2308 Because $h \leq \sqrt{X}/10$ and $Q \geq X$, $m \geq 1$, and all relevant rows below have
2309 size bounded below by a constant times X. A GOOD root Q requires the
2310 THREE forms Q, Q-h+1, Q-2h to be prime. Their nonzero pair differences
2311 are h-1, 2h, h+1. Leading coefficients are 1, so they are primitive
2312 and $\nu(\ell) \geq 1$. Uniform three-form sieving in $Q \leq 3X$ bounds the count
2313 by $O(X * (\log \log X)^2 / (\log X)^3)$.
2314 Otherwise m is odd composite. Write $m = a * q$, with $a \geq 3$ odd and q an odd
2315 prime. A GOOD child q requires primality of
2316 $Q = a * q + 2h$, $P = a * q + h + 1$, $g = N - q = (2a - 1) * q + 2h$. (S)
2317 For $q \leq \sqrt{X}$ fix q and sieve in $a \leq 3X/q$ the three forms
2318 $q * a + 2h$, $q * a + h + 1$, $2q * a + 2h - q$.
2319 If q divides 2h, Q would be a multiple of q smaller than Q, impossible.
2320 If q divides h+1, the same holds for P. Thus parameters violating
2321 these coprimality conditions contribute no actual pair. In the remaining
2322 case all three forms are primitive (their odd constant in the third
2323 form also excludes a common factor 2). Their pair determinants are
2324 $q * (1 - h)$, $-q * (q + 2h)$, $-q * (q + 2)$,
2325 all nonzero. Delta is polynomially bounded in X, uniformly in $h \leq H$.
2326 For $\ell \mid q$, the first form has a root modulo ℓ ; only the one odd
2327 prime q may have $\nu(q) = 0$. The improved $r = 3$ sieve bound gives
2328 $O(X * (\log \log X)^2 / (q * (\log X)^3))$.
2329 The sum over prime $q \leq \sqrt{X}$ is therefore
2330 $O(X * (\log \log X)^3 / (\log X)^3)$.
2331 For $q > \sqrt{X}$, fix $a \leq 3\sqrt{X}$ and sieve in $q \leq 3X/a$ using FOUR forms:
2332 q, $a * q + 2h$, $a * q + h + 1$, $(2a - 1) * q + 2h$.
2333 If $\gcd(a, 2h) > 1$ or $\gcd(a, h + 1) > 1$ or $\gcd(2a - 1, 2h) > 1$, one of Q,P,g
2334 has a fixed divisor between 2 and 2h, whereas the relevant prime value
2335 is $\geq cX > 2h$. Hence there are no actual pairs for that parameter.
2336 For the remaining parameters the four forms are primitive. Determinants
2337 involving the first form are 2h, h+1, 2h. The other three are
2338 $a * (1 - h)$, $2h * (1 - a)$, $h + 1 - 2a$.
2339 The first two are nonzero because $h \geq 3$ and $a \geq 3$. If $h + 1 - 2a = 0$, then
2340 $P = a * (q + 2)$ is composite; equivalently $\gcd(a, h + 1) = a > 1$ already excluded
2341 this case. Thus no zero-determinant case is being passed to the sieve.
2342 The first form ensures $\nu(\ell) \geq 1$ at every prime. Delta remains bounded
2343 by a fixed power of X. The $r = 4$ bound is
2344 $O(X * (\log \log X)^3 / (a * (\log X)^4))$.
2345 Summing $a \leq 3\sqrt{X}$ costs $O(\log X)$. This proves the same bound as in the
2346 small-q range, for each fixed h. All actual prime values exceed 4 in
2347 this range, and both sieve variable limits are $\geq \sqrt{X}$. Adding the
2348 root-GOOD bound and summing over $h \leq H$ proves the statement.
2349
2350 HOSTILE CHECK:
2351 The extra prime $P = N/2 + 1$ is essential for the extra sieve dimension.
2352 Ignoring its primality would lose one $\log X$ and would not prove the
2353 nearby-root prefix result below. It is available because all roots are
2354 being compared for the SAME N in the h=1 family. For h=1, P and Q
2355 coincide and this dimension gain disappears; that case is handled by
2356 COMPLETE-H1-FIRST-FRONT-RARE-SUCCESS instead. The other possible
2357 proportional pair at $h + 1 = 2a$ is not ignored: it has no prime solutions.
2358 No typical prime-gap estimate is assumed in this counting lemma.
2359 DEPENDENCIES: Uniform few-form sieve consumer, elementary identities.
2360 RELATION TO GOLDBACH: Counts successful radius-one traversals only.
2361 RELATION TO p0: Enables an actual same-N comparison with nearby roots.
2362 WHAT REMAINS OPEN: Extend complete-front estimates to deeper generations.
2363 NOVELTY: NOT YET AUDITED.
2364
2365 RESULT-ID: COMPLETE-FIRST-FRONT-NO-GO-GROWING-SAME-N-PREFIX-20260906
2366 STATUS: PROVED.
2367 EXACT STATEMENT:
2368 Let $K = K(X) \geq 1$ be integers with
2369 $K = o(\log X / (\log \log X)^3)$.
2370 For all but $o(X / \log X)$ primes P in $[X, 2X]$, set $N = 2(P - 1)$. Let $Q_0 = P$,
2371 $Q_1, Q_2, \dots$ be the successive primes starting at P. Simultaneously for
2372 EVERY $0 \leq k \leq K$, Q_k is BAD and EVERY prime factor of $N - Q_k$ is BAD.
2373 Equivalently none of these complete stopped traversals succeeds at
2374 radius 0 or 1. Each Q_k is exactly $p_k(N)$; these roots refer to the SAME N.
2375 This is a relative-density-one theorem in the prime-parametrized h=1
2376 family, not a density-one theorem among all even N.
2377 One admissible concrete prefix is $\text{floor}(\log X / (\log \log X)^5)$.
2378 A quantitative upper bound for the exceptional proportion, up to an
2379 absolute constant for sufficiently large X, is
2380 $(\log \log X)^2 / \log X + \sqrt{K * (\log \log X)^3 / \log X}$.
2381
2382 PROOF:
2383 For K in the stated range, PNT ensures that there are more than K primes
2384 between 2X and 3X for large X. Thus $Q_K \leq 3X$ for every P in $[X, 2X]$.
2385 Summing $Q_K - P$ over these starting primes counts each consecutive prime
2386 gap inside $[X, 3X]$ at most K times. The sum of all such gaps is at most
2387 2X. Consequently

2388 #[P in [X,2X] prime: $Q_K - P > H - 1$] $\leq 2KX/(H-1)$. (G)
2389 This uses only the sum of gaps, not a bound on individual gaps or a
2390 second moment. If $Q_K - P \leq H - 1$, every $k \geq 1$ has $h_k = Q_k - P + 1 \leq H$ and the
2391 pair (P, Q_k) lies in SAME-N-NEARBY-ROOT-FIRST-FRONT-SIEVE. Hence the
2392 number of starting P admitting ANY successful radius-one traversal
2393 among $k=1, \dots, K$ with $Q_K - P \leq H - 1$ is at most
2394 $O(H * X * (\log \log X)^3 / (\log X)^3)$.
2395 There is no extra K factor: that lemma already counts all eligible
2396 pairs P, Q and all offsets $h \leq H$, and a union is bounded by that count.
2397 The $k=0$ exception is $O(X * (\log \log X)^2 / (\log X)^2)$, by the preceding
2398 canonical theorem. Divide the sum of these three bounds by
2399 $\# \{P \text{ in } [X, 2X]\} - X / \log X$. The exceptional proportion is
2400 $O((\log \log X)^2 / \log X + K * \log X / H$
2401 $+ H * (\log \log X)^3 / (\log X)^2)$.
2402 Choose H equal to an integer within 1 of
2403 $\sqrt{K * (\log X)^3 / (\log \log X)^3}$.
2404 It tends to infinity and lies below $\sqrt{X}/10$ for large X; the rounding
2405 and $H-1$ in (G) change only the constant. The last two terms are both
2406 of the asserted square-root order, and tend to zero by the assumption
2407 on K. This proves the quantitative and qualitative assertions.
2408 For good starting P the complements are positive: $Q_k = P + O(H)$, so
2409 $N - Q_k = P - 2 - O(H) > 1$. No root divides N, since each root is prime in
2410 $(N/2, N)$, strictly above the diagonal. If a child q divided N then
2411 q would divide its root Q_k , impossible because $q < Q_k$. Thus the BAD
2412 conclusions have the active, positive-composite meaning, as required.
2413
2414 HOSTILE CHECK:
2415 This gives COMPLETE first-front BADness, not merely a selected CRT
2416 subtree or many small BAD children. It therefore repairs one limited
2417 part of the old full-BFS no-go with a different, analytic proof.
2418 It proves neither arbitrary depth nor failure of the complete traversal.
2419 The relative family restriction is indispensable in this proof: the
2420 additional prime $P = N/2 + 1$ supplies a sieve dimension for nearby roots.
2421 Markov's estimate in (G) is over prime starts P, not over all integer
2422 midpoints weighted by their containing prime gap. Confusing those two
2423 sampling measures would be an invalid extension to all even N.
2424 The theorem does not assert that the factor supports or deeper dynamics
2425 of Q_0 and Q_1 agree; it asserts a shared entire radius-one failure event.
2426
2427 DEPENDENCIES:
2428 FKL Lemma 5.1 via the fully checked consumers above; PNT; exact gap sum.
2429 No Goldbach-failure hypothesis, no lower-bound prime-tuples conjecture,
2430 no Scott--Bozek classification, no iteration of an envelope theorem.
2431 RELATION TO GOLDBACH:
2432 An unconditional obstruction to using a growing nearby-root prefix with
2433 one factorization generation as a typical solver in this infinite family.
2434 It does not obstruct multigeneration traversal or general Goldbach pairs.
2435 RELATION TO p_0 :
2436 A substantive same-N negative comparison: even an entire complete first
2437 front is asymptotically BAD for p_0 and a growing family of its neighbours.
2438 The empirical exceptional behavior must, if real, be attributed to
2439 something stronger than ubiquitous shallow success. No universal
2440 p_0 -specific reachability advantage is proved here.
2441 WHAT REMAINS OPEN:
2442 In particular, no corresponding complete radius-two theorem is proved.
2443 The original question about every fixed complete BAD BFS depth remains
2444 open beyond radius one.
2445 NOVELTY: NOT YET AUDITED.
2446
2447 RESULT-ID: COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS-20260906
2448 STATUS: PROVED (new full-generation extension; proof audited case by case).
2449 EXACT STATEMENT:
2450 Let P range over primes in $[X, 2X]$, $N = 2(P-1)$, canonical root P. The
2451 proportion of these P for which ANY vertex at directed distance ≤ 2
2452 from P is GOOD tends to zero as X tends to infinity. In particular
2453 there are infinitely many canonical $h=1$ roots whose COMPLETE radius-two
2454 neighbourhood consists of BAD vertices.
2455 More quantitatively, with $L = \log X$ and $l = \log \log X$, the number of exceptional
2456 P is $O((X/L) * (l^3/L)^{(1/5)})$. This bound is asymptotic with an absolute
2457 constant; it is not an asserted numerical proportion at small X.
2458 No conclusion is made about distance three or eventual traversal failure.
2459
2460 PROOF:
2461 Success at distance ≤ 1 is already counted by $O(X * l^{-2}/L^{-2})$. Consider a
2462 shortest successful path $P \rightarrow q \rightarrow r$ with q BAD and r GOOD. Put
2463 $a = (P-2)/q$, $b = (N-q)/r$, $g = N-r$.
2464 Then q, r, P, g are odd primes, a, b are odd integers > 3 , $q! = r$, and
2465 $P = a * q^2$,
2466 $b * r = (2a-1) * q^2$,
2467 $g = 2a * q^2 - r$. (T)
2468 We have $q < (2/3)X$, $r < (4/3)X$, $P > X$, and $g > 2N/3 > cX$.
2469 Fix $0 < \eta < 1/8$, initially independent of X. Remove all P for which
2470 $P-2$ has a prime divisor q in
2471 $[X^{(1/2-\eta)}, X^{(1/2+\eta)}]$. (BAND)
2472 For a fixed prime q here, the one-form FKL bound applied to $q * a^2$,
2473 $a < 2X/q$, gives $O(X/(q * L))$, uniformly for $\eta < 1/8$. Indeed its singular
2474 series is $(1-1/q)^{-(-1) < 3/2}$, its B is $\log(q)/q$, and $\log(2X/q) > 3L/8$.
2475 Partial summation using $\pi(t) = O(t/\log t)$ gives
2476 $\sum_{q \text{ in BAND prime}} 1/q = O(\eta + 1/L)$.
2477 Consequently at most $O((\eta + 1/L) * X/L)$ starting primes are removed.
2478 This is a count of a factor occurrence, not an independence assertion.
2479
2480 Every remaining path has either $q < X^{(1/2-\eta)}$ or $q > X^{(1/2+\eta)}$.
2481 Split also at $r = \sqrt{X}$. In each of the following four cases we obtain
2482 an arithmetic progression with at least a power of X available for
2483 sieving. The product of the fixed parameters is $\leq 8X^{(1-\eta)}$.
2484 All stated sieve-variable upper limits are therefore $> cX^\eta$.
2485
2486 CASE LL: $q < X^{(1/2-\eta)}$, $r \leq \sqrt{X}$. Fix the two distinct primes q, r.
2487 The conditions are $P \equiv 2 \pmod q$ and $2P \equiv 2 + q \pmod r$. They specify one
2488 class A modulo $q * r$. Choose $1 \leq A \leq q * r$ and write $P = A + q * r * t$. Since
2489 $q * r < X$ and $P > X$, $t \geq 1$ and $t \leq 2X/(q * r)$.
2490 Sieve the TWO forms P and $g = 2P - 2 - r$. Their determinant is
2491 $-(q * r) * (r + 2)$, nonzero. If either is imprimitive, its fixed divisor is
2492 at most $2q * r = o(X)$, smaller than its actual prime value, so that class
2493 has no solutions and can be discarded. Their common leading primes

2494 are only q, r ; away from those primes there is a root of the first
 2495 form (with the harmless factor 2 handled by its odd leading coefficient).
 2496 Thus their singular series is $O(1)$, since at most two odd primes can
 2497 have $\nu=0$, each contributing a bounded local factor. FKL bounds this
 2498 case for each q, r by
 2499 $O(X^{1/2}/(q^*r^*(\eta^*L)^2))$.
 2500 Summing reciprocals of q, r over primes $\leq 2X$ gives $O(L^{-2})$. The total is
 2501 $O(X^{1/3}/(\eta^*L^2))$.
 2502
 2503 CASE LC: $q < X^{1/2-\eta}$, $r > \sqrt{X}$. Fix the prime q and the odd integer
 2504 $b = (N-q)/r < 4\sqrt{X}$. The congruences are $P \equiv 2 \pmod{q}$ and
 2505 $2P \equiv 2q \pmod{b}$. They are compatible only if $\gcd(q, b) = 1$: otherwise
 2506 $q|b$ would imply $q|2$. Because b is odd, the compatible case gives one
 2507 class $A \pmod{q^*b}$. Write $P = A + q^*b^*t$, $1 < A \leq q^*b$, so $t \geq 1$ and
 2508 $t < 2X/(q^*b)$. The THREE prime forms are
 2509 $P = A + q^*b^*t$,
 2510 $r = (2A - 2q)/b + 2q^*t$,
 2511 $g = 2P - 2 - r$.
 2512 All coefficients are positive. Their leading coefficients are q^*b ,
 2513 $2q$, $2q^*(b-1)$. For large X each is smaller than its actual prime value:
 2514 $q^*b < 4X^{1-\eta}$, $P, g > cX$, and $2q < \sqrt{X} < r$. Therefore an imprimitive
 2515 form has no actual prime solutions and is discarded.
 2516 To check the determinants, regard $P = (b^*r + q^*2)/2$ and
 2517 $g = (b-1)^*r + q$ as affine forms in r . The three relevant nonzero constants
 2518 are q^*2 , q , and $q^*2(b-1)$. The last cannot vanish: q is odd and
 2519 $2(b-1)$ even. Changing to the t parameter preserves nonvanishing.
 2520 The leading coefficient $2q$ of r gives a root at every odd prime except
 2521 q ; at 2, the leading coefficient q^*b of P is odd. Hence $S = O(L^{-2})$.
 2522 The contribution for fixed q, b is
 2523 $O(X^{1/2}/(q^*b^*(\eta^*L)^3))$.
 2524 The reciprocal prime sum costs $O(1)$, and the reciprocal b sum $O(L)$.
 2525 This case contributes $O(X^{1/3}/(\eta^*L^3))$.
 2526
 2527 CASE CL: $q > X^{1/2+\eta}$, $r < \sqrt{X}$. Fix odd $a = (P-2)/q < 2X^{1/2-\eta}$
 2528 and the prime r . The condition $r | ((2a-1)q^*2)$ forces $\gcd(r, 2a-1) = 1$;
 2529 otherwise the odd prime r would divide 2. Thus $q = A + r^*t$ is a unique
 2530 progression with $1 < A \leq r$, $t \geq 1$, $t < 2X/(a^*r)$.
 2531 Sieve the THREE forms q , $P = a + q^*2$, $g = 2a + q^*2 - r$. Before substituting
 2532 $q = A + r^*t$, their pair determinants are 2, $2-r$, $-a^*(r+2)$, all nonzero.
 2533 Their t -coefficients are $r, a^*r, 2a^*r$. They are smaller than their actual
 2534 prime values because $q > \sqrt{X} > r$ and $a^*r < 2X^{1-\eta}$. Discard any
 2535 imprimitive forms, which therefore have no solutions. The first form
 2536 has a root at every prime except possibly r , so $S = O(L^{-2})$.
 2537 The count for each a, r is $O(X^{1/2}/(a^*r^*(\eta^*L)^3))$. Sum over a and
 2538 prime r to get $O(X^{1/3}/(\eta^*L^2))$.
 2539
 2540 CASE CC: $q > X^{1/2+\eta}$, $r > \sqrt{X}$. Fix odd $a < 2X^{1/2-\eta}$ and
 2541 $b < 4\sqrt{X}$. The relation $b^*r = (2a-1)q^*2$ forces $\gcd(b, 2a-1) = 1$, since
 2542 this odd $\gcd$ divides 2. Thus $q = A + b^*t$ for a unique $1 < A \leq b$. For large
 2543 X , $q > X^{1/2+\eta} > b$, so $t \geq 1$, and $t < 2X/(a^*b)$. Let
 2544 $R = ((2a-1)^*A + 2)/b$, an integer. The FOUR prime forms in t are
 2545 $q = A + b^*t$,
 2546 $r = R + (2a-1)^*t$,
 2547 $P = a^*A + 2 + a^*b^*t$,
 2548 $g = 2a^*A + 2 - R + [2a^*b - 2a + 1]^*t$.
 2549 All coefficients are positive. The coefficients of q, r are $<$ their
 2550 actual prime values since $b < q$ and $2a-1 < r$ for large X . The coefficients
 2551 of P, g are $< 16X^{1-\eta} = o(X)$, also smaller than their actual values.
 2552 Therefore nonprimitive forms again give no solutions. Their six
 2553 pairwise determinants, in the order $(q, P), (q, r), (q, g), (P, r), (P, g), (r, g)$,
 2554 are
 2555 $2b, 2, 2(b-1), 2(1-a), -2[a(b-1)+1], -2$.
 2556 They are all nonzero for $a, b \geq 3$. Moreover $\gcd(b, 2a-1) = 1$, so at least
 2557 one of the first two forms has invertible leading coefficient modulo
 2558 every prime. Hence $\nu(\ell) \geq 1$ for all ℓ and $S = O(L^{-3})$.
 2559 The count is $O(X^{1/3}/(a^*b^*(\eta^*L)^4))$. Two harmonic sums over a, b
 2560 cost $O(L^2)$, giving $O(X^{1/3}/(\eta^*L^4))$.
 2561
 2562 In all four cases, the representatives, coefficients and determinants
 2563 are bounded by fixed powers of X ; $B_{\text{FKL}} = O(1)$ uniformly. For fixed η
 2564 the sieve exponential is bounded, and adding the four cases gives
 2565 $O(X^{1/3}/(\eta^*L^2))$. Adding BAND and radius-one exceptions, and
 2566 dividing by X/L , gives
 2567 $O(\eta^*L/L^{1/2}/L^{1/3}/(\eta^*L^4))$. (R2)
 2568 For each fixed η the last three terms tend to zero. Letting η
 2569 decrease to zero proves the qualitative relative-density-one claim.
 2570 For the quantitative claim choose $\eta = (L^{1/3}/L)^{1/5}$. Then $\eta < 1/8$
 2571 for large X , η^*L tends to infinity much faster than 1, and all the
 2572 coefficient-vs-prime comparisons above remain valid. The FKL hypotheses
 2573 and its bounded exponential factor hold uniformly, since every sieve
 2574 variable limit is $> cX^\eta$ and $B_{\text{FKL}} = O(1) = o(\eta^*L)$. Factors $\eta^*(-r)$
 2575 from the denominator were already retained explicitly. Thus (R2)
 2576 is $O(L^{1/3}/L^{1/5})$, as claimed.
 2577 After excluding success at distances 0, 1, 2, every vertex in this complete
 2578 radius-two neighbourhood is active with positive composite complement:
 2579 activity propagates along edges, and children of BAD rows are $< N/3$.
 2580 This is exactly the asserted stopped-graph conclusion.
 2581
 2582 HOSTILE CHECK:
 2583 The proof counts a shortest successful path, but bounds the UNION of
 2584 all such paths. That is sufficient to exclude every GOOD vertex in the
 2585 complete radius-two frontier. It is not a selected-path construction.
 2586 The exceptional BAND covers every possible first child in that size
 2587 range, even if it is not on a shortest path. Removing it is an upper
 2588 bound, hence safe. Its estimate uses a one-form prime progression bound,
 2589 not an assumption that shifted-prime factors behave independently.
 2590 The four switches cover all remaining q, r , and none silently applies
 2591 an asymptotic in a progression with only $O(1)$ terms. The positive $t \geq 1$
 2592 checks avoid adding one unsieved term for every parameter pair.
 2593 The equations are for $h=1$. Replacing 2 by $2h$ in (T) can introduce
 2594 nontrivial common divisors in the progression moduli; this proof is
 2595 NOT automatically a theorem for neighbouring roots or general h .
 2596 A third generation is not obtained by repeating this four-case argument:
 2597 three selected smaller parameters can already have product $> X$ on a
 2598 region of nonvanishing size. That requires a new counting input.
 2599 DEPENDENCIES: FKL Lemma 5.1, including its $k=1$ case; Chebyshev upper

2600 prime count; PNT for relative prime density. No lower prime-tuple bound.
 2601 RELATION TO GOLDBACH: Rules out a universal depth-two canonical solver
 2602 on the $h=1$ family, and shows depth-two success is rare there. It leaves
 2603 eventual success and all ordinary Goldbach representations untouched.
 2604 RELATION TO p_0 : A negative theorem about complete depth two at p_0 . The
 2605 same- N nearby-prefix theorem currently remains proved only at depth one.
 2606 WHAT REMAINS OPEN: Complete radius three, arbitrary fixed radius, and
 2607 any positive restriction forbidding a closed BAD reachable set.
 2608 NOVELTY: NOT YET AUDITED.
 2609
 2610 RESULT-ID: MULTIGENERATION-SUCCESS-PATH-PRODUCT-BARRIER-20260906
 2611 STATUS: PROVED.
 2612 EXACT STATEMENT:
 2613 Fix ϵ, δ in $(0,1)$. Let
 2614 $D(X) = \text{floor}((1-\delta) \cdot \log \log X / \log \log \log X)$.
 2615 For all but $o(X/\log X)$ primes P in $[X, 2X]$, $N=2(P-1)$, every successful
 2616 stopped path $P \rightarrow q_1 \rightarrow \dots \rightarrow q_d$ with $1 \leq d \leq D(X)$ satisfies
 2617 $\text{product}_i(i=1..d) q_i > X^{(1-\epsilon)}$,
 2618 after deleting cycles if necessary. Equivalently no simple successful
 2619 path of those lengths has product $\leq X^{(1-\epsilon)}$. The root is also
 2620 BAD outside $o(X/\log X)$ exceptions.
 2621 Consequently, if success occurs by depth D , any simple successful path
 2622 of length $d \leq D$ contains some label $> X^{((1-\epsilon)/d)}$.
 2623 This is a growing-depth restriction on actual successful paths with
 2624 moving prime labels, not on an alphabet held fixed through an iteration.
 2625
 2626 PROOF:
 2627 A successful stopped path has only BAD vertices before its GOOD endpoint.
 2628 Deleting a cycle shortens it and decreases the product of its labels,
 2629 so it is enough to count simple paths. The labels $q_1, \dots, q_d$ are then
 2630 distinct odd primes $< N/3 < 2X$. Put $M = \text{product } q_i$. The edge conditions are
 2631 $P \equiv 2 \pmod{q_1}$,
 2632 $2P \equiv 2 + q_i(i-1) \pmod{q_i}$, $i=2, \dots, d$.
 2633 By CRT they specify exactly one residue class $A \pmod{M}$. Choose
 2634 $1 \leq A \leq M$ and write $P = A + M \cdot t$. Since $M \leq X^{(1-\epsilon)}$, $t \geq 1$ and
 2635 $t \leq 2X/M$. The final GOOD condition says that
 2636 $g = 2P - 2 - q_d$
 2637 is prime. Sieve the two forms $A + M \cdot t$ and $2A - 2 - q_d + 2M \cdot t$.
 2638 They have determinant $-M(2 + q_d)$, which is nonzero. If the first
 2639 form is imprimitive, any common prime divisor belongs to $\{q_i\}$;
 2640 then a prime P in this class would equal that q_i , impossible since
 2641 all path labels are $< N/3 < P$. If the second form is imprimitive, its
 2642 odd fixed prime divisor also belongs to $\{q_i\}$; but $g > 2N/3$ is larger
 2643 than all q_i , also impossible. Thus every contributing class has two
 2644 primitive forms. Their Delta is $2M^3(q_d+2)$, bounded by a fixed power
 2645 of X . Apply the general $r=2$ sieve consumer, retaining its l^2 singular-
 2646 series bound, with $z = 2X/M \geq 2X^\epsilon$. Uniformly in d and the labels
 2647 this counts at most
 2648 $O_\epsilon(X^{(1-\epsilon)} (\log \log X)^{2/(M(\log X)^2)})$.
 2649 There is no dependence on d in that constant: there are still only TWO
 2650 prime forms, and Delta and the FKL B parameter have uniform bounds.
 2651 Let $H_X = \sum_{q \leq 2X} \text{prime} 1/q$. Dropping restrictions on distinctness and
 2652 on the product only increases the reciprocal sum, so for a given d
 2653 $\sum_{\text{eligible } q_1, \dots, q_d} 1/(q_1 \dots q_d) \leq H_X^d$.
 2654 The elementary Euler-product bound proved above gives
 2655 $H_X \leq \log \log X + O(1)$. Summing $d=1, \dots, D$ costs at most $2H_X^D$ for large X .
 2656 The total exception bound, relative to $X/\log X$, is therefore
 2657 $O_\epsilon(X^{(1-\epsilon)} (\log \log X)^{2H_X^D/\log X})$.
 2658 Now $H_X^D \leq (\log X)^{(1-\delta) + o(1)}$, so this tends to zero.
 2659 The prime root-complement case adds $O(X/(\log X)^2)$ exceptions, already
 2660 proved above. This establishes the result and its maximum-label corollary.
 2661
 2662 HOSTILE CHECK:
 2663 No claim is made that a typical complete BFS remains within this product
 2664 budget. The theorem constrains successful paths only; large-product
 2665 paths remain possible, and are the precise missing class in this count.
 2666 The product excludes the initial prime P , which would otherwise make
 2667 the condition vacuous. It includes the final GOOD label, and simple
 2668 paths are required to use pairwise coprime prime moduli. Cycle deletion
 2669 justifies that reduction for the existence question.
 2670 This is not an iteration of H1-ALL-BRANCH-INJECTIVE-ESCAPE. It replaces
 2671 that mechanism with one simultaneous CRT modulus and a prime-pair
 2672 upper bound; it has its own explicit size restriction $M \leq X^{(1-\epsilon)}$.
 2673 At radius two the preceding four-switch theorem also treats large-
 2674 product paths except for a negligible band. No analogous full treatment
 2675 has been proved here for growing D or even for radius three.
 2676 DEPENDENCIES: CRT; the uniform two-form sieve consumer; PNT only to
 2677 interpret density among prime P . No prime-tuple lower bound.
 2678 RELATION TO GOLDBACH: Gives a quantitative necessary size condition on
 2679 shallow successful paths, not an obstruction to all successful paths.
 2680 RELATION TO p_0 : Nonlocal path constraint in the $h=1$ family. It is not
 2681 proved to distinguish p_0 from p_1 for the same N .
 2682 WHAT REMAINS OPEN: Count paths whose label product is large, especially
 2683 at three or more generations.
 2684 NOVELTY: NOT YET AUDITED.
 2685
 2686 FAIL-ID: AUTOMATIC-THIRD-GENERATION-SIEVE-SWITCH-20260906
 2687 STATUS: KILLED AS AN INFERENCE; no third-generation theorem is refuted.
 2688 CLAIM / ROUTE: Repeat the four-switch proof of the radius-two theorem
 2689 without a new argument to obtain every fixed complete BAD radius.
 2690 WHY IT LOOKED PROMISING: Each edge offers a choice between fixing its
 2691 prime label and fixing its cofactor, and at depth two the smaller
 2692 choices give a long enough progression outside one negligible band.
 2693 EXACT FAILURE:
 2694 At depth three, all three labels may have size about $X^{0.4}$, with their
 2695 row cofactors about $X^{0.6}$. The product of even the smaller choices then
 2696 has size about $X^{1.2}$. The progression parameter need not have a growing
 2697 interval to sieve. A whole interval of exponent choices has this
 2698 problem; it is not confined to a shrinking band around $\sqrt{x}$ that
 2699 can be discarded as in the radius-two proof.
 2700 COUNTEREXAMPLE / OBSTRUCTION:
 2701 This is a parameter-range obstruction to the proposed proof, not an
 2702 assertion that prime paths in that range have been constructed in
 2703 positive density. Those paths must be bounded arithmetically; size
 2704 constraints alone do not eliminate them. Common factors in some moduli
 2705 could help particular patterns but have not been controlled uniformly.

```

2706 WHAT SURVIVES: The complete radius-two theorem and the growing-depth
2707 path-product barrier. Neither requires a fixed factor alphabet.
2708 DO NOT REPEAT: Do not promote the heuristic  $\log\log(N)$  branching factor
2709 and a  $1/\log N$  endpoint-prime probability to an independence theorem;
2710 do not apply a prime-tuple asymptotic with a bounded sieve variable.
2711 POSSIBLE REPLACEMENT: A simultaneous estimate for the three-generation
2712 bilinear congruences, or a different parametrization that resolves the
2713 large-product region with all degeneracies and remainder terms controlled.
2714
2715 AUDIT / PROVENANCE NOTES 2026-09-06
2716 The result named SAME-N-GROWING-ROOTS-COMPLETE-FIRST-FRONT and the later
2717 COMPLETE-FIRST-FRONT-NO-GO-GROWING-SAME-N-PREFIX are the same theorem,
2718 with the latter splitting off SAME-N-NEARBY-ROOT-FIRST-FRONT-SIEVE as a
2719 separate lemma. They must not be counted as two independent advances.
2720 Both versions are retained as incremental proof records.
2721 The older failure-log statement about a prime-tuple/codegree wall is not
2722 contradicted: the present proofs only need prime-tuple UPPER bounds to
2723 show rare shallow success. They provide no pointwise lower prime count.
2724 The old automatic envelope-iteration failure likewise remains valid;
2725 the new radius-two proof uses four explicit arithmetic parametrizations,
2726 not iteration with an enlarged smoothness envelope.
2727 A selective search in the master, Biblia and post-CRD failure log found
2728 no earlier complete-radius density theorem supplying this consumer.
2729 That search is not a literature novelty audit.
2730
2731 EXPERIMENT-ID: COMPLETE-FRONT-AND-RADIUS-TWO-AUDIT-20260906
2732 STATUS: VERIFIED COMPUTATION; exact exhaustive finite scans.
2733 METHOD:
2734 A full smallest-prime-factor sieve through 4,000,100; integer-only
2735 factor extraction; exact prime table lookups. No probable-prime tests.
2736 Scan A: all prime starts in each of [100,1000), [1000,10000),
2737 [10000,100000), [100000,1000000), [1000000,2000000). For SAME  $N=2(P-1)$ ,
2738 compare roots of ranks 0,1,2,5. Classify success at depth 0, at depth 1,
2739 or absence of success through depth 1. The last category does NOT
2740 assert success at depth 2.
2741 Scan B: all prime starts in [1000,10000), [10000,100000),
2742 [100000,200000), [200000,1000000). Classify actual shortest success
2743 at depth 0,1,2, or no success through depth 2. Enumerate all successful
2744 length-two paths after excluding earlier success, and check (T), the
2745 three necessary gcd conditions, and all six CC determinants exactly.
2746 There are 43,150 such paths in these domains; every check passed.
2747 No claim of a global-depth BFS census is made. All four switching regimes
2748 occur in the finite scan, so omitting any of them would omit real paths.
2749 The finite prime interval [1000000,2000000) has 70,435 starts. Of these,
2750 14,498 have the complete radius-two BAD property. This is about 20.6%,
2751 not close to the limiting density one; the proven bounds are asymptotic
2752 and are not being calibrated to this numerical range.
2753 The first complete radius-two BAD certificate in the scanned domain
2754 starting at  $P=1000$  is  $P=1091$ ,  $N=2180$ :
2755  $N-P=1089=3^2\cdot 11^2$ ;
2756  $N-3=2177=7\cdot 311$ ;
2757  $N-11=2169=3^2\cdot 241$ ;
2758  $N-7=2173=41\cdot 53$ ;
2759  $N-241=1939=7\cdot 277$ ;
2760  $N-311=1869=3\cdot 7\cdot 89$ .
2761 Every row at distances 0,1,2 is composite. The repeated label 3 is
2762 already at distance 1; it is not counted as a new distance-2 vertex.
2763 The certificate concerns finite depth only, not a closed BAD set.
2764 For the six historical  $N=128,1718,1862,1928,2200,6142$ , the maximal low
2765 closed BAD sets were also reconstructed exactly, and their two least
2766 labels satisfy  $b^2 < N$  and coprime rows. This is a targeted audit of the
2767 height refinement, not a new census of autonomous components.
2768 Full reproducible diagnostic code and exact output follow. These are
2769 finite verification records, not the proof of any asymptotic theorem.
2770 import numpy as np
2771 from math import isqrt, gcd
2772 from collections import Counter
2773 import json
2774 L=4000100
2775 spf=np.zeros(L+1, dtype=np.int32)
2776 for p in range(2, isqrt(L)+1):
2777     if spf[p]==0:
2778         z=spf[p*p::p]
2779         z[z==0]=p
2780 primes=np.flatnonzero(spf[2:]==0)+2
2781 isprime=np.zeros(L+1, dtype=np.bool_); isprime[primes]=True
2782
2783 def pf(n):
2784     out=[]
2785     while n>1:
2786         p=int(spf[n]) or n
2787         out.append(p)
2788         while n%p==0: n//=p
2789     return out
2790
2791 def first(N,Q):
2792     m=N-Q
2793     if isprime[m]: return 0, []
2794     qs=pf(m)
2795     good=[q for q in qs if isprime[N-q]]
2796     return (1 if good else 2), good
2797 out={}
2798 for lo,hi in [(100,1000), (1000,10000), (10000,100000), (100000,1000000), (1000000,2000000)]:
2799     inds=np.flatnonzero((primes>=lo)&(primes<hi))
2800     hist=Counter(); examples={}
2801     for idx in inds:
2802         P=int(primes[idx]); N=2*(P-1)
2803         v=tuple(first(N,int(primes[idx+k]))[0] for k in [0,1,2,5])
2804         hist[v]+=1
2805         examples.setdefault(v,N)
2806     out[f'{lo}<=P<{hi}']={ 'prime_count':len(inds), 'root_depth_capped_2_counts': [sum(c for v,c in hist.items() if v[j]==d) for d in [0,1,2]] for j
        ↪ in range(4)], 'all_four_no_success_depth_0_or_1': sum(c for v,c in hist.items() if
        ↪ min(v)==2), 'first_witness_all_four': examples.get((2,2,2,2)) }
2807 checks={}
2808 for N in [128,1718,1862,1928,2200,6142]:

```



```

2809 C=set(int(p) for p in primes if p<N/3 and N%p and not isprime[N-p])
2810 while True:
2811     D={p for p in C if set(pf(N-p))<=C}
2812     if D==C:break
2813     C=D
2814     a,b=sorted(C)[:2]
2815     assert b*b<N and gcd(N-a,N-b)==1
2816     checks[N]={ 'closed_size':len(C), 'a':a, 'b':b, 'b_squared':b*b, 'row_supports': [pf(N-a),pf(N-b)] }
2817 out['closed_pair_checks']=checks
2818 out['conventions']='Exact exhaustive sieve and trial extraction via SPF. Depth 2 means no success at depth 0 or 1, and is not an actual
↳ success-depth assertion. Roots k=0,1,2,5 refer to SAME N=2(P-1).'
```

```

2819 print(json.dumps(out,indent=2))
2820 {
2821   "100<=P<1000": {
2822     "prime_count": 143,
2823     "root_depth_capped_2_counts": [
2824       [
2825         27,
2826         79,
2827         37
2828       ],
2829       [
2830         59,
2831         50,
2832         34
2833       ],
2834       [
2835         44,
2836         65,
2837         34
2838       ],
2839       [
2840         42,
2841         57,
2842         44
2843       ]
2844     ],
2845     "all_four_no_success_depth_0_or_1": 4,
2846     "first_witness_all_four": 692
2847   },
2848   "1000<=P<10000": {
2849     "prime_count": 1061,
2850     "root_depth_capped_2_counts": [
2851       [
2852         170,
2853         517,
2854         374
2855       ],
2856       [
2857         306,
2858         394,
2859         361
2860       ],
2861       [
2862         293,
2863         398,
2864         370
2865       ],
2866       [
2867         288,
2868         409,
2869         364
2870       ]
2871     ],
2872     "all_four_no_success_depth_0_or_1": 52,
2873     "first_witness_all_four": 2024
2874   },
2875   "10000<=P<100000": {
2876     "prime_count": 8363,
2877     "root_depth_capped_2_counts": [
2878       [
2879         1019,
2880         3662,
2881         3682
2882       ],
2883       [
2884         1777,
2885         2937,
2886         3649
2887       ],
2888       [
2889         1754,
2890         2915,
2891         3694
2892       ],
2893       [
2894         1774,
2895         2930,
2896         3659
2897       ]
2898     ],
2899     "all_four_no_success_depth_0_or_1": 704,
2900     "first_witness_all_four": 20276
2901   },
2902   "100000<=P<1000000": {
2903     "prime_count": 68906,
2904     "root_depth_capped_2_counts": [
2905       [
2906         6945,
2907         27567,
2908         34394
2909       ],
2910       [
2911         12469,
2912         22233,
```

```

2913         34204
2914     ],
2915     [
2916         11973,
2917         22621,
2918         34312
2919     ],
2920     [
2921         12174,
2922         22872,
2923         33860
2924     ]
2925 ],
2926 "all_four_no_success_depth_0_or_1": 8124,
2927 "first_witness_all_four": 200096
2928 },
2929 "1000000<=P<2000000": {
2930     "prime_count": 70435,
2931     "root_depth_capped_2_counts": [
2932         [
2933             6702,
2934             26908,
2935             36825
2936         ],
2937         [
2938             11667,
2939             21869,
2940             36899
2941         ],
2942         [
2943             11365,
2944             22330,
2945             36740
2946         ],
2947         [
2948             11264,
2949             22327,
2950             36844
2951         ]
2952     ],
2953     "all_four_no_success_depth_0_or_1": 9390,
2954     "first_witness_all_four": 2000064
2955 },
2956 "closed_pair_checks": {
2957     "128": {
2958         "closed_size": 10,
2959         "a": 3,
2960         "b": 5,
2961         "b_squared": 25,
2962         "row_supports": [
2963             [
2964                 5
2965             ],
2966             [
2967                 3,
2968                 41
2969             ]
2970         ]
2971     },
2972     "1718": {
2973         "closed_size": 55,
2974         "a": 3,
2975         "b": 5,
2976         "b_squared": 25,
2977         "row_supports": [
2978             [
2979                 5,
2980                 7
2981             ],
2982             [
2983                 3,
2984                 571
2985             ]
2986         ]
2987     },
2988     "1862": {
2989         "closed_size": 55,
2990         "a": 3,
2991         "b": 5,
2992         "b_squared": 25,
2993         "row_supports": [
2994             [
2995                 11,
2996                 13
2997             ],
2998             [
2999                 3,
3000                 619
3001             ]
3002         ]
3003     },
3004     "1928": {
3005         "closed_size": 41,
3006         "a": 3,
3007         "b": 5,
3008         "b_squared": 25,
3009         "row_supports": [
3010             [
3011                 5,
3012                 7,
3013                 11
3014             ],
3015             [
3016                 3,
3017                 641
3018             ]

```

```

3019     ]
3020 },
3021 "2200": {
3022     "closed_size": 2,
3023     "a": 3,
3024     "b": 13,
3025     "b_squared": 169,
3026     "row_supports": [
3027         [
3028             13
3029         ],
3030         [
3031             3
3032         ]
3033     ],
3034 },
3035 "6142": {
3036     "closed_size": 66,
3037     "a": 3,
3038     "b": 5,
3039     "b_squared": 25,
3040     "row_supports": [
3041         [
3042             7,
3043             877
3044         ],
3045         [
3046             17,
3047             19
3048         ]
3049     ]
3050 }
3051 },
3052 "conventions": "Exact exhaustive sieve and trial extraction via SPF. Depth 2 means no success at depth 0 or 1, and is not an actual success-depth
↪ assertion. Roots k=0,1,2,5 refer to SAME N=2(P-1)."
3053 }
3054 exec(open('diagnostic_20260906.py').read().split('out={}')[0])
3055 from fractions import Fraction
3056
3057 def radius2(N,P):
3058     if isprime[N-P]:return 0,0,[]
3059     qs=pf(N-P)
3060     if any(isprime[N-q] for q in qs):return 1,0,[]
3061     successes=[]
3062     cases=Counter()
3063     for q in qs:
3064         a=(P-2)//q
3065         for r in pf(N-q):
3066             if not isprime[N-r]:continue
3067             b=(N-q)//r
3068             assert q|r and a>=3 and b>=3 and a%2==b%2==1
3069             assert b*r==(2*a-1)*q+2 and P==a*q+2
3070             g=N-r
3071             successes.append((q,r,a,b,g))
3072             assert gcd(q,b)==1
3073             assert gcd(r,2*a-1)==1
3074             assert gcd(b,2*a-1)==1
3075             A=(-2*pow(2*a-1,-1,b))%b
3076             if A==0:A=b
3077             R=((2*a-1)*A+2)//b
3078             forms=[(b,A),(2*a-1,R),(a*b,a*A+2),(2*a*b-2*a+1,2*a*A+2-R)]
3079             pairs=[(0,2),(0,1),(0,3),(2,1),(2,3),(1,3)]
3080             expected=[2*b,2*(b-1),2*(1-a),-2*(a*(b-1)+1),-2]
3081             actual=[forms[i][0]*forms[j][1]-forms[j][0]*forms[i][1] for i,j in pairs]
3082             assert actual==expected and all(actual)
3083     return (2 if successes else 3),len(successes),successes
3084
3085 out={}; cert=None; totalpaths=0
3086 for lo,hi in [(1000,10000),(10000,100000),(100000,200000),(1000000,2000000)]:
3087     ps=primes[(primes>=lo)&(primes<hi)]
3088     c=Counter(); pathcount=0; regimes=Counter(); smallest={}
3089     for pp in ps:
3090         P=int(pp);N=2*(P-1)
3091         d,n,rows=radius2(N,P);c[d]+=1;pathcount+=n
3092         smallest.setdefault(d,N)
3093         for q,r,a,b,g in rows:
3094             regime=('L' if q*q<=lo else 'C')+('L' if r*r<=lo else 'C')
3095             regimes[regime]+=1
3096         if d==3 and cert is None:
3097             frontier1=pf(P-2)
3098             frontier2=sorted(set(r for q in frontier1 for r in pf(N-q)))
3099             cert={'P':P,'N':N,'root_row':pf(P-2),'first_rows':{q:pf(N-q) for q in frontier1},'second_rows':{r:pf(N-r) for r in frontier2}}
3100             assert all(not isprime[N-v] and N-v>1 for v in [P]+frontier1+frontier2)
3101             out['f{lo}<=P<{hi}']={'prime_count':len(ps),'exact_success_depth_0_1_2_or_no_success_through_2':[c[j] for j in range(4)],'number_of_depth2_suc
↪ cessful_paths_after_excluding_depth0_1':pathcount,'parameter_regimes_at_sqrt_lo':dict(regimes),'first_N_in_each_depth_category':smallest}
3102     totalpaths+=pathcount
3103 out['certificate_complete_BAD_radius_two']=cert
3104 out['all_radius2_identity_and_six_CC_determinant_checks']=f'passed for {totalpaths} actual successful length-two paths'
3105 print(json.dumps(out,indent=2))
3106 {
3107     "1000<=P<10000": {
3108         "prime_count": 1061,
3109         "exact_success_depth_0_1_2_or_no_success_through_2": [
3110             170,
3111             517,
3112             251,
3113             123
3114         ],
3115         "number_of_depth2_successful_paths_after_excluding_depth0_1": 397,
3116         "parameter_regimes_at_sqrt_lo": {
3117             "CL": 85,
3118             "CC": 70,
3119             "LL": 109,
3120             "LC": 133
3121         },

```

```

3122     "first_N_in_each_depth_category": {
3123         "1": 2016,
3124         "2": 2024,
3125         "0": 2040,
3126         "3": 2180
3127     }
3128 },
3129 "10000<=P<100000": {
3130     "prime_count": 8363,
3131     "exact_success_depth_0_1_2_or_no_success_through_2": [
3132         1019,
3133         3662,
3134         2363,
3135         1319
3136     ],
3137     "number_of_depth2_successful_paths_after_excluding_depth0_1": 3770,
3138     "parameter_regimes_at_sqrt_lo": {
3139         "LL": 1321,
3140         "LC": 1123,
3141         "CL": 746,
3142         "CC": 580
3143     },
3144     "first_N_in_each_depth_category": {
3145         "2": 20012,
3146         "0": 20016,
3147         "1": 20120,
3148         "3": 20324
3149     }
3150 },
3151 "100000<=P<200000": {
3152     "prime_count": 8392,
3153     "exact_success_depth_0_1_2_or_no_success_through_2": [
3154         936,
3155         3481,
3156         2543,
3157         1432
3158     ],
3159     "number_of_depth2_successful_paths_after_excluding_depth0_1": 4103,
3160     "parameter_regimes_at_sqrt_lo": {
3161         "LL": 1835,
3162         "CC": 426,
3163         "LC": 1056,
3164         "CL": 786
3165     },
3166     "first_N_in_each_depth_category": {
3167         "1": 200004,
3168         "3": 200084,
3169         "2": 200096,
3170         "0": 200304
3171     }
3172 },
3173 "1000000<=P<2000000": {
3174     "prime_count": 70435,
3175     "exact_success_depth_0_1_2_or_no_success_through_2": [
3176         6702,
3177         26908,
3178         22327,
3179         14498
3180     ],
3181     "number_of_depth2_successful_paths_after_excluding_depth0_1": 34880,
3182     "parameter_regimes_at_sqrt_lo": {
3183         "LL": 16192,
3184         "CL": 6598,
3185         "LC": 8591,
3186         "CC": 3499
3187     },
3188     "first_N_in_each_depth_category": {
3189         "1": 2000004,
3190         "3": 2000064,
3191         "0": 2000076,
3192         "2": 2000160
3193     }
3194 },
3195 "certificate_complete_BAD_radius_two": {
3196     "P": 1091,
3197     "N": 2180,
3198     "root_row": [
3199         3,
3200         11
3201     ],
3202     "first_rows": {
3203         "3": [
3204             7,
3205             311
3206         ],
3207         "11": [
3208             3,
3209             241
3210         ]
3211     },
3212     "second_rows": {
3213         "3": [
3214             7,
3215             311
3216         ],
3217         "7": [
3218             41,
3219             53
3220         ],
3221         "241": [
3222             7,
3223             277
3224         ],
3225         "311": [
3226             3,
3227             7,

```

```

3228         89
3229     }
3230 }
3231 },
3232 "all_radius2_identity_and_six_CC_determinant_checks": "passed for 43150 actual successful length-two paths"
3233 }
3234
3235 EXTERNAL-INPUT: PINTZ-GOLDBACH-EXCEPTIONAL-SET
3236 STATUS: Exact theorem statement and hypotheses checked in primary paper.
3237 Janos Pintz, A new explicit formula in the additive theory of primes with
3238 applications II. The exceptional set in Goldbach's problem, Theorem 1,
3239 arXiv:1804.09084v2 (30 April 2018), printed p.2.
3240 https://arxiv.org/pdf/1804.09084
3241 Writing  $E(Y)$  for the number of even  $n \leq Y$  not expressible as two primes,
3242 the theorem states  $E(Y) < Y^{0.72}$  for  $Y$  above an ineffective constant.
3243 Only the consequence  $E(4X) = o(X/\log X)$  is used below. This run checked
3244 the statement, not the entire 45-page proof and its prerequisite papers.
3245 The exponent is an inherited external theorem, not improved in this run.
3246
3247 RESULT-ID: GOLDBACH-TRUE-WITH-COMPLETE-SHALLOW-BAD-20260906
3248 STATUS: PROVED, using the preceding external exceptional-set theorem.
3249 EXACT STATEMENT:
3250 For relative density one of primes  $P$  in  $[X, 2X]$ ,  $N=2(P-1)$  simultaneously:
3251 (a) has an ordinary binary Goldbach representation;
3252 (b) has no GOOD vertex through depth two from its canonical root  $P$ ;
3253 (c) has no GOOD vertex through depth one from any of the first  $K+1$ 
3254 roots, whenever  $K = o(\log X / (\log \log X)^3)$ .
3255 For fixed  $\epsilon, \delta$  in  $(0, 1)$ , the growing-depth path-product barrier
3256 may also be imposed simultaneously. In particular there are infinitely
3257 many genuine Goldbach numbers with all these shallow BAD properties.
3258 PROOF:
3259 The map  $P \rightarrow 2(P-1)$  is injective and its values lie below  $4X$ . Hence the
3260 number of starting primes violating (a) is at most  $E(4X) = O(X^{0.72})$ ,
3261 which is  $o(X/\log X)$ . The exceptions to (b), (c), and the optional
3262 path-product assertion are each  $o(X/\log X)$  by the proved results above.
3263 The union of these finitely many exceptional sets remains  $o(X/\log X)$ ,
3264 whereas PNT gives  $\sim X/\log X$  prime starts. This proves the statement.
3265 HOSTILE CHECK:
3266 This does not claim that their complete canonical traversals eventually
3267 succeed: the Goldbach pair in (a) might be outside the reachable set.
3268 It does prove that a complete shallow BAD neighbourhood is compatible
3269 with ordinary Goldbach truth on an asymptotically full part of this
3270 prime-parametrized family. No theorem requiring global failure is being
3271 applied to such ordinary successful  $N$ .
3272 DEPENDENCIES: The new shallow-front theorems plus the stated external
3273 exceptional-set bound. No pointwise Goldbach assertion is smuggled in.
3274 RELATION TO  $p_0$ : Reinforces the distinction between generic shallow BADness
3275 and whatever deeper canonical-root phenomenon the computations suggest.
3276 WHAT REMAINS OPEN: Eventual canonical reachability, and binary Goldbach
3277 for every even  $N$ . NOVELTY: NOT YET AUDITED.
3278
3279 FINAL HOSTILE AUDIT 2026-09-06
3280 1. The radius-two proof's weakest step was availability of a long sieve
3281 variable. All four parametrizations explicitly have fixed-parameter
3282 product  $\leq 8X^{1-\epsilon}$ . The only removed first-factor range has mass
3283  $O(\epsilon^{1/\log X})$  in the prime-start sample.  $\epsilon \log X$  grows for the
3284 quantitative choice, so all variable-length and coefficient-size
3285 comparisons used by the sieve remain valid.
3286 2. Primitive forms and nonzero determinants were checked separately in
3287 every use, including the nearby-root degeneracy  $h+1=2a$ . An upper
3288 prime-tuples bound was never silently used for coincident forms.
3289 3. Source uniformity comes from FKL's explicit B condition. A generic
3290 theorem with constants depending arbitrarily on coefficients would
3291 not justify the switching proofs or their growing prefixes.
3292 4. The sample throughout the analytic continuation is  $P$  prime, with
3293  $N=2(P-1)$ . It is not the uniform-even- $N$  sample of the earlier transfer
3294 theorem. No unproved change of sampling measure has been made.
3295 5. Shallow BADness is a union-over-all-successful-paths conclusion,
3296 hence controls the complete BFS radius in question. It does not
3297 control its remaining radii or the final closed reachable set.
3298 6. The  $h=1$  radius-two switching result is not automatically shared by
3299  $p_1$  for the same  $N$ ; no  $p_0$  advantage may be inferred from the absence
3300 of that extension. The full first-front theorem IS explicitly shared
3301 by a growing same- $N$  nearby prefix.
3302 7. Numerical scans verify exact identities and finite classifications;
3303 they do not verify the limiting densities. At the scanned scales,
3304 many depth-two successes still occur. The upper bounds were not
3305 fitted to the data and no finite threshold was claimed.
3306 8. The height refinement is a necessary condition shared by any closed
3307 active BAD set. It neither excludes arbitrary moving support nor
3308 repairs the killed automatic envelope iteration.
3309 9. Literature novelty has not been audited. The mathematical proofs
3310 have been locally stress-tested in this run, but have not received
3311 a separate external adversarial review.
3312 Reproduction note: in the diagnostic section, save the first new Python
3313 block as diagnostic_20260906.py and the second as audit_radius2_20260906.py;
3314 the intervening JSON blocks are their exact outputs. Both use numpy.
3315 No additional diagnostic files are required as permanent deliverables.
3316
3317 =====
3318 STRONGEST NEW RESULTS
3319 =====
3320 1. COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS: almost every prime start  $P$  in
3321 the  $h=1$  family has its entire radius-two neighbourhood BAD, with
3322 exceptional proportion  $O((\log \log X)^3 / \log X)^{(1/5)}$ . Four explicit
3323 sieve switches control both generations, including large factors.
3324 2. SAME-N-GROWING-ROOTS-COMPLETE-FIRST-FRONT: for almost all such  $P$ ,
3325 every root of rank  $\leq K = o(\log X / (\log \log X)^3)$  has its entire first
3326 front BAD. This is a same- $N$  comparison, not a selected CRT tree.
3327 3. MULTIGENERATION-SUCCESS-PATH-PRODUCT-BARRIER: up to depth
3328  $(1-\delta)\log \log X / \log \log \log X$ , successful simple paths must have
3329 product of nonroot labels  $> X^{1-\epsilon}$ , outside  $o(X/\log X)$  starts.
3330 4. TWO-SMALLEST-ROWS-HEIGHT: every active closed BAD set has two least
3331 labels below  $\sqrt{N}$  with coprime rows, yielding a height constraint
3332 on their actual joint support without a pure-row assumption.
3333 The first three statements coexist with ordinary Goldbach truth for

```

3334 relative density one of the $h=1$ prime starts, by a known exceptional-set
3335 bound. That external Goldbach bound has not been improved here.
3336
3337 =====
3338 CURRENT FRONTIER
3339 =====
3340 No proof of binary Goldbach, no pointwise Goldbach threshold, and no
3341 proof of universal canonical traversal success. The new analytic results
3342 settle full shallow BADness through radius two in the $h=1$ family in a
3343 relative-density-one sense. They do not yet settle every fixed radius.
3344 The radius-three large-product parameter region is not controlled by
3345 the current one-variable sieve switching. Closed BAD reachable sets
3346 with moving large support remain unexcluded. No upper support bound
3347 contradicts the new or inherited necessary height bounds.
3348
3349 =====
3350 p_0 STATUS
3351 =====
3352 No nontrivial property proved for every p_0 with a demonstrated same- N
3353 failure at p_1 . A substantial SAME- N negative theorem was proved: complete
3354 first-front BADness is shared by p_0 and a growing nearby prefix in the
3355 $h=1$ family. Complete radius-two BADness was proved asymptotically for
3356 p_0 there, but no conclusion comparing the radius-two laws of p_0 and p_1
3357 is justified yet. The empirical eventual-traversal exceptionality of
3358 p_0 remains unexplained.
3359
3360 =====
3361 EXACT NEXT QUESTION
3362 =====
3363 Is the number of primes P in $[X, 2X]$ for which the canonical root of
3364 $N=2(P-1)$ reaches a GOOD vertex within THREE edges $o(X/\log X)$?
3365 A proof must control the complete third frontier, especially successful
3366 paths with large product of labels for which all naive choices of
3367 fixed labels or cofactors leave too short a sieve variable.
3368
3369 =====
3370 CONTINUATION 2026-09-06B
3371 =====
3372 The previous run's end sections are archival. This continuation audits
3373 its radius-two theorem and investigates a deeper SAME- N comparison.
3374 No pointwise Goldbach or eventual p_0 -reachability theorem is assumed.
3375
3376 RESULT-ID: AUDIT-RADIUS-TWO-INPUT-20260906B
3377 STATUS: AUDITED; previous theorem retained as PROVED.
3378 The four parametrizations in COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS were
3379 rechecked directly. The first and second cofactors are odd and ≥ 3 ;
3380 $q|r$ follows from activity; after removing the first-factor band each
3381 fixed-parameter product is $\leq 8X^{1-\epsilon}$. Prime values exceed their
3382 corresponding positive leading coefficients for sufficiently large X ,
3383 including $\epsilon = ((\log \log X)^3 / \log X)^{1/5}$. Therefore discarded
3384 imprimitive forms truly have no actual prime solutions. The six CC
3385 determinants agree with the written formulas. No fixed-depth/global-
3386 failure scope change is needed.
3387 The exact FKL Lemma 5.1 was reopened in the primary author paper,
3388 <https://arxiv.org/pdf/0904.0473>, printed pp.12--13. Its rank/size
3389 hypotheses and explicit B term also permit FIVE fixed linear forms;
3390 no new external theorem is required for that extension.
3391 The potential new obstacle at later roots is real: the equations now
3392 contain $2h$ rather than 2 , so $\gcd(b, 2a-1)$ can be a nontrivial divisor
3393 of h . The following lemma controls its total contribution instead of
3394 assuming it away.
3395
3396 RESULT-ID: WEIGHTED-COMPATIBLE-GCD-AVERAGE-20260906B
3397 STATUS: PROVED.
3398 EXACT STATEMENT:
3399 For $A, B \geq 2$ and an odd positive integer h , define
3400 $W_h(A, B) = \sum_{1 \leq a \leq A, 1 \leq b \leq B; \gcd(b, 2a-1) | h}$
3401 $\quad \gcd(b, 2a-1) / (a \cdot b)$.
3402 Then, with an absolute constant C ,
3403 $W_h(A, B) \leq C \cdot \log(2A) \cdot \log(2B) \cdot \sigma_{-1}(h)$,
3404 where $\sigma_{-1}(h) = \sum_{e|h} 1/e$. Consequently, for $H \geq 1$,
3405 $\sum_{1 \leq h \leq H, h \text{ odd}} W_h(A, B)$
3406 $\leq C \cdot \zeta(2) \cdot H \cdot \log(2A) \cdot \log(2B)$.
3407 Restricting a, b to odd integers ≥ 3 preserves these upper bounds.
3408 Also, for any odd prime r ,
3409 $\sum_{1 \leq a \leq A; 2a \equiv 1 \pmod r} 1/a \leq C \cdot \log(2A)/r$,
3410 and
3411 $\sum_{h \leq H} \sum_{r|h, r \text{ prime}} 1/r \leq H \cdot \sum_r 1/r^2 \leq H \cdot \zeta(2)$.
3412
3413 PROOF:
3414 For $d = \gcd(b, 2a-1)$, the elementary identity $d = \sum_e d \phi(e)$ holds.
3415 If $d|h$, all divisors e of d divide h . Dropping the condition $d|h$ after
3416 expanding gives
3417 $W_h \leq \sum_{e|h} \phi(e)$
3418 $\quad \left[\sum_{b \leq B, e|b} 1/b \right]$
3419 $\quad \left[\sum_{a \leq A, 2a \equiv 1 \pmod e} 1/a \right]$.
3420 All e here are odd. The first bracket is $\leq (1 + \log B)/e$. For $e \geq 3$,
3421 the least positive solution of $2a \equiv 1 \pmod e$ is $(e+1)/2$. The subsequent
3422 solutions are spaced e apart. Hence its reciprocal sum is at most
3423 $2/e + (1/e) \cdot \log(2A)$, by comparison with the integral of $1/(e \cdot t + (e+1)/2)$.
3424 For $e=1$ the ordinary harmonic bound gives the same assertion with an
3425 absolute constant. Thus the second bracket is $\leq C \cdot \log(2A)/e$.
3426 Since $\phi(e) \leq e$, this proves the first bound. Finally
3427 $\sum_{h \leq H} \sigma_{-1}(h)$
3428 $= \sum_{e \leq H} (1/e) \cdot \text{floor}(H/e) \leq H \cdot \sum_{e \geq 1} 1/e^2 \leq H \cdot \zeta(2)$.
3429 The prime- r variants follow from the same progression estimate and
3430 by summing $\text{floor}(H/r)/r$. All quantities are nonnegative, so the
3431 additional parity and lower-end restrictions can be dropped safely.
3432
3433 HOSTILE CHECK:
3434 The hypothesis $\gcd(b, 2a-1) | h$ is essential to the stated per- h estimate;
3435 there is no claimed uniform $O(\log A \log B)$ bound for the unrestricted
3436 $\gcd$ sum. The proof does not replace a moving divisor by a fixed one.
3437 It sums every compatible divisor and retains its exact multiplicative
3438 weight before averaging h . No prime distribution or Goldbach premise
3439 is used in this lemma.

3440 DEPENDENCIES: gcd=sum phi, elementary harmonic estimates.
 3441 IMPLICATION: Compensates for the d-fold increase in progression length
 3442 when $b+r=(2a-1)*q+2h$ has $\gcd(b,2a-1)=d>1$. This is a real consumer
 3443 of the compatibility relation dh, not a collision-to-new-support claim.
 3444 RELATION TO GOLDBACH: Counting tool only, not a positivity theorem.
 3445 RELATION TO p0: Makes deeper same-N nearby-root comparisons possible.
 3446 WHAT REMAINS OPEN: Verify every accompanying prime-form and exceptional
 3447 progression case in that comparison. NOVELTY: NOT YET AUDITED.
 3448
 3449 RESULT-ID: DISTINCT-PRIME-FORMS-DEGENERACY-FILTER-20260906B
 3450 STATUS: PROVED; elementary lemma used to audit five-form switches.
 3451 EXACT STATEMENT:
 3452 Suppose a parametrized arithmetic configuration requires k integer
 3453 linear forms $L_i(t)=u_i*t+v_i$, with $u_i>0$, to take pairwise DISTINCT
 3454 prime values, each greater than u_i . If one form is imprimitive, or
 3455 if a pair of primitive forms has zero determinant $u_i*v_j-u_j*v_i$,
 3456 then the configuration has no solutions.
 3457 PROOF:
 3458 If $d=\gcd(u_i,v_i)>1$, it divides every value and satisfies $d\leq u_i\leq L_i(t)$;
 3459 no such value is prime. If two primitive coefficient vectors are
 3460 proportional with positive ratio lambda, Bezout for each vector shows
 3461 both lambda and $1/\lambda$ are positive integers. Hence $\lambda=1$ and
 3462 the forms are identical. Their values cannot be distinct. This proves
 3463 both exclusions. A zero constant term causes no exception to Bezout.
 3464 HOSTILE CHECK: The coefficient-versus-value comparison and DISTINCT
 3465 prime-value requirement must be proved from the actual configuration.
 3466 Neither is automatic for arbitrary linear forms. The filter excludes
 3467 parameter choices with no actual solutions; it does not count their
 3468 solutions by an incorrectly high-dimensional sieve.
 3469 DEPENDENCIES: Bezout and primality. NOVELTY: NOT YET AUDITED.
 3470
 3471 SIEVE EXTENSION 2026-09-06B:
 3472 UNIFORM-FEW-FORMS-SIEVE-CONSUMER applies unchanged to $k=5$ (as well as
 3473 $k=2,3,4$): the primary FKL lemma permits every fixed k. Write
 3474 $L=\log X$, $l=\log\log X$. If coefficients and nonzero determinants are
 3475 bounded by fixed powers of X , $\log z\leq c\eta*L$, $z\leq X^C$, and
 3476 $L^{-(1/4)}\leq \eta\leq 1/8$, then $B_{FKL}=O(1)=o(\eta*L)$. Thus the sieve count
 3477 is $O(z*S/(\eta*L)^k)$, with an absolute constant when $k\leq 5$ and c,C
 3478 are fixed. The earlier Euler-product proof gives $S=O(l^{-(k-1)})$ when
 3479 $nu(ell)>1$ at every prime. The same bound survives at most TWO
 3480 exceptional odd primes where nu may vanish, since each extra local
 3481 factor is bounded by a fixed power of $3/2$. No fixed-coefficient-only
 3482 asymptotic is substituted for the uniform FKL statement.
 3483
 3484 RESULT-ID: SAME-N-RADIUS-TWO-PAIR-COUNT-20260906B
 3485 STATUS: PROVED.
 3486 EXACT STATEMENT:
 3487 Let $L=\log X$ and $l=\log\log X$. Assume X sufficiently large,
 3488 $3\leq H\leq L^2$, $L^{-(1/4)}\leq \eta\leq 1/8$.
 3489 Let $T_2(X,H)$ be the number of pairs of primes (P,Q) with
 3490 $X\leq P\leq 2X$, $P<Q$, $h=Q-P+1\leq H$,
 3491 such that in the graph for $N=2(P-1)$, the stopped traversal rooted at Q
 3492 has a GOOD vertex at distance ≤ 2 . Then, with an absolute constant,
 3493 $T_2(X,H) \ll H*X*[1*(\eta+1/L)/L^2 + 1^4/(\eta^5*L^3)]$. (PAIR2)
 3494 In particular, taking $\eta=1^{(1/2)}/L^{(1/6)}$,
 3495 $T_2(X,H) \ll H*X*1^{(3/2)}/L^{(13/6)}$. (PAIR2*)
 3496 These count pairs (P,Q) , not paths with multiplicity. No rank restriction
 3497 or consecutiveness assumption on Q is imposed.
 3498
 3499 PROOF:
 3500 All offsets h in the count are odd and ≥ 3 . Also $Q\leq 2X+H\leq 3X$. The
 3501 known same-N first-front pair sieve counts success at distances 0 or 1
 3502 by $O(H*X*1^{3/3}/L^3)$, which is absorbed by (PAIR2). Hence consider a
 3503 shortest successful path $Q \rightarrow q \rightarrow r$ with Q and q BAD and r GOOD.
 3504 Write
 3505 $m=N-Q=Q-2h=a*q$,
 3506 $N-q=b*r$,
 3507 $g=N-r$.
 3508 Then a,b are odd integers ≥ 3 , and
 3509 $Q=a*q+2h$, $P=a*q+h+1$,
 3510 $b*r=(2a-1)*q+2h$, $g=2a*q+2h-r$. (E)
 3511 The prime values q,r,P,Q,g are pairwise distinct: $q\neq r$ by activity,
 3512 $q,r\leq N/3$; $P=(N+2)/2$; $Q=P+h-1=P+o(X)$; and $g\geq 2N/3>Q$ for large X .
 3513 Thus $g>Q>P>\max(q,r)$, apart from the already separately ordered q,r .
 3514 In particular every tuple of variable prime forms below really must
 3515 have distinct values.
 3516
 3517 BAND REMOVAL:
 3518 For each h remove any pair whose root complement has a prime divisor
 3519 q in $[X^{(1/2-\eta)}, X^{(1/2+\eta)}]$.
 3520 For fixed q , the two forms $q*a+2h$ and $q*a+h+1$ must be prime.
 3521 Here $q>2H$ for large X , so both forms are primitive. Their determinant
 3522 $q*(1-h)$ is nonzero. Except at $ell=q$ there is a root modulo every prime;
 3523 the two-form singular series is $O(1)$. Sieving $a\leq 3X/q$, whose logarithm
 3524 is at least $3L/8$, gives $O(X*1/(q*L^2))$. By partial summation from the
 3525 Chebyshev bound $\pi(t)\ll t/\log t$, the reciprocal sum over this band is
 3526 $O(\eta+1/L)$. Sum $h\leq H$ to bound the removed PAIRS by
 3527 $O(H*X*1*(\eta+1/L)/L^2)$. (B2)
 3528 Removing all such pairs only enlarges the exceptional set; the divisor
 3529 need not actually belong to the successful path being counted.
 3530
 3531 For a remaining pair choose any shortest successful length-two path.
 3532 Its q is either below $X^{(1/2-\eta)}$ or above $X^{(1/2+\eta)}$. Split also at
 3533 $r=\sqrt{X}$. Use the following four counts to bound the existence of such
 3534 a path. Parameters outside the indicated arithmetic compatibility
 3535 conditions have no actual solutions.
 3536
 3537 CASE LL: $q<X^{(1/2-\eta)}$, $r\leq\sqrt{X}$. Fix h and distinct odd primes q,r .
 3538 The edge conditions give
 3539 $Q=2h$ modulo q , $2Q=2h+q$ modulo r .
 3540 They specify one class A modulo $q*r$. Choose $1\leq A\leq q*r$ and write
 3541 $Q=A+q*r*t$, $P=Q-h+1$, $g=2Q-2h-r$.
 3542 Here $t\geq 1$ and $t\leq 3X/(q*r)$, since $q*r\leq X$ and $Q>X$. These are THREE
 3543 integer prime forms with positive leading coefficients $q*r, q*r, 2q*r$.
 3544 All coefficients are $o(X)$, uniformly for $\eta>L^{-(1/4)}$, and smaller
 3545 than their corresponding actual prime values. By the degeneracy filter,

```

3546 nonprimitive forms or zero pair determinants imply no actual solution
3547 and are discarded. (Directly the determinants are proportional to
3548  $h-1$ ,  $2h+r$ , and  $r+2$ , all nonzero.) Every prime outside  $\{q,r\}$  has an
3549 invertible leading coefficient; at 2,  $q*r$  is odd. Thus  $S=O(l^2)$ .
3550 The uniform three-form bound is
3551  $O(X^{1/2}/(q*r*(\eta*L)^3))$ .
3552 Summing reciprocals of the two prime parameters costs  $O(l^2)$ , giving
3553  $O(X^{1/4}/(\eta^3*L^3))$  for each  $h$ .
3554
3555 CASE LC:  $q < X^{(1/2-\eta)}$ ,  $r > \sqrt{X}$ . Fix  $h$ , prime  $q$ , and odd
3556  $b=(N-q)/r < 4\sqrt{X}$ . The congruences are
3557  $Q \equiv 2h \pmod{q}$ ,  $2Q \equiv 2h+q \pmod{b}$ .
3558 They require  $\gcd(q,b)=1$ . Indeed  $q|b$  would imply  $q|2h$  by (E), and
3559 then  $q|Q=a*q+2h$ , impossible since  $Q$  is prime and  $q < Q$ .
3560 Thus there is a unique class  $A$  modulo  $q*b$ . Write
3561  $Q=A+q*b*t$ ,  $P=Q-h+1$ ,
3562  $r=(2A-2h-q)/b+2q*t$ ,  $g=2Q-2h-r$ .
3563 The FOUR positive leading coefficients are  $q*b, q*b, 2q, 2q*(b-1)$ .
3564 We have  $q*b < 4X^{(1-\eta)}$ , and  $2q < \sqrt{X}$  for large  $X$ . Thus every
3565 coefficient is smaller than its actual prime value. The degeneracy
3566 filter removes all nonprimitive or coincident forms without losing a
3567 possible solution. The coefficient  $2q$  of  $r$  is invertible at every
3568 odd prime except  $q$ , while  $Q$  has odd leading coefficient at 2.
3569 Consequently  $S=O(l^3)$ . The variable satisfies
3570  $1 < t < 3X/(q*b)$ ,  $\log(3X/(q*b)) > \eta*L-O(1)$ .
3571 The four-form count is  $O(X^{1/3}/(q*b*(\eta*L)^4))$ . Sum the prime  $q$ 
3572 reciprocals and integer  $b$  reciprocals to obtain
3573  $O(X^{1/4}/(\eta^4*L^3))$  for each  $h$ .
3574
3575 CASE CL:  $q > X^{(1/2+\eta)}$ ,  $r < \sqrt{X}$ . Fix  $h$ , odd
3576  $a=(Q-2h)/q < 3X^{(1/2-\eta)}$ ,
3577 and the prime  $r$ . If  $\gcd(r,2a-1)=1$ , the equation (E) gives one class
3578  $q=A+r*t$ ,  $1 < A < r$ , with  $t \geq 1$  and  $t < 3X/(a*r)$ . Sieve the FOUR forms
3579  $q$ ,  $Q=a*q+2h$ ,  $P=a*q+h+1$ ,  $g=2a*q+2h-r$ .
3580 Their  $t$ -coefficients are  $r, a*r, a*r, 2a*r$ , smaller than the actual prime
3581 values because  $a*r < 3X^{(1-\eta)}$  and  $q > \sqrt{X} > r$ . The degeneracy filter
3582 again applies. Outside  $\ell=r$  the first form has an invertible leading
3583 coefficient, giving  $S=O(l^3)$ . Summing the bound
3584  $O(X^{1/3}/(a*r*(\eta*L)^4))$ 
3585 over integer  $a$  and prime  $r$  gives  $O(X^{1/4}/(\eta^4*L^3))$  for each  $h$ .
3586
3587 CL EXCEPTION (must not be omitted): If  $r|2a-1$ , compatibility forces
3588  $r|2h$  and hence  $r|h$ . The progression condition on  $q$  then degenerates.
3589 For these pairs it suffices to DROP  $q,a$  and count the necessary prime
3590 conditions directly:  $P, Q=P+h-1$ , and  $g=2P-2-r$  are all prime for some
3591 odd prime  $r|h$ . These three forms are primitive and their determinants
3592 are nonzero ( $h-1, -(r+2), -(2h+r)$ ). At every prime there is a root of
3593 the form  $P$ , so  $S=O(l^2)$ . Since  $P < 2X$ , the count for each  $h,r$  is
3594  $O(X^{1/2}/L^3)$ . Sum over  $h < H$  and  $r|h$ :
3595  $\text{sum}_{(h \leq H)} \omega(h) < \text{sum}_{(r \leq H \text{ prime})} \text{floor}(H/r) < H * \log \log(3H)$ .
3596 Hence this exceptional branch contributes
3597  $O(H * X^{1/2} * \log \log(3H) / L^3) = O(H * X^{1/3} / L^3)$ .
3598 This branch counts PAIRS directly, not all paths with multiplicity.
3599 That is sufficient for the stated union/existence problem.
3600
3601 CASE CC:  $q > X^{(1/2+\eta)}$ ,  $r > \sqrt{X}$ . Fix  $h$ , odd
3602  $a < 3X^{(1/2-\eta)}$ ,  $b < 4\sqrt{X}$ .
3603 Put  $c=2a-1$  and  $d=\gcd(b,c)$ . By (E),  $d|2h$ , hence  $d|h$  because  $d$  is odd.
3604 No assumption  $d=1$  is made. Set  $b'=b/d$ ,  $c'=c/d$ ,  $h'=h/d$ , so
3605  $\gcd(b',c')=1$ ,  $b'*r=c'*q+2h'$ .
3606 There is exactly one class  $q=A+b'*t$ , with  $1 < A < b'$ . Put
3607  $R=(c'*A+2h')/b'$ , an integer. Then the FIVE prime forms in  $t$  are
3608  $q=A+b'*t$ ,
3609  $r=R+c'*t$ ,
3610  $Q=a*A+2h+a*b'*t$ ,
3611  $P=a*A+h+1+a*b'*t$ ,
3612  $g=2a*A+2h-R+(2a*b'-c')*t$ .
3613 The last leading coefficient is positive since  $b' \geq 3$  and
3614  $2a*b'-c'=[2a*(b-1)+1]/d > 0$ . Moreover  $q > b' > b$  and  $r > c' > c$  for
3615 large  $X$ ; the  $Q,P,g$  coefficients are  $O(a*b)=O(X^{(1-\eta)})$  and are
3616 smaller than their actual prime values. Thus the degeneracy filter
3617 rigorously excludes every imprimitive or proportional case. For all
3618 remaining parameter choices  $\Delta$  is nonzero and polynomially bounded
3619 in  $X$ . The coprime coefficients  $b',c'$  guarantee  $\nu(\ell) > 1$  at every
3620 prime. Therefore  $S=O(l^4)$ .
3621 Here  $t \geq 1$  and
3622  $t < 3X/(a*b')=3dX/(a*b)$ ,
3623 whose logarithm is  $> \eta*L-O(1)$ . The five-form bound for fixed  $h,a,b$  is
3624  $O(d*X^{1/4}/(a*b*(\eta*L)^5))$ .
3625 Crucially, the  $d$  in this count is retained. Summing all compatible  $a,b$ 
3626 using WEIGHTED-COMPATIBLE-GCD-AVERAGE gives at most
3627  $O(X^{1/4} * \sigma_{-1}(h) / (\eta^5 * L^3))$ 
3628 for this  $h$ , because the two logarithmic harmonic sums are  $O(l^2)$ .
3629 Sum  $h < H$ . The identity
3630  $\text{sum}_{(h \leq H)} \sigma_{-1}(h) < \zeta(2) * H$ 
3631 makes the TOTAL CC bound
3632  $O(H * X^{1/4} / (\eta^5 * L^3))$ ,
3633 with no growing extra divisor-count factor.
3634
3635 In every case the representatives, positive leading coefficients and
3636 constant terms have absolute values bounded by fixed powers of  $X$ .
3637 After nondegenerate primitive filtering,  $\Delta$  has the same property.
3638 The variable logarithms are  $> c*\eta*L$ , while  $B_{FKL}=O(l)=o(\eta*L)$ ,
3639 uniformly in the stipulated  $\eta$  interval. Actual prime values in the
3640 sieve exceed the number of forms (fixed  $q,r$  are not sieved as variables).
3641 Thus all FKL applications are uniform, including  $k=5$ . Add LL, LC, CL,
3642 the CL exception, CC, BAND, and earlier-depth successes. Since  $\eta < 1$ ,
3643 the resulting total is (PAIR2).
3644 For  $\eta = l^{(1/2)}/L^{(1/6)}$ , the hypotheses hold for sufficiently large  $X$ ,
3645 and both main terms in the brackets are  $O(l^{(3/2)}/L^{(13/6)})$ . The
3646  $1/L$  part of BAND is smaller. This proves (PAIR2*).
3647
3648 HOSTILE CHECK:
3649 1.  $\gcd(q,b)=1$  in LC follows from ROOT primality; it is not an independent
3650 CRT-programming assumption. The different  $\gcd$  in CC can be  $> 1$  and
3651 is handled by reducing the modulus and then averaging its full cost.

```


3652 2. The CL zero-coefficient congruence branch r|h is treated separately.
3653 Simply writing an inverse of $2a-1$ modulo r would omit real cases.
3654 3. Primitive/proportional filtering uses actual distinct prime values
3655 and coefficient-size bounds, verified above in all four cases. A
3656 generic promise that forms are 'different' would not suffice.
3657 4. The count concerns existence of SOME shortest successful path and
3658 therefore controls the COMPLETE radius-two frontier. It does not
3659 prescribe a path or claim the neighborhoods themselves are identical.
3660 5. The prime $P=N/2+1$ is an additional sieve condition for the later root
3661 Q . Without it every case loses a logarithm; this proof cannot be
3662 exported unchanged to the uniform-even- N sample.
3663 6. This is an explicit repair of the changing-offset/gcd obstruction
3664 mentioned in the prior radius-two theorem, not its automatic reuse.
3665 DEPENDENCIES: FKL Lemma 5.1, $k \leq 5$; compatible-gcd harmonic average;
3666 Chebyshev prime upper bound; the audited radius-one pair count.
3667 RELATION TO GOLDBACH: Upper bound on shallow successful traversal pairs.
3668 RELATION TO p0: Supplies a deeper SAME- N comparison with nearby roots.
3669 WHAT REMAINS OPEN: Radius three and eventual rooted reachability.
3670 NOVELTY: NOT YET AUDITED.
3671
3672 RESULT-ID: SAME-N-GROWING-ROOTS-COMPLETE-RADIUS-TWO-20260906B
3673 STATUS: PROVED.
3674 EXACT STATEMENT:
3675 Write $L = \log X$ and $l = \log \log X$. Let $K = K(X)$ be a positive integer with
3676 $K = o(L^{(1/6)/L^{(3/2)}})$.
3677 For all but $o(X/L)$ primes P in $[X, 2X]$, set $N = 2(P-1)$ and let Q_k be the
3678 $(k+1)$ -st prime at or above $N/2$. Then EVERY root Q_k , $0 \leq k \leq K$, has a
3679 COMPLETE stopped radius-two neighborhood consisting of BAD vertices.
3680 An upper bound for the exceptional proportion among primes P is
3681 $O(L^{(1/3)/L^{(1/5)}} + \sqrt{K} L^{(3/2)/L^{(1/6)}})$. (PREFIX2)
3682 All implied constants are absolute, for X sufficiently large along
3683 any K satisfying the displayed hypothesis. A concrete admissible
3684 choice is $K = \text{floor}((\log X)^{(1/8)})$.
3685
3686 HYPOTHESES / SCOPE:
3687 The sample is primes P , with $N = 2(P-1)$; it is not all even N . BADness is
3688 asserted for every vertex reachable within at most TWO edges, including
3689 the root itself, with expansion stopped at GOOD vertices. The growing
3690 prefix is for the SAME N . There is no assertion of eventual failure,
3691 closedness, or a p0-specific invariant.
3692
3693 PROOF:
3694 For large P the integer $P-1$ is composite and there is no integer between
3695 $P-1$ and P , so $Q_0 = P$. The previous audited canonical radius-two theorem
3696 bounds starts with a GOOD vertex within two edges from Q_0 by
3697 $O((X/L) * L^{(1/3)/L^{(1/5)}})$. (C0)
3698
3699 We use an elementary aggregate gap bound. List primes as s_i . For each
3700 $P = s_i$ in $[X, 2X]$, $Q_k = s_{(i+K)}$. The prime number theorem, together with
3701 $K = o(L^{(1/6)/L^{(3/2)}})$, ensures $s_{(i+K)} \leq 3X$ for every such i when X is
3702 large, since $\pi(3X) - \pi(2X) \gg X/L > K$. Expand
3703 $Q_k - P = \sum_{j=i}^{i+K-1} (s_{(j+1)} - s_j)$.
3704 After summing over the eligible i , each consecutive prime gap occurs
3705 at most K times. All participating gaps lie in $[X, 3X]$, up to endpoints
3706 already lying inside that interval; their total length is at most $2X$.
3707 Consequently
3708 $\sum_P (P \text{ prime in } [X, 2X]) (Q_k - P) \leq 2KX$.
3709 For $H \geq 3$, the number of starts with $Q_k - P > H-1$ is therefore at most
3710 $2KX / (H-1) = O(KX/H)$. (GAP)
3711 No assertion about a uniform short interval containing K primes has
3712 been made; the long-gap starts have been counted explicitly.
3713
3714 For every remaining start, all $1 \leq k \leq K$ satisfy
3715 $Q_k > P$, $Q_k - P + 1 \leq H$.
3716 If any such root has success within two edges, the pair (P, Q_k) is
3717 included in SAME- N -RADIUS-TWO-PAIR-COUNT. Counting all pairs only
3718 increases the number of exceptional starts. Thus (PAIR2*) gives
3719 $O(H * X * L^{(3/2)/L^{(13/6)}})$. (LATER)
3720 provided $H \leq L^2$. There is NO further factor K in (LATER), since that
3721 pair count already includes every prime Q at the allowed offset.
3722
3723 Take H to be an integer within 1 of
3724 $H_* = \sqrt{K} * L^{(13/12)/L^{(3/4)}}$.
3725 Then H_* tends to infinity, and the bound on K gives
3726 $H_* = o(L^{(7/6)/L^{(3/2)}}) < L^2$,
3727 so the pair theorem applies. Divide (C0), (GAP), and (LATER) by
3728 $\pi(2X) - \pi(X)$ asymptotic to X/L . The last two resulting bounds are
3729 $O(K * L/H + H * L^{(3/2)/L^{(7/6)}})$
3730 $= O(\sqrt{K} * L^{(3/2)/L^{(1/6)}})$.
3731 This proves (PREFIX2), which tends to zero under the stated condition.
3732 For $K = \text{floor}(L^{(1/8)})$, the required ratio is at most
3733 $L^{(3/2)/L^{(1/24)}}$, and tends to zero. The proof is complete.
3734
3735 HOSTILE CHECK:
3736 1. This does not carry the earlier, larger radius-one prefix
3737 $K = o(L^{1/3})$ to radius two. The new sufficient range is shorter.
3738 2. The extra primality condition $P = N/2+1$ in the pair count is retained;
3739 that count is not being applied to a generic root at a generic N .
3740 3. Both a root being directly GOOD and success at the first frontier
3741 are included in the exceptional set. Therefore outside it the whole
3742 radius-two BFS, rather than selected BAD paths, is BAD.
3743 4. The gap bound is a double count of actual consecutive prime gaps;
3744 it does not replace a mean gap by a pointwise gap estimate.
3745 5. H is chosen only after the uniform pair estimate has been proved
3746 for the full range $3 \leq H \leq L^2$. Hence no hidden dependence of a
3747 sieve constant on an arbitrarily growing H is introduced.
3748 6. The theorem compares shallow BADness only. It gives no coupling of
3749 later BFS generations and no equality of eventual success rates.
3750 DEPENDENCIES: SAME-N-RADIUS-TWO-PAIR-COUNT-20260906B; audited
3751 COMPLETE-H1-RADIUS-TWO-RARE-SUCCESS; PNT and elementary gap summation.
3752 IMPLICATION: A deeper SAME- N negative result for explanations based on
3753 universally shallow success of p0: asymptotically almost every member
3754 of this family has complete radius-two BADness simultaneously at p0
3755 and a growing nearby prefix.
3756 RELATION TO GOLDBACH: This is compatible with Goldbach being true for
3757 all the starts. It is not a lower-bound argument for prime pairs.

```

3758 RELATION TO p0: A property previously established at radius two only
3759 for p0 is now proved to be shared by nearby roots at the SAME N.
3760 WHAT REMAINS OPEN: Radius three, large-product successful paths,
3761 eventual reachability, and any pointwise p0-specific distinction.
3762 NOVELTY: NOT YET AUDITED.
3763
3764 COROLLARY-ID: GOLDBACH-TRUE-WITH-SAME-N-PREFIX-RADIUS-TWO-BAD-20260906B
3765 STATUS: PROVED, using the previously source-checked global exceptional
3766 set theorem, not an assumption of rooted success.
3767 For all but  $o(X/\log X)$  primes  $P$  in  $[X, 2X]$ ,  $N=2(P-1)$  has a Goldbach
3768 representation while every root of the prefix in the preceding theorem
3769 has a complete radius-two BAD neighborhood.
3770 PROOF: Pintz's checked bound  $E(Y) < Y^{0.72}$  for sufficiently large  $Y$  gives
3771 at most  $E(4X) = o(X/\log X)$  possible Goldbach counterexamples in this
3772 injectively parametrized family. Unite that exceptional set with the
3773 one from (PREFIX2). No root reachability conclusion follows merely
3774 from existence of a Goldbach pair elsewhere in the graph.
3775 SOURCE: J. Pintz, A new explicit formula in the additive theory of
3776 primes with applications II, primary https://arxiv.org/pdf/1804.09084,
3777 Theorem 1 (statement previously checked; its long proof is not
3778 reproved in this checkpoint). Ineffective threshold is harmless here.
3779
3780 RESULT-ID: SAME-N-RADIUS-TWO-ADVERSARIAL-SCAN-20260906B
3781 STATUS: VERIFIED COMPUTATION; finite evidence, not an asymptotic test.
3782 EXACT DOMAIN:
3783 All primes  $P$  in  $[1000, 2000]$ ,  $[10000, 20000]$ ,  $[100000, 200000]$ , and
3784  $[500000, 1000000]$ : respectively 135, 1033, 8392, 36960 starts. For each
3785 start set  $N=2(P-1)$ , and examine the SAME-N roots of ranks 0,1,2,5.
3786 No starts excluded. All distinct prime factors of every expanded row
3787 are included. The stopped search checks success at distances 0,1,2.
3788 Category 3 means no success by distance 2, not success at distance 3.
3789 Exact smallest-prime-factor sieve through 4,000,100 suffices for every
3790 integer factored or tested. Arithmetic tests below concern successful
3791 length-two paths from ranks 1,2,5 only, after excluding roots with ANY
3792 success at distance 0 or 1. Thus they are actual shortest paths.
3793
3794 MAIN DIAGNOSTICS:
3795 70,836 shortest successful length-two paths were reconstructed from
3796 the reduced five-form parametrization. Every affine identity and every
3797 compatibility dlh held;  $\gcd(q,b)=1$  held for all of them. Of these,
3798 15,761 had  $d=\gcd(b,2a-1) > 1$ . There were 12,256 paths with  $r|(2a-1)$ ,
3799 and in all such cases  $r|h$ , as required. At the illustrative split
3800  $\eta=0.1$ , 1,180 paths with  $d > 1$  were in the CC size regime, and 2,996
3801  $r|(2a-1)$  paths were in the CL size regime. This finite  $\eta$  is for
3802 case diagnostics only; these small  $X$  need not meet the theorem's
3803 asymptotic  $\eta$  bounds.
3804 For 16,752 reconstructions all five actual prime values exceeded
3805 their positive leading coefficients; in every such reconstruction
3806 all five forms were primitive and all ten determinants nonzero.
3807 The remaining reconstructions were NOT declared counterexamples:
3808 the size hypothesis of the degeneracy filter was not being imposed
3809 outside the large-factor regime.
3810
3811 REAL NONCOPRIME WITNESS:
3812  $P=1423$ ,  $N=2844$ ,  $p_1=Q=1427$ ,  $h=5$ ,  $q=109$ ,  $r=547$ ,
3813  $a=13$ ,  $b=5$ ,  $g=2297$ . The path  $Q \rightarrow 109 \rightarrow 547$  succeeds at depth two.
3814 Here  $d=\gcd(5,25)=5$ , not 1. The reduced variable  $t=108$  gives
3815  $q=t+1$ ,  $r=5t+7$ ,  $Q=13t+23$ ,  $P=13t+19$ ,  $g=21t+29$ .
3816 This certifies why reusing the canonical  $\gcd=1$  proof would miss an
3817 actual nearby-root configuration. The compatible-gcd averaging lemma
3818 has a necessary arithmetic consumer.
3819
3820 REAL DEGENERATE-CONGRUENCE WITNESS:
3821  $P=1277$ ,  $N=2552$ ,  $p_5=Q=1297$ ,  $h=21$ ,  $q=251$ ,  $r=3$ ,
3822  $a=5$ ,  $b=767$ ,  $g=2549$ . Then  $2a-1=9$  is divisible by  $r=3$  and  $h=21$ 
3823 is divisible by  $r$ . The congruence for  $q$  modulo  $r$  does not specify a
3824 unique class. The separate CL-exception sieve includes this case.
3825
3826 SAME-N POINTWISE NON-DOMINANCE AT RADIUS TWO:
3827 At  $N=2180$ , the previously verified  $p_0=1091$  certificate has its complete
3828 radius-two neighborhood BAD;  $p_1=1093$  is directly GOOD since 1087 is
3829 prime. In the other direction,  $N=2024$  has  $p_0=1013$  with a successful
3830 length-two path  $1013 \rightarrow 337 \rightarrow 7$  (2024-7=2017 prime), while  $p_1=1019$ 
3831 has complete radius-two BADness. Its full certificate is:
3832 2024-1019=3*5*67;
3833 2024-3=43*47, 2024-5=3*673, 2024-67=19*103;
3834 2024-43=7*283, 2024-47=3*659, 2024-673=7*193,
3835 2024-19=5*401, 2024-103=17*113.
3836 Every displayed factor is prime, and every radius<=2 row is composite.
3837 Thus neither radius-two BADness event includes the other pointwise for
3838  $p_0$  and  $p_1$ . This is stronger evidence of a precise scope limit than
3839 simply comparing average percentages. No eventual-reachability claim
3840 is made for either example.
3841
3842 At the largest interval, category counts [depth0,depth1,depth2,BAD<=2]
3843 were  $p_0=[3604,14500,11540,7316]$ ,  $p_1=[6406,11779,11426,7349]$ ,
3844  $p_2=[6242,11976,11366,7376]$ ,  $p_5=[6297,12160,11217,7286]$ .
3845 722 starts had complete radius-two BADness at all four selected roots.
3846 The finite proportions are far from the asymptotic density-one limit;
3847 no numerical convergence rate or eventual advantage is inferred.
3848 The complete diagnostic source and output follow, preserving all
3849 metrics and first witnesses without a second report artifact.
3850
3851 BEGIN DIAGNOSTIC SOURCE audit_same_n_radius2_20260906b.py
3852 import numpy as np
3853 from math import isqrt, gcd
3854 from collections import Counter
3855 from itertools import combinations
3856 import json
3857 LIMIT=4000100
3858 spf=np.zeros(LIMIT+1, dtype=np.int32)
3859 for p in range(2, isqrt(LIMIT)+1):
3860     if spf[p]==0:
3861         v=spf[p*p:p:p]
3862         v[v==0]=p
3863 primes=np.flatnonzero(spf[2:]==0)+2

```

```

3864 isprime=np.zeros(LIMIT+1,dtype=np.bool_)
3865 isprime[primes]=True
3866
3867 def pf(n):
3868     out=[]
3869     while n>1:
3870         p=int(spf[n]) or n
3871         out.append(p)
3872         while n%p==0:n//=p
3873     return out
3874
3875 def depth2(N,Q):
3876     m=N-Q
3877     assert m>1 and N%Q
3878     if isprime[m]:return 0,[],[]
3879     qs=pf(m)
3880     if any(isprime[N-q] for q in qs):return 1,qs,[]
3881     paths=[]
3882     for q in qs:
3883         for r in pf(N-q):
3884             if isprime[N-r]:paths.append((q,r))
3885     return (2 if paths else 3),qs,paths
3886
3887 out={'domain':'Exhaustive primes P in each stated half-open dyadic interval; N=2(P-1); same-N roots k=0,1,2,5. Category 3 means no GOOD within
↪ distance 2, not actual success at distance 3. All prime factors included, no size cutoffs. Exact SPF sieve through 4,000,100.',
↪ 'intervals':{}}
3888 audit=Counter();witness={}
3889 ks=[0,1,2,5]
3890 for X in [1000,10000,100000,500000]:
3891     inds=np.flatnonzero((primes>X)&(primes<2*X))
3892     hist=Counter();firstexamples={}
3893     for idx in inds:
3894         P=int(primes[idx]);N=2*(P-1);cats=[]
3895         for k in ks:
3896             Q=int(primes[idx+k]);h=Q-P+1
3897             depth,qs,paths=depth2(N,Q);cats.append(depth)
3898             if k==0:continue
3899             for q,r in paths:
3900                 a=(N-Q)//q;b=(N-q)//r;c=2*a-1;d=gcd(b,c);g=N-r
3901                 audit['shortest_success_length2_paths']+=1
3902                 assert Q==a*q+2*h and P==a*q+h+1
3903                 assert b*r==c*q+2*h and g==2*a*q+2*h-r
3904                 assert gcd(q,b)==1 and h%d==0
3905                 assert len({q,r,P,Q,g})==5
3906                 audit['identities_activity_compatibility_checked']+=1
3907                 bp=b//d;cp=c//d;hp=h//d
3908                 A=((-2*hp)*pow(cp,-1,bp))%bp if bp>1 else 0
3909                 if A==0:A=bp
3910                 R=(cp*A+2*hp)//bp
3911                 assert (cp*A+2*hp)%bp==0
3912                 assert (q-A)%bp==0
3913                 t=(q-A)//bp
3914                 forms=[(bp,A),(cp,R),(a*bp,a*A+2*h),(a*bp,a*A+h+1),(2*a*bp-cp,2*a*A+2*h-R)]
3915                 vals=[u*t+v for u,v in forms]
3916                 assert vals==[q,r,Q,P,g]
3917                 assert all(u>0 for u,v in forms)
3918                 audit['reduced_five_form_reconstruction_checked']+=1
3919                 if all(value>u for value,(u,v) in zip(vals,forms)):
3920                     assert all(gcd(u,v)==1 for u,v in forms)
3921                     assert all(u*y-u*v!=0 for (u,v),(w,y) in combinations(forms,2))
3922                     audit['full_filter_hypotheses_and_conclusion_checked']+=1
3923                 rec={'X':X,'P':P,'N':N,'k':k,'Q':Q,'h':h,'q':q,'r':r,'a':a,'b':b,'g':g,'d':d,'reduced_forms':forms,'t':t}
3924                 if d>1:
3925                     audit['d_greater_than_one']+=1
3926                     witness.setdefault('noncoprime_CC_equation',rec)
3927                     if q>X*0.6 and r>isqrt(X):
3928                         audit['noncoprime_CC_regime_eta_point1']+=1
3929                         witness.setdefault('noncoprime_CC_regime_eta_point1',rec)
3930                     if c%r==0:
3931                         assert h%r==0
3932                         audit['degenerate_r_divides_2a_minus1']+=1
3933                         witness.setdefault('CL_exception_equation',rec)
3934                     if q>X*0.6 and r<=isqrt(X):
3935                         audit['CL_exception_regime_eta_point1']+=1
3936                         witness.setdefault('CL_exception_regime_eta_point1',rec)
3937                 key=tuple(cats);hist[key]+=1;firstexamples.setdefault(key,P)
3938     out['intervals'][str(X)]=[
3939         'prime_count':len(inds),
3940         'root_categories_0_1_2_bad_radius2':{str(k):sum(n for v,n in hist.items() if v[j]==d) for d in range(4)} for j,k in enumerate(ks)},
3941         'all_four_radius2_BAD':sum(n for v,n in hist.items() if min(v)==3),
3942         'first_P_all_four_radius2_BAD':firstexamples.get((3,3,3,3)),
3943         'p0_bad_radius2_pi_success_within2':sum(n for v,n in hist.items() if v[0]==3 and v[1]<3),
3944         'pi_bad_radius2_p0_success_within2':sum(n for v,n in hist.items() if v[1]==3 and v[0]<3)
3945     ]
3946 out['audit']=dict(audit);out['witnesses']=witness
3947 print(json.dumps(out,indent=2))
3948 END DIAGNOSTIC SOURCE
3949 BEGIN DIAGNOSTIC OUTPUT
3950 {
3951     "domain": "Exhaustive primes P in each stated half-open dyadic interval; N=2(P-1); same-N roots k=0,1,2,5. Category 3 means no GOOD within
↪ distance 2, not actual success at distance 3. All prime factors included, no size cutoffs. Exact SPF sieve through 4,000,100.",
3952     "intervals": {
3953         "1000": {
3954             "prime_count": 135,
3955             "root_categories_0_1_2_bad_radius2": {
3956                 "0": [
3957                     26,
3958                     68,
3959                     26,
3960                     15
3961                 ],
3962                 "1": [
3963                     46,
3964                     44,
3965                     34,

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3966         11
3967     ],
3968     "2": [
3969         37,
3970         56,
3971         29,
3972         13
3973     ],
3974     "5": [
3975         34,
3976         57,
3977         31,
3978         13
3979     ]
3980 },
3981 "all_four_radius2_BAD": 1,
3982 "first_P_all_four_radius2_BAD": 1217,
3983 "p0_bad_radius2_p1_success_within2": 13,
3984 "p1_bad_radius2_p0_success_within2": 9
3985 },
3986 "100000": {
3987     "prime_count": 1033,
3988     "root_categories_0_1_2_bad_radius2": {
3989         "0": [
3990             137,
3991             469,
3992             278,
3993             149
3994         ],
3995         "1": [
3996             235,
3997             362,
3998             305,
3999             131
4000         ],
4001         "2": [
4002             227,
4003             383,
4004             281,
4005             142
4006         ],
4007         "5": [
4008             236,
4009             360,
4010             272,
4011             165
4012         ]
4013     },
4014     "all_four_radius2_BAD": 11,
4015     "first_P_all_four_radius2_BAD": 10559,
4016     "p0_bad_radius2_p1_success_within2": 105,
4017     "p1_bad_radius2_p0_success_within2": 87
4018 },
4019 "1000000": {
4020     "prime_count": 8392,
4021     "root_categories_0_1_2_bad_radius2": {
4022         "0": [
4023             936,
4024             3481,
4025             2543,
4026             1432
4027         ],
4028         "1": [
4029             1656,
4030             2802,
4031             2436,
4032             1498
4033         ],
4034         "2": [
4035             1611,
4036             2820,
4037             2455,
4038             1506
4039         ],
4040         "5": [
4041             1661,
4042             2851,
4043             2419,
4044             1461
4045         ]
4046     },
4047     "all_four_radius2_BAD": 129,
4048     "first_P_all_four_radius2_BAD": 100931,
4049     "p0_bad_radius2_p1_success_within2": 898,
4050     "p1_bad_radius2_p0_success_within2": 964
4051 },
4052 "5000000": {
4053     "prime_count": 36960,
4054     "root_categories_0_1_2_bad_radius2": {
4055         "0": [
4056             3604,
4057             14500,
4058             11540,
4059             7316
4060         ],
4061         "1": [
4062             6406,
4063             11779,
4064             11426,
4065             7349
4066         ],
4067         "2": [
4068             6242,
4069             11976,
4070             11366,
4071             7376

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4072     ],
4073     "5": [
4074         6297,
4075         12160,
4076         11217,
4077         7286
4078     ],
4079 },
4080 "all_four_radius2_BAD": 722,
4081 "first_P_all_four_radius2_BAD": 501617,
4082 "p0_bad_radius2_pi_success_within2": 4538,
4083 "pi_bad_radius2_p0_success_within2": 4571
4084 }
4085 },
4086 "audit": {
4087     "shortest_success_length2_paths": 70836,
4088     "identities_activity_compatibility_checked": 70836,
4089     "reduced_five_form_reconstruction_checked": 70836,
4090     "full_filter_hypotheses_and_conclusion_checked": 16752,
4091     "d_greater_than_one": 15761,
4092     "degenerate_r_divides_2a_minus1": 12256,
4093     "CL_exception_regime_eta_point1": 2996,
4094     "noncoprime_CC_regime_eta_point1": 1180
4095 },
4096 "witnesses": {
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4246 }
4247 END DIAGNOSTIC OUTPUT
4248
4249 HISTORICAL CHECK 2026-09-06B:
4250 The original post-CRD failure log was searched for common-divisor,
4251 local-offset, and support-height routes. Its Tail Hard-Edge Universality
4252 entry and All-root collision identity do not imply the present theorem:
4253 they establish identities without a BFS consumer. Here the compatibility
4254 weight is consumed by a uniform prime-form count, and complete success
4255 is bounded by a union over ALL possible first two edges. No assertion
4256 that a gcd collision forces new support is used. The log's h=1
4257 root-factor descent still has its stated scope; the present proof does
4258 not presume that descent persists after an offset or support change.
4259
4260 RESULT-ID: COFACTOR-CHAIN-EXACT-MODULUS-20260906B
4261 STATUS: PROVED; affine identity archival in substance, exact modulus
4262 and averaging consumer recorded explicitly for the new sieve use.
4263 EXACT STATEMENT:
4264 Let  $x, h$  be positive integers,  $q_0 = x + h$ , and let  $a_1, \dots, a_d$  be odd
4265 integers  $\geq 3$ . Put  $A_0 = C_0 = 1$  and, for  $1 \leq j \leq d$ ,
4266  $A_j = a_1 \dots a_j$ ,  $C_j = 2 * A_{(j-1)} - C_{(j-1)}$ .
4267 The recurrence  $a_j * q_j = 2x - q_{(j-1)}$  is equivalent to (CF)
4268  $A_j * q_j = C_j * x + (-1)^j * h$ . (CF)
4269 All  $q_j$  are integers if and only if (CM)
4270  $C_d * x = (-1)^{(d+1)} * h$  modulo  $A_d$ . (CM)
4271 Writing  $e = \gcd(C_d, A_d)$ , this condition is soluble precisely when  $e|h$ ;
4272 if soluble, it specifies exactly one residue class of  $x$  modulo  $A_d/e$ .
4273 Every  $q_j$  is an integer affine form in the resulting progression
4274 variable, with positive leading coefficient  $C_j * A_d / (A_j * e)$ .
4275 In particular for  $h=1$  the compatible modulus is exactly  $A_d$ , since
4276  $e=1$  is necessary. This is not assumed for later same- $N$  roots.
4277
4278 PROOF:
4279 Induction in the recurrence gives (CF), with the initial identity
4280  $q_0 = x + h$ . Since every  $a_j \geq 3$ ,  $C_{-1} = 1$ , and induction gives
4281  $0 < C_j \leq 2 * A_{(j-1)}$  for all  $j \geq 1$ .
4282 Indeed the subtraction term  $C_{(j-1)}$  is positive and, for  $j \geq 2$ ,
4283  $C_{(j-1)} \leq 2 * A_{(j-2)} < 2 * A_{(j-1)}$ . Thus all coefficients are positive.

```

4284 For $i < j$, reduction modulo A_i in the recurrence for C shows
4285 $C_j = (-1)^{(j-i)} C_i$ modulo A_i ,
4286 because $A_{(j-1)}$ is divisible by A_i whenever $j > i$. Consequently (CM)
4287 implies $C_i \cdot x + (-1)^i \cdot h = 0$ modulo A_i for EVERY $i \leq d$. This proves
4288 that final integrality implies all earlier integrality; the converse
4289 is immediate. The solvability and modulus assertions are the elementary
4290 linear-congruence theorem after dividing by e .
4291 For $x = x_0 + (A_d/e) \cdot t$, divisibility of each earlier numerator for all
4292 integer t implies A_i divides $C_i \cdot A_d/e$. Alternatively C_i agrees
4293 up to sign with C_d modulo A_i and $C_d \cdot A_d/e$ is divisible by A_i .
4294 Hence every stated leading coefficient is a positive integer. The
4295 constants are integers by the choice of x_0 . This proves everything.
4296
4297 HOSTILE CHECK:
4298 One cannot simply divide every earlier modulus A_j by the final gcd e :
4299 e may not divide A_j . What is integral is $C_j \cdot A_d / (A_j \cdot e)$, by the
4300 congruence just proved. The reduced modulus need not equal the product
4301 of separately reduced edge moduli. At $h=1$ the coprimality conclusion
4302 comes from the numerator ± 1 , not from hypothetical independence of
4303 cofactors. No prime or BAD hypothesis is needed for the algebra.
4304 HISTORICAL RELATION: The master identifies
4305 G-CRD-PATH-AFFINE-COFACTOR-NORMAL-FORM as a surviving exact transport
4306 lemma. FAIL-V48-DIRECT-REFLECTED-TAG-INHERITANCE-001 warns against
4307 preserving an old divisibility tag along a path; (CF) transports the
4308 changing term instead. FAIL-V51-ALTERNATE-PATH-CONGRUENCE-AMPLIFIES-
4309 DEEP-POWER-001 does not apply: no extra valuation is inferred from an
4310 invertible common factor. The consumer below counts prime solutions.
4311 DEPENDENCIES: Elementary induction and linear congruences.
4312 NOVELTY: NOT YET AUDITED; do not claim novelty of affine transport.
4313
4314 RESULT-ID: ORDERED-COFACTOR-HARMONIC-SIMPLEX-20260906B
4315 STATUS: PROVED; elementary counting lemma.
4316 EXACT STATEMENT:
4317 For every integer $d \geq 1$ and $Y \geq 1$,
4318 $\sum (a_1, \dots, a_d) \leq 3$; product $a_j \leq Y$ $1/(\text{product } a_j)$
4319 $\leq (\max(\log Y - d \cdot \log 2, 0))^d / d!$ $\leq (\log Y)^d / d!$
4320 (the last expression used for $Y \geq 1$). Restricting the a_j to odd
4321 integers preserves the bound. The tuples are ORDERED; their factorial
4322 saving is not an unjustified division by permutations.
4323 PROOF:
4324 For each integer $a \geq 3$,
4325 $1/a \leq \int_a^\infty (a-1)^{-a} dt/t$.
4326 The boxes $\text{product}_j [a_j - 1, a_j]$ have disjoint interiors. If product a_j
4327 is at most Y , every point of its box has all coordinates ≥ 2 and
4328 product coordinates $\leq Y$. Integrate $\text{product}_j dt_j/t_j$ over the union
4329 of boxes, then over this larger region. In variables $u_j = \log t_j - \log 2$
4330 it is the simplex $u_j \geq 0$, $\sum u_j \leq \log Y - d \log 2$. Its volume is exactly
4331 the claimed expression. A negative right endpoint gives an empty
4332 region. This proves the bound with no symmetry assumption.
4333 IMPLICATION: Cancels the factorial cost of sieving $d+O(1)$ prime forms
4334 when summing over all short-product cofactor sequences.
4335 NOVELTY: NOT YET AUDITED; standard volume method.
4336
4337 RESULT-ID: GROWING-FORMS-UNIFORM-SIEVE-CONSUMER-20260906B
4338 STATUS: PROVED from the same primary FKL Lemma 5.1.
4339 EXACT STATEMENT USED:
4340 Fix $\epsilon > 0$. Let $L = \log X$, $l = \log \log X$, and
4341 $2 < k \leq 2 \cdot l / \log l$, $X^\epsilon \leq z \leq 3X$.
4342 Consider k primitive positive-leading-coefficient integer linear forms
4343 with coefficients and constants of absolute value $\leq 10X^2$, and all
4344 pairwise determinants nonzero. If their root count $\text{nu}(\text{ell}) \geq 1$ at every
4345 prime ell , then the number of positive $t \leq z$ with every form prime and
4346 $> k$ is at most
4347 $2^{-(k+1)} k! \cdot z \cdot (C \cdot l)^{-(k-1)} / (\epsilon \cdot l \cdot L)^k$ (GF)
4348 for an absolute C and sufficiently large X depending on ϵ .
4349 For any fixed k the same assertion holds with $(C \cdot l)^k$ without the
4350 $\text{nu} \geq 1$ assumption. The constants are uniform in the varying forms.
4351
4352 PROOF:
4353 Let Δ be the absolute product of all leading coefficients and all
4354 pairwise determinants. It is a nonzero integer, with
4355 $\log(3 \cdot \Delta) = O(k^2 \cdot L)$.
4356 Outside primes dividing Δ , the roots of the k forms are distinct,
4357 so $\text{nu}(\text{ell}) = k$. As in the earlier Euler-product proof,
4358 $\sum_{\text{ell}} (\text{ell} | \Delta) \log \text{ell} / \text{ell} = O(\log \log(3 \cdot \Delta)) = O(1)$,
4359 $\prod_{\text{ell}} (\text{ell} | \Delta) (1 - 1/\text{ell})^{-(k-1)} \leq C \cdot \log \log(3 \cdot \Delta) \leq C' \cdot l$.
4360 The first estimate follows by splitting at $\log(3 \cdot \Delta)$, using the
4361 Chebyshev bound below that cutoff, and summing $\log \text{ell}$ over divisors
4362 above it. The second is the already proved elementary Euler bound.
4363 Thus $B_{\text{FKL}} \leq k \cdot \sum_{\text{ell}} (\text{ell} | \Delta) \log \text{ell} / \text{ell} = O(k \cdot l)$.
4364 If the singular series S is zero there are no such prime values $> k$.
4365 Otherwise its factors outside Δ are at most 1 by
4366 $1 - k/\text{ell} \leq (1 - 1/\text{ell})^k$, and at primes dividing Δ the condition $\text{nu} \geq 1$
4367 gives factors $\leq (1 - 1/\text{ell})^{-(k-1)}$. Therefore $S \leq (C \cdot l)^{-(k-1)}$.
4368 For fixed k without $\text{nu} \geq 1$ the exponent k gives a valid upper bound.
4369 Both FKL hypotheses hold uniformly: $k = O(l / \log l) = o(L/l)$ and
4370 $B = O(k \cdot l) = o(L)$, while $\log z \geq \epsilon \cdot l \cdot L$. Its explicit exponential
4371 error is $\exp(O(k^2 \cdot l / (\epsilon \cdot l \cdot L))) = 1 + o(1)$, hence ≤ 2 for large X .
4372 Substitution gives (GF). No unspecified fixed- k constant has been
4373 allowed to grow with the number of generations.
4374
4375 SOURCE-AUDIT:
4376 The exact growing- k statement and its error term were reopened in
4377 Ford-Konyagin--Luca, Prime chains and Pratt trees, Lemma 5.1, primary
4378 <https://arxiv.org/pdf/0904.0473>, printed pp.12--13. Their following
4379 Lemma 5.2 confirms the same Euler-product method, but the argument
4380 above checks it directly for the present forms. This does not invoke
4381 their more specialized averaging theorem for Pratt-chain forms.
4382 HOSTILE CHECK: The $k!$ remains explicit and will only be cancelled by
4383 the ordered-simplex bound; ignoring it without that bound would reduce
4384 the provable depth. All determinants are required nonzero before B
4385 is estimated; repeating a form could make B infinite.
4386
4387 RESULT-ID: MULTIGENERATION-COFACTOR-PRODUCT-BARRIER-20260906B
4388 STATUS: PROVED.
4389 EXACT STATEMENT:

4390 Fix ϵ in $(0,1)$ and c in $(0,1)$. Write $L=\log X$, $l=\log\log X$, and
 4391 $D=\text{floor}(c\cdot l/\log l)$.
 4392 For all but $o(X/L)$ primes P in $[X,2X]$, $N=2(P-1)$, the root P is BAD
 4393 and every simple successful stopped path
 4394 $P=q_0 \rightarrow q_1 \rightarrow \dots \rightarrow q_d$, $1 \leq d \leq D$,
 4395 has
 4396 $\text{product}_j(j=1..d) a_j > X^{(1-\epsilon)}$,
 4397 where $a_j=(N-q_j(j-1))/q_j$ is the integer cofactor of the actual edge.
 4398 In fact the exceptional count is $< X \cdot L^{(-2+c+o(1))}$.
 4399 This is complementary to, not a restatement of, the earlier barrier
 4400 on product q_j : a large prime child has a small cofactor.
 4401
 4402 The following uniform pair version is part of the theorem. For
 4403 $3 \leq H \leq L^2$, the number of pairs of primes (P,Q) with
 4404 $X \leq P \leq 2X$, $P < Q$, $h=Q-P+1 \leq H$,
 4405 for which the Q -rooted stopped traversal at $N=2(P-1)$ has a simple
 4406 successful path of length $\leq D$ with cofactor product $< X^{(1-\epsilon)}$
 4407 (or Q is directly GOOD), is
 4408 $< H \cdot X \cdot L^{(-3+c+o(1))}$. (CFPAIR)
 4409 The $o(1)$ is uniform over the stated H range, depending only on fixed
 4410 ϵ, c . No fixed alphabet, support-size bound, or closed-core
 4411 hypothesis is used.
 4412
 4413 PROOF:
 4414 Put $x=P-1$, so $N=2x$ and $Q=x+h$. For a successful stopped path of positive
 4415 length, all earlier vertices are BAD. Activity propagates to all
 4416 children: if a prime q divides both N and $N-u$, it also divides u ;
 4417 for prime active u this is impossible. All labels are odd, and proper
 4418 prime factors of a composite odd row are $< N/3$. Thus the path labels
 4419 $q_1, \dots, q_d$ are $< N/3$, while $g=N-q_d > 2N/3 > Q$ for large X . For a simple
 4420 path, the required prime values $q_1, \dots, q_d, P, Q, g$ are all distinct,
 4421 except that $Q=P$ when $h=1$; in that case we include that form only once.
 4422 The cofactors are odd integers ≥ 3 .
 4423
 4424 Fix d and an ordered tuple of cofactors $a_1, \dots, a_d$ with
 4425 $A=A_d=\text{product}_j a_j < X^{(1-\epsilon)}$.
 4426 Use C_j from COFACTOR-CHAIN-EXACT-MODULUS. If $e=\gcd(C_d, A)$ does not
 4427 divide h there are no integer paths. Otherwise let x_0 in $[1, A/e]$
 4428 represent its unique compatible class, and set
 4429 $M=A/e$, $x=x_0+M \cdot t$.
 4430 All $q_j=(C_j \cdot x + (-1)^j \cdot h)/A_j$ are integer affine forms in t , with
 4431 positive leading coefficients $C_j \cdot M/A_j$. In particular the last
 4432 coefficient is C_d/e , coprime to M . Also $g=2x-q_d$ has positive leading
 4433 coefficient $(2A-C_d)/e$: indeed $C_d \leq 2A_{(d-1)} \leq 2A/3$.
 4434 Since $x \geq X-1$, $M \leq A < X^{(1-\epsilon)}$, and $h \leq L^2$, the progression variable
 4435 satisfies $1 \leq t \leq 2X/M = 2eX/A$ for large X . Every required prime value is
 4436 larger than its leading coefficient. For q_j , subtracting that
 4437 coefficient gives
 4438 $[C_j \cdot (x-M) + (-1)^j \cdot h]/A_j > 0$,
 4439 because $C_j \geq 1$ and $x-M \geq h$. For g the same assertion follows from
 4440 $(2-C_d/A) \cdot (x-M) - (-1)^d \cdot h/A > 0$. For P, Q it is immediate.
 4441 All q_j also exceed $X/(2A) \geq X^{\epsilon}/2$: the first row $x-h$ is $> X/2$
 4442 and all subsequent rows are $> 2N/3$; divide by $a_j \leq A$. Hence all variable
 4443 prime values exceed $k=d+2$ or $d+3$ when X is sufficiently large.
 4444
 4445 Discard any imprimitive or zero-determinant form tuple. It has no
 4446 actual solution by DISTINCT-PRIME-FORMS-DEGENERACY-FILTER, whose
 4447 coefficient and distinctness hypotheses were just verified. For all
 4448 remaining tuples, the two coprime leading coefficients M and C_d/e
 4449 ensure $\nu(\ell) \geq 1$ at every prime. The coefficients and constants of
 4450 P, Q, q_j, g are bounded in absolute value by $10X^2$, uniformly in j, d, h :
 4451 $C_j \leq 2A_{(j-1)}$, $x_0 \leq A/e$, and $h \leq L^2$ suffice. Thus the growing-form
 4452 sieve applies, with $\log(2eX/A) \geq \epsilon \log X$.
 4453
 4454 CANONICAL CASE $h=1$:
 4455 Compatibility forces $e=1$. There are $k=d+2$ forms: P , all q_j , and g .
 4456 For each cofactor tuple the sieve gives at most
 4457 $2^{-(d+3)} \cdot (d+2)! \cdot (2X/A) \cdot (C+1)^{-(d+1)} / (\epsilon \log X)^{-(d+2)}$.
 4458 Sum $1/A$ over all ordered cofactor tuples of product at most
 4459 $X^{(1-\epsilon)}$. The harmonic-simplex lemma bounds this by $L^d/d!$.
 4460 Therefore the total at depth d is at most
 4461 $C_{\epsilon} \cdot \epsilon \log X \cdot (X/L^2)^{(d+3)} \cdot 3^d \cdot (C_{\epsilon} \log X)^{-(d+2)}$. (CF0d)
 4462 This bound is deliberately loose in a fixed power of 1 ; the key
 4463 factorial ratio $(d+2)!/d!$ is only quadratic in d .
 4464
 4465 LATER-ROOT PAIRS:
 4466 For $h \geq 3$ there are $k=d+3$ forms, including the extra prime $P=x+1$ and
 4467 $Q=x+h$ separately. The fixed tuple, fixed compatible h bound has the
 4468 same shape with $d+3$ in place of $d+2$, and the progression length carries
 4469 its full factor e :
 4470 $2^{-(d+4)} \cdot (d+3)! \cdot (2eX/A) \cdot (C+1)^{-(d+2)} / (\epsilon \log X)^{-(d+3)}$.
 4471 For the chosen cofactor tuple e is fixed and depends on that tuple,
 4472 not on h . Sum over all compatible odd $3 \leq h \leq H$. Crucially,
 4473 $\sum_{h \leq H} (e|h) e \leq e \cdot \text{floor}(H/e) \leq H$.
 4474 This includes every compatible gcd with no growing divisor loss.
 4475 Now sum $1/A$ using the ordered-simplex bound. The depth- d pair count is
 4476 at most
 4477 $C_{\epsilon} \cdot \epsilon \log X \cdot (H \cdot X/L^3)^{(d+3)} \cdot 3^d \cdot (C_{\epsilon} \log X)^{-(d+2)}$. (CFPd)
 4478 It was essential to average the actual reduced modulus; setting $e=1$
 4479 for later roots would have omitted valid paths.
 4480
 4481 SUM OVER DEPTH:
 4482 For $D=\text{floor}(c \cdot l/\log l)$,
 4483 $\log[\sum_d(d=1..D) (d+3)^3 \cdot (C_{\epsilon} \log X)^{-(d+2)}]$
 4484 $\leq (D+2) \cdot \log(C_{\epsilon} \log X) + O(\log(D+3)) = (c+o(1)) \cdot l$.
 4485 Thus that sum is $L^{-(c+o(1))}$. Summing (CF0d) and (CFPd) proves the
 4486 claimed bounds for positive path lengths. Direct canonical success
 4487 has the previously proved $O(X/L^2)$ count. Direct later-root success
 4488 requires $P, P+h-1, P-h-1$ prime; these are three primitive forms with
 4489 nonzero determinants for $h \geq 3$ and $\nu \geq 1$. The same fixed-three-form
 4490 sieve gives $O(H \cdot X \cdot l^{-2}/L^3)$, absorbed by (CFPAIR).
 4491 Finally $X \cdot L^{(-2+c+o(1))} = o(X/L)$ because $c < 1$. This proves the theorem.
 4492
 4493 HOSTILE CHECK:
 4494 1. The bound controls all cofactor sequences of short product, not just
 4495 a chosen transcript. It remains a restriction on successful paths,

4496 not a proof of full BFS failure.

4497 2. A simple path is used to ensure distinct variable prime forms. The

4498 proof does not sieve repeated forms as independent prime conditions.

4499 3. The growing factorial in FKL is cancelled by an explicitly proved

4500 volume bound for ORDERED tuples, not by an illicit d! symmetry.

4501 4. For later roots, reduced earlier coefficients are integers because

4502 of final-to-earlier congruence compatibility, not because e divides

4503 every partial product A_j .

4504 5. The e factor is averaged before the cofactor tuples are summed.

4505 No uncontrolled average of a moving gcd is hidden in the estimate.

4506 6. Short cofactor product gives a polynomially long progression. When

4507 the product is larger, the proof makes no estimate for those paths.

4508 7. This uses the full explicit growing-k FKL bound; a fixed-k big-O

4509 sieve assertion alone would not justify D tending to infinity.

4510 DEPENDENCIES: Exact cofactor modulus; harmonic-simplex bound; growing

4511 prime-form sieve; activity; no prime-pair lower bound or global failure.

4512 RELATION TO GOLDBACH: A restriction on shallow successful routes. A

4513 closed BAD core has no successful path, so this theorem alone cannot

4514 contradict its existence.

4515 RELATION TO p0: The pair theorem prepares a same-N extension. The h=1

4516 coprimality is arithmetically special but not a success mechanism.

4517 WHAT REMAINS OPEN: Paths with both large label product and large

4518 cofactor product; complete radius three; eventual reachability.

4519 NOVELTY: NOT YET AUDITED.

4520

4521 RESULT-ID: MULTIGENERATION-LABEL-PRODUCT-PAIR-COUNT-20260906B

4522 STATUS: PROVED; same-N extension of the previous label-product barrier.

4523 EXACT STATEMENT:

4524 For fixed ϵ , c in $(0,1)$, $D = \text{floor}(c \cdot l / \log l)$, $3 \leq H \leq L^2$,

4525 the number of prime pairs $P < Q$, P in $[X, 2X]$, $h = Q - P + 1 \leq H$, such that

4526 Q is GOOD or a simple successful path from Q of length $d \leq D$ has

4527 $M = \text{product}_{(j=1..d)} q_j < X^{1-\epsilon}$, is

4528 $\leq H \cdot X \cdot L^{-(3+c+o(1))}$,

4529 uniformly over H . The analogous canonical count is

4530 $\leq X \cdot L^{-(2+c+o(1))}$, as in the previously audited theorem.

4531

4532 PROOF OF THE PAIR EXTENSION:

4533 Fix h and a tuple of distinct odd prime labels $q_1, \dots, q_d$ with the

4534 specified product bound. Put $x = P - 1$. The actual edges impose

4535 $x \equiv h \pmod{q_1}$,

4536 $2x \equiv q_j - 1 \pmod{q_j}$ for $j \geq 2$.

4537 CRT gives one class $x \equiv x_0 + M \cdot t$, $1 \leq t \leq M$, and

4538 $1 \leq t \leq 2X/M$ for actual starts. The three necessary prime forms are

4539 $P = x + 1$, $Q = x + h$, $g = 2x - q_d$.

4540 Their positive leading coefficients are $M, M, 2M$ and are smaller than

4541 actual prime values. The determinants, up to sign, are

4542 $M \cdot (h-1)$, $M \cdot (q_d + 2)$, $M \cdot (q_d + 2h)$,

4543 all nonzero. Imprimitve forms have no actual prime solutions since

4544 the values exceed the leading coefficients. Their Delta is bounded by

4545 a fixed power of X uniformly in d, h : only three forms are present.

4546 The fixed-three-form sieve with the general singular-series bound

4547 $O(l^3)$ gives $O_\epsilon(X^{1/3} / (M \cdot L^3))$. All path primes are $< 2X$.

4548 Sum ordered tuples, dropping distinctness and the product cutoff only

4549 in the positive reciprocal sum. It is at most

4550 $(\sum_{q \leq 2X} 1/q)^d \leq (1 + O(1))^{-d}$.

4551 Sum $d \leq D$ and $h \leq H$. The result is at most

4552 $O_\epsilon(X^{1/3} / L^3) \cdot (1 + O(1))^{-D}$

4553 $\leq H \cdot X \cdot L^{-(3+c+o(1))}$.

4554 Direct GOOD roots are included by the three-form count already proved

4555 in the cofactor theorem. For $h=1$ $P=Q$, so the canonical proof correctly

4556 uses only TWO prime forms and has the stated one-logarithm weaker count.

4557 HOSTILE CHECK: No extra prime condition is counted at $h=1$. The varying

4558 path length does not enter the Delta bound because the modulus M itself

4559 is bounded; no infinite-rank sieve is invoked here.

4560 NOVELTY: NOT YET AUDITED.

4561

4562 RESULT-ID: SAME-N-GROWING-DEPTH-DUAL-PRODUCT-BARRIER-20260906B

4563 STATUS: PROVED.

4564 EXACT STATEMENT:

4565 Fix ϵ , κ in $(0,1)$, c in $(0,1)$, and $\kappa \geq 0$ with $c + \kappa < 1$. Put

4566 $L = \log X$, $l = \log \log X$,

4567 $D = \text{floor}(c \cdot l / \log l)$, $K = \text{floor}(L^\kappa)$.

4568 For all but $o(X/L)$ primes P in $[X, 2X]$, let $N = 2(P-1)$. Then every

4569 SAME-N root $p_k(N)$, $0 \leq k \leq K$, is BAD, and every simple successful

4570 stopped path from any one of those roots, of length $1 \leq d \leq D$, satisfies

4571 BOTH

4572 $\text{product}_{(j=1..d)} q_j > X^{1-\epsilon}$,

4573 $\text{product}_{(j=1..d)} [(N - q_j - 1) / q_j] > X^{1-\epsilon}$. (DUAL)

4574 The exceptional proportion among primes P is at most

4575 $L^{-(1-c-\kappa)/2+o(1)}$.

4576 Here q_0 is the selected root, and the products exclude q_0 as a label

4577 but include the cofactor of its first edge. The result is simultaneous

4578 over the entire root prefix and over every eligible simple path.

4579

4580 PROOF:

4581 The canonical union of the two small-product exceptions and direct

4582 success has size $\leq X \cdot L^{-(2+c+o(1))}$. For later roots, the union of the

4583 two pair exceptions just proved has size $\leq H \cdot X \cdot L^{-(3+c+o(1))}$, uniformly

4584 for $3 \leq H \leq L^2$. These are counts of all possible later-root pairs, so

4585 there will be no extra factor K when using them for a root prefix.

4586 Take H to be an integer within 1 of

4587 $L^{((3-c+\kappa)/2)}$.

4588 The inequality $c + \kappa < 1$ guarantees $H \leq L^2$ for large X . The same

4589 consecutive-gap summation proved in SAME-N-GROWING-ROOTS-COMPLETE-

4590 RADIUS-TWO gives at most $O(KX/H)$ starts with $p_K - P > H - 1$. The use of PNT

4591 to place all p_K below $3X$ remains valid because K is a fixed power of

4592 $\log X$. Outside these long-gap starts, every later prefix root has

4593 $h \leq H$, so all its exceptions are counted by the pair union.

4594 Divide the total exceptional count by $\pi(2X) - \pi(X)$ asymptotic to X/L .

4595 It is bounded by

4596 $L^{-(1+c+o(1))} + O(K \cdot L/H) + H \cdot X \cdot L^{-(2+c+o(1))}$

4597 $\leq L^{-(1-c-\kappa)/2+o(1)}$.

4598 The exponent is negative. Every root and every successful simple path

4599 outside this exceptional set has (DUAL), proving the assertion.

4600

4601 HOSTILE CHECK:

4602 1. This is a restriction on successful paths, not complete BADness
 4603 through the growing depth D. Large products remain possible.
 4604 2. Both products are for the ACTUAL same path. Their simultaneous
 4605 lower bounds are obtained by removing the union of the two failure
 4606 sets; no independence is assumed.
 4607 3. The depth-prefix tradeoff $c+\kappa \leq 1$ is explicit. It is not legitimate
 4608 to let both parameters approach 1 with this proof.
 4609 4. The cofactor proof had to handle the changing gcd of a moving
 4610 product; the label proof had to retain the extra prime P. Neither
 4611 is an automatic $h=1$ -to- $h>1$ substitution.
 4612 5. A shortest successful path is simple, so the theorem supplies a
 4613 necessary condition for success by D. A closed BAD core has no
 4614 successful path and therefore satisfies the conclusion vacuously.
 4615 6. The proof does not claim that the scalar data C_j or A_j distinguish
 4616 p_0 . The SAME-N conclusion proves the asymptotic obstruction is shared.
 4617 DEPENDENCIES: Both product pair bounds; PNT; prime-gap double count.
 4618 RELATION TO GOLDBACH: Constrains increasingly long routes but supplies
 4619 no contradiction to arbitrary closed support or to large-product success.
 4620 RELATION TO p_0 : A multigeneration property previously available only
 4621 for p_0 is now shared by a growing SAME-N nearby prefix, with a second,
 4622 complementary arithmetic product constraint added.
 4623 WHAT REMAINS OPEN: The middle range where both products are large,
 4624 complete radius three, and eventual escape from a closed BAD set.
 4625 NOVELTY: NOT YET AUDITED.
 4626
 4627 COROLLARY / EXACT RESIDUAL WINDOW:
 4628 At any successful path the identity
 4629 $(\text{product } q_j) * (\text{product } a_j) = \text{product}_j (1..d) (N - q_j(j-1))$
 4630 is exact. Since $N \leq 4X$, (DUAL) implies, on the same typical starts,
 4631 $X^{(1-\epsilon)} < \text{product } q_j < 4^d * X^{(d-1+\epsilon)}$,
 4632 and the same upper and lower bounds for product a_j . At depth three,
 4633 the unexcluded label-product range is therefore between approximately
 4634 $X^{(1-\epsilon)}$ and $X^{(2+\epsilon)}$. It still includes the region with
 4635 all three labels of size $X^{0.4}$. Nothing here resolves the existing
 4636 FAIL-ID AUTOMATIC-THIRD-GENERATION-SIEVE-SWITCH-20260906.
 4637
 4638 FAIL-ID: SUCCESS-PRODUCT-BARRIER-TO-CLOSED-CORE-HEIGHT-20260906B
 4639 STATUS: KILLED AS AN INFERENCE; both input theorems survive.
 4640 CLAIM / ROUTE:
 4641 Combine the growing-depth product barriers with TWO-SMALLEST-ROWS-
 4642 HEIGHT to rule out a closed active BAD core reached from p_0 .
 4643 WHY IT LOOKED PROMISING:
 4644 One input controls products along moving paths; the other forces
 4645 complexity in a core's two least rows. Both refer to large arithmetic
 4646 objects, so it is tempting to regard their bounds as opposing capacities.
 4647 EXACT FAILURE:
 4648 The product barriers quantify over SUCCESSFUL paths, which a closed
 4649 BAD core has none of. Applying them inside such a core yields only a
 4650 vacuously true statement. The height theorem is a lower complexity
 4651 bound, not a conflicting upper bound. The new cofactor theorem does
 4652 not alter this logical domain mismatch.
 4653 In its sieve proof the final prime $g=N-q_d$ is a crucial EXTRA prime
 4654 condition. Dropping it to count paths ending at BAD vertices loses a
 4655 factor $\log X$: for canonical starts k changes from $d+2$ to $d+1$. The
 4656 resulting crude bound is of size $(X/L)*L^{(c+o(1))}$, not $o(X/L)$. Thus
 4657 the written proof itself identifies the missing consumer, rather than
 4658 merely failing to state one.
 4659 WHAT SURVIVES:
 4660 Both complete radius-two density theorems, both path-product barriers,
 4661 and the pointwise two-smallest-row height restriction. None assumes a
 4662 fixed prime alphabet through the path.
 4663 DO NOT REPEAT:
 4664 Do not apply a successful-endpoint sieve to a failed closed core;
 4665 do not treat two lower bounds as a lower-versus-upper contradiction.
 4666 POSSIBLE REPLACEMENT:
 4667 A bound for BAD path endpoints or moving support that supplies an
 4668 independent upper restriction in a genuinely closed failed world.
 4669
 4670 RESULT-ID: MULTIGENERATION-COFACTOR-ADVERSARIAL-SCAN-20260906B
 4671 STATUS: VERIFIED COMPUTATION; no asymptotic conclusion inferred.
 4672 DOMAIN AND METHOD:
 4673 Exhaustive primes P in [1000,2000) and [10000,20000), respectively
 4674 135 and 1033 starts. For each, $N=2(P-1)$, same-N roots $k=0,1,2,5$.
 4675 Enumerate every SIMPLE stopped successful path of lengths 1 through 4;
 4676 these need not be shortest paths if another branch succeeds earlier.
 4677 Stop expansion at GOOD vertices, and exclude repeated labels within
 4678 one path. Exact SPF sieve through 1,000,100. No factor-size exclusions.
 4679 The diagnostic's depth 4 is fixed independently of asymptotic $D(X)$.
 4680 The small-product diagnostic uses the exact integer test product¹⁰
 4681 $\leq X^9$, i.e. $\epsilon=1/10$, so floating-point comparisons do not decide
 4682 membership.
 4683
 4684 CHECKS AND RESULTS:
 4685 22,562 actual successful paths passed every affine-identity, exact-
 4686 modulus, reduced-coefficient-integrality, and gcd-compatibility check.
 4687 For all canonical paths the final gcd e was exactly 1. For 2,320 paths
 4688 with small cofactor product, the progression variable was ≥ 1 , all
 4689 actual prime values exceeded their leading coefficients, every form
 4690 was primitive, and every pair determinant was nonzero.
 4691 There were 4,767 depth-three-or-four paths where e failed to divide
 4692 at least one earlier partial product A_j . Thus the proof of integer
 4693 reduced coefficients is indispensable; division of each A_j by e
 4694 would be an actual error.
 4695 Maximum observed e was 85, at $P=10343$, $N=20684$, $Q=p_5=10427$, $h=85$,
 4696 path $Q \rightarrow 13 \rightarrow 7 \rightarrow 29 \rightarrow 3$. Its cofactor product is
 4697 11,437,574,202,585. This large-product path is outside the cofactor
 4698 sieve domain, and its progression variable is $t=0$. It is evidence of
 4699 the exact large-modulus obstruction, not a counterexample to the bound.
 4700
 4701 NONREDUNDANCY WITNESS:
 4702 $P=1279$, $N=2556$, $Q=p_5=1301$, $h=23$ admits
 4703 $1301 \rightarrow 251 \rightarrow 461 \rightarrow 419$, $N=419=2137$ prime.
 4704 All three edge cofactors equal 5. Hence $A=125$ while the label product
 4705 is 48,482,909. The new cofactor-product count includes this type of
 4706 configuration even though the earlier label-product count does not.
 4707 Its exact forms at $t=10$ are

```

4708 P=125t+29, Q=125t+51, q1=25t+1, q2=45t+11,
4709 g3=41t+9, g=209t+47.
4710 An actual finite example inside an asymptotically sparse region is
4711 expected and does not contradict the theorem.
4712
4713 At X=10000, among all depth-three-or-four successful paths, 14,200
4714 had both products large, 1046 had only the label product small, and
4715 58 had only the cofactor product small. These finite multiplicity
4716 counts identify regions tested by the two arguments; they are not
4717 counts of exceptional starts, convergence evidence, or proof that the
4718 residual region has positive asymptotic density.
4719
4720 BEGIN DIAGNOSTIC SOURCE audit_cofactor_paths_20260906b.py
4721 import numpy as np
4722 from math import isqrt,gcd,prod
4723 from collections import Counter
4724 from itertools import combinations
4725 import json
4726 LIMIT=1000100
4727 spf=np.zeros(LIMIT+1,dtype=np.int32)
4728 for p in range(2,isqrt(LIMIT)+1):
4729     if spf[p]==0:
4730         v=spf[p*p:p];v[v==0]=p
4731 primes=np.flatnonzero(spf[2:]==0)+2
4732 ip=np.zeros(LIMIT+1,dtype=np.bool_);ip[primes]=True
4733
4734 def pf(n):
4735     out=[]
4736     while n>1:
4737         p=int(spf[n]) or n;out.append(p)
4738         while n%p==0:n//=p
4739     return out
4740
4741 stats=Counter();domains={};witness={};max_e=(0,None)
4742 for X in [1000,10000]:
4743     inds=np.flatnonzero((primes>=X)&(primes<2*X))
4744     local=Counter()
4745     for idx in inds:
4746         P=int(primes[idx]);x=P-1;N=2*x
4747         for k in [0,1,2,5]:
4748             Q=int(primes[idx+k]);h=Q-x
4749             def visit(path,aa):
4750                 global max_e
4751                 u=path[-1];d=len(aa)
4752                 if ip[N-u]:
4753                     if d==0:
4754                         local['direct_GOOD_roots']+=1;return
4755                     local[f'success_paths_depth_{d}']+=1
4756                     AP=[1];CP=[1]
4757                     for a in aa:
4758                         CP.append(2*AP[-1]-CP[-1]);AP.append(AP[-1]*a)
4759                     A=AP[-1];C=CP[-1];e=gcd(A,C);mod=A//e
4760                     assert h%e==0
4761                     if k==0:assert e==1
4762                     residue=((-1)**(d+1)*(h//e)*pow(C//e,-1,mod))%mod if mod>1 else 0
4763                     if residue==0:residue=mod
4764                     assert (x-residue)%mod==0
4765                     t=(x-residue)//mod
4766                     forms=[(mod,residue+1)]
4767                     values=[P]
4768                     if k:
4769                         forms.append((mod,residue+h));values.append(Q)
4770                     for j in range(1,d+1):
4771                         assert AP[j]*path[j]==CP[j]*x+(-1)**j*h
4772                         assert CP[j]*mod%AP[j]==0
4773                         assert (CP[j]*residue+(-1)**j*h)%AP[j]==0
4774                         forms.append((CP[j]*mod//AP[j],(CP[j]*residue+(-1)**j*h)//AP[j]))
4775                         values.append(path[j])
4776                     last_u,last_v=forms[-1]
4777                     forms.append((2*mod-last_u,2*residue-last_v));values.append(N-u)
4778                     assert all(z>0 for z,b in forms)
4779                     assert [z*t+b for z,b in forms]==values
4780                     assert gcd(mod,C//e)==1 and len(set(values))==len(forms)
4781                     stats['all_affine_and_modulus_checks']+=1
4782                     M=prod(path[1:]);small_A=M**10<=X**9;small_M=M**10<=X**9
4783                     if small_A:
4784                         assert t>=1 and all(v>z for v,(z,b) in zip(values,forms))
4785                         assert all(gcd(z,b)==1 for z,b in forms)
4786                         assert all(z*d-w*b!=0 for (z,b),(w,d) in combinations(forms,2))
4787                         stats['short_cofactor_product_full_filter_checks']+=1
4788                         rec={'X':X,'P':P,'N':N,'k':k,'Q':Q,'h':h,'path':path,'cofactors':aa,'label_product':M,'cofactor_product':A,'C_d':C,'e':e,'modulus':mod,'resid':
4789                             ue:'residue','t':t,'prime_forms':forms,'prime_values':values}
4789                         if e>max_e[0]:max_e=(e,rec)
4790                         if d>=3:
4791                             local[f'depth3or4_small_label_{small_M}_small_cofactor_{small_A}']+=1
4792                             if small_A and not small_M:witness.setdefault('large_labels_small_cofactors_depth3or4',rec)
4793                             if e>1 and any(z%e for z in AP[1:-1]):
4794                                 stats['e_does_not_divide_all_earlier_partial_products_depth3or4']+=1
4795                                 witness.setdefault('reduced_earlier_coefficients_require_proof',rec)
4796                             return
4797                         if d==4:return
4798                         for q in pf(N-u):
4799                             if q in path:continue
4800                             visit(path+[q],aa+[(N-u)//q])
4801             visit([Q],[])
4802     domains[str(X)]={'prime_starts':len(inds),'counts':dict(local)}
4803 print(json.dumps({'domain':'Exhaustive primes in [X,2X), X=1000,10000; roots k=0,1,2,5; all SIMPLE stopped successful paths through length 4, not
↳ only shortest paths. No size exclusions. Exact SPF through 1,000,100. Small-product flag is exact product^10<=X^9, i.e. epsilon=1/10. This
↳ diagnostic depth is fixed independently of the asymptotic
↳ D(X), 'intervals':domains,'audit':dict(stats),'witnesses':witness,'maximum_e':max_e,indent=2}))
4804 END DIAGNOSTIC SOURCE
4805 BEGIN DIAGNOSTIC OUTPUT
4806 {
4807     "domain": "Exhaustive primes in [X,2X), X=1000,10000; roots k=0,1,2,5; all SIMPLE stopped successful paths through length 4, not only shortest
↳ paths. No size exclusions. Exact SPF through 1,000,100. Small-product flag is exact product^10<=X^9, i.e. epsilon=1/10. This diagnostic depth
↳ is fixed independently of the asymptotic D(X).",

```

```

4808 "intervals": {
4809   "1000": {
4810     "prime_starts": 135,
4811     "counts": {
4812       "success_paths_depth_1": 270,
4813       "success_paths_depth_2": 394,
4814       "success_paths_depth_3": 582,
4815       "depth3or4_small_label_False_small_cofactor_False": 1313,
4816       "direct_GOOD_roots": 143,
4817       "success_paths_depth_4": 791,
4818       "depth3or4_small_label_True_small_cofactor_False": 54,
4819       "depth3or4_small_label_False_small_cofactor_True": 6
4820     }
4821   },
4822   "10000": {
4823     "prime_starts": 1033,
4824     "counts": {
4825       "success_paths_depth_4": 9541,
4826       "depth3or4_small_label_False_small_cofactor_False": 14200,
4827       "success_paths_depth_2": 3342,
4828       "success_paths_depth_3": 5763,
4829       "depth3or4_small_label_True_small_cofactor_False": 1046,
4830       "success_paths_depth_1": 1879,
4831       "direct_GOOD_roots": 835,
4832       "depth3or4_small_label_False_small_cofactor_True": 58
4833     }
4834   }
4835 },
4836 "audit": {
4837   "all_affine_and_modulus_checks": 22562,
4838   "short_cofactor_product_full_filter_checks": 2320,
4839   "e_does_not_divide_all_earlier_partial_products_depth3or4": 4767
4840 },
4841 "witnesses": {
4842   "reduced_earlier_coefficients_require_proof": {
4843     "X": 1000,
4844     "P": 1013,
4845     "N": 2024,
4846     "k": 1,
4847     "Q": 1019,
4848     "h": 7,
4849     "path": [
4850       1019,
4851       3,
4852       43,
4853       283
4854     ],
4855     "cofactors": [
4856       335,
4857       47,
4858       7
4859     ],
4860     "label_product": 36507,
4861     "cofactor_product": 110215,
4862     "C_d": 30821,
4863     "e": 7,
4864     "modulus": 15745,
4865     "residue": 1012,
4866     "t": 0,
4867     "prime_forms": [
4868       [
4869         15745,
4870         1013
4871       ],
4872       [
4873         15745,
4874         1019
4875       ],
4876       [
4877         47,
4878         3
4879       ],
4880       [
4881         669,
4882         43
4883       ],
4884       [
4885         4403,
4886         283
4887       ],
4888       [
4889         27087,
4890         1741
4891       ]
4892     ],
4893     "prime_values": [
4894       1013,
4895       1019,
4896       3,
4897       43,
4898       283,
4899       1741
4900     ]
4901   },
4902   "large_labels_small_cofactors_depth3or4": {
4903     "X": 1000,
4904     "P": 1279,
4905     "N": 2556,
4906     "k": 5,
4907     "Q": 1301,
4908     "h": 23,
4909     "path": [
4910       1301,
4911       251,
4912       461,
4913       419

```

```

4914 ],
4915 "cofactors": [
4916     5,
4917     5,
4918     5
4919 ],
4920 "label_product": 48482909,
4921 "cofactor_product": 125,
4922 "C_d": 41,
4923 "e": 1,
4924 "modulus": 125,
4925 "residue": 28,
4926 "t": 10,
4927 "prime_forms": [
4928     [
4929         125,
4930         29
4931     ],
4932     [
4933         125,
4934         51
4935     ],
4936     [
4937         25,
4938         1
4939     ],
4940     [
4941         45,
4942         11
4943     ],
4944     [
4945         41,
4946         9
4947     ],
4948     [
4949         209,
4950         47
4951     ]
4952 ],
4953 "prime_values": [
4954     1279,
4955     1301,
4956     251,
4957     461,
4958     419,
4959     2137
4960 ]
4961 }
4962 },
4963 "maximum_e": [
4964     85,
4965     {
4966         "X": 10000,
4967         "P": 10343,
4968         "N": 20684,
4969         "k": 5,
4970         "Q": 10427,
4971         "h": 85,
4972         "path": [
4973             10427,
4974             13,
4975             7,
4976             29,
4977             3
4978         ],
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4981             2953,
4982             713,
4983             6885
4984         ],
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4987         "C_d": 3317803385,
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4989         "modulus": 134559696501,
4990         "residue": 10342,
4991         "t": 0,
4992         "prime_forms": [
4993             [
4994                 134559696501,
4995                 10343
4996             ],
4997             [
4998                 134559696501,
4999                 10427
5000             ],
5001             [
5002                 170544609,
5003                 13
5004             ],
5005             [
5006                 91076481,
5007                 7
5008             ],
5009             [
5010                 377318817,
5011                 29
5012             ],
5013             [
5014                 39032981,
5015                 3
5016             ],
5017             [
5018                 269080360021,
5019                 20681

```

```

5020     ]
5021 ],
5022 "prime_values": [
5023     10343,
5024     10427,
5025     13,
5026     7,
5027     29,
5028     3,
5029     20681
5030 ]
5031 }
5032 ]
5033 }
5034 END DIAGNOSTIC OUTPUT
5035
5036 FINAL HOSTILE AUDIT 2026-09-06B:
5037 The radius-two pair proof retains the two independent degeneracies:
5038 gcd(b,2a-1)|h in CC and r|(2a-1), r|h in CL. Both occur in actual
5039 successful paths. The cofactor-chain proof uses final integrality to
5040 justify all earlier affine forms, rather than assuming reductions
5041 commute with partial products. The growing-k sieve constants, the
5042 factorial sum, the h-average of the exact gcd, and the prime-gap prefix
5043 consumer are written explicitly. Parameters epsilon,c,kappa in the
5044 dual theorem are FIXED; no uniformity as they approach their endpoints
5045 is claimed. The only asymptotic variable in that theorem is X.
5046 No independent external referee audit or literature novelty audit has
5047 been performed. Computational checks test algebra and scope, not the
5048 asymptotic sieve theorem. All original corpus files are unchanged.
5049
5050 =====
5051 STRONGEST NEW RESULTS
5052 =====
5053 1. SAME-N-GROWING-DEPTH-DUAL-PRODUCT-BARRIER: for fixed c+kappa<1,
5054 almost every prime P in [X,2X] has, simultaneously for the same-N
5055 roots k<=floor((log X)^kappa), no successful simple path of length
5056 <=floor(c*loglog X/logloglog X) whose LABEL product or COFACTOR
5057 product is <=X^(1-epsilon). The new cofactor argument uses an exact
5058 moving modulus, growing-form sieve, and an ordered-simplex bound.
5059 2. SAME-N-GROWING-ROOTS-COMplete-RADIUS-TWO: almost every such P has
5060 COMPLETE radius-two BADness simultaneously for all same-N roots
5061 k<K=o((log X)^(1/6)/(loglog X)^(3/2)). This extends the previous
5062 canonical radius-two theorem to nearby roots, with all gcd cases.
5063 3. Exact same-N non-dominance: at N=2180 p0 is BAD through radius two
5064 while p1 is directly GOOD; at N=2024 p0 succeeds at depth two while
5065 p1 has complete radius-two BADness. This refutes either universal
5066 inclusion of the two shallow BADness events, not eventual success.
5067 All novelty claims remain NOT YET AUDITED.
5068
5069 =====
5070 CURRENT FRONTIER
5071 =====
5072 No proof of binary Goldbach, no universal p0 success theorem, and no
5073 exclusion of moving closed BAD support. Complete radius three remains
5074 uncontrolled. The new count removes another class of large-label paths,
5075 namely those with small cofactor product, but the region where BOTH
5076 products are large remains. The successful-path barriers have no
5077 closed-core consumer: inside a BAD closed core their hypotheses never
5078 occur. The two-smallest-row height restriction remains a compatible
5079 lower bound, with no contradictory upper bound. The Scott--Bozek
5080 classification issue was not used or advanced in this continuation.
5081
5082 =====
5083 p0 STATUS
5084 =====
5085 A stronger SAME-N no-go was obtained: complete radius-two BADness and
5086 the growing-depth dual product restriction are shared by nearby roots.
5087 No nontrivial property holding for every p0 and failing at a nearby
5088 same-N root with an eventual-reachability consequence was proved.
5089 Finite shallow success can favor either p0 or p1. The empirical
5090 exceptionality of eventual p0 traversal remains unexplained.
5091
5092 =====
5093 EXACT NEXT QUESTION
5094 =====
5095 Is the number of primes P in [X,2X] admitting the following system
5096 o(X/log X)?
5097 P=a*q^2,
5098 b*r=(2a-1)*q^2,
5099 c*s=2*a*q^2-r,
5100 g=2*a*q^2-s prime,
5101 where q,r,s are distinct odd primes, a,b,c are odd integers >=3,
5102 and BOTH q*r*s>X^(9/10) and a*b*c>X^(9/10).
5103 Count distinct starts P, not paths with multiplicity. This is the
5104 explicit remaining region for canonical stopped success at depth three.
5105 Together with the proved small-product barriers and the existing
5106 radius-two estimate, an affirmative answer would establish complete
5107 canonical radius-three BADness for almost all h=1 starts. It would
5108 still be an asymptotic traversal theorem, not a Goldbach proof.
5109
5110 =====
5111 CONTINUATION 2026-09-07: RETURN FROM ARTICLE PREPARATION TO RESEARCH
5112 =====
5113
5114 The 2026-09-06B frontier above remains the starting point. No recent
5115 asymptotic result is being used critically in the following exact arguments.
5116 Research resumes in this same raw TXT file. The article is a separate,
5117 completed deliverable and is not the research checkpoint.
5118
5119 RESULT-ID: SAME-N-SUPPORT-WITNESS-RANK-CORRECTION-20260907
5120 STATUS: SCOPE-CORRECTED / VERIFIED COMPUTATION.
5121 EXACT STATEMENT:
5122 For N=212, the primes at or above N/2 begin 107,109,113. Consequently
5123 113 is p2, not p1. A temporary p1 label during article preparation was
5124 incorrect. The support comparison at N=212 is valid as p0 versus p2:
5125 PF(212-107)={3,5,7}, PF(212-113)={3,11}, PF(212-3)={11,19}.

```

5126 A genuine same-N p0 versus p1 replacement is N=344:
 5127 p0=173, p1=179,
 5128 344-173=171=3²*19,
 5129 344-179=165=3*5*11,
 5130 344-3=341=11*31.
 5131 Thus the least first factor 3 has a second row disjoint from the p0
 5132 factor set {3,19}, but meeting the p1 factor set {3,5,11} at 11.
 5133 Both searches have first-success depth 2 and inclusive work W=5.
 5134 The common successful tail is 3 ->31, with 344-31=313 prime.
 5135 For p0 the FIFO test order is 173,3,19,11,31; for p1 it is
 5136 179,3,5,11,31. All indicated primalities and all root ranks were checked
 5137 by exact trial division, including the absence of intervening primes.
 5138 DEPENDENCIES / PROVENANCE:
 5139 The underlying least-factor escape is already PCRD-H1-1 in the master,
 5140 not a newly discovered general theorem. The new point is the exact
 5141 same-N witness and correction of the previously mislabelled index.
 5142 HOSTILE CHECK:
 5143 This distinguishes initial-support retention, not eventual success.
 5144 Both roots here even have the SAME first-success depth and work.
 5145 Neither this witness nor the minimum-factor lemma implies universal p0
 5146 success. The same-N radius-two witnesses N=2024 and N=2180 retain their
 5147 previously stated p0/p1 ranks after explicit enumeration.
 5148 NOVELTY: NOT YET AUDITED.
 5149
 5150 RESULT-ID: EXACT-SCAN-WORK-CONVENTION-AUDIT-20260907
 5151 STATUS: VERIFIED COMPUTATION / SOURCE-CORRECTION.
 5152 DOMAIN:
 5153 Exact full FIFO BFS, ascending distinct prime factors, duplicate suppression
 5154 on discovery; every even $4 \leq N \leq 10^7$. No work/depth cutoff, random sampling,
 5155 probable-prime test or incomplete factorization.
 5156 RESULT:
 5157 4,999,999 canonical inputs; zero failures. Inclusive work W counts the
 5158 successful vertex. Mean $W=7.052697210539442$, maximum $W=151$ first at
 5159 $N=3542948$; maximum success depth 10 first at $N=11822$.
 5160 Through $N \leq 10^6$: mean $W=5.804359608719217$, max $W=133$ at $N=976904$.
 5161 The old article's displayed step statistics use W, but its pseudocode
 5162 counted expanded composite rows E. On success $W=E+1$; on failure $W=E$.
 5163 This is a counting-convention repair, not a change to success verdicts.
 5164 Root-rank ablation $k=0, \dots, 6$ through $N \leq 200000$ reproduces failures
 5165 $[0, 2, 4, 5, 3, 2, 1]$, with eligible composite-complement denominators
 5166 $[73394, 81448, 81043, 80909, 80788, 80980, 80829]$.
 5167 The complete upper-half composite-root census through $N \leq 20000$ has
 5168 4,041,839 eligible roots, 301 failures at the six known hosts, and
 5169 2261 excluded unit-complement rows. Host multiplicities are
 5170 128:8, 1718:68, 1862:63, 1928:57, 2200:4, 6142:101.
 5171 Independent trial division verifies all 999 canonical inputs through
 5172 2000, all three named record inputs, and ten exact example roots with
 5173 explicit prime ranks. This is NOT a second independent full-range scan.
 5174 SOURCE:
 5175 Reproduction source and raw compressed records are in the completed
 5176 p0_zenodo_release archive. These regenerated records do not replace
 5177 proof and do not extend the original canonical numerical bound.
 5178
 5179 RESULT-ID: PRIME-LIFT-FULL-BFS-CONJUGACY-20260907
 5180 STATUS: PROVED.
 5181 EXACT STATEMENT:
 5182 Let $N \geq 4$ be even, d a prime divisor of N , $x=N/d$, and $m=x-1 \geq 2$.
 5183 Suppose $R=N-m=(d-1)x+1$ is prime. Use full factor BFS from any admissible
 5184 prime, including the inactive start d ; remove duplicate visits, use FIFO
 5185 and enqueue all distinct prime factors in increasing order.
 5186 Then $R > N/2$, $R \leq N-2$, and:
 5187 (A) If m is prime, R succeeds immediately with $(\text{depth}, W) = (0, 1)$, while
 5188 d succeeds via $d \rightarrow m$ with $(\text{depth}, W) = (1, 2)$.
 5189 (B) If m is composite, the two searches have identical verdicts. If they
 5190 succeed, their first-success depths and inclusive work counts agree.
 5191 In fact their entire ordered test transcripts agree after replacing
 5192 the initial d by R . All complete stopped frontiers after the initial
 5193 root are the same. On failure the reachable sets differ only by
 5194 replacing d with R .
 5195 At graph level in (B), deletion of the harmless self-loop $d \rightarrow d$ and
 5196 renaming that root gives the same complete reachable stopped graph.
 5197
 5198 PROOF:
 5199 Since $d \geq 2$, $R=N-N/d+1 \geq N/2+1$. The assumption $m \geq 2$ gives $R \leq N-2$.
 5200 Also $R-d=(d-1)(x-1)=(d-1)m > 0$. If $d|m$, then $d|N-m=R$, contradicting
 5201 primality of $R > d$. Hence d does not divide m .
 5202 If a prime q divides both $m=x-1$ and $N=dx$, it must divide d , since
 5203 q cannot divide x and $x-1$. This would mean $q=d$, already excluded.
 5204 Therefore every prime factor of m is active, i.e. coprime to N .
 5205 We have the EXACT row relation
 5206 $N-d=d*m, \quad N-R=m,$
 5207 so $PF(N-d)=\{d\}$ disjoint-union $PF(m)$.
 5208 The self-loop at d is ignored by duplicate suppression. Thus, after
 5209 processing the first BAD row, both queues contain exactly the same
 5210 ordered set $PF(m)$, each at distance 1.
 5211 Activity is preserved: if p is an active prime and $q|N-p$ also divides
 5212 N , then $q|p$, so $q=p$, contradicting p not dividing N . Consequently no
 5213 later vertex can equal the inactive prime d .
 5214 Nor can a later prime parent p have an edge to R in the composite- m
 5215 case. Since $R > N/2$, a positive $N-p < N$ divisible by R must equal R ;
 5216 this would give $p=N-R=m$, which is composite. Thus the root R is never
 5217 encountered again from any prime vertex, including any terminal row.
 5218 The two discovered sets therefore differ only in their initial root;
 5219 this difference cannot affect any later duplicate test. Inductively,
 5220 FIFO removes the same next prime at the same distance, tests the same
 5221 row, and enqueues the same new factors. This proves (B), including
 5222 complete-frontier equality and the graph statement. No assumption of
 5223 success is used in the induction.
 5224 If m is prime, the R -root test is immediately successful. The d -root
 5225 row has factors d and m , with $m! \equiv d$ as established above. After ignoring
 5226 d 's self-loop, m is tested at depth 1 and is GOOD because $N-m=R$ is
 5227 prime. This proves (A).
 5228
 5229 HOSTILE CHECK:
 5230 The initial d is INACTIVE. The active-root theorem cannot be invoked
 5231 at that initial vertex; the proof explicitly separates its self-loop

5232 and proves that all other children are active. The conjugacy is for
 5233 composite m ; applying its depth equality to prime m would be false.
 5234 The absence of an incoming edge to R is proved, rather than inferred
 5235 from a selected BAD path. Entire BFS queues are compared, not one path.
 5236 The theorem does not assert that either search succeeds.
 5237 DEPENDENCIES: Elementary divisibility, primality and finite full BFS only.
 5238 RELATION TO GOLDBACH:
 5239 An exact equivalence of two algorithmic reachability questions, not
 5240 an independent source of a Goldbach representation.
 5241 RELATION TO p_0 :
 5242 Taking $d=2$ gives $x=P-1$, $N=2(P-1)$, $R=P$. If $P \geq 5$ is prime, P is the
 5243 canonical root. Therefore every non-direct canonical $h=1$ traversal
 5244 is, for ALL generations, exactly the traversal from prime 2 after the
 5245 stated initial-root replacement. An invariant of the resulting stopped
 5246 transition graph which ignores that replacement and its self-loop
 5247 cannot distinguish these two starts.
 5248 This is a restricted $h=1$ theorem; root 2 is not a nearby upper root.
 5249 It does not claim equivalence between p_0 and p_k for $k \geq 1$.
 5250 WHAT REMAINS OPEN:
 5251 Whether the prime-lift condition with $d=2$ excludes eventual failure.
 5252 The same question for all prime divisors d is FALSE, as the following
 5253 exact example shows. Thus lift primality alone is not a general closer.
 5254 NOVELTY: NOT YET AUDITED. Targeted corpus search found the $h=1$
 5255 least-factor/shift-2 lemmas but no prior full-BFS conjugacy entry.
 5256
 5257 FAIL-ID: PRIME-LIFT-PRIMALITY-FORCES-SUCCESS-20260907
 5258 STATUS: KILLED.
 5259 CLAIM / ROUTE:
 5260 If the affine lift $R=(d-1)N/d+1$ of a prime divisor $d|N$ is prime, then
 5261 full factor search from d (equivalently its lift R) must succeed.
 5262 WHY IT LOOKED PROMISING:
 5263 Lift primality automatically removes inactive factors from $PF(N/d-1)$,
 5264 and for $d=2$ gives exactly the canonical $h=1$ initial condition.
 5265 EXACT COUNTEREXAMPLE:
 5266 $N=1862$, $d=7$, $x=266$, $m=265=5 \cdot 53$, $R=1597$ prime.
 5267 Both complete BFS runs fail, each after $W=22$ tests. Their common active
 5268 nonroot reachable set U has the 21 labels listed by the rows below.
 5269 For every p in U , $N-p$ is composite and has all factors in U . The root
 5270 1597 also has all its factors in U . This is a COMPLETE closure
 5271 certificate, independent of any claim about terminal-SCC classification.
 5272 $1862-3 = 1859 = 11 \cdot 13^2$
 5273 $1862-5 = 1857 = 3 \cdot 619$
 5274 $1862-11 = 1851 = 3 \cdot 617$
 5275 $1862-13 = 1849 = 43^2$
 5276 $1862-17 = 1845 = 3^2 \cdot 5 \cdot 41$
 5277 $1862-41 = 1821 = 3 \cdot 607$
 5278 $1862-43 = 1819 = 17 \cdot 107$
 5279 $1862-47 = 1815 = 3 \cdot 5 \cdot 11^2$
 5280 $1862-53 = 1809 = 3^3 \cdot 67$
 5281 $1862-67 = 1795 = 5 \cdot 359$
 5282 $1862-83 = 1779 = 3 \cdot 593$
 5283 $1862-107 = 1755 = 3^3 \cdot 5 \cdot 13$
 5284 $1862-113 = 1749 = 3 \cdot 11 \cdot 53$
 5285 $1862-167 = 1695 = 3 \cdot 5 \cdot 113$
 5286 $1862-179 = 1683 = 3^2 \cdot 11 \cdot 17$
 5287 $1862-251 = 1611 = 3^2 \cdot 179$
 5288 $1862-359 = 1503 = 3^2 \cdot 167$
 5289 $1862-593 = 1269 = 3^3 \cdot 47$
 5290 $1862-607 = 1255 = 5 \cdot 251$
 5291 $1862-617 = 1245 = 3 \cdot 5 \cdot 83$
 5292 $1862-619 = 1243 = 11 \cdot 113$
 5293 $1862-1597 = 265 = 5 \cdot 53$
 5294
 5295 The inactive root has $1862-7=1855=7 \cdot 5 \cdot 53$, so only its own duplicate
 5296 self-loop is added. Primality and all rows were independently checked
 5297 by trial division. The prime-divisor/lift enumeration was exhaustive
 5298 only over divisors of the six known hosts, not over all N : this was the
 5299 only admissible prime lift in that small diagnostic.
 5300 For the SAME N , $p_0=937$ succeeds along $937 \rightarrow 37 \rightarrow 73$, since
 5301 $1862-937=5^2 \cdot 37$, $1862-37=5^2 \cdot 73$, and $1862-73=1789$ prime.
 5302 Thus a successful canonical search can have a separate branch into U ;
 5303 reaching a BAD closed component does not imply full rooted failure.
 5304 WHAT SURVIVES:
 5305 The exact conjugacy and the special $d=2$ question. The counterexample
 5306 has $d=7$ and does not contradict canonical $h=1$ success.
 5307 DO NOT REPEAT:
 5308 Do not infer success merely from a prime affine lift or the absence of
 5309 additional inactive factors. Those conditions hold in this example.
 5310 POSSIBLE REPLACEMENT:
 5311 A restriction using $d=2$ essentially, or a pointwise theorem controlling
 5312 access from $PF(N/2-1)$ to moving closed active support.
 5313
 5314
 5315 RESULT-ID: LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE-20260907
 5316 STATUS: PROVED.
 5317 EXACT STATEMENT:
 5318 Let $N \geq 4$ be even, $R \geq N/2$ an admissible prime root, $m=N-R$ composite,
 5319 $h=R-N/2$, and $q=\min PF(m)$. Suppose $h < q$. In the COMPLETE stopped factor
 5320 search from R , either:
 5321 (i) a GOOD vertex occurs at depth at most 2, or
 5322 (ii) at least THREE DISTINCT nonroot primes occur by depth 3.
 5323 In particular, if the full rooted traversal fails, its nonroot reachable
 5324 active set has at least three elements. No active closed BAD alphabet
 5325 of size at most two can contain the whole first factor set $PF(m)$.
 5326 This is a pointwise statement, with no asymptotic or fixed-envelope premise.
 5327
 5328 PROOF:
 5329 Since m is composite and $R \geq N/2$ is prime, $R > N/2$, R is odd and active,
 5330 m is odd, $q \geq 3$, and h is a positive integer. Every subsequent prime is
 5331 active, by the activity lemma. A child t of an odd composite BAD row
 5332 $N-p$ has $t < N/3$: its cofactor is an odd integer at least 3.
 5333 Assume there is no GOOD vertex at depth ≤ 2 . Put $S=PF(m)$.
 5334 If $|S| \geq 3$, the conclusion already holds at depth 1.
 5335 If $|S|=2$, consider its least element q . For any r in S ,
 5336 $r|N-q$ implies $r|(2h-q)$, since $N=2m+2h$.
 5337 But $h < q$ implies $0 < |2h-q| < q \leq r$; the zero case is excluded because q

5338 is odd. Thus $PF(N-q)$ is disjoint from S . The row $N-q$ is composite
 5339 under our assumption and has at least one prime factor, reached at
 5340 depth 2 outside S . Together with S this gives three distinct labels.
 5341 It remains to treat $S=\{q\}$. Then $m=q^u$ for an integer $u \geq 2$. The BAD
 5342 row $N-q$ has no q factor, by activity. If it has two distinct factors,
 5343 we already have at least three nonroot labels at depth ≤ 2 .
 5344 Otherwise $N-q=b^v$ for a prime $b \neq q$ and $v \geq 2$. The vertex b occurs
 5345 at depth 2, is BAD by assumption, and $b < N/3$. It cannot divide its
 5346 own row. If no third label occurred by depth 3, the entire factor set
 5347 of $N-b$ would consequently be $\{q\}$, so $N-b=q^w$ with $w \geq 2$.
 5348 If $w \leq u$, then $b=N-q^w=N-q^u=R > N/2$, contradicting $b < N/3$.
 5349 If $w > u$, then $q^w = q^u m$. But
 5350 $q^u m = (q-2)m - 2h \geq m - 2h \geq q^{2-2h} > 0$,
 5351 since $q \geq 3$ and $h < q$. This contradicts $q^w = N - b < N$.
 5352 Thus $N-b$ must introduce a third distinct nonroot prime by depth 3.
 5353 Every case has been covered.
 5354 For the closed-alphabet corollary, a set containing $PF(m)$ and closed
 5355 under all BAD rows contains every nonroot descendant. If it had at
 5356 most two elements, the proved alternative would force a GOOD vertex,
 5357 contrary to BAD closure. No terminal-core assumption is used.
 5358
 5359 HOSTILE CHECK:
 5360 The assertion is about COMPLETE rows, not a selected-factor path.
 5361 In the singleton case the strict inequality $q^u m > N$ is the fragile
 5362 arithmetic step, and is explicitly justified by $h < q$ and $u \geq 2$.
 5363 Dropping the small-gap premise is false: at $N=2200$, $R=1471$ prime,
 5364 $m=729=3^6$, $h=371$, $q=3$,
 5365 $2200-3=2197=13^3$, $2200-13=2187=3^7$.
 5366 The full search is BAD and has exactly the two nonroot labels $\{3,13\}$.
 5367 This uses the existing $N=2200$ two-core certificate, not a new EAC.
 5368 Crucially the conclusion does NOT exclude a two-vertex TERMINAL core
 5369 reached after additional transient labels; it only excludes a whole
 5370 nonroot reachable alphabet of size ≤ 2 . This avoids the open
 5371 Scott--Bozek classification issue and does not repair that citation.
 5372
 5373 DEPENDENCIES / HISTORICAL AUDIT:
 5374 The $|S| \geq 2$ step is an explicit consumer of the existing minimum-factor
 5375 escape corollary (master cor: minfactor, PCRD-Hi-1 at $h=1$). The singleton
 5376 step uses a gap between consecutive powers of q and the upper-root
 5377 geometry. The historical C-MONOMIAL-TWO-EDGE-RIGIDITY and its novelty
 5378 corrections concern a square-map/high-floor mechanism; no claim is made
 5379 that pure-power path arguments as such are new. No Matveev estimate,
 5380 height threshold, growing sieve or global Goldbach failure is assumed.
 5381 RELATION TO GOLDBACH:
 5382 A small, unconditional restriction on a possible failed traversal.
 5383 It does not rule out larger alphabets or prove a prime-pair existence.
 5384 RELATION TO p_0 :
 5385 For every canonical $h=1$ root with composite first complement it applies
 5386 automatically, since $q \geq 3$. It ALSO applies to every nearby or other
 5387 upper root satisfying its stated $h < q$ premise. Thus it is not a universal
 5388 p_0 -only invariant and supplies no explanation of eventual p_0 reliability.
 5389 WHAT REMAINS OPEN:
 5390 A quantitative lower bound growing with N on the complete moving
 5391 reachable support, or any new arithmetic mechanism excluding its closure.
 5392 The old automatic envelope-iteration obstruction remains in force.
 5393 NOVELTY: NOT YET AUDITED.
 5394
 5395
 5396 RESULT-ID: PRIME-LIFT-AND-LOW-GAP-EXHAUSTIVE-AUDIT-20260907
 5397 STATUS: VERIFIED COMPUTATION.
 5398 DOMAIN / METHOD:
 5399 Exact smallest-prime-factor sieve; FIFO queue; ascending distinct factors;
 5400 duplicates suppressed on enqueue; no work or depth cutoff for full BFS.
 5401 For every even N from 4 through 1,000,000 (499,999 inputs), run BFS from 2.
 5402 For every prime divisor $d|N$ with $m=N/d-1 \geq 2$ and $R=N-m$ prime, compare the
 5403 full queue, depths, verdict and number of composite rows in the two starts
 5404 d and R , including the separate direct-complement case. This is exhaustive
 5405 in that domain, not a random sample or a scan of selected paths.
 5406 COUNTS:
 5407 41,536 admissible $h=1$ inputs, of which 36,971 have composite $P-2$.
 5408 265,847 admissible prime-divisor lifts: 200,602 composite-complement lifts
 5409 and 65,245 direct-complement lifts. All claimed conjugacies and direct-case
 5410 corrections passed. Exactly one lift failed: $(N,d,R)=(1862,7,1597)$, the
 5411 complete certificate recorded above. No $h=1$ input failed in this range.
 5412 The ONLY root-2 failures in this finite range are
 5413 $N=2^a+2$, $2 \leq a \leq 19$, each $W=1$,
 5414 and $N=128$ ($W=9$), $N=1718$ ($W=29$).
 5415 The power-of-two family is rigorously a self-loop: $N=2^a$. For $a=1$,
 5416 $N=4$ succeeds directly; thus that endpoint is deliberately excluded.
 5417 The statement that the list is exhaustive beyond the scanned range is NOT
 5418 claimed. In particular a finite list of known active cores was not assumed
 5419 complete by the scan. At $N=6142$, root 2 actually succeeds along
 5420 $2 \rightarrow 307 \rightarrow 389 \rightarrow 11$, with final complement 6131 prime:
 5421 $6142-2=2^2+5 \cdot 307$, $6142-307=3 \cdot 5 \cdot 389$,
 5422 $6142-389=11 \cdot 523$. Inclusive BFS work $W=12$.
 5423 Thus the mere existence of a known closed BAD component at the SAME N does
 5424 not imply failure from 2 or from any other specified root.
 5425
 5426 SECOND DOMAIN / METRIC:
 5427 Every even $4 \leq N \leq 2000000$ and each admissible upper rank $k=0, \dots, 6$;
 5428 retain exactly composite $m=N-p_k$ with $h=p_k-N/2 \leq PF(m)$.
 5429 If no GOOD vertex occurs at depths 0,1,2, expand ALL these rows and count
 5430 the distinct nonroot labels discovered at depths 1,2,3, without expanding
 5431 depth-3 rows for the count. Early exit when a GOOD vertex at ≤ 2 is found
 5432 is permitted only for deciding that the input is outside the BAD2 class.
 5433 Counts (rank: eligible low-gap inputs, complete BAD through depth 2):
 5434 0: 38715, 7135
 5435 1: 22044, 4503
 5436 2: 14356, 3136
 5437 3: 11229, 2495
 5438 4: 9285, 2096
 5439 5: 7915, 1808
 5440 6: 6790, 1516
 5441 Every input passed the proved lower bound of three distinct nonroot labels.
 5442 The observed minimum was FIVE labels, first at $(N,R,k)=(4422,2213,0)$.
 5443 This stronger minimum is EMPIRICAL ONLY. Exact complete first rows:

5444 4422-2213=47²;
 5445 4422-47=5⁴*7;
 5446 4422-5=7*631;
 5447 4422-7=5*883.
 5448 Hence depths are {47}:1, {5,7}:2, {631,883}:3. The last vertex 883 is
 5449 GOOD (4422-883=3539 prime); 631 is BAD (3791=17*223).
 5450 Thus this is complete BADness THROUGH depth 2, not through depth 3.
 5451 Full BFS succeeds at depth 3 after W=6 tests.
 5452
 5453 INDEPENDENT HOSTILE CHECK:
 5454 Trial division independently verified the N=4422 complete rows and ranks,
 5455 the two nontrivial root-2 failure reachable sets, the N=6142 successful
 5456 path, and the N=1862 lift closure certificate. This is an independent
 5457 certificate audit, not a second exhaustive million-input scan.
 5458 The three-label theorem is not made stronger by the observed five-label
 5459 minimum. The existing master already treats three-vertex closed cores as
 5460 an open coupled prime-power/support system; empirical absence is not a
 5461 classification theorem. No Scott--Bozek uniqueness claim is used.
 5462 REPRODUCTION:
 5463 The retained C++ source audit_prime_lifts_20260907.cpp uses an SPF array
 5464 through 1,001,000, enough for all roots and rows in the stated ranges.
 5465 Its raw result is audit_prime_lifts_20260907.out. All counts, domains,
 5466 exceptions and the direction-changing witnesses are preserved here so
 5467 this checkpoint remains usable without those scratch diagnostics.
 5468 RELATION TO p0 / GOLDBACH:
 5469 Confirms exact finite behaviour and the theorem's breadth across nearby
 5470 roots; proves neither universal canonical success nor binary Goldbach.
 5471 NOVELTY: NOT YET AUDITED.
 5472
 5473
 5474 RESULT-ID: CANONICAL-FIRST-SUPPORT-CYCLE-REALIZATION-20260907
 5475 STATUS: PROVED; an explicit consumer of the surviving partial-CRT method.
 5476 EXACT STATEMENT:
 5477 For every ordered list $q_1, \dots, q_l$ of distinct odd primes, $l \geq 2$, there
 5478 exists a FIXED positive integer h and infinitely many even integers N
 5479 such that, with $P=p_0(N)=N/2+h$ and $m=N-P$:
 5480 (a) each q_i divides m , so every listed label lies at depth 1;
 5481 (b) every q_i is active and BAD;
 5482 (c) $q_1 \rightarrow q_2 \rightarrow \dots \rightarrow q_l \rightarrow q_1$ is a directed cycle of the actual
 5483 factorization graph induced on the complete first support PF(m).
 5484 There may be additional edges, factors and GOOD children. No assertion
 5485 is made that PF(m) equals the prescribed list, that the induced cycle
 5486 is a terminal component, or that the COMPLETE BFS fails.
 5487 Consequently canonical roots have first-factor induced subgraphs with
 5488 arbitrarily long directed BAD cycles and arbitrarily large strongly
 5489 connected subgraphs. The strong component may have additional vertices
 5490 and edges, and need not be closed in the full factorization graph.
 5491
 5492 HYPOTHESES / QUANTIFIERS:
 5493 The list is fixed before constructing h or the arithmetic progression.
 5494 The infinitude is in N for each such fixed list and fixed h . It is not
 5495 uniform in growing l , not a density assertion, and not a statement about
 5496 the $h=1$ subfamily. The construction does NOT prescribe all cofactors.
 5497
 5498 PROOF:
 5499 Set $M = \text{product } q_i$ and write indices cyclically, $q_0 = q_l$. Choose h by
 5500 $2h = q_{i-1} \pmod{q_i}$, for $i=1, \dots, l$.
 5501 The Chinese remainder theorem applies because the odd primes q_i are
 5502 distinct and 2 is invertible modulo each q_i . It determines one residue
 5503 class h modulo M . None of its residues is zero: q_{i-1} is a distinct
 5504 prime, hence is not divisible by q_i . In particular $\gcd(2h, M)=1$.
 5505 Choose any fixed positive representative h of that class.
 5506
 5507 We will choose a large odd prime P with $P=2h \pmod{M}$, and set
 5508 $N=2(P-h)$, $m=P-2h$. To enforce canonicity, for EVERY even integer j
 5509 with $2 \leq j \leq h$ choose a new prime $\text{ell}_j > h$, outside $\{q_1, \dots, q_l\}$, with
 5510 all these new primes distinct, and impose $P \equiv j \pmod{\text{ell}_j}$. Add $P=1$
 5511 $\pmod{2}$. These congruences have pairwise coprime moduli. Their common
 5512 class A modulo $L=2M \cdot \text{product } \text{ell}_j$ is reduced: its residues are nonzero
 5513 modulo every q_i , every ell_j (because $0 < j \leq \text{ell}_j$), and 2.
 5514 Dirichlet's theorem therefore supplies infinitely many primes $P=A$
 5515 $\pmod{L}$. Discard finitely many small P so that $P > 2h+M$, $P-h > 2$,
 5516 $P-j > \text{ell}_j$ for every chosen j , and all inequalities below are strict.
 5517 For odd $1 \leq j \leq h$, $P-j$ is even and > 2 . For even j , ell_j is a proper
 5518 factor of $P-j$. Thus EVERY integer in $[P-h, P)$ is composite. It follows
 5519 that P is exactly the least prime at or above $N/2 = P-h$: $P=p_0(N)$.
 5520
 5521 By $P=2h \pmod{M}$, all q_i divide $m=P-2h$. The root complement is positive
 5522 and divisible by M with $l \geq 2$, hence composite; each q_i is reached in
 5523 the complete first factor set. Modulo q_i , $N=2P-2h=2h \pmod{q_i}$, which
 5524 is nonzero; therefore all q_i are active. For each i ,
 5525 $N - q_i = 2h - q_i \equiv 0 \pmod{q_{i+1}}$.
 5526 For sufficiently large P , $N - q_i > q_{i+1}$; so this row is composite and
 5527 its COMPLETE factorization includes q_{i+1} . This gives every claimed
 5528 BAD cycle edge, including the closing edge. Increasing P changes N ,
 5529 so infinitely many distinct N are obtained. Every label of the cycle
 5530 is in PF(m); therefore the cycle is genuinely in that induced subgraph,
 5531 not merely somewhere deeper in the reachable graph. This proves the
 5532 statement without a prime-tuple conjecture or a bounded-BFS assertion.
 5533
 5534 EXTERNAL INPUT AUDIT:
 5535 Only Dirichlet's theorem for one FIXED reduced arithmetic progression
 5536 is used: if $\gcd(A, L)=1$, there are infinitely many primes $P=A \pmod{L}$.
 5537 Verified in Andrew V. Sutherland, MIT 18.785, Fall 2021, Lecture 18,
 5538 Theorem 18.1, page 1:
 5539 <https://math.mit.edu/classes/18.785/2021fa/LectureNotes18.pdf>
 5540 No simultaneous primality of different linear forms is requested.
 5541
 5542 HOSTILE CHECK / NECESSARY SMALL-GAP BARRIER:
 5543 For ANY upper root $R=N/2+h$ with $m=N-R$ and $S=PF(m)$, an internal edge
 5544 $u \rightarrow v$ with u, v in S obeys $v \mid (2h-u)$, the old shadow identity. Since R
 5545 is prime and $m < R$, no v in S divides $2h$; in particular internal self-
 5546 loops are impossible. If a is the least label on a directed cycle and
 5547 b is its next label, then $b > a$. A nonzero multiple of b equal to $2h-a$
 5548 cannot be negative: if $2h < a$ its absolute value is $< a < b$. Consequently
 5549 $2h-a > b$ and $h > a$. Thus a first-support cycle necessarily has a label

5550 below h. In particular the existing $h=1$ DAG theorem is untouched.
 5551 The CRT construction is compatible with this necessary bound: its h
 5552 cannot lie below every prescribed cycle label, because its congruences
 5553 would then give a strictly decreasing edge at every step of the cycle.
 5554 The fragile step in canonical CRT programming is primitivity of the
 5555 P progression; it was checked for EACH old and new prime modulus above.
 5556 Using the q_i themselves as arbitrary prime-gap blockers without that
 5557 check would be unjustified. Proper-factor thresholds were also imposed,
 5558 so a small shifted prime equal to its assigned divisor is excluded.
 5559
 5560 TWO-CYCLE SPECIALIZATION:
 5561 For distinct odd primes a, b in S , the two edges $a \rightarrow b$ and $b \rightarrow a$ occur
 5562 if and only if $2h = a+b \pmod{ab}$, equivalently
 5563 $h = (a+b)/2 \pmod{ab}$.
 5564 Indeed each edge supplies one of the two congruences, and their CRT
 5565 combination is $2h = a+b \pmod{ab}$; division by 2 is legitimate modulo odd
 5566 ab . The smallest positive representative is $(a+b)/2 \pmod{ab}$. All positive
 5567 solutions are $h = (a+b)/2 + t \cdot ab$ with integers $t \geq 0$. For $a=3, b=11$ one can
 5568 therefore choose the small fixed value $h=7$. The general construction
 5569 then gives infinitely many canonical first-support $3 \leq h \leq 11$ cycles.
 5570
 5571 HISTORICAL RELATION:
 5572 This does not revive the killed reading of FBTU as complete-BFS
 5573 universality. Cite the existing FAIL-PO-FINITE-BAD-TRAVERSAL together
 5574 with its binding correction AUDIT-FBTU-PARTIAL-VERSUS-COMplete-20260905
 5575 and the Biblia iteration-108 cofactor-leakage warning. The construction
 5576 is a concrete, fully audited partial-factor consumer of what survived.
 5577 It adds an explicit FIRST-SUPPORT cyclic obstruction to a proposed
 5578 acyclicity explanation; it is not a new general finite-local no-go.
 5579 RELATION TO GOLDBACH:
 5580 No Goldbach failure follows. Extra branches can and do reach GOOD.
 5581 RELATION TO p_0 :
 5582 An all- p_0 acyclicity or uniform cycle-size bound is false. The $h=1$
 5583 acyclicity property and its full-BFS conjugacy with root 2 remain valid,
 5584 but neither property extends automatically to the whole canonical family.
 5585 WHAT REMAINS OPEN:
 5586 Any obstruction to complete BAD closure, with all moving factors retained.
 5587 NOVELTY: NOT YET AUDITED; no novelty claim for CRT programming itself.
 5588
 5589
 5590 RESULT-ID: SAME-N-FIRST-SUPPORT-CYCLE-NON-DOMINANCE-20260907
 5591 STATUS: VERIFIED COMPUTATION with exact hand-checkable certificates.
 5592 EXACT STATEMENT:
 5593 Whether the first-support induced factorization subgraph is acyclic can
 5594 favor p_0 or p_1 at the SAME N . It is not a universal separator of the roots.
 5595 EXAMPLE A (p_0 acyclic, p_1 cyclic):
 5596 $N=344$, $p_0=173$ ($h=1$), $p_1=179$ ($h=7$).
 5597 $S_0 = PF(171) = \{3, 19\}$; its induced graph has no edges:
 5598 $344-3=11 \cdot 31$, $344-19=5 \cdot 2 \cdot 13$.
 5599 $S_1 = PF(165) = \{3, 5, 11\}$; its induced edges are
 5600 $3 \rightarrow 11$, $5 \rightarrow 3$, $11 \rightarrow 3$, because
 5601 $344-3=11 \cdot 31$, $344-5=3 \cdot 113$, $344-11=3 \cdot 2 \cdot 37$.
 5602 Thus $3 \leq h \leq 11$ is a BAD cycle inside S_1 . As already recorded above,
 5603 BOTH full searches succeed at depth 2 with $W=5$. The cycle does not
 5604 force even a larger work count here.
 5605 EXAMPLE B (p_0 cyclic, p_1 acyclic):
 5606 $N=1268$, $p_0=641$ ($h=7$), $p_1=643$ ($h=9$).
 5607 The integers 634, ..., 640 are composite, and 641 and 643 are primes;
 5608 hence the ranks are exact. The first complements are
 5609 $1268-641=627=3 \cdot 11 \cdot 19$; $1268-643=625=5^4$.
 5610 In $S_0 = \{3, 11, 19\}$, the BAD vertices 3 and 11 form a two-cycle:
 5611 $1268-3=1265=5 \cdot 11 \cdot 23$; $1268-11=1257=3 \cdot 419$.
 5612 The remaining child 19 is GOOD, since $1268-19=1249$ prime. Therefore
 5613 the COMPLETE stopped induced subgraph on S_0 has exactly edges
 5614 $3 \rightarrow 11$ and $11 \rightarrow 3$. The p_0 search succeeds at depth 1, $W=4$, via 19.
 5615 In $S_1 = \{5\}$, there is no internal edge: $1268-5=1263=3 \cdot 421$.
 5616 Thus p_1 's first-support induced graph is acyclic while p_0 's is cyclic.
 5617
 5618 HOSTILE CHECK:
 5619 All roots, prime factors, prime endpoints and root ranks were verified
 5620 by trial division. The induced graph uses all internal edges; GOOD
 5621 vertices contribute no outgoing edges in the stopped graph. In B the
 5622 cycle labels are BAD even though another first child is GOOD. This is
 5623 precisely why a prescribed BAD cycle must not be confused with complete
 5624 first-front BADness or full BFS failure. These are exact examples,
 5625 not an exhaustive claim about first occurrences.
 5626 DEPENDENCIES: Elementary arithmetic and the fixed graph definition.
 5627 RELATION TO p_0 / GOLDBACH:
 5628 Same- N non-dominance of this structural diagnostic. Neither example is
 5629 a failed canonical search or a Goldbach counterexample. The earlier
 5630 performance non-dominance examples $N=2024$ and $N=2180$ remain separate.
 5631 NOVELTY: NOT YET AUDITED.
 5632
 5633
 5634 RESULT-ID: FIXED-GAP-CANONICAL-VALUATION-EMBEDDING-20260907
 5635 STATUS: PROVED; precise extension/consumer of CRT-TRANSCRIPT-EXACT-SCOPE.
 5636 EXACT STATEMENT:
 5637 Let $N_0 \geq 4$ be even and let $R_0 \geq N_0/2$ be an admissible prime upper root
 5638 with COMPOSITE $m_0 = N_0 - R_0$. Set $h_0 = R_0 - N_0/2$ and $S = PF(m_0)$.
 5639 There exist infinitely many even N with canonical prime root
 5640 $P = p_0(N) = N/2 + h_0$, $m = N - P$ composite, such that for every q in S ,
 5641 $v_q(m) = v_q(m_0)$,
 5642 and for every p, q in S ,
 5643 $v_q(N-p) = v_q(N_0-p)$.
 5644 All primes of S occur at depth 1 in the COMPLETE search from P .
 5645 Every such N sufficiently large has additional first factors outside S .
 5646
 5647 GRAPH FORM OF THE CONCLUSION:
 5648 Construct the first-support graph on $PF(N-R)$, labelling each vertex q
 5649 by its prime value and by $v_q(N-R)$, and labelling every internal edge
 5650 $p \rightarrow q$ by $v_q(N-p) > 0$. This agrees with the internal edges of the stopped
 5651 factorization graph: an internal first-support edge cannot come from
 5652 a prime complement. On the original subset S , the new canonical graph
 5653 has EXACTLY the original labelled vertices, edge valuations and NONEDGES.
 5654 The numerical gap h_0 is also preserved. Additional vertices outside S
 5655 and edges incident to them remain unrestricted. GOOD/BAD flags at a

5656 vertex with no internal edges are not asserted to be preserved.

5657

5658 PROOF:

5659 Because m_0 is composite, $R_0 > N_0/2$, h_0 is a positive integer, R_0 is odd,

5660 and m_0 is odd. Moreover $R_0 > m_0$, so R_0 is distinct from every prime in S .

5661 For each q in S set

5662 $t_q = 1 + \max(v_q(m_0), \max_{\{p \in S\}} v_q(N_0 - p))$,

5663 and put $L = \text{product}_{\{q \in S\}} q^{t_q}$. These numbers are finite, L is odd,

5664 and $\gcd(R_0, L) = 1$. Require a new prime P to satisfy $P \equiv R_0 \pmod{L}$.

5665 Keep h_0 FIXED and define $N = 2(P - h_0)$, $m = P - 2h_0$. Then

5666 $m \equiv m_0 \pmod{L}$, $N \equiv N_0 \pmod{L}$,

5667 since $m_0 = R_0 - 2h_0$ and $N_0 = 2(R_0 - h_0)$. If a positive integer n is congruent

5668 to a modulo $q^{t_q}(v_q(a)+1)$, then $v_q(n) = v_q(a)$: division by q^{t_q} leaves

5669 the same nonzero residue modulo q . By the definition of t_q this gives

5670 all the asserted root and row valuations, including their zero values.

5671

5672 To make P canonical with the very SAME h_0 , add $P \equiv 1 \pmod{2}$. For every

5673 even j in $[2, h_0]$, choose a distinct new prime ell_j h_0 not dividing L ,

5674 and add $P \equiv j \pmod{\text{ell}_j}$. The moduli are pairwise coprime. The common

5675 class is reduced: R_0 is coprime to L , P is odd, and $0 < j < \text{ell}_j$ at every

5676 new modulus. By Dirichlet's theorem there are infinitely many primes P

5677 in this class. Take $P > R_0$ and large enough that $P - h_0 > 2$ and $P - j > \text{ell}_j$

5678 for every even j used. Each odd j in $[1, h_0]$ makes $P - j$ even and > 2 ;

5679 each even j gives a proper factor ell_j . The entire interval

5680 $[P - h_0, P]$ is therefore composite. Hence $P = p_0(2(P - h_0))$, with gap h_0 .

5681 This verifies canonicity by actual compositeness certificates rather

5682 than silently treating an arbitrary upper root as canonical.

5683

5684 The exact positive valuations of m at every q in S place all S in

5685 $\text{PF}(m)$. Since $m > m_0 > 4$ and m is divisible by the original m_0 , the new

5686 root is BAD and every label in S is indeed reached at depth 1 in the

5687 complete search. For any original p, q in S , $p < m_0 < R_0$ and $q < m_0$, so

5688 $N_0 - p > N_0 - m_0 = R_0 > q$.

5689 Thus if q divides $N_0 - p$ it is a proper divisor, and the corresponding

5690 row is BAD. For the larger N the same properness holds. Positive row

5691 valuations therefore give actual stopped-graph edges on S in both

5692 worlds, with identical weights; zero valuations give identical nonedges.

5693 This proves the asserted induced-graph equality, including the case of

5694 an originally GOOD vertex whose complement prime is outside S .

5695 Finally, all exponents at primes of S in m equal those in m_0 . If there

5696 were no extra factor outside S , unique factorization would give $m = m_0$,

5697 contrary to $P > R_0$. So additional support is NECESSARY in every retained

5698 new world. Distinct primes P give distinct even N , proving infinitude.

5699

5700 HOSTILE CHECK:

5701 Primitivity uses $R_0 > m_0$. The diagonal/direct case $m_0 = R_0$ is NOT allowed:

5702 then the needed congruence $P \equiv R_0 \pmod{R_0}$ would not be reduced. The

5703 hypothesis m_0 composite removes this problem and fixes the relevant

5704 BAD-root domain. A single-pair Goldbach success among first children

5705 is not required to survive: preserving finite valuations is not the

5706 same as preserving complement primality. No factor is declared absent

5707 outside S , and no complete BFS transcript is copied. The proof itself

5708 shows that new first factors MUST appear. Consequently it cannot

5709 conflict with complete fixed-alphabet finiteness or supply an iteration

5710 with an unchanged alphabet.

5711

5712 DEPENDENCIES / HISTORICAL AUDIT:

5713 CRT, elementary exact valuations, and Dirichlet's fixed reduced-

5714 progression theorem verified above in MIT Lecture 18, Theorem 18.1.

5715 This extends the checkpoint's CRT-TRANSCRIPT-EXACT-SCOPE-20260905 from

5716 already compatible $h=1$ partial constraints to a concrete compatible

5717 table obtained from ANY actual upper BAD root, with its own h_0 held

5718 fixed. The additional step enforces the prime-free canonical interval.

5719 It is a precise consumer of what survived FAIL-PO-FINITE-VALUATION-

5720 SIGNATURE and FAIL-PO-FINITE-BAD-TRAVERSAL, subject to the later

5721 AUDIT-FBTU-PARTIAL-VERSUS-COMPLETE-20260905 correction. No duplicate

5722 claim of complete-BFS universality is introduced.

5723 RELATION TO GOLDBACH:

5724 No root failure is copied, and no prime-pair existence theorem follows.

5725 RELATION TO p_0 :

5726 A rigorous limit on what the root's finite internal factor arithmetic

5727 can distinguish. The useful consumer is the hereditary-property no-go

5728 below; the statement is not itself an eventual-success mechanism.

5729 WHAT REMAINS OPEN:

5730 Properties using the COMPLETE alphabet, external row factors, GOOD

5731 status, absolute N or root location, or later reachability can escape

5732 this embedding theorem. No conclusion about them is inferred here.

5733 NOVELTY: NOT YET AUDITED; no novelty claim for the underlying CRT method.

5734

5735

5736 RESULT-ID: HEREDITARY-FIRST-SUPPORT-PO-NO-GO-20260907

5737 STATUS: PROVED, as an exact consumer of the preceding embedding.

5738 EXACT STATEMENT:

5739 Consider a property $Q(h, G)$ of finite directed graphs G whose vertices

5740 carry prime labels q and root exponents, and whose edges carry row

5741 valuations, as in FIXED-GAP-CANONICAL-VALUATION-EMBEDDING. Suppose Q

5742 is hereditary under induced subgraphs at fixed h : whenever $Q(h, G)$

5743 holds, it holds on every induced subgraph, retaining vertex/edge labels

5744 and exponents. Q may use the numerical labels and h , but has no extra

5745 unrecorded dependence on the absolute N , root R , or outside factors.

5746 Then the following two statements are equivalent:

5747 (A) Q holds for the first-support graph of EVERY canonical BAD root p_0 .

5748 (B) Q holds for the first-support graph of EVERY admissible upper BAD

5749 prime root $R > N/2$, including all nearby roots that are BAD.

5750 Moreover, if (B) fails, (A) fails for infinitely many N with one fixed h .

5751

5752 PROOF:

5753 (B) implies (A) because canonical roots are upper roots. Conversely,

5754 assume (A) and take an arbitrary upper BAD root (N_0, R_0) . The preceding

5755 embedding supplies a canonical BAD-root graph G' at the same h_0 that

5756 contains its full original graph G_0 as an induced subgraph with all

5757 stated labels and weights unchanged. By (A), $Q(h_0, G')$ holds. Heredity

5758 then implies $Q(h_0, G_0)$. The original upper BAD root was arbitrary,

5759 so (B) follows. If $Q(h_0, G_0)$ fails for one such root, every canonical

5760 embedding G' fails Q by the contrapositive of heredity. There are

5761 infinitely many distinct N carrying those embeddings. This proves

5762 the strengthened failure conclusion as well.
5763
5764 HOSTILE CHECK / EXACT CLASS OF THE NO-GO:
5765 Heredity is essential and is not silently assumed for an arbitrary
5766 invariant. Examples inside the class include acyclicity, forbidding a
5767 fixed labelled finite cycle/pattern, a uniform upper bound on directed
5768 cycle length or strong-component size, and hereditary restrictions on
5769 internal row valuations. These properties cannot universally hold for
5770 p_0 while failing for a nearby BAD root. The theorem does not say their
5771 frequencies are equal, or compare p_0 and p_1 at every same N .
5772 OUTSIDE its scope are complete BADness (GOOD flags can change), equality
5773 of the WHOLE prime alphabet with a fixed set, completeness of all row
5774 factorizations inside that alphabet, a lower support bound, full BFS
5775 closure, or full multigeneration success. Such properties need not be
5776 hereditary under the stated restriction, and external factors are the
5777 precise information deliberately not preserved by the construction.
5778 This is therefore a mathematically specified subclass of a finite-local
5779 no-go, NOT the old overbroad claim that any finite BFS invariant fails.
5780
5781 RELATION TO p_0 / GOLDBACH:
5782 A broad, rigorous negative answer for these graph-and-valuation
5783 properties; it does not answer whether the complete canonical traversal
5784 always reaches GOOD. Same- N witnesses above illustrate both directions.
5785 for acyclicity; this theorem supplies the universal logical limitation.
5786 WHAT REMAINS OPEN:
5787 A non-hereditary complete-support or multigeneration invariant that
5788 actually forbids a closed BAD canonical reachable world.
5789 NOVELTY: NOT YET AUDITED; theorem consumer and scope are explicitly new
5790 entries here, not a claim of priority over the literature.
5791
5792
5793 RESULT-ID: CANONICAL-VALUATION-EMBEDDING-EXACT-CERTIFICATE-20260907
5794 STATUS: VERIFIED COMPUTATION.
5795 DOMAIN:
5796 One constructed example, using the noncanonical source $(NO,RO)=(344,179)$
5797 with $h_0=7$ and $S=\{3,5,11\}$. No exhaustive-range claim is made.
5798 CONSTRUCTION / VALUES:
5799 The proof gives $t_3=3, t_5=2, t_{11}=2$, hence $L=81675$.
5800 Use the progression $P=179 \pmod{L}$, $P=1 \pmod{2}$,
5801 $P=2 \pmod{13}$, $P=4 \pmod{17}$, $P=6 \pmod{19}$.
5802 Its common class is $P=499197779 \pmod{685906650}$.
5803 A prime member is $P=9415984229$, yielding $N=18831968444$ and $h=7$.
5804 The interval $[P-7,P)$ is composite: odd shifts are even >2 ; shifts
5805 2,4,6 have the proper divisors 13,17,19 respectively. Trial division
5806 also checked primality of P . Thus $P=p_0(N)$ exactly, with no reliance on
5807 a probable-prime certificate or an assumed preceding prime gap.
5808 COMPLETE ROWS NEEDED FOR THE VALUATION CERTIFICATE:
5809 $m=P-14=3*5*11*229*249199$;
5810 $N-3 = 11*13*131692087$;
5811 $N-5 = 3*89*6709*10513$;
5812 $N-11 = 3^2*17*101*239*5099$.
5813 All displayed factors are prime. On S these reproduce the original
5814 root exponents (1,1,1), and exactly the internal weighted edges
5815 $3 \rightarrow 11$ (weight 1), $5 \rightarrow 3$ (weight 1), $11 \rightarrow 3$ (weight 2).
5816 The extra root multiplier is $57066571=229*249199$, coprime to $3*5*11$.
5817 The full first support is therefore strictly larger than S , as the
5818 proof requires. Both extra first children are BAD:
5819 $N-229=5*53*653*108827$;
5820 $N-249199=5*37*199*511523$.
5821 The COMPLETE first frontier is BAD in this particular example, but
5822 that extra fact was verified here and does not follow from the embedding.
5823 The full FIFO search nevertheless succeeds at depth 2 after $W=11$ tests:
5824 $P \rightarrow 5 \rightarrow 10513$, and $N-10513=18831957931$ is prime.
5825 Primality, all displayed factors and this BFS were checked using exact
5826 trial division. The finite example illustrates preservation of the
5827 internal cycle and valuations while an unprescribed factor supplies a
5828 GOOD escape. It is not a failed canonical world.
5829 REPRODUCTION:
5830 CRT was solved with integer modular inverses, starting with class 179
5831 modulo 81675 and then combining moduli 2,13,17,19. The first retained
5832 prime in the resulting nonnegative progression is the displayed P ;
5833 that order is diagnostic only, not a minimality theorem about all N .
5834 The complete certificate is retained in this TXT; the auxiliary scratch
5835 record is `canonical_valuation_embedding_20260907.json`.
5836 NOVELTY: NOT YET AUDITED.
5837
5838
5839 AUDIT ADDENDUM: NECESSARY BOUNDARIES OF THE EMBEDDING NO-GO
5840 STATUS: VERIFIED COMPUTATION / SCOPE-CORRECTED.
5841 1. GOOD flags really are not preserved. A small exact example is
5842 $(NO,RO)=(20,11)$, $h_0=1$, $m_0=9=3^2$, $S=\{3\}$.
5843 The original child 3 is GOOD because $20-3=17$ prime. Take $P=173$,
5844 $N=344$ and $h=1$. Then $P=11 \pmod{27}$, $m=171=3^2*19$ has the same
5845 3-adic root exponent, and $344-3=341=11*31$ has the same 3-adic row
5846 valuation zero as 17. The entire induced graph on the labelled subset
5847 $\{3\}$ (one vertex of root weight 2, no internal edge) is unchanged,
5848 but that vertex is now BAD. Thus adding GOOD/BAD flags to the theorem
5849 without new hypotheses is false. This is a concrete boundary of the
5850 retained theorem, not a counterexample to it.
5851 2. Arbitrary incompatible edge data are still impossible. On labels
5852 $\{3,5,7\}$, simultaneous edges $5 \rightarrow 3$ and $7 \rightarrow 3$ would require $N=2$ and $N=1$
5853 modulo 3. No N realizes them. The fixed-gap theorem starts from an
5854 actual upper-root graph, so its compatibility is guaranteed and its
5855 primitivity is then proved. The cycle-realization theorem prescribes
5856 one incoming predecessor per target and permits extra edges; it does
5857 not assert universality of every directed graph on arbitrary labels.
5858 3. The low-gap support LOWER bound proved earlier is not hereditary
5859 under deleting vertices. It therefore escapes the hereditary no-go,
5860 as do the complete-support height restrictions. This logical escape
5861 alone does not turn them into a contradiction for large moving support.
5862 4. None of these arguments assumes global Goldbach failure. The
5863 failure counterexample (1862,1597) is a rooted failure in an ordinary
5864 Goldbach-successful N . Its use kills only the unconditional prime-lift
5865 success claim, not a theorem restricted to a global failed world.
5866 5. No recent asymptotic estimate was needed in this continuation.
5867 The checkpoint's earlier growing-root/growing-depth sieve claims have

```

5868 not been strengthened or reclassified by the elementary results here.
5869
5870 =====
5871 STRONGST NEW RESULTS
5872 =====
5873 1. FIXED-GAP-CANONICAL-VALUATION-EMBEDDING and its HEREDITARY-FIRST-
5874 SUPPORT-PO-NO-GO: exact preservation of the original induced first-
5875 support graph, root exponents and internal row valuations at the
5876 SAME numerical h, inside infinitely many canonical worlds. This
5877 rules out a precisely specified class of universal p0 separators.
5878 Complete support and eventual reachability are explicitly outside it.
5879 2. PRIME-LIFT-FULL-BFS-CONJUGACY: for  $N=d*x$  and prime lift
5880  $R=(d-1)*x+1$  with composite  $x-1$ , the full BFS from inactive d and
5881 from R are identical after renaming their initial root. At  $d=2$ 
5882 this is an exact all-generation equivalence for canonical  $h=1$ .
5883 The tempting general success consequence is FALSE at
5884  $(N,d,R)=(1862,7,1597)$ , with a complete 22-vertex closure certificate.
5885 3. LOW-GAP-COMplete-THREE-LABEL-ALTERNATIVE: if  $h<\min PF(N-R)$ , either
5886 GOOD occurs by depth 2 or at least three distinct nonroot labels
5887 occur by depth 3. A whole closed BAD nonroot alphabet of size  $\leq 2$ 
5888 is excluded in that domain. This also applies to nearby roots.
5889 4. Canonical first-support BAD cycles can have arbitrary prescribed
5890 length. Same-N acyclicity reverses between p0 and p1 at  $N=344$  and
5891  $N=1268$ . Exact root-rank corrections, million-input lift checks and
5892 the explicit canonical valuation-embedding example are recorded above.
5893
5894 =====
5895 CURRENT FRONTIER
5896 =====
5897 Binary Goldbach is not proved, and no universal p0 success theorem was
5898 obtained. The new embedding result explains why inherited finite graph
5899 patterns and their internal valuations cannot provide certain universal
5900 p0 distinctions; it does NOT control the extra factors whose presence
5901 prevents the transplant from remaining closed. The exact conjugacy makes
5902 root 2 an equivalent starting point in the  $h=1$  subfamily, but gives no
5903 mechanism forcing that root to succeed when  $P=N/2+1$  is prime.
5904 A possible large, moving, completely closed active BAD reachable set is
5905 still unexcluded. Earlier support-height bounds remain lower constraints,
5906 not contradictory upper bounds. The earlier asymptotic frontier is also
5907 unchanged: the large-label-product AND large-cofactor-product region at
5908 radius three is not controlled here. Shallow BADness is compatible with
5909 later success, and none of it is a global Goldbach counterexample.
5910
5911 =====
5912 p0 STATUS
5913 =====
5914 Found a nontrivial all-generation equivalence for canonical  $h=1$ , but its
5915 partner is root 2, not a nearby upper root, and it does not cover all p0.
5916 Found a precise hereditary graph/valuation no-go and additional same-N
5917 structural non-dominance witnesses. The small-gap support restriction is
5918 shared by nearby roots satisfying the same premise. Canonical first-
5919 support acyclicity holds at  $h=1$  but fails for other h, even infinitely
5920 often with arbitrarily long cycles. No convincing explanation of the
5921 empirical eventual reliability of p0 has been established.
5922
5923 =====
5924 EXACT NEXT QUESTION
5925 =====
5926 Can complete factor search from 2 fail at  $N=2(P-1)$  with  $P>5$  prime
5927 and  $P-2$  composite? Equivalently, can there exist such a P and a finite
5928 set C of odd primes below  $2(P-1)/3$  such that
5929  $PF(P-2)$  is contained in C,
5930 and for EVERY q in C the integer  $2(P-1)-q$  is composite with its ENTIRE
5931 prime support contained in C?
5932 This is the exact  $d=2$  obstruction left by the proved conjugacy. A
5933 solution must use complete support or another feature not preserved
5934 by the canonical valuation embedding. Showing selected BAD cycles,
5935 reproducing finite internal valuations, or merely increasing a support-
5936 height lower bound does not answer it. A counterexample would refute
5937 canonical  $h=1$  traversal success, not by itself binary Goldbach.
5938
5939
5940 =====
5941 CONTINUATION IN LOCAL WORKSPACE -- 2026-09-07
5942 =====
5943 SOURCE AND SCOPE:
5944 The preceding checkpoint was copied intact from the supplied Downloads
5945 file goldbach_research_checkpoint_2026-09-05 (3).txt. Its latest entries
5946 already extend through 2026-09-07 and supersede the earlier master where
5947 they explicitly correct it. This continuation uses the documents as
5948 research data, not as instructions. All new research files are in nowe.
5949 Only one independent subagent was used. No PDF or TeX output is produced.
5950 The immediate focus is complete-support restrictions, rather than another
5951 transplant of selected edges or an unsupported all-N conclusion.
5952
5953 FAIL-ID: LOW-GAP-FIVE-LABEL-OVERGENERALIZATION-20260907C
5954 STATUS: REFUTED by an exact counterexample; the earlier three-label
5955 theorem and the stated rank-0-through-6 finite experiment are unaffected.
5956 CLAIM / ROUTE:
5957 Strengthen LOW-GAP-COMplete-THREE-LABEL-ALTERNATIVE by replacing THREE
5958 with FIVE for every upper prime root with  $h<\min PF(N-r)$ .
5959 WHY IT LOOKED PROMISING:
5960 The old experiment found no fewer than five nonroot labels by depth 3
5961 under complete BADness through depth 2, at ranks 0,...,6 and  $N\leq 200000$ .
5962 EXACT COUNTEREXAMPLE:
5963  $N=44670$ ,  $r=22469$ ,  $m=N-r=22201=149^2$ ,  $h=134<149$ .
5964  $N-149=44521=211^2$ .
5965  $N-211=44459=23*1933$ .
5966  $N-23=44647$  prime;  $N-1933=42737$  prime.
5967 All displayed prime labels, the root and the two prime complements are
5968 prime by independent trial division through the integer square root.
5969 Thus complete BADness holds at depths 0,1,2. The distinct nonroot
5970 labels discovered through depth 3 are exactly {149,211,23,1933}:
5971 depth 1 is {149}, depth 2 is {211}, depth 3 is {23,1933}.
5972 The complete FIFO search, ascending factors and testing on dequeue,
5973 succeeds at depth 3 after four tests r,149,211,23. No branch is omitted.

```

5974 The exact upper rank is 12, since the primes at or above $N/2=22335$
5975 through r are
5976 22343, 22349, 22367, 22369, 22381, 22391, 22397,
5977 22409, 22433, 22441, 22447, 22453, 22469.
5978 This explains why the earlier rank-restricted experiment missed it.
5979 WHAT SURVIVES:
5980 The proved bound THREE. The possibility of a universal FOUR under the
5981 same low-gap hypotheses remains under investigation. FIVE for p_0 itself,
5982 or for ranks at most six outside the tested range, is not decided by
5983 this noncanonical example. No five-label theorem for those roots is
5984 claimed. The observed five-label minimum was already explicitly empirical.
5985 RELATION TO p_0 / GOLDBACH:
5986 An exact boundary of a candidate strengthening, not a failed canonical
5987 traversal. This N has many Goldbach representations, including those above.
5988 REPRODUCTION:
5989 Discovered by the single independent agent in a targeted singleton-
5990 first-support scan, with scratch source agent_scan_singles.c.js. The
5991 displayed arithmetic is a self-contained certificate; the domain and
5992 results of any larger scan are recorded separately after verification.
5993 NOVELTY: no literature-priority claim.
5994
5995
5996 RESULT-ID: LOW-GAP-SHARP-TWO-STEP-RETURN-BARRIER-20260907C
5997 STATUS: PROVED, elementary pointwise statement.
5998 EXACT STATEMENT:
5999 Let N be even and $r \geq N/2$ a prime upper root with composite complement
6000 $m=N-r$. Put $h=r-N/2$ and $q=\min \text{PF}(m)$, and assume $0 < h < q$.
6001 Suppose q and an odd prime b are BAD and form an actual two-cycle:
6002 b divides $N-q$, and q divides $N-b$.
6003 Then
6004 $b=2h+(2j+1)q$ for an integer $j \geq 1$;
6005 in particular $b \geq 3q+2h$.
6006 The numerical lower bound is sharp, both for canonical roots and for
6007 noncanonical upper roots. It bounds a two-cycle through the minimum
6008 first child; it is not a no-cycle theorem for the whole reached graph.
6009
6010 PROOF:
6011 The BAD root is odd, m is odd, $r \geq N/2$, and $q \geq 3$. Every child of an
6012 active prime is active: if t divides both $N-p$ and N then t divides p ,
6013 forcing $t=p$ and contradicting activity of p . Since r is an upper
6014 nondiagonal prime, it is active; so q, b are active and $b \neq q$.
6015 Because q divides m , $N=2m+2h$ is congruent to $2h$ modulo q .
6016 The return edge $b \rightarrow q$ gives $b=2h+kq$ for an integer k . The primes b, q
6017 are odd, so k is odd. Since $0 < 2h < 2q$ and $b > 0$, $k \leq -3$ is impossible.
6018 If $k=-1$, then $b=2h-q < q$ and the other edge $q \rightarrow b$ gives
6019 b divides $N-q=2m+b$.
6020 Thus b divides m , contradicting the minimality of q .
6021 If $k=1$, then $b=q+2h$, and
6022 $N-q=2m+2h-q=2(m+2h)-b=2r-b$.
6023 Hence b divides $2r$, so $b=r$ because b is odd and r is prime.
6024 This is impossible: a prime factor b of the odd composite row $N-q$
6025 satisfies $b < N/3 < r$. Thus $k \geq 3$, proving the assertion.
6026
6027 SHARPNESS / EXACT CERTIFICATES:
6028 Canonical case: $N=80$, $r=p_0=41$, $h=1$, $m=39=3 \cdot 13$, $q=3$.
6029 $N-3=77=7 \cdot 11$; $N-11=69=3 \cdot 23$.
6030 Thus $b=11=3q+2h$ is an actual BAD two-cycle partner.
6031 The child 13 is GOOD ($80-13=67$ prime), so this cycle does not cause
6032 full BFS failure or even complete first-front BADness.
6033 Noncanonical case: $N=1638$, $r=p_1=823$, $p_0=821$, $h=4$,
6034 $m=815=5 \cdot 163$, $q=5$.
6035 $N-5=1633=23 \cdot 71$; $N-23=1615=5 \cdot 17 \cdot 19$.
6036 Here $b=23=3q+2h$ again. The root ranks are exact: 819 is composite,
6037 820 and 822 are even, and 821 and 823 are prime.
6038 All displayed factors, roots and prime complements were checked by
6039 integer trial division / exact sieve lookup.
6040
6041 HISTORICAL AUDIT:
6042 The underlying odd parent lattice $p=2h \cdot \text{ell} \cdot q$ is already present in
6043 the Biblia, proposition prop:v73-parent-lattice (near line 45279 in
6044 the supplied file). It is not claimed as new. This consumer removes
6045 the two closest positive candidates in the small-gap setting:
6046 $\text{ell}=1$ by complete first-support minimality, $\text{ell}=1$ by primality of
6047 the upper root. An exact-string search for $3q+2h$ and its reordered
6048 forms in the master, Biblia, core, failure log and input checkpoint
6049 found no earlier version. This is a selective corpus check, not a
6050 literature priority audit.
6051 RELATION TO p_0 :
6052 The bound applies automatically at $h=1$ but is shared by every upper
6053 prime root satisfying $h < q$. Its sharp p_1 example prevents interpreting
6054 its sharpness as a p_0 -only signature. It is an actual two-step return
6055 restriction, but it does not force a GOOD child or prevent longer cycles.
6056 RELATION TO GOLDBACH: no prime-pair existence conclusion.
6057 NOVELTY: not externally audited; no priority claim.
6058
6059
6060 RESULT-ID: LOW-GAP-SQUARE-ROW-NO-RETURN-20260907C
6061 STATUS: PROVED; independent elementary lemma supplied by the one agent
6062 and checked by the root agent.
6063 EXACT STATEMENT:
6064 Under the same root, gap and minimum-factor hypotheses, suppose
6065 $N-q=b^2$ for a prime b .
6066 Then q does not divide $N-b$, and
6067 $(b-1)^2 < N-b < b^2$.
6068 PROOF:
6069 Since m is composite and q is its least prime factor, $m \geq q^2$.
6070 Consequently $b^2=2m+2h-q \geq q^2$, hence $b \geq q$.
6071 If q divided $N-b$, then $b=N=2h \pmod{q}$. The equation $b^2=N-q$
6072 also gives $b^2=N-b \pmod{q}$. Since $b \neq q$ is prime, this implies
6073 $b \equiv 1 \pmod{q}$ and $2h \equiv 1 \pmod{q}$. The only even integer strictly
6074 between 0 and $2q$ in this residue class is $q+1$. Thus $2h=q+1$.
6075 But then $b^2=2m+1 \equiv 3 \pmod{4}$, impossible because m is odd.
6076 Finally $N-b=b^2+q-b < b^2$, and subtraction of $(b-1)^2$ leaves
6077 $q+b-1 > 0$. This proves the strict interval and the no-return assertion.
6078
6079 COMPLETE-SUPPORT CONSEQUENCE:

6080 If $PF(m)=\{q\}$, $N=q=b^2$, and no GOOD vertex occurs through depth 2,
6081 then either at least FOUR distinct nonroot primes are discovered by
6082 depth 3, or the next COMPLETE row is
6083 $N=b=c^w$ with a new prime $c!=q, b$ and an ODD exponent $w>=3$.
6084 Indeed q is excluded by the lemma, b by activity. Two distinct factors
6085 of $N=b$ introduce two new labels. Otherwise it is a prime power; its
6086 exponent is at least two by BADness and cannot be even because $N=b$
6087 lies strictly between consecutive integer squares.
6088 Thus a three-label square bottleneck would satisfy the exact system
6089 $2q^u+2h=q+b^2=b+c^w$,
6090 $u>=2$, $w>=3$ odd, $0<h<q$, q^u+2h prime.
6091 This system has NOT been excluded. It is an obstruction to completing
6092 a universal FOUR-label proof, not a claimed counterexample to FOUR.
6093 The child c need not be BAD because its row is at depth 3.
6094 RELATION TO p0: shared under the stated gap condition.
6095 NOVELTY: no literature-priority claim.
6096
6097
6098 RESULT-ID: LOW-GAP-MONOMIAL-RETURN-EXPONENT-DROP-20260907C
6099 STATUS: PROVED, as a consumer of the preceding two elementary lemmas.
6100 EXACT STATEMENT:
6101 Assume $m=q^u$, $u>=2$, $0<h<q$, $r=q^u+2h$ prime, $N=2q^u+2h$,
6102 and $N-q=b^v$ for a prime b and integer $v>=2$.
6103 If q divides $N-b$ (a BAD return $q->b->q$), necessarily
6104 $3<v<=u-1$, hence $u>=4$.
6105 More precisely $(3q+2h)^u <= 2q^u+2h-q$.
6106 In particular neither $m=q^2$ nor $m=q^3$ can have such a return when
6107 the complete row $N-q$ has only one distinct prime factor.
6108 PROOF:
6109 q is the minimum first factor automatically. Since $N-q$ is composite,
6110 q is BAD. The condition $q|N-b$ also makes b BAD: $b<N/3$ and $N-b>q$,
6111 so q is a proper factor. Apply the sharp return bound $b>=3q+2h$.
6112 If $v>u$, then
6113 $b^v >= (3q+2h)^u > (3q)^u >= 9q^u$
6114 $> 2q^u+2h-q=N-q$,
6115 where $2h-q<=q^u$. This is impossible. Thus $v<=u-1$.
6116 The square-row lemma excludes $v=2$. The remaining inequalities follow.
6117 HOSTILE CHECK:
6118 The conclusion concerns a direct return to q , not absence of all
6119 later returns or full BAD closure. The assumption that $N-q$ is a
6120 COMPLETE prime power is essential; choosing one divisor of a
6121 multifactor row does not satisfy it. At $N=80$ a squarefree multifactor
6122 row has a return at the sharp size threshold, as recorded above.
6123 No root minimality within the upper primes was used.
6124 WHAT REMAINS OPEN:
6125 Longer return paths, branching rows, and singleton-support roots with
6126 $u>=4$ remain possible. Even an acyclic initial segment need not succeed.
6127 NOVELTY: no literature-priority claim.
6128
6129 ADDITIONAL HOSTILE BOUNDARY OF THE RETURN LEMMA:
6130 The sharpened lattice applies to RECIPROCAL edges. It is not a correction
6131 asserting that all parents of q exceed $3q+2h$. At $N=344$, $r=173$, $h=1$,
6132 $m=171=3^2 \cdot 19$, $q=3$, the label $b=5=q+2h$ is a genuine parent of q ,
6133 since $344-5=3 \cdot 113$, but $q->b$ is absent since $344-3=11 \cdot 31$.
6134 The prime-root contradiction needs the second divisibility $b|(N-q)$.
6135
6136
6137 RESULT-ID: H1-SINGLETON-SQUARE-ROW-RESIDUE-FILTER-20260907C
6138 STATUS: PROVED; necessary conditions only.
6139 EXACT STATEMENT:
6140 If $r=q^u+2$ is prime, $u>=2$, $N=2q^u+2$, and $N-q=b^2$ with b prime,
6141 then u is odd and at least three, and $q=23 \pmod{24}$.
6142 PROOF:
6143 If $q=3$, then $N-q=2 \cdot 3^u-1 \equiv 2 \pmod{3}$, which cannot be a square.
6144 For $q \neq 3$, even u gives $r=q^u+2 \equiv 0 \pmod{3}$, with $r>3$, contradicting
6145 primality. Thus u is odd and $u>=3$. The primality of r now excludes
6146 $q=1 \pmod{3}$, so $q \equiv 2 \pmod{3}$. Since an odd integer to an odd power
6147 is congruent to itself modulo 8, $b^2=2q^u+2-q \equiv q+2 \pmod{8}$.
6148 Odd squares are 1 modulo 8; hence $q \equiv 7 \pmod{8}$, giving $q=23 \pmod{24}$.
6149 SCOPE:
6150 This uses $h=1$ and complete pure-power rows. It is not asserted for all
6151 p_0 or all h , and supplies neither existence nor impossibility of this
6152 sparse configuration. It merely reduces the unresolved square bottleneck.
6153
6154
6155 EXPERIMENT-ID: LOW-GAP-RETURN-AND-COMPLETE-SUPPORT-AUDIT-20260907C
6156 STATUS: VERIFIED COMPUTATION; exact finite exhaustive domain.
6157 DOMAIN:
6158 Every even $4<N \leq 200000$, each upper rank $k=0, \dots, 12$, retaining exactly
6159 composite $m=N-p_k$ and $0<h=p_k N/2 < q = \min PF(m)$.
6160 METHOD:
6161 An independently written exact linear smallest-prime-factor sieve through
6162 201000. All rows have arguments within that range. Enumerate every factor
6163 b of the COMPLETE BAD row $N-q$ and test whether q divides $N-b$; check the
6164 sharp return barrier. Check all prime-square rows and the consecutive-
6165 square interval. For singleton monomial returns check exponent drop.
6166 For support counts, expand complete depths 0,1,2 only when these depths
6167 are all BAD; count all distinct nonroot labels discovered through depth 3.
6168 Suppress repeated labels. Do not expand newly discovered depth-3 rows.
6169 COUNTS:
6170 134898 eligible (N,k) inputs; 28289 completely BAD through depth 2.
6171 10249 reciprocal $q->b->q$ incidences, all satisfying $b>=3q+2h$.
6172 1360 incidences attain equality, so the sharp boundary is repeatedly
6173 realized in the finite domain. There are 97 eligible prime-square rows;
6174 every no-return and interval check passed.
6175 There are ZERO singleton monomial returns in this finite domain; thus
6176 the exponent-drop condition received no nonvacuous returned-path test
6177 here. Its status rests on the proof, not that vacuous check.
6178 Per rank: k , eligible, complete BAD through 2, least label count through 3:
6179 0 38715 7135 5
6180 1 22044 4503 5
6181 2 14356 3136 6
6182 3 11229 2495 6
6183 4 9285 2096 6
6184 5 7915 1808 6
6185 6 6790 1516 6

6186 7 5824 1318 6
 6187 8 4930 1144 6
 6188 9 4191 965 6
 6189 10 3670 836 6
 6190 11 3186 698 6
 6191 12 2763 639 4
 6192 The unique four-label case in THIS DOMAIN is (N,k)=(44670,12), whose
 6193 complete certificate was recorded above. No three-label case occurred.
 6194 The rank-0-through-6 eligible/BAD2 counts independently reproduce all
 6195 seven counts from the supplied checkpoint. This is a check on definitions
 6196 and implementation, not a promotion of any finite minimum to a theorem.
 6197 REPRODUCTION: nowe/return_barrier_audit.cjs; no random or probable-prime tests.
 6198
 6199 SEPARATE AGENT DIAGNOSTIC, RETAINED HERE IN FULL NUMERICAL SCOPE:
 6200 All odd-prime powers $m=q^u$, $u \geq 2$, and ALL integers $1 < h < q$ such that
 6201 $N=2m+2h < 10000000$ and $r=m+2h$ is prime, with no root-rank restriction.
 6202 The exact SPF scan in agent_scan_singles.cjs found 47490 eligible roots,
 6203 25413 completely BAD through depth 2, including 11 with a singleton
 6204 factor set for N-q. The minimum count through depth 3 is FOUR and
 6205 occurs on exactly three inputs:
 6206 (N,r,q,u,h)=(44670,22469,149,2,134),
 6207 (241428,121019,347,2,305),
 6208 (2037338,1019257,1009,2,588).
 6209 For the latter two, the complete first rows are
 6210 241428-121019=347²;
 6211 241428-347=491²;
 6212 241428-491=479*503;
 6213 2037338-1019257=1009²;
 6214 2037338-1009=1427²;
 6215 2037338-1427=3*678637.
 6216 The root agent independently verified all displayed prime factors and
 6217 roots by trial division; it did not repeat the full ten-million-range
 6218 singleton scan. That scan covers ONLY singleton initial support and is
 6219 not an all-root or complete closed-core census.
 6220
 6221
 6222 RESULT-ID: CLOSED-SUPPORT-MULTIPLICATIVE-SUBGROUP-TEST-20260907C
 6223 STATUS: PROVED elementary necessary condition; investigated consumer
 6224 does not yield a general p0 distinction.
 6225 EXACT STATEMENT:
 6226 Let C be a nonempty set of active primes with every N-p composite and
 6227 PF(N-p) contained in C. In the unit group modulo N let H_C be the
 6228 subgroup generated by the residue classes of all primes in C.
 6229 Then -1 belongs to H_C. If an upper prime root r has PF(N-r) contained
 6230 in C and N-r>1, then r belongs to H_C as well.
 6231 PROOF:
 6232 Choose p in C. Complete factor closure makes N-p an element of H_C
 6233 modulo N. Since p is a unit in H_C, (N-p)*p⁻¹=-1 modulo N belongs
 6234 to H_C. If m=N-r has its full support in C, then m belongs to H_C,
 6235 so r=m belongs to H_C. This proves both claims.
 6236 CAUTION:
 6237 Excluding r from H_C excludes failure contained in THIS C only. To
 6238 deduce actual success by this test one must use an envelope containing
 6239 every possible failed nonroot vertex, for example the maximal closed
 6240 low BAD set. Checking only one terminal component is insufficient.
 6241
 6242 FAIL-ID: SUBGROUP-SEPARATION-OF-P0-AS-GENERAL-EXPLANATION-20260907C
 6243 STATUS: NO GENERAL SEPARATION; necessary condition and occasional use survive.
 6244 ROUTE:
 6245 Test whether the group generated by all failed low primes is a proper
 6246 subgroup that excludes p0, thereby explaining the known exceptional hosts.
 6247 EXACT DIAGNOSTIC:
 6248 For EACH of the six historical hosts reconstruct C by starting with
 6249 all active BAD primes p<N/3 and repeatedly deleting rows having a prime
 6250 factor outside the current set, until stabilization. This is complete
 6251 factor pruning and supplies the maximal closed low BAD set at that N.
 6252 Generate H_C by finite multiplication modulo N. Counts are:
 6253

N	C	H_C	phi(N)	index
128	10	64	64	1
1718	55	858	858	1
1862	55	756	756	1
1928	41	960	960	1
2200	2	40	800	20
6142	66	2952	2952	1

 6260 At five of the six hosts the group is the ENTIRE unit group; consequently
 6261 no multiplicative character modulo N that is trivial on C can separate
 6262 p0, or any other active root, there.
 6263 At N=2200, C={3,13}, the group has order 40 and does exclude p0=1103.
 6264 All upper prime members of H_C with complement>1 are
 6265 1471,1523,1693,2083,2161.
 6266 Exactly 1471,1693,2083,2161 fail. The remaining member 1523 succeeds
 6267 DIRECTLY, since 2200-1523=677 is prime. Thus even in the proper-group
 6268 case membership is NOT sufficient for failure.
 6269 The full group at N=2200 is
 6270 {1,3,9,13,27,39,81,117,163,169,243,351,489,507,677,679,729,733,
 6271 959,1053,1147,1241,1467,1471,1521,1523,1693,1711,1849,1957,
 6272 2031,2037,2083,2119,2161,2173,2187,2191,2197,2199} modulo 2200.
 6273 Additional diagnostic: of the 301 failed upper roots at these SIX N,
 6274 the only roots satisfying $h < \min \text{PF}(N-r)$ are
 6275 (N,r)=(1718,877) and (6142,3089).
 6276 They are noncanonical. This confirms that the low-gap premise itself
 6277 does not force eventual success. For N=1718 the midpoint 859 is prime
 6278 and is p0; its h=0 diagonal success must not be mislabeled h=1.
 6279 REPRODUCTION: nowe/structural_probe.py. These are six targeted cases,
 6280 not an exhaustive search for all hosts or a classification of bad cores.
 6281 HISTORICAL AUDIT:
 6282 The Biblia already contains the stronger valuation-matrix relation
 6283 $\text{product_j } p_j \cdot (E_{ij} - \delta_{ij}) = -1 \pmod{N}$ and its determinant/resonance
 6284 consumer near lines 35135 onward. The subgroup lemma is an elementary
 6285 consequence of that known row relation; it is NOT claimed as new theory.
 6286 The new work here is the direct test and its negative explanatory result.
 6287 WHAT SURVIVES:
 6288 A sufficient exclusion test outside a verified full BAD-envelope group.
 6289 Smaller moduli dividing N and their characters cannot recover information
 6290 lost when H_C is already the entire unit group modulo N. More structured
 6291 valuation or additive constraints were not tested by this argument.

6292 No inference is made that all possible residue methods are impossible.
6293
6294
6295 RESULT-ID: LOW-GAP-THREE-LABEL-EXACT-OBSTRUCTION-SPLIT-20260907C
6296 STATUS: PROVED finite-support reduction; the resulting Diophantine
6297 systems are unresolved, so this is NOT a four-label theorem.
6298 EXACT STATEMENT:
6299 Fix a prime upper root r with composite $m=N-r$, $0<h<q=\min PF(m)$,
6300 and no GOOD vertex through depth 2. If the complete nonroot support
6301 discovered through depth 3 has at most three labels, then it has
6302 exactly three, and at least one of the following holds:
6303
6304 (A) There is a closed active BAD set C of exactly three primes that
6305 contains $PF(m)$; the whole nonroot reachable set is C .
6306
6307 (B) There are distinct odd primes q, b, c and integers $u, v \geq 2$,
6308 $a \geq 0, w \geq 1, a+w \geq 2$ satisfying the COMPLETE row equations
6309 $m = q^u$,
6310 $N = q^b v$,
6311 $N = q^a c^w$,
6312 $N = 2q^u + 2h$, $0 < h < q$, $r = q^u + 2h$ prime.
6313 The row $N-c$ is not constrained; c is first discovered at depth 3.
6314 If $a > 0$, then $3 < v < u-1$. If $v = 2$, then $a = 0$ and w is odd ≥ 3 .
6315
6316 Conversely, (A) in the stated low-gap root domain yields no GOOD and
6317 exactly three discovered nonroot labels by depth 3; and a system (B)
6318 with the listed hypotheses yields no GOOD through depth 2 and exactly
6319 three labels through depth 3. Cases (A) and (B) may overlap.
6320
6321 PROOF OF EXHAUSTIVENESS:
6322 Let $S = PF(m)$. Minimum-factor escape gives $PF(N-q)$ disjoint from S :
6323 any common prime t would divide $2h-q$, a nonzero integer of absolute
6324 value $< q \leq t$. The old LOW-GAP-COMPLETE-THREE-LABEL-ALTERNATIVE proves
6325 that at least three labels occur. If $|S| \geq 3$, expansion of the BAD
6326 row at q adds an outside label by depth 2, giving at least four.
6327 Therefore $|S| \leq 2$ in a three-label case.
6328 If $|S| = 2$, $PF(N-q)$ must be a singleton $\{b\}$, or at least four labels
6329 are already present. The three labels S union $\{b\}$ are all reached
6330 by depth 2. Since no GOOD occurs there, all three rows are expanded,
6331 and no child outside this set is discovered by depth 3. Thus it is
6332 a complete closed active BAD set C , giving (A).
6333 If $|S| = 1$, write $m = q^u$, $u \geq 2$, and $T = PF(N-q)$. If $|T| \geq 3$, at least
6334 four labels occur. If $|T| = 2$, then $\{q\}$ union T consists of three
6335 BAD labels reached by depth 2; expanding all three again proves (A).
6336 Finally $|T| = 1$ gives $N = q^b v$, $v \geq 2$. Activity gives $b \neq q$. Under the
6337 three-label count the entire support of $N-b$ lies in $\{q, c\}$, where c
6338 is the sole new label. The earlier three-label theorem ensures c
6339 really occurs. Hence $N = q^a c^w$, $w \geq 1$, with $a+w \geq 2$ by BADness.
6340 The sharp monomial return lemma applies when $a > 0$. If $v = 2$, the
6341 square-row lemma gives $a = 0$ and excludes an even w . This proves (B).
6342 For the converse in (A), closure and BADness prevent any GOOD or
6343 fourth label, while the old three-label theorem forces three labels
6344 by depth 3. In (B), the complete rows show exactly the successive
6345 fronts $\{q\}$, $\{b\}$, $\{c\}$, with a possible repeated q in the last row.
6346 The root and q, b are BAD; the status of c is immaterial through depth 2.
6347
6348 HOSTILE CHECK:
6349 Case (A) is a statement about the entire reachable set, not just a
6350 terminal SCC of size three. No classification of three-vertex cores
6351 or of two-vertex terminal cores is assumed. In (B), requiring $N-c$
6352 to be BAD or supported on $\{q, b, c\}$ would be an extra, unjustified
6353 condition. The reduction explicitly retains the full cofactors in
6354 both expanded rows. A selected path alone would not establish it.
6355 RELATION TO p_0 :
6356 The reduction is shared by all low-gap roots; in the $h=1$ square
6357 subcase one can add u odd ≥ 3 and $q \equiv 23 \pmod{24}$. It does not prove
6358 that even one of the reduced configurations exists for p_0 , nor that
6359 all are impossible. The five-label counterexample with $r=p_{12}$ has
6360 FOUR labels and therefore lies outside this three-label obstruction.
6361 NOVELTY: no literature priority claim; this is a precise consumer of
6362 the existing three-label theorem and the new reciprocal-row lemmas.
6363
6364
6365 =====
6366 END-OF-RUN CHECKPOINT -- 2026-09-07C
6367 =====
6368 STRONGEST RESULTS OF THIS LOCAL CONTINUATION:
6369 1. Sharp reciprocal-edge bound $b \geq 3q+2h$ for a low-gap return
6370 $q \rightarrow b \rightarrow q$ through the minimum first factor. Complete proof and sharp
6371 p_0 and p_1 examples are recorded. The root's primality excludes
6372 the candidate $b = q+2h$; minimum-factor arithmetic excludes $b = 2h-q$.
6373 2. For $m = q^u$ and complete prime-power row $N = q^b v$, a return forces
6374 $3 < v < u-1$. Thus $u=2$ and $u=3$ prohibit this immediate monomial return.
6375 The separate square-row lemma forces the next row between two
6376 consecutive squares and coprime to q .
6377 3. The unrestricted low-gap FIVE-label strengthening is false at
6378 $N=44670$, $r=p_{12}=22469$. Both depth-3 endpoints are GOOD. The old
6379 rank-0-through-6 experiment remains correct within its exact domain.
6380 4. A precise reduction of the open FOUR-label strengthening to a full
6381 three-label closed world or an explicit complete prime-power system.
6382 5. A modular-subgroup explanation was tested and found uninformative
6383 at five of the six known hosts, where the failed support generates
6384 the whole unit group. The necessary subgroup lemma is retained.
6385
6386 VALIDATION:
6387 One subagent supplied an independent counterexample search and proof
6388 audit; no additional agents were created. The root checked the exact
6389 certificates independently and ran the new return/support audit across
6390 134898 eligible rank-0-through-12 inputs. All 10249 reciprocal-edge
6391 checks passed; 1360 attain equality. Proofs are elementary and use no
6392 new external sieve, prime-tuple, height or classification theorem.
6393 The supplied TXT's complete previous contents remain above this run.
6394
6395 CURRENT LIMIT:
6396 Neither binary Goldbach nor universal p_0 reachability is proved.
6397 The new return restrictions use the primality of an upper root and a

6398 small numerical gap; they are not properties unique to the canonical
6399 selector. No general bound of FOUR, let alone FIVE, on the depth-3
6400 nonroot support has been proved. The earlier asymptotic radius-three
6401 large-product obstruction and moving closed-support obstruction are
6402 unchanged. Finite counts are not all-N theorems or literature novelty.
6403
6404 EXACT NEXT QUESTIONS:
6405 Local: can either configuration in LOW-GAP-THREE-LABEL-EXACT-
6406 OBSTRUCTION-SPLIT occur? In its singleton square branch the remaining
6407 equations are $2q^u + 2h = q + b^2 = b + c^w$ with $w \text{ odd} \geq 3$; in $h=1$ also u is
6408 $\text{odd} \geq 3$ and $q \equiv 23 \pmod{24}$. These equations still need an actual
6409 Diophantine exclusion or a checked example.
6410 Global: the $h=1$ full-BFS question from the previous checkpoint remains
6411 the same: can a complete closed active BAD set contain $PF(P-2)$ when
6412 $N=2(P-1)$ and P is prime? Excluding only immediate two-cycles, or only
6413 three-label worlds, does not settle this moving-support problem.
6414
6415 FINAL INDEPENDENT AUDIT:
6416 The same single subagent checked the strengthened reciprocal bound,
6417 the exponent-drop consumer, and the exhaustive three-label split,
6418 including both converses and the unconstrained status of the depth-3
6419 prime c . No correction was needed. This is an internal independent
6420 proof check, not external peer review or a novelty audit.

Appendix C

Exact 2026-09-07E Rank-Spectrum Checkpoint

Lossless source transcript of the standalone 07E rank-spectrum checkpoint. It is preserved as source data; later 07G corrections override it where explicitly stated.

```
1 RANK-SPECTRUM / NONCANONICAL-ROOT STRESS CHECKPOINT -- 2026-09-07E
2 STATUS: research checkpoint; no Goldbach proof; novelty not externally audited unless explicitly noted.
3
4 1. NEW EXACT RESULT: ALL-UPPER-ROOT FIXED POINT
5 For even N and any upper prime root  $r \geq N/2$  with composite complement  $m = N - r > 1$ ,
6  $r$  fails  $\Leftrightarrow PF(N-r)$  subset  $B_N$ ,
7 where  $B_N$  is the largest active closed BAD set below  $N/3$  from the existing CRD fixed-point filtration.
8 Reason: every factor of an odd composite row  $N-p$  has cofactor  $\geq 3$ , hence every child is  $< N/3$ ; failure therefore yields a closed BAD low set.
   $\hookrightarrow$  Converse is closure.
9 STATUS: PROVED (elementary generalization of existing canonical fixed-point theorem).
10
11 2. EXACT RANK FAILURE FORMULA
12 Put  $x = N/2$  and  $\pi(x^-) =$  number of primes strictly below  $x$ . If  $m = N - p_k$ , then
13  $k = \pi(N-m) - \pi(x^-) - 1$ .
14 On the composite-complement domain,
15  $p_k$  fails  $\Leftrightarrow \text{rad}(m) \mid R_N$ ,
16 where  $R_N = \text{product}_{\{q \in B_N\}} q$ .
17 Thus the composite failure-rank spectrum is exactly the prime-counting image of  $B_N$ -smooth prime-backed mirrors.
18
19 3. NEW EXACT RESULT: RAMANUJAN UNIT-ROW OBSTRUCTION
20 Let  $R_n$  be the  $n$ th Ramanujan prime. Standard Ramanujan-prime fact:  $R_n$  is prime and
21  $\pi(R_n) - \pi(R_n/2) = n$ .
22 For every fixed  $k \geq 1$  set  $P = R_{k+1}$  and  $N = P+1$ . Then  $N$  is even and exactly  $k+1$  primes lie in  $(P/2, P]$ , equivalently in  $[N/2, N)$ . Therefore
23  $p_k(N) = P - N + 1$ ,
24 so  $N - p_k = 1$ : a degenerate unit-row failure.
25 Hence NO fixed noncanonical rank  $k \geq 1$  can be universally successful for the raw algorithm.
26 By Bertrand,  $p_0$  can never be  $N-1$ : for  $P = N-1 \geq 5$  prime, Bertrand applied to  $(P-1)/2$  gives an earlier prime in  $[N/2, P)$ ;  $N=4$  is direct. Thus  $p_0$  is the
   $\hookrightarrow$  unique fixed rank not forced to have a unit-row failure.
27 STATUS: PROVED using standard Ramanujan-prime theorem / Bertrand. Application novelty not audited.
28
29 Guaranteed Ramanujan witnesses for the composite-safe candidates:
30  $k=10$ :  $R_{11}=101$ ,  $N=102$ 
31  $k=14$ :  $R_{15}=151$ ,  $N=152$ 
32  $k=26$ :  $R_{27}=307$ ,  $N=308$ 
33  $k=28$ :  $R_{29}=347$ ,  $N=348$ 
34  $k=30$ :  $R_{31}=367$ ,  $N=368$ 
35  $k=62$ :  $R_{63}=823$ ,  $N=824$ 
36  $k=63$ :  $R_{64}=827$ ,  $N=828$ 
37  $k=71$ :  $R_{72}=983$ ,  $N=984$ 
38  $k=74$ :  $R_{75}=1021$ ,  $N=1022$ 
39 The exact scan also found additional unit rows for some ranks.
40
41 4. FULL FIFO STRESS SCAN -- ALL EVEN  $N \leq 10^7$ 
42 Implementation: exact linear SPF sieve, full distinct-factor FIFO BFS, duplicate suppression, inclusive W, no depth/work cutoff, no probable-prime
   $\hookrightarrow$  tests. Unit rows excluded from the composite-failure count and counted separately. Regression:  $k=0$  reproduces published  $p_0$  statistics exactly:
   $\hookrightarrow$  mean  $W=7.052697211$ , max  $W=151$  at  $N=3542948$ , max depth=10 at  $N=11822$ .
43
44 rank eligible composite unit composite_fail meanW maxW@N maxDepth@N
45 0 4999999 4016277 0 0 7.052697211 151@3542948 10@11822
46 10 4999949 4305357 1 0 7.667464808 138@8708888 8@4298
47 14 4999925 4303563 2 0 7.668315425 157@8765846 9@2214218
48 20 4999884 4303761 2 1 7.670092946 175@8817218 9@1928
49 26 4999855 4303279 2 0 7.669350611 155@6435428 9@7202
50 28 4999835 4302322 5 0 7.668785470 163@9751754 9@11420
51 30 4999819 4302484 3 0 7.665228081 152@7644278 10@44246
52 62 4999588 4302844 1 0 7.669281949 160@7713518 9@1928
53 63 4999586 4302932 1 0 7.661838000 149@4531018 9@11822
54 71 4999511 4301412 3 0 7.662366579 159@9534652 8@14198
55 74 4999489 4302377 1 0 7.665502814 146@9751304 9@11822
56
57 Interpretation: all nine noncanonical candidate ranks retain ZERO nondegenerate/composite failures through  $10^7$ , but they are not easier than  $p_0$ .
   $\hookrightarrow$  Their mean work is higher ( $\sim 7.66-7.67$  vs  $7.05$ ), and several have max work exceeding the canonical record 151 in this same range. Rank 30
   $\hookrightarrow$  attains depth 10.
```

```

58 Control k=20 has the known genuine composite failure at N=1862.
59
60 5. FULL-WORLD / ESCAPE-MARGIN STRESS -- ALL EVEN  $N \leq 10^6$ 
61 Root-generic analogue of the project's escape margin  $G_{\text{esc}}$ : expand the entire reachable BAD world and count all GOOD child edges. Failure iff
   $\hookrightarrow G_{\text{esc}}=0$ ;  $G_{\text{esc}}=1$  is graph-theoretically one escape edge away from closure.
62 Every one of  $k=10,14,26,28,30,62,63,71,74$  has MANY  $G_{\text{esc}}=1$  cases despite zero composite failures.
63 Largest observed one-escape worlds among these ranks:
64 k10: world 34 at  $N=2078$ ,  $r=1103$ 
65 k14: world 34 at  $N=2078$ ,  $r=1129$ 
66 k26: world 66 at  $N=14198$ ,  $r=7349$ 
67 k28: world 65 at  $N=44246$ ,  $r=22409$ 
68 k30: world 65 at  $N=44246$ ,  $r=22441$ 
69 k62: world 65 at  $N=44246$ ,  $r=22741$ 
70 k63: world 64 at  $N=14198$ ,  $r=7673$ 
71 k71: world 71 at  $N=14198$ ,  $r=7727$ 
72 k74: world 65 at  $N=44246$ ,  $r=22871$ 
73
74 Exact sole escape edges:
75  $N=2078$ : for  $k=10,14,26,28,30,62,63$  the sole escape is  $179 \rightarrow 211$ , because  $2078-211=1867$  prime.
76  $N=14198$ : for  $k=26,63,71$  (and several others, including  $p_0$  in its own world) the sole escape is  $71 \rightarrow 277$ , because  $14198-277=13921$  prime.
77  $N=44246$ : for  $k=28,30,62,74$  (also  $k_{26}$  and control  $k_{20}$ ) the sole escape is  $9941 \rightarrow 2287$ , because  $44246-2287=41959$  prime.
78 At  $N=44246$ ,  $k=30$  succeeds only at depth 10 with  $W=64$  despite having exactly one escape edge in a 65-vertex BAD world.
79
80 MULTI-RANK NEAR-FAILURE SEARCH,  $N \leq 10^6$ :
81 Maximum number of the nine candidate ranks simultaneously having  $G_{\text{esc}}=1$  is 7.
82  $N=2078$ :  $k=10,14,26,28,30,62,63$  all have the SAME sole escape  $179 \rightarrow 211$ .
83  $N=1718$ : seven candidate ranks have  $G_{\text{esc}}=1$  (with several different sole edges).
84 Thus zero-failure ranks are not separated from failure by a large margin; several can sit on the same one-edge bottleneck.
85
86 6. HARD-PO INPUT CROSS-STRESS
87 Selected examples:
88  $N=11822$  (canonical depth record 10):  $k_{26}$  depth9  $W_{62}$ ;  $k_{63}$  depth9  $W_{58}$ ;  $k_{74}$  depth9  $W_{57}$ .
89  $N=976904$  (old canonical work record 133):  $k_{62}$  depth7  $W_{104}$ ;  $k_{63}$  depth7  $W_{96}$ ;  $k_{28}$  depth7  $W_{74}$ .
90  $N=3542948$  (canonical  $W=151$  record):  $k_{14}$  depth7  $W_{127}$ ;  $k_{26}$  depth7  $W_{130}$ ;  $k_{10}$  depth6  $W_{94}$ .
91 No candidate failure is produced; several remain comparably hard.
92
93 7. SAFE-RANK SPECTRUM ON THE SIX KNOWN EAC HOSTS
94 Using the six historical hosts 128,1718,1862,1928,2200,6142 and exact basin membership, for ranks  $0 < k \leq 118$  the composite-failure-free indices (on
   $\hookrightarrow$  every host where the rank exists) are exactly
95  $\{0,10,14,26,28,30,62,63,71,74\}$ .
96 There is no exact periodicity of this finite safe/fail indicator with period  $m \leq 60$ .
97 The neighbors of the safe indices frequently fail, e.g.  $k_9$  and  $k_{11}$  around  $k_{10}$ ;  $k_{13}$  and  $k_{15}$  around  $k_{14}$ ;  $k_{25}/k_{27}$  around  $k_{26}$ . This looks like
   $\hookrightarrow$  irregular holes in shifted-smooth mirror spectra, not a congruence class in  $k$ .
98
99 Important finite-range caveat: for large  $k$  some small EAC hosts do not even possess  $p_k$  below  $N$ , so finite- $X$  'zero failure' becomes increasingly
   $\hookrightarrow$  vacuous. The common full six-host rank range is only  $k=0, \dots, 12$ ; in that range the only composite-safe indices are 0 and 10. Among the five
   $\hookrightarrow$  larger hosts the common range extends to  $k=118$ .
100
101 8. EXACT GENERAL FORM OF THE RANK SPECTRUM
102 For  $N=2x$ , define
103  $M_{\text{fail}}(N) = \{m: 1 < m < x, N-m \text{ prime}, \text{rad}(m) | R_N\}$ ,
104 where  $R_N = \text{product}_{\{q \text{ in } B_N\}} q$ .
105 Then
106  $F_{\text{comp}}(N) = \{ \pi(N-m) - \pi(x^-) - 1 : m \text{ in } M_{\text{fail}}(N) \}$ .
107 The raw failure set adds the unit term  $m=1$  when  $N-1$  is prime.
108 Thus a fixed rank  $k$  is nondegenerately safe for every  $N$  iff  $k$  never occurs in the prime-counting image of a  $B_N$ -smooth prime-backed mirror over all
   $\hookrightarrow N$ .
109 No simpler rank-only formula is presently known.
110
111 9. CURRENT INTERPRETATION / FRONTIER
112 - Strict raw universality: every  $k \geq 1$  is KILLED by the Ramanujan unit-row theorem. Only  $k=0$  can even be a universal-success candidate.
113 - Composite-only universality  $H_{k^*}$ : still open for  $k=0$  and for the finite-survivor noncanonical ranks;  $k=20$  is refuted at  $N=1862$ .
114 - Stress data strongly argues that  $k=10$  etc are not protected by a large escape margin or easy traversal; they can be one edge from failure and as
   $\hookrightarrow$  hard as  $p_0$ .
115 - Best structural object remains the shifted-smooth mirror / basin rank spectrum, not  $k$  modulo a fixed integer.
116 - Next serious proof target: simultaneous  $B_N$ -smooth mirrors at small prime-root gaps ( $S$ -unit spacing / support-overlap constraints), or a
   $\hookrightarrow$  construction showing some currently composite-safe fixed  $k$  eventually hits a genuine EAC basin.
117
118 END CHECKPOINT

```

Appendix D

Exact 2026-09-07F Rank-Mirror Checkpoint

Lossless source transcript of the standalone 07F rank-mirror checkpoint. The 07G audit in Appendix E is binding when it narrows or corrects this source.

```
1 RANK-MIRROR SURJECTIVITY / DANGEROUS-MIRROR TRANSPLANT CHECKPOINT -- 2026-09-07F
2 STATUS: research checkpoint; no Goldbach proof; no universal composite-success theorem; novelty not externally audited unless explicitly stated.
3
4 STARTING POINT
5 This continuation starts from rank_spectrum_checkpoint_2026-09-07E.txt. The main question is whether the apparent composite-failure-free ranks
  ↳ k=10,14,26,28,30,62,63,71,74 possess a rank-specific arithmetic protection, or whether their finite survival is only an accident of the global
  ↳ BAD-basin landscape.
6
7 1. NEW EXACT THEOREM: FIXED-RANK MIRROR SURJECTIVITY
8 STATUS: PROVED, using only the Prime Number Theorem for unboundedness of prime counts in half intervals. Application novelty NOT AUDITED.
9
10 For odd m>=3 and an odd prime P>m define N=P+m and
11   kappa_m(P) = #{q prime : N/2 <= q < P}.
12 Then P=p_{kappa_m(P)}(N).
13
14 THEOREM.
15 (a) For every odd m>=3 and every integer k>=0, there exists an odd prime P>m such that kappa_m(P)=k. Hence there exists an even N=P+m with
16   p_k(N)=P,   N-p_k(N)=m.
17 (b) For m=1 the same statement holds exactly for k>=1, while k=0 is impossible.
18
19 PROOF.
20 Fix odd m>=3 and list consecutive primes P_0<P_1<... above m. For P_0=nextprime(m), the interval [(P_0+m)/2,P_0] lies strictly inside (m,P_0), so
  ↳ kappa_m(P_0)=0. Passing from P_j to P_{j+1}, the left endpoint moves right and may delete primes, while the right endpoint gains only the
  ↳ single new prime P_j; therefore
21   kappa_m(P_{j+1}) <= kappa_m(P_j)+1.
22 By the Prime Number Theorem,
23   kappa_m(P) = pi(P)-pi((P+m)/2)+O(1) ~ P/(2 log P) -> infinity
24 along primes P. An integer sequence starting at 0, unbounded above, and whose upward jumps are at most 1 must attain every nonnegative integer.
  ↳ This proves (a).
25 For m=1 start at P=3: N=4 and kappa_1(3)=1. The same argument gives every k>=1. The case k=0,m=1 is impossible (equivalently p_0 can never have unit
  ↳ complement).
26
27 COROLLARY: COMPLETE RANGE OF ROOT COMPLEMENTS.
28 Let C_k={N-p_k(N) : p_k(N)<N}. Then
29   C_0 = {3,5,7,...} (all odd integers >=3),
30   C_k = {1,3,5,7,...} for every fixed k>=1.
31 Thus, except for the single value 1, every fixed rank has exactly the same possible root-complement values as p_0.
32
33 COROLLARY: EXACT FIRST-ROW UNIVERSALITY.
34 For every fixed k>=0, every finite set S of odd primes, and every prescribed positive exponent vector (e_q)_{q in S} with m=prod q^{e_q} >=3, there
  ↳ is an N for which the p_k root row is EXACTLY m, with exactly that support and exactly those valuations. No CRT is needed. Therefore no
  ↳ nontrivial property depending only on the value/factorization/radical/Omega/largest prime factor of N-p_k can distinguish a fixed rank.
35
36 COROLLARY: UNIT-ROW OBSTRUCTION WITHOUT RAMANUJAN PRIMES.
37 For every k>=1 there exists N with N-p_k=1, so every fixed noncanonical rank is killed as a raw universal-success selector. This supersedes the 07E
  ↳ Ramanujan-prime route as a stronger and simpler existence theorem. p_0 remains the unique fixed rank not forced to a unit-row failure.
38
39 2. NEW EXACT NEGATIVE RESULT: (RANK, MIRROR) DOES NOT DETERMINE VERDICT
40 STATUS: VERIFIED COMPUTATION with self-contained exact certificates.
41
42 Even fixing BOTH the rank k and the exact complement m does not determine eventual success. Same (k,m), different N, opposite verdicts:
43
44 (A) k=7, m=27=3^3.
45   N=116, p7=89: SUCCESS depth 1 because 116-3=113 prime.
46   N=128, p7=101: FAILURE; exact 9-test closed transcript begins 101->3->5->41->29->11->13->23->7 and closes inside the known basin.
47
48 (B) k=20, m=799=17*47.
49   N=1862, p20=1063: FAILURE, W=22.
50   N=1886, p20=1087: SUCCESS depth 2, W=5 via 1087->17->7 and 1886-7=1879 prime.
51 The first row is literally identical: PF(m)={17,47} in both worlds.
52
53 (C) k=48, m=729=3^6.
54   N=2200, p48=1471: FAILURE via the twin core 3<->13, W=3.
```

```

55 N=2180, p48=1451: SUCCESS depth 4, W=10 via 1451->3->7->53->709, with 2180-709=1471 prime.
56 N=2210, p48=1481: SUCCESS depth 1.
57
58 CONSEQUENCE.
59 There is no verdict function depending only on (k, N-p_k), even with the complete integer factorization of the root complement. The indispensable
  ↪ variable is the ambient N-dependent landscape/B_N.
60
61 3. ADVERSARIAL DANGEROUS-MIRROR TRANSPLANT STRESS
62 STATUS: VERIFIED COMPUTATION, exact SPF/BFS; finite domain N<=200000.
63
64 Source adversarial corpus: all 301 known failing upper roots on the six EAC hosts 128,1718,1862,1928,2200,6142. Their root rows contain 214
  ↪ DISTINCT composite complements m (max m=3053).
65
66 For each dangerous mirror m and each candidate fixed rank
67   K={10,14,26,28,30,62,63,71,74},
68 enumerate EVERY realization N=P+m<=200000 with P=p_k(N), preserving the exact integer root row m and therefore its entire factorization with
  ↪ multiplicity. Then run exact full FIFO factor BFS.
69
70 RESULT:
71   total candidate transplant cases = 4035
72   genuine failures = 0
73   one-escape full-world cases = 147
74
75 Per rank (cases, failures, one-escape cases):
76 k=10: 458, 0, 20
77 k=14: 441, 0, 23
78 k=26: 442, 0, 16
79 k=28: 453, 0, 15
80 k=30: 424, 0, 19
81 k=62: 462, 0, 16
82 k=63: 452, 0, 15
83 k=71: 451, 0, 15
84 k=74: 452, 0, 8
85
86 Thus the nine candidate ranks survive 4035 exact transplants of the actual root complements that caused every known noncanonical failure, including
  ↪ 147 cases whose complete reachable BAD world has exactly one escape edge.
87
88 The harness is sensitive rather than vacuous. Control ranks recover genuine failures under the same transplant protocol:
89 k=0: 0 failures among 358 transplant realizations
90 k=7: 1 failures among 430 transplant realizations; first={'m': 27, 'N': 128, 'p': 101, 'W': 9}
91 k=8: 5 failures among 434 transplant realizations; first={'m': 25, 'N': 128, 'p': 103, 'W': 9}
92 k=9: 1 failures among 445 transplant realizations; first={'m': 21, 'N': 128, 'p': 107, 'W': 9}
93 k=11: 3 failures among 454 transplant realizations; first={'m': 15, 'N': 128, 'p': 113, 'W': 9}
94 k=20: 1 failures among 439 transplant realizations; first={'m': 799, 'N': 1862, 'p': 1063, 'W': 22}
95 k=48: 3 failures among 446 transplant realizations; first={'m': 517, 'N': 1718, 'p': 1201, 'W': 31}
96
97 The exact initial row therefore does not carry the known failure mechanism with it when transplanted to another ambient N. This directly supports
  ↪ the selector-landscape decomposition: failure is an incidence between the mirror support and B_N, not an intrinsic property of the mirror.
98
99 4. STRONGEST TRANSPLANT NEAR-FAILURES
100 Each candidate has one-escape transplanted examples. Largest one-escape BAD worlds in the transplant suite:
101 k=10: {'m': 975, 'N': 2078, 'p': 1103, 'bad_world': 34, 'edge': [179, 211]}
102 k=14: {'m': 531, 'N': 1258, 'p': 727, 'bad_world': 25, 'edge': [11, 29]}
103 k=26: {'m': 833, 'N': 2026, 'p': 1193, 'bad_world': 20, 'edge': [5, 47]}
104 k=28: {'m': 841, 'N': 2078, 'p': 1237, 'bad_world': 34, 'edge': [179, 211]}
105 k=30: {'m': 819, 'N': 2078, 'p': 1259, 'bad_world': 34, 'edge': [179, 211]}
106 k=62: {'m': 1391, 'N': 3748, 'p': 2357, 'bad_world': 32, 'edge': [7, 29]}
107 k=63: {'m': 585, 'N': 2078, 'p': 1493, 'bad_world': 34, 'edge': [179, 211]}
108 k=71: {'m': 411, 'N': 1862, 'p': 1451, 'bad_world': 24, 'edge': [23, 613]}
109 k=74: {'m': 309, 'N': 1718, 'p': 1409, 'bad_world': 30, 'edge': [103, 19]}
110
111 Notably the dangerous-mirror transplant reproduces the N=2078 shared bottleneck for k=10,28,30,63: the sole escape is 179->211. These are exact
  ↪ full-world near-failures, not shallow heuristics.
112
113 5. NEW EXACT RANK/SPACING LEMMA: BASIN-COPRIME GAPS FORCE DISJOINT SUPPORT
114 STATUS: PROVED, elementary.
115 Let r<s be two genuine failing upper roots for the same N, with mirrors u=N-r and v=N-s, and let R_N=prod_{q in B_N}q. Then
116   gcd(u,v)=gcd(u,s-r).
117 Hence
118   PF(u) intersect PF(v) = PF(gcd(u,s-r)).
119 In particular, if gcd(s-r,R_N)=1, then gcd(u,v)=1 and the two failing mirrors have disjoint prime supports. More generally, for a family of
  ↪ pairwise B_N-coprime root gaps, all mirror supports are pairwise disjoint, so the number of such failing roots is at most |B_N| (or at most
  ↪ floor(|B_N|/a) if each mirror has at least a distinct prime factors).
120
121 6. FIXED-GAP S-UNIT PAIR BOUND
122 STATUS: PROVED as a direct consumer of COPRIME-SUPPORTED-AFFINE-COUNT from the 07 checkpoint.
123 Fix a finite prime alphabet S, s=|S|, and d>0. The number of ordered positive S-smooth pairs (u,v) with u-v=d is at most
124   tau(d) * 2^(9s+16).
125 Proof: g=gcd(u,v) divides d. Write u=gU, v=gV. Then U,V are coprime S-smooth and U-V=d/g. For each divisor g|d the existing affine S-unit theorem
  ↪ bounds (U,V) by 2^(9s+16); sum over divisors g. Therefore, for a fixed basin alphabet and fixed root gap, only finitely many mirror-pair shapes
  ↪ can support two failures. The bound is enormous and is NOT a closure theorem, but it is the first direct S-unit consumer aimed at rank-spectrum
  ↪ spacing.
126
127 7. CONDITIONAL CONSEQUENCE OF FINITE EAC HOSTS
128 STATUS: exact implication; hypothesis OPEN.
129 If the set of even N carrying an EAC is finite, then all sufficiently large fixed ranks are universally safe on the genuine composite-complement
  ↪ domain. Indeed any genuine root failure forces an EAC host, while each fixed host contains only finitely many upper roots.
130
131 Under the specific conjectural hypothesis that the six known hosts are the ONLY EAC hosts globally, the largest upper-root rank available on them
  ↪ is 360 (at N=6142). Therefore
132   H_k* holds for EVERY k>=361,
133 where H_k* means universal success restricted to N-p_k>1 composite.
134 For 0<k<360, the exact six-host spectrum has 188 distinct failing ranks and 173 composite-safe ranks.
135 The 173 safe ranks are:
136   0,10,14,26,28,30,62,63,71,74,119,123,132,133,134,136,137,138,139,142,143,144,145,146,147,148,150,151,153,154,156,157,158,159,160,161,162,164,165
  ↪   ,167,168,169,172,175,176,177,180,181,182,186,187,188,190,191,192,193,194,195,196,197,199,201,202,203,204,206,207,208,209,210,211,212,213,214
  ↪   ,215,216,217,218,220,221,222,223,224,225,226,227,228,229,231,232,233,234,235,236,239,240,241,242,243,245,247,249,251,254,255,256,257,258,259
  ↪   ,260,261,263,266,267,268,269,270,272,275,276,277,278,279,281,282,285,286,288,289,290,291,292,294,295,296,297,298,299,300,301,302,303,304,306
  ↪   ,307,309,310,312,315,317,318,319,321,323,326,327,328,331,332,334,335,337,340,342,345,346,348,350,352,354,356,357,359
137 This is CONDITIONAL on the global six-EAC conjecture and must not be stated as a theorem about the actual integers.
138 Raw universality remains false for every k>=1 because of the unit-row theorem above.
139
140 8. INTERPRETATION
141 The new theorem sharply narrows what a putative "magic index" can mean.

```

```

142 - Rank k places essentially NO restriction on the root row: every odd composite mirror, every finite support, and every exponent pattern occurs at
    ↪ every fixed rank.
143 - Even (k, exact mirror m) does not determine verdict.
144 - The known dangerous mirrors can be transplanted thousands of times into the currently zero-composite-failure ranks without importing failure.
145 - Thus a rank-only congruence law or a first-row factorization law cannot explain the zero-failure indices.
146 - The only exact failure criterion remains ambient:
147     p_k fails ⇔ PF(N-p_k) subset B_N.
148
149 9. CURRENT FRONTIER
150 The most informative next question is no longer "find a formula in k". It is:
151
152   For fixed k, can the rank-realization curve
153       kappa_m(P)=k, N=P+m
154   intersect the moving EAC/B_N-smooth condition
155       rad(m) | R_N
156   for some composite m?
157
158 For k=20 the answer is yes at (N,m)=(1862,799). For k=10,14,26,28,30,62,63,71,74 no such intersection is known; exact stress to N<=10^7 and the
    ↪ dangerous-mirror transplant suite found none. The rank-mirror theorem proves this cannot be explained by an excluded class of mirrors.
159
160 No Goldbach proof. No universal p0 theorem. No universal composite-success theorem for the survivor ranks. Novelty of the rank-mirror formulation
    ↪ and its applications has not been externally audited.
161
162 REPRODUCTION FILES
163 - rank_mirror_transplant_stress_20260907.py
164 - rank_mirror_transplant_stress_20260907.json
165
166 END CHECKPOINT -- 2026-09-07F

```

Appendix E

Exact 2026-09-07D–H Continuation Delta

Lossless continuation of `goldbach_research_checkpoint_2026-09-07(9).txt`, source lines 6421–7687. Lines 1–6420 are exactly the transcript already preserved in Appendix B and are therefore not duplicated.

```
1
2
3 =====
4 CONTINUATION -- 2026-09-07D
5 =====
6 Same mathematical definitions and TXT checkpoint. At most one subagent
7 is used in this continuation. New claims below do not supersede the
8 earlier caveats about complete closure or universal p0 success.
9
10 RESULT-ID: TWICE-POWER-OF-THREE-MINUS-ONE-NOT-PERFECT-POWER-20260907D
11 STATUS: PROVED by an elementary factorization argument.
12 EXACT STATEMENT:
13 For every integer  $u \geq 1$ ,  $2 \cdot 3^u - 1$  is not a proper integer power  $A^k$ ,
14 with integers  $A \geq 2$  and  $k \geq 2$ . A need not be prime.
15 PROOF:
16 Suppose  $2 \cdot 3^u - 1 = A^k$ . The number is odd and is 2 modulo 3, so A is
17 odd,  $A \equiv 2 \pmod 3$ , and k is odd. Choose an odd prime  $\ell$  dividing k
18 and put  $B = A^{k/\ell}$ . Then  $B \equiv 3 \pmod 3$  is odd,  $B \equiv -1 \pmod 3$ , and
19  $(B+1)(B^{(\ell-1)} - B^{(\ell-2)} + \dots - B + 1) = 2 \cdot 3^u$ .
20 The second factor F is odd and greater than one; hence F is a
21 positive power of 3. Modulo 3 each of its  $\ell$  terms is 1, so
22  $F \equiv \ell \pmod 3$ , forcing  $\ell = 3$ . Thus
23  $B^2 - B + 1$  is a positive power of 3.
24 Writing  $B = 3t - 1$  gives  $B^2 - B + 1 = 3(3t(t-1) + 1)$ , with exactly one
25 factor of 3. Therefore  $B^2 - B + 1 = 3$ , implying  $B = 2$  or  $B = -1$ .
26 Both contradict B odd and  $B \equiv 3 \pmod 3$ . This proves the statement.
27 HOSTILE CHECK:
28 This excludes every proper power, not just a prime power. It does
29 not say the number is composite:  $u=1,2,3$  give 5,17,53, all prime.
30 No primality of  $3^u + 2$  is used. The result was derived by the one
31 subagent and independently checked by the root agent.
32 NOVELTY: elementary lemma/application, not a literature priority claim.
33
34
35 RESULT-ID: LOW-GAP-SINGLETON-UPWARD-FOUR-LABEL-ALTERNATIVE-20260907D
36 STATUS: PROVED; consumer of complete prime supports.
37 EXACT STATEMENT:
38 Let  $r = q^u + 2h$  be prime,  $u \geq 2$ , q odd prime,  $0 < h < q$ ,  $N = 2q^u + 2h$ .
39 Suppose  $N - q$  is composite, has at least two distinct prime factors,
40 and EVERY prime factor of  $N - q$  exceeds q. Then either a GOOD vertex
41 is reached by depth 2 from r, or at least FOUR distinct nonroot
42 prime labels occur by depth 3 in the complete stopped search.
43 PROOF:
44 The first frontier is {q}. If  $PF(N - q)$  has at least three elements,
45 the claim holds by depth 2. Otherwise write  $PF(N - q) = \{b, c\}$ ,  $q < b < c$ .
46 Assume no GOOD vertex occurs through depth 2 and at most three
47 nonroot labels are discovered through depth 3. All q, b, c are BAD
48 and their COMPLETE rows must be supported on {q, b, c}.
49 The row  $N - b$  cannot have b as a factor by activity. It cannot have
50 c as a factor either: c already divides  $N - q$ , so this would give
51  $c \mid (b - q)$ , with  $0 < b - q < c$ . Thus  $N - b = q^v$  for an integer  $v \geq 2$ .
52 If  $v < u$ , then  $b = N - q^v = N - q^u + r = N/2$ , contradicting  $b < N/3$ ,
53 since b is a factor of the odd composite row  $N - q$ .
54 If  $v \geq u + 1$ , then  $q^v = q \cdot q^u \geq N$ , because
55  $q \cdot q^u = N - (q - 2)q^u - 2h \geq q^u - 2h > 0$ .
56 This contradicts  $q^v = N - b < N$ . Both cases are impossible.
57 SCOPE:
58 The proof uses COMPLETE factors and their ordering, not one chosen
59 large child. The assumption that all factors of  $N - q$  exceed q is
60 essential to the displayed three-label exclusion and is not silently
61 assumed for a general singleton first support.
62
```

```

63 COROLLARY-ID: H1-POWER-OF-THREE-COMPLETE-FOUR-LABEL-ALTERNATIVE-20260907D
64 STATUS: PROVED; pointwise strengthening of the prior three-label result.
65 EXACT STATEMENT:
66 For  $u \geq 2$  with  $P=3^u+2$  prime and  $N=2(P-1)$ , the canonical search from
67  $p_0=P$  either succeeds by depth 2 or discovers at least FOUR distinct
68 nonroot labels by depth 3.
69 PROOF:
70  $P-1$  is even and composite, so  $P$  is exactly  $p_0$  and  $h=1$ . The first
71 complement is  $3^u$ . If  $N-3$  is prime, success occurs at depth 1.
72 Otherwise the preceding non-perfect-power lemma implies that its
73 complete factorization has at least two distinct primes. None is 3,
74 because  $N-3=2 \cdot 3^u - 1 \equiv 2 \pmod{3}$ ; all its factors therefore exceed 3.
75 The upward four-label theorem applies with  $q=3, h=1$ .
76 RELATION TO  $p_0$ :
77 This is a real restriction in a specific canonical family, not a
78 universal four-label theorem or an explanation for all  $p_0$ . The more
79 general lemma is shared by other roots satisfying its explicit
80 complete-factor hypotheses. Branches returning through small primes
81 outside those hypotheses remain unexcluded.
82 RELATION TO GOLDBACH:
83 Success or branching. FOUR does not rule out a larger closed BAD world.
84 The three-label obstruction from the previous run is now excluded
85 for this  $q=3, h=1$  family, but the global  $h=1$  problem remains open.
86
87 EXACT NONVACUITY CERTIFICATE FOR THE BAD-THROUGH-TWO CASE:
88  $u=14$ ,  $P=4782971$  prime,  $N=9565940$ ,  $N-P=3^{14}$ .
89  $N-3=9565937=1063 \cdot 8999$ .
90  $N-1063=9564877=7 \cdot 43 \cdot 739$ .
91  $N-8999=9556941=3 \cdot 17 \cdot 73 \cdot 151$ .
92 Every listed factor is prime, and all rows at depths 0,1,2 are
93 composite. Thus nine distinct nonroot labels appear through depth 3:
94  $\{3, 1063, 8999, 7, 43, 739, 17, 73, 151\}$ .
95 The prime root and factors were checked by exact integer trial division.
96 The lower bound FOUR is not claimed sharp in this restricted family.
97
98
99 RESULT-ID: LOW-GAP-MONOMIAL-RETURN-ORDER-AT-LEAST-THREE-20260907D
100 STATUS: PROVED; strengthens the previous  $3 \leq v \leq u-1$  to  $4 \leq v \leq u-1$ .
101 Let  $q, b$  be odd primes,  $u, v \geq 2$ ,  $0 < \chi_q$ ,  $r = q^u + 2h$  prime,
102  $N = 2q^u + 2h$ ,  $N - q = b^v$ , and  $q \mid (N - b)$ . Then
103  $d = \text{ord}_q(b) = \text{ord}_q(2h) \geq 3$ ,  $d \mid (v-1)$ ,  $d \mid (q-1)$ .
104 Consequently  $4 \leq v \leq u-1$  and  $u \geq 5$ . If  $v=4$  then  $q \equiv 1 \pmod{3}$ ;
105 if  $v=5$  then  $q \equiv 1 \pmod{4}$ . No such monomial return exists for  $u \leq 4$ .
106 PROOF:
107 The return gives  $b \equiv 2h \pmod{q}$ , while the complete first row gives
108  $b^v \equiv 2h \pmod{q}$ . Hence  $b^{v-1} \equiv 1 \pmod{q}$ .
109 If  $b \equiv 1 \pmod{q}$ , low gap forces  $2h \equiv q+1$  and  $b^v = 2q^u + 1$ .
110 This is  $3 \pmod{4}$ , so  $v$  is odd. The positive integer
111  $F = (b^v - 1)/(b - 1) = 1 + b + \dots + b^{v-1}$ 
112 is odd and divides  $2q^u$ , hence is a pure  $q$ -power. Its  $q$ -adic
113 valuation is exactly the valuation of  $v$ . Thus  $F \leq v$ , whereas
114 the displayed sum is strictly greater than  $v$ : contradiction.
115 If  $b \not\equiv 1 \pmod{q}$ , then  $2h \equiv q-1$  and  $b^v = 2q^u - 1$ . Reduction modulo  $q$ 
116 forces  $v$  odd. Now  $F = (b^v + 1)/(b + 1)$  is again an odd pure  $q$ -power
117 of valuation exactly that of  $v$ , so  $F \leq v$ . However its alternating
118 sum is at least  $b^{v-2} + (b-1) \geq 2 \cdot 3^{v-2} + v > v$  for  $v \geq 3$ .
119 This is another contradiction. Orders 1 and 2 are excluded.
120 Use the earlier sharp reciprocal bound and exponent-drop lemma
121 to obtain  $v \leq u-1$ . This is where primality of the upper root enters.
122 ELEMENTARY VALUATION JUSTIFICATION:
123 For odd prime  $q$  and  $x \equiv 1 \pmod{q}$ , valuation $_q(x^{n-1})$  equals
124 valuation $_q(x-1) + \text{valuation}_q(n)$ . Write  $n = q^t \cdot s$ ,  $q$  not dividing  $s$ .
125 The geometric quotient for exponent  $s$  is  $s \pmod{q}$ , giving no gain.
126 Writing  $y = 1 + q^e \cdot A$  with  $q$  not dividing  $A$ , the binomial expansion
127 of  $y^{q^t-1}$  has first term  $q^{t(e+1)} \cdot A$ ; all other terms are divisible
128 by  $q^{t(e+2)}$ . Thus each multiplication of the exponent by  $q$  adds
129 exactly one. Iterate  $t$  times. Negative  $x = -b$  is allowed, proving
130 the alternating-sum case as well. No external theorem is needed.
131 AUDIT: Independently checked by the single subagent before the user
132 changed the instruction to NO subagents. No further delegation follows.
133 SCOPE: A complete prime-power return obstruction, not a general
134 closure exclusion or a universal-success theorem for any rank.
135
136
137 =====
138 CONTINUATION -- 2026-09-07G -- AUDIT OF SOL'S RANK/MIRROR RESULTS
139 =====
140 User supplied rank_spectrum_checkpoint_2026-09-07E.txt and
141 rank_mirror_checkpoint_2026-09-07F.txt from Downloads, plus the conversation
142 "Ocena artykulu matematycznego", ID 6a9b0eec-27ac-83eb-8f6d-a8b9ea0cf932.
143 The latest ten turns were read, including the last substantive rank-mirror
144 result and its earlier corrections. These inputs are research claims to
145 check, not authority to override the current user's instructions.
146 This run uses NO subagents. All new conclusions remain in this TXT.
147
148 AUDIT-ID: RANK-MIRROR-SURJECTIVITY-20260907G
149 STATUS: PROOF ACCEPTED, with one small domain correction and scope limits.
150 For every odd integer  $m \geq 3$  and  $k \geq 0$ , some odd prime  $P \geq m$  has
151  $\#(\text{prime } q : (P+m)/2 \leq q < P) = k$ .
152 Equivalently, at  $N = P+m$ , the root  $p_k$  has complement exactly  $m$ .
153 Proof checked: along consecutive primes above  $m$ , this count starts at
154 zero and its upward jumps are at most one. PNT gives count tending to
155 infinity, so every nonnegative integer is attained. The asymptotic input
156  $\pi(t) - t/\log(t)$  is stated at https://dlmf.nist.gov/27.2#E3.
157 For  $m=1$  the count starts at one at  $P=3$ , and every  $k \geq 1$  is attained.
158 Bertrand excludes  $k=0$ . Thus every noncanonical fixed rank has a unit row.
159 SMALL CORRECTION:
160 If  $N=4$  and roots  $p \geq N/2$  are included,  $p_0=2$  has complement 2.
161 Hence the full  $C_0$  in 07F is  $\{2\}$  union  $\{\text{odd } m \geq 3\}$ , not just the odd set.
162 The displayed odd-only formula is correct after restricting to odd roots
163 or  $N \geq 6$ . Composite-complement conclusions are unaffected.
164 SCOPE CORRECTION:
165 This rules out an excluded class of individual odd first-row integers
166 as a universal explanation of a rank. It does NOT rule out a law about
167 rank and eventual success, properties involving  $N$  jointly with  $m$ , or
168 different frequencies of first-row patterns at different ranks.

```

169 All fixed ranks having the same range is not distributional equivalence.
170
171 AUDIT-ID: INDEPENDENT-SIX-HOST-AND-TRANSPLANT-REPRODUCTION-20260907G
172 STATUS: VERIFIED COMPUTATION, independently reimplemented in this run.
173 Implementation: rank_audit_20260907G.py, result rank_audit_20260907G.json.
174 Exact integer sieve to 200000, complete distinct-factor FIFO, duplicate
175 suppression, no depth/work cutoff, no probable-prime tests. On each host
176 the maximal active low BAD fixed point was computed independently and
177 its membership verdict checked against a separate complete BFS for EVERY
178 upper root having composite complement.
179 Host N: upper roots below N; size of maximal low BAD set; failed roots:
180 128: 13; 10; 8
181 1718: 119; 55; 68
182 1862: 126; 55; 63
183 1928: 131; 41; 57
184 2200: 143; 2; 4
185 6142: 361; 66; 101
186 Total 301 failed roots, 214 distinct composite mirrors, 188 distinct
187 failed ranks. Exactly 173 indices in 0..360 avoid these six spectra.
188 The exact 173-index list agrees with the uploaded 07F, entry for entry.
189 Within 0..118 the survivors are exactly
190 {0,10,14,26,28,30,62,63,71,74}.

191
192 The ENTIRE specified transplant suite was also independently rerun:
193 all 214 dangerous m, every P>m with even N=P+m<=200000, retain the
194 indicated exact rank, test full BFS and explore the entire stopped BAD
195 world for GOOD child edges. Every count in 07F was reproduced.
196 Rank: cases / failures / worlds with exactly one GOOD child edge:
197 0: 358 / 0 / 10
198 7: 430 / 1 / 18
199 8: 434 / 5 / 17
200 9: 445 / 1 / 21
201 11: 454 / 3 / 18
202 20: 439 / 1 / 20
203 48: 446 / 3 / 13
204 10: 458 / 0 / 20
205 14: 441 / 0 / 23
206 26: 442 / 0 / 16
207 28: 453 / 0 / 15
208 30: 424 / 0 / 19
209 62: 462 / 0 / 16
210 63: 452 / 0 / 15
211 71: 451 / 0 / 15
212 74: 452 / 0 / 8
213 Nine candidate ranks together: 4035 cases, zero failures, 147 one-edge
214 worlds. This is an audit of the supplied experiment, not a new range record.
215

216 All same-(rank,mirror) opposite-verdict certificates in 07F were checked:
217 (7,27): N116 success versus N128 failure;
218 (20,799): N1862 failure versus N1886 success;
219 (48,729): N2200 failure versus N2180 and N2210 success.
220 Their exact complete transcripts are in the result JSON. These establish
221 that (k,m) does not determine verdict, even with all first-row exponents.
222 The root clone at N1144 was checked too: roots 577=p0, 997=p62,
223 1123=p82 all have initial support {3,7}, success depth 7 and W=16.
224 The post-root tested labels are
225 3,7,163,379,109,5,17,23,67,19,59,359,31,157,53;
226 the final complement is 1091 prime. BOTH root complements must be
227 composite in the full-transcript clone theorem; equal supports alone
228 do not equate a direct GOOD root with a BAD root having that support.
229

230 EXTERNAL-CENSUS PROVENANCE:
231 Bozek's paper reports an EAC census through 10^8 with precisely the six
232 hosts above, in its section 4. Source <https://arxiv.org/pdf/1909.09900>.
233 Combining that reported census with the elementary active-BAD/EAC
234 equivalence transfers the six-host failure spectrum to $N \leq 10^8$.
235 This uses the author's computation; the present run did not scan all
236 $N \leq 10^8$ (or repeat the uploaded 10^7 work/depth statistics).
237 The global assertion that these are the only hosts is still unproved.
238
239

240 RESULT-ID: EFFECTIVE-FINITE-RANK-MIRROR-FIBRES-20260907G
241 STATUS: PROVED using an explicitly identified prime-counting theorem.
242 Let $m \geq 1$ be odd, $P > m$ an odd prime, $N = P + m$ and $P = p_k(N)$, $k \geq 0$.
243 Set $c(m) = \text{ceil}((m-1)/4)$, $n = k + c(m) + 2$. Then
244 $P < R_n < 4n * \log(4n)$,
245 $N < m + 4n * \log(4n)$.
246 Here R_n is the n th Ramanujan prime. Thus for EVERY fixed pair (k,m)
247 there are only finitely many realizations, with an effective upper bound.
248 For $m \geq 3$ that finite set is nonempty by the audited surjectivity theorem.
249 The apparent tension disappears: surjectivity over ranks does not imply
250 infinitely many realizations at a fixed rank.
251 PROOF:
252 Put $H(P) = \pi(P) - \pi(P/2)$. The interval $(P/2, P]$ contains $H(P)$ primes.
253 To obtain $[(P+m)/2, P)$, remove P and the primes in
254 $(P/2, (P+m)/2)$.
255 The latter interval contains exactly $(m-1)/2$ integers. For $P \geq 5$ all
256 these primes are odd, so there are at most $c(m)$ of them. The only
257 remaining case $P=3, m=1$ removes no integers and satisfies the same bound.
258 Therefore $k \geq H(P) - 1 - c(m)$. If $P \geq R_n$, the defining property of R_n
259 gives $H(P) \geq n = k + c(m) + 2$, hence $k \geq k+1$, a contradiction.
260 EXTERNAL INPUT:
261 J. Sondow, Ramanujan Primes and Bertrand's Postulate, Theorem 2:
262 $R_n < 4n * \log(4n)$ for every $n \geq 1$.
263 Primary source <https://arxiv.org/pdf/0907.5232>, pages 2-3 of the PDF.
264 Only this established bound and the definition of R_n are imported;
265 the rank/mirror argument above is the present elementary application.
266 Logarithms are natural. No EAC census or twin-core classification enters.
267 BOUNDARY CHECK:
268 The deletion is an OPEN interval at both ends; if $(P+m)/2$ is prime it
269 must remain in the rank count. Removing exactly $(m-1)/2$ integers respects
270 this. The prime P itself contributes the separate subtraction of one.
271
272 COROLLARY-ID: DANGEROUS-MIRROR-TRANSPLANTS-GLOBALLY-EXHAUSTED-20260907G
273 STATUS: PROVED EFFECTIVE EXHAUSTION plus the independent exact computation.
274 For every odd $m \leq 3053$ and every $k \leq 74$, $n = 74 + 763 + 2 = 839$. Thus

```

275 P<R_839<3356*log(3356)<30204,
276 N<33257.
277 The deliberately loose final numerical inequality needs no floating
278 rounding: e>5/2 and (5/2)^9>3356 give log(3356)<9.
279 The audited scan to N<=200000 therefore covers EVERY realization over
280 ALL N of the 214 dangerous mirrors at the nine candidate ranks
281 K={10,14,26,28,30,62,63,71,74}.
282 Consequently, the total is globally EXACTLY 4035 realizations, with
283 ZERO failed traversals and exactly 147 one-GOOD-edge worlds.
284 This upgrades the domain of the 07F transplant statement: it is a
285 complete finite set, not merely a test that might acquire further
286 realizations above 200000. The control-rank totals displayed above are
287 also globally exhaustive, since every control rank is at most 74.
288 SCOPE:
289 This is not universal safety of these ranks. It excludes failure only
290 for this FIXED list of 214 exact integers m. New mirrors, changed
291 exponents, and new closed BAD structures are not covered. It does not
292 prove the global six-host conjecture. This global finite-fibre conclusion
293 does not rely on the external EAC census through 10^8.
294 RESEARCH CONSEQUENCE:
295 Increasing the N cutoff while keeping this mirror list and these ranks
296 cannot add any case at all. Further transplant research must vary the
297 exact mirrors or the ranks; continued scanning of these same fibres is
298 mathematically exhausted.
299
300 COROLLARY-ID: EVERY-FIXED-RANK-HAS-ONLY-FINITELY-MANY-UNIT-ROWS-20260907G
301 STATUS: PROVED.
302 For m=1, any unit row at rank k satisfies
303 N<1+R_(k+2)<1+4*(k+2)*log(4*(k+2)).
304 Combined with surjectivity: every k>=1 has a NONEMPTY FINITE set of
305 unit-row inputs, while p0 has none. This is an exact distinction in
306 raw all-input admissibility. It provides no distinction in eventual
307 success after discarding finitely many inputs or restricting to
308 composite complements. Do not turn this finite degeneracy into a
309 proof that only p0 can be asymptotically or nondegenerately universal.
310
311
312 RESULT-ID: COMPLETE-BOUNDED-MIRROR-SAFETY-CENSUS-20260907G
313 STATUS: VERIFIED EXACT COMPUTATION WITH PROVED GLOBAL SEARCH BOUND.
314 The previous finite-fibre theorem permits a wider census than the old
315 dangerous-mirror list. Enumerate EVERY odd composite integer m<=3053,
316 and EVERY realization at each rank in {0,10,14,26,28,30,62,63,71,74},
317 over ALL even N. It suffices to test odd primes m<P<30204, with N=P+m.
318 Implementation/result: rank_fibre_closure_20260907G.py / corresponding JSON.
319 There are 1090 such distinct composite integers m, and each occurs at
320 each listed rank, consistent with surjectivity.
321 Rank: total composite cases / failures / largest realized N / max depth / max W:
322 0: 1882 / 0 / 6114 / 7 / 34
323 10: 2219 / 0 / 6282 / 8 / 30
324 14: 2244 / 0 / 6354 / 8 / 39
325 26: 2205 / 0 / 6552 / 7 / 39
326 28: 2242 / 0 / 6570 / 8 / 35
327 30: 2236 / 0 / 6582 / 7 / 27
328 62: 2274 / 0 / 7144 / 9 / 32
329 63: 2275 / 0 / 7146 / 8 / 33
330 71: 2271 / 0 / 7296 / 7 / 40
331 74: 2282 / 0 / 7324 / 7 / 36
332 The largest N for each rank has m=3053; root values and depth/work
333 witnesses are recorded exactly in the JSON. This is an exhaustive
334 bounded-mirror statement for all N, NOT a new large-N BFS record.
335
336 The same enumeration gives the COMPLETE global unit-row input sets:
337 k=0: empty
338 k=10: {102}
339 k=14: {140,152}
340 k=26: {284,308}
341 k=28: {314,318,332,338,348}
342 k=30: {354,360,368}
343 k=62: {824}
344 k=63: {828}
345 k=71: {972,978,984}
346 k=74: {1022}
347 The stated analytic bound proves there can be no further unit rows
348 at arbitrarily large N for these ranks. This is stronger than simply
349 counting unit rows up to a chosen experimental cutoff.
350 HARNESS CHECK:
351 An initial unit-only diagnostic accidentally included P=2,N=3. This
352 violates the odd-prime/even-N domain and was removed before accepting
353 these results. The final code explicitly starts at P>=3 and asserts
354 P odd, N even, N>=4, and an empty canonical unit set. Composite-case
355 counts were unaffected. All 214-mirror transplant counts were asserted
356 equal to the independent preceding audit.
357
358 AUDIT-ID: SCOTT-BOZEK-PARITY-GAP-REAPPEARS-IN-SOL-CONVERSATION-20260907G
359 STATUS: DEPENDENCY NOT ACCEPTED AS A COMPLETE GLOBAL CLASSIFICATION.
360 The older conversation uses uniqueness of the two-vertex EAC to exclude
361 all canonical two-vertex terminal cores and to remove the 'all children
362 above q' hypothesis from an h=1 singleton FOUR lemma. This relies on
363 Bozek's Theorem 3.7. Its displayed proof paraphrases Scott as giving at
364 most one solution to a^x-b^y=c, with five exceptions, without the parity
365 qualification. Scott's original publisher abstract instead bounds
366 solutions with y ODD (five exceptions) and y EVEN separately.
367 Source https://www.sciencedirect.com/science/article/pii/S0022314X83710413
368 DOI https://doi.org/10.1006/jnth.1993.1041.
369 The trivial solution (x,y)=(1,1) and a proposed solution with even y
370 belong to different parity classes. Those two separate uniqueness
371 statements do not themselves exclude their coexistence.
372 Thus the quoted reduction leaves a branch, precisely the one already
373 quarantined in the older project checkpoint. This is an identified gap
374 in the supplied proof dependency, not a counterexample to the claimed
375 two-vertex classification. No complete repair has been proved here.
376 Neither the rank/mirror finite-fibre theorem nor any of the exact
377 finite computations above uses that classification. They remain valid.
378 The previously proved upward FOUR lemma and its q=3,h=1 corollary also
379 do not depend on it. The unconditional all-q singleton strengthening
380 from the Sol conversation is not promoted to PROVED in this checkpoint.

```

```

381
382
383 CERTIFICATE-BLOCK: SIX-HOST-EXACT-ARITHMETIC-20260907G
384 STATUS: exact certificates from the independent audit, plain TXT.
385
386 N=128; B_N=[3, 5, 7, 11, 13, 17, 23, 29, 37, 41]
387 Genuine failed upper-root ranks: [2, 3, 4, 5, 7, 8, 9, 11]
388 Complete rows of every vertex in B_N:
389 128-3=125=5^3
390 128-5=123=3*41
391 128-7=121=11^2
392 128-11=117=3^2*13
393 128-13=115=5*23
394 128-17=111=3*37
395 128-23=105=3*5*7
396 128-29=99=3^2*11
397 128-37=91=7*13
398 128-41=87=3*29
399 Rank 0 and rank 10:
400 k=0, r=67, m=61=61, factors outside B_N=[61], success depth=0, W=1
401 k=10, r=109, m=19=19, factors outside B_N=[19], success depth=0, W=1
402
403 N=1718; B_N=[3, 5, 7, 11, 13, 17, 29, 31, 37, 41, 43, 47, 53, 59, 67, 73, 79, 89, 101, 127, 137, 149, 163, 167, 181, 191, 197, 199, 211, 233, 239,
    ↪ 241, 251, 281, 283, 311, 313, 349, 353, 383, 389, 409, 431, 443, 449, 463, 479, 491, 503, 509, 523, 557, 563, 569, 571]
404 Genuine failed upper-root ranks: [2, 3, 4, 8, 11, 13, 17, 21, 22, 23, 25, 29, 31, 34, 36, 37, 39, 41, 43, 46, 47, 48, 49, 50, 51, 52, 54, 55, 56,
    ↪ 57, 59, 60, 65, 69, 70, 73, 75, 77, 79, 81, 82, 83, 85, 86, 87, 89, 90, 92, 93, 94, 95, 96, 97, 99, 101, 102, 103, 104, 106, 107, 109, 110,
    ↪ 112, 113, 114, 115, 116, 118]
405 Complete rows of every vertex in B_N:
406 1718-3=1715=5*7^3
407 1718-5=1713=3*571
408 1718-7=1711=29*59
409 1718-11=1707=3*569
410 1718-13=1705=5*11*31
411 1718-17=1701=3^5*7
412 1718-29=1689=3*563
413 1718-31=1687=7*241
414 1718-37=1681=41^2
415 1718-41=1677=3*13*43
416 1718-43=1675=5^2*67
417 1718-47=1671=3*557
418 1718-53=1665=3^2*5*37
419 1718-59=1659=3*7*79
420 1718-67=1651=13*127
421 1718-73=1645=5*7*47
422 1718-79=1639=11*149
423 1718-89=1629=3^2*181
424 1718-101=1617=3*7^2*11
425 1718-127=1591=37*43
426 1718-137=1581=3*17*31
427 1718-149=1569=3*523
428 1718-163=1555=5*311
429 1718-167=1551=3*11*47
430 1718-181=1537=29*53
431 1718-191=1527=3*509
432 1718-197=1521=3^2*13^2
433 1718-199=1519=7^2*31
434 1718-211=1507=11*137
435 1718-233=1485=3^3*5*11
436 1718-239=1479=3*17*29
437 1718-241=1477=7*211
438 1718-251=1467=3^2*163
439 1718-281=1437=3*479
440 1718-283=1435=5*7*41
441 1718-311=1407=3*7*67
442 1718-313=1405=5*281
443 1718-349=1369=37^2
444 1718-353=1365=3*5*7*13
445 1718-383=1335=3*5*89
446 1718-389=1329=3*443
447 1718-409=1309=7*11*17
448 1718-431=1287=3^2*11*13
449 1718-443=1275=3*5^2*17
450 1718-449=1269=3^3*47
451 1718-463=1255=5*251
452 1718-479=1239=3*7*59
453 1718-491=1227=3*409
454 1718-503=1215=3^5*5
455 1718-509=1209=3*13*31
456 1718-523=1195=5*239
457 1718-557=1161=3^3*43
458 1718-563=1155=3*5*7*11
459 1718-569=1149=3*383
460 1718-571=1147=31*37
461 Rank 0 and rank 10:
462 k=0, r=859, m=859=859, factors outside B_N=[859], success depth=0, W=1
463 k=10, r=937, m=781=11*71, factors outside B_N=[71], success depth=2, W=6
464
465 N=1862; B_N=[3, 5, 11, 13, 17, 29, 41, 43, 47, 53, 59, 67, 83, 97, 101, 107, 113, 127, 131, 151, 167, 179, 181, 191, 211, 233, 251, 257, 263, 269,
    ↪ 307, 311, 347, 353, 359, 383, 401, 421, 431, 443, 487, 499, 503, 509, 547, 557, 569, 577, 587, 593, 599, 601, 607, 617, 619]
466 Genuine failed upper-root ranks: [1, 3, 4, 5, 6, 8, 13, 20, 23, 25, 27, 33, 36, 39, 40, 42, 44, 46, 47, 48, 50, 52, 53, 54, 56, 58, 59, 60, 64, 66,
    ↪ 68, 69, 70, 75, 77, 79, 80, 81, 82, 84, 87, 88, 89, 92, 93, 94, 96, 97, 99, 100, 101, 103, 105, 106, 108, 109, 111, 112, 116, 118, 121, 122,
    ↪ 124]
467 Complete rows of every vertex in B_N:
468 1862-3=1859=11*13^2
469 1862-5=1857=3*619
470 1862-11=1851=3*617
471 1862-13=1849=43^2
472 1862-17=1845=3^2*5*41
473 1862-29=1833=3*13*47
474 1862-41=1821=3*607
475 1862-43=1819=17*107
476 1862-47=1815=3*5*11^2
477 1862-53=1809=3^3*67
478 1862-59=1803=3*601
479 1862-67=1795=5*359

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```

480 1862-83=1779=3*593
481 1862-97=1765=5*353
482 1862-101=1761=3*587
483 1862-107=1755=3^3*5*13
484 1862-113=1749=3*11*53
485 1862-127=1735=5*347
486 1862-131=1731=3*577
487 1862-151=1711=29*59
488 1862-167=1695=3*5*113
489 1862-179=1683=3^2*11*17
490 1862-181=1681=41^2
491 1862-191=1671=3*557
492 1862-211=1651=13*127
493 1862-233=1629=3^2*181
494 1862-251=1611=3^2*179
495 1862-257=1605=3*5*107
496 1862-263=1599=3*13*41
497 1862-269=1593=3^3*59
498 1862-307=1555=5*311
499 1862-311=1551=3*11*47
500 1862-347=1515=3*5*101
501 1862-353=1509=3*503
502 1862-359=1503=3^2*167
503 1862-383=1479=3*17*29
504 1862-401=1461=3*487
505 1862-421=1441=11*131
506 1862-431=1431=3^3*53
507 1862-443=1419=3*11*43
508 1862-487=1375=5^3*11
509 1862-499=1363=29*47
510 1862-503=1359=3^2*151
511 1862-509=1353=3*11*41
512 1862-547=1315=5*263
513 1862-557=1305=3^2*5*29
514 1862-569=1293=3*431
515 1862-577=1285=5*257
516 1862-587=1275=3*5^2*17
517 1862-593=1269=3^3*47
518 1862-599=1263=3*421
519 1862-601=1261=13*97
520 1862-607=1255=5*251
521 1862-617=1245=3*5*83
522 1862-619=1243=11*113
523 Rank 0 and rank 10:
524 k=0, r=937, m=925=5^2*37, factors outside B_N=[37], success depth=2, W=6
525 k=10, r=1009, m=853=853, factors outside B_N=[853], success depth=0, W=1
526
527 N=1928; B_N=[3, 5, 7, 11, 13, 17, 29, 43, 53, 71, 73, 79, 101, 103, 113, 131, 163, 173, 179, 211, 227, 233, 239, 251, 269, 311, 317, 353, 373, 383,
    ↪ 401, 443, 449, 461, 467, 487, 509, 563, 599, 619, 641]
528 Genuine failed upper-root ranks: [1, 2, 3, 8, 11, 15, 16, 18, 22, 23, 29, 33, 35, 36, 38, 45, 46, 47, 50, 52, 55, 56, 60, 61, 64, 65, 67, 73, 75,
    ↪ 76, 78, 80, 82, 85, 91, 92, 93, 95, 98, 99, 102, 104, 107, 108, 110, 111, 113, 117, 118, 120, 124, 125, 126, 127, 128, 129, 130]
529 Complete rows of every vertex in B_N:
530 1928-3=1925=5^2*7*11
531 1928-5=1923=3*641
532 1928-7=1921=17*113
533 1928-11=1917=3^3*71
534 1928-13=1915=5*383
535 1928-17=1911=3*7^2*13
536 1928-29=1899=3^2*211
537 1928-43=1885=5*13*29
538 1928-53=1875=3*5^4
539 1928-71=1857=3*619
540 1928-73=1855=5*7*53
541 1928-79=1849=43^2
542 1928-101=1827=3^2*7*29
543 1928-103=1825=5^2*73
544 1928-113=1815=3*5*11^2
545 1928-131=1797=3*599
546 1928-163=1765=5*353
547 1928-173=1755=3^3*5*13
548 1928-179=1749=3*11*53
549 1928-211=1717=17*101
550 1928-227=1701=3^5*7
551 1928-233=1695=3*5*113
552 1928-239=1689=3*563
553 1928-251=1677=3*13*43
554 1928-269=1659=3*7*79
555 1928-311=1617=3*7^2*11
556 1928-317=1611=3^2*179
557 1928-353=1575=3^2*5^2*7
558 1928-373=1555=5*311
559 1928-383=1545=3*5*103
560 1928-401=1527=3*509
561 1928-443=1485=3^3*5*11
562 1928-449=1479=3*17*29
563 1928-461=1467=3^2*163
564 1928-467=1461=3*487
565 1928-487=1441=11*131
566 1928-509=1419=3*11*43
567 1928-563=1365=3*5*7*13
568 1928-599=1329=3*443
569 1928-619=1309=7*11*17
570 1928-641=1287=3^2*11*13
571 Rank 0 and rank 10:
572 k=0, r=967, m=961=31^2, factors outside B_N=[31], success depth=2, W=4
573 k=10, r=1031, m=897=3*13*23, factors outside B_N=[23], success depth=2, W=9
574
575 N=2200; B_N=[3, 13]
576 Genuine failed upper-root ranks: [48, 79, 129, 141]
577 Complete rows of every vertex in B_N:
578 2200-3=2197=13^3
579 2200-13=2187=3^7
580 Rank 0 and rank 10:
581 k=0, r=1103, m=1097=1097, factors outside B_N=[1097], success depth=0, W=1
582 k=10, r=1187, m=1013=1013, factors outside B_N=[1013], success depth=0, W=1

```

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583
584 N=6142; B_N=[3, 5, 7, 13, 17, 19, 31, 43, 67, 71, 73, 97, 107, 109, 157, 199, 227, 229, 233, 283, 311, 317, 337, 353, 409, 467, 541, 557, 613, 617,
↳ 647, 673, 727, 769, 787, 811, 827, 877, 947, 997, 1039, 1063, 1087, 1117, 1129, 1229, 1237, 1249, 1297, 1303, 1327, 1399, 1483, 1487, 1543,
↳ 1553, 1579, 1613, 1627, 1669, 1693, 1777, 1787, 1823, 1867, 2011]
585 Genuine failed upper-root ranks: [2, 3, 8, 12, 15, 19, 21, 24, 25, 31, 32, 36, 41, 46, 50, 51, 53, 56, 64, 68, 70, 72, 75, 78, 84, 88, 93, 95, 98,
↳ 103, 104, 125, 127, 131, 135, 140, 149, 152, 155, 163, 166, 170, 171, 173, 174, 178, 179, 183, 184, 185, 189, 198, 200, 205, 219, 230, 237,
↳ 238, 244, 246, 248, 250, 252, 253, 262, 264, 265, 271, 273, 274, 280, 283, 284, 287, 293, 305, 308, 311, 313, 314, 316, 320, 322, 324, 325,
↳ 329, 330, 333, 336, 338, 339, 341, 343, 344, 347, 349, 351, 353, 355, 358, 360]
586 Complete rows of every vertex in B_N:
587 6142-3=6139=7*877
588 6142-5=6137=17*19^2
589 6142-7=6135=3*5*409
590 6142-13=6129=3^3*227
591 6142-17=6125=5^3*7^2
592 6142-19=6123=3*13*157
593 6142-31=6111=3^2*7*97
594 6142-43=6099=3*19*107
595 6142-67=6075=3^5*5^2
596 6142-71=6071=13*467
597 6142-73=6069=3*7*17^2
598 6142-97=6045=3*5*13*31
599 6142-107=6035=5*17*71
600 6142-109=6033=3*2011
601 6142-157=5985=3^2*5*7*19
602 6142-199=5943=3*7*283
603 6142-227=5915=5*7*13^2
604 6142-229=5913=3^4*73
605 6142-233=5909=19*311
606 6142-283=5859=3^3*7*31
607 6142-311=5831=7^3*17
608 6142-317=5825=5^2*233
609 6142-337=5805=3^3*5*43
610 6142-353=5789=7*827
611 6142-409=5733=3^2*7^2*13
612 6142-467=5675=5^2*227
613 6142-541=5601=3*1867
614 6142-557=5585=5*1117
615 6142-613=5529=3*19*97
616 6142-617=5525=5^2*13*17
617 6142-647=5495=5*7*157
618 6142-673=5469=3*1823
619 6142-727=5415=3*5*19^2
620 6142-769=5373=3^3*199
621 6142-787=5355=3^2*5*7*17
622 6142-811=5331=3*1777
623 6142-827=5315=5*1063
624 6142-877=5265=3^4*5*13
625 6142-947=5195=5*1039
626 6142-997=5145=3*5*7^3
627 6142-1039=5103=3^3*6*7
628 6142-1063=5079=3*1693
629 6142-1087=5055=3*5*337
630 6142-1117=5025=3*5^2*67
631 6142-1129=5013=3^3*2*557
632 6142-1229=4913=17^3
633 6142-1237=4905=3^2*5*109
634 6142-1249=4893=3*7*233
635 6142-1297=4845=3*5*17*19
636 6142-1303=4839=3*1613
637 6142-1327=4815=3^2*5*107
638 6142-1399=4743=3^2*17*31
639 6142-1483=4659=3*1553
640 6142-1487=4655=5*7^2*19
641 6142-1543=4599=3^2*7*73
642 6142-1553=4589=13*353
643 6142-1579=4563=3^3*13^2
644 6142-1613=4529=7*647
645 6142-1627=4515=3*5*7*43
646 6142-1669=4473=3^2*7*71
647 6142-1693=4449=3*1483
648 6142-1777=4365=3^2*5*97
649 6142-1787=4355=5*13*67
650 6142-1823=4319=7*617
651 6142-1867=4275=3^2*5^2*19
652 6142-2011=4131=3^5*17
653 Rank 0 and rank 10:
654 k=0, r=3079, m=3063=3*1021, factors outside B_N=[1021], success depth=2, W=6
655 k=10, r=3181, m=2961=3^2*7*47, factors outside B_N=[47], success depth=2, W=9
656
657 Opposite verdicts at exactly the same rank and mirror:
658 N=116, r=89=p_7, m=27, success=True, depth=1, W=2
659 depth 0: 116-89=27=3^3
660 depth 1: 116-3=113=113
661 N=128, r=101=p_7, m=27, success=False, depth=None, W=9
662 depth 0: 128-101=27=3^3
663 depth 1: 128-3=125=5^3
664 depth 2: 128-5=123=3*41
665 depth 3: 128-41=87=3*29
666 depth 4: 128-29=99=3^2*11
667 depth 5: 128-11=117=3^2*13
668 depth 6: 128-13=115=5*23
669 depth 7: 128-23=105=3*5*7
670 depth 8: 128-7=121=11^2
671 N=1862, r=1063=p_20, m=799, success=False, depth=None, W=22
672 depth 0: 1862-1063=799=17*47
673 depth 1: 1862-17=1845=3^2*5*41
674 depth 1: 1862-47=1815=3*5*11^2
675 depth 2: 1862-3=1859=11*13^2
676 depth 2: 1862-5=1857=3*619
677 depth 2: 1862-41=1821=3*607
678 depth 2: 1862-11=1851=3*617
679 depth 3: 1862-13=1849=43^2
680 depth 3: 1862-619=1243=11*113
681 depth 3: 1862-607=1255=5*251
682 depth 3: 1862-617=1245=3*5*83

```

```

683 depth 4: 1862-43=1819=17*107
684 depth 4: 1862-113=1749=3*11*53
685 depth 4: 1862-251=1611=3*2*179
686 depth 4: 1862-83=1779=3*593
687 depth 5: 1862-107=1755=3*3*5*13
688 depth 5: 1862-53=1809=3*3*67
689 depth 5: 1862-179=1683=3*2*11*17
690 depth 5: 1862-593=1269=3*3*47
691 depth 6: 1862-67=1795=5*359
692 depth 7: 1862-359=1503=3*2*167
693 depth 8: 1862-167=1695=3*5*113
694 N=1886, r=1087=p_20, m=799, success=True, depth=2, W=5
695 depth 0: 1886-1087=799=17*47
696 depth 1: 1886-17=1869=3*7*89
697 depth 1: 1886-47=1839=3*613
698 depth 2: 1886-3=1883=7*269
699 depth 2: 1886-7=1879=1879
700 N=2200, r=1471=p_48, m=729, success=False, depth=None, W=3
701 depth 0: 2200-1471=729=3^6
702 depth 1: 2200-3=2197=13^3
703 depth 2: 2200-13=2187=3^7
704 N=2180, r=1451=p_48, m=729, success=True, depth=4, W=10
705 depth 0: 2180-1451=729=3^6
706 depth 1: 2180-3=2177=7*311
707 depth 2: 2180-7=2173=41*53
708 depth 2: 2180-311=1869=3*7*89
709 depth 3: 2180-41=2139=3*23*31
710 depth 3: 2180-53=2127=3*709
711 depth 3: 2180-89=2091=3*17*41
712 depth 4: 2180-23=2157=3*719
713 depth 4: 2180-31=2149=7*307
714 depth 4: 2180-709=1471=1471
715 N=2210, r=1481=p_48, m=729, success=True, depth=1, W=2
716 depth 0: 2210-1481=729=3^6
717 depth 1: 2210-3=2207=2207
718
719 Fixed dangerous-mirror set (all 214 exact integers):
720 [9, 15, 21, 25, 27, 39, 45, 49, 51, 55, 63, 75, 81, 85, 91, 95, 99, 105, 111, 117, 121, 129, 135, 141, 145, 147, 153, 155, 159, 165, 169, 175, 187,
↳ 189, 195, 205, 215, 219, 225, 231, 235, 237, 243, 245, 249, 255, 259, 261, 265, 267, 273, 279, 285, 289, 291, 295, 301, 303, 309, 315, 319,
↳ 321, 335, 339, 351, 357, 363, 369, 375, 381, 385, 399, 405, 411, 415, 423, 425, 429, 435, 441, 453, 459, 469, 477, 481, 485, 489, 495, 501,
↳ 505, 507, 517, 519, 525, 531, 535, 543, 555, 559, 561, 565, 567, 573, 583, 585, 595, 603, 609, 615, 625, 633, 637, 639, 645, 649, 657, 663,
↳ 665, 675, 685, 697, 699, 705, 711, 715, 729, 735, 749, 753, 765, 771, 777, 799, 819, 825, 833, 835, 837, 841, 845, 865, 867, 871, 885, 891,
↳ 895, 909, 921, 933, 945, 951, 957, 963, 975, 995, 1029, 1065, 1083, 1139, 1209, 1349, 1391, 1413, 1491, 1505, 1521, 1539, 1575, 1581, 1623,
↳ 1625, 1635, 1649, 1685, 1701, 1785, 1805, 1853, 1911, 1941, 1989, 2015, 2043, 2219, 2223, 2261, 2289, 2295, 2345, 2375, 2433, 2451, 2471, 2499,
↳ 2511, 2535, 2601, 2613, 2625, 2631, 2675, 2709, 2769, 2795, 2799, 2835, 2841, 2883, 2889, 2925, 2951, 2975, 3033, 3053]
721
722 END EXACT CERTIFICATE BLOCK.
723
724
725 RESULT-ID: LAST-RANK-MIRROR-REALIZATION-AT-THE-UPPER-CORNER-20260907G
726 STATUS: PROVED; a refinement of the effective finite-fibre result.
727 Fix odd  $M \geq 3$  and  $K \geq 0$ . Among ALL odd  $m$  with  $1 \leq m \leq M$ , odd prime
728 roots  $P \geq m$ , and ranks  $k \leq K$  at  $N = P + m$ , the largest possible  $N$  is
729 attained with  $m = M$  and  $k = K$ .
730 More precisely, define  $L_K(M)$  as the largest prime  $P \geq M$  for which
731  $\kappa_M(P) \leq K$ . Then  $\kappa_M(L_K(M)) = K$ , and every such  $N$  satisfies
732  $N \leq M + L_K(M)$ , with equality at that upper-corner realization.
733 PROOF:
734 Surjectivity makes the set defining  $L_K(M)$  nonempty; the effective
735 finite-fibre bound makes it finite. At the next prime after  $L_K(M)$ ,
736  $\kappa_M$  is greater than  $K$ . Its upward jump is at most one, so its
737 value at  $L_K(M)$  must be exactly  $K$ .
738 For any admitted  $P \leq M$ ,  $N = P + m \leq 2M + L_K(M)$ .
739 For  $P > M$  and  $m \leq M$ , moving the left interval endpoint to the right gives
740  $\kappa_M(P) \geq \kappa_M(P)$ .
741 Thus  $\kappa_M(P) \leq K$  implies  $\kappa_M(P) \leq K$  and  $P \leq L_K(M)$ .
742 Consequently  $N = P + m \leq L_K(M) + M$ . Equality is supplied by  $m = M, k = K$ .
743 Allowing a boundary root  $P = N/2$  adds only  $P = m$  prime,  $N = 2m \leq 2M$ ,
744 so it cannot change this maximum.
745 This concerns maximum  $N$ , not minimum  $N$ , and does not assert that
746  $\kappa_M(P)$  is monotone in  $P$ . Downward jumps are explicitly allowed.
747
748 EXACT SPECIALIZATION:
749  $M = 3053$ ,  $K = 74$ :  $L_K(M) = 4271$  and the exact maximum  $N$  is 7324.
750 The endpoint has  $m3053 = 43 \cdot 71$  and  $\text{root}4271 = p74$  at  $N7324$ .
751 The next prime is 4273, giving  $\kappa_{3053}(4273) = 75$ .
752 An independent sieve checked ALL 2681 primes  $P$  with  $4271 < P < 30204$ :
753 the minimum  $\kappa_{3053}(P)$  is 75. The largest prime checked is 30203.
754 For  $P \geq 30204$  the preceding analytic bound proves  $\kappa_{3053}(P) \geq 75$ .
755 Hence the endpoint is globally certified, not inferred from observing
756 the next prime alone. Script/result: rank_last_fibre_20260907G.py/JSON.
757
758 CONSEQUENCE FOR THE RESEARCH DESIGN:
759 All the 214-mirror transplants at ranks  $\leq 74$  necessarily occur by  $N7324$ .
760 The same is true even when every odd mirror  $\leq 3053$  and every rank  $0 \leq 74$ 
761 is allowed. Thus choosing a much larger  $N$  cutoff cannot independently
762 stress these fixed mirrors. New tests of universal rank safety need new
763 mirror integers or genuinely new closed BAD structures.
764 This improves experimental scope control; it is not a proof that the
765 first-prime boundary always selects a successful support.
766
767
768 CURRENT INTERPRETATION / HANDOFF -- 2026-09-07G
769 1. The main rank/mirror proof and the 07F transplant counts survived
770 independent audit. The claimed universal twin-core classification
771 was not accepted as a dependency because of the parity gap above.
772 2. The strongest new result is effective finiteness of every (rank,m)
773 realization set, plus its exact upper-corner maximum. This turns the
774 supplied transplant test into a complete global classification of
775 those finite fibres. The larger small-mirror census has 22130 total
776 composite cases over the ten selected ranks, all successful.
777 3. There is still no theorem saying  $p_{10}$  or any other surviving
778 noncanonical rank always succeeds for composite complements.
779 The finite survivor list was selected from the same six known
780 failure hosts; repeating tests inside that landscape is not new

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781 independent evidence for a rank-specific arithmetic law.
782 4. p_0 is special as the least upper prime, equivalently the selector
783 of the greatest available mirror, and as the only fixed rank with
784 no unit row. Post-root transitions are the common N -dependent
785 graph. None of this alone proves that p_0 always escapes its BAD set.
786 5. It remains possible that many ranks are universally safe on the
787 composite domain. Finite EAC hosts would imply that every sufficiently
788 large rank is safe there. The converse is not asserted: the existence
789 of a closed low BAD component does not, without more work, guarantee
790 a failing upper root at a specified rank or any upper rank.
791 6. Most useful next target: a necessary arithmetic condition on complete
792 closed BAD sets that interacts with the maximal prime-backed mirror;
793 alternatively, construct a genuinely new complete failure host and
794 inspect its whole rank spectrum. Prescribing selected CRT edges or
795 varying N with one of the exhausted exact mirrors cannot replace this.
796 7. No Goldbach proof, no universal p_0 theorem, no universal safe-rank
797 classification, and no external novelty claim is made in this run.
798
799
800 CARRYOVER-D-SPARSE-RESULTS-PRESERVED-20260907G
801 Historical results from the already completed single-agent run D, preserved here after the new rank work. No agent was used in run G. The later
 $\hookrightarrow$ general $v \geq 4$ correction remains in force.
802
803 RESULT 2: $h=1$ MONOMIAL RETURN NEEDS $v \geq 11$ AND $u \geq 15$.
804 Assume q, b distinct odd primes, $u, v \geq 2$,
805 $r = q^u + 2$ prime, $N = 2q^u + 2$,
806 $N - q = b^v$, and q divides $N - b$.
807 Then $q \geq 5$, $q \equiv 2 \pmod{3}$, u is odd, and
808 q divides $2^{v-1} - 1$,
809 either $v = 11$, $q = 11$, $u = 5 \bmod 22$ and $u \geq 27$,
810 or $v \geq 13$ and $u \geq v + 1$ (hence $u \geq 15$).
811 The earlier exact size barrier $(3q+2)^v < 2q^u + 2 - q$ also remains.
812
813 Proof. Result1 excludes $q = 3$. For $q \neq 3$, even u would make r
814 divisible by 3; odd u and $q \equiv 1 \pmod{3}$ would do likewise. Thus
815 u is odd and $q \equiv 2 \pmod{3}$. The return condition gives $b \equiv 2 \pmod{q}$.
816 Since $b^v = N - q \equiv 2 \pmod{q}$, q divides $2^{v-1} - 1$.
817 The earlier sharp return/exponent lemma gives $3 \leq v \leq u - 1$.
818 Here is an exact elimination of $v = 3, \dots, 10$ and $v = 12$,
819 with a residue restriction on the sole remaining $v = 11$ branch:
820 v factorization of $2^{v-1} - 1$ surviving q after $q \geq 5, q \equiv 2 \pmod{3}$
821 3 3 none
822 4 7 none
823 5 3*5 5
824 6 31 none
825 7 3^2*7 none
826 8 127 none
827 9 3*5*17 5,17
828 10 7*73 none
829 11 3*11*31 11
830 12 23*89 23,89
831
832 For $v = 5, q = 5$: $b \equiv 2 \pmod{5}$ implies $b^5 \equiv 2^5 \equiv 7 \pmod{25}$, while
833 $2*5^u - 3 \equiv 22 \pmod{25}$ ($u \geq 2$), contradiction.
834 For $v = 9, q = 5$: as u is odd, $5^u \bmod 7$ is 5, 6, 3 for $u = 1, 3, 5 \bmod 6$.
835 The row $2*5^u - 3$ is respectively 0, 2, 3 mod 7. Ninth powers are
836 cubes and are only 0, 1, 6 mod 7, so only $u \equiv 1 \pmod{6}$ remains. But that
837 makes $r = 5^u + 2$ divisible by 7, with $r > 7$, contradiction.
838 For $v = 9, q = 17$: as u is odd, $17^u \equiv 3^u \pmod{7}$ lies in $\{3, 6, 5\}$.
839 Thus $2*17^u - 15$ lies in $\{5, 4, 2\} \bmod 7$, none a cube. Contradiction.
840 For $v = 12, q = 89$: $q \equiv 1 \pmod{8}$ and u odd, so $N - q = q^u + 2 \equiv 3 \pmod{8}$,
841 which is not a square. For $v = 12, q = 23$: $q \equiv 7 \pmod{16}$ and u odd,
842 so $N - q = 2*7^u - 7 \equiv 9 \pmod{16}$. An odd twelfth power is a fourth power
843 and is $\equiv 1 \pmod{16}$. Contradiction.
844
845 At $v = 11, q = 11$, testing modulo 23 gives a necessary $u \equiv 5 \pmod{22}$.
846 Consequently $u \geq 27$, but $v = 11$ remains a possible exponent.
847 No exclusion of $v = 11$ was found. In particular $v \geq 13$ alone is
848 NOT a proved conclusion. Globally $u \geq 15$: $u < 13$ would force
849 $v < 12$, leaving only $v = 11$, already requiring $u \geq 27$.
850
851 The modulus 23 calculation is finite and exact: odd $u = 1, 3, \dots, 21$
852 give $11^u \bmod 23$ respectively
853 11, 20, 5, 7, 19, 22, 17, 10, 14, 15, 21.
854 Subtracting 9 after doubling, the row can be 0, 1, -1 mod 23 only at
855 $u = 5$, where it equals 1. Fermat gives these as the only eleventh-
856 power residues mod 23 (for nonzero b); $b = 23$ would give 0 and is
857 also absent in the table. Since 11 has order 22 mod 23 this proves
858 the congruence $u \equiv 5 \pmod{22}$. (The displayed powers are verified by
859 the small computation saved alongside these notes.)
860
861 RESULT 3: ODD- v $h=1$ RETURN HAS THE STRONGER BARRIER $b \geq 25q + 2$,
862 AND NECESSARILY $u \geq v + 6$.
863 Under Result2 hypotheses and v odd, u is odd and $q \geq 5$. Modulo 8,
864 odd powers of odd numbers preserve their residue, hence
865 $b \equiv 2q^u + 2 \equiv q^u \pmod{8}$.
866 Modulo 3, the row b^v is 1 and v is odd, so $b \equiv q^u \pmod{3}$.
867 Therefore $b \equiv q^u \pmod{24}$. The return gives $b = 2 + kq$ with integer k ,
868 and $\gcd(q, 24) = 1$, so $k \equiv 1 \pmod{24}$. The earlier sharp return theorem
869 already forces $k \geq 3$ ($k = 1$ would make the root divisible by b).
870 Consequently $k \geq 25$ and $b \geq 25q + 2$.
871
872 For the exponent consumer, $u > v$ and both are odd. Suppose $u - v \leq 4$.
873 The complete row equation and $q \geq 5$ give
874 $25^v * q^v < b^v = 2q^u + 2 < 3q^u$,
875 so $25^v < 3q^{u-v} \leq 3q^4$.
876 But q divides $2^{v-1} - 1$, hence $q < 2^{v-1}$, yielding
877 $25^v < 3q^4 < 3*16^{v-1} < 16^v < 25^v$,
878 a contradiction. Thus $u - v \geq 5$, and the even parity of $u - v$
879 improves this to $u \geq v + 6$. No sharpness example is currently known.
880 This special barrier uses $h = 1$ and a full monomial row with odd v ;
881 it is not a replacement for the general $b \geq 3q + 2h$ theorem.
882
883 RESULT 4: EXACT 2-ADIC FILTER FOR EVERY EVEN MONOMIAL ROW AT $h = 1$.
884 Assume $r = q^u + 2$ prime, $u \geq 2$, $N = 2q^u + 2$, and $N - q = b^v$ for odd

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885 primes q,b with even v. No return assumption is needed here.
886 Result1 excludes q=3. Hence u is odd and q=2mod3. Put s=nu2(v).
887 Then exactly
888   nu2(q+1)=nu2(b^2-1)+s-1.
889 In particular q=-1 mod (3*2^(s+2)).
890
891 Proof. Since u is odd, q^u+1=(q+1)*A with A odd (an alternating
892 sum of u odd terms). Therefore
893   b^v-1=2q^u-q+1=2(q^u+1)-(q+1)=(q+1)*(2A-1),
894 and the last factor is odd. This gives the left valuation.
895 Write v=2^s*t with t odd. The odd geometric factor removes t
896 without changing nu2. Next factor
897   b^(2^s)-1=(b^2-1)*product_{j=1,...,s-1}(b^(2^j)+1).
898 Each factor in the product is 2mod8, contributing exactly one
899 power of 2. Thus the right valuation is nu2(b^2-1)+s-1.
900 An odd square is 1mod8, so nu2(b^2-1)>=3. Combine this with
901 q=-1mod3 to obtain the stated congruence. The previous square-
902 row filter q=23mod24 is exactly s=1. At v=12,s=2 it demands
903 q=47mod48, excluding both 23 and 89 in Result2's return table.
904
905 FINITE TARGETED SEARCH (NOT A FULL N OR BFS CENSUS):
906 For every odd prime q<=1000000 and every u>=2 with
907   2q^u+2<=10^30,
908 retaining q=3 or (q=2mod3 and u odd), inspect N-q=2q^u-q+2
909 for EVERY proper integer-power representation x^v, x>=3,v>=2.
910 75,205 retained (q,u) pairs; ZERO proper-power rows occurred.
911 This is stronger within its domain than a prime-base check; no
912 root primality test or probable-prime decision enters the zero count.
913 Exact integer powers and corrective integer-root loops decide each
914 equality. Floating roots only seed the exact loops. All candidate
915 exponents are covered by the exact bound 3^v<=N-q.
916 Source: agent_d_sparse.cjs; result: agent_d_sparse_result.json.
917 No conclusion is made for q>10^6 or N>10^30, even at h=1.
918 In particular this finite absence is not a theorem excluding (B).
919
920 SECOND TARGETED SEARCH: ALL LOW GAPS WITH PRIME-POWER FIRST ROWS.
921 For every odd prime q<=10^6, u>=2 and all 1<=h<q such that
922   N=2q^u+2h<=10^18, directly generate all integer powers b^v in
923 the interval [2q^u-q+2, 2q^u+q-2], additionally respecting N<=10^18.
924 Keep b prime, and then r=q^u+2h prime. For each retained input,
925 remove all factors q from N-b, then test whether the remaining
926 cofactor is a prime power c^w (or prime c when a>0).
927 Counts: 145750 (q,u) pairs; 4338 full prime-power N-q rows before
928 root primality; 588 retained prime roots; ZERO returns q|N-b;
929 ZERO systems (B). The largest retained N is 1995448234470,
930 at r=997724937149,q=998861,u=2,b=1412603,v=2,h=819914.
931 Most rows have u=v=2; nine retained rows fall outside that pair.
932 Primality throughout the final run is exact trial division by a
933 sieve of all primes<=10^6, with an error if insufficient for any
934 accepted candidate. The run completed without that error. Initial
935 floating integer-root estimates are corrected by exact integer
936 powers before every equality/range test. No probable-prime test
937 contributes to these final counts. This targets only singleton
938 initial support and sparse first rows, not a full N or BFS census.
939 Source/result: agent_d_lowgap_sparse.cjs,
940 agent_d_lowgap_sparse_result.json. No three-label counterexample
941 was found, and general low-gap FOUR remains unproved.
942
943
944
945 LATEST STATE -- 2026-09-07G (after historical D carryover)
946 The active direction is now the audited rank/mirror programme, not an
947 instruction to resume the old singleton branch. Run G used no subagents.
948 New proved result: for fixed odd m and rank k, every realization P>m
949 satisfies P<R_k(k+ceil((m-1)/4)+2). Combined with Sondow's explicit bound,
950 this makes the exact realization set effectively finite.
951 The last-realization theorem places the maximum N for m<=M,k<=K at
952 m=M,k=K. For M3053,K74 it is exactly N7324,P4271.
953 Therefore all 4035 dangerous-mirror transplants in the supplied 07F are
954 globally exhaustive; none fails. A wider complete all-N census for every
955 odd composite m<=3053 gives 22130 cases at the ten selected ranks,
956 again no failures. None of these claims covers larger mirror integers.
957 The seven-page Scott source and its publisher abstract were located;
958 the unqualified twin-EAC uniqueness dependency still has the identified
959 odd/even exponent gap. No all-q singleton FOUR or canonical two-core
960 exclusion was promoted from that dependency. The rank results do not
961 use it. Global p0 success and universal composite safety of other ranks
962 remain open. Main proofs, numerical counts and complete six-host
963 arithmetic certificates are included above in this same TXT.
964 END CURRENT CHECKPOINT -- 2026-09-07G
965
966
967 RESEARCH RUN 2026-09-07H -- SAME-N PRIME PREFIXES AND FACTOR SUPPORT
968 User scope: continue the mathematical investigation; zero subagents;
969 write results as plain TXT only. No PDF or TeX output is being produced.
970 The earlier source documents remain mathematical data, not instructions.
971
972 RESULT-ID: SAME-N-PREFIX-VALUATION-REALIZATION-20260907H
973 STATUS: PROVED, using an external theorem on consecutive congruent primes.
974 This is a new application within this checkpoint, not a claim that the
975 underlying theorem on primes is new.
976
977 Statement. Fix K>=0, a finite set T of odd primes, and arbitrary prescribed
978 nonnegative integers e_l for l in T. There are infinitely many even N
979 such that, writing p_j for the j-th prime at or above N/2, ZERO-based,
980   p_0=N/2+1,
981   v_l(N-p_j)=e_l for every l in T and every 0<=j<=K.
982 All K+1 complements can additionally be required to be composite.
983 Moreover, along an infinite subfamily there are FIXED integers
984   0=d_0<d_1<...<d_K
985 such that p_j=p_0+d_j and h_j=p_j-N/2=1+d_j throughout that subfamily.
986 The d_j are supplied existentially; this does not allow arbitrary chosen
987 prime-gap patterns such as a prescribed twin-prime constellation.
988
989 External input, checked in the primary paper:
990 Tristan Freiberg, A note on the theorem of Maynard and Tao,

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991 arXiv:1311.5319, Theorem 1.2, <https://arxiv.org/pdf/1311.5319>
 992 For a reduced residue a modulo q , with $q \geq 4$, and any fixed length $L \geq 3$,
 993 infinitely many strings of L consecutive primes are all a modulo q
 994 and have diameter bounded by a constant depending only on q, a, L .
 995 Consecutive means in the sequence of ALL primes. The bounded diameter
 996 strengthens Shiu's earlier result. This particular paper has ONE author,
 997 Tristan Freiberg; it is not the separate Banks--Freiberg--Turnage--Butterbaugh
 998 paper. We use only the stated existence and bounded-diameter conclusion.
 999
 1000 Proof of the application.
 1001 For l in T impose on a the congruence
 1002 $a \equiv 2 + l \cdot e_l \pmod{l^{e_l+1}}$, if $e_l \geq 1$;
 1003 $a \equiv 1 \pmod{l}$, if $e_l = 0$.
 1004 These residues are units modulo l and force $v_l(a-2) = e_l$ in the
 1005 congruence sense. Add $a \equiv 1 \pmod{4}$. CRT gives a reduced class a modulo
 1006 $M = 4 \cdot \text{product}_l(1 \text{ in } T) \cdot l^{e_l+1}$.
 1007 If no positive exponent was prescribed, add one new odd prime outside T
 1008 with prescribed exponent 1. This guarantees eventual compositeness
 1009 without changing any of the requested valuations.
 1010 Apply Freiberg with $L = \max(3, K+1)$, and retain the first $K+1$ primes
 1011 $p_0 < \dots < p_K$ in each string. Set $N = 2(P_0 - 1)$. Since $P_0 - 1$ is an even
 1012 integer greater than 2, P_0 is exactly $p_0(N)$, and the consecutive
 1013 p_j are exactly $p_j(N)$. If B bounds the string diameter, then
 1014 $m_j = N - p_j = P_0 - 2 - (p_j - P_0) \geq P_0 - 2 - B$,
 1015 so every m_j is positive and tends to infinity as P_0 tends to infinity.
 1016 Since $p_j = P_0 = a$ modulo M , we have $m_j \equiv a - 2 \pmod{M}$. This proves all
 1017 specified valuations simultaneously at the SAME N . An imposed positive
 1018 exponent supplies a proper prime factor once m_j exceeds that prime,
 1019 which proves the compositeness assertion. Finally there are only
 1020 finitely many increasing integer offset vectors with diameter $\leq B$.
 1021 One vector occurs infinitely often, proving the fixed-offset assertion.
 1022
 1023 A slightly more general form, used in the proof, is useful to retain:
 1024 for every even $M \geq 4$, every reduced class a modulo M and fixed K ,
 1025 infinitely many such N have
 1026 $N \equiv 2a - 2 \pmod{M}$,
 1027 $N - p_j \equiv a - 2 \pmod{M}$,
 1028 $h_j \equiv 1 \pmod{M}$,
 1029 for all $j \leq K$, and have fixed offsets on an infinite subfamily.
 1030
 1031 Exact scope. These are prescribed valuations at a FIXED FINITE list of
 1032 prime labels, not full factorizations. The uncontrolled cofactors are
 1033 positive and unbounded. Neither full factor closure, a common FIFO
 1034 transcript, identical GOOD/BAD verdicts below the root, nor eventual
 1035 success of p_0 or any other rank follows. The construction shows that
 1036 finite valuation data alone cannot distinguish p_0 from every p_j at
 1037 the same N , even with all these initial complements composite. It does
 1038 not say such matching is typical, or exclude criteria using h_j as an
 1039 integer, growing prime-label sets, or full N -dependent factorization.
 1040
 1041 SAVE POINT H1: theorem and proof recorded before further computation.
 1042
 1043 RESULT-ID: EARLY-RANK-FIXED-SUPPORT-SEPARATION-20260907H
 1044 STATUS: PROVED. Quantitative restriction on COMPLETE factorizations,
 1045 without any assumption about BAD closure or the BFS verdict.
 1046
 1047 Let S be a fixed nonempty finite set of odd primes, $s = |S|$, and put
 1048 $A(S) = 1.4 \cdot 30^{-(s+3)} \cdot s^{-(9/2)} \cdot \text{product}_{q \in S} \log q$,
 1049 $c(S) = (\log(3)/\exp(1))^{A(S)/6}$.
 1050 All logarithms are natural. Let N be sufficiently large and even. If
 1051 $N/2 < p_i < p_j \leq 2N/3$
 1052 and both $m_i = N - p_i$ and $m_j = N - p_j$ have ALL prime factors in S , then
 1053 $j - i > c(S) \cdot N^{(19/40)/(\log N)^{A(S)}}$. (H2-rank)
 1054 The indices are the usual ZERO-based ranks among primes $\geq N/2$.
 1055 The constant is not intended to be numerically useful.
 1056
 1057 In particular, for every fixed ϵ with $0 < \epsilon < 19/40$,
 1058 for all sufficiently large even N there is AT MOST ONE S -smooth
 1059 complement among
 1060 $N - p_0, \dots, N - p_{\text{floor}(N^{(19/40-\epsilon)})}$. (H2-prefix)
 1061 This includes every fixed prefix $p_0, \dots, p_K$. It is a pointwise eventual
 1062 statement, not just a density estimate. The excluded S -empty case is
 1063 trivial: its only smooth positive integer is 1, while these complements
 1064 are eventually $\geq N/3$.
 1065
 1066 Proof of complete-support separation.
 1067 The verified rational real-field Matveev estimate recorded earlier in
 1068 MATVEEV-IN-BMS-9.4 applies to
 1069 $\Lambda = m_i/m_j - 1 = (p_j - p_i)/m_j > 0$.
 1070 For q in S write $b_q = v_q(m_i) - v_q(m_j)$. Since $m_i, m_j < N$ and $q \geq 3$,
 1071 $|b_q| \leq \log N / \log 3$.
 1072 At least one b_q is nonzero. Omit zero coefficients and then enlarge
 1073 back to S ; $\log q > 1$ for every odd prime makes this enlargement harmless.
 1074 The already verified estimate gives
 1075 $\log \Lambda > -A(S) \cdot (1 + \log(\log N / \log 3))$,
 1076 and, since $m_j \geq N/3$,
 1077 $p_j - p_i > (N/3) \cdot (\log(3)/\exp(1))^{A(S)/(\log N)^{A(S)}}$. (H2-distance)
 1078 No coprimality between the two complements is needed.
 1079
 1080 External prime-gap input, checked in the original paper:
 1081 R. C. Baker, G. Harman, J. Pintz, The difference between consecutive
 1082 primes, II, Proc. London Math. Soc. 83 (2001), 532-562, Theorem 1.
 1083 <https://www.cs.umd.edu/~gasarch/BLOGPAPERS/BakerHarmanPintz.pdf>
 1084 For all sufficiently large real x , $[x - x^{0.525}, x]$ contains a prime.
 1085 We need only this established exponent, not an optimal current exponent.
 1086
 1087 If $u < v$ are adjacent primes in $[N/2, 2N/3]$, apply that theorem at
 1088 $x = v - 1/2$. Its prime is at most u , so
 1089 $v - u \leq 1/2 + (v - 1/2)^{0.525} \leq 2 \cdot N^{0.525}$.
 1090 Summing the $j - i$ adjacent gaps yields
 1091 $p_j - p_i \leq 2 \cdot (j - i) \cdot N^{0.525}$.
 1092 Combine with H2-distance and use $1 - 0.525 = 19/40$ to prove H2-rank.
 1093
 1094 For completeness, the growing prefix in H2-prefix lies in the required
 1095 interval for sufficiently large N . For $N/2 < x \leq 2N/3$, the same external
 1096 theorem applied at $x + 2 \cdot N^{0.525}$ supplies a prime in $(x, x + 2 \cdot N^{0.525}]$,

1097 provided this right endpoint is $\leq N$ and N is sufficiently large.
 1098 Iteration, starting at $N/2$, therefore gives
 1099 $p_j \leq N/2 + 2 \cdot (j+1) \cdot N^{0.525}$
 1100 until the bound reaches $2N/3$. For $j \leq N^{(19/40 - \epsilon)}$ it stays below
 1101 $2N/3$ for sufficiently large N , because the additional term is $o(N)$.
 1102 If two complements in this prefix were S-smooth, H2-rank would imply
 1103 $N^{(19/40 - \epsilon)} > c(S) \cdot N^{(19/40)} / (\log N)^{A(S)}$,
 1104 which is false for all sufficiently large N . This proves the result.
 1105 The harmless possibility $p_0 = N/2$ when $N/2$ is prime is covered too.
 1106
 1107 Effectivity and attribution. The analytic threshold exists; no usable
 1108 numerical value of it was computed in this run. Matveev's input is the
 1109 same classical smooth-number separation mechanism already used for
 1110 closed BAD labels in this checkpoint. The new consumer is a quantitative
 1111 separation of RANKS of two smooth first complements, even without any
 1112 closed graph, and the resulting growing prime prefix. It must not be
 1113 presented as a new theorem on general gaps between smooth numbers.
 1114
 1115 RESULT-ID: FINITE-VIEW-MATCHING-WITH-COFACTOR-ESCAPE-20260907H
 1116 STATUS: PROVED by combining H1 and H2.
 1117 Fix a finite valuation prescription T, e_1 and any $K \geq 1$. H1 realizes it
 1118 at all first $K+1$ roots simultaneously for infinitely many common N .
 1119 For ANY additional fixed finite alphabet S of odd primes, H2 says that,
 1120 after finitely many exceptions, at least K of those $K+1$ complements
 1121 contain a prime outside S . The exponents of the additional primes in
 1122 S need not be bounded. Thus agreeing on every inspected valuation is
 1123 compatible with inevitable escape from every fixed full-factor alphabet.
 1124 For $S=T$ the escape would follow trivially from fixed exponents and
 1125 growing complements; allowing an arbitrary larger S with unrestricted
 1126 exponents is the substantive extra statement.
 1127
 1128 Consequence for the research programme, with quantifiers kept explicit.
 1129 The mechanism applies to p_0 and other early ranks alike. It neither
 1130 identifies the exceptional possible S-smooth member nor forces p_0 to
 1131 be outside S . A failing basin B_N can change and grow with N , so taking
 1132 $S=B_N$ separately at each N is NOT a valid use of this fixed- S eventual
 1133 theorem. Earlier fixed-alphabet closed-BAD finiteness already rules out
 1134 any permanent bounded alphabet for such basins; H2 does not bypass that
 1135 obstruction or prove success for a class of ranks.
 1136
 1137 SAVE POINT H2: separation proof and scope recorded before experiments.
 1138
 1139 COROLLARY: A FIXED PRIME-LABEL PROJECTION ALSO FAILS TO IDENTIFY p_0 .
 1140 Fix a finite list T of prime labels and K . For the common N constructed
 1141 in H1, project each rooted graph onto its root plus the active labels
 1142 in T : keep precisely the edges whose child label is in T , and retain
 1143 their exact prime exponents. Keep GOOD/BAD flags for the labels in T
 1144 as well, if desired. For sufficiently large N all roots are outside T
 1145 and all their complements are composite. Under relabelling of the root,
 1146 these $K+1$ projected graphs are identical. The root-to- T multiplicities
 1147 are identical by H1; every other retained row and flag depends only on
 1148 the same N and the same label q in T , not on the chosen root.
 1149 This is a statement about a FIXED-LABEL projection, not a fixed-depth
 1150 ball of the full graph: even the first full BFS layer can contain new
 1151 prime labels tending to infinity with N . Thus it does not transfer a
 1152 full BFS success or failure from one root to another.
 1153
 1154 ELEMENTARY CONSUMER: LOW-GAP ROOTS HAVE DISJOINT EARLIER FIRST SUPPORTS.
 1155 Status: proved elementary consequence of the already archived identity
 1156 that two odd parents of a common factor q differ by a multiple of $2q$.
 1157 No novelty is claimed for that identity.
 1158 Suppose $N/2 < r < s < N$ are prime upper roots with positive odd complements.
 1159 If q divides both $N-r$ and $N-s$, then
 1160 $2q$ divides $s-r$, and hence $h_s = h_r + 2q = 2q + 1$.
 1161 Consequently, if $s - N/2 \leq 2 \cdot \min PF(N-s)$, its full initial support is
 1162 disjoint from that of EVERY earlier upper root $r > N/2$. A possible
 1163 boundary root $r = N/2$ prime causes no exception: its complement r is
 1164 larger than $N-s$, so their prime supports cannot meet.
 1165 The numerical factor 2 is sharp even for adjacent roots:
 1166 $N=1152, p_0=577, p_1=587,$
 1167 $N-p_0=575=5^2 \cdot 23, N-p_1=565=5 \cdot 113,$
 1168 $h_0=1, h_1=11=2 \cdot 5+1, p_1-p_0=10=2 \cdot 5.$
 1169 Another equality example is $N=80, p_0=41, p_2=47$ with common factor 3.
 1170
 1171 This explains a constraint in the H1 construction: sharing a prescribed
 1172 small factor forces later roots away from the centre by at least twice
 1173 that factor. More precisely, its fixed offsets are multiples of M , so
 1174 $h_j \geq 1+j \cdot M$. The construction does not put all these roots in the earlier
 1175 $h < \min PF(m)$ regime. Conversely, disjointness of initial supports does not
 1176 force success. The already certified failed roots $(N, r) = (1718, 877)$ and
 1177 $(6142, 3089)$ have $h=18 < \min PF(m)$, and therefore have the disjointness just
 1178 proved. Disjoint first supports can still feed closed BAD components.
 1179
 1180 EXACT FINITE WITNESSES AND AUDIT, RUN H
 1181 Source code is itself saved as a plain TXT:
 1182 `nowe/rank_prefix_probe_20260907H_code.txt`
 1183 Full output with factorization and FIFO certificates:
 1184 `nowe/rank_prefix_probe_20260907H_results.txt`
 1185 Method: complete integer smallest-prime-factor sieve through 2,000,000;
 1186 full sorted-factor FIFO BFS, and an independent greatest-closed-BAD
 1187 fixed point for $N=1862$. No probable-prime tests or delegated work.
 1188
 1189 Consecutive-prime strings were enumerated with $P_0 \leq 1,000,001$, equivalently
 1190 $N=2(P_0-1) \leq 2,000,000$, requiring every selected complement composite.
 1191 1. Residue 5 modulo 18, length 3: exactly 51 strings in this domain.
 1192 First witness: $N=40364; p_0=20183, p_1=20201, p_2=20219.$
 1193 $m_0=20181=3 \cdot 7 \cdot 31^2,$
 1194 $m_1=20163=3 \cdot 11 \cdot 13 \cdot 47,$
 1195 $m_2=20145=3 \cdot 5 \cdot 17 \cdot 79.$
 1196 All have $v_3=1$, h values 1, 19, 37, and all succeed at depth 1 via
 1197 $3 \cdot 40361=40364$. The three complete supports are different.
 1198 2. Residue 7 modulo 50, length 2: exactly 8 strings in this domain.
 1199 First witness: $N=63812; p_0=31907, p_1=31957.$
 1200 $m_0=31905=3^2 \cdot 5 \cdot 709, m_1=31855=5 \cdot 23 \cdot 277.$
 1201 Both have $v_5=1$. The p_0 run reaches GOOD at depth 1, $W=2$;
 1202 the p_1 run reaches GOOD at depth 2, $W=5$. Their FIFO lists are

1203 [31907,3] and [31957,5,23,277,3], respectively.
1204 Both end at the verified prime pair 3+63809=63812.
1205
1206 Opposite verdicts despite an identical prescribed valuation at the SAME N:
1207 N=1862,
1208 p_0=937, m_0=925=5²*37: SUCCESS, depth 2, W=6;
1209 p_118=1787, m_118=75=3*5²: FAIL, W=22.
1210 Both complements have v_5=2. The successful path is 937 -> 37 -> 73,
1211 using 1862-37=1825=5²*73 and 1862-73=1789 prime. The failing support
1212 {3,5} lies in the independently recomputed closed basin B_1862, whose
1213 complete factor rows are already in run G. The new TXT output also
1214 contains EVERY row visited in this failing 22-vertex run. Thus the
1215 different verdict does not rest on discarding an untested cofactor.
1216
1217 The elementary overlap inequality was additionally checked on the six
1218 known host values 128,1718,1862,1928,2200,6142: 98,367 upper-root pairs,
1219 36,058 common-prime incidences, and 7 composite roots satisfying the
1220 stated low-gap disjointness condition. All exact assertions passed.
1221 The pair domain is N/2<r<s<N-1, excluding unit complements and a possible
1222 boundary root. Boundary-root disjointness was covered in the proof.
1223 This is an illustration on six hosts, not a new all-N host census.
1224
1225 ARCHIVE CROSS-CHECK AND SCOPE
1226 A text search found no earlier Shiu/Freiberg application in this workspace.
1227 The old 19/40 exponent also occurs in an unrelated conditional abc
1228 calculation in the Biblia; that is not the H2 rank-separation theorem.
1229 The old flow notes already contain the 2q shared-child spacing identity;
1230 the low-gap statement above is explicitly only its present consumer.
1231 Neither external-source attribution nor a mathematical novelty claim
1232 has been inferred from the absence of a text-search hit.
1233
1234 The current Goldbug OEIS page was checked during source reconnaissance:
1235 <https://oeis.org/A306746>
1236 Its definition includes an additional condition max(label)>N/4 and its
1237 displayed sequence omits N=2200, unlike the six EAC hosts in this file.
1238 Consequently its reported next-term bound above 5*10⁸ has NOT been
1239 used to upgrade the all-EAC census or to certify any additional ranks.
1240
1241 LATEST STATE -- 2026-09-07H
1242 Ocena po tej rundzie:
1243 1. p_0 nie jest odizolowane przez dowolnie zadane, lecz skonczone dane
1244 o malych czynnikach pierwszych. Te same dane mozna zrealizowac przy
1245 jednym N dla p_0,...,p_K, dla kazdego ustalonego K, nieskonczenie
1246 wiele razy. Dotyczy to nawet dokladnych wykladnikow i projekcji
1247 grafu na ustalony skonczony zbior etykiet.
1248 2. Istnieje wspolna, udowodniona geometria wielu poczatkowych rang:
1249 ich pelne dopelnienia nie moga dwukrotnie korzystac wylacznie z tego
1250 samego ustalonego alfabetu pierwszego, gdy N jest dostatecznie duze.
1251 Mozna objac nawet prefiks do N[~](0.475-epsilon). Jest to wspolna
1252 wlasnosc arytmetyczna, NIE dowod wspolnej klasy rang zawsze udanych.
1253 3. Dla rzeczywistego wyjasnienia sukcesu potrzebne pozostaje powiazanie
1254 wyboru pierwszego korzenia z PELNYM, zmieniajacym sie wraz z N
1255 zbiorem BAD. Wlasnie tej zaleznosci nie kontroluja powyzsze twierdzenia.
1256 Nie wybraly one p_0 jako jedynego mozliwego wyjatku ani nie wykluczily
1257 porazek dla kandydackich rang 10,14,26,28,30,62,63,71,74.
1258 4. Najbardziej uzasadniony dalszy cel to ograniczenie jednoczesnych
1259 pelnych faktoryzacji w zmiennym B_N, wraz z rzeczywistym pozeniem
1260 pierwszego korzenia. Dalsze samo dopasowywanie skonczonych reszt,
1261 wykladnikow lub sztywnych alfabetow nie usuwa wykazanej przeszkody.
1262
1263 Run H used ZERO subagents. All new artifacts are TXT. Important proofs
1264 were saved at H1 and H2 before the finite experiments; complete current
1265 conclusions and reproducible examples are now saved here as well.
1266 No universal p_0 theorem or Goldbach proof has been obtained.
1267 END CURRENT CHECKPOINT -- 2026-09-07H

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